



Dashboard Summary for UnityCloud

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Affirmed Networks Inc.

A Microsoft Company

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AMF PROCEDURE DURATION STATISTICS

Description

This statistic shows the duration of the procedure in seconds

Panels

\$MmProcedure

Description

Mm Procedure Durations

Expression

{{procedure}}

```
sum by (procedure) (amfcc_mm_procedure_duration_count{procedure=~"$MmProcedure"})
```

\$MmProcedure | SUCCESS

Description

Duration of Successful MM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_mm_procedure_duration_bucket{procedure=~"$MmProcedure",status="SUCCESS"})
```

\$MmProcedure | FAILURE

Description

Duration of Failed MM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_mm_procedure_duration_bucket{procedure=~"$MmProcedure",status="FAILURE"})
```

\$HoProcedure

Description

HO Procedure Durations

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_ho_procedure_duration_count{procedure=~"$HoProcedure"})
```

\$HoProcedure | SUCCESS

Description

Duration of Successful HO procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_ho_procedure_duration_bucket{procedure=~"$HoProcedure",status="SUCCESS"})
```

\$HoProcedure | FAILURE

Description

Duration of failed HO procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_ho_procedure_duration_bucket{procedure=~"$HoProcedure",status="FAILURE"})
```

\$GnbProcedure

Description

Gnb Procedure Durations

Expression

{{procedure}}

```
sum by (procedure)
(amfcc_gnb_procedure_duration_count{procedure=~"$GnbProcedure"})
```

\$GnbProcedure | SUCCESS

Description

Duration of Successful Gnb procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_gnb_procedure_duration_bucket{procedure=~"$GnbProcedure",status="SUCCESS"})
```

\$GnbProcedure | FAILURE

Description

Duration of failed Gnb procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(amfcc_gnb_procedure_duration_bucket{procedure=~"$GnbProcedure",status="FAILURE"})
```

AMF-CC Dashboard

Description

This dashboard lists the metrics on pod AMF-CC

Panels

Initial Registration Procedure Success %

Description

This panel list the Initial Registration Procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_initial_registration_success{pod="$PodName"}))/(sum(amfcc_mm_initial_registration_attempt{pod="$PodName"})))*100
```

Inter AMF(Source) Initial Registration Procedure Success %

Description

This panel list the Inter AMF(Source) Initial Registration Procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_inter_old_amf_initial_registration_success{pod="$PodName"}))/(sum(amfcc_mm_inter_old_amf_initial_registration_attempt{pod="$PodName"})))*100
```

Inter AMF(Target) Initial Registration Procedure Success %

Description

This panel list the Inter AMF(Target) Initial Registration Procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_inter_new_amf_initial_registration_success{pod="$PodName"}))/(sum(amfcc_mm_inter_new_amf_initial_registration_attempt{pod="$PodName"})))*100
```

Initial Registration Procedure Stats

Description

This panel lists the initial registration procedure stats

Expression

amfcc_mm_initial_registration_attempt

```
sum(amfcc_mm_initial_registration_attempt{pod="$PodName"})
```

amfcc_mm_initial_registration_success

```
sum(amfcc_mm_initial_registration_success{pod="$PodName"})
```

amfcc_mm_initial_registration_failure

```
sum(amfcc_mm_initial_registration_failure{pod="$PodName"})
```

Inter AMF(Source) Initial Registration Procedure Stats

Description

This panel lists the Inter AMF(Source) Initial registration procedure stats

Expression

amfcc_mm_inter_old_amf_initial_registration_attempt

```
sum(amfcc_mm_inter_old_amf_initial_registration_attempt{pod="$PodName"})
```

amfcc_mm_inter_old_amf_initial_registration_success

```
sum(amfcc_mm_inter_old_amf_initial_registration_success{pod="$PodName"})
```

amfcc_mm_inter_old_amf_initial_registration_failure

```
sum(amfcc_mm_inter_old_amf_initial_registration_failure{pod="$PodName"})
```

Inter AMF(Target) Initial Registration Procedure Stats

Description

This panel lists the Inter AMF(Target) Initial registration procedure stats

Expression

amfcc_mm_inter_new_amf_initial_registration_attempt

```
sum(amfcc_mm_inter_new_amf_initial_registration_attempt{pod="$PodName"})
```

amfcc_mm_inter_new_amf_initial_registration_success

```
sum(amfcc_mm_inter_new_amf_initial_registration_success{pod="$PodName"})
```

amfcc_mm_inter_new_amf_initial_registration_failure

```
sum(amfcc_mm_inter_new_amf_initial_registration_failure{pod="$PodName"})
```

Intra Moblity Registration Procedure Success %

Description

This panel lists the mobility registration procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_intra_mobility_registration_success{pod="$PodName"}))/(sum(amfcc_mm_intra_mobility_registration_attempt{pod="$PodName"})))*100
```

Inter AMF(Source) Moblity Registration Procedure Success %

Description

This panel lists the Inter AMF(Source) mobility registration procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_inter_old_amf_mobility_registration_success{pod="$PodName"}))/ (sum(amfcc_mm_inter_old_amf_mobility_registration_attempt{pod="$PodName"}))) * 100
```

Inter AMF(Target) Moblity Registration Procedure Success %

Description

This panel lists the Inter AMF(Target) mobility registration procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_inter_new_amf_mobility_registration_success{pod="$PodName"}))/ (sum(amfcc_mm_inter_new_amf_mobility_registration_attempt{pod="$PodName"}))) * 100
```

Intra Mobility Registration Procedure Stats

Description

This panel lists the mobility registration procedure stats

Expression

amfcc_mm_intra_mobility_registration_attempt

```
sum(amfcc_mm_intra_mobility_registration_attempt{pod="$PodName"})
```

amfcc_mm_intra_mobility_registration_success

```
sum(amfcc_mm_intra_mobility_registration_success{pod="$PodName"})
```

amfcc_mm_intra_mobility_registration_failure

```
sum(amfcc_mm_intra_mobility_registration_failure{pod="$PodName"})
```

Inter AMF(Source) Mobility Registration Procedure Stats

Description

This panel lists the Inter AMF(Source) mobility registration procedure stats

Expression

amfcc_mm_inter_old_amf_mobility_registration_attempt

```
sum(amfcc_mm_inter_old_amf_mobility_registration_attempt{pod="$PodName"})
```

amfcc_mm_inter_old_amf_mobility_registration_success

```
sum(amfcc_mm_inter_old_amf_mobility_registration_success{pod="$PodName"})
```

amfcc_mm_inter_old_amf_mobility_registration_failure

```
sum(amfcc_mm_inter_old_amf_mobility_registration_failure{pod="$PodName"})
```

Inter AMF(Target) Mobility Registration Procedure Stats

Description

This panel lists the Inter AMF(Target) mobility registration procedure stats

Expression

amfcc_mm_inter_new_amf_mobility_registration_attempt

```
sum(amfcc_mm_inter_new_amf_mobility_registration_attempt{pod="$PodName"})
```

amfcc_mm_inter_new_amf_mobility_registration_success

```
sum(amfcc_mm_inter_new_amf_mobility_registration_success{pod="$PodName"})
```

amfcc_mm_inter_new_amf_mobility_registration_failure

```
sum(amfcc_mm_inter_new_amf_mobility_registration_failure{pod="$PodName"})
```

Periodic Registration Procedure Success %

Description

This panel lists the periodic registration procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_periodic_registration_success{pod="$PodName"}))/(sum(amfcc_mm_perio
dic_registration_attempt{pod="$PodName"})))*100
```

N26 AMF(Source) Moblity Registration Procedure Success %

Description

This panel lists the N26 AMF(Source) mobility registration procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_inter_eps_old_amf_mobility_registration_success{pod="$PodName"}))/(
sum(amfcc_mm_inter_eps_old_amf_mobility_registration_attempt{pod="$PodName"})))*10
0
```

N26 AMF(Target) Moblity Registration Procedure Success %

Description

This panel lists the N26 AMF(Target) mobility registration procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_inter_eps_new_amf_mobility_registration_success{pod="$PodName"}))/(
sum(amfcc_mm_inter_eps_new_amf_mobility_registration_attempt{pod="$PodName"})))*10
0
```

Periodic Registration Procedure Stats

Description

This panel lists the periodic registration procedure stats

Expression

amfcc_mm_periodic_registration_attempt

```
sum(amfcc_mm_periodic_registration_attempt{pod="$PodName"})
```

amfcc_mm_periodic_registration_success

```
sum(amfcc_mm_periodic_registration_success{pod="$PodName"})
```

amfcc_mm_periodic_registration_failure

```
sum(amfcc_mm_periodic_registration_failure{pod="$PodName"})
```

N26 AMF(Source) Mobility Registration Procedure Stats

Description

This panel lists the N26 AMF(Source) mobility registration procedure stats

Expression

amfcc_mm_inter_eps_old_amf_mobility_registration_attempt

```
sum(amfcc_mm_inter_eps_old_amf_mobility_registration_attempt{pod="$PodName"})
```

amfcc_mm_inter_eps_old_amf_mobility_registration_success

```
sum(amfcc_mm_inter_eps_old_amf_mobility_registration_success{pod="$PodName"})
```

amfcc_mm_inter_eps_old_amf_mobility_registration_failure

```
sum(amfcc_mm_inter_eps_old_amf_mobility_registration_failure{pod="$PodName"})
```


N26 AMF(Target) Mobility Registration Procedure Stats

Description

This panel lists the N26 AMF(Target) mobility registration procedure stats

Expression

amfcc_mm_inter_eps_new_amf_mobility_registration_attempt

```
sum(amfcc_mm_inter_eps_new_amf_mobility_registration_attempt{pod="$PodName"})
```

amfcc_mm_inter_eps_new_amf_mobility_registration_success

```
sum(amfcc_mm_inter_eps_new_amf_mobility_registration_success{pod="$PodName"})
```

amfcc_mm_inter_eps_new_amf_mobility_registration_failure

```
sum(amfcc_mm_inter_eps_new_amf_mobility_registration_failure{pod="$PodName"})
```

Auth Procedure Success %

Description

This panel lists the authentication procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_auth_success{pod="$PodName"}))/(sum(amfcc_mm_auth_attempt{pod="$PodName"})))*100
```

DeRegistration Procedure Success %

Description

This panel lists the deregistration procedure success rate

Expression

De Registration Success %

```
((sum(amfcc_n1_network_deregistration_accept{pod="$PodName"}))/(sum(amfcc_n1_ue_de
registration_request{pod="$PodName"})))*100
```

Service Request Procedure Success %

Description

This panel lists the service request procedure success rate

Expression

Registration Success %

```
((sum(amfcc_mm_service_request_success{pod="$PodName"}))/(sum(amfcc_mm_service_req
uest_attempt{pod="$PodName"})))*100
```

Auth Procedure Stats

Description

This panel lists the authentication procedure stats

Expression

amfcc_mm_auth_attempt

```
sum(amfcc_mm_auth_attempt{pod="$PodName"})
```

amfcc_mm_auth_success

```
sum(amfcc_mm_auth_success{pod="$PodName"})
```

amfcc_mm_auth_failure

```
sum(amfcc_mm_auth_failure{pod="$PodName"})
```

DeRegistration Procedure Stats

Description

This panel lists the initial deregistration procedure stats

Expression

amfcc_mm_ue_deregistration_attempt

```
sum(amfcc_mm_ue_deregistration_attempt{pod="$PodName"})
```

amfcc_mm_ue_deregistration_success

```
sum(amfcc_mm_ue_deregistration_success{pod="$PodName"})
```

amfcc_mm_ue_deregistration_failure

```
sum(amfcc_mm_ue_deregistration_failure{pod="$PodName"})
```

amfcc_mm_nwk_deregistration_attempt

```
sum(amfcc_mm_nwk_deregistration_attempt{pod="$PodName"})
```

Service Request/Paging Procedure Stats

Description

This panel lists the service request procedure stats

Expression

amfcc_service_request

```
sum(amfcc_mm_service_request_attempt{pod="$PodName"})
```

amfcc_service_request_success

```
sum(amfcc_mm_service_request_success{pod="$PodName"})
```

amfcc_service_request_failure

```
sum(amfcc_service_request_failure{pod="$PodName"})
```

amfcc_mm_paging_attempt-{{initiator}}

```
sum(amfcc_mm_paging_attempt{pod="$PodName"}) by (initiator)
```

amfcc_mm_paging_first_retry_attempt-{{initiator}}

```
sum(amfcc_mm_paging_first_retry_attempt{pod="$PodName"}) by (initiator)
```

amfcc_mm_paging_second_retry_attempt-{{initiator}}

```
sum(amfcc_mm_paging_second_retry_attempt{pod="$PodName"}) by (initiator)
```

amfcc_mm_paging_third_retry_attempt-{{initiator}}

```
sum(amfcc_mm_paging_third_retry_attempt{pod="$PodName"}) by (initiator)
```

amfcc_mm_paging_failure-{{initiator}}

```
sum(amfcc_mm_paging_failure{pod="$PodName"}) by (initiator)
```

Xn Handover Procedure Success %

Description

This panel lists the Xn handover procedure success rate

Expression

Registration Success %

```
((sum(amfcc_xn_handover_success{pod="$PodName"}))/sum(amfcc_xn_handover_attempt{pod="$PodName"}))*100
```

Intra AMF N2 Handover Procedure Success %

Description

This panel lists the Intra AMF N2 handover procedure success rate

Expression

Registration Success %

```
((sum(amfcc_n2_intra_handover_success{pod="$PodName"}))/sum(amfcc_n2_intra_handover_attempt{pod="$PodName"}))*100
```

N26 AMF(Source) Handover Procedure Success %

Description

This panel lists the N26 AMF(Source) handover procedure success rate

Expression

Registration Success %

```
((sum(amfcc_n26_inter_eps_source_amf_handover_success{pod="$PodName"}))/sum(amfcc_n26_inter_eps_source_amf_handover_attempt{pod="$PodName"}))*100
```

Xn Handover Procedure Stats

Description

This panel lists the Xn handover procedure stats

Expression

amfcc_xn_handover_registration_attempt

```
sum(amfcc_xn_handover_attempt{pod="$PodName"})
```

amfcc_xn_handover_success

```
sum(amfcc_xn_handover_success{pod="$PodName"})
```

amfcc_xn_handover_failure

```
sum(amfcc_xn_handover_failure{pod="$PodName"})
```

Intra AMF N2 Handover Procedure Stats**Description**

This panel lists the Intra AMF N2 handover procedure stats

Expression**amfcc_n2_intra_handover_attempt**

```
sum(amfcc_n2_intra_handover_attempt{pod="$PodName"})
```

amfcc_n2_intra_handover_success

```
sum(amfcc_n2_intra_handover_success{pod="$PodName"})
```

amfcc_n2_intra_handover_failure

```
sum(amfcc_n2_intra_handover_failure{pod="$PodName"})
```

amfcc_n2_intra_handover_cancel

```
sum(amfcc_n2_intra_handover_cancel{pod="$PodName"})
```

N26 AMF(Source) Handover Procedure Stats**Description**

This panel lists the N26 AMF(Source) handover procedure stats

Expression

amfcc_n26_inter_eps_source_amf_handover_attempt

```
sum(amfcc_n26_inter_eps_source_amf_handover_attempt{pod="$PodName"})
```

amfcc_n26_inter_eps_source_amf_handover_success

```
sum(amfcc_n26_inter_eps_source_amf_handover_success{pod="$PodName"})
```

amfcc_n26_inter_eps_source_amf_handover_failure

```
sum(amfcc_n26_inter_eps_source_amf_handover_failure{pod="$PodName"})
```

Inter AMF(Source) N2 Handover Procedure Success %

Description

This panel lists the Inter AMF(Source) N2 handover procedure success rate

Expression

Registration Success %

```
((sum(amfcc_n2_inter_handover_src_success{pod="$PodName"}))/(sum(amfcc_n2_inter_handover_src_attempt{pod="$PodName"})))*100
```

Inter AMF(Target) N2 Handover Procedure Success %

Description

This panel lists the Inter AMF(Target) N2 handover procedure success rate

Expression

Registration Success %

```
((sum(amfcc_n2_inter_target_amf_handover_success{pod="$PodName"}))/ (sum(amfcc_n2_inter_target_amf_handover_attempt{pod="$PodName"})))*100
```

N26 AMF(Target) Handover Procedure Success %

Description

This panel lists the N26 AMF(Target) handover procedure success rate

Expression

Registration Success %

```
((sum(amfcc_n26_inter_eps_target_amf_handover_success{pod="$PodName"}))/ (sum(amfcc_n26_inter_eps_target_amf_handover_attempt{pod="$PodName"})))*100
```

Inter AMF(Source) N2 Handover Procedure Stats

Description

This panel lists the Inter AMF(Source) N2 handover procedure stats

Expression

amfcc_n2_inter_source_amf_handover_attempt

```
sum(amfcc_n2_inter_source_amf_handover_attempt{pod="$PodName"})
```

amfcc_n2_inter_source_amf_handover_success

```
sum(amfcc_n2_inter_source_amf_handover_success{pod="$PodName"})
```

amfcc_n2_inter_source_amf_handover_failure

```
sum(amfcc_n2_inter_source_amf_handover_failure{pod="$PodName"})
```

Inter AMF(Target) N2 Handover Procedure Stats

Description

This panel lists the Inter AMF(Target) N2 handover procedure stats

Expression

amfcc_n2_inter_target_amf_handover_attempt

```
sum(amfcc_n2_inter_target_amf_handover_attempt{pod="$PodName"})
```

amfcc_n2_inter_target_amf_handover_success

```
sum(amfcc_n2_inter_target_amf_handover_success{pod="$PodName"})
```

amfcc_n2_inter_target_amf_handover_failure

```
sum(amfcc_n2_inter_target_amf_handover_failure{pod="$PodName"})
```

N26 AMF(Target) Handover Procedure Stats

Description

This panel lists the N26 AMF(Target) handover procedure stats

Expression

amfcc_n26_inter_eps_target_amf_handover_attempt

```
sum(amfcc_n26_inter_eps_target_amf_handover_attempt{pod="$PodName"})
```

amfcc_n26_inter_eps_target_amf_handover_success

```
sum(amfcc_n26_inter_eps_target_amf_handover_success{pod="$PodName"})
```

amfcc_n26_inter_eps_target_amf_handover_failure

```
sum(amfcc_n26_inter_eps_target_amf_handover_failure{pod="$PodName"})
```

gNB initiated Modify Procedure Success %

Description

This panel lists the gNB initiated Modify procedure success rate

Expression

Registration Success %

```
((sum(amfcc_gnb_init_modify_success{pod="$PodName"}))/(sum(amfcc_gnb_init_modify_attempt{pod="$PodName"})))*100
```

gNB initiated Modify Procedure Stats

Description

This panel lists the gNB initiated Modify procedure stats

Expression

amfcc_gnb_init_modify_attempt

```
sum(amfcc_gnb_init_modify_attempt{pod="$PodName"})
```

amfcc_gnb_init_modify_success

```
sum(amfcc_gnb_init_modify_success{pod="$PodName"})
```

amfcc_gnb_init_modify_failure

```
sum(amfcc_gnb_init_modify_failure{pod="$PodName"})
```

Initial Registration Message Stats

Description

This panel lists the initial registration message stats

Expression

initialRegistrationRequest

```
sum(amfcc_n1_init_registration_request{pod="$PodName"})
```

initialRegistrationAccept

```
sum(amfcc_n1_init_registration_accept{pod="$PodName"})
```

initialRegistrationComplete

```
sum(amfcc_n1_init_registration_complete{pod="$PodName"})
```

amfcc_n1_init_registration_reject

```
sum(amfcc_n1_init_registration_reject{pod="$PodName"})
```

Mobility Registration Message Stats**Description**

This panel lists the mobility registration message stats

Expression**mobilityRegistrationRequest**

```
sum(amfcc_n1_mobility_registration_request{pod="$PodName"})
```

mobilityRegistrationAccept

```
sum(amfcc_n1_mobility_registration_accept{pod="$PodName"})
```

mobilityRegistrationComplete

```
sum(amfcc_n1_mobility_registration_complete{pod="$PodName"})
```

amfcc_n1_mobility_registration_reject

```
sum(amfcc_n1_mobility_registration_reject{pod="$PodName"})
```

Periodic Registration Message Stats

Description

This panel lists the periodic registration message stats

Expression

periodicRegistrationRequest

```
sum(amfcc_n1_periodic_registration_request{pod="$PodName"})
```

periodicRegistrationAccept

```
sum(amfcc_n1_periodic_registration_accept{pod="$PodName"})
```

periodicRegistrationComplete

```
sum(amfcc_n1_periodic_registration_complete{pod="$PodName"})
```

amfcc_n1_periodic_registration_reject

```
sum(amfcc_n1_periodic_registration_reject{pod="$PodName"})
```

Service Request Message Stats

Description

This panel lists the service request message stats

Expression

serviceRequest

```
sum(amfcc_n1_service_request{pod="$PodName"})
```

serviceAccept

```
sum(amfcc_n1_service_accept{pod="$PodName"})
```

amfcc_n1_service_reject

```
sum(amfcc_n1_service_reject{pod="$PodName"})
```

amfcc_n1_configuration_update_command

```
sum(amfcc_n1_configuration_update_command{pod="$PodName"})
```

amfcc_n1_configuration_update_complete

```
sum(amfcc_n1_configuration_update_complete{pod="$PodName"})
```

Deregistration Message Stats

Description

This panel lists the deregistration message stats

Expression**deregistrationRequest**

```
sum(amfcc_n1_ue_deregistration_request{pod="$PodName"})
```

deregistrationAccept

```
sum(amfcc_n1_network_deregistration_accept{pod="$PodName"})
```

amfcc_n1_network_deregistration_request

```
sum(amfcc_n1_network_deregistration_request{pod="$PodName"})
```

amfcc_n1_ue_deregistration_accept

```
sum(amfcc_n1_ue_deregistration_accept{pod="$PodName"})
```

Authentication and Security Message Stats

Description

This panel lists the authentication and security message stats

Expression

authenticationRequest

```
sum(amfcc_n1_auth_request{pod="$PodName"})
```

authenticationResponse

```
sum(amfcc_n1_auth_response{pod="$PodName"})
```

securityModeCommand

```
sum(amfcc_n1_security_mode_command{pod="$PodName"})
```

securityModeComplete

```
sum(amfcc_n1_security_mode_complete{pod="$PodName"})
```

amfcc_n1_auth_failure

```
sum(amfcc_n1_auth_failure{pod="$PodName"})
```

amfcc_n1_auth_reject

```
sum(amfcc_n1_auth_reject{pod="$PodName"})
```

amfcc_n1_security_mode_reject

```
sum(amfcc_n1_security_mode_reject{pod="$PodName"})
```

Ausf Message Stats**Description**

This panel lists the AUSF message stats

Expression**ueAuthInfoReq**

```
sum(amfcc_nausf_ueau_request{pod="$PodName"})
```

ueAuthInfoResp-{{result}}

```
sum(amfcc_nausf_ueau_response{pod="$PodName"}) by (result)
```

ueAuthInfoConfirmReq

```
sum(amfcc_nausf_ueau_confirm_request{pod="$PodName"})
```

ueAuthInfoConfirmResp-{{result}}

```
sum(amfcc_nausf_ueau_confirm_response{pod="$PodName"}) by (result)
```

amfcc_nausf_ueau_eap_session_request

```
sum(amfcc_nausf_ueau_eap_session_request{pod="$PodName"})
```

amfcc_nausf_ueau_eap_session_response-{{result}}

```
sum(amfcc_nausf_ueau_eap_session_response{pod="$PodName"}) by (result)
```

UL/DL Nas Transport Message Stats

Description

This panel lists the uplink and downlink NAS message stats

Expression

downlinkNasTransport

```
sum(amfcc_n1_downlink_nas_transport{pod="$PodName"})
```

uplinkNasTransport

```
sum(amfcc_n1_uplink_nas_transport{pod="$PodName"})
```

UDM Message Stats

Description

This panel lists the UDM message stats

Expression

amfcc_nudm_sdm_get_request

```
amfcc_nudm_sdm_get_request{pod="$PodName"}
```

amfcc_nudm_sdm_get_response-{{result}}

```
sum(amfcc_nudm_sdm_get_response{pod="$PodName"}) by (result)
```


amfcc_nudm_uecm_registration_3gpp_request

```
amfcc_nudm_uecm_registration_3gpp_request{pod="$PodName"}
```

amfcc_nudm_uecm_registration_3gpp_response-{{result}}

```
sum(amfcc_nudm_uecm_registration_3gpp_response{pod="$PodName"}) by (result)
```

amfcc_nudm_uecm_deregistration_3gpp_request

```
sum(amfcc_nudm_uecm_deregistration_3gpp_request{pod="$PodName"})
```

amfcc_nudm_uecm_deregistration_3gpp_response-{{result}}

```
sum(amfcc_nudm_uecm_deregistration_3gpp_response{pod="$PodName"}) by (result)
```

amfcc_nudm_sdm_get_nssai_request

```
sum(amfcc_nudm_sdm_get_nssai_request{pod="$PodName"})
```

amfcc_nudm_sdm_get_nssai_response-{{result}}

```
sum(amfcc_nudm_sdm_get_nssai_response{pod="$PodName"}) by (result)
```

amfcc_nudm_sdm_subscription_request

```
sum(amfcc_nudm_sdm_subscription_request{pod="$PodName"})
```

amfcc_nudm_sdm_subscription_response-{{result}}

```
sum(amfcc_nudm_sdm_subscription_response{pod="$PodName"}) by (result)
```

amfcc_nudm_sdm_change_notification_request

```
sum(amfcc_nudm_sdm_change_notification_request{pod="$PodName"})
```

amfcc_nudm_sdm_change_notification_response-{{result}}

```
sum(amfcc_nudm_sdm_change_notification_response{pod="$PodName"}) by (result)
```

amfcc_nudm_uecm_deregistration_notification

```
sum(amfcc_nudm_uecm_deregistration_notification{pod="$PodName"})
```

NSSF Message Stats**Description**

This panel lists the NSSF message stats

Expression**amfcc_nssf_nsselection_request_for_registration**

```
sum(amfcc_nssf_nsselection_request_for_registration{pod="$PodName"})
```

amfcc_nssf_nsselection_response_for_registration-{{result}}

```
sum(amfcc_nssf_nsselection_response_for_registration{pod="$PodName"}) by (result)
```

amfcc_nssf_nsselection_request_for_pdu_session_establishment

```
sum(amfcc_nssf_nsselection_request_for_pdu_session_establishment{pod="$PodName"})
```

amfcc_nssf_nsselection_response_for_pdu_session_establishment-{{result}}

```
sum(amfcc_nnssf_nsselection_response_for_pdu_session_establishment{pod="$PodName"}  
  ) by (result)
```

Inter AMF Message Stats

Description

This panel lists the Inter AMF message stats

Expression

amfcc_client_namf_comm_ue_context_transfer_request

```
sum(amfcc_client_namf_comm_ue_context_transfer_request{pod="$PodName"})
```

amfcc_client_namf_comm_ue_context_transfer_response-{{result}}

```
sum(amfcc_client_namf_comm_ue_context_transfer_response{pod="$PodName"}) by  
(result)
```

amfcc_client_namf_comm_ue_registration_status_update_request

```
sum(amfcc_client_namf_comm_ue_registration_status_update_request{pod="$PodName"})
```

amfcc_client_namf_comm_ue_registration_status_update_response-{{result}}

```
sum(amfcc_client_namf_comm_ue_registration_status_update_response{pod="$PodName"})  
by (result)
```

amfcc_namf_comm_ue_context_transfer_request

```
sum(amfcc_namf_comm_ue_context_transfer_request{pod="$PodName"})
```

amfcc_namf_comm_ue_context_transfer_response-{{result}}

```
sum(amfcc_namf_comm_ue_context_transfer_response{pod="$PodName"}) by (result)
```

amfcc_namf_comm_ue_registration_status_update_request

```
sum(amfcc_namf_comm_ue_registration_status_update_request{pod="$PodName"})
```

amfcc_namf_comm_ue_registration_status_update_response-{{result}}

```
sum(amfcc_namf_comm_ue_registration_status_update_response{pod="$PodName"}) by  
(result)
```

amfcc_client_namf_comm_create_ue_context_request

```
sum(amfcc_client_namf_comm_create_ue_context_request{pod="$PodName"})
```

amfcc_client_namf_comm_create_ue_context_response-{{result}}

```
sum(amfcc_client_namf_comm_create_ue_context_response{pod="$PodName"}) by (result)
```

amfcc_client_namf_comm_release_ue_context_request

```
sum(amfcc_client_namf_comm_release_ue_context_request{pod="$PodName"})
```

amfcc_client_namf_comm_release_ue_context_response-{{result}}

```
sum(amfcc_client_namf_comm_release_ue_context_response{pod="$PodName"}) by  
(result)
```

amfcc_namf_comm_n2_info_notification

```
sum(amfcc_namf_comm_n2_info_notification{pod="$PodName"})
```

amfcc_namf_comm_n2_info_notification_response-{{result}}

```
sum(amfcc_namf_comm_n2_info_notification_response{pod="$PodName"}) by (result)
```

amfcc_client_namf_comm_n1n2_message_transfer

```
sum(amfcc_client_namf_comm_n1n2_message_transfer_request{pod="$PodName"})
```

amfcc_client_namf_comm_n1n2_message_transfer_response-{{result}}

```
sum(amfcc_client_namf_comm_n1n2_message_transfer_response{pod="$PodName"}) by  
(result)
```

amfcc_namf_comm_create_ue_context_request

```
sum(amfcc_namf_comm_create_ue_context_request{pod="$PodName"})
```

amfcc_namf_comm_create_ue_context_response-{{result}}

```
sum(amfcc_namf_comm_create_ue_context_response{pod="$PodName"}) by (result)
```

amfcc_namf_comm_release_ue_context_request

```
sum(amfcc_namf_comm_release_ue_context_request{pod="$PodName"})
```

amfcc_namf_comm_release_ue_context_response-{{result}}

```
sum(amfcc_namf_comm_release_ue_context_response{pod="$PodName"}) by (result)
```

amfcc_client_namf_comm_n2_info_notification

```
sum(amfcc_client_namf_comm_n2_info_notification{pod="$PodName"})
```

amfcc_client_namf_comm_n2_info_notification_response-{{result}}

```
sum(amfcc_client_namf_comm_n2_info_notification_response{pod="$PodName"}) by (result)
```

PCF Message Stats

Description

This panel lists the PCF message stats

Expression

amfcc_npcf_am_policy_control_create_request

```
sum(amfcc_npcf_am_policy_control_create_request{pod="$PodName"})
```

amfcc_npcf_am_policy_control_create_response-{{result}}

```
sum(amfcc_npcf_am_policy_control_create_response{pod="$PodName"}) by (result)
```

amfcc_npcf_am_policy_control_update_request

```
sum(amfcc_npcf_am_policy_control_update_request{pod="$PodName"})
```

amfcc_npcf_am_policy_control_update_response-{{result}}

```
sum(amfcc_npcf_am_policy_control_update_response{pod="$PodName"}) by (result)
```

amfcc_npcf_am_policy_control_delete_request

```
sum(amfcc_npcf_am_policy_control_delete_request{pod="$PodName"})
```

amfcc_npcf_am_policy_control_delete_response-{{result}}

```
sum(amfcc_npcf_am_policy_control_delete_response{pod="$PodName"}) by (result)
```

amfcc_npcf_am_policy_control_update_notify_request

```
sum(amfcc_npcf_am_policy_control_update_notify_request{pod="$PodName"})
```

amfcc_npcf_am_policy_control_update_notify_response-{{result}}

```
sum(amfcc_npcf_am_policy_control_update_notify_response{pod="$PodName"}) by  
(result)
```

amfcc_npcf_ue_policy_control_create_request

```
sum(amfcc_npcf_ue_policy_control_create_request{pod="$PodName"})
```

amfcc_npcf_ue_policy_control_create_response-{{result}}

```
sum(amfcc_npcf_ue_policy_control_create_response{pod="$PodName"}) by (result)
```

amfcc_npcf_ue_policy_control_update_request

```
sum(amfcc_npcf_ue_policy_control_update_request{pod="$PodName"})
```

amfcc_npcf_ue_policy_control_update_response-{{result}}

```
sum(amfcc_npcf_ue_policy_control_update_response{pod="$PodName"}) by (result)
```

amfcc_npcf_ue_policy_control_update_notify_request

```
sum(amfcc_npcf_ue_policy_control_update_notify_request{pod="$PodName"})
```

amfcc_npcf_ue_policy_control_update_notify_response-{{result}}

```
sum(amfcc_npcf_ue_policy_control_update_notify_response{pod="$PodName"}) by (result)
```

amfcc_npcf_ue_policy_control_delete_request

```
sum(amfcc_npcf_ue_policy_control_delete_request{pod="$PodName"})
```

amfcc_npcf_ue_policy_control_delete_response-{{result}}

```
sum(amfcc_npcf_ue_policy_control_delete_response{pod="$PodName"}) by (result)
```

SMF Selection Success %

Description

This panel lists the SMF selection success rate

Expression

SMF Selection Success %

```
((sum(amfcc_sm_smf_selection_success{pod="$PodName"}))/(sum(amfcc_sm_smf_selection_attempt{pod="$PodName"})))*100
```

SMF Selection Procedure Stats

Description

This panel lists the SMF selection procedure stats

Expression

amfcc_sm_smf_selection_attempt

```
(sum(amfcc_sm_smf_selection_attempt{pod="$PodName"}))
```

amfcc_sm_smf_selection_success


```
sum(amfcc_sm_smf_selection_success{pod="$PodName"})
```

amfcc_sm_smf_selection_failure-{{result}}

```
sum(amfcc_sm_smf_selection_failure{pod="$PodName"}) by (result)
```

PDU Session Create Message Stats

Description

This panel lists the PDU session create message stats

Expression

createSmContextRequest

```
sum(amfcc_nsmf_pdusession_create_sm_context_request{pod="$PodName"})
```

createSmContextResponse-{{result}}

```
sum(amfcc_nsmf_pdusession_create_sm_context_response{pod="$PodName"}) by (result)
```

N1N2 Transfer Message Stats

Description

This panel lists the N1N2 transfer message stats

Expression

n1n2MessageTransferRequest-{{msg_type}}

```
sum(amfcc_namf_comm_n1n2_message_transfer_request{pod="$PodName"}) by (msg_type)
```

n1n2MessageTransferResponse-{{msg_type}}-{{result}}

```
sum(amfcc_namf_comm_n1n2_message_transfer_response{pod="$PodName"}) by (msg_type, result)
```

PDU Session Update Message Stats

Description

This panel lists the PDU session update message stats

Expression

updateSmContextRequest-{{msg_type}}

```
sum(amfcc_nsmf_pdusession_update_sm_context_request{pod="$PodName"}) by  
(msg_type)
```

updateSmContextResponse-{{msg_type}}-{{result}}

```
sum(amfcc_nsmf_pdusession_update_sm_context_response{pod="$PodName"}) by  
(msg_type, result)
```

PDU Session Release Message Stats

Description

This panel lists the PDU session release message stats

Expression

releaseSmContextRequest

```
sum(amfcc_nsmf_pdusession_release_sm_context_request{pod="$PodName"})
```

releaseSmContextResponse-{{result}}

```
sum(amfcc_nsmf_pdusession_release_sm_context_response{pod="$PodName"}) by (result)
```

PDU Session Notify Message Stats

Description

This panel lists the PDU session notify message stats

Expression

releaseSmContextRequest

```
sum(amfcc_nsmf_pdusession_notify_sm_context_status_request_cb{pod="$PodName"})
```

releaseSmContextResponse-{{result}}

```
sum(amfcc_nsmf_pdusession_notify_sm_context_status_response{pod="$PodName"}) by  
(result)
```

PDU Session Retrieve Message Stats

Description

This panel lists the PDU session retrieve message stats

Expression

amfcc_nsmf_pdusession_retrieve_sm_context_request

```
sum(amfcc_nsmf_pdusession_retrieve_sm_context_request{pod="$PodName"})
```

amfcc_nsmf_pdusession_retrieve_sm_context_response-{{result}}

```
sum(amfcc_nsmf_pdusession_retrieve_sm_context_response{pod="$PodName"}) by  
(result)
```

LCS NI-LR Procedure Success %

Description

This panel lists the LCS NI-LR Procedure success rate

Expression

LCS NI-LR Procedure Success %

```
((sum(amfcc_lcs_network_induced_location_request_success{pod="$PodName"}))/(sum(am  
fcc_lcs_network_induced_location_request_attempt{pod="$PodName"})))*100
```

LCS MT-LR Procedure Success %

Description

This panel lists the LCS MT-LR Procedure success rate

Expression

LCS MT-LR Procedure Success %

```
((sum(amfcc_lcs_mobile_terminated_location_request_success{pod="$PodName"}))/(sum(amfcc_lcs_mobile_terminated_location_request_attempt{pod="$PodName"})))*100
```

LCS NI-LR Procedure Stats

Description

This panel lists the LCS network induced location request procedure stats

Expression

amfcc_lcs_network_induced_location_request_attempt

```
sum(amfcc_lcs_network_induced_location_request_attempt{pod="$PodName"})
```

amfcc_lcs_network_induced_location_request_success

```
sum(amfcc_lcs_network_induced_location_request_success{pod="$PodName"})
```

amfcc_lcs_network_induced_location_request_failure

```
sum(amfcc_lcs_network_induced_location_request_failure{pod="$PodName"})
```

LCS MT-LR Procedure Stats

Description

This panel lists the LCS mobile terminated location request procedure stats

Expression

amfcc_lcs_mobile_terminated_location_request_attempt

```
sum(amfcc_lcs_mobile_terminated_location_request_attempt{pod="$PodName"})
```

amfcc_lcs_mobile_terminated_location_request_success

```
sum(amfcc_lcs_mobile_terminated_location_request_success{pod="$PodName"})
```

amfcc_lcs_mobile_terminated_location_request_failure

```
sum(amfcc_lcs_mobile_terminated_location_request_failure{pod="$PodName"})
```

GMLC Message Stats

Description

This panel lists the GMLC message stats

Expression

gmlcEventNotifyReq

```
sum(amfcc_namf_location_event_notification{pod="$PodName"})
```

gmlcEventNotifyResp-{{result}}

```
sum(amfcc_namf_location_event_notification_response{pod="$PodName"}) by (result)
```

gmlcProvidePosInfoReq

```
sum(amfcc_namf_location_provide_positioning_info_request{pod="$PodName"})
```

gmlcProvidePosInfoResp-{{result}}

```
sum(amfcc_namf_location_provide_postioning_info_response{pod="$PodName"}) by (result)
```

LMF Message Stats

Description

This panel lists the LMF message stats

Expression

ImfDetermineLocReq

```
sum(amfcc_nlmf_location_determine_location_request{pod="$PodName"})
```

ImfDetermineLocResp-{{result}}

```
sum(amfcc_nlmf_location_determine_location_response{pod="$PodName"}) by (result)
```

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Summary

Dashboard Name	Panel Name
AMF Master Dashboard	Registered subscribers
AMF Master Dashboard	Number of Pods
AMF Master Dashboard	Number of connected gNB
AMF Master Dashboard	Number of active TNLA
AMF Master Dashboard	Registered subscribers - Connected
AMF Master Dashboard	Number of registered TAI
AMF Master Dashboard	Registered subscribers - Idle

Dashboard Name	Panel Name
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	gNB/Tnla Count
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	Initial Registration Attempt
AMF Master Dashboard	Initial Registration Success %
AMF Master Dashboard	Initial Registration Success
AMF Master Dashboard	Intra AMF Mobility Reg Attempt
AMF Master Dashboard	Intra AMF Mobility Reg Success %
AMF Master Dashboard	Intra AMF Mobility Reg Success
AMF Master Dashboard	Nodal MM Initial Registration Success %
AMF Master Dashboard	Network MM Initial Registration Success %
AMF Master Dashboard	Nodal Intra AMF Mobility Reg Success %
AMF Master Dashboard	Network Intra AMF Mobility Reg Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	Inter AMF(Source) Initial Reg Attempt
AMF Master Dashboard	Inter AMF(Source) Initial Reg Success %
AMF Master Dashboard	Inter AMF(Source) Initial Reg Success
AMF Master Dashboard	Inter AMF(Target) Initial Reg Attempt
AMF Master Dashboard	Inter AMF(Target) Initial Reg Success %
AMF Master Dashboard	Inter AMF(Target) Initial Reg Success
AMF Master Dashboard	Nodal Inter AMF(Source) Initial Reg Success %
AMF Master Dashboard	Network Inter AMF(Source) Initial Reg Success %
AMF Master Dashboard	Nodal Inter AMF(Target) Initial Reg Success %

Dashboard Name	Panel Name
AMF Master Dashboard	Network Inter AMF(Target) Initial Reg Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	Inter AMF(Source) Mobility Reg Attempt
AMF Master Dashboard	Inter AMF(Source) Mobility Reg Success %
AMF Master Dashboard	Inter AMF(Source) Mobility Reg Success
AMF Master Dashboard	Inter AMF(Target) Mobility Reg Attempt
AMF Master Dashboard	Inter AMF(Target) Mobility Reg Success %
AMF Master Dashboard	Inter AMF(Target) Mobility Reg Success
AMF Master Dashboard	Nodal Inter AMF(Source) Mobility Reg Success %
AMF Master Dashboard	Network Inter AMF(Source) Mobility Reg Success %
AMF Master Dashboard	Nodal Inter AMF(Target) Mobility Reg Success %
AMF Master Dashboard	Network Inter AMF(Target) Mobility Reg Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	N26(Source) Mobility Reg Attempt
AMF Master Dashboard	N26(Source) Mobility Reg Success %
AMF Master Dashboard	N26(Source) Mobility Reg Success
AMF Master Dashboard	N26(Target) Mobility Reg Attempt
AMF Master Dashboard	N26(Target) Mobility Reg Success %
AMF Master Dashboard	N26(Target) Mobility Reg Success
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	UE DeRegistration Attempt
AMF Master Dashboard	UE DeRegistration Success %
AMF Master Dashboard	UE DeRegistration Success
AMF Master Dashboard	Network Initiated DeReg Attempt

Dashboard Name	Panel Name
AMF Master Dashboard	Network Initiated DeReg Success %
AMF Master Dashboard	Network Initiated DeReg Success
AMF Master Dashboard	Nodal UE initiated DeRegistration Success %
AMF Master Dashboard	Network UE DeRegistration Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	Service Request Attempt
AMF Master Dashboard	Service Request Success %
AMF Master Dashboard	Service Request Success
AMF Master Dashboard	Periodic Registration Attempt
AMF Master Dashboard	Periodic Registration Success %
AMF Master Dashboard	Periodic Registration Success
AMF Master Dashboard	Nodal Service Request Success %
AMF Master Dashboard	Network Service Request Success %
AMF Master Dashboard	Nodal Periodic Registration Success %
AMF Master Dashboard	Network Periodic Registration Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	Xn Handover Attempt
AMF Master Dashboard	Xn Handover Success %
AMF Master Dashboard	Xn Handover Success
AMF Master Dashboard	Intra AMF N2 Handover Attempt
AMF Master Dashboard	Intra AMF N2 Handover Success %
AMF Master Dashboard	Intra AMF N2 Handover Success
AMF Master Dashboard	Nodal Xn Handover Success %
AMF Master Dashboard	Network Xn Handover Success %
AMF Master Dashboard	Nodal Intra AMF N2 Handover Success %

Dashboard Name	Panel Name
AMF Master Dashboard	Network Intra AMF N2 Handover Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	Inter AMF(Source) N2 HO Attempt
AMF Master Dashboard	Inter AMF(Source) N2 HO Success %
AMF Master Dashboard	Inter AMF(Source) N2 HO Success
AMF Master Dashboard	Inter AMF(Target) N2 HO Attempt
AMF Master Dashboard	Inter AMF(Target) N2 HO Success %
AMF Master Dashboard	Inter AMF(Target) N2 HO Success
AMF Master Dashboard	Nodal Inter AMF (Source) N2 Handover Success %
AMF Master Dashboard	Network Inter AMF (Source) N2 Handover Success %
AMF Master Dashboard	Nodal Inter AMF (Target) N2 Handover Success %
AMF Master Dashboard	Network Inter AMF (Target) N2 Handover Success %
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	N26 AMF(Source) HO Attempt
AMF Master Dashboard	N26 AMF(Source) HO Success %
AMF Master Dashboard	N26 AMF(Source) HO Success
AMF Master Dashboard	N26 AMF(Target) HO Attempt
AMF Master Dashboard	N26 AMF(Target) HO Success %
AMF Master Dashboard	N26 AMF(Target) HO Success
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	SMF Selection Attempt
AMF Master Dashboard	SMF Selection Success %
AMF Master Dashboard	SMF Selection Success

Dashboard Name	Panel Name
AMF Master Dashboard	PDU Resource Setup Attempt
AMF Master Dashboard	PDU Resource Setup Success %
AMF Master Dashboard	PDU Resource Setup Success
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	AN Release Attempt
AMF Master Dashboard	AN Release Success %
AMF Master Dashboard	AN Release Success
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	NI-LR Procedure Attempt
AMF Master Dashboard	NI-LR Procedure Success %
AMF Master Dashboard	NI-LR Procedure Success
AMF Master Dashboard	
AMF Master Dashboard	MT-LR Procedure Attempt
AMF Master Dashboard	MT-LR Procedure Success %
AMF Master Dashboard	MT-LR Procedure Success
AMF Master Dashboard	
AMF Master Dashboard	
AMF Master Dashboard	AMF Call Rate Per Second
AMF-MME Combo Master Dashboard	Subscriber and Radio Status
AMF-MME Combo Master Dashboard	Total Subscribers
AMF-MME Combo Master Dashboard	AMF-MME Subscribers
AMF-MME Combo Master Dashboard	
AMF-MME Combo Master Dashboard	Total Radios
AMF-MME Combo Master Dashboard	
AMF-MME Combo Master Dashboard	Total Subscribers - Connected

Dashboard Name	Panel Name
AMF-MME Combo Master Dashboard	Total Radios
AMF-MME Combo Master Dashboard	Total Subscribers - Idle
AMF-MME Combo Master Dashboard	Pod Status
AMF-MME Combo Master Dashboard	Running Pods
AMF-MME Combo Master Dashboard	TPS
AMF-MME Combo Master Dashboard	AMF TPS
AMF-MME Combo Master Dashboard	MME TPS
AMF-MME Combo Master Dashboard	KPIs
AMF-MME Combo Master Dashboard	AMF KPIs
AMF-MME Combo Master Dashboard	MME KPIs
MME Master Dashboard	Attached Subscribers
MME Master Dashboard	Number of Pods
MME Master Dashboard	Number of connected eNB
MME Master Dashboard	Attached Subscribers - Connected
MME Master Dashboard	Number of registered TAI
MME Master Dashboard	Attached Subscribers - Idle
MME Master Dashboard	S11 Active Path
MME Master Dashboard	S10 Active Path
MME Master Dashboard	S6 Active Paths
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	

Dashboard Name	Panel Name
MME Master Dashboard	
MME Master Dashboard	Attach Attempt Total Count
MME Master Dashboard	Attach Success %
MME Master Dashboard	Attach Success Total Count
MME Master Dashboard	Detach UE Attempt Total Count
MME Master Dashboard	Detach UE Success %
MME Master Dashboard	Detach UE Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	Service Request Attempt
MME Master Dashboard	Service Request Success %
MME Master Dashboard	Service Request Success
MME Master Dashboard	Intra-MME TAU Attempt Total Count
MME Master Dashboard	Intra-MME TAU Success %
MME Master Dashboard	Intra-MME TAU Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	Periodic TAU Attempt Total Count
MME Master Dashboard	Periodic TAU Success %
MME Master Dashboard	Periodic TAU Success Total Count
MME Master Dashboard	GUTI Reallocation Attempt Total Count
MME Master Dashboard	GUTI Reallocation Success %
MME Master Dashboard	GUTI Reallocation Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	Standalone Security Attempt Total Count
MME Master Dashboard	Standalone Security Success %
MME Master Dashboard	Standalone Security Success Total Count
MME Master Dashboard	Inter Old MME TAU Attempt Total Count
MME Master Dashboard	Inter Old MME TAU Success %

Dashboard Name	Panel Name
MME Master Dashboard	Inter Old MME TAU Success Total Count
MME Master Dashboard	
MME Master Dashboard	Inter New MME TAU Attempt Total Count
MME Master Dashboard	Inter New MME TAU Success %
MME Master Dashboard	Inter New MME TAU Success Total Count
MME Master Dashboard	
MME Master Dashboard	Inter RAT Old MME TAU Attempt Total Count
MME Master Dashboard	Inter RAT Old MME TAU Success %
MME Master Dashboard	Inter RAT Old MME TAU Success Total Count
MME Master Dashboard	
MME Master Dashboard	Inter RAT New MME TAU Attempt Total Count
MME Master Dashboard	Inter RAT New MME TAU Success %
MME Master Dashboard	Inter RAT New MME TAU Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	X2 Handover Attempt
MME Master Dashboard	X2 Handover Success %
MME Master Dashboard	X2 Handover Success
MME Master Dashboard	X2 Handover with SGW change Attempt
MME Master Dashboard	X2 Handover with SGW change Success %
MME Master Dashboard	X2 Handover Success
MME Master Dashboard	
MME Master Dashboard	S1 Handover Attempt
MME Master Dashboard	S1 Handover Success %
MME Master Dashboard	S1 Handover Success
MME Master Dashboard	
MME Master Dashboard	Inter Source MME S1 Handover Attempt
MME Master Dashboard	Inter Source MME S1 Handover Success %
MME Master Dashboard	Inter Source MME S1 Handover Success

Dashboard Name	Panel Name
MME Master Dashboard	
MME Master Dashboard	Inter Target MME S1 Handover Attempt
MME Master Dashboard	Inter Target MME S1 Handover Success %
MME Master Dashboard	Inter Target MME S1 Handover Success
MME Master Dashboard	
MME Master Dashboard	Inter RAT Source MME Handover Attempt
MME Master Dashboard	Inter RAT Source MME Handover Success %
MME Master Dashboard	Inter RAT Source MME Handover Success
MME Master Dashboard	
MME Master Dashboard	Inter RAT Target MME Handover Attempt
MME Master Dashboard	Inter RAT Target MME Handover Success %
MME Master Dashboard	Inter RAT Target MME Handover Success
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	Default Bearer Activation during Attach with request-type initial attempt Total Count
MME Master Dashboard	Default Bearer Activation during Attach with request-type initial Success %
MME Master Dashboard	Default Bearer Activation during Attach with request-type Initial Success Total Count
MME Master Dashboard	Activate Default Bearer PDN Connectivity with request-type initial request Attempt Total Count
MME Master Dashboard	Activate Default Bearer PDN Connectivity with request-type initial request Success %
MME Master Dashboard	Activate Default Bearer PDN Connectivity with request-type initial request Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	Bearer Resource Allocation Attempt Total Count
MME Master Dashboard	Bearer Resource Allocation Success %
MME Master Dashboard	Bearer Resource Allocation Success Total Count

Dashboard Name	Panel Name
MME Master Dashboard	UE initiated Dedicated Bearer Modification Attempt Total Count
MME Master Dashboard	UE initiated Dedicated Bearer Modification Success %
MME Master Dashboard	UE initiated Dedicated Bearer Modification Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	PGW initiated Default Bearer Modification Attempt Total Count
MME Master Dashboard	PGW Initiated Default Bearer Modification Success %
MME Master Dashboard	PGW Initiated Default Bearer Modification Success Total Count
MME Master Dashboard	PGW initiated Dedicated Bearer Creation Attempt Total Count
MME Master Dashboard	PGW Initiated Dedicated Bearer Creation Success %
MME Master Dashboard	PGW Initiated Dedicated Bearer Creation Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	UE Initiated Bearer Resource Modification Attempt Total Count
MME Master Dashboard	UE Initiated Bearer Resource Modification Success %
MME Master Dashboard	UE Initiated Bearer Resource Modification Success Total Count
MME Master Dashboard	
MME Master Dashboard	Default Bearer Modification Attempt Total Count
MME Master Dashboard	Default Bearer Modification Success %
MME Master Dashboard	Default Bearer Modification Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	SGW Relocation Attempt Total Count
MME Master Dashboard	SGW Relocation Success %
MME Master Dashboard	SGW Relocation Success Total Count
MME Master Dashboard	UE Initiated Dedicated EPS Bearer Deactivation Total Count
MME Master Dashboard	UE Initiated Dedicated EPS Bearer Deactivation Success %
MME Master Dashboard	UE Initiated Dedicated EPS Bearer Deactivation Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	GW Initiated Dedicated EPS Bearer Deactivation Attempt Total Count

Dashboard Name	Panel Name
MME Master Dashboard	GW Initiated Dedicated EPS Bearer Deactivation Success %
MME Master Dashboard	GW Initiated Dedicated EPS Bearer Deactivation Success Total Count
MME Master Dashboard	MME Initiated Dedicated EPS Bearer Deactivation Total Count
MME Master Dashboard	MME Initiated Dedicated EPS Bearer Deactivation Success %
MME Master Dashboard	MME Initiated Dedicated EPS Bearer Deactivation Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	UE Initiated PDN Disconnect Attempt Total Count
MME Master Dashboard	UE Initiated PDN Disconnect Success %
MME Master Dashboard	UE Initiated PDN Disconnect Success Total Count
MME Master Dashboard	GW Initiated PDN Disconnect Attempt Total Count
MME Master Dashboard	GW Initiated PDN Disconnect Success %
MME Master Dashboard	GW Initiated PDN Disconnect Success Total Count
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	
MME Master Dashboard	Intra MME Configuration Transfer Attempt Total Count
MME Master Dashboard	Intra MME Configuration Transfer Success %
MME Master Dashboard	Intra MME Configuration Transfer Success Total Count
MME Master Dashboard	Inter MME Configuration Transfer Attempt Total Count
MME Master Dashboard	Inter MME Configuration Transfer Success %
MME Master Dashboard	Inter MME Configuration Transfer Success Total Count
MME Master Dashboard	
MME Master Dashboard	DNS Request Rate Per Pod
MME Master Dashboard	DNS Success %
MME Master Dashboard	DNS Query Total Count
MME Master Dashboard	

AMF Master Dashboard

Description

This dashboard lists the common metrics on fed-amf

Panels

Registered subscribers

Description

This panel lists the registered subscribers stats

Expression

Registered subscribers

```
amf_registered_subscribers{namespace="$AMFnamespace"}
```

Number of Pods

Description

This panel lists the number of AMF specific pods in the fed-amf deployment

Expression

{{pod_type}}

```
ready_pods{namespace="$AMFnamespace", pod_type != "diameter-base-group1", pod_type  
!= "mme-cc", pod_type != "mme-ta", pod_type != "mme-s1" }
```

Number of connected gNB

Description

This panel lists the number of connected gNBs

Expression

```
amf_connected_gnb{namespace="$AMFnamespace"}
```

Number of active TNLA

Description

This panel lists the number of active TNLA

Expression

```
amf_active_tnla{namespace="$AMFnamespace"}
```

Registered subscribers - Connected

Description

This panel lists the number of registered subscribers in connected state

Expression

Registered subscribers

```
amf_registered_subscribers_connected{namespace="$AMFnamespace"}
```

Number of registered TAI

Description

This panel lists the number of registered TAI

Expression

```
amf_registered_tai{namespace="$AMFnamespace"}
```

Registered subscribers - Idle

Description

This panel lists the number of registered subscribers in idle state

Expression

Registered subscribers

```
amf_registered_subscribers_idle{namespace="$AMFnamespace"}
```

Description

Description

This panel lists the amf stats

Description

This panel lists the number of overall gNB stats

gNB/Tnla Count

Description

This panel lists the gNB/Tnla count

Expression

amf_connected_gnb

```
sum(amf_connected_gnb{namespace="$AMFnamespace"})
```

amf_active_tnla

```
sum(amf_active_tnla{namespace="$AMFnamespace"})
```

Description

Description

This panel lists the mobility procedure stats

Description

This panel lists the initial registration procedure stats

Description

This panel lists the Intra AMF Mobility registration procedure stats

Initial Registration Attempt

Description

This panel lists the initial registration procedure attempts

Expression

```
sum(amfcc_mm_initial_registration_attempt{namespace="$AMFnamespace"})
```

Initial Registration Success %

Description

This panel lists the initial registration procedure success rate

Expression

```
((sum(amfcc_mm_initial_registration_success{namespace="$AMFnamespace"}))/(sum(amfcc_mm_initial_registration_attempt{namespace="$AMFnamespace"})))*100
```

Initial Registration Success

Description

This panel lists the initial registration procedure success count

Expression

```
sum(amfcc_mm_initial_registration_success{namespace="$AMFnamespace"})
```

Intra AMF Mobility Reg Attempt

Description

This panel lists the Intra AMF mobility registration procedure attempts

Expression

```
sum(amfcc_mm_intra_mobility_registration_attempt{namespace="$AMFnamespace"})
```

Intra AMF Mobility Reg Success %

Description

This panel lists the Intra AMF mobility registration procedure success rate

Expression

```
(sum(amfcc_mm_intra_mobility_registration_success{namespace="$AMFnamespace"})/sum(
amfcc_mm_intra_mobility_registration_attempt{namespace="$AMFnamespace"}))*100
```

Intra AMF Mobility Reg Success

Description

This panel lists the Intra AMF mobility registration procedure success count

Expression

```
sum(amfcc_mm_intra_mobility_registration_success{namespace="$AMFnamespace"})
```

Nodal MM Initial Registration Success %

Description

This panel lists the Nodal MM initial registration procedure success rate

Expression

```
((sum(amfcc_mm_initial_registration_success{namespace="$AMFnamespace"})) /
(sum(amfcc_mm_initial_registration_success{namespace="$AMFnamespace"}) +
(sum(amfcc_mm_initial_registration_failure{source=~"amf|db",
namespace="$AMFnamespace"}) OR vector(0)))) * 100
```

Network MM Initial Registration Success %

Description

This panel lists the Network MM initial registration procedure success rate

Expression

```
((sum(amfcc_mm_initial_registration_success{namespace="$AMFnamespace"})) /
(sum(amfcc_mm_initial_registration_success{namespace="$AMFnamespace"}) +
(sum(amfcc_mm_initial_registration_failure{source=~"amf|db|nrf|udm|ausf|peerAmf|gn
b|smf|pcf|ue|peerMme", namespace="$AMFnamespace"}) OR vector(0)))) * 100
```

Nodal Intra AMF Mobility Reg Success %

Description

This panel lists the Nodal Intra AMF mobility registration procedure success rate

Expression

```
(sum(amfcc_mm_intra_mobility_registration_success{namespace="$AMFnamespace"})/
(sum(amfcc_mm_intra_mobility_registration_success{namespace="$AMFnamespace"}) +
(sum(amfcc_mm_intra_mobility_registration_failure{source=~"amf|db",
namespace="$AMFnamespace"}) or vector(0))))*100
```

Network Intra AMF Mobility Reg Success %

Description

This panel lists the Network Intra AMF mobility registration procedure success rate

Expression

```
(sum(amfcc_mm_intra_mobility_registration_success{namespace="$AMFnamespace"})/
(sum(amfcc_mm_intra_mobility_registration_success{namespace="$AMFnamespace"}) +
(sum(amfcc_mm_intra_mobility_registration_failure{source=~"amf|db|nrf|udm|ausf|peer
rAmf|smf|gnb|pcf|ue|peerMme", namespace="$AMFnamespace"}) or vector(0))))*100
```

Description

Description

This panel lists the Inter AMF(Source) Initial registration procedure stats

Description

This panel lists the Inter AMF(Target) Initial registration procedure stats

Inter AMF(Source) Initial Reg Attempt

Description

This panel lists the Inter AMF(Source) Initial registration procedure attempts

Expression

```
sum(amfcc_mm_inter_old_amf_initial_registration_attempt{namespace="$AMFnamespace"}
)
```

Inter AMF(Source) Initial Reg Success %

Description

This panel lists the Inter AMF(Source) Initial registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_old_amf_initial_registration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_inter_old_amf_initial_registration_attempt{namespace="$AMFnamespace"})))*100
```

Inter AMF(Source) Initial Reg Success

Description

This panel lists the Inter AMF(Source) Initial registration procedure success count

Expression

```
sum(amfcc_mm_inter_old_amf_initial_registration_success{namespace="$AMFnamespace"})
```

Inter AMF(Target) Initial Reg Attempt

Description

This panel lists the Inter AMF(Target) Initial registration procedure attempts

Expression

```
sum(amfcc_mm_inter_new_amf_initial_registration_attempt{namespace="$AMFnamespace"})
```

Inter AMF(Target) Initial Reg Success %

Description

This panel lists the Inter AMF(Target) Initial registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_new_amf_initial_registration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_inter_new_amf_initial_registration_attempt{namespace="$AMFnamespace"})))*100
```


Inter AMF(Target) Initial Reg Success

Description

This panel lists the Inter AMF(Target) Initial registration procedure success count

Expression

```
sum(amfcc_mm_inter_new_amf_initial_registration_success{namespace="$AMFnamespace"}
)
```

Nodal Inter AMF(Source) Initial Reg Success %

Description

This panel lists the Nodal Inter AMF(Source) Initial registration procedure success rate

Expression

```
(sum(amfcc_mm_inter_old_amf_initial_registration_success{namespace="$AMFnamespace"}
))/
(sum(amfcc_mm_inter_old_amf_initial_registration_success{namespace="$AMFnamespace"}
) + (sum(amfcc_mm_inter_old_amf_initial_registration_failure{source=~"amf|db",
namespace="$AMFnamespace"}) or vector(0))))*100
```

Network Inter AMF(Source) Initial Reg Success %

Description

This panel lists the Network Inter AMF(Source) Initial registration procedure success rate

Expression

```
(sum(amfcc_mm_inter_old_amf_initial_registration_success{namespace="$AMFnamespace"}
))/
(sum(amfcc_mm_inter_old_amf_initial_registration_success{namespace="$AMFnamespace"}
) +
(sum(amfcc_mm_inter_old_amf_initial_registration_failure{source=~"amf|db|nrf|udm|a
usf|peerAmf|smf|gnb|pcf|ue|peerMme", namespace="$AMFnamespace"}) or
vector(0))))*100
```

Nodal Inter AMF(Target) Initial Reg Success %

Description

This panel lists the Nodal Inter AMF(Target) Initial registration procedure success rate

Expression

```
(sum(amfcc_mm_inter_new_amf_initial_registration_success{namespace="$AMFnamespace"}))/
(sum(amfcc_mm_inter_new_amf_initial_registration_success{namespace="$AMFnamespace"}
) + (sum(amfcc_mm_inter_new_amf_initial_registration_failure{source=~"amf|db",
namespace="$AMFnamespace"}) or vector(0))))*100
```

Network Inter AMF(Target) Initial Reg Success %

Description

This panel lists the Network Inter AMF(Target) Initial registration procedure success rate

Expression

```
(sum(amfcc_mm_inter_new_amf_initial_registration_success{namespace="$AMFnamespace"}))/
(sum(amfcc_mm_inter_new_amf_initial_registration_success{namespace="$AMFnamespace"}
) +
(sum(amfcc_mm_inter_new_amf_initial_registration_failure{source=~"amf|db|nrf|udm|a
usf|peerAmf|smf|gnb|pcf|ue|peerMme", namespace="$AMFnamespace"}) or
vector(0))))*100
```

Description**Description**

This panel lists the Inter AMF(Source) Mobility registration procedure stats

Description

This panel lists the Inter AMF(Target) Mobility registration procedure stats

Inter AMF(Source) Mobility Reg Attempt

Description

This panel lists the Inter AMF(Source) Mobility registration procedure attempts

Expression

```
sum(amfcc_mm_inter_old_amf_mobility_registration_attempt{namespace="$AMFnamespace"})
```

Inter AMF(Source) Mobility Reg Success %

Description

This panel lists the Inter AMF(Source) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="$AMFnamespace"}))/sum(amfcc_mm_inter_old_amf_mobility_registration_attempt{namespace="$AMFnamespace"}))*100
```

Inter AMF(Source) Mobility Reg Success

Description

This panel lists the Inter AMF(Source) Mobility registration procedure success count

Expression

```
sum(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="$AMFnamespace"})
```

Inter AMF(Target) Mobility Reg Attempt

Description

This panel lists the Inter AMF(Target) Mobility registration procedure attempts

Expression

```
sum(amfcc_mm_inter_new_amf_mobility_registration_attempt{namespace="$AMFnamespace"})
```

Inter AMF(Target) Mobility Reg Success %

Description

This panel lists the Inter AMF(Target) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_new_amf_mobility_registration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_inter_new_amf_mobility_registration_attempt{namespace="$AMFnamespace"}))))*100
```

Inter AMF(Target) Mobility Reg Success

Description

This panel lists the Inter AMF(Target) Mobility registration procedure success count

Expression

```
sum(amfcc_mm_inter_new_amf_mobility_registration_success{namespace="$AMFnamespace"})
```

Nodal Inter AMF(Source) Mobility Reg Success %

Description

This panel lists the Nodal Inter AMF(Source) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="$AMFnamespace"})) + (sum(amfcc_mm_inter_old_amf_mobility_registration_failure{source=~"amf|db",namespace="$AMFnamespace"} OR vector(0))))) * 100
```

Network Inter AMF(Source) Mobility Reg Success %

Description

This panel lists the Network Inter AMF(Source) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="$AMFnamespace"})) + (sum(amfcc_mm_inter_old_amf_mobility_registration_failure{source=~"amf|db|nrf|udm|
```

```
ausf|peerAmf|gnb|smf|peerMme|pcf|ue", namespace="$AMFnamespace")) OR  
vector(0))) * 100
```

Nodal Inter AMF(Target) Mobility Reg Success %

Description

This panel lists the Nodal Inter AMF(Target) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_new_amf_mobility_registration_success{namespace="$AMFnamespac  
e"}))/((sum(amfcc_mm_inter_new_amf_mobility_registration_success{namespace="$AMFna  
mespace"})) +  
(sum(amfcc_mm_inter_new_amf_mobility_registration_failure{source=~"amf|db", namespa  
ce="$AMFnamespace"}) OR vector(0)))) * 100
```

Network Inter AMF(Target) Mobility Reg Success %

Description

This panel lists the Network Inter AMF(Target) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_new_amf_mobility_registration_success{namespace="$AMFnamespac  
e"}))/((sum(amfcc_mm_inter_new_amf_mobility_registration_success{namespace="$AMFna  
mespace"})) +  
(sum(amfcc_mm_inter_new_amf_mobility_registration_failure{source=~"amf|db|nrf|udm|  
ausf|peerAmf|smf|gnb|peerMme|pcf|ue", namespace="$AMFnamespace"}) OR  
vector(0)))) * 100
```

Description

Description

This panel lists the N26(Source) Mobility registration procedure stats

Description

This panel lists the N26(Target) Mobility registration procedure stats

N26(Source) Mobility Reg Attempt

Description

This panel lists the N26(Source) Mobility registration procedure attempts

Expression

```
sum(amfcc_mm_inter_eps_old_amf_mobility_registration_attempt{namespace="$AMFnamespace"})
```

N26(Source) Mobility Reg Success %

Description

This panel lists the N26(Source) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_inter_eps_old_amf_mobility_registration_success{namespace="$AMFnamespace"}))/(sum(amfcc_mm_inter_eps_old_amf_mobility_registration_attempt{namespace="$AMFnamespace"})))*100
```

N26(Source) Mobility Reg Success

Description

This panel lists the N26(Source) Mobility registration procedure success count

Expression

```
sum(amfcc_mm_inter_eps_old_amf_mobility_registration_success{namespace="$AMFnamespace"})
```

N26(Target) Mobility Reg Attempt

Description

This panel lists the N26(Target) Mobility registration procedure attempts

Expression

```
sum(amfcc_mm_n26_inter_eps_new_amf_mobility_registration_attempt{namespace="$AMFnamespace"})
```

N26(Target) Mobility Reg Success %

Description

This panel lists the N26(Target) Mobility registration procedure success rate

Expression

```
((sum(amfcc_mm_n26_inter_eps_new_amf_mobility_registration_success{namespace="$AMFnamespace"}))/sum(amfcc_mm_n26_inter_eps_new_amf_mobility_registration_attempt{namespace="$AMFnamespace"}))*100
```

N26(Target) Mobility Reg Success

Description

This panel lists the N26(Target) Mobility registration procedure success count

Expression

```
sum(amfcc_mm_n26_inter_eps_new_amf_mobility_registration_success{namespace="$AMFnamespace"})
```

Description**Description**

This panel lists the UE initiated deregistration procedure stats

Description

This panel lists the deregistration procedure stats

UE DeRegistration Attempt

Description

This panel lists the UE initiated deregistration procedure attempts

Expression

```
sum(amfcc_mm_ue_deregistration_attempt{namespace="$AMFnamespace"})
```

UE DeRegistration Success %

Description

This panel lists the UE initiated deregistration procedure success rate

Expression

```
((sum(amfcc_mm_ue_deregistration_success{namespace="$AMFnamespace"}))/(sum(amfcc_mm_ue_deregistration_attempt{namespace="$AMFnamespace"})))*100
```

UE DeRegistration Success

Description

This panel lists the UE initiated deregistration procedure success count

Expression

```
sum(amfcc_mm_ue_deregistration_success{namespace="$AMFnamespace"})
```

Network Initiated DeReg Attempt

Description

This panel lists the network initiated deregistration procedure attempts

Expression

```
sum(amfcc_mm_nwk_deregistration_attempt{namespace="$AMFnamespace"})
```

Network Initiated DeReg Success %

Description

This panel lists the Network initiated deregistration procedure success rate

Expression

```
((sum(amfcc_mm_nwk_deregistration_attempt{namespace="$AMFnamespace"}))/(sum(amfcc_mm_nwk_deregistration_attempt{namespace="$AMFnamespace"})))*100
```

Network Initiated DeReg Success

Description

This panel lists the network initiated deregistration procedure attempts

Expression

```
sum(amfcc_mm_nwk_deregistration_attempt{namespace="$AMFnamespace"})
```

Nodal UE initiated DeRegistration Success %

Description

This panel lists the Nodal UE initiated deregistration procedure success rate

Expression

```
((sum(amfcc_mm_ue_deregistration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_ue_deregistration_success{namespace="$AMFnamespace"}) + (sum(amfcc_mm_ue_deregistration_failure{source=~"amf|db",namespace="$AMFnamespace"}) OR vector(0)))*100
```

Network UE DeRegistration Success %

Description

This panel lists the Network UE initiated deregistration procedure success rate

Expression

```
((sum(amfcc_mm_ue_deregistration_success{namespace="$AMFnamespace"}))/((sum(amfcc_mm_ue_deregistration_success{namespace="$AMFnamespace"}) + (sum(amfcc_mm_ue_deregistration_failure{source=~"amf|db|nrf|udm|ausf|peerAmf|gnb|smf|peerMme|pcf|ue",namespace="$AMFnamespace"}) OR vector(0)))*100
```

Description

Description

This panel lists the service request procedure stats

Description

This panel lists the periodic registration procedure stats

Service Request Attempt

Description

This panel lists the service request procedure attempts

Expression

```
sum(amfcc_mm_service_request_attempt{namespace="$AMFnamespace"})
```

Service Request Success %

Description

This panel lists the service request procedure success rate

Expression

```
(sum(amfcc_mm_service_request_success{namespace="$AMFnamespace"})/sum(amfcc_mm_service_request_attempt{namespace="$AMFnamespace"}))*100
```

Service Request Success

Description

This panel lists the service request procedure success count

Expression

```
sum(amfcc_mm_service_request_success{namespace="$AMFnamespace"})
```

Periodic Registration Attempt

Description

This panel lists the periodic registration procedure attempts

Expression

```
sum(amfcc_mm_periodic_registration_attempt{namespace="$AMFnamespace"})
```

Periodic Registration Success %

Description

This panel lists the periodic registration procedure success rate

Expression

```
(sum(amfcc_mm_periodic_registration_success{namespace="$AMFnamespace"})/sum(amfcc_mm_periodic_registration_attempt{namespace="$AMFnamespace"}))*100
```

Periodic Registration Success

Description

This panel lists the periodic registration procedure success count

Expression

```
sum(amfcc_mm_periodic_registration_success{namespace="$AMFnamespace"})
```

Nodal Service Request Success %

Description

This panel lists the Nodal Service request procedure success rate

Expression

```
(sum(amfcc_mm_service_request_success{namespace="$AMFnamespace"})/((sum(amfcc_mm_service_request_success{namespace="$AMFnamespace"})) + (sum(amfcc_mm_service_request_failure{source=~"amf|db", namespace="$AMFnamespace"})or vector(0)))) * 100
```

Network Service Request Success %

Description

This panel lists the Network Service request procedure success rate

Expression

```
(sum(amfcc_mm_service_request_success{namespace="$AMFnamespace"})/((sum(amfcc_mm_service_request_success{namespace="$AMFnamespace"})) +
```

```
(sum(amfcc_mm_service_request_failure{source=~"amf|db|nrf|udm|ausf|peerAmf|smf|gnb|peerMme|pcf|ue", namespace="$AMFnamespace"})or vector(0))) * 100
```

Nodal Periodic Registration Success %

Description

This panel lists the Nodal periodic registration procedure success rate

Expression

```
(sum(amfcc_mm_periodic_registration_success{namespace="$AMFnamespace"})/
((sum(amfcc_mm_periodic_registration_success{namespace="$AMFnamespace"}) +
(sum(amfcc_mm_periodic_registration_failure{source=~"amf|db",
namespace="$AMFnamespace"})or vector(0)))) * 100
```

Network Periodic Registration Success %

Description

This panel lists the Network periodic registration procedure success rate

Expression

```
(sum(amfcc_mm_periodic_registration_success{namespace="$AMFnamespace"})/
((sum(amfcc_mm_periodic_registration_success{namespace="$AMFnamespace"}) +
(sum(amfcc_mm_periodic_registration_failure{source=~"amf|db|nrf|udm|ausf|peerAmf|smf|gnb|peerMme|pcf|ue", namespace="$AMFnamespace"})or vector(0)))) * 100
```

Description

Description

This panel lists the Handover procedure stats

Description

This panel lists the Xn handover procedure stats

Description

This panel lists the intra AMF n2 handover procedure stats

Xn Handover Attempt

Description

This panel lists the Xn handover procedure attempts

Expression

```
sum(amfcc_xn_handover_attempt{namespace="$AMFnamespace"})
```

Xn Handover Success %

Description

This panel lists the Xn handover procedure success rate

Expression

```
((sum(amfcc_xn_handover_success{namespace="$AMFnamespace"}))/(sum(amfcc_xn_handover_attempt{namespace="$AMFnamespace"})))*100
```

Xn Handover Success

Description

This panel lists the Xn handover procedure success count

Expression

```
sum(amfcc_xn_handover_success{namespace="$AMFnamespace"})
```

Intra AMF N2 Handover Attempt

Description

This panel lists the Intra AMF N2 handover procedure attempts

Expression

```
sum(amfcc_n2_intra_handover_attempt{namespace="$AMFnamespace"})
```

Intra AMF N2 Handover Success %

Description

This panel lists the intra AMF n2 handover procedure success rate

Expression

```
((sum(amfcc_n2_intra_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_intra_handover_attempt{namespace="$AMFnamespace"}))))*100
```

Intra AMF N2 Handover Success

Description

This panel lists the Intra AMF N2 handover procedure success count

Expression

```
sum(amfcc_n2_intra_handover_success{namespace="$AMFnamespace"})
```

Nodal Xn Handover Success %

Description

This panel lists the Nodal Xn Handover procedure success rate

Expression

```
((sum(amfcc_xn_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_xn_handover_success{namespace="$AMFnamespace"})) +  
(sum(amfcc_xn_handover_failure{source=~"amf|db",namespace="$AMFnamespace"}) OR  
vector(0))))*100
```

Network Xn Handover Success %

Description

This panel lists the Network Xn Handover procedure success rate

Expression

```
((sum(amfcc_xn_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_xn_handover_success{namespace="$AMFnamespace"})) +
```

```
((sum(amfcc_xn_handover_failure{source=~"amf|db|smf|udm|targetGnb|ue",namespace="$AMFnamespace"}) OR vector(0))))*100
```

Nodal Intra AMF N2 Handover Success %

Description

This panel lists the Nodal Intra AMF N2 Handover procedure success rate

Expression

```
((sum(amfcc_n2_intra_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_intra_handover_success{namespace="$AMFnamespace"}) + (sum(amfcc_n2_intra_handover_failure{source=~"amf|db",namespace="$AMFnamespace"}) OR vector(0)))))*100
```

Network Intra AMF N2 Handover Success %

Description

This panel lists the Network Intra AMF N2 Handover procedure success rate

Expression

```
((sum(amfcc_n2_intra_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_intra_handover_success{namespace="$AMFnamespace"}) + (sum(amfcc_n2_intra_handover_failure{source=~"amf|db|smf|udm|targetGnb|sourceGnb|ue",namespace="$AMFnamespace"}) OR vector(0)))))*100
```

Description

Description

This panel lists the Inter AMF(Source) N2 handover procedure stats

Description

This panel lists the Inter AMF(Target) N2 handover procedure stats

Inter AMF(Source) N2 HO Attempt

Description

This panel lists the Inter AMF(Source) N2 handover procedure attempts

Expression

```
sum(amfcc_n2_inter_source_amf_handover_attempt{namespace="$AMFnamespace"})
```

Inter AMF(Source) N2 HO Success %

Description

This panel lists the Inter AMF(Source) N2 handover procedure success rate

Expression

```
((sum(amfcc_n2_inter_source_amf_handover_success{namespace="$AMFnamespace"}))/(sum(amfcc_n2_inter_source_amf_handover_attempt{namespace="$AMFnamespace"})))*100
```

Inter AMF(Source) N2 HO Success

Description

This panel lists the Inter AMF(Source) N2 handover procedure success count

Expression

```
sum(amfcc_n2_inter_source_amf_handover_success{namespace="$AMFnamespace"})
```

Inter AMF(Target) N2 HO Attempt

Description

This panel lists the Inter AMF(Target) N2 handover procedure attempts

Expression

```
sum(amfcc_n2_inter_target_amf_handover_attempt{namespace="$AMFnamespace"})
```

Inter AMF(Target) N2 HO Success %

Description

This panel lists the Inter AMF(Target) N2 handover procedure success rate

Expression

```
((sum(amfcc_n2_inter_target_amf_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_inter_target_amf_handover_attempt{namespace="$AMFnamespace"})))*100
```

Inter AMF(Target) N2 HO Success

Description

This panel lists the Inter AMF(Target) N2 handover procedure success count

Expression

```
sum(amfcc_n2_inter_target_amf_handover_success{namespace="$AMFnamespace"})
```

Nodal Inter AMF (Source) N2 Handover Success %

Description

This panel lists the Nodal Inter Amf (Source) N2 Handover procedure success rate

Expression

```
((sum(amfcc_n2_inter_source_amf_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_inter_source_amf_handover_success{namespace="$AMFnamespace"}) + (sum(amfcc_n2_inter_source_amf_handover_failure{source=~"amf|db", namespace="$AMFnamespace"}) OR vector(0))))*100
```

Network Inter AMF (Source) N2 Handover Success %

Description

This panel lists the Network Inter AMF (Source) N2 Handover procedure success rate

Expression

```
((sum(amfcc_n2_inter_source_amf_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_inter_source_amf_handover_success{namespace="$AMFnamespace"}) + (sum(amfcc_n2_inter_source_amf_handover_failure{source=~"amf|db|smf|udm|sourceGnb|targetGnb|peerAmf|ue", namespace="$AMFnamespace"}) OR vector(0))))*100
```

Nodal Inter AMF (Target) N2 Handover Success %

Description

This panel lists the Nodal Inter AMF (Target) N2 Handover procedure success rate

Expression

```
((sum(amfcc_n2_inter_target_amf_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_inter_target_amf_handover_success{namespace="$AMFnamespace"}) + (sum(amfcc_n2_inter_target_amf_handover_failure{source=~"amf|db",namespace="$AMFnamespace"}) OR vector(0)))*100
```

Network Inter AMF (Target) N2 Handover Success %

Description

This panel lists the Network Inter AMF (Target) N2 Handover procedure success rate

Expression

```
((sum(amfcc_n2_inter_target_amf_handover_success{namespace="$AMFnamespace"}))/((sum(amfcc_n2_inter_target_amf_handover_success{namespace="$AMFnamespace"}) + (sum(amfcc_n2_inter_target_amf_handover_failure{source=~"amf|db|smf|udm|sourceGnb|targetGnb|peerAmf|ue",namespace="$AMFnamespace"}) OR vector(0)))*100
```

Description**Description**

This panel lists the N26 Source AMF handover procedure stats

Description

This panel lists the N26 target AMF handover procedure stats

N26 AMF(Source) HO Attempt

Description

This panel lists the N26 AMF(Source) handover procedure attempts

Expression

```
sum(amfcc_n26_inter_eps_source_amf_handover_attempt{namespace="$AMFnamespace"})
```

N26 AMF(Source) HO Success %

Description

This panel lists the N26 AMF(Source) handover procedure success rate

Expression

```
((sum(amfcc_n26_inter_eps_source_amf_handover_success{namespace="$AMFnamespace"}))  
/(sum(amfcc_n26_inter_eps_source_amf_handover_attempt{namespace="$AMFnamespace"}))  
)*100
```

N26 AMF(Source) HO Success

Description

This panel lists the N26 AMF(Source) handover procedure success count

Expression

```
sum(amfcc_n26_inter_eps_source_amf_handover_success{namespace="$AMFnamespace"})
```

N26 AMF(Target) HO Attempt

Description

This panel lists the N26 AMF(Target) handover procedure attempts

Expression

```
sum(amfcc_n26_inter_eps_target_amf_handover_attempt{namespace="$AMFnamespace"})
```

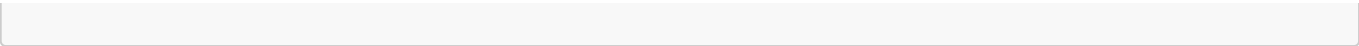
N26 AMF(Target) HO Success %

Description

This panel lists the N26 AMF(Target) handover procedure success rate

Expression

```
((sum(amfcc_n26_inter_eps_target_amf_handover_success{namespace="$AMFnamespace"}))  
/(sum(amfcc_n26_inter_eps_target_amf_handover_attempt{namespace="$AMFnamespace"}))  
)*100
```



N26 AMF(Target) HO Success

Description

This panel lists the N26 AMF(Target) handover procedure success count

Expression

```
sum(amfcc_n26_inter_eps_target_amf_handover_success{namespace="$AMFnamespace"})
```

Description

Description

This panel lists the PDU session procedure stats

Description

This procedure lists the SMF selections stats

Description

This procedure lists the SMF selections stats

SMF Selection Attempt

Description

This panel lists the SMF selection procedure attempts

Expression

```
sum(amfcc_sm_smf_selection_attempt{namespace="$AMFnamespace"})
```

SMF Selection Success %

Description

This panel lists the SMF selection procedure success rate

Expression

```
(sum(amfcc_sm_smf_selection_success{namespace="$AMFnamespace"})/sum(amfcc_sm_smf_selection_attempt{namespace="$AMFnamespace"}))*100
```

SMF Selection Success

Description

This panel lists the SMF selection procedure success count

Expression

```
sum(amfcc_sm_smf_selection_success{namespace="$AMFnamespace"})
```

PDU Resource Setup Attempt

Description

This panel lists the PDU Resource Setup Request procedure attempts

Expression

```
sum( amfn2_n2_pdu_session_resource_setup_request{namespace="$AMFnamespace"})
```

PDU Resource Setup Success %

Description

This panel lists the PDU Resource Setup procedure success rate

Expression

```
(sum(amfn2_n2_pdu_session_resource_setup_response{namespace="$AMFnamespace"})/sum(amfn2_n2_pdu_session_resource_setup_request{namespace="$AMFnamespace"}))*100
```

PDU Resource Setup Success

Description

This panel lists the PDU Resource Setup procedure success count

Expression

```
sum(amfn2_n2_pdu_session_resource_setup_response{namespace="$AMFnamespace"})
```

Description

Description

This panel lists the AN release procedure stats

AN Release Attempt

Description

This panel lists the AN release procedure attempts

Expression

```
sum(amfn2_n2_ue_context_release_command{namespace="$AMFnamespace"})
```

AN Release Success %

Description

This panel lists the AN release procedure success rate

Expression

```
(sum(amfn2_n2_ue_context_release_complete{namespace="$AMFnamespace"})/sum(amfn2_n2_ue_context_release_command{namespace="$AMFnamespace"}))*100
```

AN Release Success

Description

This panel lists the AN release procedure success count

Expression

```
sum(amfn2_n2_ue_context_release_complete{namespace="$AMFnamespace"})
```

Description

Description

This panel lists the LCS procedure stats

Description

This procedure lists the NI-LR Procedure Stats

Description

This procedure lists the MT-LR Procedure Stats

NI-LR Procedure Attempt**Description**

This panel lists the LCS NI-LR procedure attempts

Expression

```
sum(amfcc_lcs_network_induced_location_request_attempt{namespace="$AMFnamespace"})
```

NI-LR Procedure Success %**Description**

This panel lists the LCS NI-LR procedure success rate

Expression

```
(sum(amfcc_lcs_network_induced_location_request_success{namespace="$AMFnamespace"}  
)/sum(amfcc_lcs_network_induced_location_request_attempt{namespace="$AMFnamespace"}  
))*100
```

NI-LR Procedure Success**Description**

This panel lists the LCS NI-LR procedure success count

Expression

```
sum(amfcc_lcs_network_induced_location_request_success{namespace="$AMFnamespace"})
```

MT-LR Procedure Attempt

Description

This panel lists the LCS MT-LR procedure attempts

Expression

```
sum(amfcc_lcs_mobile_terminated_location_request_attempt{namespace="$AMFnamespace"})
```

MT-LR Procedure Success %

Description

This panel lists the LCS MT-LR procedure success rate

Expression

```
(sum(amfcc_lcs_mobile_terminated_location_request_success{namespace="$AMFnamespace"})/sum(amfcc_lcs_mobile_terminated_location_request_attempt{namespace="$AMFnamespace"}))*100
```

MT-LR Procedure Success

Description

This panel lists the LCS MT-LR procedure success count

Expression

```
sum(amfcc_lcs_mobile_terminated_location_request_success{namespace="$AMFnamespace"})
```

Description

Description

This panel lists the AMF Procedure Rate

AMF Call Rate Per Second

Description

This panel lists the amf call rate per sec

Expression

AMF Procedure Rate

```
(sum((rate(amfcc_mm_initial_registration_attempt{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfcc_mm_ue_deregistration_attempt{namespace="$AMFnamespace"}[5m])) or
vector(0)) + sum((rate(amfcc_sm_smf_selection_attempt{namespace="$AMFnamespace"}
[5m])) or vector(0)) + sum((rate(amfn2_n2_ue_context_release_command{}[5m])) or
vector(0)) +
sum((rate(amfcc_mm_service_request_attempt{namespace="$AMFnamespace"}[5m])) or
vector(0)) + sum((rate(amfcc_xn_handover_attempt{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfcc_mm_intra_mobility_registration_attempt{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_n2_intra_handover_attempt{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_mm_inter_old_amf_initial_registration_attempt{namespace="$AMFnames
pace"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_new_amf_initial_registration_attempt_attempt{namespace="$
AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_old_amf_mobility_registration_attempt{namespace="$AMFname
space"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_new_amf_mobility_registration_attempt{namespace="$AMFname
space"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_eps_old_amf_mobility_registration_attempt{namespace="$AMF
namespace"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_n26_inter_eps_new_amf_mobility_registration_attempt{namespace="
$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_periodic_registration_attempt{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfcc_n2_inter_source_amf_handover_attempt{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_n2_inter_target_amf_handover_attempt{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_n26_inter_eps_target_amf_handover_attempt{namespace="$AMFnamespace
"}[5m])) or vector(0)) +
sum((rate(amfcc_n26_inter_eps_source_amf_handover_attempt{namespace="$AMFnamespace
"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_paging_attempt{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_paging_first_retry_attempt{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_mm_paging_second_retry_attempt{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_mm_paging_third_retry_attempt{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_lcs_network_induced_location_request_attempt{namespace="$AMFnamesp
ace"}[5m])) or vector(0))) +
```

```
sum((rate(amfcc_lcs_mobile_terminated_location_request_attempt{namespace="$AMFname
space"}[5m])) or vector(0))
```

AMF http2 TPS

```
(sum((rate(amfcc_nsmf_pdusection_create_sm_context_response{namespace="$AMFnamespa
ce"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_n1n2_message_transfer_request{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusection_update_sm_context_response{namespace="$AMFnamespac
e"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusection_notify_sm_context_status_request_cb{namespace="$AM
Fnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusection_release_sm_context_response{namespace="$AMFnamespa
ce"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusection_retrieve_sm_context_response{namespace="$AMFnamesp
ace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_ebi_assignment_request{namespace="$AMFnamespace"}[5m]))
or vector(0)) + sum((rate(amfcc_nausf_ueau_response{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_nausf_ueau_confirm_response{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_nausf_ueau_eap_session_response{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfcc_nudm_uecm_registration_3gpp_response{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_nudm_uecm_deregistration_3gpp_response{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_nudm_sdm_get_response{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_nudm_sdm_get_nssai_response{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_nudm_sdm_subscription_response{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_nudm_sdm_change_notification_request{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_nnssf_nsselection_response_for_registration{namespace="$AMFnamespa
ce"}[5m])) or vector(0)) +
sum((rate(amfcc_nnssf_nsselection_response_for_pdu_session_establishment{namespace
="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_client_namf_comm_ue_context_transfer_response{namespace="$AMFnames
pace"}[5m])) or vector(0)) +
sum((rate(amfcc_client_namf_comm_ue_registration_status_update_response{namespace=
"$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_ue_context_transfer_request{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_ue_registration_status_update_request{namespace="$AMFnam
espace"}[5m])) or vector(0)) +
sum((rate(amfcc_client_namf_comm_create_ue_context_request{namespace="$AMFnamespac
```

```

e"}[5m])) or vector(0)) +
sum((rate(amfcc_client_namf_comm_release_ue_context_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_nudm_uecm_deregistration_notification{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_n2_info_notification{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_client_namf_comm_n1n2_message_transfer_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_create_ue_context_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_comm_release_ue_context_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_client_namf_comm_n2_info_notification{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_location_event_notification_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_namf_location_provide_positioning_info_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_nlmf_location_determine_location_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_am_policy_control_create_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_am_policy_control_update_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_am_policy_control_delete_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_am_policy_control_update_notify_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_ue_policy_control_create_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_ue_policy_control_update_response{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_ue_policy_control_update_notify_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_npcf_ue_policy_control_delete_response{namespace="$AMFnamespace"}[5m])) or vector(0)))

```

AMF n1/n2 TPS

```

(sum((rate(amfcc_n1_init_registration_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_init_registration_complete{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_mobility_registration_request{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_mobility_registration_complete{namespace="$AMFnamespace"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_periodic_registration_request{namespace="$AMFnamespace"}[5m]))

```

```

or vector(0)) +
sum((rate(amfcc_n1_periodic_registration_complete{namespace="$AMFnamespace"}[5m]))
or vector(0)) + sum((rate(amfcc_n1_auth_response{namespace="$AMFnamespace"}[5m]))
or vector(0)) + sum((rate(amfcc_n1_auth_failure{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfcc_n1_security_mode_complete{namespace="$AMFnamespace"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_identity_response{namespace="$AMFnamespace"}[5m]))
or vector(0)) + sum((rate(amfcc_n1_uplink_nas_transport{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfcc_n1_ue_deregistration_request{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfcc_n1_ue_deregistration_accept{namespace="$AMFnamespace"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_service_request{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfcc_n1_configuration_update_complete{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfn2_n2_initial_context_setup_request{namespace="$AMFnamespace"}[5m]))
or vector(0)) +
sum((rate(amfn2_n2_ue_context_release_request{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfn2_n2_ue_context_release_complete{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfn2_n2_ue_context_modification_request{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_setup_response{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_modify_response{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_release_response{namespace="$AMFnamespace"}
}[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_notify{namespace="$AMFnamespace"}[5m])) or
vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_modify_indication{namespace="$AMFnamespace"}
}[5m])) or vector(0)) +
sum((rate(amfn2_n2_handover_required{namespace="$AMFnamespace"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_handover_request_ack{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_handover_failure{namespace="$AMFnamespace"}[5m])) or vector(0))
+ sum((rate(amfn2_n2_handover_notify{namespace="$AMFnamespace"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_handover_cancel{namespace="$AMFnamespace"}[5m]))
or vector(0)) + sum((rate(amfn2_n2_pathswitch_request{namespace="$AMFnamespace"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_uplink_ran_status_transfer{namespace="$AMFnamespace"}[5m])) or
vector(0)))

```

AMF-MME Combo Master Dashboard

Description

Number of all the metrics that can be used to assess the health of the AMF-MME Combo federation.

Panels

Subscriber and Radio Status

Total Subscribers

Description

Number of all the Subscribers in connected state

Expression

Registered Subscribers

```
sum(amf_registered_subscribers{namespace="fed-amf-mme"} or vector(0)) +  
sum(mme_attached_subscribers{namespace="fed-amf-mme"} or vector(0))
```

AMF-MME Subscribers

Description

Number of subscribers in connected or idle state

Expression

4G Connected

```
sum(mme_attached_subscribers_connected{namespace="fed-amf-mme"} or vector(0))
```

4G Idle

```
sum(mme_attached_subscribers_idle{namespace="fed-amf-mme"} or vector(0))
```

5G Connected

```
sum(amf_registered_subscribers_connected{namespace="fed-amf-mme"} or vector(0))
```

5G Idle

```
sum(amf_registered_subscribers_idle{namespace="fed-amf-mme"} or vector(0))
```

Total Radios

Description

Number of connected eNB

Expression

```
(sum(mme_connected_enb{namespace="fed-amf-mme"}) or vector(0)) +  
(sum(amf_connected_gnb{namespace="fed-amf-mme"}) or vector(0))
```

Total Subscribers - Connected

Description

Number of subscribers in connected state

Expression

Attached Subscribers

```
(sum(mme_attached_subscribers_connected{namespace="fed-amf-mme"}) or vector(0)) +  
(sum(amf_registered_subscribers_connected{namespace="fed-amf-mme"}) or vector(0))
```

Total Radios

Description

Number of connected radios.

Expression

eNB

```
sum(mme_connected_enb{namespace="fed-amf-mme"})
```

gNB

```
sum(amf_connected_gnb{namespace="fed-amf-mme"})
```

N3IWF Non-3GPP RAN

```
sum(amf_connected_n3iwf{namespace="fed-amf-mme"})
```

Total Subscribers - Idle

Description

Number of subscribers in idle state

Expression

Attached Subscribers

```
(sum(mme_attached_subscribers_idle{namespace="fed-amf-mme"}) or vector(0)) +  
(sum(amf_registered_subscribers_idle{namespace="fed-amf-mme"}) or vector(0))
```

Pod Status

Running Pods

Description

Number of all the running pods in the fed-amf-mme deployment

Expression

{{pod_type}}

```
sum by (pod_type) (ready_pods{namespace="fed-amf-mme"})
```

TPS

AMF TPS

Expression

AMF N8 TPS

```
(sum((rate(amfcc_nudm_uecm_registration_3gpp_request{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_nudm_uecm_registration_3gpp_response{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_nudm_uecm_registration_update_3gpp_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nudm_uecm_registration_update_3gpp_response{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nudm_uecm_deregistration_notification{namespace="fed-amf-mme"}
[5m])) or vector(0)) + sum((rate(amfcc_nudm_sdm_get_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) + sum((rate(amfcc_nudm_sdm_get_response{namespace="fed-
amf-mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nudm_sdm_get_nssai_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_nudm_sdm_get_nssai_response{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_nudm_sdm_subscription_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_nudm_sdm_subscription_response{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nudm_sdm_change_notification_request{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_nudm_sdm_change_notification_response{namespace="fed-amf-mme"}
[5m])) or vector(0)) + sum((rate(amfcc_nudm_sdm_info_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) + sum((rate(amfcc_nudm_sdm_info_response{namespace="fed-
amf-mme"}[5m])) or vector(0)))
```

AMF N1/N2 TPS

```
(sum((rate(amfcc_n1_init_registration_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_init_registration_complete{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_mobility_registration_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_mobility_registration_complete{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_periodic_registration_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_periodic_registration_complete{namespace="fed-amf-
mme"}[5m])) or vector(0)) + sum((rate(amfcc_n1_auth_response{namespace="fed-amf-
mme"}[5m])) or vector(0)) + sum((rate(amfcc_n1_auth_failure{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_n1_security_mode_complete{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_identity_response{namespace="fed-amf-mme"}[5m]))
or vector(0)) + sum((rate(amfcc_n1_uplink_nas_transport{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_n1_ue_deregistration_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_n1_ue_deregistration_accept{namespace="fed-amf-mme"}
[5m])) or vector(0)) + sum((rate(amfcc_n1_service_request{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_n1_configuration_update_complete{namespace="fed-amf-mme"}[5m])) or
```



```

vector(0)) + sum((rate(amfn2_n2_initial_context_setup_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfn2_n2_ue_context_release_request{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_ue_context_release_complete{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfn2_n2_ue_context_modification_request{namespace="fed-amf-mme"}[5m]))
or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_setup_response{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_modify_response{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_release_response{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_notify{namespace="fed-amf-mme"}[5m])) or
vector(0)) +
sum((rate(amfn2_n2_pdu_session_resource_modify_indication{namespace="fed-amf-mme"}
[5m])) or vector(0)) + sum((rate(amfn2_n2_handover_required{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfn2_n2_handover_request_ack{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_handover_failure{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_handover_notify{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_handover_cancel{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfn2_n2_pathswitch_request{namespace="fed-amf-mme"}[5m]))
or vector(0)) + sum((rate(amfn2_n2_uplink_ran_status_transfer{namespace="fed-amf-
mme"}[5m])) or vector(0)))

```

AMF N11 TPS

```

(sum((rate(amfcc_nsmf_pdusession_create_sm_context_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_create_sm_context_response{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_update_sm_context_request{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_update_sm_context_response{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_notify_sm_context_status_request_cb{namespace="fed
-amf-mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_notify_sm_context_status_response{namespace="fed-
amf-mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_release_sm_context_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_release_sm_context_response{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_retrieve_sm_context_request{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_nsmf_pdusession_retrieve_sm_context_response{namespace="fed-amf-
mme"}[5m])) or vector(0)))

```

AMF N12

```
(sum((rate(amfcc_nausf_ueau_request{namespace="fed-amf-mme"}[5m])) or vector(0)) +  
sum((rate(amfcc_nausf_ueau_response{namespace="fed-amf-mme"}[5m])) or vector(0)) +  
sum((rate(amfcc_nausf_ueau_confirm_request{namespace="fed-amf-mme"}[5m])) or  
vector(0)) + sum((rate(amfcc_nausf_ueau_confirm_response{namespace="fed-amf-mme"}  
[5m])) or vector(0)) +  
sum((rate(amfcc_nausf_ueau_eap_session_request{namespace="fed-amf-mme"}[5m])) or  
vector(0)) + sum((rate(amfcc_nausf_ueau_eap_session_response{namespace="fed-amf-  
mme"}[5m])) or vector(0)))
```

MME TPS**Expression****S1 Rx TPS**

```
(sum((rate(mmes1_s1_incoming_message_total{namespace="fed-amf-mme"}[5m])) or  
vector(0)))
```

S1 Tx TPS

```
(sum((rate(mmes1_s1_outgoing_message_total{namespace="fed-amf-mme"}[5m])) or  
vector(0)))
```

S11 Rx TPS

```
(sum((rate(gtpc_agent_in_packet_total{namespace="fed-amf-mme", interface="S11-  
MME"}[5m])) or vector(0)))
```

S11 Tx TPS

```
(sum((rate(gtpc_agent_out_packet_total{namespace="fed-amf-mme", interface="S11-  
MME"}[5m])) or vector(0)))
```

S6 Rx TPS

```
(sum((rate(diameter_peer_response_received_total{namespace="fed-amf-mme"}[5m])) or vector(0)) + sum((rate(diameter_peer_request_received_total{namespace="fed-amf-mme"}[5m])) or vector(0)))
```

S6 Tx TPS

```
(sum((rate(diameter_peer_request_sent_total{namespace="fed-amf-mme"}[5m])) or vector(0)) + sum((rate(diameter_peer_response_sent_total{namespace="fed-amf-mme"}[5m])) or vector(0)))
```

S10 Rx TPS

```
(sum((rate(gtpc_agent_in_packet_total{namespace="fed-amf-mme", interface="S10"}[5m])) or vector(0)))
```

S10 Tx TPS

```
(sum((rate(gtpc_agent_out_packet_total{namespace="fed-amf-mme", interface="S10"}[5m])) or vector(0)))
```

N26 Rx TPS

```
(sum((rate(gtpc_agent_in_packet_total{namespace="fed-amf-mme", interface="N26"}[5m])) or vector(0)))
```

N26 Tx TPS

```
(sum((rate(gtpc_agent_out_packet_total{namespace="fed-amf-mme", interface="N26"}[5m])) or vector(0)))
```

KPIs**AMF KPIs****Expression**

Registration

```
(sum((rate(amfcc_mm_initial_registration_success{namespace="fed-amf-mme"}[5m])) or
vector(0)) +
sum((rate(amfcc_mm_intra_mobility_registration_success{namespace="fed-amf-mme"}
[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_old_amf_initial_registration_success{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_old_amf_mobility_registration_success{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_inter_new_amf_initial_registration_success{namespace="fed-amf-
mme"}[5m])) or vector(0)) + sum((rate(
```

```
amfcc_mm_inter_new_amf_mobility_registration_success{namespace="fed-amf-mme"}[5m])) or vector(0)) +
sum((rate( amfcc_mm_periodic_registration_success{namespace="fed-amf-mme"}[5m])) or vector(0)))
```

N26(Target) Mobility Registration

```
sum((rate(amfcc_mm_inter_eps_new_amf_mobility_registration_success{namespace="fed-
amf-mme"}[5m])) or vector(0))
```

N26(Source) Mobility Registration

```
sum((rate(amfcc_mm_inter_eps_old_amf_mobility_registration_success{namespace="fed-
amf-mme"}[5m])) or vector(0))
```

Paging

```
(sum((rate(amfcc_mm_paging_attempt{namespace="fed-amf-mme"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_paging_first_retry_attempt{namespace="fed-amf-mme"}[5m])) or
vector(0)) + sum((rate(amfcc_mm_paging_second_retry_attempt{namespace="fed-amf-
mme"}[5m])) or vector(0)) +
sum((rate(amfcc_mm_paging_third_retry_attempt{namespace="fed-amf-mme"}[5m])) or
vector(0)))
```

MME KPIs**Expression****Attach**

```
(sum(rate(mmecc_emm_attach_eps_success{namespace="fed-amf-mme"}[1m])) or vector(0))
```

Paging

```
(sum(rate(mmecc_emm_eps_paging_attempt{namespace="fed-amf-mme"}[1m])) or vector(0))
```

MME Master Dashboard

Description

This dashboard contains metrics that can be used to assess the health of the MME federation.

Panels

Attached Subscribers

Description

This panel lists the attached subscribers stats

Expression

Attached Subscribers

```
sum(mme_attached_subscribers{namespace="$MMEnamespace"}) or vector(0)
```

Number of Pods

Description

This panel lists the number of MME specific pods in the fed-mme deployment

Expression

{{pod_type}}

```
ready_pods{namespace="$MMEnamespace", pod_type != "amf-ta", pod_type != "amf-n2",  
pod_type != "amf-cc", pod_type != "nrf-agent", pod_type != "amf-n2"}
```

Number of connected eNB

Description

This panel lists the number of connected eNB

Expression

```
sum(mme_connected_enb{namespace="$MMENamespace"}) or vector(0)
```

Attached Subscribers - Connected

Description

This panel lists the number of attached subscribers in connected state

Expression

Attached Subscribers

```
sum(mme_attached_subscribers_connected{namespace="$MMENamespace"}) or vector(0)
```

Number of registered TAI

Description

This panel lists the number of registered TAI

Expression

```
sum(mme_registered_tai{namespace="$MMENamespace"}) or vector(0)
```

Attached Subscribers - Idle

Description

This panel lists the number of attached subscribers in idle state

Expression

Attached Subscribers

```
sum(mme_attached_subscribers_idle{namespace="$MMENamespace"}) or vector(0)
```

S11 Active Path

Description

This panel lists the number of S11 Path present

Expression

```
sum(gtpc_agent_path{namespace="$MMENamespace", interface="S11-MME"})
```

S10 Active Path

Description

This panel lists the number of S10 Path present

Expression

```
sum(gtpc_agent_path{namespace="$MMENamespace", interface="S10"})
```

S6 Active Paths

Description

This panel lists the number of S10 Path present

Expression

{{destination_host}}

```
count(count by(destination_host)
(diameter_peer_connection_established{namespace="$MMENamespace"}))
```

Description

Description

This dashboard contains metrics that can be used to assess the health of the MME pods.

Description

This panel lists the mobility procedure stats

Description

Metrics for Attach procedure including attempts and success rate.

Description

Metrics for Detach procedure including attempts and success rate.

Attach Attempt Total Count

Description

The number of attempted EPS attach procedures.

Expression

```
sum(mmecc_emm_attach_eps_attempt{namespace="$MMEnamespace"})
```

Attach Success %

Description

The success rate (%) of EPS Attach procedures.

Expression

```
((sum(mmecc_emm_attach_eps_success{namespace="$MMEnamespace"}))/(sum(mmecc_emm_attach_eps_attempt{namespace="$MMEnamespace"})))*100
```

Attach Success Total Count

Description

The number of successfully completed EPS Attach procedures.

Expression

```
sum(mmecc_emm_attach_eps_success{namespace="$MMEnamespace"})
```

Detach UE Attempt Total Count

Description

The number of Detach procedures initiated by the UE.

Expression

```
sum(mmecc_emm_eps_detach_ue_attempt{namespace="$MMEnamespace"})
```

Detach UE Success %

Description

The success rate (%) of Detach procedures initiated by the UE.

Expression

```
(sum(mmecc_emm_eps_detach_ue_success{namespace="$MMEnamespace"})/sum(mmecc_emm_eps_detach_ue_attempt{namespace="$MMEnamespace"}))*100
```

Detach UE Success Total Count

Description

The number of successfully completed Detach procedures initiated by the UE.

Expression

```
sum(mmecc_emm_eps_detach_ue_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Service Request procedure stats

Description

This panel lists the Intra-MME TAU procedure stats

Service Request Attempt

Description

This panel lists the Service Request procedure attempts

Expression

```
sum(mmecc_emm_service_request_attempt{namespace="$MMENamespace"})
```

Service Request Success %

Description

This panel lists the Service Request procedure success rate

Expression

```
(sum(mmecc_emm_service_request_success{namespace="$MMENamespace"})/sum(mmecc_emm_service_request_attempt{namespace="$MMENamespace"}))*100
```

Service Request Success

Description

This panel lists the Service Request procedure success count

Expression

```
sum(mmecc_emm_service_request_success{namespace="$MMENamespace"})
```

Intra-MME TAU Attempt Total Count

Description

The number of Intra-MME TAU procedures.

Expression

```
sum(mmecc_emm_intra_mme_tau_attempt{namespace="$MMENamespace"})
```

Intra-MME TAU Success %

Description

The success rate (%) of Intra-MME TAU.

Expression

```
(sum(mmecc_emm_intra_mme_tau_success{namespace="$MMEnamespace"})/sum(mmecc_emm_intra_mme_tau_attempt{namespace="$MMEnamespace"}))*100
```

Intra-MME TAU Success Total Count

Description

The number of successfully completed Intra-MME TAU procedures.

Expression

```
sum(mmecc_emm_intra_mme_tau_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Periodic TAU procedure stats

Description

This panel lists the GUTI Reallocation procedure stats

Periodic TAU Attempt Total Count

Description

The number of Periodic TAU procedures.

Expression

```
sum(mmecc_emm_periodic_tau_attempt{namespace="$MMEnamespace"})
```

Periodic TAU Success %

Description

The success rate (%) of Periodic TAU.

Expression

```
(sum(mmecc_emm_periodic_tau_success{namespace="$MMEnamespace"})/sum(mmecc_emm_periodic_tau_attempt{namespace="$MMEnamespace"}))*100
```

Periodic TAU Success Total Count

Description

The number of successfully completed Periodic TAU procedures.

Expression

```
sum(mmecc_emm_periodic_tau_success{namespace="$MMEnamespace"})
```

GUTI Reallocation Attempt Total Count

Description

The number of GUTI Reallocation procedures.

Expression

```
sum(mmecc_emm_guti_reallocation_attempt{namespace="$MMEnamespace"})
```

GUTI Reallocation Success %

Description

The success rate (%) of GUTI Reallocation

Expression

```
(sum(mmecc_emm_guti_reallocation_success{namespace="$MMEnamespace"})/sum(mmecc_emm_guti_reallocation_attempt{namespace="$MMEnamespace"}))*100
```

GUTI Reallocation Success Total Count

Description

The number of successfully completed GUTI Reallocation procedures.

Expression

```
sum(mmecc_emm_guti_reallocation_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Standalone Security procedure stats

Description

This panel lists the Inter Old MME TAU procedure stats

Standalone Security Attempt Total Count

Description

The number of Standalone Security procedures.

Expression

```
sum(mmecc_emm_standalone_security_attempt{namespace="$MMEnamespace"})
```

Standalone Security Success %

Description

The success rate (%) of Standalone Security

Expression

```
(sum(mmecc_emm_standalone_security_success{namespace="$MMEnamespace"})/sum(mmecc_emm_standalone_security_attempt{namespace="$MMEnamespace"}))*100
```

Standalone Security Success Total Count

Description

The number of successfully completed Standalone Security procedures.

Expression

```
sum(mmecc_emm_standalone_security_success{namespace="$MMEnamespace"})
```

Inter Old MME TAU Attempt Total Count

Description

The number of Inter Old MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_old_mme_tau_attempt{namespace="$MMEnamespace"})
```

Inter Old MME TAU Success %

Description

The success rate (%) of Inter Old MME TAU.

Expression

```
(sum(mmecc_emm_inter_old_mme_tau_success{namespace="$MMEnamespace"})/sum(mmecc_emm_inter_old_mme_tau_attempt{namespace="$MMEnamespace"}))*100
```

Inter Old MME TAU Success Total Count

Description

The number of successfully completed Inter Old MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_old_mme_tau_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter New MME TAU procedure stats

Inter New MME TAU Attempt Total Count

Description

The number of Inter New MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_new_mme_tau_attempt{namespace="$MMEnamespace"})
```

Inter New MME TAU Success %

Description

The success rate (%) of Inter New MME TAU.

Expression

```
(sum(mmecc_emm_inter_new_mme_tau_success{namespace="$MMEnamespace"})/sum(mmecc_emm_inter_new_mme_tau_attempt{namespace="$MMEnamespace"}))*100
```

Inter New MME TAU Success Total Count

Description

The number of successfully completed Inter New MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_new_mme_tau_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter RAT Old MME TAU procedure stats

Inter RAT Old MME TAU Attempt Total Count

Description

The number of Inter RAT Old MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_rat_old_mme_tau_attempt{namespace="$MMEnamespace"})
```

Inter RAT Old MME TAU Success %

Description

The success rate (%) of Inter RAT Old MME TAU.

Expression

```
(sum(mmecc_emm_inter_rat_old_mme_tau_success{namespace="$MMEnamespace"})/sum(mmecc_emm_inter_rat_old_mme_tau_attempt{namespace="$MMEnamespace"}))*100
```

Inter RAT Old MME TAU Success Total Count

Description

The number of successfully completed Inter RAT Old MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_rat_old_mme_tau_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter RAT New MME TAU procedure stats

Inter RAT New MME TAU Attempt Total Count

Description

The number of Inter RAT New MME TAU procedures.

Expression

```
sum(mmecc_emm_inter_rat_new_mme_tau_attempt{namespace="$MMEnamespace"})
```

Inter RAT New MME TAU Success %

Description

The success rate (%) of Inter RAT New MME TAU.

Expression

```
(sum(mmecc_emm_inter_rat_new_mme_tau_success{namespace="$MMEnamespace"})/sum(mmecc_emm_inter_rat_new_mme_tau_attempt{namespace="$MMEnamespace"}))*100
```

Inter RAT New MME TAU Success Total Count

Description

The number of successfully completed Inter RAT New MME TAU procedures.

Expression


```
sum(mmecc_emm_inter_rat_new_mme_tau_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Handover procedure stats

Description

This panel lists the X2 Handover procedure stats

Description

This panel lists the X2 Handover with SGW change procedure stats

X2 Handover Attempt

Description

This panel lists the X2 Handover procedure attempts

Expression

```
sum(mmecc_x2_handover_attempt{namespace="$MMEnamespace"})
```

X2 Handover Success %

Description

This panel lists the X2 Handover procedure success rate

Expression

```
((sum(mmecc_x2_handover_success{namespace="$MMEnamespace"}))/(sum(mmecc_x2_handover_attempt{namespace="$MMEnamespace"})))*100
```

X2 Handover Success

Description

This panel lists the X2 Handover procedure success count

Expression

```
sum(mmecc_x2_handover_success{namespace="$MMEnamespace"})
```

X2 Handover with SGW change Attempt

Description

This panel lists the X2 Handover with SGW change procedure attempts

Expression

```
sum(mmecc_x2_handover_inter_sgw_attempt{namespace="$MMEnamespace"})
```

X2 Handover with SGW change Success %

Description

This panel lists the X2 Handover with SGW change procedure success rate

Expression

```
((sum(mmecc_x2_handover_inter_sgw_success{namespace="$MMEnamespace"}))/(sum(mmecc_x2_handover_inter_sgw_attempt{namespace="$MMEnamespace"})))*100
```

X2 Handover Success

Description

This panel lists the X2 Handover with SGW change procedure success count

Expression

```
sum(mmecc_x2_handover_inter_sgw_success{namespace="$MMEnamespace"})
```

Description

This panel lists the S1 Handover procedure stats

S1 Handover Attempt

Description

This panel lists the S1 Handover procedure attempts

Expression

```
sum(mmecc_s1_intra_handover_attempt{namespace="$MMEnamespace"})
```

S1 Handover Success %

Description

This panel lists the S1 Handover procedure success rate

Expression

```
((sum(mmecc_s1_intra_handover_success{namespace="$MMEnamespace"}))/(sum(mmecc_s1_intra_handover_attempt{namespace="$MMEnamespace"})))*100
```

S1 Handover Success

Description

This panel lists the S1 Handover procedure success count

Expression

```
sum(mmecc_s1_intra_handover_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter Source MME S1 Handover procedure stats

Inter Source MME S1 Handover Attempt

Description

This panel lists the Inter Source MME S1 Handover procedure attempts

Expression

```
sum(mmecc_s1_inter_source_mme_handover_attempt{namespace="$MMEnamespace"})
```

Inter Source MME S1 Handover Success %

Description

This panel lists the Inter Source MME S1 Handover procedure success rate

Expression

```
((sum(mmecc_s1_inter_source_mme_handover_success{namespace="$MMEnamespace"}))/(sum(mmecc_s1_inter_source_mme_handover_attempt{namespace="$MMEnamespace"})))*100
```

Inter Source MME S1 Handover Success

Description

This panel lists the Inter Source MME S1 Handover procedure success count

Expression

```
sum(mmecc_s1_inter_source_mme_handover_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter Target MME S1 Handover procedure stats

Inter Target MME S1 Handover Attempt

Description

This panel lists the Inter Target MME S1 Handover procedure attempts

Expression

```
sum(mmecc_s1_inter_target_mme_handover_attempt{namespace="$MMEnamespace"})
```

Inter Target MME S1 Handover Success %

Description

This panel lists the Inter Target MME S1 Handover procedure success rate

Expression

```
((sum(mmecc_s1_inter_target_mme_handover_success{namespace="$MMEnamespace"}))/sum(mmecc_s1_inter_target_mme_handover_attempt{namespace="$MMEnamespace"}))*100
```

Inter Target MME S1 Handover Success

Description

This panel lists the Inter Target MME S1 Handover procedure success count

Expression

```
sum(mmecc_s1_inter_target_mme_handover_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter RAT Source MME Handover procedure stats

Inter RAT Source MME Handover Attempt

Description

This panel lists the Inter RAT Source MME Handover procedure attempts

Expression

```
sum(mmecc_inter_rat_source_mme_handover_attempt{namespace="$MMEnamespace"})
```

Inter RAT Source MME Handover Success %

Description

This panel lists the Inter RAT Source MME Handover procedure success rate

Expression

```
((sum(mmecc_inter_rat_source_mme_handover_success{namespace="$MMEnamespace"}))/sum(mmecc_inter_rat_source_mme_handover_attempt{namespace="$MMEnamespace"}))*100
```

Inter RAT Source MME Handover Success

Description

This panel lists the Inter RAT Source MME Handover procedure success count

Expression

```
sum(mmecc_inter_rat_source_mme_handover_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Inter RAT Target MME Handover procedure stats

Inter RAT Target MME Handover Attempt

Description

This panel lists the Inter RAT Target MME Handover procedure attempts

Expression

```
sum(mmecc_inter_rat_target_mme_handover_attempt{namespace="$MMEnamespace"})
```

Inter RAT Target MME Handover Success %

Description

This panel lists the Inter RAT Target MME Handover procedure success rate

Expression

```
((sum(mmecc_inter_rat_target_mme_handover_success{namespace="$MMEnamespace"}))/(sum(mmecc_inter_rat_target_mme_handover_attempt{namespace="$MMEnamespace"})))*100
```

Inter RAT Target MME Handover Success

Description

This panel lists the Inter RAT Target MME Handover procedure success count

Expression

```
sum(mmecc_inter_rat_target_mme_handover_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Session procedure stats

Description

This panel lists the Activate Default Bearer Attach procedure stats

Description

Metrics for Activate Default Bearer PDN Connectivity including attempts and success rate.

Default Bearer Activation during Attach with request-type initial attempt Total Count

Description

The number of Default Bearer Activation during Attach with request-type initial attempt procedures.

Expression

```
sum(mmecc_esm_activate_default_bearer_attach_initial_attempt{namespace="$MMEnamespace"})
```

Default Bearer Activation during Attach with request-type initial Success %

Description

The success rate (%) of Default Bearer Activation during Attach with request-type initial Success.

Expression

```
(sum(mmecc_esm_activate_default_bearer_attach_initial_success{namespace="$MMEnamespace"})/sum(mmecc_esm_activate_default_bearer_attach_initial_attempt{namespace="$MMEnamespace"}))*100
```

Default Bearer Activation during Attach with request-type Initial Success Total Count

Description

The number of successfully completed Default Bearer Activation during Attach with request-type initial procedures.

Expression

```
sum(mmecc_esm_activate_default_bearer_attach_initial_success{namespace="$MMEnamespace"})
```

Activate Default Bearer PDN Connectivity with request-type initial request Attempt Total Count

Description

The number of Activate Default Bearer PDN Connectivity with request-type initial request attempt procedures.

Expression

```
sum(mmecc_esm_activate_default_bearer_pdncon_initial_attempt{namespace="$MMEnamespace"})
```

Activate Default Bearer PDN Connectivity with request-type initial request Success %

Description

The success rate (%) of Activate Default Bearer PDN Connectivity with request-type initial request.

Expression

```
(sum(mmecc_esm_activate_default_bearer_pdncon_initial_success{namespace="$MMEnamespace"})/sum(mmecc_esm_activate_default_bearer_pdncon_initial_attempt{namespace="$MMEnamespace"}))*100
```

Activate Default Bearer PDN Connectivity with request-type initial request Success Total Count

Description

The number of successfully completed Activate Default Bearer PDN Connectivity procedures with request-type initial request.

Expression

```
sum(mmecc_esm_activate_default_bearer_pdncon_initial_success{namespace="$MMEnamespace"})
```

Description

This panel lists the Bearer Resource Allocation procedure stats

Description

Metrics for UE initiated Dedicated Bearer Modification including attempts and success rate.

Bearer Resource Allocation Attempt Total Count

Description

The number of Bearer Resource Allocation attempt procedures.

Expression

```
sum(mmecc_esm_bearer_resource_alloc_ue_attempt{namespace="$MMENamespace"})
```

Bearer Resource Allocation Success %

Description

The success rate (%) of Bearer Resource Allocation Success.

Expression

```
(sum(mmecc_esm_bearer_resource_alloc_ue_success{namespace="$MMENamespace"})/sum(mmecc_esm_bearer_resource_alloc_ue_attempt{namespace="$MMENamespace"}))*100
```

Bearer Resource Allocation Success Total Count

Description

The number of successfully completed Bearer Resource Allocation procedures.

Expression

```
sum(mmecc_esm_bearer_resource_alloc_ue_success{namespace="$MMENamespace"})
```

UE initiated Dedicated Bearer Modification Attempt Total Count

Description

The number of UE initiated Dedicated Bearer Modification attempt procedures.

Expression

```
sum(mmecc_esm_modify_dedicated_bearer_ue_attempt{namespace="$MMEnamespace"})
```

UE initiated Dedicated Bearer Modification Success %

Description

The success rate (%) of UE initiated Dedicated Bearer Modification.

Expression

```
(sum(mmecc_esm_modify_dedicated_bearer_ue_success{namespace="$MMEnamespace"})/sum(mmecc_esm_modify_dedicated_bearer_ue_attempt{namespace="$MMEnamespace"}))*100
```

UE initiated Dedicated Bearer Modification Success Total Count

Description

The number of successfully completed UE initiated Dedicated Bearer Modification.

Expression

```
sum(mmecc_esm_modify_dedicated_bearer_ue_success{namespace="$MMEnamespace"})
```

Description

Metrics for PGW initiated Default Bearer Modification including attempts and success rate.

Description

Metrics for PGW initiated Dedicated Bearer Creation including attempts and success rate.

PGW initiated Default Bearer Modification Attempt Total Count

Description

The number of PGW initiated Default Bearer Modification attempt procedures.

Expression

```
sum(mmecc_esm_modify_default_bearer_pgw_attempt{namespace="$MMEnamespace"})
```

PGW Initiated Default Bearer Modification Success %

Description

The success rate (%) of PGW initiated Default Bearer Modification.

Expression

```
(sum(mmecc_esm_modify_default_bearer_pgw_success{namespace="$MMEnamespace"})/sum(mmecc_esm_modify_default_bearer_pgw_attempt{namespace="$MMEnamespace"}))*100
```

PGW Initiated Default Bearer Modification Success Total Count

Description

The number of successfully completed PGW initiated Default Bearer Modification.

Expression

```
sum(mmecc_esm_modify_default_bearer_pgw_success{namespace="$MMEnamespace"})
```

PGW initiated Dedicated Bearer Creation Attempt Total Count

Description

The number of PGW initiated Dedicated Bearer Creation attempt procedures.

Expression

```
sum(mmecc_esm_activate_dedicated_bearer_attempt_total{namespace="$MMEnamespace"})
```

PGW Initiated Dedicated Bearer Creation Success %

Description

The success rate (%) of PGW initiated Dedicated Bearer Creation.

Expression

```
(sum(mmecc_esm_activate_dedicated_bearer_success_total{namespace="$MMEnamespace"})/sum(mmecc_esm_activate_dedicated_bearer_attempt_total{namespace="$MMEnamespace"}))*100
```

PGW Initiated Dedicated Bearer Creation Success Total Count

Description

The number of successfully completed PGW initiated Dedicated Bearer Creation.

Expression

```
sum(mmecc_esm_activate_dedicated_bearer_success_total{namespace="$MMENamespace"})
```

Description

Metrics for UE Initiated Bearer Resource Modification including attempts and success rate.

Description

Metrics for Default Bearer Modification including attempts and success rate.

UE Initiated Bearer Resource Modification Attempt Total Count

Description

The number of UE Initiated Bearer Resource Modification attempt procedures.

Expression

```
sum(mmecc_esm_bearer_resource_mod_ue_attempt{namespace="$MMENamespace"})
```

UE Initiated Bearer Resource Modification Success %

Description

The success rate (%) of UE Initiated Bearer Resource Modification.

Expression

```
(sum(mmecc_esm_bearer_resource_mod_ue_success{namespace="$MMENamespace"})/sum(mmecc_esm_bearer_resource_mod_ue_attempt{namespace="$MMENamespace"}))*100
```

UE Initiated Bearer Resource Modification Success Total Count

Description

The number of successfully completed UE Initiated Bearer Resource Modification.

Expression

```
sum(mmecc_esm_bearer_resource_mod_ue_success{namespace="$MMENamespace"})
```

Default Bearer Modification Attempt Total Count**Description**

The number of Default Bearer Modification attempt procedures.

Expression

```
sum(mmecc_esm_modify_default_bearer_ue_attempt{namespace="$MMENamespace"})
```

Default Bearer Modification Success %**Description**

The success rate (%) of Default Bearer Modification.

Expression

```
(sum(mmecc_esm_modify_default_bearer_ue_success{namespace="$MMENamespace"})/sum(mmecc_esm_modify_default_bearer_ue_attempt{namespace="$MMENamespace"}))*100
```

Default Bearer Modification Success Total Count**Description**

The number of successfully completed Default Bearer Modification.

Expression

```
sum(mmecc_esm_modify_default_bearer_ue_success{namespace="$MMENamespace"})
```

Description

Metrics for SGW Relocation including attempts and success rate.

Description

Metrics for UE Initiated Dedicated EPS Bearer Deactivation including attempts and success rate.

SGW Relocation Attempt Total Count

Description

The number of SGW Relocation attempt procedures.

Expression

```
sum(mmecc_esm_sgw_relocation_attempt{namespace="$MMEnamespace"})
```

SGW Relocation Success %

Description

The success rate (%) of SGW Relocation.

Expression

```
(sum(mmecc_esm_sgw_relocation_success{namespace="$MMEnamespace"})/sum(mmecc_esm_sgw_relocation_attempt{namespace="$MMEnamespace"}))*100
```

SGW Relocation Success Total Count

Description

The number of successfully completed SGW Relocation.

Expression

```
sum(mmecc_esm_sgw_relocation_success{namespace="$MMEnamespace"})
```

UE Initiated Dedicated EPS Bearer Deactivation Total Count

Description

The number of UE Initiated Dedicated EPS Bearer Deactivation attempt procedures.

Expression

```
sum(mmecc_esm_deactivate_dedicated_bearer_ue_attempt{namespace="$MMEnamespace"})
```

UE Initiated Dedicated EPS Bearer Deactivation Success %

Description

The success rate (%) of UE Initiated Dedicated EPS Bearer Deactivation.

Expression

```
(sum(mmecc_esm_deactivate_dedicated_bearer_ue_success{namespace="$MMEnamespace"})/  
sum(mmecc_esm_deactivate_dedicated_bearer_ue_attempt{namespace="$MMEnamespace"}))*  
100
```

UE Initiated Dedicated EPS Bearer Deactivation Success Total Count

Description

The number of successfully completed UE Initiated Dedicated EPS Bearer Deactivation.

Expression

```
sum(mmecc_esm_deactivate_dedicated_bearer_ue_success{namespace="$MMEnamespace"})
```

Description

Metrics for GW Initiated Dedicated EPS Bearer Deactivation including attempts and success rate.

Description

Metrics for MME Initiated Dedicated EPS Bearer Deactivation including attempts and success rate.

GW Initiated Dedicated EPS Bearer Deactivation Attempt Total Count

Description

The number of GW Initiated Dedicated EPS Bearer Deactivation attempt procedures.

Expression

```
sum(mmecc_esm_deactivate_dedicated_bearer_gw_attempt{namespace="$MMEnamespace"})
```

GW Initiated Dedicated EPS Bearer Deactivation Success %

Description

The success rate (%) of GW Initiated Dedicated EPS Bearer Deactivation.

Expression

```
(sum(mmecc_esm_deactivate_dedicated_bearer_gw_success{namespace="$MMEnamespace"})/  
sum(mmecc_esm_deactivate_dedicated_bearer_gw_attempt{namespace="$MMEnamespace"}))*  
100
```

GW Initiated Dedicated EPS Bearer Deactivation Success Total Count

Description

The number of successfully completed GW Initiated Dedicated EPS Bearer Deactivation.

Expression

```
sum(mmecc_esm_deactivate_dedicated_bearer_gw_success{namespace="$MMEnamespace"})
```

MME Initiated Dedicated EPS Bearer Deactivation Total Count

Description

The number of MME Initiated Dedicated EPS Bearer Deactivation attempt procedures.

Expression

```
sum(mmecc_esm_deactivate_dedicated_bearer_mme_attempt{namespace="$MMEnamespace"})
```

MME Initiated Dedicated EPS Bearer Deactivation Success %

Description

The success rate (%) of MME Initiated Dedicated EPS Bearer Deactivation.

Expression

```
(sum(mmecc_esm_deactivate_dedicated_bearer_mme_success{namespace="$MMEnamespace"})  
/sum(mmecc_esm_deactivate_dedicated_bearer_mme_attempt{namespace="$MMEnamespace"}))  
)*100
```


MME Initiated Dedicated EPS Bearer Deactivation Success Total Count

Description

The number of successfully completed MME Initiated Dedicated EPS Bearer Deactivation.

Expression

```
sum(mmecc_esm_deactivate_dedicated_bearer_mme_success{namespace="$MMEnamespace"})
```

Description

Metrics for UE Initiated PDN Disconnect including attempts and success rate.

Description

Metrics for GW Initiated PDN Disconnect including attempts and success rate.

UE Initiated PDN Disconnect Attempt Total Count

Description

The number of UE Initiated PDN Disconnect attempt procedures.

Expression

```
sum(mmecc_esm_deactivate_default_bearer_pdndiscon_attempt{namespace="$MMEnamespace"})
```

UE Initiated PDN Disconnect Success %

Description

The success rate (%) of UE Initiated PDN Disconnect.

Expression

```
(sum(mmecc_esm_deactivate_default_bearer_pdndiscon_success{namespace="$MMEnamespace"})/sum(mmecc_esm_deactivate_default_bearer_pdndiscon_attempt{namespace="$MMEnamespace"}))*100
```

UE Initiated PDN Disconnect Success Total Count

Description

The number of successfully completed UE Initiated PDN Disconnect.

Expression

```
sum(mmecc_esm_deactivate_default_bearer_pdndiscon_success{namespace="$MMENamespace"})
```

GW Initiated PDN Disconnect Attempt Total Count**Description**

The number of GW Initiated PDN Disconnect attempt procedures.

Expression

```
sum(mmecc_esm_deactivate_default_bearer_gw_attempts{namespace="$MMENamespace"})
```

GW Initiated PDN Disconnect Success %**Description**

The success rate (%) of GW Initiated PDN Disconnect.

Expression

```
(sum(mmecc_esm_deactivate_default_bearer_gw_success{namespace="$MMENamespace"})/sum(mmecc_esm_deactivate_default_bearer_gw_attempts{namespace="$MMENamespace"}))*100
```

GW Initiated PDN Disconnect Success Total Count**Description**

The number of successfully completed GW Initiated PDN Disconnect.

Expression

```
sum(mmecc_esm_deactivate_default_bearer_gw_success{namespace="$MMENamespace"})
```

Description

This panel lists the Configuration Transfer procedure stats

Description

Metrics for Intra MME Configuration Transfer including attempts and success rate.

Description

Metrics for Inter MME Configuration Transfer including attempts and success rate.

Intra MME Configuration Transfer Attempt Total Count

Description

The number of Intra MME Configuration Transfer attempt procedures.

Expression

```
sum(mmecc_s1_intra_mme_config_transfer_attempt{namespace="$MMENamespace"})
```

Intra MME Configuration Transfer Success %

Description

The success rate (%) of Intra MME Configuration Transfer.

Expression

```
(sum(mmecc_s1_intra_mme_config_transfer_success{namespace="$MMENamespace"})/sum(mmecc_s1_intra_mme_config_transfer_attempt{namespace="$MMENamespace"}))*100
```

Intra MME Configuration Transfer Success Total Count

Description

The number of Intra MME Configuration Transfer success procedures.

Expression

```
sum(mmecc_s1_intra_mme_config_transfer_success{namespace="$MMENamespace"})
```

Inter MME Configuration Transfer Attempt Total Count

Description

The number of Inter MME Configuration Transfer attempt procedures.

Expression

```
sum(mmecc_s1_inter_mme_config_transfer_attempt{namespace="$MMEnamespace"})
```

Inter MME Configuration Transfer Success %

Description

The success rate (%) of Inter MME Configuration Transfer.

Expression

```
(sum(mmecc_s1_inter_mme_config_transfer_success{namespace="$MMEnamespace"})/sum(mmecc_s1_inter_mme_config_transfer_attempt{namespace="$MMEnamespace"}))*100
```

Inter MME Configuration Transfer Success Total Count

Description

The number of Inter MME Configuration Transfer success procedures.

Expression

```
sum(mmecc_s1_inter_mme_config_transfer_success{namespace="$MMEnamespace"})
```

DNS Request Rate Per Pod

Description

The number of REST query requests received from other pods in the federation. This is displayed per DNS-Agent pod to show relative activity of each.

Expression

{{pod}}

```
round(sum by (pod)(rate(dnsagent_query_request{namespace="$MMEnamespace"}[10s])))
```

DNS Success %

Description

The percentage of REST query responses that were successful. A successful response has an HTTP response code of 200.

Expression

```
(sum(dnsagent_query_response_total{namespace="$MMEnamespace",status="200"})/sum(dnsagent_query_response_total{namespace="$MMEnamespace"}))*100
```

DNS Query Total Count

Description

The total number of REST query requests received from other pods in the federation by DNS-Agent pods.

Expression

```
sum(dnsagent_query_request_total{namespace="$MMEnamespace"})
```

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AMF-N2 Dashboard

Description

AMF-N2 Dashboard

Panels

Description

This panel lists the message stats for NGAP procedures

Description

This panel lists the message stats for Interface Management procedures

NG Setup Stats

Description

This panel lists the message stats for NG Setup procedure

Expression

amfn2_n2_ng_setup_request

```
sum(amfn2_n2_ng_setup_request{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ng_setup_response

```
sum(amfn2_n2_ng_setup_response{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ng_setup_failure

```
sum(amfn2_n2_ng_setup_failure{globalrannodeid="$GlobalRanNodeId"})
```

NG Reset Stats

Description

This panel lists the message stats for NG Reset procedure

Expression

amfn2_n2_ng_reset_rx

```
sum(amfn2_n2_ng_reset_rx{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ng_reset_ack_rx

```
sum(amfn2_n2_ng_reset_ack_rx{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ng_reset_tx

```
sum(amfn2_n2_ng_reset_tx{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ng_reset_ack_tx

```
sum(amfn2_n2_ng_reset_ack_tx{globalrannodeid="$GlobalRanNodeId"})
```

RAN/AMF Config Update Stats

Description

This panel lists the message stats for RAN/AMF Config Update procedure

Expression

amfn2_ran_config_update

```
sum(amfn2_ran_config_update{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_ran_config_update_ack

```
sum(amfn2_ran_config_update_ack{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_ran_config_update_failure

```
sum(amfn2_ran_config_update_failure{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_amf_config_update

```
sum(amfn2_amf_config_update{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_amf_config_update_ack

```
sum(amfn2_amf_config_update_ack{globalrannodeid="$GlobalRanNodeId"})
```


amfn2_amf_config_update_failure

```
sum(amfn2_amf_config_update_failure{globalrannodeid="$GlobalRanNodeId"})
```

Description

This panel lists the message stats for NGAP procedures

Description

This panel lists the message stats for NAS procedures

Description

This panel lists the message stats for NAS procedures

Initial UE Stats

Description

This panel lists the Initial Ue message stats

Expression

initialUeMessage

```
sum(amfn2_n2_initial_ue_message{globalrannodeid="$GlobalRanNodeId"})
```

UL/DL NAS Transport Stats

Description

This panel lists stats for Uplink and Down link Nas transport message

Expression

uplinkNasTransport

```
sum(amfn2_n2_uplink_nas_transport{globalrannodeid="$GlobalRanNodeId"})
```

downlinkNasTransport

```
sum(amfn2_n2_downlink_nas_transport{globalrannodeid="$GlobalRanNodeId"})
```

Error/Unsupported Stats

Description

This panel lists stats for Error/Unsupported message

Expression

amfn2_n2_message_unsupported_uplink

```
sum(amfn2_n2_message_unsupported_uplink{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_message_unsupported_downlink

```
sum(amfn2_n2_message_unsupported_downlink{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_message_error_uplink

```
sum(amfn2_n2_message_error_uplink{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_message_error_downlink

```
sum(amfn2_n2_message_error_downlink{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_error_indication_rx

```
sum(amfn2_n2_error_indication_rx{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_error_indication_tx

```
sum(amfn2_n2_error_indication_tx{globalrannodeid="$GlobalRanNodeId"})
```

Description

This panel lists the message stats for NGAP procedures

Description

This panel lists the UE Context Management Procedure stats

Initial Context Setup Stats

Description

This panel lists the Initial Context Setup stats

Expression

initialContextSetupRequest

```
sum(amfn2_n2_initial_context_setup_request{globalrannodeid="$GlobalRanNodeId"})
```

initialContextSetupResponse

```
sum(amfn2_n2_initial_context_setup_response{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_initial_context_setup_failure

```
sum(amfn2_n2_initial_context_setup_failure{globalrannodeid="$GlobalRanNodeId"})
```

UE Context Release Stats

Description

This panel lists the UE Context Release stats

Expression

ueContextReleaseCommand

```
sum(amfn2_n2_ue_context_release_command{globalrannodeid="$GlobalRanNodeId"})
```

ueContextReleaseComplete

```
sum(amfn2_n2_ue_context_release_complete{globalrannodeid="$GlobalRanNodeId"})
```

ueContextReleaseRequest

```
sum(amfn2_n2_ue_context_release_request{globalrannodeid="$GlobalRanNodeId"})
```

UE Context Modification Stats

Description

This panel lists the UE Context Modification stats

Expression

amfn2_n2_ue_context_modification_request

```
sum(amfn2_n2_ue_context_modification_request{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ue_context_modification_response

```
sum(amfn2_n2_ue_context_modification_response{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_ue_context_modification_failure

```
sum(amfn2_n2_ue_context_modification_failure{globalrannodeid="$GlobalRanNodeId"})
```

Description

This panel lists the message stats for NGAP procedures

Description

This panel lists the PDU Session Management Procedure stats

Pdu Session Setup Stats

Description

This panel lists the stats for PDU Session setup

Expression

pduSessionResourceSetupRequest

```
sum(amfn2_n2_pdu_session_resource_setup_request{globalrannodeid="$GlobalRanNodeId"})
```

pduSessionResourceSetupResponse

```
sum(amfn2_n2_pdu_session_resource_setup_response{globalrannodeid="$GlobalRanNodeId"})
```

Pdu Session Modify Stats

Description

This panel lists the stats for PDU Session modify

Expression

pduSessionResourceModifyRequest

```
sum(amfn2_n2_pdu_session_resource_modify_request{globalrannodeid="$GlobalRanNodeId"})
```

pduSessionResourceModifyResponse

```
sum(amfn2_n2_pdu_session_resource_modify_response{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_pdu_session_resource_modify_indication

```
sum(amfn2_n2_pdu_session_resource_modify_indication{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_pdu_session_resource_modify_confirm

```
sum(amfn2_n2_pdu_session_resource_modify_confirmv{globalrannodeid="$GlobalRanNodeId"})
```

Pdu Session Release Stats

Description

This panel lists the stats for PDU Session release

Expression

pduSessionResourceReleaseRequest

```
sum(amfn2_n2_pdu_session_resource_release_command{globalrannodeid="$GlobalRanNodeId"})
```

pduSessionResourceReleaseResponse

```
sum(amfn2_n2_pdu_session_resource_release_response{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_pdu_session_resource_notify

```
sum(amfn2_n2_pdu_session_resource_notify{globalrannodeid="$GlobalRanNodeId"})
```

Description

This panel lists the message stats for NGAP procedures

Description

This panel lists the UE Mobility Management Procedure stats

Path Switch Stats

Description

This panel lists the Path Switch message stats

Expression

pathSwitchRequest

```
sum(amfn2_n2_pathswitch_request{globalrannodeid="$GlobalRanNodeId"})
```

pathSwitchRequestAck

```
sum(amfn2_n2_pathswitch_request_ack{globalrannodeid="$GlobalRanNodeId"})
```

Handover Stats**Description**

This panel lists the Handover message stats

Expression**amfn2_n2_handover_required**

```
sum(amfn2_n2_handover_required{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_command

```
sum(amfn2_n2_handover_command{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_prep_failure

```
sum(amfn2_n2_handover_prep_failure{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_request

```
sum(amfn2_n2_handover_request{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_request_ack

```
sum(amfn2_n2_handover_request_ack{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_failure

```
sum(amfn2_n2_handover_failure{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_notify

```
sum(amfn2_n2_handover_notify{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_cancel

```
sum(amfn2_n2_handover_cancel{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_handover_cancel_ack

```
sum(amfn2_n2_handover_cancel_ack{globalrannodeid="$GlobalRanNodeId"})
```

Description

This panel lists the message stats for NGAP procedures

Paging Stats

Description

This panel lists the Paging message stats

Expression

amfn2_n2_paging

```
sum(amfn2_n2_paging{globalrannodeid="$GlobalRanNodeId"})
```

RAN Status/Config Transfer Stats

Description

This panel lists the RAN Status/Config Transfer stats

Expression

amfn2_n2_uplink_ran_status_transfer

```
sum(amfn2_n2_uplink_ran_status_transfer{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_downlink_ran_status_transfer

```
sum(amfn2_n2_downlink_ran_status_transfer{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_uplink_ran_config_transfer

```
sum(amfn2_n2_uplink_ran_config_transfer{globalrannodeid="$GlobalRanNodeId"})
```

amfn2_n2_downlink_ran_config_transfer

```
sum(amfn2_n2_downlink_ran_config_transfer{globalrannodeid="$GlobalRanNodeId"})
```

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- [Summary](#)
- [Capture Utility](#)

Summary

Dashboard Name	Panel Name
Capture Utility	Capture Manager Reconciliation Time

Capture Utility

Description

Dashboard for Capture Utility Metrics

Panels

Capture Manager Reconciliation Time

Description

Shows the reconciliation time taken to process a Custom Resouce

Expression

0.5

```
histogram_quantile(0.5,
sum(rate(controller_runtime_reconcile_time_seconds_bucket[10m])) by (le))
```

0.9

```
histogram_quantile(0.9,
sum(rate(controller_runtime_reconcile_time_seconds_bucket[10m])) by (le))
```

0.95

```
histogram_quantile(0.95,
sum(rate(controller_runtime_reconcile_time_seconds_bucket[10m])) by (le))
```

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- [Summary](#)
- [cert-manager](#)

Summary

Dashboard Name	Panel Name
cert-manager	Certificates
cert-manager	Time left before Certificate expires
cert-manager	Certificate Ready Status
cert-manager	Time left before Certificate expires

cert-manager

Description

Certmanager deployed in Kubernetes and scraped by Prometheus

Panels

Certificates

Description

This is the row for certificates

Time left before Certificate expires

Description

Time left before Certificate expires

Expression

`{{exported_namespace}} / {{name}}`

```
sort_desc(
  sum(certmanager_certificate_expiration_timestamp_seconds{exported_namespace=~"$Namespace"} - time()) BY (name,exported_namespace) )
```

Certificate Ready Status

Description

This gives the total number of certificates in three conditions- ready(true), not ready(false) and unknown status

Expression

`{{condition}}`

```
sum(certmanager_certificate_ready_status) BY (condition)
```

Time left before Certificate expires

Description

Time left before certificate expires

Expression

{{exported_namespace}} / {{name}}

```
sum(certmanager_certificate_expiration_timestamp_seconds{exported_namespace=~"$Namespace"} - time()) BY (name,exported_namespace)
```

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- [Summary](#)
- [DNS-Agent Dashboard](#)

Summary

Dashboard Name	Panel Name
DNS-Agent Dashboard	DNS Success %
DNS-Agent Dashboard	DNS Dynamic Cache Heap Usage
DNS-Agent Dashboard	Dns Dynamic Cache Nodes
DNS-Agent Dashboard	DNS Errors

DNS-Agent Dashboard

Description

This dashboard lists the metrics on pod DNS Agent.

Panels

DNS Success %

Description

This panel lists the DNS request success stats.

Expression

```
(sum(dnsagent_query_response{status="200",pod="$PodName"})/sum(dnsagent_query_response{pod="$PodName"}))*100
```

DNS Dynamic Cache Heap Usage

Description

This panel lists the DNS Dynamic Cache Heap Usage.

Expression

```
1 sum(dnsagent_bind_cachestats_heap_mem_total{pod="$PodName"})
2 sum(dnsagent_bind_cachestats_heap_mem_in_use{pod="$PodName"})
```

Dns Dynamic Cache Nodes

Description

This panel lists the DNS Dynamic Cache Nodes.

Expression

```
sum(dnsagent_bind_cachestats_cache_nodes{pod="$PodName"})
```

DNS Errors

Description

This panel lists the DNS Errors.

Expression

NXDOMAIN

```
sum(dnsagent_bind_resolver_nxdomain{pod="$PodName"})
```

SERVFAIL

```
sum(dnsagent_bind_resolver_servfail{pod="$PodName"})
```

FORMERR

```
sum(dnsagent_bind_resolver_formerr{pod="$PodName"})
```

Bad Cookie Rcode

```
sum(dnsagent_bind_resolver_bad_cookie_rcode{pod="$PodName"})
```

Bad Edns Version

```
sum(dnsagent_bind_resolver_bad_edns_version{pod="$PodName"})
```

Timeout

```
sum(dnsagent_bind_resolver_query_timeout{pod="$PodName"})
```

Retry

```
sum(dnsagent_bind_resolver_retry{pod="$PodName"})
```

Truncated

```
sum(dnsagent_bind_resolver_truncated{pod="$PodName"})
```

Refused

```
sum(dnsagent_bind_resolver_refused{pod="$PodName"})
```

Other

```
sum(dnsagent_bind_resolver_other_error{pod="$PodName"})
```

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-
- [Summary](#)
 - [ElasticSearch Dashboard](#)

Summary

Dashboard Name	Panel Name
ElasticSearch Dashboard	KPI
ElasticSearch Dashboard	Shards
ElasticSearch Dashboard	Active primary shards
ElasticSearch Dashboard	Active shards
ElasticSearch Dashboard	Initializing shards
ElasticSearch Dashboard	Relocating shards
ElasticSearch Dashboard	Delayed shards
ElasticSearch Dashboard	Unassigned shards
ElasticSearch Dashboard	JVM Garbage Collection
ElasticSearch Dashboard	Translog
ElasticSearch Dashboard	Breakers
ElasticSearch Dashboard	CPU and Memory
ElasticSearch Dashboard	Disk and Network
ElasticSearch Dashboard	Documents
ElasticSearch Dashboard	Times
ElasticSearch Dashboard	Caches
ElasticSearch Dashboard	Segments

ElasticSearch Dashboard

Description

ElasticSearch cluster stats

Panels

KPI

Description

KPI

Shards

Description

Shards

Active primary shards

Description

The number of primary shards in your cluster. This is an aggregate total across all indices.

Expression

```
elasticsearch_cluster_health_active_primary_shards
```

Active shards

Description

Aggregate total of all shards across all indices, which includes replica shards

Expression

```
elasticsearch_cluster_health_active_shards
```

Initializing shards

Description

Count of shards that are being freshly created

Expression

```
elasticsearch_cluster_health_initializing_shards
```

Relocating shards

Description

The number of shards that are currently moving from one node to another node.

Expression

```
elasticsearch_cluster_health_relocating_shards
```

Delayed shards

Description

Shards delayed to reduce reallocation overhead

Expression

```
elasticsearch_cluster_health_delayed_unassigned_shards
```

Unassigned shards

Description

The number of shards that exist in the cluster state, but cannot be found in the cluster itself

Expression

```
elasticsearch_cluster_health_unassigned_shards
```

JVM Garbage Collection

Description

JVM Garbage Collection

Translog

Description

Translog

Breakers

Description

Breakers

CPU and Memory

Description

CPU and Memory

Disk and Network

Description

Disk and Network

Documents

Description

Documents

Times

Description

Times

Caches

Description

Caches

Segments

Description

Segments

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- [Summary](#)
- [Etcd Server](#)

Summary

Dashboard Name	Panel Name
Etcd Server	Etcd has a leader?
Etcd Server	The number of leader changes seen
Etcd Server	The total number of failed proposals seen

Dashboard Name	Panel Name
Etcd Server	The total number of consensus proposals committed
Etcd Server	proposals_applied_total
Etcd Server	proposals_failed_total
Etcd Server	Proposals pending
Etcd Server	backend_commit_duration_seconds
Etcd Server	peer_received_bytes_total
Etcd Server	peer_round_trip_time_seconds
Etcd Server	peer_sent_failures_total
Etcd Server	client_grpc_received_bytes_total
Etcd Server	RPC Rate
Etcd Server	Active Streams
Etcd Server	DB Size
Etcd Server	Disk Sync Duration
Etcd Server	Memory
Etcd Server	Client Traffic In
Etcd Server	Client Traffic Out
Etcd Server	Peer Traffic In
Etcd Server	Peer Traffic Out
Etcd Server	Raft Proposals
Etcd Server	Total Leader Elections Per Day
Etcd Server	Disks operations
Etcd Server	Network
Etcd Server	Snapshot duration

Etcd Server

Description

Etcd Dashboard for Prometheus metrics scraper

Panels

Etcd has a leader?

Description

Informes if the etcd has a leader or not.

Expression

```
max(etcd_server_has_leader)
```

The number of leader changes seen

Description

Informes the number of times etcd leader has changed.

Expression

```
max(etcd_server_leader_changes_seen_total)
```

The total number of failed proposals seen

Description

Normally related to two issues: temporary failures related to a leader election or longer downtime caused by a loss of quorum in the cluster.

Expression

```
max(etcd_server_leader_changes_seen_total)
```

The total number of consensus proposals committed

Description

The total number of consensus proposals committed.

Expression

total number of consensus proposals committed

```
sum(rate(etcd_server_proposals_committed_total[5m]))
```

total number of consensus proposals applied

```
sum(rate(etcd_server_proposals_applied_total[5m]))
```

proposals applied total

Description

Records the total number of consensus proposals applied. The etcd server applies every committed proposal asynchronously.

Expression

```
etcd_server_proposals_applied_total
```

proposals failed total

Description

Normally related to two issues: temporary failures related to a leader election or longer downtime caused by a loss of quorum in the cluster.

Expression

```
etcd_server_proposals_failed_total
```

Proposals pending

Description

Indicates how many proposals are queued to commit. Rising pending proposals suggests there is a high client load or the member cannot commit proposals.

Expression**Proposals pending**

```
sum(etcd_server_proposals_pending)
```

backend commit duration seconds

Description

A backend_commit is called when etcd commits an incremental snapshot of its most recent changes to disk. backend_commit_duration_seconds informs the duration of each commit

Expression

```
etcd_disk_backend_commit_duration_seconds_bucket
```

peer received bytes total

Description

Number of bytes received from peers.

Expression

```
etcd_network_peer_received_bytes_total
```

peer round trip time seconds

Description

Round trip time for peer messages.

Expression

```
etcd_network_peer_round_trip_time_seconds_bucket
```

peer sent failures total

Description

Number of failed messages destined to peers.

Expression

```
etcd_network_peer_sent_failures_total
```

client grpc received bytes total

Description

Number of grpc bytes received from the clients.

Expression

```
etcd_network_client_grpc_received_bytes_total
```

RPC Rate

Description

Rate at which RPC requests are processed.

Expression

RPC Rate

```
sum(rate(grpc_server_started_total{grpc_type="unary"}[5m]))
```

RPC Failed Rate

```
sum(rate(grpc_server_handled_total{grpc_type="unary",grpc_code!="OK"}[5m]))
```

Active Streams

Description

Number of active streams server is maintaining(mostly includes watch and lease streams).

Expression

Watch Streams

```
sum(grpc_server_started_total{grpc_service="etcdserverpb.Watch",grpc_type="bidi_stream"}) -  
sum(grpc_server_handled_total{grpc_service="etcdserverpb.Watch",grpc_type="bidi_stream"})
```

Lease Streams

```
sum(grpc_server_started_total{grpc_service="etcdserverpb.Lease",grpc_type="bidi_stream"}) -  
sum(grpc_server_handled_total{grpc_service="etcdserverpb.Lease",grpc_type="bidi_stream"})
```

DB Size

Description

Size of the etcd database in bytes.

Expression

{{instance}} DB Size

```
etcd_debugging_mvcc_db_total_size_in_bytes
```

Disk Sync Duration

Description

Disk Sync Duration.

Expression

{{instance}} WAL fsync

```
histogram_quantile(0.99,  
sum(rate(etcd_disk_wal_fsync_duration_seconds_bucket[5m])) by (instance, le))
```

{{instance}} DB fsync

```
histogram_quantile(0.99,  
sum(rate(etcd_disk_backend_commit_duration_seconds_bucket[5m])) by (instance, le))
```

Memory

Description

Resident Memory in bytes.

Expression

{{instance}} Resident Memory

```
process_resident_memory_bytes
```

Client Traffic In

Description

Client Traffic coming in bytes.

Expression

{{instance}} Client Traffic In

```
rate(etcd_network_client_grpc_received_bytes_total[5m])
```

Client Traffic Out

Description

Client Traffic going out in bytes.

Expression

{{instance}} Client Traffic Out

```
rate(etcd_network_client_grpc_sent_bytes_total[5m])
```

Peer Traffic In

Description

Peer traffic coming in bytes.

Expression

{{instance}} Peer Traffic In

```
sum(rate(etcd_network_client_grpc_received_bytes_total[5m])) by (instance)
```

Peer Traffic Out

Description

Peer traffic going out in bytes.

Expression

{{instance}} Peer Traffic Out

```
sum(rate(etcd_network_client_grpc_sent_bytes_total[5m])) by (instance)
```

Raft Proposals

Description

Total number of raft proposals.

Expression

Proposal Failure Rate

```
sum(rate(etcd_server_proposals_failed_total[5m]))
```

Proposal Pending Total

```
sum(etcd_server_proposals_pending)
```

Proposal Commit Rate

```
sum(rate(etcd_server_proposals_committed_total[5m]))
```

Proposal Apply Rate

```
sum(rate(etcd_server_proposals_applied_total[5m]))
```

Total Leader Elections Per Day

Description

Total Leader Elections Per Day.

Expression

{{instance}} Total Leader Elections Per Day

```
changes(etcd_server_leader_changes_seen_total[1d])
```

Disks operations

Description

The latency distributions of commit called by backend.

Expression

The latency distributions of fsync called by wal

```
sum(rate(etcd_disk_wal_fsync_duration_seconds_sum[1m]))
```

The latency distributions of commit called by backend

```
sum(rate(etcd_disk_backend_commit_duration_seconds_sum[1m]))
```

Network

Description

The total number of bytes sent to grpc clients.

Expression

The total number of bytes received by grpc clients

```
sum(rate(etcd_network_client_grpc_received_bytes_total[1m]))
```

The total number of bytes sent to grpc clients

```
sum(rate(etcd_network_client_grpc_sent_bytes_total[1m]))
```

Snapshot duration

Description

The total latency distributions of save called by snapshot.

Expression

The total latency distributions of save called by snapshot

```
sum(rate(etcd_debugging_snap_save_total_duration_seconds_sum[1m]))
```

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- [Summary](#)
- [Fluentd Logging](#)

Summary

Dashboard Name	Panel Name
Fluentd Logging	Container logs
Fluentd Logging	Syslogs
Fluentd Logging	Audit logs
Fluentd Logging	LI logs
Fluentd Logging	Buffer Overview
Fluentd Logging	Buffer Stage
Fluentd Logging	Buffer Queue
Fluentd Logging	Buffer Space
Fluentd Logging	Buffer Retries
Fluentd Logging	Buffer Errors

Fluentd Logging

Description

fluentd deployed in kubernetes and scraped by prometheus

Panels

Container logs

Description

Container Logs

Syslogs

Description

Syslogs

Audit logs

Description

Audit Logs

LI logs

Description

LI logs

Buffer Overview

Description

General details

Buffer Stage

Description

Buffer Stage

Buffer Queue

Description

Buffer Queue

Buffer Space

Description

Buffer Space

Buffer Retries

Description

Buffer Retries

Buffer Errors

Description

Buffer Errors

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- [Summary](#)
- [AO5GC Dashboards](#)
- [Grafana Dashboard](#)

Summary

Dashboard Name	Panel Name
Grafana Dashboard	Grafana Dashboards count
Grafana Dashboard	Grafana Total Page response
Grafana Dashboard	Grafana Proxy Response Status
Grafana Dashboard	Panel Title

AO5GC Dashboards

Description

Dashboards for monitoring cpu and memory of multiple clusters

Grafana Dashboard

Description

Dashboard for Grafana

Panels

Grafana Dashboards count

Description

This panel description is about grafana Dashboards Count.

Expression

```
grafana_stat_totals_dashboard
```

Grafana Total Page response

Description

This panel description is about Grafana Total Page response.

Expression

```
grafana_page_response_status_total
```

Grafana Proxy Response Status

Description

This panel description is about Grafana Proxy response status.

Expression

```
grafana_proxy_response_status_total
```

Panel Title

Description

This panel description is about Grafana Api total response status.

Expression

```
grafana_api_response_status_total
```

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- [Summary](#)
- [Interface Manager Dashboard](#)

- [Interface Manager Debug Dashboard](#)

Summary

Dashboard Name	Panel Name
Interface Manager Dashboard	Number of External Interfaces
Interface Manager Dashboard	Number of IPv4 VLAN Subnets
Interface Manager Dashboard	Number of IPv4 Loopback Subnets
Interface Manager Dashboard	Number of VLAN Subinterfaces
Interface Manager Dashboard	Number of IPv6 VLAN Subnets
Interface Manager Dashboard	Number of IPv6 Loopback Subnets
Interface Manager Debug Dashboard	Number of Attempts to Add or Update a Row
Interface Manager Debug Dashboard	Number of Successful Attempts to Add or Update a Row
Interface Manager Debug Dashboard	Number of Failed Attempts to Add or Update a Row
Interface Manager Debug Dashboard	Number of Attempts to Delete a Row
Interface Manager Debug Dashboard	Number of Successful Attempts to Delete a Row
Interface Manager Debug Dashboard	Number of Failed Attempts to Delete a Row
Interface Manager Debug Dashboard	Number of Attempts to Get all Rows
Interface Manager Debug Dashboard	Number of Successful Attempts to Get all Rows
Interface Manager Debug Dashboard	Number of Failed Attempts to Get all Rows
Interface Manager Debug Dashboard	Number of Configuration Updates Received

Interface Manager Dashboard

Description

Interface Manager Dashboard

Panels

Number of External Interfaces

Description

Number of external interfaces currently reported to Interface Manager

Expression

```
sum(intfmgr_external_interfaces)
```

Number of IPv4 VLAN Subnets

Description

Number of IPv4 vlan subnets on the vlan subinterfaces in network context

Expression

```
sum(intfmgr_ipv4_vlan_subnets{net_ctxt=~"$net_ctxt", vlan_id=~"$vlan_id"})
```

Number of IPv4 Loopback Subnets

Description

Number of IPv4 loopback subnets in network context

Expression

```
sum(intfmgr_ipv4_loopback_subnets{net_ctxt=~"$net_ctxt"})
```

Number of VLAN Subinterfaces

Description

Number of vlan subinterfaces in a network context

Expression

```
sum(intfmgr_vlan_subinterfaces{net_ctxt=~"$net_ctxt", vlan_id=~"$vlan_id"})
```

Number of IPv6 VLAN Subnets

Description

Number of IPv6 vlan subnets on the vlan subinterfaces in network context

Expression

```
sum(intfmgr_ipv6_vlan_subnets{net_ctxt=~"$net_ctxt", vlan_id=~"$vlan_id"})
```

Number of IPv6 Loopback Subnets

Description

Number of IPv6 loopback subnets in network context

Expression

```
sum(intfmgr_ipv6_loopback_subnets{net_ctxt=~"$net_ctxt"})
```

Interface Manager Debug Dashboard

Description

Interface Manager Debug Dashboard

Panels

Number of Attempts to Add or Update a Row

Description

Number of attempts to add or update a row in the etcd DB table

Expression

```
sum(intfmgr_etcd_put_row_attempts{etcd_table=~"$etcd_table"})
```

Number of Successful Attempts to Add or Update a Row

Description

Number of successful attempts to add or update a row in the etcd DB table

Expression

```
sum(intfmgr_etcd_put_row_successes{etcd_table=~"$etcd_table"})
```

Number of Failed Attempts to Add or Update a Row

Description

Number of failed attempts to add or update a row in the etcd DB etcd_table due to an http_status_code error

Expression

```
sum(intfmgr_etcd_put_row_failures{etcd_table=~"$etcd_table",  
http_status_code=~"$http_status_code"})
```

Number of Attempts to Delete a Row

Description

Number of attempts to delete a row in the etcd DB table

Expression

```
sum(intfmgr_etcd_delete_row_attempts{etcd_table=~"$etcd_table"})
```

Number of Successful Attempts to Delete a Row

Description

Number of successful attempts to delete a row in the etcd DB table

Expression

```
sum(intfmgr_etcd_delete_row_successes{etcd_table=~"$etcd_table"})
```

Number of Failed Attempts to Delete a Row

Description

Number of failed attempts to delete a row in the etcd DB etcd_table due to an http_status_code error

Expression

```
sum(intfmgr_etcd_delete_row_failures{etcd_table=~"$etcd_table",  
http_status_code=~"$http_status_code"})
```

Number of Attempts to Get all Rows

Description

Number of attempts to get all rows in the etcd DB table

Expression

```
sum(intfmgr_etcd_get_table_attempts{etcd_table=~"$etcd_table"})
```

Number of Successful Attempts to Get all Rows

Description

Number of attempts to get all rows in the etcd DB table

Expression

```
sum(intfmgr_etcd_get_table_successes{etcd_table=~"$etcd_table"})
```

Number of Failed Attempts to Get all Rows

Description

Number of failed attempts to get all rows in the etcd DB etcd_table due to an http_status_code error

Expression

```
sum(intfmgr_etcd_get_table_failures{etcd_table=~"$etcd_table",  
http_status_code=~"$http_status_code"})
```

Number of Configuration Updates Received

Description

Number of configuration updates, (i.e. commits), received

Expression

```
sum(intfmgr_configuration_updates)
```

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Summary

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Istio Control Plane Dashboard

Description

Istio Control Plane Dashboard version 1.11.4

Deployed Versions Panels

Panels

Pilot Versions

Expression

{{ tag }}

```
sum(istio_build{component="pilot"}) by (tag)
```

Resource Usage Panels

Memory

Expression

Virtual Memory

```
process_virtual_memory_bytes{service="istiod"}
```

Resident Memory

```
process_resident_memory_bytes{service="istiod"}
```

heap sys

```
go_memstats_heap_sys_bytes{service="istiod"}
```

heap alloc

```
go_memstats_heap_alloc_bytes{service="istiod"}
```

Alloc

```
go_memstats_alloc_bytes{service="istiod"}
```

Heap in-use

```
go_memstats_heap_inuse_bytes{service="istiod"}
```

Stack in-use

```
go_memstats_stack_inuse_bytes{service="istiod"}
```

Discovery (container)

```
container_memory_working_set_bytes{container=~"discovery", pod=~"istiod-.*|istio-pilot-.*"}
```

Sidecar (container)

```
container_memory_working_set_bytes{container=~"istio-proxy", pod=~"istiod-.*|istio-pilot-.*"}
```

Goroutines

Expression

Number of Goroutines

```
go_goroutines{service="istiod"}
```

Pilot Push Information Panels

Pilot Pushes

Description

Shows the rate of pilot pushes

Expression

Cluster

```
sum(irate(pilot_xds_pushes{type="cds"}[1m]))
```

Endpoints

```
sum(irate(pilot_xds_pushes{type="eds"}[1m]))
```

Listeners

```
sum(irate(pilot_xds_pushes{type="lds"}[1m]))
```

Routes

```
sum(irate(pilot_xds_pushes{type="rds"}[1m]))
```

Secrets

```
sum(irate(pilot_xds_pushes{type="sds"}[1m]))
```

Nametables

```
sum(irate(pilot_xds_pushes{type="nds"}[1m]))
```

Pilot Errors

Description

Captures a variety of pilot errors

Expression

Rejected CDS Configs

```
sum(pilot_xds_cds_reject{service="istiod"}) or  
(absent(pilot_xds_cds_reject{service="istiod"}) - 1)
```

Rejected EDS Configs

```
sum(pilot_xds_eds_reject{service="istiod"}) or  
(absent(pilot_xds_eds_reject{service="istiod"}) - 1)
```

Rejected RDS Configs

```
sum(pilot_xds_rds_reject{service="istiod"}) or  
(absent(pilot_xds_rds_reject{service="istiod"}) - 1)
```

Rejected LDS Configs

```
sum(pilot_xds_lds_reject{service="istiod"}) or  
(absent(pilot_xds_lds_reject{service="istiod"}) - 1)
```

Write Timeouts

```
sum(rate(pilot_xds_write_timeout{service="istiod"}[1m]))
```

Internal Errors

```
sum(rate(pilot_total_xds_internal_errors{service="istiod"}[1m]))
```

Config Rejection Rate

```
sum(rate(pilot_total_xds_rejects{service="istiod"}[1m]))
```

Push Context Errors

```
sum(rate(pilot_xds_push_context_errors{service="istiod"}[1m]))
```

Push Timeouts

```
sum(rate(pilot_xds_write_timeout{service="istiod"}[1m]))
```

Proxy Push Time

Description

Shows the total time it takes to push a config update to a proxy

Expression

p50

```
histogram_quantile(0.5, sum(rate(pilot_proxy_convergence_time_bucket[1m])) by (le))
```

p90

```
histogram_quantile(0.9, sum(rate(pilot_proxy_convergence_time_bucket[1m])) by (le))
```

p99

```
histogram_quantile(0.99, sum(rate(pilot_proxy_convergence_time_bucket[1m])) by (le))
```

p99.9

```
histogram_quantile(0.999, sum(rate(pilot_proxy_convergence_time_bucket[1m])) by (le))
```

Conflicts

Expression

Inbound Listeners

```
pilot_conflict_inbound_listener{service="istiod"}
```

Outbound Listeners (http over current tcp)

```
pilot_conflict_outbound_listener_http_over_current_tcp{service="istiod"}
```

Outbound Listeners (tcp over current tcp)

```
pilot_conflict_outbound_listener_tcp_over_current_tcp{service="istiod"}
```

Outbound Listeners (tcp over current http)

```
pilot_conflict_outbound_listener_tcp_over_current_http{service="istiod"}
```

ADS Monitoring

Expression

Virtual Services

```
avg(pilot_virt_services{service="istiod"})
```

Services

```
avg(pilot_services{service="istiod"})
```

Connected Endpoints {{pod}}

```
sum(pilot_xds{service="istiod"}) by (pod)
```

Envoy Information Panels

Envoy Details

Description

Shows details about Envoy proxies in the mesh

Expression

XDS Connections

```
sum(irate(envoy_cluster_upstream_cx_total{cluster_name="xds-grpc"}[1m]))
```

XDS Connection Failures

```
sum(irate(envoy_cluster_upstream_cx_connect_fail{cluster_name="xds-grpc"}[1m]))
```

Envoy Restarts

```
sum(increase(envoy_server_hot_restart_epoch[1m]))
```

XDS Active Connections

Expression

XDS Active Connections

```
sum(envoy_cluster_upstream_cx_active{cluster_name="xds-grpc"})
```

XDS Requests Size

Description

Shows the size of XDS requests and responses

Expression

XDS Response Bytes Max


```
max(rate(envoy_cluster_upstream_cx_rx_bytes_total{cluster_name="xds-grpc"}[1m]))
```

XDS Response Bytes Average

```
quantile(0.5, rate(envoy_cluster_upstream_cx_rx_bytes_total{cluster_name="xds-grpc"}[1m]))
```

XDS Request Bytes Max

```
max(rate(envoy_cluster_upstream_cx_tx_bytes_total{cluster_name="xds-grpc"}[1m]))
```

XDS Request Bytes Average

```
quantile(.5, rate(envoy_cluster_upstream_cx_tx_bytes_total{cluster_name="xds-grpc"}[1m]))
```

Webhooks

Configuration Validation

Expression

Validations (Success)

```
sum(rate(galley_validation_passed[1m]))
```

Validation (Failure)

```
sum(rate(galley_validation_failed[1m]))
```

Sidecar Injection

Description

Expression

Injections (Success)

```
sum(rate(sidecar_injection_success_total[1m]))
```

Injections (Failure)

```
sum(rate(sidecar_injection_failure_total[1m]))
```

Istio Galley Dashboard

Description

Istio Galley Dashboard

Panels

Galley Versions

Description

Istio Galley component versions

Expression

{{ tag }}

```
sum(istio_build{component="galley"}) by (tag)
```

Resource Usage Panels

Memory

Description

Galley memory statistics

Expression

Virtual Memory

```
process_virtual_memory_bytes{job="galley"}
```

Resident Memory

```
process_resident_memory_bytes{job="galley"}
```

heap sys

```
go_memstats_heap_sys_bytes{job="galley"}
```

heap alloc

```
go_memstats_heap_alloc_bytes{job="galley"}
```

Alloc

```
go_memstats_alloc_bytes{job="galley"}
```

Heap in-use

```
go_memstats_heap_inuse_bytes{job="galley"}
```

Stack in-use

```
go_memstats_stack_inuse_bytes{job="galley"}
```

Total (kis)

```
sum(container_memory_usage_bytes{container_name=~"galley", pod_name=~"istio-galley-.*"})
```

CPU

Description

Galley spu statistics

Expression

Total (k8s)

```
sum(rate(container_cpu_usage_seconds_total{container_name=~"galley",  
pod_name=~"istio-galley-.*"}[1m]))
```

{{ container_name }} (k8s)

```
sum(rate(container_cpu_usage_seconds_total{container_name=~"galley",  
pod_name=~"istio-galley-.*"}[1m])) by (container_name)
```

galley (self-reported)

```
irate(process_cpu_seconds_total{job="galley"}[1m])
```

Disk

Description

Galley disk storage statistics

Expression

Open FDs (galley)

```
process_open_fds{job="galley"}
```

{{ container_name }}

```
container_fs_usage_bytes{container_name=~"galley", pod_name=~"istio-galley-.*"}
```

Goroutines

Description

Galley Go software statistics

Expression

goroutines_total

```
go_goroutines{job="galley"}
```

clients_total

```
galley_mcp_source_clients_total
```

avg_goroutines_per_client

```
go_goroutines{job="galley"}/galley_mcp_source_clients_total
```

Runtime Panels

Event Rates

Description

Galley event statistics

Expression

Strategy Change Events

```
sum(rate(galley_runtime_strategy_on_change_total[1m])) * 60
```

Processed Events

```
sum(rate(galley_runtime_processor_events_processed_total[1m])) * 60
```

Snapshot Published

```
sum(rate(galley_runtime_processor_snapshots_published_total[1m])) * 60
```

Timer Rates

Description

Galley timer statistics

Expression

Max Time Reached

```
sum(rate(galley_runtime_strategy_timer_max_time_reached_total[1m])) * 60
```

Quiesce Reached

```
sum(rate(galley_runtime_strategy_timer_quiesce_reached_total[1m])) * 60
```

Timer Resets

```
sum(rate(galley_runtime_strategy_timer_resets_total[1m])) * 60
```

Events Per Snapshot

Description

Galley event statistics per snapshot

Expression

P50

```
histogram_quantile(0.50, sum by (le)  
(galley_runtime_processor_snapshot_events_total_bucket))
```

P90

```
histogram_quantile(0.90, sum by (le)  
(galley_runtime_processor_snapshot_events_total_bucket))
```

P95

```
histogram_quantile(0.95, sum by (le)
(galley_runtime_processor_snapshot_events_total_bucket))
```

P99

```
histogram_quantile(0.99, sum by (le)
(galley_runtime_processor_snapshot_events_total_bucket))
```

State Type Instances

Description

State Type Instances

Expression

{{ typeURL }}

```
sum by (typeURL) (galley_runtime_state_type_instances_total)
```

Validation Panels

Validation Webhook Certificate

Description

Galley validation webhook certificate statistics

Expression

Key Updates

```
galley_validation_cert_key_updates{job="galley"}
```

Key Update Errors: {{ error }}

```
galley_validation_cert_key_update_errors{job="galley"}
```

Resource Validation

Description

Resource Validation statistics

Expression

Passed: {{ group }}/{{ version }}/{{resource}}

```
sum(galley_validation_passed{job="galley"}) by (group, version, resource)
```

Failed: {{ group }}/{{ version }}/{{resource}} ({{ reason}})

```
sum(galley_validation_failed{job="galley"}) by (group, version, resource, reason)
```

Validation HTTP Errors

Description

Galley Validation HTTP Errors

Expression

{{ status }}

```
sum(galley_validation_http_error{job="galley"}) by (status)
```

Kubernetes Source Panels

Source Event Rate

Description

Galley source event rates

Expression

Success

```
rate(galley_source_kube_event_success_total[1m]) * 60
```


Error

```
rate(galley_source_kube_event_error_total[1m]) * 60
```

Kubernetes Object Conversion Successes

Description

Kubernetes Object Conversion Successes

Expression

`{{apiVersion}}",group="{{group}}",kind="{{kind}}"`

```
rate(galley_source_kube_dynamic_converter_success_total[1m]) * 60
```

Kubernetes Object Conversion Failures

Description

Kubernetes Object Conversion Failures

Expression

Error

```
rate(galley_source_kube_dynamic_converter_failure_total[1m]) * 60
```

Mesh Configuration Protocol Panels

Connected Clients

Description

Galley connected clients

Expression

Clients

```
sum(galley_mcp_source_clients_total)
```

Request ACKs

Description

Galley request acknowledgements

Expression

```
sum by(collection)(irate(galley_mcp_source_request_acks_total[1m]) * 60)
```

Request NACKs

Description

Galley request nacks for Mesh Configuration Protocol

Expression

```
rate(galley_mcp_source_request_nacks_total[1m]) * 60
```

Istio Mesh Dashboard

Description

Istio Mesh Dashboard version 1.11.4

Panels

Global Request Volume

Expression

```
round(sum(irate(istio_requests_total{reporter="source"}[1m])), 0.001)
```

Global Success Rate (non-5xx responses)

Expression

```
sum(rate(istio_requests_total{reporter="source", response_code!~"5.*"}[1m])) /  
sum(rate(istio_requests_total{reporter="source"}[1m]))
```

4xxs

Expression

```
sum(irate(istio_requests_total{reporter="source", response_code=~"4.*"}[1m]))
```

5xxs

Expression

```
sum(irate(istio_requests_total{reporter="source", response_code=~"5.*"}[1m]))
```

Virtual Services

Expression

```
max(pilot_k8s_cfg_events{type="VirtualService", event="add"}) -  
(max(pilot_k8s_cfg_events{type="VirtualService", event="delete"}) or max(up * 0))
```

Destination Rules

Expression

```
max(pilot_k8s_cfg_events{type="DestinationRule", event="add"}) -  
(max(pilot_k8s_cfg_events{type="DestinationRule", event="delete"}) or max(up * 0))
```

Gateways

Expression

```
max(pilot_k8s_cfg_events{type="Gateway", event="add"}) -  
(max(pilot_k8s_cfg_events{type="Gateway", event="delete"}) or max(up * 0))
```

Service Entries

Expression

```
max(pilot_k8s_cfg_events{type="ServiceEntry", event="add"}) -
(max(pilot_k8s_cfg_events{type="ServiceEntry", event="delete"}) or max(up * 0))
```

HTTP/GRPC Workloads

Expression

```
{{ destination_workload }}.{{ destination_workload_namespace }}
```

```
label_join(sum(rate(istio_requests_total{reporter="destination",
response_code="200"}[1m])) by (destination_workload,
destination_workload_namespace, destination_service), "destination_workload_var",
".", "destination_workload", "destination_workload_namespace")
```

```
{{ destination_workload }}.{{ destination_workload_namespace }}
```

```
label_join((histogram_quantile(0.50,
sum(rate(istio_request_duration_milliseconds_bucket{reporter="destination"}[1m]))
by (le, destination_workload, destination_workload_namespace)) / 1000) or
histogram_quantile(0.50,
sum(rate(istio_request_duration_seconds_bucket{reporter="destination"}[1m])) by
(le, destination_workload, destination_workload_namespace)),
"destination_workload_var", ".", "destination_workload",
"destination_workload_namespace")
```

```
{{ destination_workload }}.{{ destination_workload_namespace }}
```

```
label_join((histogram_quantile(0.90,
sum(rate(istio_request_duration_milliseconds_bucket{reporter="destination"}[1m]))
by (le, destination_workload, destination_workload_namespace)) / 1000) or
histogram_quantile(0.90,
sum(rate(istio_request_duration_seconds_bucket{reporter="destination"}[1m])) by
(le, destination_workload, destination_workload_namespace)),
"destination_workload_var", ".", "destination_workload",
"destination_workload_namespace")
```

```
{{ destination_workload }}.{{ destination_workload_namespace }}
```

```
label_join((histogram_quantile(0.99,
sum(rate(istio_request_duration_milliseconds_bucket{reporter="destination"}[1m]))
```

```
by (le, destination_workload, destination_workload_namespace)) / 1000) or
histogram_quantile(0.99,
sum(rate(istio_request_duration_seconds_bucket{reporter="destination"}[1m])) by
(le, destination_workload, destination_workload_namespace)),
"destination_workload_var", ".", "destination_workload",
"destination_workload_namespace")
```

```
{{ destination_workload }}.{{ destination_workload_namespace }}
```

```
label_join((sum(rate(istio_requests_total{reporter="destination",
response_code!~"5.*"}[1m])) by (destination_workload,
destination_workload_namespace) /
sum(rate(istio_requests_total{reporter="destination"}[1m])) by
(destination_workload, destination_workload_namespace)),
"destination_workload_var", ".", "destination_workload",
"destination_workload_namespace")
```

TCP Workloads

Expression

```
{{ destination_workload }}
```

```
label_join(sum(rate(istio_tcp_received_bytes_total{reporter="source"}[1m])) by
(destination_workload, destination_workload_namespace, destination_service),
"destination_workload_var", ".", "destination_workload",
"destination_workload_namespace")
```

```
{{ destination_workload }}
```

```
label_join(sum(rate(istio_tcp_sent_bytes_total{reporter="source"}[1m])) by
(destination_workload, destination_workload_namespace, destination_service),
"destination_workload_var", ".", "destination_workload",
"destination_workload_namespace")
```

Istio Components by Version

Expression

```
{{ component }}: {{ tag }}
```

```
sum(istio_build) by (component, tag)
```

Istio Mixer Dashboard

Description

Istio Mixer Dashboard

Panels

Mixer Versions

Description

Istio Mixer Versions

Expression

{{ tag }}

```
sum(istio_build{component="mixer"}) by (tag)
```

Memory

Description

Istio memory statistics

Expression

Virtual Memory ({{ job }})

```
sum(process_virtual_memory_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

Resident Memory ({{ job }})

```
sum(process_resident_memory_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

heap sys ({{ job }})

```
sum(go_memstats_heap_sys_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

heap alloc {{{ job }}}

```
sum(go_memstats_heap_alloc_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

Alloc {{{ job }}}

```
sum(go_memstats_alloc_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

Heap in-use {{{ job }}}

```
sum(go_memstats_heap_inuse_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

Stack in-use {{{ job }}}

```
sum(go_memstats_stack_inuse_bytes{job=~"istio-telemetry|istio-policy"}) by (job)
```

{{ service }} total (k8s)

```
sum(label_replace(container_memory_usage_bytes{container_name=~"mixer|istio-proxy", pod_name=~"istio-telemetry-.*|istio-policy-.*"}, "service", "$1", "pod_name", "(istio-telemetry|istio-policy)-.*")) by (service)
```

{{ service }} - {{ container_name }} (k8s)

```
sum(label_replace(container_memory_usage_bytes{container_name=~"mixer|istio-proxy", pod_name=~"istio-telemetry-.*|istio-policy-.*"}, "service", "$1", "pod_name", "(istio-telemetry|istio-policy)-.*")) by (container_name, service)
```

CPU

Description

Istio cpu statistics

Expression

{{ service }} total (k8s)

```
label_replace(sum(rate(container_cpu_usage_seconds_total{container_name=~"mixer|istio-proxy", pod_name=~"istio-telemetry-.*|istio-policy-.*"}[1m])) by (pod_name), "service", "$1", "pod_name", "(istio-telemetry|istio-policy)-.*")
```

{{ service }} - {{ container_name }} (k8s)

```
label_replace(sum(rate(container_cpu_usage_seconds_total{container_name=~"mixer|istio-proxy", pod_name=~"istio-telemetry-.*|istio-policy-.*"}[1m])) by (container_name, pod_name), "service", "$1", "pod_name", "(istio-telemetry|istio-policy)-.*")
```

{{ job }} (self-reported)

```
sum(irate(process_cpu_seconds_total{job=~"istio-telemetry|istio-policy"}[1m])) by (job)
```

Disk

Description

Istio disk storage statistics

Expression

Open FDs ({{ job }})

```
sum(process_open_fds{job=~"istio-telemetry|istio-policy"}) by (job)
```

{{ service }} - {{ container_name }}

```
sum(label_replace(container_fs_usage_bytes{container_name=~"mixer|istio-proxy", pod_name=~"istio-telemetry-.*|istio-policy-.*"}, "service", "$1", "pod_name", "(istio-telemetry|istio-policy)-.*")) by (container_name, service)
```


Goroutines

Description

Istio Go software statistics

Expression

Number of Goroutines ({{ job }})

```
sum(go_goroutines{job=~"istio-telemetry|istio-policy"}) by (job)
```

Incoming Requests

Description

Istio gRPC incoming request statistics

Expression

mixer (Total)

```
sum(rate(grpc_io_server_completed_rpcs[1m]))
```

mixer ({{ grpc_server_method }})

```
sum(rate(grpc_io_server_completed_rpcs[1m])) by (grpc_server_method)
```

Response Durations

Description

Istio gRPC response duration statistics

Expression

{{ grpc_server_method }} 0.5

```
histogram_quantile(0.5, sum(rate(grpc_io_server_server_latency_bucket{}}[1m])) by (grpc_server_method, le))
```

{{ grpc_server_method }} 0.9

```

    histogram_quantile(0.9, sum(rate(grpc_io_server_server_latency_bucket{}[1m])) by
    (grpc_server_method, 1e))

```

{{ grpc_server_method }} 0.99

```

    histogram_quantile(0.99, sum(rate(grpc_io_server_server_latency_bucket{}[1m])) by
    (grpc_server_method, 1e))

```

Server Error Rate (5xx responses)

Description

Istio gRPC Server Error Rate (5xx responses) for 1m

Expression

Mixer {{ grpc_method }}

```

    sum(rate(grpc_server_handled_total{grpc_code=~"Unknown|Unimplemented|Internal|Data
    Loss"}[1m])) by (grpc_method)

```

Non-successes (4xxs)

Description

Istio gRPC Non-successes (4xxs) for 1m

Expression

Mixer {{ grpc_method }}

```

    sum(irate(grpc_server_handled_total{grpc_code!="OK",grpc_service=~".*Mixer"}[1m]))
    by (grpc_method)

```

Adapter Dispatch Count

Description

Istio Mixer Adapter Dispatch Count for 1m

Expression

{{ adapter }}

```
sum(irate(mixer_runtime_dispatches_total{adapter=~"$adapter"}[1m])) by (adapter)
```

Adapter Dispatch Duration

Description

Istio Mixer Adapter Dispatch Duration

Expression

{{ adapter }} - p50

```
histogram_quantile(0.5,  
sum(irate(mixer_runtime_dispatch_duration_seconds_bucket{adapter=~"$adapter"}  
[1m])) by (adapter, 1e))
```

{{ adapter }} - p90

```
histogram_quantile(0.9,  
sum(irate(mixer_runtime_dispatch_duration_seconds_bucket{adapter=~"$adapter"}  
[1m])) by (adapter, 1e))
```

{{ adapter }} - p99

```
histogram_quantile(0.99,  
sum(irate(mixer_runtime_dispatch_duration_seconds_bucket{adapter=~"$adapter"}  
[1m])) by (adapter, 1e))
```

Rules

Description

Istio mixer configuration rule statistics

Expression

Rules

```
scalar(topk(1, max(mixer_config_rule_config_count) by (configID)))
```

Config Errors

```
scalar(topk(1, max(mixer_config_rule_config_error_count) by (configID)))
```

Match Errors

```
scalar(topk(1, max(mixer_config_rule_config_match_error_count) by (configID)))
```

Unsatisfied Actions

```
scalar(topk(1, max(mixer_config_unsatisfied_action_handler_count) by (configID)))
```

Instances in Latest Config

Description

Istio Mixer Configuration Instances in Latest Config

Expression

Instances

```
scalar(topk(1, max(mixer_config_instance_config_count) by (configID)))
```

Handlers in Latest Config

Description

Istio Mixer handlers in Latest Config

Expression

Handlers

```
scalar(topk(1, max(mixer_config_handler_config_count) by (configID)))
```

Attributes in Latest Config

Description

Istio Mixer Attributes in Latest Config

Expression

Attributes

```
scalar(topk(1, max(mixer_config_attribute_count) by (configID)))
```

\$adapter Adapter Panels

Dispatch Count By Handler

Description

Istio Mixer Dispatch Count By Handler

Expression

{{ handler }} (error: {{ error }})

```
label_replace(irate(mixer_runtime_dispatches_total{adapter="$adapter"}  
[1m]), "handler", "$1 ($3)", "handler", "(.*)\\.(.*)\\.(.*)")
```

Dispatch Duration By Handler

Description

Istio Mixer Dispatch Duration By Handler

Expression

p50 - {{ handler_short }} (error: {{ error }})

```
label_replace(histogram_quantile(0.5,  
sum(rate(mixer_runtime_dispatch_duration_seconds_bucket{adapter="$adapter"}[1m]))  
by (handler, error, le)), "handler_short", "$1 ($3)", "handler", "(.*)\\.(.*)\\.(.*)")
```

p90 - {{ handler_short }} (error: {{ error }})

```
label_replace(histogram_quantile(0.9,
sum(irate(mixer_runtime_dispatch_duration_seconds_bucket{adapter="$adapter"}[1m]))
by (handler, error, le)), "handler_short", "$1 ($3)", "handler", "(.*)\\.(.*)\\.(.*)")
```

p99 - {{ handler_short }} (error: {{ error }})

```
label_replace(histogram_quantile(0.99,
sum(irate(mixer_runtime_dispatch_duration_seconds_bucket{adapter="$adapter"}[1m]))
by (handler, error, le)), "handler_short", "$1 ($3)", "handler", "(.*)\\.(.*)\\.(.*)")
```

Istio Performance Dashboard

Description

Istio Performance Dashboard version 1.11.4

Performance Dashboard Notes Panels

Memory and Data Rates Panels

Panels

Bytes transferred / sec

Expression

istio-ingressgateway

```
sum(irate(istio_response_bytes_sum{source_workload="istio-ingressgateway",
reporter="source"}[1m]))
```

istio-proxy

```
sum(irate(istio_response_bytes_sum{source_workload_namespace!="istio-system",
reporter="source"}[1m])) +
```

```
sum(irate(istio_request_bytes_sum{source_workload_namespace!="istio-system",
reporter="source"}[1m]))
```

Istio Component Versions Panels

Istio Components by Version

Expression

```
{{ component }}: {{ tag }}
```

```
sum(istio_build) by (component, tag)
```

Istiod Resource Usage Panels

Memory

Expression

Virtual Memory

```
process_virtual_memory_bytes{service="istiod"}
```

Resident Memory

```
process_resident_memory_bytes{service="istiod"}
```

heap sys

```
go_memstats_heap_sys_bytes{service="istiod"}
```

heap alloc

```
go_memstats_heap_alloc_bytes{service="istiod"}
```

Alloc

```
go_memstats_alloc_bytes{service="istiod"}
```

Heap in-use

```
go_memstats_heap_inuse_bytes{service="istiod"}
```

Stack in-use

```
go_memstats_stack_inuse_bytes{service="istiod"}
```

Total (k8s)

```
sum(container_memory_working_set_bytes{container=~"discovery|istio-proxy",  
pod=~"istiod-.*"})
```

{{ container }} (k8s)

```
container_memory_working_set_bytes{container=~"discovery|istio-proxy",  
pod=~"istiod-.*"}
```

Goroutines

Expression

Number of Goroutines

```
go_goroutines{service="istiod"}
```

Istio Pilot Dashboard

Description

Istio Pilot Dashboard

Panels

Pilot Versions

Description

Istio Pilot Versions

Expression

{{ tag }}

```
sum(istio_build{component="pilot"}) by (tag)
```

Memory

Description

Memory Statistics

Expression

Virtual Memory

```
process_virtual_memory_bytes{job="pilot"}
```

Resident Memory

```
process_resident_memory_bytes{job="pilot"}
```

heap sys

```
go_memstats_heap_sys_bytes{job="pilot"}
```

heap alloc

```
go_memstats_heap_alloc_bytes{job="pilot"}
```

Alloc

```
go_memstats_alloc_bytes{job="pilot"}
```

Heap in-use

```
go_memstats_heap_inuse_bytes{job="pilot"}
```

Stack in-use

```
go_memstats_stack_inuse_bytes{job="pilot"}
```

Total (k8s)

```
sum(container_memory_usage_bytes{container_name=~"discovery|istio-proxy",  
pod_name=~"istio-pilot-.*"})
```

{{ container_name }} (k8s)

```
container_memory_usage_bytes{container_name=~"discovery|istio-proxy",  
pod_name=~"istio-pilot-.*"}
```

CPU

Description

CPU Statistics

Expression

Total (k8s)

```
sum(rate(container_cpu_usage_seconds_total{container_name=~"discovery|istio-  
proxy", pod_name=~"istio-pilot-.*"}[1m]))
```

{{ container_name }} (k8s)

```
sum(rate(container_cpu_usage_seconds_total{container_name=~"discovery|istio-proxy", pod_name=~"istio-pilot-.*"}[1m])) by (container_name)
```

pilot (self-reported)

```
irate(process_cpu_seconds_total{job="pilot"}[1m])
```

Disk

Description

Disk storage statistics

Expression

Open FDs (pilot)

```
process_open_fds{job="pilot"}
```

{{ container_name }}

```
container_fs_usage_bytes{container_name=~"discovery|istio-proxy", pod_name=~"istio-pilot-.*"}
```

Goroutines

Description

Go Routines statistics

Expression

Number of Goroutines

```
go_goroutines{job="pilot"}
```

Updates

Description

xDS gRPC Update Successes

Expression

XDS GRPC Successes

```
sum(irate(envoy_cluster_update_success{cluster_name="xds-grpc"}[1m]))
```

Failures

Description

xDS gRPC update failures

Expression

XDS GRPC

```
round(sum(rate(envoy_cluster_update_attempt{cluster_name="xds-grpc"}[1m])) -  
sum(rate(envoy_cluster_update_success{cluster_name="xds-grpc"}[1m])))
```

Active Connections

Description

xDS gRPC active connections

Expression

Pilot (XDS GRPC)

```
sum(envoy_cluster_upstream_cx_active{cluster_name="xds-grpc"})
```

Conflicts

Description

Inbound and Outbound listener conflicts

Expression

Inbound Listeners

```
pilot_conflict_inbound_listener{job="pilot"}
```

Outbound Listeners (http over current tcp)

```
pilot_conflict_outbound_listener_http_over_current_tcp{job="pilot"}
```

Outbound Listeners (tcp over current tcp)

```
pilot_conflict_outbound_listener_tcp_over_current_tcp{job="pilot"}
```

Outbound Listeners (tcp over current http)

```
pilot_conflict_outbound_listener_tcp_over_current_http{job="pilot"}
```

ADS Monitoring

Description

xDS Statistics

Expression

Virtual Services

```
pilot_virt_services{job="pilot"}
```

Services

```
pilot_services{job="pilot"}
```

Rejected CDS Configs - {{ node }}: {{ err }}

```
label_replace(sum(pilot_xds_cds_reject{job="pilot"}) by (node, err), "node", "$1",  
"node", ".*~.*~(.*?)~.*")
```

Rejected EDS Configs

```
pilot_xds_eds_reject{job="pilot"}
```

Connected Endpoints

```
pilot_xds{job="pilot"}
```

Write Timeouts

```
rate(pilot_xds_write_timeout{job="pilot"}[1m])
```

Push Timeouts

```
rate(pilot_xds_push_timeout{job="pilot"}[1m])
```

Pushes ({{ type }})

```
rate(pilot_xds_pushes{job="pilot"}[1m])
```

Push Errors ({{ type }})

```
rate(pilot_xds_push_errors{job="pilot"}[1m])
```

Rejected CDS Configs

Description

xDS cDS configuration rejects

Expression

{{ node }} ({{ err }})

```
label_replace(sum(pilot_xds_cds_reject{job="pilot"}) by (node, err), "node", "$1",  
"node", ".*~.*~(.*)~.*")
```

Rejected EDS Configs

Description

xDS eDS rejected configurations by node

Expression

```
{{ node }} ({{err}})
```

```
label_replace(sum(pilot_xds_eds_reject{job="pilot"}) by (node, err), "node", "$1",  
"node", ".*~.*~(.*)~.*")
```

Rejected LDS Configs

Description

xDS LDS rejected configurations by node

Expression

```
{{ node }} ({{err}})
```

```
label_replace(sum(pilot_xds_lds_reject{job="pilot"}) by (node, err), "node", "$1",  
"node", ".*~.*~(.*)~.*")
```

Rejected RDS Configs

Description

xDS rDS rejected configurations by node

Expression

```
{{ node }} ({{err}})
```

```
label_replace(sum(pilot_xds_rds_reject{job="pilot"}) by (node, err), "node", "$1",  
"node", ".*~.*~(.*)~.*")
```

EDS Instances

Description

xDS eDS instance count

Expression

{{ cluster }}

```
sum(pilot_xds_eds_instances{job="pilot"}) by (cluster)
```

Istio Service Dashboard

Description

Istio Service Dashboard version 1.11.4

General Panels

Client Request Volume

Expression

```
round(sum(irate(istio_requests_total{reporter=~"$qrep",destination_service=~"$service"}[5m])), 0.001)
```

Client Success Rate (non-5xx responses)

Expression

```
sum(irate(istio_requests_total{reporter=~"$qrep",destination_service=~"$service",response_code!~"5.*"}[5m])) /  
sum(irate(istio_requests_total{reporter=~"$qrep",destination_service=~"$service"}[5m]))
```

Client Request Duration

Expression

P50


```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",destination_service=~"$service"}[1m])) by (le)) / 1000) or histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",destination_service=~"$service"}[1m])) by (le))
```

P90

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",destination_service=~"$service"}[1m])) by (le)) / 1000) or histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",destination_service=~"$service"}[1m])) by (le))
```

P99

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",destination_service=~"$service"}[1m])) by (le)) / 1000) or histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",destination_service=~"$service"}[1m])) by (le))
```

TCP Received Bytes**Expression**

```
sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
destination_service=~"$service"}[1m]))
```

Server Request Volume**Expression**

```
round(sum(irate(istio_requests_total{reporter="destination",destination_service=~"$service"}[5m])), 0.001)
```

Server Success Rate (non-5xx responses)**Expression**

```
sum(irate(istio_requests_total{reporter="destination",destination_service=~"$service",response_code!~"5.*"}[5m])) /  
sum(irate(istio_requests_total{reporter="destination",destination_service=~"$service"}[5m]))
```

Server Request Duration

Expression

P50

```
(histogram_quantile(0.50,  
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",destination_service=~"$service"}[1m])) by (le)) / 1000) or histogram_quantile(0.50,  
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",destination_service=~"$service"}[1m])) by (le))
```

P90

```
(histogram_quantile(0.90,  
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",destination_service=~"$service"}[1m])) by (le)) / 1000) or histogram_quantile(0.90,  
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",destination_service=~"$service"}[1m])) by (le))
```

P99

```
(histogram_quantile(0.99,  
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",destination_service=~"$service"}[1m])) by (le)) / 1000) or histogram_quantile(0.99,  
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",destination_service=~"$service"}[1m])) by (le))
```

TCP Sent Bytes

Expression

```
sum(irate(istio_tcp_sent_bytes_total{reporter=~"$qrep",  
destination_service=~"$service"}[1m]))
```

Client Workloads Panels

Incoming Requests By Source And Response Code

Expression

{{ source_workload }}.{{ source_workload_namespace }} : {{ response_code }} (🔒mTLS)

```
round(sum(irate(istio_requests_total{connection_security_policy="mutual_tls",destination_service=~"$service",reporter=~"$qrep",source_workload=~"$srcwl",source_workload_namespace=~"$srcns"}[5m])) by (source_workload, source_workload_namespace, response_code), 0.001)
```

{{ source_workload }}.{{ source_workload_namespace }} : {{ response_code }}

```
round(sum(irate(istio_requests_total{connection_security_policy!="mutual_tls",destination_service=~"$service",reporter=~"$qrep",source_workload=~"$srcwl",source_workload_namespace=~"$srcns"}[5m])) by (source_workload, source_workload_namespace, response_code), 0.001)
```

Incoming Success Rate (non-5xx responses) By Source

Expression

{{ source_workload }}.{{ source_workload_namespace }} (🔒mTLS)

```
sum(irate(istio_requests_total{reporter=~"$qrep",connection_security_policy="mutual_tls",destination_service=~"$service",response_code!~"5.*",source_workload=~"$srcwl",source_workload_namespace=~"$srcns"}[5m])) by (source_workload, source_workload_namespace) / sum(irate(istio_requests_total{reporter=~"$qrep",connection_security_policy="mutual_tls",destination_service=~"$service",source_workload=~"$srcwl",source_workload_namespace=~"$srcns"}[5m])) by (source_workload, source_workload_namespace)
```

{{ source_workload }}.{{ source_workload_namespace }}

```
sum(irate(istio_requests_total{reporter=~"$qrep",connection_security_policy!="mutual_tls",destination_service=~"$service",response_code!~"5.*",source_workload=~"$srcwl",source_workload_namespace=~"$srcns"}[5m])) by (source_workload, source_workload_namespace) / sum(irate(istio_requests_total{reporter=~"$qrep",
```

```
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[5m])) by
(source_workload, source_workload_namespace)
```

Incoming Request Duration By Source

Description

Expression

{{source_workload}}.{{source_workload_namespace}} P50 (1mTLS)

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P90 (1mTLS)

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P95 (1mTLS)

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.95,
```

```
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P99 (90mTLS)

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P50

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P90

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P95

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P99

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le)) / 1000) or
histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

Incoming Request Size By Source**Expression****{{source_workload}}.{{source_workload_namespace}} P50 (mTLS)**

```
histogram_quantile(0.50, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P90 (mTLS)

```
histogram_quantile(0.90, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
```

```
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P95 (mTLS)

```
histogram_quantile(0.95, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P99 (mTLS)

```
histogram_quantile(0.99, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P50

```
histogram_quantile(0.50, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P90

```
histogram_quantile(0.90, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P95

```
histogram_quantile(0.95, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P99

```

histogram_quantile(0.99, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

Response Size By Source

Expression

{{source_workload}}.{{source_workload_namespace}} P50 (mTLS)

```

histogram_quantile(0.50, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

{{source_workload}}.{{source_workload_namespace}} P90 (mTLS)

```

histogram_quantile(0.90, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

{{source_workload}}.{{source_workload_namespace}} P95 (mTLS)

```

histogram_quantile(0.95, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

{{source_workload}}.{{source_workload_namespace}} P99 (mTLS)

```

histogram_quantile(0.99, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```


{{source_workload}}.{{source_workload_namespace}} P50

```

histogram_quantile(0.50, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

{{source_workload}}.{{source_workload_namespace}} P90

```

histogram_quantile(0.90, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

{{source_workload}}.{{source_workload_namespace}} P95

```

histogram_quantile(0.95, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

{{source_workload}}.{{source_workload_namespace}} P99

```

histogram_quantile(0.99, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace, le))

```

Bytes Received from Incoming TCP Connection**Expression**

{{ source_workload }}.{{ source_workload_namespace}} (mTLS)

```

round(sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace), 0.001)

```

```
{{ source_workload }}.{{ source_workload_namespace }}
```

```
round(sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_service=~"$service",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace), 0.001)
```

Bytes Sent to Incoming TCP Connection

Expression

```
{{ source_workload }}.{{ source_workload_namespace }} (🔗mTLS)
```

```
round(sum(irate(istio_tcp_sent_bytes_total{connection_security_policy="mutual_tls"
, reporter=~"$qrep", destination_service=~"$service", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace), 0.001)
```

```
{{ source_workload }}.{{ source_workload_namespace }}
```

```
round(sum(irate(istio_tcp_sent_bytes_total{connection_security_policy!="mutual_tls"
, reporter=~"$qrep", destination_service=~"$service", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace), 0.001)
```

Service Workloads Panels

Incoming Requests By Destination Workload And Response Code

Expression

```
{{ destination_workload }}.{{ destination_workload_namespace }} : {{ response_code }} (🔗mTLS)
```

```
round(sum(irate(istio_requests_total{connection_security_policy="mutual_tls",desti
nation_service=~"$service",reporter="destination",destination_workload=~"$dstwl",d
estination_workload_namespace=~"$dstns"}[5m])) by (destination_workload,
destination_workload_namespace, response_code), 0.001)
```

```
{{ destination_workload }}.{{ destination_workload_namespace }} : {{ response_code }}
```

```
round(sum(irate(istio_requests_total{connection_security_policy!="mutual_tls",
destination_service=~"$service", reporter="destination",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[5m])) by
(destination_workload, destination_workload_namespace, response_code), 0.001)
```

Incoming Success Rate (non-5xx responses) By Destination Workload

Expression

{{ destination_workload }}.{{ destination_workload_namespace }} (⚠️mTLS)

```
sum(irate(istio_requests_total{reporter="destination",
connection_security_policy="mutual_tls",
destination_service=~"$service", response_code!~"5.*",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[5m])) by
(destination_workload, destination_workload_namespace) /
sum(irate(istio_requests_total{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[5m])) by
(destination_workload, destination_workload_namespace)
```

{{ destination_workload }}.{{ destination_workload_namespace }}

```
sum(irate(istio_requests_total{reporter="destination",
connection_security_policy!="mutual_tls",
destination_service=~"$service", response_code!~"5.*",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[5m])) by
(destination_workload, destination_workload_namespace) /
sum(irate(istio_requests_total{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[5m])) by
(destination_workload, destination_workload_namespace)
```

Incoming Request Duration By Service Workload

Description

Expression

{{ destination_workload }}.{{ destination_workload_namespace }} P50 (⚠️mTLS)

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
```

```

connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P90 (🔗mTLS)

```

(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P95 (🔗mTLS)

```

(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P99 (🔗mTLS)

```

(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",

```

```
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P50

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P90

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P95

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P99

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le)) / 1000) or
histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

Incoming Request Size By Service Workload

Expression

{{ destination_workload }}.{{ destination_workload_namespace }} P50 (mTLS)

```
histogram_quantile(0.50,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P90 (mTLS)

```
histogram_quantile(0.90,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P95 (mTLS)

```
histogram_quantile(0.95,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P99 (mTLS)

```

histogram_quantile(0.99,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P50

```

histogram_quantile(0.50,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P90

```

histogram_quantile(0.90,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P95

```

histogram_quantile(0.95,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P99

```

histogram_quantile(0.99,
sum(irate(istio_request_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, le))

```

Response Size By Service Workload

Expression

{{ destination_workload }}.{{ destination_workload_namespace }} P50 (🔗mTLS)

```

    histogram_quantile(0.50,
    sum(irate(istio_response_bytes_bucket{reporter="destination",
    connection_security_policy="mutual_tls", destination_service=~"$service",
    destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
    (destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P90 (🔗mTLS)

```

    histogram_quantile(0.90,
    sum(irate(istio_response_bytes_bucket{reporter="destination",
    connection_security_policy="mutual_tls", destination_service=~"$service",
    destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
    (destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P95 (🔗mTLS)

```

    histogram_quantile(0.95,
    sum(irate(istio_response_bytes_bucket{reporter="destination",
    connection_security_policy="mutual_tls", destination_service=~"$service",
    destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
    (destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P99 (🔗mTLS)

```

    histogram_quantile(0.99,
    sum(irate(istio_response_bytes_bucket{reporter="destination",
    connection_security_policy="mutual_tls", destination_service=~"$service",
    destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
    (destination_workload, destination_workload_namespace, le))

```

{{ destination_workload }}.{{ destination_workload_namespace }} P50

```

    histogram_quantile(0.50,
    sum(irate(istio_response_bytes_bucket{reporter="destination",

```



```
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, 1e))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P90

```
histogram_quantile(0.90,
sum(irate(istio_response_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, 1e))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P95

```
histogram_quantile(0.95,
sum(irate(istio_response_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, 1e))
```

{{ destination_workload }}.{{ destination_workload_namespace }} P99

```
histogram_quantile(0.99,
sum(irate(istio_response_bytes_bucket{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace, 1e))
```

Bytes Received from Incoming TCP Connection

Expression

{{ destination_workload }}.{{ destination_workload_namespace }} ( mTLS)

```
round(sum(irate(istio_tcp_received_bytes_total{reporter="destination",
connection_security_policy="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace), 0.001)
```

{{ destination_workload }}.{{ destination_workload_namespace }}

```
round(sum(irate(istio_tcp_received_bytes_total{reporter="destination",
connection_security_policy!="mutual_tls", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace), 0.001)
```

Bytes Sent to Incoming TCP Connection

Expression

{{ destination_workload }}.{{destination_workload_namespace }} (mTLS)

```
round(sum(irate(istio_tcp_sent_bytes_total{connection_security_policy="mutual_tls"
, reporter="destination", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace), 0.001)
```

{{ destination_workload }}.{{destination_workload_namespace }}

```
round(sum(irate(istio_tcp_sent_bytes_total{connection_security_policy!="mutual_tls"
, reporter="destination", destination_service=~"$service",
destination_workload=~"$dstwl", destination_workload_namespace=~"$dstns"}[1m])) by
(destination_workload, destination_workload_namespace), 0.001)
```

Istio Workload Dashboard

Description

Istio Workload Dashboard version 1.11.4

General Panels

Incoming Request Volume

Expression

```
round(sum(irate(istio_requests_total{reporter=~"$qrep",destination_workload_namesp
ace=~"$namespace",destination_workload=~"$workload"}[5m])), 0.001)
```

Incoming Success Rate (non-5xx responses)

Expression

```
sum(irate(istio_requests_total{reporter=~"$qrep",destination_workload_namespace=~"$namespace",destination_workload=~"$workload",response_code!~"5.*"}[5m])) /
sum(irate(istio_requests_total{reporter=~"$qrep",destination_workload_namespace=~"$namespace",destination_workload=~"$workload"}[5m]))
```

Request Duration

Expression

P50

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",destination_workload=~"$workload", destination_workload_namespace=~"$namespace"}[1m])) by
(1e)) / 1000) or histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",destination_workload=~"$workload", destination_workload_namespace=~"$namespace"}[1m])) by (1e))
```

P90

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",destination_workload=~"$workload", destination_workload_namespace=~"$namespace"}[1m])) by
(1e)) / 1000) or histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",destination_workload=~"$workload", destination_workload_namespace=~"$namespace"}[1m])) by (1e))
```

P99

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",destination_workload=~"$workload", destination_workload_namespace=~"$namespace"}[1m])) by
(1e)) / 1000) or histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",destination_workload=~"$workload", destination_workload_namespace=~"$namespace"}[1m])) by (1e))
```

TCP Server Traffic

Expression

```
sum(irate(istio_tcp_sent_bytes_total{reporter=~"$qrep",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload"}
[1m])) + sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload"}
[1m]))
```

TCP Client Traffic

Expression

```
sum(irate(istio_tcp_sent_bytes_total{reporter=~"$qrep",
source_workload_namespace=~"$namespace", source_workload=~"$workload"}[1m])) +
sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
source_workload_namespace=~"$namespace", source_workload=~"$workload"}[1m]))
```

Inbound Workloads Panels

Incoming Requests By Source And Response Code

Expression

{{ source_workload }}.{{ source_workload_namespace }} : {{ response_code }} ( mTLS)

```
round(sum(irate(istio_requests_total{connection_security_policy="mutual_tls",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload",
reporter=~"$qrep", source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}
[5m])) by (source_workload, source_workload_namespace, response_code), 0.001)
```

{{ source_workload }}.{{ source_workload_namespace }} : {{ response_code }}

```
round(sum(irate(istio_requests_total{connection_security_policy!="mutual_tls",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload",
reporter=~"$qrep", source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}
[5m])) by (source_workload, source_workload_namespace, response_code), 0.001)
```

Incoming Success Rate (non-5xx responses) By Source

Expression

{{ source_workload }}.{{ source_workload_namespace }} ( mTLS)

```
sum(irate(istio_requests_total{reporter=~"$qrep",
connection_security_policy="mutual_tls",
destination_workload_namespace=~"$namespace",
destination_workload=~"$workload",response_code!~"5.*", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[5m])) by (source_workload,
source_workload_namespace) / sum(irate(istio_requests_total{reporter=~"$qrep",
connection_security_policy="mutual_tls",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[5m])) by
(source_workload, source_workload_namespace)
```

{{ source_workload }}.{{ source_workload_namespace }}

```
sum(irate(istio_requests_total{reporter=~"$qrep",
connection_security_policy!="mutual_tls",
destination_workload_namespace=~"$namespace",
destination_workload=~"$workload",response_code!~"5.*", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[5m])) by (source_workload,
source_workload_namespace) / sum(irate(istio_requests_total{reporter=~"$qrep",
connection_security_policy!="mutual_tls",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[5m])) by
(source_workload, source_workload_namespace)
```

Incoming Request Duration By Source

Description

Expression

{{source_workload}}.{{source_workload_namespace}} P50 ( mTLS)

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P90 (🔗mTLS)

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P95 (🔗mTLS)

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P99 (🔗mTLS)

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P50

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P90

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P95

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le)) / 1000) or histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P99

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e)) / 1000) or histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

Incoming Request Size By Source

Expression

{{source_workload}}.{{source_workload_namespace}} P50 (🔗 mTLS)

```
histogram_quantile(0.50, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P90 (🔗 mTLS)

```
histogram_quantile(0.90, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P95 (🔗 mTLS)

```
histogram_quantile(0.95, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P99 (🔗 mTLS)


```
histogram_quantile(0.99, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P50

```
histogram_quantile(0.50, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P90

```
histogram_quantile(0.90, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P95

```
histogram_quantile(0.95, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P99

```
histogram_quantile(0.99, sum(irate(istio_request_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

Expression

{{source_workload}}.{{source_workload_namespace}} P50 (🔗mTLS)

```
histogram_quantile(0.50, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P90 (🔗mTLS)

```
histogram_quantile(0.90, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P95 (🔗mTLS)

```
histogram_quantile(0.95, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P99 (🔗mTLS)

```
histogram_quantile(0.99, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, le))
```

{{source_workload}}.{{source_workload_namespace}} P50

```
histogram_quantile(0.50, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
```

```
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P90

```
histogram_quantile(0.90, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P95

```
histogram_quantile(0.95, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

{{source_workload}}.{{source_workload_namespace}} P99

```
histogram_quantile(0.99, sum(irate(istio_response_bytes_bucket{reporter=~"$qrep",
connection_security_policy!="mutual_tls", destination_workload=~"$workload",
destination_workload_namespace=~"$namespace", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace, 1e))
```

Bytes Received from Incoming TCP Connection

Expression

{{ source_workload }}.{{ source_workload_namespace }} (mTLS)

```
round(sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
connection_security_policy="mutual_tls",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace), 0.001)
```

{{ source_workload }}.{{ source_workload_namespace }}

```
round(sum(irate(istio_tcp_received_bytes_total{reporter=~"$qrep",
connection_security_policy!="mutual_tls",
destination_workload_namespace=~"$namespace", destination_workload=~"$workload",
source_workload=~"$srcwl", source_workload_namespace=~"$srcns"}[1m])) by
(source_workload, source_workload_namespace), 0.001)
```

Bytes Sent to Incoming TCP Connection

Expression

{{ source_workload }}.{{ source_workload_namespace }} (🔗mTLS)

```
round(sum(irate(istio_tcp_sent_bytes_total{connection_security_policy="mutual_tls"
, reporter=~"$qrep", destination_workload_namespace=~"$namespace",
destination_workload=~"$workload", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace), 0.001)
```

{{ source_workload }}.{{ source_workload_namespace }}

```
round(sum(irate(istio_tcp_sent_bytes_total{connection_security_policy!="mutual_tls"
, reporter=~"$qrep", destination_workload_namespace=~"$namespace",
destination_workload=~"$workload", source_workload=~"$srcwl",
source_workload_namespace=~"$srcns"}[1m])) by (source_workload,
source_workload_namespace), 0.001)
```

Outbound Services Panels

Outgoing Requests By Destination And Response Code

Expression

{{ destination_service }} : {{ response_code }} (🔗mTLS)

```
round(sum(irate(istio_requests_total{destination_principal=~"spiffe.*",
source_workload_namespace=~"$namespace", source_workload=~"$workload",
reporter="source", destination_service=~"$dstsvc"}[5m])) by (destination_service,
response_code), 0.001)
```

{{ destination_service }} : {{ response_code }}

```
round(sum(irate(istio_requests_total{destination_principal!~"spiffe.*",
source_workload_namespace=~"$namespace", source_workload=~"$workload",
reporter="source", destination_service=~"$dstsvc"}[5m])) by (destination_service,
response_code), 0.001)
```

Outgoing Success Rate (non-5xx responses) By Destination

Expression

{{ destination_service }} (⚠️ mTLS)

```
sum(irate(istio_requests_total{reporter="source",
connection_security_policy="mutual_tls", source_workload_namespace=~"$namespace",
source_workload=~"$workload", response_code!~"5.*", destination_service=~"$dstsvc"}
[5m])) by (destination_service) /
sum(irate(istio_requests_total{reporter="source",
connection_security_policy="mutual_tls", source_workload_namespace=~"$namespace",
source_workload=~"$workload", destination_service=~"$dstsvc"}[5m])) by
(destination_service)
```

{{ destination_service }}

```
sum(irate(istio_requests_total{reporter="source",
connection_security_policy!="mutual_tls", source_workload_namespace=~"$namespace",
source_workload=~"$workload", response_code!~"5.*", destination_service=~"$dstsvc"}
[5m])) by (destination_service) /
sum(irate(istio_requests_total{reporter="source",
connection_security_policy!="mutual_tls", source_workload_namespace=~"$namespace",
source_workload=~"$workload", destination_service=~"$dstsvc"}[5m])) by
(destination_service)
```

Outgoing Request Duration By Destination

Description

Expression

{{ destination_service }} P50 (⚠️ mTLS)

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
```

```
(destination_service, le)) / 1000) or histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P90 (mTLS)

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le)) / 1000) or histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P95 (mTLS)

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le)) / 1000) or histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P99 (mTLS)

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le)) / 1000) or histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P50

```
(histogram_quantile(0.50,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le)) / 1000) or histogram_quantile(0.50,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P90

```
(histogram_quantile(0.90,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le)) / 1000) or histogram_quantile(0.90,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P95

```
(histogram_quantile(0.95,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le)) / 1000) or histogram_quantile(0.95,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P99

```
(histogram_quantile(0.99,
sum(irate(istio_request_duration_milliseconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
```

```
(destination_service, le)) / 1000) or histogram_quantile(0.99,
sum(irate(istio_request_duration_seconds_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

Outgoing Request Size By Destination

Expression

{{ destination_service }} P50 (🕒mTLS)

```
histogram_quantile(0.50, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P90 (🕒mTLS)

```
histogram_quantile(0.90, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P95 (🕒mTLS)

```
histogram_quantile(0.95, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P99 (🕒mTLS)

```
histogram_quantile(0.99, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, le))
```

{{ destination_service }} P50


```

histogram_quantile(0.50, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P90

```

histogram_quantile(0.90, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P95

```

histogram_quantile(0.95, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P99

```

histogram_quantile(0.99, sum(irate(istio_request_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

Response Size By Destination

Expression

{{ destination_service }} P50 (mTLS)

```

histogram_quantile(0.50, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P90 (mTLS)

```

histogram_quantile(0.90, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P95 (mTLS)

```

histogram_quantile(0.95, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P99 (mTLS)

```

histogram_quantile(0.99, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P50

```

histogram_quantile(0.50, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P90

```

histogram_quantile(0.90, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))

```

{{ destination_service }} P95

```

histogram_quantile(0.95, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",

```

```
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))
```

{{ destination_service }} P99

```
histogram_quantile(0.99, sum(irate(istio_response_bytes_bucket{reporter="source",
connection_security_policy!="mutual_tls", source_workload=~"$workload",
source_workload_namespace=~"$namespace", destination_service=~"$dstsvc"}[1m])) by
(destination_service, 1e))
```

Bytes Sent on Outgoing TCP Connection

Expression

{{ destination_service }} ( mTLS)

```
round(sum(irate(istio_tcp_received_bytes_total{connection_security_policy="mutual_
tls", reporter="source", source_workload_namespace=~"$namespace",
source_workload=~"$workload", destination_service=~"$dstsvc"}[1m])) by
(destination_service), 0.001)
```

{{ destination_service }}

```
round(sum(irate(istio_tcp_received_bytes_total{connection_security_policy!="mutual_
tls", reporter="source", source_workload_namespace=~"$namespace",
source_workload=~"$workload", destination_service=~"$dstsvc"}[1m])) by
(destination_service), 0.001)
```

Bytes Received from Outgoing TCP Connection

Expression

{{ destination_service }} ( mTLS)

```
round(sum(irate(istio_tcp_sent_bytes_total{reporter="source",
connection_security_policy="mutual_tls", source_workload_namespace=~"$namespace",
source_workload=~"$workload", destination_service=~"$dstsvc"}[1m])) by
(destination_service), 0.001)
```

{{ destination_service }}

```
round(sum(irate(istio_tcp_sent_bytes_total{reporter="source",
connection_security_policy!="mutual_tls", source_workload_namespace=~"$namespace",
source_workload=~"$workload", destination_service=~"$dstsvc"}[1m])) by
(destination_service), 0.001)
```

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- [Summary](#)
- [Jaeger](#)

Summary

Dashboard Name	Panel Name
Jaeger	Jaeger Collector-Spans Received/sec
Jaeger	Jaeger Collector-Average in-queue latency (sec)
Jaeger	Jaeger Collector-Batch Size (spans)
Jaeger	Jaeger Collector-Queue Length (Spans)
Jaeger	Jaeger Collector-Spans Dropped/sec
Jaeger	Jaeger Collector-Span save latency (sec)
Jaeger	Jaeger Query-Average request latency (milliseconds)
Jaeger	Jaeger Query-Requests/min (All requests)
Jaeger	Jaeger Query-jaeger_tracer_sampler_queries_total
Jaeger	Jaeger Query-jaeger_query_responses_sum

Jaeger

Description

Dashboard for monitoring jaeger running in a k8s environment. Works with 1.9+. Feedbacks? Please send to luong.vo@employmentthero.com

Jaeger Collector Panels

Spans Received/sec

Expression

```
sum(rate(jaeger_collector_spans_received_total[5m]))
```

Average in-queue latency (sec)

Expression

```
jaeger_collector_in_queue_latency_sum / jaeger_collector_in_queue_latency_count
```

Batch Size (spans)

Expression

```
jaeger_collector_batch_size
```

Queue Length (Spans)

Expression

```
jaeger_collector_queue_length
```

Spans Dropped/sec

Expression

```
rate(jaeger_collector_spans_dropped_total[5m])
```

Span save latency (sec)

Expression

```
jaeger_collector_save_latency_sum / jaeger_collector_save_latency_count
```

Jaeger Query Panels

Average request latency (milliseconds)

Expression

jaeger_query_latency_sum

Requests/min (All requests)

Expression

All requests (/min)

sum(rate(jaeger_query_requests_total[5m]))*60

2xx Requests (/min)

sum(rate(jaeger_query_requests_total{status_code=~"2xx"}[5m]))*60

Non-2xx Requests (/min)

sum(rate(jaeger_query_requests_total{status_code!="2xx"}[5m]))*60

jaeger tracer sampler queries total

Expression

All requests (/min)

jaeger_tracer_sampler_queries_total

jaeger query responses sum

Expression

All requests (/min)

jaeger_query_responses_sum

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- [Summary](#)
- [Kafka Overview](#)

Summary

Dashboard Name	Panel Name
Kafka Overview	CPU Usage
Kafka Overview	JVM Memory Used
Kafka Overview	Time spent in GC
Kafka Overview	Messages In Per Topic
Kafka Overview	Bytes In Per Topic
Kafka Overview	Bytes Out Per Topic

Kafka Overview

Description

Kafka resource usage and throughput

Panels

CPU Usage

Description

This panel description is about CPU Usage

Expression

{{instance}}

```
rate(process_cpu_seconds_total{job="kafka-headless"}[2m])
```

JVM Memory Used

Description

This panel description is about JVM memory Used.

Expression

{{instance}}

```
sum without(area)(jvm_memory_used_bytes{job="kafka-headless"})
```

Time spent in GC

Description

Time Spent in GC

Expression

{{instance}}

```
sum without(gc)(rate(jvm_gc_collection_seconds_sum{job="kafka-headless"}[5m]))
```

Messages In Per Topic

Description

Messages in per topic

Expression

{{topic}}

```
sum without(instance)
(rate(kafka_server_brokertopicmetrics_messagesin_total{job="kafka-
headless",topic!=""}[5m]))
```

Bytes In Per Topic

Description

Bytes in per topic

Expression

{{topic}}

```
sum without(instance)
(rate(kafka_server_brokertopicmetrics_bytesin_total{job="kafka-
```



```
headless",topic!=""}[5m]))
```

Bytes Out Per Topic

Description

Bytes out per topic

Expression

{{topic}}

```
sum without(instance)
(rate(kafka_server_brokertopicmetrics_bytesout_total{job="kafka-
headless",topic!=""}[5m]))
```

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- [Summary](#)
- [Kibana](#)

Summary

Dashboard Name	Panel Name
Kibana	Uptime
Kibana	Samples ingested (rate-5m)

Kibana

Description

Kibana Dashboard

Panels

Uptime

Description

Uptime

Expression

```
kibana_millis_uptime
```

Samples ingested (rate-5m)

Description

Samples ingested

Expression

OS Load 1m

```
kibana_os_load1
```

OS Load 5m

```
kibana_os_load5
```

OS Load 15m

```
kibana_os_load15
```

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- [Summary](#)
- [Certificates Expiration \(X509 Certificate Exporter\)](#)

Summary

Dashboard Name	Panel Name
Certificates Expiration (X509 Certificate Exporter)	Overview
Certificates Expiration (X509 Certificate Exporter)	Total Certificates
Certificates Expiration (X509 Certificate Exporter)	Expired

Dashboard Name	Panel Name
Certificates Expiration (X509 Certificate Exporter)	Expiring within \$critical_threshold days
Certificates Expiration (X509 Certificate Exporter)	Expiring within \$warning_threshold days
Certificates Expiration (X509 Certificate Exporter)	Exporters
Certificates Expiration (X509 Certificate Exporter)	Exporter Errors
Certificates Expiration (X509 Certificate Exporter)	Expiration
Certificates Expiration (X509 Certificate Exporter)	Kubernetes Secrets (time left < \$list_threshold days)
Certificates Expiration (X509 Certificate Exporter)	Host Files (time left < \$list_threshold days)
Certificates Expiration (X509 Certificate Exporter)	Charts
Certificates Expiration (X509 Certificate Exporter)	Top Issuers
Certificates Expiration (X509 Certificate Exporter)	Top Namespaces (Kubernetes Secrets)
Certificates Expiration (X509 Certificate Exporter)	Top Instances (Host Paths)
Certificates Expiration (X509 Certificate Exporter)	Kubernetes Secrets : Shortest Validity Period
Certificates Expiration (X509 Certificate Exporter)	Host Paths : Shortest Validity Period
Certificates Expiration (X509 Certificate Exporter)	Kubernetes Secrets : Longest Validity Period
Certificates Expiration (X509 Certificate Exporter)	Host Paths : Longest Validity Period
Certificates Expiration (X509 Certificate Exporter)	Exporters
Certificates Expiration (X509 Certificate Exporter)	Reporting Exporters
Certificates Expiration (X509 Certificate Exporter)	Exporters with Errors
Certificates Expiration (X509 Certificate Exporter)	Error Rate
Certificates Expiration (X509 Certificate Exporter)	Cumulative Errors
Certificates Expiration (X509 Certificate Exporter)	Top Exporters by Error Rate
Certificates Expiration (X509 Certificate Exporter)	Top Exporters by Cumulative Errors

Certificates Expiration (X509 Certificate Exporter)

Description

Unified dashboard for checking certificates expiration: Kubernetes Secrets, certificate files on nodes, or on any server.

Panels

Overview

Total Certificates

Expression

```
count(x509_cert_not_after)
```

Expired

Expression

```
sum(((x509_cert_not_after - time()) / 86400) < bool 0)
```

Expiring within \$critical threshold days

Expression

```
sum(0 < ((x509_cert_not_after - time()) / 86400) < bool $critical_threshold)
```

Expiring within \$warning threshold days

Expression

```
sum(0 < ((x509_cert_not_after - time()) / 86400) < bool $warning_threshold)
```

Exporters

Expression

```
count(x509_read_errors)
```

Exporter Errors

Expression

```
sum(x509_read_errors)
```

Expiration

Kubernetes Secrets (time left < \$list threshold days)

Description

Because of a missing feature in Grafana, critical and warning thresholds from dashboard variables will not affect coloration of the Time Left column in this table.

Thresholds are to be set manually in the Overrides settings for this widget.

Please vote or contribute to issue : <https://github.com/grafana/grafana/issues/922>

Expression

```
sort(((x509_cert_not_after{secret_name!=""} - time()) / 86400) < $list_threshold)
```

Host Files (time left < \$list threshold days)

Description

Because of a missing feature in Grafana, critical and warning thresholds from dashboard variables will not affect coloration of the Time Left column in this table.

Thresholds are to be set manually in the Overrides settings for this widget.

Please vote or contribute to issue : <https://github.com/grafana/grafana/issues/922>

Expression

```
sort(((x509_cert_not_after{filepath!=""} - time()) / 86400) < $list_threshold)
```

Charts

Top Issuers

Expression

```
topk(10, sort_desc(count by (issuer_CN) (x509_cert_not_after)))
```

Top Namespaces (Kubernetes Secrets)

Expression

```
topk(10, sort_desc(count by (secret_namespace)
(x509_cert_not_after{secret_namespace!=""})))
```

Top Instances (Host Paths)

Expression

```
topk(10, sort_desc(count by (instance) (x509_cert_not_after{filepath!=""})))
```

Kubernetes Secrets : Shortest Validity Period

Expression

```
bottomk(10, (x509_cert_not_after{secret_name!=""} - x509_cert_not_before) / 86400)
```

Host Paths : Shortest Validity Period

Expression

```
bottomk(10, (x509_cert_not_after{filepath!=""} - x509_cert_not_before) / 86400)
```

Kubernetes Secrets : Longest Validity Period

Expression

```
topk(10, (x509_cert_not_after{secret_name!=""} - x509_cert_not_before) / 86400)
```

Host Paths : Longest Validity Period

Expression

```
topk(10, (x509_cert_not_after{filepath!=""} - x509_cert_not_before) / 86400)
```

Exporters

Reporting Exporters

Expression

exporters

```
count(x509_read_errors)
```

Exporters with Errors

Expression

exporters with errors

```
sum (x509_read_errors > bool 0)
```

Error Rate

Expression

error rate

```
sum(rate(x509_read_errors[15m]))
```

Cumulative Errors

Expression

errors

```
sum(x509_read_errors)
```

Top Exporters by Error Rate

Expression

```
topk(10, rate(x509_read_errors[6h]))
```

Top Exporters by Cumulative Errors

Expression

```
topk(10, x509_read_errors)
```

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MME PROCEDURE DURATION STATISTICS	\$MmProcedure Count
MME PROCEDURE DURATION STATISTICS	[\$MmProcedure
MME PROCEDURE DURATION STATISTICS	[\$MmProcedure
MME PROCEDURE DURATION STATISTICS	\$MmProcedure Count
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MME PROCEDURE DURATION STATISTICS	\$SmProcedure count
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Dashboard Name	Panel Name
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MME-CC Dashboard	
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MME PROCEDURE DURATION STATISTICS

Description

This statistic shows the duration of the procedure in seconds

Panels

\$MmProcedure Count

Description

Mm Procedure Durations

Expression

{{procedure}}

```
sum by (procedure) (mmecc_mm_procedure_duration_count{procedure=~"$MmProcedure"})
```

\$MmProcedure | SUCCESS

Description

Duration of Successful MM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(mmecc_mm_procedure_duration_bucket{procedure=~"$MmProcedure",status="SUCCESS"})
```

\$MmProcedure | FAILURE

Description

Duration of Failed MM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(mmecc_mm_procedure_duration_bucket{procedure=~"$MmProcedure",status="FAILURE"})
```

\$MmProcedure Count

Description

Mm Procedure Durations

Expression

{{procedure}}

```
sum by (procedure) (mmecc_mm_procedure_duration_count{procedure=~"$MmProcedure"})
```

\$MmProcedure | SUCCESS

Description

Duration of Successful MM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(mmecc_mm_procedure_duration_bucket{procedure=~"$MmProcedure",status="SUCCESS"})
```

\$MmProcedure | FAILURE

Description

Duration of Failed MM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(mmecc_mm_procedure_duration_bucket{procedure=~"$MmProcedure",status="FAILURE"})
```

\$SmProcedure count

Description

SM Procedure Durations

Expression

{{procedure}}

```
sum by (procedure) (mmecc_sm_procedure_duration_count{procedure=~"$SmProcedure"})
```

\$SmProcedure | SUCCESS

Description

Duration of Successful SM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(mmecc_sm_procedure_duration_bucket{procedure=~"$SmProcedure",status="SUCCESS"})
```

\$SmProcedure | FAILURE

Description

Duration of Failed SM procedures in seconds.

Expression

{{le}}

```
sum by (procedure,le)
(mmecc_sm_procedure_duration_bucket{procedure=~"$SmProcedure",status="FAILURE"})
```

MME-CC Dashboard

Description

This dashboard lists the metrics on pod MME CC.

Panels

Success % for Attach Procedure

Description

This panel lists the Attach procedure success stats.

Expression

Attach Success %

```
((sum(mmecc_emm_attach_eps_success{pod="$PodName"}))/(sum(mmecc_emm_attach_eps_att
empt{pod="$PodName"})))*100
```

Success % for Detach Procedure

Description

This panel lists the Detach procedure success stats.

Expression

Detach Success %

```
((sum(mmecc_emm_eps_detach_ue_success{pod="$PodName"}))/(sum(mmecc_emm_eps_detach_
ue_attempt{pod="$PodName"})))*100
```

Success % for Service Request Procedure

Description

This panel lists the Service Request procedure success stats.

Expression

Service Request Success %

```
((sum(mmecc_emm_service_request_success{pod="$PodName"}))/(sum(mmecc_emm_service_request_attempt{pod="$PodName"})))*100
```

Attach Procedure Stats

Description

This panel lists the Attach procedure stats.

Expression

mmecc_emm_attach_eps_attempt

```
sum(mmecc_emm_attach_eps_attempt{pod="$PodName"})
```

mmecc_emm_attach_eps_success

```
sum(mmecc_emm_attach_eps_success{pod="$PodName"})
```

mmecc_emm_attach_eps_failure

```
sum(mmecc_emm_attach_eps_failure{pod="$PodName"})
```

mmecc_emm_attach_eps_ignore

```
sum(mmecc_emm_attach_eps_ignore{pod="$PodName"})
```

Detach Procedure Stats

Description

This panel lists the Detach procedure stats.

Expression

mmecc_emm_eps_detach_ue_attempt

```
sum(mmecc_emm_eps_detach_ue_attempt{pod="$PodName"})
```

mmecc_emm_eps_detach_ue_success

```
sum(mmecc_emm_eps_detach_ue_success{pod="$PodName"})
```

Service Request Procedure Stats

Description

This panel lists the Service Request procedure stats.

Expression

mmecc_emm_service_request_attempt

```
sum(mmecc_emm_service_request_attempt{pod="$PodName"})
```

mmecc_emm_service_request_success

```
sum(mmecc_emm_service_request_success{pod="$PodName"})
```

mmecc_emm_service_request_failure

```
sum(mmecc_emm_service_request_failure{pod="$PodName"})
```

Success % for Intra-MME TAU Procedure

Description

This panel lists the Intra-MME normal TAU procedure success stats.

Expression

Intra-MME TAU Success %

```
((sum(mmecc_emm_intra_mme_tau_success{pod="$PodName"}))/(sum(mmecc_emm_intra_mme_tau_attempt{pod="$PodName"})))*100
```

Success % for Periodic TAU Procedure

Description

This panel lists the Periodic TAU procedure success stats.

Expression

Periodic TAU Success %

```
((sum(mmecc_emm_periodic_tau_success{pod="$PodName"}))/(sum(mmecc_emm_periodic_tau_attempt{pod="$PodName"})))*100
```

Intra-MME TAU Procedure Stats

Description

This panel lists the Intra-MME TAU procedure stats.

Expression

mmecc_emm_intra_mme_tau_attempt

```
sum(mmecc_emm_intra_mme_tau_attempt{pod="$PodName"})
```

mmecc_emm_intra_mme_tau_success

```
sum(mmecc_emm_intra_mme_tau_success{pod="$PodName"})
```

mmecc_emm_intra_mme_tau_failure

```
sum(mmecc_emm_intra_mme_tau_failure{pod="$PodName"})
```

mmecc_emm_intra_mme_tau_ignore

```
sum(mmecc_emm_intra_mme_tau_ignore{pod="$PodName"})
```

Periodic TAU Procedure Stats

Description

This panel lists the Periodic TAU procedure stats.

Expression

mmecc_emm_periodic_tau_attempt

```
sum(mmecc_emm_periodic_tau_attempt{pod="$PodName"})
```

mmecc_emm_periodic_tau_success

```
sum(mmecc_emm_periodic_tau_success{pod="$PodName"})
```

mmecc_emm_periodic_tau_failure

```
sum(mmecc_emm_periodic_tau_failure{pod="$PodName"})
```

Success % for Paging procedure

Description

This panel lists the success stats for Paging procedure.

Expression

Paging Success %

```
((sum(mmecc_emm_eps_paging_attempt{pod="$PodName"}) -  
sum(mmecc_emm_eps_paging_failure{pod="$PodName"}))/(sum(mmecc_emm_eps_paging_attem  
pt{pod="$PodName"})))*100
```

Success % for GUTI Reallocation procedure

Description

This panel lists the success stats for GUTI Reallocation procedure.

Expression

GUTI Reallocation Success %

```
((sum(mmecc_emm_guti_reallocation_success{pod="$PodName"}))/(sum(mmecc_emm_guti_reallocation_attempt{pod="$PodName"})))*100
```

Paging Procedure Stats

Description

This panel lists the Paging procedure stats.

Expression

mmecc_emm_eps_paging_attempt-{{initiator}}

```
sum(mmecc_emm_eps_paging_attempt{pod="$PodName"}) by (initiator)
```

mmecc_emm_eps_paging_first_retry_attempt-{{initiator}}

```
sum(mmecc_emm_eps_paging_first_retry_attempt{pod="$PodName"}) by (initiator)
```

mmecc_emm_eps_paging_second_retry_attempt-{{initiator}}

```
sum(mmecc_emm_eps_paging_second_retry_attempt{pod="$PodName"}) by (initiator)
```

mmecc_emm_eps_paging_third_retry_attempt-{{initiator}}

```
sum(mmecc_emm_eps_paging_third_retry_attempt{pod="$PodName"}) by (initiator)
```

mmecc_emm_eps_paging_fourth_retry_attempt-{{initiator}}

```
sum(mmecc_emm_eps_paging_fourth_retry_attempt{pod="$PodName"}) by (initiator)
```

mmecc_emm_eps_paging_failure-{{initiator}}

```
sum(mmecc_emm_eps_paging_failure{pod="$PodName"}) by (initiator)
```

Procedure Stats for GUTI Reallocation

Description

This panel lists the stats for GUTI Reallocation procedure.

Expression

mmecc_emm_guti_reallocation_attempt

```
sum(mmecc_emm_guti_reallocation_attempt{pod="$PodName"})
```

mmecc_emm_guti_reallocation_success

```
sum(mmecc_emm_guti_reallocation_success{pod="$PodName"})
```

mmecc_emm_guti_reallocation_failure

```
sum(mmecc_emm_guti_reallocation_failure{pod="$PodName"})
```

Success % for Standalone Security procedure

Description

This panel lists the success stats for Standalone Security procedure.

Expression

Standalone Security Success %

```
((sum(mmecc_emm_standalone_security_success{pod="$PodName"}))/(sum(mmecc_emm_standalone_security_attempt{pod="$PodName"})))*100
```

Procedure Stats for Standalone Security

Description

This panel lists the stats for Standalone Security procedure.

Expression

mmecc_emm_standalone_security_attempt

```
sum(mmecc_emm_standalone_security_attempt{pod="$PodName"})
```

mmecc_emm_standalone_security_success

```
sum(mmecc_emm_standalone_security_success{pod="$PodName"})
```

mmecc_emm_standalone_security_failure

```
sum(mmecc_emm_standalone_security_failure{pod="$PodName"})
```

Success % for Inter Old MME TAU Procedure

Description

This panel lists the success stats for Inter Old MME TAU procedure.

Expression

Inter Old MME TAU Success %

```
((sum(mmecc_emm_inter_old_mme_tau_success{pod="$PodName"}))/(sum(mmecc_emm_inter_old_mme_tau_attempt{pod="$PodName"})))*100
```

Procedure Stats for Inter Old MME TAU

Description

This panel lists the stats for Inter Old MME TAU procedure.

Expression

mmecc_emm_inter_old_mme_tau_attempt

```
sum(mmecc_emm_inter_old_mme_tau_attempt{pod="$PodName"})
```

mmecc_emm_inter_old_mme_tau_success

```
sum(mmecc_emm_inter_old_mme_tau_success{pod="$PodName"})
```

mmecc_emm_inter_old_mme_tau_failure

```
sum(mmecc_emm_inter_old_mme_tau_failure{pod="$PodName"})
```

Success % for Inter New MME TAU Procedure

Description

This panel lists the success stats for Inter New MME TAU procedure.

Expression

Inter New MME TAU Success %

```
((sum(mmecc_emm_inter_new_mme_tau_success{pod="$PodName"}))/(sum(mmecc_emm_inter_new_mme_tau_attempt{pod="$PodName"})))*100
```

Procedure Stats for Inter New MME TAU

Description

This panel lists the stats for Inter New MME TAU procedure.

Expression

mmecc_emm_inter_new_mme_tau_attempt

```
sum(mmecc_emm_inter_new_mme_tau_attempt{pod="$PodName"})
```

mmecc_emm_inter_new_mme_tau_success

```
sum(mmecc_emm_inter_new_mme_tau_success{pod="$PodName"})
```

mmecc_emm_inter_new_mme_tau_failure

sum(mmecc_emm_inter_new_mme_tau_failure{pod="\$PodName"})

mmecc_emm_inter_new_mme_tau_success_restrict_dcnr

sum(mmecc_emm_inter_new_mme_tau_success_restrict_dcnr{pod="\$PodName"})

mmecc_emm_inter_new_mme_tau_ignore

sum(mmecc_emm_inter_new_mme_tau_ignore{pod="\$PodName"})

Success % for Inter RAT Old MME TAU Procedure

Description

This panel lists the success stats for Inter RAT Old MME TAU procedure.

Expression

Inter RAT Old MME TAU Success %

((sum(mmecc_emm_inter_rat_old_mme_tau_success{pod="\$PodName"}))/(sum(mmecc_emm_inter_rat_old_mme_tau_attempt{pod="\$PodName"})))*100

Procedure Stats for Inter RAT Old MME TAU

Description

This panel lists the stats for Inter RAT Old MME TAU procedure.

Expression

mmecc_emm_inter_rat_old_mme_tau_attempt

sum(mmecc_emm_inter_rat_old_mme_tau_attempt{pod="\$PodName"})

mmecc_emm_inter_rat_old_mme_tau_success

```
sum(mmecc_emm_inter_rat_old_mme_tau_success{pod="$PodName"})
```

mmecc_emm_inter_rat_old_mme_tau_failure

```
sum(mmecc_emm_inter_rat_old_mme_tau_failure{pod="$PodName"})
```

Success % for Inter RAT New MME TAU Procedure**Description**

This panel lists the success stats for Inter RAT New MME TAU procedure.

Expression**Inter RAT New MME TAU Success %**

```
((sum(mmecc_emm_inter_rat_new_mme_tau_success{pod="$PodName"}))/(sum(mmecc_emm_inter_rat_new_mme_tau_attempt{pod="$PodName"})))*100
```

Procedure Stats for Inter RAT New MME TAU**Description**

This panel lists the stats for Inter RAT New MME TAU procedure.

Expression**mmecc_emm_inter_rat_new_mme_tau_attempt**

```
sum(mmecc_emm_inter_rat_new_mme_tau_attempt{pod="$PodName"})
```

mmecc_emm_inter_rat_new_mme_tau_success

```
sum(mmecc_emm_inter_rat_new_mme_tau_success{pod="$PodName"})
```

mmecc_emm_inter_rat_new_mme_tau_failure

```
sum(mmecc_emm_inter_rat_new_mme_tau_failure{pod="$PodName"})
```

Success % for MME initiated Detach Procedure when GW deactivates last PDN

Description

This panel lists the success stats for MME initiated Detach procedure when GW deactivates the last PDN.

Expression

Detach Success %

```
((sum(mmecc_emm_eps_detach_mme_success_gw{pod="$PodName"}))/(sum(mmecc_emm_eps_detach_mme_attempt_gw{pod="$PodName"})))*100
```

Success % for MME Initiated Detach Procedure when MRT expires

Description

This panel lists the success stats for MME initiated Detach procedure when MRT expires.

Expression

Detach Success %

```
((sum(mmecc_emm_eps_detach_mme_success_mrt{pod="$PodName"}))/(sum(mmecc_emm_eps_detach_mme_attempt_mrt{pod="$PodName"})))*100
```

Procedure Stats for MME Initiated Detach when GW deactivates the last PDN

Description

This panel lists the stats for MME initiated Detach procedure when GW deactivates the last PDN.

Expression

mmecc_emm_eps_detach_mme_attempt_gw

```
sum(mmecc_emm_eps_detach_mme_attempt_gw{pod="$PodName"})
```

mmecc_emm_eps_detach_mme_success_gw

```
sum(mmecc_emm_eps_detach_mme_success_gw{pod="$PodName"})
```

Procedure Stats For MME Initiated Detach when MRT Expires

Description

This panel lists the stats for MME initiated Detach procedure when MRT expires.

Expression

mmecc_emm_eps_detach_mme_attempt_mrt

```
sum(mmecc_emm_eps_detach_mme_attempt_mrt{pod="$PodName"})
```

mmecc_emm_eps_detach_mme_success_mrt

```
sum(mmecc_emm_eps_detach_mme_success_mrt{pod="$PodName"})
```

Attach Message Stats

Description

This panel lists the Attach message stats

Expression

Attach Request

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="attach-request"})
```

Attach Accept

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="attach-accept"})
```

Attach Complete

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="attach-complete"})
```

Attach Reject

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="attach-reject"})
```

UE Initiated Detach Request Message Stats

Description

This panel lists the UE initiated Detach message stats

Expression

Detach Request

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="detach-request"})
```

Detach Accept

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="detach-accept"})
```

Network Initiated Detach Message Stats

Description

This panel lists the network initiated detach message stats

Expression

Detach Request

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="detach-request"})
```

Detach Accept

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="detach-accept"})
```

Service Request Message Stats

Description

This panel lists the Service Request message stats

Expression

Service Request

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="service-request"})
```

Service Accept

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="service-accept"})
```

Service Reject

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="service-reject"})
```

Tracking Area Update Message Stats

Description

This panel lists the Tracking Area Update message stats

Expression

Tracking Area Update Request

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="tracking-area-update-request"})
```

Tracking Area Update Accept

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="tracking-area-update-accept"})
```

Tracking Area Update Complete

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="tracking-area-update-complete"})
```

Tracking Area Update Reject

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="tracking-area-update-reject"})
```

GUTI Reallocation Command Message Stats

Description

This panel lists the GUTI Reallocation Command message stats

Expression

GUTI Reallocation Command

```
sum(mmecc_nas_outgoing_message{pod="$PodName", msg_type="guti-reallocation-command"})
```

GUTI Reallocation Complete

```
sum(mmecc_nas_incoming_message{pod="$PodName", msg_type="guti-reallocation-complete"})
```

Success % for Default Bearer Activation during Attach with request-type initial request

Description

This panel lists the success stats for Default Bearer Activation during Attach with request-type initial request.

Expression

Success % for Default Bearer Activation during Attach with request-type initial request

```
((sum(mmecc_esm_activate_default_bearer_attach_initial_success{pod="$PodName"}))/  
sum(mmecc_esm_activate_default_bearer_attach_initial_attempt{pod="$PodName"}))*10  
0
```

Success % for PDN connection procedures with request-type initial request

Description

This panel lists the success stats for PDN connection procedures with request-type initial request.

Expression

Success % for PDN connection procedures with request-type initial request

```
((sum(mmecc_esm_activate_default_bearer_pdncon_initial_success{pod="$PodName"}))/  
sum(mmecc_esm_activate_default_bearer_pdncon_initial_attempt{pod="$PodName"}))*10  
0
```

Stats for Default Bearer Activation during Attach with request-type initial request

Description

This panel lists the stats for Default Bearer Activation during Attach with request-type initial request.

Expression

mmecc_esm_activate_default_bearer_attach_initial_attempt

```
sum(mmecc_esm_activate_default_bearer_attach_initial_attempt{pod="$PodName"})
```

mmecc_esm_activate_default_bearer_attach_initial_success

```
sum(mmecc_esm_activate_default_bearer_attach_initial_success{pod="$PodName"})
```

mmecc_esm_activate_default_bearer_attach_failure

```
sum(mmecc_esm_activate_default_bearer_attach_failure{pod="$PodName"})
```

Stats for PDN connection procedures with request-type initial request

Description

This panel lists the stats for PDN connection procedures with request-type initial request.

Expression

mmecc_esm_activate_default_bearer_pdncon_initial_attempt

```
sum(mmecc_esm_activate_default_bearer_pdncon_initial_attempt{pod="$PodName"})
```

mmecc_esm_activate_default_bearer_pdncon_initial_success

```
sum(mmecc_esm_activate_default_bearer_pdncon_initial_success{pod="$PodName"})
```

mmecc_esm_activate_default_bearer_pdncon_failure

```
sum(mmecc_esm_activate_default_bearer_pdncon_failure{pod="$PodName"})
```

Success % for Bearer Resource Allocation procedures**Description**

This panel lists the success stats for Bearer Resource Allocation procedures.

Expression**Success % for Bearer Resource Allocation procedures**

```
((sum(mmecc_esm_bearer_resource_alloc_ue_success{pod="$PodName"}))/(sum(mmecc_esm_bearer_resource_alloc_ue_attempt{pod="$PodName"})))*100
```

Success % for Dedicated EPS Bearer Activation procedures**Description**

This panel lists the success stats for Dedicated EPS Bearer Activation procedures.

Expression**Success % for Dedicated EPS Bearer Activation procedures**

```
((sum(mmecc_esm_activate_dedicated_bearer_success{pod="$PodName"}))/(sum(mmecc_esm_activate_dedicated_bearer_attempt{pod="$PodName"})))*100
```

Stats for Bearer Resource Allocation procedures.

Description

This panel lists the stats for Bearer Resource Allocation procedures.

Expression

mmecc_esm_bearer_resource_alloc_ue_attempt

```
sum(mmecc_esm_bearer_resource_alloc_ue_attempt{pod="$PodName"})
```

mmecc_esm_bearer_resource_alloc_ue_success

```
sum(mmecc_esm_bearer_resource_alloc_ue_success{pod="$PodName"})
```

mmecc_esm_bearer_resource_alloc_ue_failure

```
sum(mmecc_esm_bearer_resource_alloc_ue_failure{pod="$PodName"})
```

Stats for Dedicated EPS Bearer Activation procedures.

Description

This panel lists the stats for Dedicated EPS Bearer Activation procedures.

Expression

mmecc_esm_activate_dedicated_bearer_attempt

```
sum(mmecc_esm_activate_dedicated_bearer_attempt{pod="$PodName"})
```

mmecc_esm_activate_dedicated_bearer_success

```
sum(mmecc_esm_activate_dedicated_bearer_success{pod="$PodName"})
```

mmecc_esm_activate_dedicated_bearer_failure

```
sum(mmecc_esm_activate_dedicated_bearer_failure{pod="$PodName"})
```

Success % for GW Initiated Dedicated EPS Bearer Deactivation procedures

Description

This panel lists the success stats for GW initiated Dedicated EPS Bearer Deactivation procedures.

Expression

Success % for GW initiated Dedicated EPS Bearer Deactivation procedures

```
((sum(mmecc_esm_deactivate_dedicated_bearer_gw_success{pod="$PodName"})))/(sum(mmecc_esm_deactivate_dedicated_bearer_gw_attempt{pod="$PodName"}))*100
```

Success % for MME Initiated Dedicated EPS Bearer Deactivation procedures

Description

This panel lists the success stats for MME initiated Dedicated EPS Bearer Deactivation procedures.

Expression

Success % for MME initiated Dedicated EPS Bearer Deactivation procedures

```
((sum(mmecc_esm_deactivate_dedicated_bearer_mme_success{pod="$PodName"})))/(sum(mmecc_esm_deactivate_dedicated_bearer_mme_attempt{pod="$PodName"}))*100
```

Success % for UE Initiated Dedicated EPS Bearer Deactivation procedures

Description

This panel lists the success stats for UE initiated Dedicated EPS Bearer Deactivation procedures.

Expression

Success % for UE initiated Dedicated EPS Bearer Deactivation procedures

```
((sum(mmecc_esm_deactivate_dedicated_bearer_ue_success{pod="$PodName"})))/(sum(mmecc_esm_deactivate_dedicated_bearer_ue_attempt{pod="$PodName"}))*100
```

Stats for GW Initiated Dedicated EPS Bearer Deactivation procedures.

Description

This panel lists the stats for GW initiated Dedicated EPS Bearer Deactivation procedures.

Expression

mmecc_esm_deactivate_dedicated_bearer_gw_attempt

```
sum(mmecc_esm_deactivate_dedicated_bearer_gw_attempt{pod="$PodName"})
```

mmecc_esm_deactivate_dedicated_bearer_gw_success

```
sum(mmecc_esm_deactivate_dedicated_bearer_gw_success{pod="$PodName"})
```

Stats for MME Initiated Dedicated EPS Bearer Deactivation procedures.

Description

This panel lists the stats for MME initiated Dedicated EPS Bearer Deactivation procedures.

Expression

mmecc_esm_deactivate_dedicated_bearer_mme_attempt

```
sum(mmecc_esm_deactivate_dedicated_bearer_mme_attempt{pod="$PodName"})
```

mmecc_esm_deactivate_dedicated_bearer_mme_success

```
sum(mmecc_esm_deactivate_dedicated_bearer_mme_success{pod="$PodName"})
```

Stats for UE Initiated Dedicated EPS Bearer Deactivation procedures.

Description

This panel lists the stats for UE initiated Dedicated EPS Bearer Deactivation procedures.

Expression

mmecc_esm_deactivate_dedicated_bearer_ue_attempt

```
sum(mmecc_esm_deactivate_dedicated_bearer_ue_attempt{pod="$PodName"})
```

mmecc_esm_deactivate_dedicated_bearer_ue_success

```
sum(mmecc_esm_deactivate_dedicated_bearer_ue_success{pod="$PodName"})
```

Success % for UE Initiated Bearer Resource Modification procedures

Description

This panel lists the success stats for UE initiated Bearer Resource Modification procedures.

Expression

Success % for UE initiated Bearer Resource Modification procedures

```
((sum(mmecc_esm_bearer_resource_mod_ue_success{pod="$PodName"}))/(sum(mmecc_esm_bearer_resource_mod_ue_attempt{pod="$PodName"})))*100
```

Success % for UE Initiated PDN Disconnect procedures

Description

This panel lists the success stats for UE initiated PDN Disconnect procedures.

Expression

Success % for UE initiated PDN Disconnect procedures

```
((sum(mmecc_esm_deactivate_default_bearer_pdndiscon_success{pod="$PodName"}))/(sum(mmecc_esm_deactivate_default_bearer_pdndiscon_attempt{pod="$PodName"})))*100
```

Success % for GW Initiated PDN Disconnect procedures

Description

This panel lists the success stats for GW initiated PDN Disconnect procedures.

Expression

Success % for GW initiated PDN Disconnect procedures

```
((sum(mmecc_esm_deactivate_default_bearer_gw_success{pod="$PodName"}))/(sum(mmecc_esm_deactivate_default_bearer_gw_attempts{pod="$PodName"})))*100
```

Stats for UE Initiated Bearer Resource Modification procedures.

Description

This panel lists the stats for UE initiated Bearer Resource Modification procedures.

Expression

mmecc_esm_bearer_resource_mod_ue_attempt

```
sum(mmecc_esm_bearer_resource_mod_ue_attempt{pod="$PodName"})
```

mmecc_esm_bearer_resource_mod_ue_success

```
sum(mmecc_esm_bearer_resource_mod_ue_success{pod="$PodName"})
```

mmecc_esm_bearer_resource_mod_ue_failure

```
sum(mmecc_esm_bearer_resource_mod_ue_failure{pod="$PodName"})
```

Stats for UE Initiated PDN Disconnect procedures.

Description

This panel lists the stats for UE initiated PDN Disconnect procedures.

Expression

mmecc_esm_deactivate_default_bearer_pdndiscon_attempt

```
sum(mmecc_esm_deactivate_default_bearer_pdndiscon_attempt{pod="$PodName"})
```

mmecc_esm_deactivate_default_bearer_pdndiscon_success

```
sum(mmecc_esm_deactivate_default_bearer_pdndiscon_success{pod="$PodName"})
```

mmecc_esm_deactivate_default_bearer_pdndiscon_failure

```
sum(mmecc_esm_deactivate_default_bearer_pdndiscon_failure{pod="$PodName"})
```

Stats for GW Initiated PDN Disconnect procedures.

Description

This panel lists the stats for GW initiated PDN Disconnect procedures.

Expression

mmecc_esm_deactivate_default_bearer_gw_attempts

```
sum(mmecc_esm_deactivate_default_bearer_gw_attempts{pod="$PodName"})
```

mmecc_esm_deactivate_default_bearer_gw_success

```
sum(mmecc_esm_deactivate_default_bearer_gw_success{pod="$PodName"})
```

Success % for SGW Relocation procedures

Description

This panel lists the success stats for SGW Relocation procedures.

Expression

Success % for SGW Relocation procedures

```
((sum(mmecc_esm_sgw_relocation_success{pod="$PodName"}))/(sum(mmecc_esm_sgw_relocation_attempt{pod="$PodName"})))*100
```

Success % for Intra-MME TAU procedures

Description

This panel lists the success stats for Intra-MME TAU procedures.

Expression

Success % for Intra-MME TAU procedures

```
((sum(mmecc_esm_active_tau_success{pod="$PodName"}))/(sum(mmecc_esm_active_tau_attempt{pod="$PodName"})))*100
```

Success % for Service Request procedures

Description

This panel lists the success stats for Service Request procedures.

Expression

Success % for Service Request procedures

```
((sum(mmecc_esm_service_request_success{pod="$PodName"}))/(sum(mmecc_esm_service_request_attempt{pod="$PodName"})))*100
```

Stats for SGW Relocation procedures.

Description

This panel lists the stats for SGW Relocation procedures.

Expression

mmecc_esm_sgw_relocation_attempt

```
sum(mmecc_esm_sgw_relocation_attempt{pod="$PodName"})
```

mmecc_esm_sgw_relocation_success

```
sum(mmecc_esm_sgw_relocation_success{pod="$PodName"})
```

Stats for Intra-MME TAU procedures.

Description

This panel lists the stats for Intra-MME TAU procedures.

Expression

mmecc_esm_active_tau_attempt

```
sum(mmecc_esm_active_tau_attempt{pod="$PodName"})
```

mmecc_esm_active_tau_success

```
sum(mmecc_esm_active_tau_successs{pod="$PodName"})
```

Stats for Service Request procedures.

Description

This panel lists the stats for Service Request procedures.

Expression

mmecc_esm_service_request_attempt

```
sum(mmecc_esm_service_request_attempt{pod="$PodName"})
```

mmecc_esm_service_request_success

```
sum(mmecc_esm_service_request_success{pod="$PodName"})
```

Success % for UE Initiated Default Bearer Modification procedures

Description

This panel lists the success stats for UE initiated Default Bearer Modification procedures.

Expression

Success % for UE initiated Default Bearer Modification procedures

```
((sum(mmecc_esm_modify_default_bearer_ue_success{pod="$PodName"}))/(sum(mmecc_esm_modify_default_bearer_ue_attempt{pod="$PodName"})))*100
```

Success % for PGW Initiated Default Bearer Modification procedures

Description

This panel lists the success stats for PGW initiated Default Bearer Modification procedures.

Expression

Success % for PGW initiated Default Bearer Modification procedures

```
((sum(mmecc_esm_modify_default_bearer_pgw_success{pod="$PodName"}))/(sum(mmecc_esm_modify_default_bearer_pgw_attempt{pod="$PodName"})))*100
```

Stats for UE Initiated Default Bearer Modification procedures.

Description

This panel lists the UE initiated Default Bearer Modification procedure stats.

Expression

mmecc_esm_modify_default_bearer_ue_attempt

```
sum(mmecc_esm_modify_default_bearer_ue_attempt{pod="$PodName"})
```

mmecc_esm_modify_default_bearer_ue_success

```
sum(mmecc_esm_modify_default_bearer_ue_success{pod="$PodName"})
```

mmecc_esm_modify_default_bearer_ue_failure

```
sum(mmecc_esm_modify_default_bearer_ue_failure{pod="$PodName"})
```

Stats for PGW Initiated Default Bearer Modification procedures.

Description

This panel lists the stats for PGW initiated Default Bearer Modification procedures.

Expression

mmecc_esm_modify_default_bearer_pgw_attempt

```
sum(mmecc_esm_modify_default_bearer_pgw_attempt{pod="$PodName"})
```

mmecc_esm_modify_default_bearer_pgw_success

```
sum(mmecc_esm_modify_default_bearer_pgw_success{pod="$PodName"})
```

mmecc_esm_modify_default_bearer_pgw_failure

```
sum(mmecc_esm_modify_default_bearer_pgw_failure{pod="$PodName"})
```

Success % for UE Initiated Dedicated Bearer Modification procedures

Description

This panel lists the success stats for UE initiated Dedicated Bearer Modification procedures.

Expression

Success % for UE initiated Dedicated Bearer Modification procedures

```
((sum(mmecc_esm_modify_dedicated_bearer_ue_success{pod="$PodName"}))/(sum(mmecc_esm_modify_dedicated_bearer_ue_attempt{pod="$PodName"})))*100
```

Success % for PGW Initiated Dedicated Bearer Modification procedures

Description

This panel lists the success stats for PGW initiated Dedicated Bearer Modification procedures.

Expression

Success % for PGW initiated Dedicated Bearer Modification procedures

```
((sum(mmecc_esm_modify_dedicated_bearer_pgw_success{pod="$PodName"}))/(sum(mmecc_esm_modify_dedicated_bearer_pgw_attempt{pod="$PodName"})))*100
```

Stats for UE Initiated Dedicated Bearer Modification procedures.

Description

This panel lists the UE initiated Dedicated Bearer Modification procedure stats.

Expression

mmecc_esm_modify_dedicated_bearer_ue_attempt

```
sum(mmecc_esm_modify_dedicated_bearer_ue_attempt{pod="$PodName"})
```

mmecc_esm_modify_dedicated_bearer_ue_success

```
sum(mmecc_esm_modify_dedicated_bearer_ue_success{pod="$PodName"})
```

mmecc_esm_modify_dedicated_bearer_ue_failure

```
sum(mmecc_esm_modify_dedicated_bearer_ue_failure{pod="$PodName"})
```

Stats for PGW Initiated Dedicated Bearer Modification procedures.

Description

This panel lists the PGW initiated Dedicated Bearer Modification procedure stats.

Expression

mmecc_esm_modify_dedicated_bearer_pgw_attempt

```
sum(mmecc_esm_modify_dedicated_bearer_pgw_attempt{pod="$PodName"})
```

mmecc_esm_modify_dedicated_bearer_pgw_success

```
sum(mmecc_esm_modify_dedicated_bearer_pgw_success{pod="$PodName"})
```

mmecc_esm_modify_dedicated_bearer_pgw_failure

```
sum(mmecc_esm_modify_dedicated_bearer_pgw_failure{pod="$PodName"})
```

Success % for ENB Initiated E-RAB Modification Indication procedures

Description

This panel lists the success stats for ENB initiated E-RAB Modification Indication procedures.

Expression

Success % for ENB initiated E-RAB Modification Indication procedures

```
((sum(mmecc_esm_erab_modification_indication_success{pod="$PodName"}))/(sum(mmecc_esm_erab_modification_indication_attempt{pod="$PodName"})))*100
```

Stats for ENB Initiated E-RAB Modification Indication procedures.

Description

This panel lists the ENB initiated E-RAB Modification Indication procedure stats.

Expression

mmecc_esm_erab_modification_indication_attempt

```
sum(mmecc_esm_erab_modification_indication_attempt{pod="$PodName"})
```

mmecc_esm_erab_modification_indication_success

```
sum(mmecc_esm_erab_modification_indication_success{pod="$PodName"})
```

mmecc_esm_erab_modification_indication_failure

```
sum(mmecc_esm_erab_modification_indication_failure{pod="$PodName"})
```

Success % for X2 Handover

Description

This panel lists the success stats for X2 Handover.

Expression

Success % for X2 Handover

```
((sum(mmecc_x2_handover_success{pod="$PodName"}))/sum(mmecc_x2_handover_attempt{pod="$PodName"}))*100
```

Stats for X2 Handover

Description

This panel lists the stats for X2 Handover.

Expression

mmecc_x2_handover_attempt

```
sum(mmecc_x2_handover_attempt{pod="$PodName"})
```

mmecc_x2_handover_success

```
sum(mmecc_x2_handover_success{pod="$PodName"})
```

mmecc_x2_handover_failure

```
sum(mmecc_x2_handover_failure{pod="$PodName"})
```

Success % for X2 Handover with SGW change

Description

This panel lists the success stats for X2 Handover with SGW change.

Expression

Success % for X2 Handover with SGW change

```
((sum(mmecc_x2_handover_inter_sgw_success{pod="$PodName"}))/sum(mmecc_x2_handover_inter_sgw_attempt{pod="$PodName"}))*100
```

Stats for X2 Handover with SGW change

Description

This panel lists the stats for X2 Handover with SGW change.

Expression

mmecc_x2_handover_inter_sgw_attempt

```
sum(mmecc_x2_handover_inter_sgw_attempt{pod="$PodName"})
```

mmecc_x2_handover_inter_sgw_success

```
sum(mmecc_x2_handover_inter_sgw_success{pod="$PodName"})
```

mmecc_x2_handover_inter_sgw_failure

```
sum(mmecc_x2_handover_inter_sgw_failure{pod="$PodName"})
```

Success % for S1 Handover

Description

This panel lists the success stats for S1 Handover.

Expression

Success % for S1 Handover

```
((sum(mmecc_s1_intra_handover_success{pod="$PodName"}))/(sum(mmecc_s1_intra_handover_attempt{pod="$PodName"})))*100
```

Stats for S1 Handover

Description

Stats for S1 Handover

Expression

mmecc_s1_intra_handover_attempt

```
sum(mmecc_s1_intra_handover_attempt{pod="$PodName"})
```

mmecc_s1_intra_handover_success

```
sum(mmecc_s1_intra_handover_success{pod="$PodName"})
```

mmecc_s1_intra_handover_failure

```
sum(mmecc_s1_intra_handover_failure{pod="$PodName"})
```

mmecc_s1_intra_handover_cancel

```
sum(mmecc_s1_intra_handover_cancel{pod="$PodName"})
```

Success % for Inter Target MME S1 Handover Procedure

Description

This panel lists the success stats for Inter Target MME S1 Handover procedure.

Expression

Inter Target MME S1 Handover Success %

```
((sum(mmecc_s1_inter_target_mme_handover_success{pod="$PodName"}))/(sum(mmecc_s1_inter_target_mme_handover_attempt{pod="$PodName"})))*100
```

Procedure Stats for Inter Target MME S1 Handover

Description

This panel lists the stats for Inter Target MME S1 Handover procedure.

Expression

mmecc_s1_inter_target_mme_handover_attempt

```
sum(mmecc_s1_inter_target_mme_handover_attempt{pod="$PodName"})
```

mmecc_s1_inter_target_mme_handover_success

```
sum(mmecc_s1_inter_target_mme_handover_success{pod="$PodName"})
```

mmecc_s1_inter_target_mme_handover_failure

```
sum(mmecc_s1_inter_target_mme_handover_failure{pod="$PodName"})
```

mmecc_s1_inter_target_mme_handover_cancel

```
sum(mmecc_s1_inter_target_mme_handover_cancel{pod="$PodName"})
```

Success % for Inter Source MME S1 Handover Procedure**Description**

This panel lists the success stats for Inter Source MME S1 Handover procedure.

Expression**Inter Source MME S1 Handover Success %**

```
((sum(mmecc_s1_inter_source_mme_handover_success{pod="$PodName"}))/(sum(mmecc_s1_inter_source_mme_handover_attempt{pod="$PodName"})))*100
```

Procedure Stats for Inter Source MME S1 Handover**Description**

This panel lists the stats for Inter Source MME S1 Handover procedure.

Expression**mmecc_s1_inter_source_mme_handover_attempt**

```
sum(mmecc_s1_inter_source_mme_handover_attempt{pod="$PodName"})
```

mmecc_s1_inter_source_mme_handover_success


```
sum(mmecc_s1_inter_source_mme_handover_success{pod="$PodName"})
```

mmecc_s1_inter_source_mme_handover_failure

```
sum(mmecc_s1_inter_source_mme_handover_failure{pod="$PodName"})
```

mmecc_s1_inter_source_mme_handover_cancel

```
sum(mmecc_s1_inter_source_mme_handover_cancel{pod="$PodName"})
```

Success % for Inter RAT Target MME Handover Procedure

Description

This panel lists the success stats for Inter RAT Target MME Handover procedure.

Expression

Inter RAT Target MME Handover Success %

```
((sum(mmecc_inter_rat_target_mme_handover_success{pod="$PodName"}))/(sum(mmecc_inter_rat_target_mme_handover_attempt{pod="$PodName"})))*100
```

Procedure Stats for Inter RAT Target MME Handover

Description

This panel lists the stats for Inter RAT Target MME Handover procedure.

Expression

mmecc_inter_rat_target_mme_handover_attempt

```
sum(mmecc_inter_rat_target_mme_handover_attempt{pod="$PodName"})
```

mmecc_inter_rat_target_mme_handover_success

```
sum(mmecc_inter_rat_target_mme_handover_success{pod="$PodName"})
```

mmecc_inter_rat_target_mme_handover_failure

sum(mmecc_inter_rat_target_mme_handover_failure{pod="\$PodName"})

mmecc_inter_rat_target_mme_handover_cancel

sum(mmecc_inter_rat_target_mme_handover_cancel{pod="\$PodName"})

Success % for Inter RAT Source MME Handover Procedure

Description

This panel lists the success stats for Inter RAT Source MME Handover procedure.

Expression

Inter RAT Source MME Handover Success %

((sum(mmecc_inter_rat_source_mme_handover_success{pod="\$PodName"}))/(sum(mmecc_inter_rat_source_mme_handover_attempt{pod="\$PodName"})))*100

Procedure Stats for Inter RAT Source MME Handover

Description

This panel lists the stats for Inter RAT Source MME Handover procedure.

Expression

mmecc_inter_rat_source_mme_handover_attempt

sum(mmecc_inter_rat_source_mme_handover_attempt{pod="\$PodName"})

mmecc_inter_rat_source_mme_handover_success

sum(mmecc_inter_rat_source_mme_handover_success{pod="\$PodName"})

mmecc_inter_rat_source_mme_handover_failure

```
sum(mmecc_inter_rat_source_mme_handover_failure{pod="$PodName"})
```

mmecc_inter_rat_source_mme_handover_cancel

```
sum(mmecc_inter_rat_source_mme_handover_cancel{pod="$PodName"})
```

Description

This panel lists the Configuration Transfer procedure stats

Success % for Intra MME Configuration Transfer

Description

This panel lists the success stats for Intra MME Configuration Transfer.

Expression

Success % for Intra MME Configuration Transfer

```
((sum(mmecc_s1_intra_mme_config_transfer_success{pod="$PodName"}))/(sum(mmecc_s1_intra_mme_config_transfer_attempt{pod="$PodName"})))*100
```

Success % for Inter MME Configuration Transfer

Description

This panel lists the success stats for Inter MME Configuration Transfer.

Expression

Success % for Inter MME Configuration Transfer

```
((sum(mmecc_s1_inter_mme_config_transfer_success{pod="$PodName"}))/(sum(mmecc_s1_inter_mme_config_transfer_attempt{pod="$PodName"})))*100
```

Stats for Intra MME Configuration Transfer

Description

This panel lists the stats for Intra MME Configuration Transfer.

Expression

mmecc_s1_intra_mme_config_transfer_attempt

```
sum(mmecc_s1_intra_mme_config_transfer_attempt{pod="$PodName"})
```

mmecc_s1_intra_mme_config_transfer_success

```
sum(mmecc_s1_intra_mme_config_transfer_success{pod="$PodName"})
```

mmecc_s1_intra_mme_config_transfer_failure

```
sum(mmecc_s1_intra_mme_config_transfer_failure{pod="$PodName"})
```

Stats for Inter MME Configuration Transfer

Description

This panel lists the stats for Inter MME Configuration Transfer.

Expression

mmecc_s1_inter_mme_config_transfer_attempt

```
sum(mmecc_s1_inter_mme_config_transfer_attempt{pod="$PodName"})
```

mmecc_s1_inter_mme_config_transfer_success

```
sum(mmecc_s1_inter_mme_config_transfer_success{pod="$PodName"})
```

mmecc_s1_inter_mme_config_transfer_failure

```
sum(mmecc_s1_inter_mme_config_transfer_failure{pod="$PodName"})
```

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MME-S1 Dashboard

Description

This dashboard lists the metrics on pod MME S1

Panels

Description

This panel lists the message stats for S1AP procedures

Description

This panel lists the message stats for Interface Management procedures

S1 Setup Stats

Description

This panel lists the message stats for S1 Setup procedure

Expression

S1 Setup Request

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="s1-setup-request"})
```

S1 Setup Response

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="s1-setup-response"})
```

S1 Setup Failure

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="s1-setup-failure"})
```

S1 Reset Stats

Description

This panel lists the message stats for S1 Reset procedure

Expression

Reset (from eNB)

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="reset"})
```

Reset Acknowledge (to eNB)

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="reset-acknowledge"})
```

Reset (to eNB)

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="reset"})
```

Reset Acknowledge (from eNB)

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="reset-acknowledge"})
```

ENB/MME Config Update Stats

Description

This panel lists the message stats for ENB/MME Config Update procedure

Expression

eNB Configuration Update

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="enb-configuration-update"})
```

eNB Configuration Update Acknowledge

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="enb-configuration-update-acknowledge"})
```

eNB Configuration Update Failure

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="enb-configuration-update-failure"})
```

MME Configuration Update

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="mme-configuration-update"})
```

MME Configuration Update Acknowledge

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="mme-configuration-update-acknowledge"})
```

MME Configuration Update Failure

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="mme-configuration-update-failure"})
```


Description

This panel lists the message stats for S1AP procedures

Description

This panel lists the message stats for NAS procedures

Description

This panel lists the message stats for Errors

Initial UE Stats

Description

This panel lists the Initial UE message stats

Expression

Initial UE Message

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="initial-ue-  
message"})
```

UL/DL NAS Transport Stats

Description

This panel lists stats for Uplink and Downlink NAS Transport message

Expression

Uplink NAS Transport

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="uplink-nas-  
transport"})
```

Downlink NAS Transport

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="downlink-nas-  
transport"})
```

Error Stats

Description

This panel lists stats for Error message

Expression

Error Indication (from eNB)

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="error-indication"})
```

Error Indication (from MME)

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="error-indication"})
```

NAS Non Delivery Indication

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="nas-non-delivery-indication"})
```

Description

This panel lists the message stats for S1AP procedures

Description

This panel lists the message stats for UE Context Management Procedures

Initial Context Setup Stats

Description

This panel lists the Initial Context Setup stats

Expression

Initial Context Setup Request

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="initial-  
context-setup-request"})
```

Initial Context Setup Response

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="initial-  
context-setup-response"})
```

Initial Context Setup Failure

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="initial-  
context-setup-failure"})
```

UE Context Release Stats

Description

This panel lists the UE Context Release stats

Expression

UE Context Release Command

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="ue-context-  
release-command"})
```

UE Context Release Complete

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="ue-context-  
release-complete"})
```

UE Context Release Request

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="ue-context-  
release-request"})
```

UE Context Modification Stats

Description

This panel lists the UE Context Modification stats

Expression

UE Context Modification Request

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="ue-context-modification-request"})
```

UE Context Modification Response

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="ue-context-modification-response"})
```

UE Context Modification Indication

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="ue-context-modification-indication"})
```

UE Context Modification Confirm

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="ue-context-modification-confirm"})
```

UE Context Modification Failure

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="ue-context-modification-failure"})
```

Description

This panel lists the message stats for S1AP procedures

Description

This panel lists the message stats for E-RAB Management Procedures

E-RAB Setup Stats

Description

This panel lists the E-RAB Setup message stats

Expression

E-RAB Setup Request

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="erab-setup-request"})
```

E-RAB Setup Response

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="erab-setup-response"})
```

E-RAB Modify Stats

Description

This panel lists the E-RAB Modify message stats

Expression

E-RAB Modify Request

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="erab-modify-request"})
```

E-RAB Modify Response

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="erab-modify-response"})
```

E-RAB Modification Indication

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="erab-  
modification-indication"})
```

E-RAB Modification Confirm

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="erab-  
modification-confirm"})
```

E-RAB Release Stats

Description

This panel lists the E-RAB Release message stats

Expression

E-RAB Release Indication

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="erab-release-  
indication"})
```

E-RAB Release Command

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="erab-release-  
command"})
```

E-RAB Release Response

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="erab-release-  
response"})
```

Description

This panel lists the message stats for S1AP procedures

Description

This panel lists the message stats for UE Mobility Management Procedures

Path Switch Stats

Description

This panel lists the Path Switch message stats

Expression

Path Switch Request

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="path-switch-request"})
```

Path Switch Request Acknowledge

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="path-switch-request-acknowledge"})
```

Path Switch Request Failure

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="path-switch-request-failure"})
```

Description

This panel lists the message stats for S1AP procedures

Paging Stats

Description

This panel lists the Paging message stats

Expression

Paging

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="paging"})
```

Description

This panel lists the S1 Handover Message stats

S1 Handover

Description

This panel lists the S1 Handover message stats

Expression

Handover Required

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="handover-required"})
```

Handover Request

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="handover-request"})
```

Handover Request Acknowledge

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="handover-request-acknowledge"})
```

Handover Notify

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="handover-notify"})
```

Handover Preparation Failure

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="handover-preparation-failure"})
```

Handover Command


```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="handover-command"})
```

Handover Cancel

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="handover-cancel"})
```

Handover Cancel Acknowledge

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="handover-cancel-acknowledge"})
```

Description

This panel lists the message stats for S1AP procedures

Description

This panel lists the Configuration Transfer Procedure stats

Configuration Transfer Messages

Description

This panel lists the Configuration Transfer message stats

Expression

eNB Configuration Transfer

```
sum(mmes1_s1_incoming_message{globalenbid="$GlobalEnbId", msg_type="enb-configuration-transfer"})
```

MME Configuration Transfer

```
sum(mmes1_s1_outgoing_message{globalenbid="$GlobalEnbId", msg_type="mme-configuration-transfer"})
```

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- [NRF-Agent Dashboard](#)

Summary

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NRF-Agent Dashboard	NF Registration Success %
NRF-Agent Dashboard	NRFA Registration Response Total Count
NRF-Agent Dashboard	NRFA Deregistration Request Total Count
NRF-Agent Dashboard	NF Deregistration Success %
NRF-Agent Dashboard	NRFA Deregistration Response Total Count
NRF-Agent Dashboard	NRFA Registration Request Stats
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NRF-Agent Dashboard	NRFA NF Registration Response Stats
NRF-Agent Dashboard	NRFA NF Deregistration Response Stats
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NRF-Agent Dashboard	
NRF-Agent Dashboard	NRFA Update Request Total Count
NRF-Agent Dashboard	NF Update Success %
NRF-Agent Dashboard	NRFA Update Response Total Count
NRF-Agent Dashboard	NRFA Heart-beat Request Total Count
NRF-Agent Dashboard	NF Heart-beat Success %
NRF-Agent Dashboard	NRFA Heart-beat Response Total Count
NRF-Agent Dashboard	NRFA Update Request Stats
NRF-Agent Dashboard	NRFA Heart-Beat Request Stats
NRF-Agent Dashboard	NRFA NF Update Response Stats

Dashboard Name	Panel Name
NRF-Agent Dashboard	NRFA NF Heart-Beat Response Stats
NRF-Agent Dashboard	
NRF-Agent Dashboard	
NRF-Agent Dashboard	NRFA Subscription Request Total Count
NRF-Agent Dashboard	NF subscription Success %
NRF-Agent Dashboard	NRFA subscription Response Total Count
NRF-Agent Dashboard	NRFA notification Total Count
NRF-Agent Dashboard	NF Notification Success %
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NRF-Agent Dashboard	NRFA Subscription Request Stats
NRF-Agent Dashboard	NRFA Notification Stats
NRF-Agent Dashboard	NRFA NF Subscription Response Stats
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NRF-Agent Dashboard	
NRF-Agent Dashboard	NRFA Discovery Request Total Count
NRF-Agent Dashboard	NF Discovery Success %
NRF-Agent Dashboard	NRFA Discovery Response Total Count
NRF-Agent Dashboard	NRFA Discovery Request Stats
NRF-Agent Dashboard	NRFA Discovery Response Stats

NRF-Agent Dashboard

Description

This dashboard displays all NRF-Agent pod related stats

Panels

Description

'''

Description

'''

Description

""

NRFA Registration Request Total Count

Description

This Panel displays NRF-A Registration Request Total Count

Expression

```
sum(nrfa_nnr_fnm_registration_request{namespace = "$Namespace"})
```

NF Registration Success %

Description

This panel displays NF registration Service success rate

Expression

```
1
((sum(nrfa_nnr_fnm_registration_response{namespace="$Namespace",status="201.created"})))/(sum(nrfa_nnr_fnm_registration_request{namespace="$Namespace"}))*100

2
(((sum(nrfa_nnr_fnm_registration_response{namespace="$Namespace",status="201.created"} or vector(0))) +
(sum(nrfa_nnr_fnm_registration_response{namespace="$Namespace",status="200.ok"} or vector(0))))
/(sum(nrfa_nnr_fnm_registration_request{namespace="$Namespace"})))*100
```

NRFA Registration Response Total Count

Description

This Panel displays NRF-A Registration Response Total Count

Expression

```
sum(nrfa_nnr_fnm_registration_response{namespace = "$Namespace"})
```

NRFA Deregistration Request Total Count

Description

This Panel displays NRF-A De-registration Request Total Count

Expression

```
sum(nrfa_nnr_fnm_deregistration_request{namespace = "$Namespace"})
```

NF Deregistration Success %

Description

This panel displays NF De-registration service success rate

Expression

```
((sum(nrfa_nnr_fnm_deregistration_response{namespace="$Namespace",  
status="204.noContent"})))/(sum(nrfa_nnr_fnm_deregistration_request{namespace="$Na  
mespace"})))*100
```

NRFA Deregistration Response Total Count

Description

This Panel displays NRF-A De-registration Response Total Count

Expression

```
sum(nrfa_nnr_fnm_deregistration_response{namespace = "$Namespace"})
```

NRFA Registration Request Stats

Description

NF Registration Request Stats

Expression

Registration Request Total Count :

```
sum(nrfa_nnr_fnm_registration_request{namespace = "$Namespace"})
```

NRFA Deregistration Request Stats

Description

NF De-registration Request Stats

Expression

Deregistration Request Total Count :

```
sum(nrfa_nnr_fnm_deregistration_request{namespace = "$Namespace"})
```

NRFA NF Registration Response Stats

Description

NF Registration Response Stats

Expression

Registration Response {{status}} Total Count :

```
sum(nrfa_nnr_fnm_registration_response{namespace = "$Namespace"}) by (status)
```

NRFA NF Deregistration Response Stats

Description

NF De-registration Response Stats

Expression

Deregistration Response {{status}} Total Count :

```
sum(nrfa_nnr_fnm_deregistration_response{namespace = "$Namespace"}) by (status)
```

Description

""

Description

""

NRFA Update Request Total Count

Description

This Panel displays NRF-A Update Request Total Count

Expression

```
sum(nrfa_nnr_fnm_update_request{namespace = "$Namespace"})
```

NF Update Success %

Description

This panel lists the NF Update Service success rate

Expression

```
((sum(nrfa_nnr_fnm_update_response{namespace="$Namespace",  
status="204.noContent"} or vector(0))) +  
(sum(nrfa_nnr_fnm_update_response{namespace="$Namespace", status="200.ok"} or  
vector(0)))) / (sum(nrfa_nnr_fnm_update_request{namespace="$Namespace"})) )*100
```

NRFA Update Response Total Count

Description

This Panel displays NRF-A Update Response Total Count

Expression

```
sum(nrfa_nnr_fnm_update_response{namespace = "$Namespace"})
```

NRFA Heart-beat Request Total Count

Description

This Panel displays NRF-A Heart-beat Request Total Count

Expression

```
sum(nrfa_nnr_fnm_heartbeat_request{namespace = "$Namespace"})
```

NF Heart-beat Success %

Description

This panel lists the Heart-beat Service success rate

Expression

```
((sum(nrfa_nnr_fnm_heartbeat_response{namespace="$Namespace",  
status="204.noContent"})) /  
(sum(nrfa_nnr_fnm_heartbeat_request{namespace="$Namespace"})))*100
```

NRFA Heart-beat Response Total Count

Description

This Panel displays NRF-A Heart-beat Response Total Count

Expression

```
sum(nrfa_nnr_fnm_heartbeat_response{namespace = "$Namespace"})
```

NRFA Update Request Stats

Description

NF Update Request Stats

Expression

Update Request Total Count :

```
sum(nrfa_nnr_fnm_update_request{namespace = "$Namespace"})
```

NRFA Heart-Beat Request Stats

Description

NF Heart-Beat Request Stats

Expression

Heart-Beat Request Total Count :


```
sum(nrfa_nnr_fnm_heartbeat_request{namespace = "$Namespace"})
```

NRFA NF Update Response Stats

Description

NF Update Response Stats

Expression

Update Response {{status}} Total Count :

```
sum(nrfa_nnr_fnm_update_response{namespace = "$Namespace"}) by (status)
```

NRFA NF Heart-Beat Response Stats

Description

NF Heart-Beat Response Stats

Expression

Heart-Beat Response {{status}} Total Count :

```
sum(nrfa_nnr_fnm_heartbeat_response{namespace = "$Namespace"}) by (status)
```

Description

""

Description

""

NRFA Subscription Request Total Count

Description

This Panel displays NRF-A Subscription Request Total Count

Expression

```
sum(nrfa_nnr_fnm_subscription_request{namespace = "$Namespace"})
```

NF subscription Success %

Description

This panel lists the NF Subscription service success rate

Expression

```
((sum(nrfa_nnr_fnm_subscription_response{namespace="$Namespace",  
status="204.noContent"} or vector(0))) +  
(sum(nrfa_nnr_fnm_subscription_response{namespace="$Namespace",  
status="201.created"} or  
vector(0))))/(sum(nrfa_nnr_fnm_subscription_request{namespace="$Namespace"}))*10  
0
```

NRFA subscription Response Total Count

Description

This Panel displays NRF-A Subscription Response Total Count

Expression

```
sum(nrfa_nnr_fnm_subscription_response{namespace = "$Namespace"})
```

NRFA notification Total Count

Description

This Panel displays NRF-A Notification Total Count

Expression

```
sum(nrfa_nnr_fnm_notification{namespace = "$Namespace"})
```

NF Notification Success %

Description

This panel lists the Notification service success rate

Expression

```
((sum(nrfa_nnr_fnm_notification_response{namespace="$Namespace",
status="204.noContent"}))) /
(sum(nrfa_nnr_fnm_notification{namespace="$Namespace"}))) * 100
```

NRFA Notification ResponseTotal Count

Description

This Panel displays NRF-A Notification Response Total Count

Expression

```
sum(nrfa_nnr_fnm_notification_response{namespace = "$Namespace"})
```

NRFA Subscription Request Stats

Description

NF Registration Request Stats

Expression

Subscription Request Total Count :

```
sum(nrfa_nnr_fnm_subscription_request{namespace = "$Namespace"})
```

NRFA Notification Stats

Description

NF Registration Request Stats

Expression

Notification Total Count :

```
sum(nrfa_nnr_fnm_notification{namespace = "$Namespace"})
```

NRFA NF Subscription Response Stats

Description

NF Subscription Response Stats

Expression

Subscription Response {{status}} Total Count :

```
sum(nrfa_nnrfa_nfm_subscription_response{namespace = "$Namespace"}) by (status)
```

NRFA NF Notification Response Stats

Description

NF Notification Response Stats

Expression

Notification Response {{status}} Total Count :

```
sum(nrfa_nnrfa_nfm_notification_response{namespace = "$Namespace"}) by (status)
```

Description

""

NRFA Discovery Request Total Count

Description

This Panel displays NRF-A Discovery Request Total Count

Expression

```
sum(nrfa_nnrfa_disc_request{namespace = "$Namespace"})
```

NF Discovery Success %

Description

This panel lists the NF Discovery service success rate

Expression

```
((sum(nrfa_nnrfdisc_response{namespace="$Namespace", status="200.ok"})) /
(sum(nrfa_nnrfdisc_request{namespace="$Namespace"})))*100
```

NRFA Discovery Response Total Count

Description

This Panel displays NRF-A Discovery response Total Count

Expression

```
sum(nrfa_nnrfdisc_response{namespace = "$Namespace"})
```

NRFA Discovery Request Stats

Description

NF Discovery Request Stats

Expression

{{target_nf_type}} Discovery Req Total Count :

```
sum(nrfa_nnrfdisc_request{namespace = "$Namespace"}) by (target_nf_type)
```

NRFA Discovery Response Stats

Description

NF Discovery Response Stats

Expression

{{target_nf_type}} Discovery Response {{status}} Total Count :

```
sum(nrfa_nnrfdisc_response{namespace = "$Namespace"}) by (target_nf_type, status)
```

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- [NRF Master Dashboard](#)

Summary

Dashboard Name	Panel Name
NRF Master Dashboard	Number of Pods
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	Registered NF Count
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF Registration Request Total Count
NRF Master Dashboard	NF Registration Response Total Count
NRF Master Dashboard	NF Registration Request Stats
NRF Master Dashboard	NF Registration Response Stats
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF De-registration Request Total Count
NRF Master Dashboard	NF De-registration Response Total Count
NRF Master Dashboard	NF De-registration Request Stats
NRF Master Dashboard	NF De-registration Response Stats
NRF Master Dashboard	NF CLI NF Delete Request Total Count
NRF Master Dashboard	NF CLI NF Delete Response Total Count
NRF Master Dashboard	NF CLI NF Delete Request Stats
NRF Master Dashboard	NF CLI NF Delete Response Stats
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF Discovery Request Stats
NRF Master Dashboard	NF Discovery Response Stats
NRF Master Dashboard	
NRF Master Dashboard	

Dashboard Name	Panel Name
NRF Master Dashboard	Options Request Stats
NRF Master Dashboard	Options Response Stats
NRF Master Dashboard	Discovery Cache Hits
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF HeartBeat Request Stats
NRF Master Dashboard	NF HeartBeat Timeout Stats
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF Update Request Stats
NRF Master Dashboard	NF Update Response Stats
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF Subscription Request/Delete Stats
NRF Master Dashboard	NF Subscription Response Stats
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	NF Notification Request Stats
NRF Master Dashboard	NF Notification Response Stats
NRF Master Dashboard	
NRF Master Dashboard	NF Oauth Token Request Stats
NRF Master Dashboard	NF Oauth Token Response Stats
NRF Master Dashboard	
NRF Master Dashboard	
NRF Master Dashboard	Initial Data Sync Request Sent
NRF Master Dashboard	Initial Data Sync Response Received
NRF Master Dashboard	Initial Data Sync Request Received
NRF Master Dashboard	Initial Data Sync Response Sent
NRF Master Dashboard	
NRF Master Dashboard	Nf Profile Sync Request Sent

Dashboard Name	Panel Name
NRF Master Dashboard	NF Profile Sync Response Received
NRF Master Dashboard	NF Profile Sync Request Received
NRF Master Dashboard	NF Profile Sync Response Sent
NRF Master Dashboard	
NRF Master Dashboard	Subscription Data Sync Request Sent
NRF Master Dashboard	Subscription Data Sync Response Received
NRF Master Dashboard	Subscription Data Sync Request Received
NRF Master Dashboard	Subscription Data Sync Response Sent
NRF Master Dashboard	
NRF Master Dashboard	Config Data Sync Request Sent
NRF Master Dashboard	Config Data Sync Response Received
NRF Master Dashboard	Config Data Sync Request Received
NRF Master Dashboard	Config Data Sync Response Sent
NRF Master Dashboard	
NRF Master Dashboard	Peer Nrf HB Request Sent
NRF Master Dashboard	Peer NRF HB Response Received
NRF Master Dashboard	Peer NRF HB Request Received
NRF Master Dashboard	Peer NRF HB Response Sent

NRF Master Dashboard

Description

NRF Fed Level Dashboard

Panels

Number of Pods

Description

This panel lists the number of NRF specific pods in the fed-nrf deployment

Expression

`{{pod_type}}`


```
ready_pods{namespace="fed-nrf"}
```

Description

Description

This panel lists the NRF stats

Registered NF Count

Description

Shows the Count of NFs being Registered

Expression

Registered {{source_nf_type}} Count :

```
sum(nrf_nf_registered) by (source_nf_type)
```

Description

Description

NF Registration Request Total Count

Description

NF Registration Request Total Count

Expression

```
sum(nfim_nnrf_registration_request)
```

NF Registration Response Total Count

Description

NF Registration Response Total Count

Expression

```
sum(nfim_nnrf_registration_response)
```

NF Registration Request Stats

Description

NF Registration Request Stats

Expression

{{source_nf_type}} RegistrationReq Total Count :

```
sum(nfim_nnrf_registration_request) by (source_nf_type)
```

NF Registration Response Stats

Description

NF Registration Response Stats

Expression

{{source_nf_type}} RegistrationResp {{status}} Total Count:

```
sum(nfim_nnrf_registration_response) by (source_nf_type, status)
```

Kavita {{status}}

Description

Description

NF De-registration Request Total Count

Description

NF De-registration Request Total Count

Expression

```
sum(nfim_nnrf_deregistration_request)
```

NF De-registration Response Total Count

Description

NF De-registration Response Total Count

Expression

```
sum(nfim_nnrf_deregistration_response)
```

NF De-registration Request Stats

Description

NF De-registration Request Stats

Expression

{{source_nf_type}} De-registrationReq Total Count :

```
sum(nfim_nnrf_deregistration_request) by (source_nf_type)
```

NF De-registration Response Stats

Description

NF De-registration Response Stats

Expression

{{source_nf_type}} De-registrationResp {{status}} Total Count

```
sum(nfim_nnrf_deregistration_response) by (source_nf_type,status)
```

NF CLI NF Delete Request Total Count

Description

NF CLI NF Delete Request Total Count

Expression

```
sum(nfim_nnr_f_cli_nf_delete_request)
```

NF CLI NF Delete Response Total Count

Description

NF CLI NF Delete Response Total Count

Expression

```
sum(nfim_nnr_f_cli_nf_delete_request)
```

NF CLI NF Delete Request Stats

Description

NF CLI NF Delete Request Stats

Expression

{{source_nf_type}} DeleteReq Total Count :

```
sum(nfim_nnr_f_cli_nf_delete_request) by (source_nf_type)
```

NF CLI NF Delete Response Stats

Description

NF CLI NF Delete Response Stats

Expression

{{source_nf_type}} CLI NF DeleteResp {{status}} Total Count:

```
sum(nfim_nnr_f_cli_nf_delete_response) by (source_nf_type, status)
```

Kavita {{status}}

Description

Description

NF Discovery Request Stats

Description

NF Discovery Request Stats

Expression

{{source_nf_type}}-DiscoveryReq Total Count :

```
sum(nfsel_nnrfdiscoveryrequest) by (source_nf_type)
```

NF Discovery Response Stats

Description

NF Discovery Response Stats

Expression

{{source_nf_type}} DiscoveryResp {{status}} Total Count :

```
sum(nfsel_nnrfdiscoveryresponse) by (source_nf_type, status)
```

Description

Description

Options Request Stats

Description

Options Request Stats

Expression

Options Request Total Count :

```
sum(nfim_nnrfoptionsrequest)
```

Options Response Stats

Description

Options Response Stats

Expression

Options Response Total Count :

```
sum(nfim_nnr_options_response)
```

Discovery Cache Hits

Description

Shows the Count of Discovery response from cache

Expression

AF Count :

```
nfsel_nnr_discovery_af_cache_response{}
```

AUSF Count :

```
nfsel_nnr_discovery_ausf_cache_response{}
```

AMF Count :

```
nfsel_nnr_discovery_amf_cache_response{}
```

BSF Count :

```
nfsel_nnr_discovery_bsf_cache_response{}
```

CBCF Count :

```
nfsel_nnr_discovery_cbcf_cache_response{}
```

CHF Count :

```
nfsel_nnrfdiscovery_chf_cache_response{}
```

EIR Count :

```
nfsel_nnrfdiscovery_eir_cache_response{}
```

GMLC Count :

```
nfsel_nnrfdiscovery_gmlc_cache_response{}
```

LMF Count :

```
nfsel_nnrfdiscovery_lmf_cache_response{}
```

N3IWF Count :

```
nfsel_nnrfdiscovery_n3iwf_cache_response{}
```

NEF Count :

```
nfsel_nnrfdiscovery_nef_cache_response{}
```

NSSF Count :

```
nfsel_nnrfdiscovery_nssf_cache_response{}
```

NWDAF Count :

```
nfsel_nnrfdiscovery_nwdaf_cache_response{}
```

PCF Count :

```
nfsel_nnrfdiscoverypcf_cache_response{}
```

PCSCF Count :

```
nfsel_nnrfdiscoverypcscf_cache_response{}
```

SMF Count :

```
nfsel_nnrfdiscoverysmf_cache_response{}
```

SMSF Count :

```
nfsel_nnrfdiscoverysmsf_cache_response{}
```

UDM Count :

```
nfsel_nnrfdiscoveryudm_cache_response{}
```

UDR Count :

```
nfsel_nnrfdiscoveryudr_cache_response{}
```

UPF Count :

```
nfsel_nnrfdiscoveryupf_cache_response{}
```

Description

Description

NF HeartBeat Request Stats

Description

NF HeartBeat Request Stats

Expression

{{source_nf_type}} HeartBeat Total Count :

```
sum(nfim_nnrfs_heartbeat) by (source_nf_type)
```

NF HeartBeat Timeout Stats

Description

NF HeartBeat Timeout Stats

Expression

{{source_nf_type}} HeartBeat Timeout Total Count :

```
sum(nfim_nnrfs_heartbeat_timeout) by (source_nf_type)
```

Description

Description

NF Update Request Stats

Description

NF Update Request Stats

Expression

{{source_nf_type}} UpdateReq Total Count :

```
sum(nfim_nnrfs_update_request) by (source_nf_type)
```

NF Update Response Stats

Description

NF Update Response Stats

Expression

{{source_nf_type}} UpdateResp-{{status}} Total Count :

```
sum(nfim_nnrfs_update_response) by (source_nf_type, status)
```

Description

Description

NF Subscription Request/Delete Stats

Description

NF Subscription Request/Delete Stats

Expression

{{requested_nf_type}} SubscriptionReq Total Count :

```
sum(nfim_nnrfs_subscription_request) by (requested_nf_type)
```

{{requested_nf_type}} SubscriptionDel Total Count :

```
sum(nfim_nnrfs_subscription_delete) by (requested_nf_type)
```

NF Subscription Response Stats

Description

NF Subscription Response Stats

Expression

{{requested_nf_type}} SubscriptionResp {{status}} Total Count :

```
sum(nfim_nnrfs_subscription_response) by (requested_nf_type, status)
```

{{requested_nf_type}} SubscriptionDelResp {{status}} Total Count :

```
sum(nfim_nnrf_subscription_delete_response) by (requested_nf_type, status)
```

Description

Description

NF Notification Request Stats

Description

NF Notification Request Stats

Expression

{{requested_nf_type}} NotificationReq Total Count :

```
sum(nfim_nnrf_notification) by (requested_nf_type)
```

NF Notification Response Stats

Description

NF Notification Response Stats

Expression

{{requested_nf_type}} NotificationResp {{status}} Total Count :

```
sum(nfim_nnrf_notification_response) by (requested_nf_type, status)
```

Description

NF Oauth Token Request Stats

Description

NF Oauth Token Request Stats

Expression

OauthTokenReq Total Count :

```
sum(oauth_nnrp_access_token_request)
```

NF Oauth Token Response Stats

Description

NF Oauth Token Response Stats

Expression

OauthResp {{status}} Total Count :

```
sum(oauth_nnrp_access_token_response) by (status)
```

Description

Description

Initial Data Sync Request Sent

Description

Initial Data Sync Request Sent

Expression

{{target_nrf_id}} Count :

```
sum(nfim_nnrp_sync_initial_data_request_tx) by (target_nrf_id)
```

Initial Data Sync Response Received

Description

Initial Data Sync Response Received

Expression

{{target_nrf_id}} {{status}} Count :

```
sum(nfim_nnrp_sync_initial_data_response_rx) by (target_nrf_id , status)
```

Initial Data Sync Request Received

Description

Initial Data Sync Request Received

Expression

{{source_nrf_id}} Count :

```
sum(nfim_nnrp_sync_initial_data_request_rx) by (source_nrf_id)
```

Initial Data Sync Response Sent

Description

Initial Data Sync Response Sent

Expression

{{source_nrf_id}} {{status}} Count :

```
sum(nfim_nnrp_sync_initial_data_response_tx) by (source_nrf_id , status)
```

Description

Nf Profile Sync Request Sent

Description

Nf Profile Sync Request Sent

Expression

{{target_nrf_id}} Count :

```
sum(nfim_nnrp_sync_nf_profile_request_tx) by (target_nrf_id)
```

NF Profile Sync Response Received

Description

NF Profile Sync Response Received

Expression

{{target_nrf_id}} {{status}} Count :

```
sum(nfim_nnrp_sync_nf_profile_response_rx) by (target_nrf_id, status)
```

NF Profile Sync Request Received

Description

NF Profile Sync Request Received

Expression

{{source_nrf_id}} Count :

```
sum(nfim_nnrp_sync_nf_profile_request_rx) by (source_nrf_id)
```

NF Profile Sync Response Sent

Description

NF Profile Sync Response Sent

Expression

{{source_nrf_id}} {{status}}Count :

```
sum(nfim_nnrp_sync_nf_profile_response_tx) by (source_nrf_id , status)
```

Description

Subscription Data Sync Request Sent

Description

Subscription Data Sync Request Sent

Expression

{{target_nrf_id}} Count :

```
sum(nfim_nnrf_sync_subscription_data_request_tx) by (target_nrf_id)
```

Subscription Data Sync Response Received

Description

Subscription Data Sync Response Received

Expression

{{target_nrf_id}} {{status}} Count :

```
sum(nfim_nnrf_sync_subscription_data_response_rx) by (target_nrf_id , status)
```

Subscription Data Sync Request Received

Description

Subscription Data Sync Request Received

Expression

{{source_nrf_id}} Count :

```
sum(nfim_nnrf_sync_subscription_data_request_rx) by (source_nrf_id)
```

Subscription Data Sync Response Sent

Description

Subscription Data Sync Response Sent

Expression

{{source_nrf_id}} {{status}}Count :

```
sum(nfim_nnrf_sync_subscription_data_response_tx) by (source_nrf_id , status)
```

Description

Config Data Sync Request Sent

Description

Config Data Sync Request Sent

Expression

{{target_nrf_id}} Count :

```
sum(nfim_nnrfsync_config_data_request_tx) by (target_nrf_id)
```

Config Data Sync Response Received

Description

Config Data Sync Response Received

Expression

{{target_nrf_id}} {{status}} Count :

```
sum(nfim_nnrfsync_config_data_response_rx) by (target_nrf_id, status)
```

Config Data Sync Request Received

Description

Config Data Sync Request Received

Expression

{{source_nrf_id}} Count :

```
sum(nfim_nnrfsync_config_data_request_rx) by (source_nrf_id)
```

Config Data Sync Response Sent

Description

Config Data Sync Response Sent

Expression

{{source_nrf_id}} {{status}}Count :


```
sum(nfim_nnrf_sync_config_data_response_tx) by (source_nrf_id , status)
```

Description

Peer Nrf HB Request Sent

Description

Peer Nrf HB Request Sent

Expression

{{target_nrf_id}} Count :

```
sum(nfim_nnrf_peer_nrf_heartbeat_request_tx) by (target_nrf_id)
```

Peer NRF HB Response Received

Description

Peer NRF HB Response Received

Expression

{{target_nrf_id}} {{status}} Count :

```
sum(nfim_nnrf_peer_nrf_heartbeat_response_rx) by (target_nrf_id , status)
```

Peer NRF HB Request Received

Description

Peer NRF HB Request Received

Expression

{{source_nrf_id}} Count :

```
sum(nfim_nnrf_peer_nrf_heartbeat_request_rx) by (source_nrf_id)
```

Peer NRF HB Response Sent

Description

Peer NRF HB Response Sent

Expression

{{source_nrf_id}} {{status}}Count :

```
sum(nfim_nnrf_peer_nrf_heartbeat_response_tx) by (source_nrf_id , status)
```

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- [NSSF Master Dashboard](#)

Summary

Dashboard Name	Panel Name
NSSF Master Dashboard	Number of Pods
NSSF Master Dashboard	Nsselection request for Registration
NSSF Master Dashboard	Network Slice Selection request for PDU Establishment

NSSF Master Dashboard

Description

NSSF Fed Level Dashboard

Panels

Number of Pods

Description

This panel lists the number of NSSF specific pods in the fed-nssf deployment.

Expression

{{pod_type}}

```
ready_pods{namespace="fed-nssf"}
```

Nsselection request for Registration

Description

Nsselection request for Registration

Expression

ss_nssf_nsselection_request_for_registration

```
sum(ss_nssf_nsselection_request_for_registration{})
```

ss_nssf_nsselection_response_for_registration

```
sum(ss_nssf_nsselection_response_for_registration{})
```

Network Slice Selection request for PDU Establishment

Description

Network Slice Selection request for PDU Establishment

Expression

ss_nssf_nsselection_request_for_pdu_session

```
sum(ss_nssf_nsselection_request_for_pdu_session{})
```

ss_nssf_nsselection_response_for_pdu_session

```
sum(ss_nssf_nsselection_response_for_pdu_session{})
```

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- [Summary](#)
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Summary

Dashboard Name	Panel Name
Open Policy Agent Runtime Metrics	Avg Response Time
Open Policy Agent Runtime Metrics	HTTP Response Time
Open Policy Agent Runtime Metrics	Total Used Memory

Open Policy Agent Runtime Metrics

Description

Open Policy Agent Runtime Metrics

Panels

Avg Response Time

Description

The average response time

Expression

```
sum(rate(http_request_duration_seconds_sum{ namespace="fed-opa"}[$interval])) /
sum(rate(http_request_duration_seconds_count{namespace="fed-opa"}[$interval]))
```

HTTP Response Time

Description

The HTTP response time

Expression

50th Quantile (Median)

```
histogram_quantile(0.5,
sum(rate(http_request_duration_seconds_bucket{namespace="fed-opa"}[$interval])) by
(1e))
```

90th Quantile

```

    histogram_quantile(0.9,
    sum(rate(http_request_duration_seconds_bucket{namespace="fed-opa"}[$interval])) by
    (1e))

```

95th Quantile

```

    histogram_quantile(0.95,
    sum(rate(http_request_duration_seconds_bucket{namespace="fed-opa"}[$interval])) by
    (1e))

```

99th Quantile

```

    histogram_quantile(0.99,
    sum(rate(http_request_duration_seconds_bucket{namespace="fed-opa"}[$interval])) by
    (1e))

```

Total Used Memory

Description

The total used memory

Expression

```

    go_memstats_sys_bytes{namespace="fed-opa"}

```

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- [Summary](#)
- [Multi NF Dashboard - Global Error](#)
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Summary

Dashboard Name	Panel Name
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Dashboard Name	Panel Name
Multi NF Dashboard - Global Error	Per Service Error Response Rate Per Minute
Multi NF Dashboard - Global Error	Per Service Error Response Rate Per Minute
Multi NF Dashboard - Global Error	Pod down list
Multi NF Dashboard - Source more details	HTTP Request sent/Response received diff Percentage - \$pod
Multi NF Dashboard - Source more details	HTTP Response Latency of Pod - "\$pod"
Multi NF Dashboard - Source more details	HTTP Response Received by "\$pod" - SUCCESS
Multi NF Dashboard - Source more details	HTTP Response Received by "\$pod" - FAILURE
Multi NF Dashboard - Source more details	HTTP FAILURE Response Per Minute by Pod/Destination/Status
Multi NF Dashboard - Source to Dest Details	Total HTTP Request Sent
Multi NF Dashboard - Source to Dest Details	Total HTTP Response Received
Multi NF Dashboard - Source to Dest Details	Total HTTP Response - SUCCESS
Multi NF Dashboard - Source to Dest Details	Total HTTP Response - FAILURE
Multi NF Dashboard - Source to Dest Details	HTTP Request Sent Per Minute
Multi NF Dashboard - Source to Dest Details	Total SUCCESS Received Per Minute
Multi NF Dashboard - Source to Dest Details	Total FAILURE Received Per Minute
Multi NF Dashboard - Source to Dest Details	HTTP FAILURE Response Per Minute by Source/Destination/Status
Multi NF Dashboard - Source to Dest Details	HTTP FAILURE Response Per Minute by Source/Destination/Status
Multi NF Dashboard - Source to Dest Details	HTTP Request sent/ Response received diff Percentage

Multi NF Dashboard - Global Error

Description

Dashboard gives global view of HTTP/Network failure in the system/cluster

Panels

Per Service Error Response Rate Per Minute

Description

Per service error response rate in the system/cluster

Expression

Namespace: {{namespace}}, Source: {{job}}, Destination: {{destination_host}}

```
topk(50,sort_desc(sum by (namespace,job,destination_host)
(irate(http_async_client_response_received{http_status_code=~"5..|4..",namespace=~
"$namespace"}[$__range]) * 60) > 1))
```

Per Service Error Response Rate Per Minute

Description

Per Service Error Response Rate Per Minute Time Graph

Expression

{{namespace}}/{{job}}/{{destination_host}}

```
sum without(http_status_code)
((sum_over_time(http_async_client_response_received{http_status_code=~"5..",namesp
ace=~"$namespace"}[$datapoints]) / ignoring(http_status_code) group_left()
sum_over_time(http_async_client_request_enqueue_success{namespace=~"$namespace"}
[$datapoints])) * 100) >= $percentage
```

{{namespace}}/{{job}}

```
sum by (namespace,job)
(irate(http_async_client_response_received{http_status_code=~"5..|4..",namespace=~
"$namespace"}[10m]) * 60) > 0
```

Pod down list

Description

List of pods down currently. Pick time range to see history of pod down alerts

Expression

{{namespace}}/{{job}}/{{pod}}

```
count(changes (
  ALERTS_FOR_STATE{alertname=~"prometheus_target_missing_alert",namespace=~"$namespace"}[$__range] ) + 1 ) by (namespace,job,pod)
```

Multi NF Dashboard - Source more details

Description

HTTP/Network failure dashboard across the NFs at the pod level

Panels

HTTP Request sent/Response received diff Percentage - \$pod

Description

HTTP Request sent/Response received difference Percentage

Expression

Source pod - {{pod}}

```
((sum by (job,pod)
(irate(http_async_client_request_enqueue_success{namespace=~"$namespace",job=~"$job",pod=~"$pod"}[$__range]) * 60) - sum by (job,pod)
(irate(http_async_client_response_received{namespace=~"$namespace",job=~"$job",pod=~"$pod"}[$__range]) * 60))/sum by (job,pod)
(irate(http_async_client_request_enqueue_success{namespace=~"$namespace",job=~"$job",pod=~"$pod"}[$__range]) * 60))*100) > 0.00001
```

HTTP Response Latency of Pod - "\$pod"

Description

HTTP response latency heatmap

Expression

{{le}}

```
sum by (le)
(irate(http_client_response_duration_seconds_bucket{namespace=~"$namespace",job=~"$job",pod=~"$pod"}[$__range])*60)
```

HTTP Response Received by "\$pod" - SUCCESS

Description

HTTP success Response received at the pod level

Expression

Status = {{http_status_code}}

```
sum by(http_status_code)
(http_async_client_response_received{namespace=~"$namespace",job=~"$job",pod=~"$pod",http_status_code=~"2.."})
```

HTTP Response Received by "\$pod" - FAILURE

Description

HTTP failed response received at the pod level

Expression

Status = {{http_status_code}}

```
sum by(http_status_code)
(http_async_client_response_received{namespace=~"$namespace",job=~"$job",pod=~"$pod",http_status_code=~"5..|4.."})
```

HTTP FAILURE Response Per Minute by Pod/Destination/Status

Description

HTTP FAILURE Response Per Minute by Pod/Destination/Status

Expression

Pod = {{pod}} , Destination = {{destination_host}}, Status = {{http_status_code}}

```
topk($top_pods,(sum by (pod,destination_host,http_status_code)
(irate(http_async_client_response_received{http_status_code=~"5..|4..",namespace=~
"$namespace",job=~"$job",pod=~"$pod",destination_host=~"$destination_host"}
[$__range]) * 60)))
```

Multi NF Dashboard - Source to Dest Details

Description

Multi NF dashboard gives HTTP/Network error details between source and destination

Panels

Total HTTP Request Sent

Description

Total HTTP request sent from source App

Expression

```
sum(http_async_client_request_enqueue_success{namespace=~"$namespace",
job=~"$job"})
```

Total HTTP Response Received

Description

Total HTTP Response Received at the source App

Expression

```
sum(http_async_client_response_received{namespace=~"$namespace",job=~"$job"})
```

Total HTTP Response - SUCCESS

Description

Total HTTP Response - SUCCESS received at the source

Expression

```
sum(http_async_client_response_received{namespace=~"$namespace",job=~"$job",http_status_code=~"2.."})
```

Total HTTP Response - FAILURE

Description

Total HTTP Response - FAILURES received at the source

Expression

```
sum(http_async_client_response_received{namespace=~"$namespace",job=~"$job",http_status_code=~"5..|4.."})
```

HTTP Request Sent Per Minute

Description

HTTP request sent per minute from the source

Expression

Request Sent

```
sum(irate(http_async_client_request_enqueue_success{namespace=~"$namespace",job=~"$job",destination_host=~"$destination_host"}[10m]) * 60)
```

Total SUCCESS Received Per Minute

Description

Total success response received at the source

Expression

```
sum(irate(http_async_client_response_received{namespace=~"$namespace",job=~"$job",destination_host=~"$destination_host",http_status_code=~"2.."}[10m]) * 60)
```

Total FAILURE Received Per Minute

Description

Total HTTP Failures received per minute at the source

Expression

Status = {{http_status_code}}

```
sum(sum_over_time(http_async_client_response_received{namespace=~"$namespace",
job=~"$job",destination_host=~"$destination_host"}[$datapoints])) by (http_status_code)
```

Status = {{http_status_code}}

```
sum by (http_status_code)
(irate(http_async_client_response_received{namespace=~"$namespace",job=~"$job",destination_host=~"$destination_host",http_status_code=~"5..|4.."}[10m]) * 60) > 1
```

HTTP FAILURE Response Per Minute by Source/Destination/Status

Description

HTTP FAILURE Response Per Minute by Source/Destination/Status

Expression

Source = {{job}} , Destination = {{destination_host}}, Status = {{http_status_code}}

```
sort_desc(topk(10,sum by (namespace,job,destination_host,http_status_code)
(irate(http_async_client_response_received{http_status_code=~"5..|4..",namespace=~"$namespace",job=~"$job",destination_host=~"$destination_host"}[10m]) * 60) > 0))
```

HTTP FAILURE Response Per Minute by Source/Destination/Status

Description

HTTP FAILURE Response Per Minute by Source/Destination/Status

Expression

Source = {{job}} , Destination = {{destination_host}}, Status = {{http_status_code}}

```
sum by (namespace,job,destination_host,http_status_code)
(irate(http_async_client_response_received{http_status_code=~"5..|4..",namespace=~
"$namespace",job=~"$job",destination_host=~"$destination_host"}[10m]) * 60) > 0
```

HTTP Request sent/ Response received diff Percentage

Description

HTTP Request sent/ Response received difference Percentage

Expression

Source: {{job}}

```
((((sum by(namespace,job)
(irate(http_async_client_request_enqueue_success{namespace=~"$namespace",
job=~"$job"}[10m]) * 60)) - (sum by(namespace,job)
(irate(http_async_client_response_received{namespace=~"$namespace", job=~"$job"}
[10m]) * 60))) / (sum by(namespace,job)
(irate(http_async_client_request_enqueue_success{namespace=~"$namespace",
job=~"$job"}[10m]) * 60))) * 100) > -0.00000001
```

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Dashboard Name	Panel Name
PFCP-C Associations Procedures Dashboard	Association Setup - Attempts
PFCP-C Associations Procedures Dashboard	Association Setup - Success %
PFCP-C Associations Procedures Dashboard	Association Setup - Successes
PFCP-C Associations Procedures Dashboard	Association Setup - Failures
PFCP-C Associations Procedures Dashboard	Association Update - Attempts

Dashboard Name	Panel Name
PFCP-C Associations Procedures Dashboard	Association Update - Success %
PFCP-C Associations Procedures Dashboard	Association Update - Successes
PFCP-C Associations Procedures Dashboard	Association Update - Failures
PFCP-C Associations Procedures Dashboard	Peer Association Update - Attempts
PFCP-C Associations Procedures Dashboard	Peer Association Update - Success %
PFCP-C Associations Procedures Dashboard	Peer Association Update - Successes
PFCP-C Associations Procedures Dashboard	Peer Association Update - Failures
PFCP-C Associations Procedures Dashboard	Association Release - Attempts
PFCP-C Associations Procedures Dashboard	Association Release - Success %
PFCP-C Associations Procedures Dashboard	Association Release - Successes
PFCP-C Associations Procedures Dashboard	Association Release - Failures
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PFCP-C Associations Procedures Dashboard	Heartbeat - Success %
PFCP-C Associations Procedures Dashboard	Heartbeat - Successes
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PFCP-C Associations Protocols Dashboard	Association Release - Requests Sent
PFCP-C Associations Protocols Dashboard	Association Release - Retransmitted Requests Sent
PFCP-C Associations Protocols Dashboard	Association Release - Responses Received
PFCP-C Associations Protocols Dashboard	Heartbeat - Requests Sent
PFCP-C Associations Protocols Dashboard	Heartbeat - Retransmitted Requests Sent
PFCP-C Associations Protocols Dashboard	Heartbeat - Responses Received
PFCP-C Session Bundling Dashboard	Failure %
PFCP-C Session Bundling Dashboard	PFCP number of bundled message sent failure - \$cause

Dashboard Name	Panel Name
PFCP-C Session Bundling Dashboard	Failure %
PFCP-C Session Bundling Dashboard	PFCP number of bundled message decode failure - \$causeR
PFCP-C Session Bundling Dashboard	Pfcp number of bundled messages sent \$bundle_trigger
PFCP-C Session Bundling Dashboard	Pfcp number of bundled messages sent \$peer_ip
PFCP-C Session Bundling Dashboard	PFCP number of bundled messages received \$peer_ip
PFCP-C Session Bundling Dashboard	\$peer_ip
PFCP-C Session Bundling Dashboard	Message Size in Bundle
PFCP-C Session Bundling Dashboard	\$peer_ip
PFCP-C Session Bundling Dashboard	Number of messages in Bundle
PFCP-C Session Bundling Dashboard	\$peer_ip
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PFCP-C Session Bundling Dashboard	\$peer_ip
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PFCP-C Sessions Procedures Dashboard	Session Establishment - Attempts
PFCP-C Sessions Procedures Dashboard	Session Establishment - Success %
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PFCP-C Sessions Procedures Dashboard	Session Modification - Failures
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PFCP-C Sessions Procedures Dashboard	Session Deletion - Success %
PFCP-C Sessions Procedures Dashboard	Session Deletion - Successes
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PFCP-C Sessions Procedures Dashboard	Session Report - Attempts
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Dashboard Name	Panel Name
PFCP-C Sessions Procedures Dashboard	Session Report - Successes
PFCP-C Sessions Procedures Dashboard	Session Report - Failures
PFCP-C Sessions Protocols Dashboard	Session Establishment - Requests Sent
PFCP-C Sessions Protocols Dashboard	Session Establishment - Retransmitted Requests Sent
PFCP-C Sessions Protocols Dashboard	Session Establishment - Responses Received
PFCP-C Sessions Protocols Dashboard	Session Modification - Requests Sent
PFCP-C Sessions Protocols Dashboard	Session Modification - Retransmitted Requests Sent
PFCP-C Sessions Protocols Dashboard	Session Modification - Responses Received
PFCP-C Sessions Protocols Dashboard	Session Deletion - Requests Sent
PFCP-C Sessions Protocols Dashboard	Session Deletion - Retransmitted Requests Sent
PFCP-C Sessions Protocols Dashboard	Session Deletion - Responses Received
PFCP-C Sessions Protocols Dashboard	Session Report - Requests Received
PFCP-C Sessions Protocols Dashboard	Session Report - Responses Sent

PFCP-C Associations Procedures Dashboard

Description

PFCP-C Associations Procedures Dashboard

Panels

Association Setup - Attempts

Description

The number of association setup attempts.

Expression

```
sum(pfcpc_association_setup_attempt{namespace=~"$namespace", job="pfcpc"})
```

Association Setup - Success %

Description

Association Setup - Success %

Expression

```
((sum(pfcg_association_setup_success{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"}))/(sum(pfcg_association_setup_attempt{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})))*100
```

Association Setup - Successes

Description

The number of association setup successes.

Expression

```
sum(pfcg_association_setup_success{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})
```

Association Setup - Failures

Description

The number of association setup failures.

Expression

```
sum(pfcg_association_setup_failure{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id", cause=~"$cause"})
```

Association Update - Attempts

Description

The number of association update attempts.

Expression

```
sum(pfcg_association_update_attempt{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})
```

Association Update - Success %

Description

Association Update - Success %

Expression

```
((sum(pfcg_association_update_success{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"}))/(sum(pfcg_association_update_attempt{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})))*100
```

Association Update - Successes

Description

The number of association update successes.

Expression

```
sum(pfcg_association_update_success{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})
```

Association Update - Failures

Description

The number of association update failures.

Expression

```
sum(pfcg_association_update_failure{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id", cause=~"$cause"})
```

Peer Association Update - Attempts

Description

The number of association update peer attempts.

Expression

```
sum(pfcg_peer_association_update_attempt{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})
```

Peer Association Update - Success %

Description

Peer Association Update - Success %

Expression

```
((sum(pfcpeer_association_update_success{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"}))/sum(pfcpeer_association_update_attempt{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"}))*100
```

Peer Association Update - Successes

Description

The number of association update peer successes.

Expression

```
sum(pfcpeer_association_update_success{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"})
```

Peer Association Update - Failures

Description

The number of association update peer failures.

Expression

```
sum(pfcpeer_association_update_failure{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id", cause=~"$cause"})
```

Association Release - Attempts

Description

The number of association release attempts.

Expression

```
sum(pfcg_association_release_attempt{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id"})
```

Association Release - Success %

Description

Association Release - Success %

Expression

```
((sum(pfcg_association_release_success{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id"}))/(sum(pfcg_association_release_attempt{namespace=~"$namespac  
e", job="pfcg", peer_id=~"$peer_id"})))*100
```

Association Release - Successes

Description

The number of association release successes.

Expression

```
sum(pfcg_association_release_success{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id"})
```

Association Release - Failures

Description

The number of association release failures.

Expression

```
sum(pfcg_association_release_failure{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id", cause=~"$cause"})
```

Heartbeat - Attempts

Description

The number of heartbeat attempts.

Expression

```
sum(pfcg_heartbeat_attempt{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id"})
```

Heartbeat - Success %

Description

Heartbeat - Success %

Expression

```
((sum(pfcg_heartbeat_success{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id"}))/(sum(pfcg_heartbeat_attempt{namespace=~"$namespace",
job="pfcg", peer_id=~"$peer_id"})))*100
```

Heartbeat - Successes

Description

The number of heartbeat successes.

Expression

```
sum(pfcg_heartbeat_success{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id"})
```

Heartbeat - Failures

Description

The number of heartbeat failures.

Expression

```
sum(pfcg_heartbeat_failure{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id", cause=~"$cause"})
```

Description

PFCP-C Associations Protocols Dashboard

Panels

Association Setup - Requests Sent

Description

The number of association setup request messages sent.

Expression

```
sum(pfcpc_association_setup_request_sent{namespace=~"$namespace", job="pfcpc", peer_ip=~"$peer_ip"})
```

Association Setup - Retransmitted Requests Sent

Description

The number of association setup retransmitted request messages sent.

Expression

```
sum(pfcpc_association_setup_retransmitted_request_sent{namespace=~"$namespace", job="pfcpc", peer_ip=~"$peer_ip"})
```

Association Setup - Responses Received

Description

The number of association setup response messages received.

Expression

```
sum(pfcpc_association_setup_response_received{namespace=~"$namespace", job="pfcpc", peer_ip=~"$peer_ip", pfcpc_cause=~"$pfcpc_cause"})
```

Association Update - Requests Sent

Description

The number of association update request messages sent.

Expression

```
sum(pfcg_association_update_request_sent{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip"})
```

Association Update - Retransmitted Requests Sent

Description

The number of association update retransmitted request messages sent.

Expression

```
sum(pfcg_association_update_retransmitted_request_sent{namespace=~"$namespace",  
job="pfcg", peer_ip=~"$peer_ip"})
```

Association Update - Responses Received

Description

The number of association update response messages received.

Expression

```
sum(pfcg_association_update_response_received{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

Association Release - Requests Sent

Description

The number of association release request messages sent.

Expression

```
sum(pfcg_association_release_request_sent{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip"})
```

Association Release - Retransmitted Requests Sent

Description

The number of association release retransmitted request messages sent.

Expression

```
sum(pfc_p_association_release_retransmitted_request_sent{namespace=~"$namespace",  
job="pfc_p", peer_ip=~"$peer_ip"})
```

Association Release - Responses Received

Description

The number of association release response messages received.

Expression

```
sum(pfc_p_association_release_response_received{namespace=~"$namespace",  
job="pfc_p", peer_ip=~"$peer_ip", pfc_p_cause=~"$pfc_p_cause"})
```

Heartbeat - Requests Sent

Description

The number of heartbeat request messages sent.

Expression

```
sum(pfc_p_heartbeat_request_sent{namespace=~"$namespace", job="pfc_p",  
peer_ip=~"$peer_ip"})
```

Heartbeat - Retransmitted Requests Sent

Description

The number of heartbeat retransmitted request messages sent.

Expression

```
sum(pfc_p_heartbeat_retransmitted_request_sent{namespace=~"$namespace", job="pfc_p",  
peer_ip=~"$peer_ip"})
```

Heartbeat - Responses Received

Description

The number of heartbeat response messages received.

Expression

```
sum(pfcg_heartbeat_response_received{namespace=~"$namespace", job="pfcg",
peer_ip=~"$peer_ip"})
```

PFCE-C Session Bundling Dashboard

Description

PFCE-C Session Bundling Statistics Dashboard

PFCE Sent Failures - \$peer_ip Panels

Panels

Failure %

Expression

```
((sum(pfcg_bundle_message_sent_failure{namespace=~"$namespace",
job="pfcg",cause=~"$cause"}))/(sum(pfcg_bundle_message_sent{namespace=~"$namespace",
job="pfcg",cause=~"$cause"})))*100
```

PFCE number of bundled message sent failure - \$cause

Description

Number of PFCE bundle messages sent failures to a particular peer(represented by peer_ip and peer_port)

Expression

```
sum(pfcg_bundle_message_sent_failure{namespace=~"$namespace",
job="pfcg",cause=~"$cause"} OR on() vector(0))
```

PFCE Receive Failures - \$peer_ip Panels

Failure %

Expression

```
((sum(pfcpl_bundle_message_received_failure{namespace=~"$namespace",
job="pfcpl",cause=~"$cause"})))/(sum(pfcpl_bundle_message_received{namespace=~"$names
pace", job="pfcpl",cause=~"$cause"})))*100
```

PFCP number of bundled message decode failure - \$causeR

Description

Number of PFCP bundle messages decode failures, cause represents failure reason(like invalid packet)

Expression

```
sum(pfcpl_bundle_message_received_failure{namespace=~"$namespace",
job="pfcpl",cause=~"$causeR"} OR on() vector(0))
```

PFCP Number of Sent Messages Per Bundle Trigger - \$peer_ip Panels

Pfcpl number of bundled messages sent \$bundle trigger

Description

Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle

Expression

_auto

```
sum(pfcpl_bundle_message_sent{namespace=~"$namespace",
job="pfcpl",bundle_trigger=~"$bundle_trigger"} OR on() vector(0))
```

PFCP Total Number of Sent and Received Messages Per Peer Panels

Pfcpl number of bundled messages sent \$peer ip

Description

Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle

Expression

_auto

```
sum(pfcg_bundle_message_sent{namespace=~"$namespace",
job="pfcg",peer_ip=~"$peer_ip"} OR on() vector(0))
```

PFCP number of bundled messages received \$peer ip

Description

Number of PFCP bundle messages received from a particular peer(represented by peer_ip and peer_port).

Expression

```
sum(pfcg_bundle_message_received{namespace=~"$namespace",
job="pfcg",peer_ip=~"$peer_ip"} OR on() vector(0))
```

PFCP Sent Message Size in Bundle Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcg_size_of_message_of_bundled_request_sent_sum{namespace=~"$namespace",
job="pfcg",peer_ip=~"$peer_ip"})/(pfcg_size_of_message_of_bundled_request_sent_cou
nt{namespace=~"$namespace", job="pfcg",peer_ip=~"$peer_ip"}))
```

Message Size in Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfcg_size_of_message_of_bundled_request_sent_bucket{namespace=~"$namespace",
job="pfcg",peer_ip=~"$peer_ip"})
```

PFCP Sent Message Count in Bundle Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfc_p_number_of_messages_in_bundled_request_sent_sum{namespace=~"$namespace",
job="pfc_p",peer_ip=~"$peer_ip"})/(pfc_p_number_of_messages_in_bundled_request_sent_
count{namespace=~"$namespace", job="pfc_p",peer_ip=~"$peer_ip"}))
```

Number of messages in Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfc_p_number_of_messages_in_bundled_request_sent_bucket{namespace=~"$namespace",
job="pfc_p",peer_ip=~"$peer_ip"})
```

PFCP Sent Message Bundling Duration Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfc_p_duration_of_bundling_in_bundled_request_sent_sum{namespace=~"$namespace",
job="pfc_p",peer_ip=~"$peer_ip"})/(pfc_p_duration_of_bundling_in_bundled_request_sen
t_count{namespace=~"$namespace", job="pfc_p",peer_ip=~"$peer_ip"}))
```

Duration of Bundling in Sent Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfc_p_duration_of_bundling_in_bundled_request_sent_bucket{namespace=~"$namespace",
```

```
job="pfcip",peer_ip=~"$peer_ip"}))
```

PFCP Received Message Size Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcip_size_of_message_of_bundled_request_received_sum{namespace=~"$namespace",
job="pfcip",peer_ip=~"$peer_ip"})/(pfcip_size_of_message_of_bundled_request_received
_count{namespace=~"$namespace", job="pfcip",peer_ip=~"$peer_ip"}))
```

Message Size in Received Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfcip_size_of_message_of_bundled_request_received_bucket{namespace=~"$namespace",
job="pfcip",peer_ip=~"$peer_ip"})
```

PFCP Received Messages Count Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcip_number_of_messages_in_bundled_request_received_sum{namespace=~"$namespace",
job="pfcip",peer_ip=~"$peer_ip"})/(pfcip_number_of_messages_in_bundled_request_recei
ved_count{namespace=~"$namespace", job="pfcip",peer_ip=~"$peer_ip"}))
```

Number of Messages in Bundle Received

Expression

{{le}}

```
sum by (procedure,le)
(pfcp_number_of_messages_in_bundled_request_received_bucket{namespace=~"$namespace", job="pfcp",peer_ip=~"$peer_ip"})
```

PFCP-C Sessions Procedures Dashboard

Description

PFCP-C Sessions Procedures Dashboard

Panels

Session Establishment - Attempts

Description

The number of session establishment attempts.

Expression

```
sum(pfcp_session_establishment_attempt{namespace=~"$namespace", job="pfcp", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Establishment - Success %

Description

Session Establishment - Success %

Expression

```
((sum(pfcp_session_establishment_success{namespace=~"$namespace", job="pfcp", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))/(sum(pfcp_session_establishment_attempt{namespace=~"$namespace", job="pfcp", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})))*100
```

Session Establishment - Successes

Description

The number of session establishment successes.

Expression

```
sum(pfcpsessionestablishment_success{namespace=~"$namespace", job="pfcpsessionestablishment_success", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Establishment - Failures

Description

The number of session establishment failures.

Expression

```
sum(pfcpsessionestablishment_failure{namespace=~"$namespace", job="pfcpsessionestablishment_failure", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Modification - Attempts

Description

The number of session modification attempts.

Expression

```
sum(pfcpsessionmodification_attempt{namespace=~"$namespace", job="pfcpsessionmodification_attempt", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Modification - Success %

Description

Session Modification - Success %

Expression

```
((sum(pfcpsessionmodification_success{namespace=~"$namespace", job="pfcpsessionmodification_success", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})) / (sum(pfcpsessionmodification_attempt{namespace=~"$namespace", job="pfcpsessionmodification_attempt", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))) * 100
```

Session Modification - Successes

Description

The number of session modification successes.

Expression

```
sum(pfcpsessionmodification_success{namespace=~"$namespace", job="pfcps",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Modification - Failures

Description

The number of session modification failures.

Expression

```
sum(pfcpsessionmodification_failure{namespace=~"$namespace", job="pfcps",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Deletion - Attempts

Description

The number of session deletion attempts.

Expression

```
sum(pfcpsessiondeletion_attempt{namespace=~"$namespace", job="pfcps",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Deletion - Success %

Description

Session Deletion - Success %

Expression

```
((sum(pfcpsessiondeletion_success{namespace=~"$namespace", job="pfcps",  
peer_id=~"$peer_id", nssai=~"$nssai",  
dnn=~"$dnn"})))/(sum(pfcpsessiondeletion_attempt{namespace=~"$namespace",  
job="pfcps", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))*100
```


Session Deletion - Successes

Description

The number of session deletion successes.

Expression

```
sum(pfcg_session_deletion_success{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Deletion - Failures

Description

The number of session deletion failures.

Expression

```
sum(pfcg_session_deletion_failure{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Report - Attempts

Description

The number of session report attempts.

Expression

```
sum(pfcg_session_report_attempt{namespace=~"$namespace", job="pfcg",
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Report - Success %

Description

Session Report - Success %

Expression

```
((sum(pfcpsessionreportsuccess{namespace=~"$namespace", job="pfcps",
peer_id=~"$peer_id", nssai=~"$nssai",
dnn=~"$dnn"})))/(sum(pfcpsessionreportattempt{namespace=~"$namespace",
job="pfcps", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})))*100
```

Session Report - Successes

Description

The number of session report successes.

Expression

```
sum(pfcpsessionreportsuccess{namespace=~"$namespace", job="pfcps",
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Report - Failures

Description

The number of session report failures.

Expression

```
sum(pfcpsessionreportfailure{namespace=~"$namespace", job="pfcps",
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

PFCP-C Sessions Protocols Dashboard

Description

PFCP-C Sessions Protocols Dashboard

Panels

Session Establishment - Requests Sent

Description

The number of session establishment request messages sent.

Expression

```
sum(pfcpsessionestablishmentrequestsent{namespace=~"$namespace", job="pfcps",  
peer_ip=~"$peer_ip"})
```

Session Establishment - Retransmitted Requests Sent

Description

The number of session establishment retransmitted request messages sent.

Expression

```
sum(pfcpsessionestablishmentretransmittedrequestsent{namespace=~"$namespace",  
job="pfcps", peer_ip=~"$peer_ip"})
```

Session Establishment - Responses Received

Description

The number of session establishment response messages received.

Expression

```
sum(pfcpsessionestablishmentresponsereceived{namespace=~"$namespace",  
job="pfcps", peer_ip=~"$peer_ip", pfcps_cause=~"$pfcps_cause"})
```

Session Modification - Requests Sent

Description

The number of session modification request messages sent.

Expression

```
sum(pfcpsessionmodificationrequestsent{namespace=~"$namespace", job="pfcps",  
peer_ip=~"$peer_ip"})
```

Session Modification - Retransmitted Requests Sent

Description

The number of session modification retransmitted request messages sent.

Expression

```
sum(pfcsp_session_modification_retransmitted_request_sent{namespace=~"$namespace",  
job="pfcsp", peer_ip=~"$peer_ip"})
```

Session Modification - Responses Received

Description

The number of session modification response messages received.

Expression

```
sum(pfcsp_session_modification_response_received{namespace=~"$namespace",  
job="pfcsp", peer_ip=~"$peer_ip", pfcsp_cause=~"$pfcsp_cause"})
```

Session Deletion - Requests Sent

Description

The number of session deletion request messages sent.

Expression

```
sum(pfcsp_session_deletion_request_sent{namespace=~"$namespace", job="pfcsp",  
peer_ip=~"$peer_ip"})
```

Session Deletion - Retransmitted Requests Sent

Description

The number of session deletion retransmitted request messages sent.

Expression

```
sum(pfcsp_session_deletion_retransmitted_request_sent{namespace=~"$namespace",  
job="pfcsp", peer_ip=~"$peer_ip"})
```

Session Deletion - Responses Received

Description

The number of session deletion response messages received.

Expression

```
sum(pfcg_session_deletion_response_received{namespace=~"$namespace", job="pfcg", peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

Session Report - Requests Received

Description

The number of session report request messages received.

Expression

```
sum(pfcg_session_report_request_received{namespace=~"$namespace", job="pfcg", peer_ip=~"$peer_ip"})
```

Session Report - Responses Sent

Description

The number of session report response messages sent.

Expression

```
sum(pfcg_session_report_response_sent{namespace=~"$namespace", job="pfcg", peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

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Summary

Dashboard Name	Panel Name
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Dashboard Name	Panel Name
PFCP-U Associations Procedures Dashboard	Peer Association Setup - Attempts
PFCP-U Associations Procedures Dashboard	Peer Association Setup - Success %
PFCP-U Associations Procedures Dashboard	Peer Association Setup - Successes
PFCP-U Associations Procedures Dashboard	Peer Association Setup - Failures
PFCP-U Associations Procedures Dashboard	Association Update - Attempts
PFCP-U Associations Procedures Dashboard	Association Update - Success %
PFCP-U Associations Procedures Dashboard	Association Update - Successes
PFCP-U Associations Procedures Dashboard	Association Update - Failures
PFCP-U Associations Procedures Dashboard	Peer Association Update - Attempts
PFCP-U Associations Procedures Dashboard	Peer Association Update - Success %
PFCP-U Associations Procedures Dashboard	Peer Association Update - Successes
PFCP-U Associations Procedures Dashboard	Peer Association Update - Failures
PFCP-U Associations Procedures Dashboard	Peer Association Release - Attempts
PFCP-U Associations Procedures Dashboard	Peer Association Release - Success %
PFCP-U Associations Procedures Dashboard	Peer Association Release - Successes
PFCP-U Associations Procedures Dashboard	Peer Association Release - Failures
PFCP-U Associations Procedures Dashboard	Peer Heartbeat - Attempts
PFCP-U Associations Procedures Dashboard	Peer Heartbeat - Success %
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PFCP-U Associations Protocols Dashboard	Association Setup - Requests Received
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PFCP-U Associations Protocols Dashboard	Heartbeat - Requests Received
PFCP-U Associations Protocols Dashboard	Heartbeat - Responses Sent
PFCP-U Associations Protocols Dashboard	Version Not Supported - Responses Received

Dashboard Name	Panel Name
PFCP-U Associations Protocols Dashboard	Version Not Supported - Responses Sent
PFCP-U Session Bundling Dashboard	Failure %
PFCP-U Session Bundling Dashboard	PFCP number of bundled message sent failure - \$cause
PFCP-U Session Bundling Dashboard	Failure %
PFCP-U Session Bundling Dashboard	PFCP number of bundled message decode failure - \$causeR
PFCP-U Session Bundling Dashboard	Pfcp number of bundled messages sent \$bundle_trigger
PFCP-U Session Bundling Dashboard	Pfcp number of bundled messages sent \$peer_ip
PFCP-U Session Bundling Dashboard	PFCP number of bundled messages received \$peer_ip
PFCP-U Session Bundling Dashboard	\$peer_ip
PFCP-U Session Bundling Dashboard	Message Size in Bundle
PFCP-U Session Bundling Dashboard	\$peer_ip
PFCP-U Session Bundling Dashboard	Number of messages in Bundle
PFCP-U Session Bundling Dashboard	\$peer_ip
PFCP-U Session Bundling Dashboard	Duration of Bundling in Sent Bundle
PFCP-U Session Bundling Dashboard	\$peer_ip
PFCP-U Session Bundling Dashboard	Message Size in Received Bundle
PFCP-U Session Bundling Dashboard	\$peer_ip
PFCP-U Session Bundling Dashboard	Number of Messages in Bundle Received
PFCP-U Sessions Procedures Dashboard	Session Establishment - Attempts
PFCP-U Sessions Procedures Dashboard	Session Establishment - Success %
PFCP-U Sessions Procedures Dashboard	Session Establishment - Successes
PFCP-U Sessions Procedures Dashboard	Session Establishment - Failures
PFCP-U Sessions Procedures Dashboard	Session Modification - Attempts
PFCP-U Sessions Procedures Dashboard	Session Modification - Success %
PFCP-U Sessions Procedures Dashboard	Session Modification - Successes
PFCP-U Sessions Procedures Dashboard	Session Modification - Failures
PFCP-U Sessions Procedures Dashboard	Session Deletion - Attempts
PFCP-U Sessions Procedures Dashboard	Session Deletion - Success %
PFCP-U Sessions Procedures Dashboard	Session Deletion - Successes

Dashboard Name	Panel Name
PFCP-U Sessions Procedures Dashboard	Session Deletion - Failures
PFCP-U Sessions Procedures Dashboard	Session Report - Attempts
PFCP-U Sessions Procedures Dashboard	Session Report - Success %
PFCP-U Sessions Procedures Dashboard	Session Report - Successes
PFCP-U Sessions Procedures Dashboard	Session Report - Failures
PFCP-U Sessions Protocols Dashboard	Session Establishment - Requests Received
PFCP-U Sessions Protocols Dashboard	Session Establishment - Responses Sent
PFCP-U Sessions Protocols Dashboard	Session Modification - Requests Received
PFCP-U Sessions Protocols Dashboard	Session Modification - Responses Sent
PFCP-U Sessions Protocols Dashboard	Session Deletion - Requests Received
PFCP-U Sessions Protocols Dashboard	Session Deletion - Responses Sent
PFCP-U Sessions Protocols Dashboard	Session Report - Requests Sent
PFCP-U Sessions Protocols Dashboard	Session Report - Retransmitted Requests Sent
PFCP-U Sessions Protocols Dashboard	Session Report - Responses Received

PFCP-U Associations Procedures Dashboard

Description

PFCP-U Associations Procedures Dashboard

Panels

Peer Association Setup - Attempts

Description

The number of association setup attempts.

Expression

```
sum(pfcpeer_association_setup_attempt{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"})
```

Peer Association Setup - Success %

Description

Peer Association Setup - Success %

Expression

```
((sum(pfcpeer_association_setup_success{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"}))/sum(pfcpeer_association_setup_attempt{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"}))*100
```

Peer Association Setup - Successes

Description

The number of association setup successes.

Expression

```
sum(pfcpeer_association_setup_success{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"})
```

Peer Association Setup - Failures

Description

The number of association setup failures.

Expression

```
sum(pfcpeer_association_setup_failure{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id", cause=~"$cause"})
```

Association Update - Attempts

Description

The number of association update attempts.

Expression

```
sum(pfcpeer_association_update_attempt{namespace=~"$namespace", job="pfcpeer", peer_id=~"$peer_id"})
```

Association Update - Success %

Description

Association Update - Success %

Expression

```
((sum(pfcg_association_update_success{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"}))/(sum(pfcg_association_update_attempt{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})))*100
```

Association Update - Successes

Description

The number of association update successes.

Expression

```
sum(pfcg_association_update_success{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id"})
```

Association Update - Failures

Description

The number of association update failures.

Expression

```
sum(pfcg_association_update_failure{namespace=~"$namespace", job="pfcg", peer_id=~"$peer_id", cause=~"$cause"})
```

Peer Association Update - Attempts

Description

The number of association update peer attempts.

Expression

```
sum(pfcpeer_association_update_attempt{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id"})
```

Peer Association Update - Success %

Description

Peer Association Update - Success %

Expression

```
((sum(pfcpeer_association_update_success{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id}))/sum(pfcpeer_association_update_attempt{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id}))*100
```

Peer Association Update - Successes

Description

The number of association update peer successes.

Expression

```
sum(pfcpeer_association_update_success{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id})
```

Peer Association Update - Failures

Description

The number of association update peer failures.

Expression

```
sum(pfcpeer_association_update_failure{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id", cause=~"$cause"})
```

Peer Association Release - Attempts

Description

The number of association release attempts.

Expression

```
sum(pfcpeer_association_release_attempt{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id"})
```

Peer Association Release - Success %

Description

Peer Association Release - Success %

Expression

```
((sum(pfcpeer_association_release_success{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id}))/sum(pfcpeer_association_release_attempt{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id}))*100
```

Peer Association Release - Successes

Description

The number of association release successes.

Expression

```
sum(pfcpeer_association_release_success{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id})
```

Peer Association Release - Failures

Description

The number of association release failures.

Expression

```
sum(pfcpeer_association_release_failure{namespace=~"$namespace", job="pfcpeer_id=~"$peer_id", cause=~"$cause"})
```

Peer Heartbeat - Attempts

Description

The number of heartbeat peer attempts.

Expression

```
sum(pfcpeer_heartbeat_attempt{namespace=~"$namespace", job="pfcpeer",  
peer_id=~"$peer_id"})
```

Peer Heartbeat - Success %

Description

Peer Heartbeat - Success %

Expression

```
((sum(pfcpeer_heartbeat_success{namespace=~"$namespace", job="pfcpeer",  
peer_id=~"$peer_id"}))/(sum(pfcpeer_heartbeat_attempt{namespace=~"$namespace",  
job="pfcpeer", peer_id=~"$peer_id"})))*100
```

Peer Heartbeat - Successes

Description

The number of heartbeat peer successes.

Expression

```
sum(pfcpeer_heartbeat_success{namespace=~"$namespace", job="pfcpeer",  
peer_id=~"$peer_id"})
```

Peer Heartbeat - Failures

Description

The number of heartbeat peer failures.

Expression

```
sum(pfcpeer_heartbeat_failure{namespace=~"$namespace", job="pfcpeer",  
peer_id=~"$peer_id", cause=~"$cause"})
```

PFCP-U Associations Protocols Dashboard

Description

PFCP-U Associations Protocols Dashboard

Panels

Association Setup - Requests Received

Description

The number of association setup request messages received.

Expression

```
sum(pfcip_association_setup_request_received{namespace=~"$namespace", job="pfcip",
peer_ip=~"$peer_ip"})
```

Association Setup - Responses Sent

Description

The number of association setup response messages sent.

Expression

```
sum(pfcip_association_setup_response_sent{namespace=~"$namespace", job="pfcip",
peer_ip=~"$peer_ip", pfcip_cause=~"$pfcip_cause"})
```

Association Update - Requests Received

Description

The number of association update request messages received.

Expression

```
sum(pfcip_association_update_request_received{namespace=~"$namespace", job="pfcip",
peer_ip=~"$peer_ip"})
```

Association Update - Responses Sent

Description

The number of association update response messages sent.

Expression

```
sum(pfcg_association_update_response_sent{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

Association Release - Requests Received**Description**

The number of association release request messages received.

Expression

```
sum(pfcg_association_release_request_received{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip"})
```

Association Release - Responses Sent**Description**

The number of association release response messages sent.

Expression

```
sum(pfcg_association_release_response_sent{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

Heartbeat - Requests Received**Description**

The number of heartbeat request messages received.

Expression

```
sum(pfcg_heartbeat_request_received{namespace=~"$namespace", job="pfcg",  
peer_ip=~"$peer_ip"})
```

Heartbeat - Responses Sent

Description

The number of heartbeat response messages sent.

Expression

```
sum(pfcg_heartbeat_response_sent{namespace=~"$namespace", job="pfcg",
peer_ip=~"$peer_ip"})
```

Version Not Supported - Responses Received

Description

The number of version not supported response messages received.

Expression

```
sum(pfcg_version_not_supported_response_received{namespace=~"$namespace",
job="pfcg", peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

Version Not Supported - Responses Sent

Description

The number of version not supported response messages sent.

Expression

```
sum(pfcg_version_not_supported_response_sent{namespace=~"$namespace", job="pfcg",
peer_ip=~"$peer_ip", pfcg_cause=~"$pfcg_cause"})
```

PFCE-U Session Bundling Dashboard

Description

PFCE-U Session Bundling Statistics Dashboard

PFCE Sent Failures - \$peer_ip Panels

Panels

Failure %

Expression

```
((sum(pfcpl_bundle_message_sent_failure{namespace=~"$namespace",  
job="pfcpl",cause=~"$cause"}))/sum(pfcpl_bundle_message_sent{namespace=~"$namespace",  
job="pfcpl",cause=~"$cause"})))*100
```

PFCP number of bundled message sent failure - \$cause

Description

Number of PFCP bundle messages sent failures to a particular peer(represented by peer_ip and peer_port)

Expression

```
sum(pfcpl_bundle_message_sent_failure{namespace=~"$namespace",  
job="pfcpl",cause=~"$cause"} OR on() vector(0))
```

PFCP Receive Failures - \$peer_ip Panels

Failure %

Expression

```
((sum(pfcpl_bundle_message_received_failure{namespace=~"$namespace",  
job="pfcpl",cause=~"$cause"}))/sum(pfcpl_bundle_message_received{namespace=~"$names  
pace", job="pfcpl",cause=~"$cause"})))*100
```

PFCP number of bundled message decode failure - \$causeR

Description

Number of PFCP bundle messages decode failures, cause represents failure reason(like invalid packet)

Expression

```
sum(pfcpl_bundle_message_received_failure{namespace=~"$namespace",  
job="pfcpl",cause=~"$causeR"} OR on() vector(0))
```

PFCP Number of Sent Messages Per Bundle Trigger - \$peer_ip Panels

Pfcp number of bundled messages sent \$bundle trigger

Description

Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle

Expression

`_auto`

```
sum(pfcp_bundle_message_sent{namespace=~"$namespace",  
job="pfcp",bundle_trigger=~"$bundle_trigger"} OR on() vector(0))
```

PFCP Total Number of Sent and Received Messages Per Peer Panels

Pfcp number of bundled messages sent \$peer ip

Description

Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle

Expression

`_auto`

```
sum(pfcp_bundle_message_sent{namespace=~"$namespace",  
job="pfcp",peer_ip=~"$peer_ip"} OR on() vector(0))
```

PFCP number of bundled messages received \$peer ip

Description

Number of PFCP bundle messages received from a particular peer(represented by peer_ip and peer_port).

Expression

```
sum(pfcp_bundle_message_received{namespace=~"$namespace",  
job="pfcp",peer_ip=~"$peer_ip"} OR on() vector(0))
```

PFCP Sent Message Size in Bundle Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfc_p_size_of_message_of_bundled_request_sent_sum{namespace=~"$namespace",
job="pfc_p",peer_ip=~"$peer_ip"})/(pfc_p_size_of_message_of_bundled_request_sent_cou
nt{namespace=~"$namespace", job="pfc_p",peer_ip=~"$peer_ip"}))
```

Message Size in Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfc_p_size_of_message_of_bundled_request_sent_bucket{namespace=~"$namespace",
job="pfc_p",peer_ip=~"$peer_ip"})
```

PFCP Sent Message Count in Bundle Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfc_p_number_of_messages_in_bundled_request_sent_sum{namespace=~"$namespace",
job="pfc_p",peer_ip=~"$peer_ip"})/(pfc_p_number_of_messages_in_bundled_request_sent_
count{namespace=~"$namespace", job="pfc_p",peer_ip=~"$peer_ip"}))
```

Number of messages in Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfc_p_number_of_messages_in_bundled_request_sent_bucket{namespace=~"$namespace",
```

```
job="pfcip",peer_ip=~"$peer_ip"}))
```

PFCP Sent Message Bundling Duration Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcip_duration_of_bundling_in_bundled_request_sent_sum{namespace=~"$namespace",
job="pfcip",peer_ip=~"$peer_ip"})/(pfcip_duration_of_bundling_in_bundled_request_sen
t_count{namespace=~"$namespace", job="pfcip",peer_ip=~"$peer_ip"}))
```

Duration of Bundling in Sent Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfcip_duration_of_bundling_in_bundled_request_sent_bucket{namespace=~"$namespace",
job="pfcip",peer_ip=~"$peer_ip"})
```

PFCP Received Message Size Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcip_size_of_message_of_bundled_request_received_sum{namespace=~"$namespace",
job="pfcip",peer_ip=~"$peer_ip"})/(pfcip_size_of_message_of_bundled_request_received
_count{namespace=~"$namespace", job="pfcip",peer_ip=~"$peer_ip"}))
```

Message Size in Received Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfcp_size_of_message_of_bundled_request_received_bucket{namespace=~"$namespace",
job="pfcp",peer_ip=~"$peer_ip"})
```

PFCP Received Messages Count Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcp_number_of_messages_in_bundled_request_received_sum{namespace=~"$namespace",
job="pfcp",peer_ip=~"$peer_ip"})/(pfcp_number_of_messages_in_bundled_request_recei
ved_count{namespace=~"$namespace", job="pfcp",peer_ip=~"$peer_ip"}))
```

Number of Messages in Bundle Received

Expression

{{le}}

```
sum by (procedure,le)
(pfcp_number_of_messages_in_bundled_request_received_bucket{namespace=~"$namespace
", job="pfcp",peer_ip=~"$peer_ip"})
```

PFCP-U Sessions Procedures Dashboard

Description

PFCP-U Sessions Procedures Dashboard

Panels

Session Establishment - Attempts

Description

The number of session establishment attempts.

Expression

```
sum(pfc_session_establishment_attempt{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Establishment - Success %

Description

Session Establishment - Success %

Expression

```
((sum(pfc_session_establishment_success{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai",  
dnn=~"$dnn"})))/(sum(pfc_session_establishment_attempt{namespace=~"$namespace",  
job="pfc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))*100
```

Session Establishment - Successes

Description

The number of session establishment successes.

Expression

```
sum(pfc_session_establishment_success{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Establishment - Failures

Description

The number of session establishment failures.

Expression

```
sum(pfc_session_establishment_failure{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Modification - Attempts

Description

The number of session modification attempts.

Expression

```
sum(pfcsp_session_modification_attempt{namespace=~"$namespace", job="pfcsp",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Modification - Success %

Description

Session Modification - Success %

Expression

```
((sum(pfcsp_session_modification_success{namespace=~"$namespace", job="pfcsp",  
peer_id=~"$peer_id", nssai=~"$nssai",  
dnn=~"$dnn"})))/(sum(pfcsp_session_modification_attempt{namespace=~"$namespace",  
job="pfcsp", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))*100
```

Session Modification - Successes

Description

The number of session modification successes.

Expression

```
sum(pfcsp_session_modification_success{namespace=~"$namespace", job="pfcsp",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Modification - Failures

Description

The number of session modification failures.

Expression

```
sum(pfcsp_session_modification_failure{namespace=~"$namespace", job="pfcsp",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Deletion - Attempts

Description

The number of session deletion attempts.

Expression

```
sum(pfcg_session_deletion_attempt{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Deletion - Success %

Description

Session Deletion - Success %

Expression

```
((sum(pfcg_session_deletion_success{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id", nssai=~"$nssai",  
dnn=~"$dnn"})))/(sum(pfcg_session_deletion_attempt{namespace=~"$namespace",  
job="pfcg", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})))*100
```

Session Deletion - Successes

Description

The number of session deletion successes.

Expression

```
sum(pfcg_session_deletion_success{namespace=~"$namespace", job="pfcg",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Deletion - Failures

Description

The number of session deletion failures.

Expression


```
sum(pfc_session_deletion_failure{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Report - Attempts

Description

The number of session report attempts.

Expression

```
sum(pfc_session_report_attempt{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Report - Success %

Description

Session Report - Success %

Expression

```
((sum(pfc_session_report_success{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai",  
dnn=~"$dnn"})))/(sum(pfc_session_report_attempt{namespace=~"$namespace",  
job="pfc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))*100
```

Session Report - Successes

Description

The number of session report successes.

Expression

```
sum(pfc_session_report_success{namespace=~"$namespace", job="pfc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Report - Failures

Description

The number of session report failures.

Expression

```
sum(pfcpsessionreportfailure{namespace=~"$namespace", job="pfcpsessionreportfailure", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

PFCP-U Sessions Protocols Dashboard

Description

PFCP-U Sessions Protocols Dashboard

Panels

Session Establishment - Requests Received

Description

The number of session establishment request messages received.

Expression

```
sum(pfcpsessionestablishmentrequestreceived{namespace=~"$namespace", job="pfcpsessionestablishmentrequestreceived", peer_ip=~"$peer_ip"})
```

Session Establishment - Responses Sent

Description

The number of session establishment response messages sent.

Expression

```
sum(pfcpsessionestablishmentresponse_sent{namespace=~"$namespace", job="pfcpsessionestablishmentresponse_sent", peer_ip=~"$peer_ip", pfcpsessioncause=~"$pfcpsessioncause"})
```

Session Modification - Requests Received

Description

The number of session modification request messages received.

Expression

```
sum(pfcsp_session_modification_request_received{namespace=~"$namespace",  
job="pfcsp", peer_ip=~"$peer_ip"})
```

Session Modification - Responses Sent

Description

The number of session modification responses messages sent.

Expression

```
sum(pfcsp_session_modification_response_sent{namespace=~"$namespace", job="pfcsp",  
peer_ip=~"$peer_ip", pfcsp_cause=~"$pfcsp_cause"})
```

Session Deletion - Requests Received

Description

The number of session deletion request messages received.

Expression

```
sum(pfcsp_session_deletion_request_received{namespace=~"$namespace", job="pfcsp",  
peer_ip=~"$peer_ip"})
```

Session Deletion - Responses Sent

Description

The number of session deletion response messages sent.

Expression

```
sum(pfcsp_session_deletion_response_sent{namespace=~"$namespace", job="pfcsp",  
peer_ip=~"$peer_ip", pfcsp_cause=~"$pfcsp_cause"})
```

Session Report - Requests Sent

Description

The number of session report request messages sent.

Expression

```
sum(pfc_session_report_request_sent{namespace=~"$namespace", job="pfc", peer_ip=~"$peer_ip"})
```

Session Report - Retransmitted Requests Sent

Description

The number of session report retransmitted request messages sent.

Expression

```
sum(pfc_session_report_retransmitted_request_sent{namespace=~"$namespace", job="pfc", peer_ip=~"$peer_ip"})
```

Session Report - Responses Received

Description

The number of session report response messages received.

Expression

```
sum(pfc_session_report_response_received{namespace=~"$namespace", job="pfc", peer_ip=~"$peer_ip", pfc_cause=~"$pfc_cause"})
```

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Summary

Dashboard Name	Panel Name
PDR Statistics	
PDR Statistics	
PDR Statistics	Total PDR Evaluations
PDR Statistics	Total PDR Matches
TDEF Proxy Base Dashboard	HTTP Proxy - Client Connections
TDEF Proxy Base Dashboard	HTTP Proxy - Transaction Rate
TDEF Proxy Base Dashboard	HTTP Proxy - Transaction Success Rate
TDEF Proxy Base Dashboard	HTTP Proxy - Transaction Failure Rate
TDEF Proxy Base Dashboard	HTTP Proxy - Transaction Success Percentage
TDEF Proxy Base Dashboard	HTTP Proxy - Average Service Time
TDEF Proxy Base Dashboard	HTTP Proxy - Client Downlink Throughput (Mbps)
TDEF Proxy Base Dashboard	HTTP Proxy - Client Uplink Throughput (Mbps)
TDEF Proxy Base Dashboard	HTTP Proxy - Server Downlink Throughput (Mbps)
TDEF Proxy Base Dashboard	HTTP Proxy - Server Uplink Throughput (Mbps)
TDEF Proxy Base Dashboard	TCP-Splicing Success Percentage
TDEF Proxy Base Dashboard	TCP-Splicing Failure Percentage
TDEF Proxy Base Dashboard	TCP-Splicing Transaction Rate

Dashboard Name	Panel Name
TDEF Proxy Base Dashboard	TCP-Splicing Client Downlink Throughput (Mbps)
TDEF Proxy Base Dashboard	TCP-Splicing Client Uplink Throughput (Mbps)
TDEF Proxy Base Dashboard	TCP-Splicing Server Downlink Throughput (Mbps)
TDEF Proxy Base Dashboard	TCP-Splicing Server Uplink Throughput (Mbps)
TDEF Proxy Base Dashboard	TCP-Splicing Client Uplink Average Throughput (Mbps)
TDEF Proxy Base Dashboard	TCP-Splicing client connections
TDEF Proxy Base Dashboard	UDP-Proxy - Transaction rate
TDEF Proxy Base Dashboard	UDP-Proxy - Transaction Success Percentage
TDEF Proxy Base Dashboard	UDP-Proxy - Upstream Throughput
TDEF Proxy Base Dashboard	UDP-Proxy - Downstream Throughput
TDEF Proxy Base Dashboard	UDP-Proxy - QUIC Transaction rate
TDEF Proxy Base Dashboard	UDP-Proxy - QUIC Upstream Throughput
TDEF Proxy Base Dashboard	UDP-Proxy - QUIC Downstream Throughput
TDEF SHIP API Dashboard	Summary Attempts
TDEF SHIP API Dashboard	Summary Successes
TDEF SHIP API Dashboard	Summary Failures
TDEF SHIP API Dashboard	HTTP OK Responses
TDEF SHIP API Dashboard	HTTP Error Responses
TDEF SHIP API Dashboard	Invalid Requests
TDEF SHIP API Dashboard	Internal Failures
TDEF SHIP API Dashboard	No Consent
TDEF SHIP Redirect Dashboard	Summary Attempts
TDEF SHIP Redirect Dashboard	Summary Successes
TDEF SHIP Redirect Dashboard	Summary Failures
TDEF SHIP Redirect Dashboard	HTTP OK Responses
TDEF SHIP Redirect Dashboard	HTTP Error Responses
TDEF SHIP Redirect Dashboard	Invalid Requests
TDEF SHIP Redirect Dashboard	Internal Failures
TDEF SHIP Redirect Dashboard	No Consent

Dashboard Name	Panel Name
TDEF Secure Entitlement Dashboard	Summary Attempts
TDEF Secure Entitlement Dashboard	Summary Successes
TDEF Secure Entitlement Dashboard	Summary Failures
TDEF Secure Entitlement Dashboard	HTTP OK Responses
TDEF Secure Entitlement Dashboard	HTTP Error Responses
TDEF Secure Entitlement Dashboard	Invalid Requests
TDEF Secure Entitlement Dashboard	Internal Failures
TDEF Secure Entitlement Dashboard	No Consent
TDEF Simple Entitlement Dashboard	Summary Attempts
TDEF Simple Entitlement Dashboard	Summary Successes
TDEF Simple Entitlement Dashboard	Summary Failures
TDEF Simple Entitlement Dashboard	HTTP OK Responses
TDEF Simple Entitlement Dashboard	HTTP Error Responses
TDEF Simple Entitlement Dashboard	Invalid Requests
TDEF Simple Entitlement Dashboard	Internal Failures
TDEF Simple Entitlement Dashboard	No Consent
TDEF Video Optimization Dashboard	HTTP Shaping - Segments

Dashboard Name	Panel Name
TDEF Video Optimization Dashboard	HTTP Proxy - Transaction Rate
TDEF Video Optimization Dashboard	HTTPS Shaping - Successes
TDEF Video Optimization Dashboard	HTTPS Shaping - Bytes
TDEF Video Optimization Dashboard	HTTPS Shaping - Opted Out by Subscriber
TDEF Video Optimization Dashboard	HTTPS Shaping - Bypasses - Disabled
TDEF Video Optimization Dashboard	UDP Shaping - Successes
TDEF Video Optimization Dashboard	UDP Shaping - Bytes
TVS Dashboard	TVS Shaped throughput (in Mbps)
TVS Dashboard	TVS-CF Session Rate
TVS Dashboard	TVS-CF Flow Rate
TVS Dashboard	TVS Shapers
TVS Dashboard	TVS-CF Total Pkt Rcvd
TVS Dashboard	TVS FLOws
UPF DPI Application Statistics Dashboard	Total packets seen from the application classified by attribute detected by DPI
UPF DPI Application Statistics Dashboard	Total bytes seen from the application classified by attribute detected by DPI
UPF DPI Application Statistics Dashboard	Uplink packets seen from the application classified by attribute detected by DPI
UPF DPI Application Statistics Dashboard	Uplink bytes seen from the application classified by attribute detected by DPI
UPF DPI Application Statistics Dashboard	Downlink packets seen from the application classified by attribute detected by DPI
UPF DPI Application Statistics Dashboard	Downlink bytes seen from the application classified by attribute detected by DPI

Dashboard Name	Panel Name
UPF DPI General Statistics Dashboard	Total connections processed by DPI detection
UPF DPI General Statistics Dashboard	Total packets on flows processed by DPI detection
UPF DPI General Statistics Dashboard	Total bytes on flows processed by DPI detection
UPF DPI General Statistics Dashboard	Uplink packets processed by DPI detection
UPF DPI General Statistics Dashboard	Uplink bytes processed by DPI detection
UPF DPI General Statistics Dashboard	Downlink packets processed by DPI detection
UPF DPI General Statistics Dashboard	Downlink bytes processed by DPI detection
UPF DPI Protocol Statistics Dashboard	Total packets seen from the protocol classified by attribute detected by DPI
UPF DPI Protocol Statistics Dashboard	Total bytes seen from the protocol classified by attribute detected by DPI
UPF DPI Protocol Statistics Dashboard	Uplink packets seen from the protocol classified by attribute detected by DPI
UPF DPI Protocol Statistics Dashboard	Uplink bytes seen from the protocol classified by attribute detected by DPI
UPF DPI Protocol Statistics Dashboard	Downlink packets seen from the protocol classified by attribute detected by DPI
UPF DPI Protocol Statistics Dashboard	Downlink bytes seen from the protocol classified by attribute detected by DPI
UPF Interface Statistics Dashboard	Rx Data Rate (Mbps)
UPF Interface Statistics Dashboard	Rx Data Rate (in Mbps)
UPF Interface Statistics Dashboard	Tx Data Rate (Mbps)
UPF Interface Statistics Dashboard	Tx Data Rate (in Mbps)
UPF Interface Statistics Dashboard	Tx Drop Rate (Mbps)
UPF Interface Statistics Dashboard	Packet Drop Data Rate (in Mbps)
UPF Interface Statistics Dashboard	Rx Pkt Rate (in Mpps)

Dashboard Name	Panel Name
UPF Interface Statistics Dashboard	Rx Pkt Rate (in Mpps)
UPF Interface Statistics Dashboard	Tx Pkt Rate (in Mpps)
UPF Interface Statistics Dashboard	Tx Pkt Rate (in Mpps)
UPF Interface Statistics Dashboard	Tx Drop Pkt Rate (in Mpps)
UPF Interface Statistics Dashboard	Drop Pkt Rate (in Mpps)
UPF NAT Module Statistics Dashboard	Total sessions created by NAT
UPF NAT Module Statistics Dashboard	Total sessions deleted by NAT
UPF NAT Module Statistics Dashboard	Total connections created by NAT
UPF NAT Module Statistics Dashboard	Total connections deleted by NAT
UPF NAT Module Statistics Dashboard	Total connection create failures due to NAT pool not found
UPF NAT Module Statistics Dashboard	Total connection create failures due to NAT pool is pending deletion
UPF NAT Module Statistics Dashboard	Total connection create failures due to NAT IP pool not configured
UPF NAT Module Statistics Dashboard	Total connection create failures due to session NAT connections quota exhaustion
UPF NAT Module Statistics Dashboard	Total connection create failures due to NAT port allocation failures
UPF NAT Module Statistics Dashboard	Total NAT port allocation success
UPF NAT Module Statistics Dashboard	Total NAT port deallocation success
UPF NAT Module Statistics Dashboard	Total NAT port allocation failures due to internal error
UPF NAT Module Statistics Dashboard	Total NAT port allocation failures due to port unavailability
UPF NAT Module Statistics Dashboard	Total NAT port allocation failures due to unsupported protocol

Dashboard Name	Panel Name
UPF NAT Module Statistics Dashboard	Total NAT port deallocation failures due to internal error
UPF NAT Module Statistics Dashboard	Total NAT port deallocation failures due to unsupported protocol
UPF NAT Pool Statistics Dashboard	Total sessions for which NAT resources were allocated from nat-pool
UPF NAT Pool Statistics Dashboard	Total sessions for which NAT resources were deallocated from nat-pool
UPF NAT Pool Statistics Dashboard	Total connections created using NAT resources from nat-pool
UPF NAT Pool Statistics Dashboard	Total connections deleted using NAT resources from nat-pool
UPF NAT Pool Statistics Dashboard	Total connection create failures due to nat port allocation failures
UPF NAT Pool Statistics Dashboard	Total connection create failures due to nat ip pool not configured
UPF NAT Pool Statistics Dashboard	Total connection create failures due to session nat connections quota exhaustion
UPF PPE Performance Dashboard	Rx Data Rate (Mbps)
UPF PPE Performance Dashboard	Rx Data Rate (in Mbps)
UPF PPE Performance Dashboard	Tx Data Rate (Mbps)
UPF PPE Performance Dashboard	Tx Data Rate (in Mbps)
UPF PPE Performance Dashboard	Rx Pkt Rate (in Mpps)
UPF PPE Performance Dashboard	Rx Pkt Rate (in Mpps)
UPF PPE Performance Dashboard	Tx Pkt Rate (in Mpps)
UPF PPE Performance Dashboard	Tx Pkt Rate (in Mpps)
UPF Session Duration	Session Create Duration
UPF Session Duration	Session Create VPP Duration
UPF Session Duration	Session DB Create Duration
UPF Session Duration	Session Modify Duration
UPF Session Duration	Session Modify VPP Duration
UPF Session Duration	Session DB Modify Duration

Dashboard Name	Panel Name
UPF Session Duration	Session Delete Duration
UPF Session Duration	Session Delete VPP Duration
UPF Session Duration	Session DB Delete Duration
UPF Session Duration	Session Create Proxy Duration
UPF Session Duration	Session Modify Proxy Duration
UPF Session Duration	Session Delete Proxy Duration
UPF Session Duration	Session Restore Proxy Duration
UPF Session Duration	Session Report PCFP Duration
UPF Session Errors	Session Create Failure
UPF Session Errors	Create Fail - No Resource Blocks
UPF Session Errors	Create Fail - UE IP Out of Range
UPF Session Errors	Create Fail - Punt Failure
UPF Session Errors	Duplicate Session Establishment Failures
UPF Session Errors	Ignored Duplicate Session Establishment Requests
UPF Session Errors	Session Modify Failure
UPF Session Errors	Modify Fail - No Resource Blocks
UPF Session Errors	Modify Fail - Session ID Out of Range
UPF Session Errors	Modify Fail - Punt Failure
UPF Session Errors	Ignored Duplicate Session Modify Requests
UPF Session Errors	Session Delete Failure
UPF Session Errors	Delete Fail - No Resource Blocks
UPF Session Errors	Delete Fail - Session ID Out of Range
UPF Session Errors	Delete Fail - Punt Failure
UPF Session Errors	Ignored Duplicate Session Delete Requests
UPF Session Errors	Session Report Request Failure
UPF Session Errors	Session Report Request Timeout
UPF Session Errors	Bad Session Report Requests
UPF Session Errors	Session Report Response Failure
UPF Session Errors	Session Restore Failure

Dashboard Name	Panel Name
UPF Session Errors	Session Local Deletes
UPF Session Errors	Local Delete Report Response Error
UPF Session Errors	Session Clear
UPF Session Errors	Idle Session Clear Request
UPF Session Errors	Session Clear Pacer Attempts
UPF Session Errors	Session Clear Pacer Completed
UPF Sessions Detailed View - Per DNN	Active Session Count
UPF Sessions Detailed View - Per DNN	UPF Active Sessions
UPF Sessions Detailed View - Per DNN	UPF Active Session Rate per pod per Sec
UPF Sessions Detailed View - Per DNN	Session Creation Request - Attempts
UPF Sessions Detailed View - Per DNN	Session Create Success %
UPF Sessions Detailed View - Per DNN	Session Creation Request - Successes
UPF Sessions Detailed View - Per DNN	Session Creation Request - Failures
UPF Sessions Detailed View - Per DNN	UPF Session Creation Rate per pod per Sec
UPF Sessions Detailed View - Per DNN	Session Modify Request - Attempts
UPF Sessions Detailed View - Per DNN	Session Modify Success %
UPF Sessions Detailed View - Per DNN	Session Modify Request - Successes
UPF Sessions Detailed View - Per DNN	Session Modify Request - Failures
UPF Sessions Detailed View - Per DNN	UPF Session Modify Rate Per Pod
UPF Sessions Detailed View - Per DNN	Session Delete Request - Attempts

Dashboard Name	Panel Name
UPF Sessions Detailed View - Per DNN	Session Delete Success %
UPF Sessions Detailed View - Per DNN	Session Delete Request - Successes
UPF Sessions Detailed View - Per DNN	Session Delete Request - Failures
UPF Sessions Detailed View - Per DNN	UPF Session Delete Rate Per Pod
UPF Sessions Detailed View - Per DNN	Session Report Request - Attempts
UPF Sessions Detailed View - Per DNN	Session Report Response - Attempts
UPF Sessions Detailed View - Per DNN	Session Report Success %
UPF Sessions Detailed View - Per DNN	Session Report Failure%
UPF Sessions Detailed View - Per DNN	UPF Session Report Request Rate Per Pod
UPF Sessions Detailed View - Per DNN	Session Restore Request - Attempts
UPF Sessions Detailed View - Per DNN	Session Restore Success %
UPF Sessions Detailed View - Per DNN	Session Restore Request - Successes
UPF Sessions Detailed View - Per DNN	Session Restore Request - Failures
UPF Sessions Detailed View - Per DNN	UPF Session Report Response Rate Per Pod
UPF Sessions Detailed View - Per DNN	UPF Session Restore Rate Per Pod
UPF Sessions Detailed View - Per PPE	Active Session Count
UPF Sessions Detailed View - Per PPE	UPF Active Sessions

Dashboard Name	Panel Name
UPF Sessions Detailed View - Per PPE	UPF Active Session Rate per pod per Sec
UPF Sessions Detailed View - Per PPE	Session Creation Request - Attempts
UPF Sessions Detailed View - Per PPE	Session Create Success %
UPF Sessions Detailed View - Per PPE	Session Creation Request - Successes
UPF Sessions Detailed View - Per PPE	Session Creation Request - Failures
UPF Sessions Detailed View - Per PPE	UPF Session Creation Rate per pod per Sec
UPF Sessions Detailed View - Per PPE	Session Modify Request - Attempts
UPF Sessions Detailed View - Per PPE	Session Modify Success %
UPF Sessions Detailed View - Per PPE	Session Modify Request - Successes
UPF Sessions Detailed View - Per PPE	Session Modify Request - Failures
UPF Sessions Detailed View - Per PPE	UPF Session Modify Rate Per Pod
UPF Sessions Detailed View - Per PPE	Session Delete Request - Attempts
UPF Sessions Detailed View - Per PPE	Session Delete Success %
UPF Sessions Detailed View - Per PPE	Session Delete Request - Successes
UPF Sessions Detailed View - Per PPE	Session Delete Request - Failures
UPF Sessions Detailed View - Per PPE	UPF Session Delete Rate Per Pod
UPF Sessions Detailed View - Per PPE	Session Report Request - Attempts
UPF Sessions Detailed View - Per PPE	Session Report Response - Attempts

Dashboard Name	Panel Name
UPF Sessions Detailed View - Per PPE	Session Report Success %
UPF Sessions Detailed View - Per PPE	Session Report Failure %
UPF Sessions Detailed View - Per PPE	UPF Session Report Request Rate Per Pod
UPF Sessions Detailed View - Per PPE	Session Restore Request - Attempts
UPF Sessions Detailed View - Per PPE	Session Restore Success %
UPF Sessions Detailed View - Per PPE	Session Restore Request - Successes
UPF Sessions Detailed View - Per PPE	Session Restore Request - Failures
UPF Sessions Detailed View - Per PPE	UPF Session Report Response Rate Per Pod
UPF Sessions Detailed View - Per PPE	UPF Session Restore Rate Per Pod
UPF Sessions Internal Errors	VPP Send Failures
UPF Sessions Internal Errors	VPP Response Failures
UPF Sessions Internal Errors	VPP Response Timeouts
UPF Sessions Internal Errors	Session DB Failures
UPF Sessions Internal Errors	Proxy Send Failures
UPF Sessions Internal Errors	Proxy Response Failures
UPF Sessions Internal Errors	Proxy Response Timeouts
UPF Sessions Overview	CPU Utilisation
UPF Sessions Overview	Average CPU
UPF Sessions Overview	Max CPU
UPF Sessions Overview	UPF Level Flows
UPF Sessions Overview	Total Flows Created
UPF Sessions Overview	Total Flow Rate
UPF Sessions Overview	Total TCP Flows

Dashboard Name	Panel Name
UPF Sessions Overview	Total UDP Flows
UPF Sessions Overview	Total CGNAT Flows
UPF Sessions Overview	Total CGNAT Flows rate
UPF Sessions Overview	Total QUIC Flows
UPF Sessions Overview	Total HTTPS Flows
UPF Sessions Overview	Total DNS Flows
UPF Sessions Overview	Total DNS flow Rate
UPF Sessions Overview	Total Proxy Flows
UPF Sessions Overview	Total DPI Detected Flows
UPF Sessions Overview	Total DPI Un-detected Flows
UPF Sessions Overview	UPF Level Total Throughput & Packet Rate
UPF Sessions Overview	Total Throughput (in Mbps)
UPF Sessions Overview	Total Rx Throughput (in Mbps)
UPF Sessions Overview	Total Tx Throughput (in Mbps)
UPF Sessions Overview	Total Packet Rate (in Mpps)
UPF Sessions Overview	Total Rx Packet Rate (in Mpps)
UPF Sessions Overview	Total Tx Packet Rate (in Mpps)
UPF Sessions Overview	Total Drops
UPF Sessions Overview	UPF Level Session Overview
UPF Sessions Overview	Session Creation Request - Attempts
UPF Sessions Overview	Session Create Success %
UPF Sessions Overview	Session Creation Request - Successes
UPF Sessions Overview	Session Creation Request - Failures
UPF Sessions Overview	UPF Session Creation Rate
UPF Sessions Overview	Session Modify Request - Attempts
UPF Sessions Overview	Session Modify Success %
UPF Sessions Overview	Session Modify Request - Successes
UPF Sessions Overview	Session Modify Request - Failures
UPF Sessions Overview	UPF Session Modify Rate
UPF Sessions Overview	Session Delete Request - Attempts

Dashboard Name	Panel Name
UPF Sessions Overview	Session Delete Success %
UPF Sessions Overview	Session Delete Request - Successes
UPF Sessions Overview	Session Delete Request - Failures
UPF Sessions Overview	UPF Session Delete Rate
UPF Sessions Overview	Active Session Count
UPF Sessions Overview	UPF Active Sessions
UPF Sessions Overview	UPF Active Session Rate
UPF Standby Session Errors	Standby Session Create Failure
UPF Standby Session Errors	Standby Create Fail - No Resource Blocks
UPF Standby Session Errors	Standby Create Fail - UE IP Out of Range
UPF Standby Session Errors	Standby Create Fail - Invalid Length
UPF Standby Session Errors	Standby Create Fail - Invalid Message
UPF Standby Session Errors	Duplicate Standby Session Create - Ignored
UPF Standby Session Errors	Duplicate Standby Session Create - Failures
UPF Standby Session Errors	Standby Session Create Fail- No Association
UPF Standby Session Errors	Standby Session Modify Failure
UPF Standby Session Errors	Standby Modify Fail - No Resource Blocks
UPF Standby Session Errors	Standby Modify Fail - Session ID Out of Range
UPF Standby Session Errors	Standby Modify Fail - Invalid Length
UPF Standby Session Errors	Standby Modify Fail - Invalid Message
UPF Standby Session Errors	Standby Session Modify Fail - No Session
UPF Standby Session Errors	Standby Modify Request Fail - Deleting Session
UPF Standby Session Errors	Standby Modify Request Fail - Session Not In Vpp
UPF Standby Session Errors	Standby Session Delete Failure
UPF Standby Session Errors	Standby Delete Fail - No Resource Blocks
UPF Standby Session Errors	Standby Delete Fail - Session ID Out of Range
UPF Standby Session Errors	Duplicate Standby Session Delete Requests - Ignored
UPF Standby Session Errors	Standby Session Delete Fail - Invalid Session
UPF Standby Session Errors	Standby Session Delete Fail - Invalid Message

Dashboard Name	Panel Name
UPF Standby Sessions Detailed View	Standby Session Count
UPF Standby Sessions Detailed View	UPF Standby Sessions
UPF Standby Sessions Detailed View	UPF Standby Session Rate per Sec
UPF Standby Sessions Detailed View	Standby Session Creation Request - Attempts
UPF Standby Sessions Detailed View	Standby Session Create Success %
UPF Standby Sessions Detailed View	Standby Session Creation Request - Successes
UPF Standby Sessions Detailed View	Standby Session Creation Request - Failures
UPF Standby Sessions Detailed View	UPF Standby Session Creation Rate per pod per Sec
UPF Standby Sessions Detailed View	Standby Session Modify Request - Attempts
UPF Standby Sessions Detailed View	Standby Session Modify Success %
UPF Standby Sessions Detailed View	Standby Session Modify Request - Successes
UPF Standby Sessions Detailed View	Standby Session Modify Request - Failures
UPF Standby Sessions Detailed View	UPF Standby Session Modify Rate Per Pod
UPF Standby Sessions Detailed View	Standby Session Delete Request - Attempts
UPF Standby Sessions Detailed View	Standby Session Delete Success %
UPF Standby Sessions Detailed View	Standby Session Delete Request - Successes
UPF Standby Sessions Detailed View	Standby Session Delete Request - Failures
UPF Standby Sessions Detailed View	UPF Standby Session Delete Rate Per Pod

Dashboard Name	Panel Name
UPF Standby Sessions Internal Errors	VPP Send Failures
UPF Standby Sessions Internal Errors	VPP Response Failures
UPF Standby Sessions Internal Errors	VPP Response Timeouts
vProbe Statistics Dashboard	vProbe Statistics Dashboard
vProbe Statistics Dashboard	Rate of Start Flow EDRs Generated
vProbe Statistics Dashboard	Start Flow EDR Sent Success
vProbe Statistics Dashboard	Rate of Start Flow EDR Errors
vProbe Statistics Dashboard	Rate of Interim Flow EDRs Generated
vProbe Statistics Dashboard	Interim Flow EDR Sent Success
vProbe Statistics Dashboard	Rate of Interim Flow EDR Errors
vProbe Statistics Dashboard	Rate of Stop Flow EDRs Generated
vProbe Statistics Dashboard	Stop Flow EDR Sent Success
vProbe Statistics Dashboard	Rate of Stop Flow EDR Errors
vProbe Statistics Dashboard	Rate of Start-Stop Flow EDRs Generated
vProbe Statistics Dashboard	Start Stop Flow EDR Sent Success
vProbe Statistics Dashboard	Rate of Start Stop Flow EDR Errors

PDR Statistics

Description

PDR Statistics Dashboard

Panels

Description

PDR Statistics

Total PDR Evaluations

Description

Total PDR Evaluations

Expression

{{pdr_name}}

```
total_pdr_evaluations
```

Total PDR Matches

Description

Total PDR Matches

Expression

{{pdr_name}}

```
total_pdr_matches
```

TDEF Proxy Base Dashboard

Description

TDEF Secure Entitlement Dashboard

Panels

HTTP Proxy - Client Connections

Description

HTTP Proxy - Client Connections

Expression

```
(sum(tdef_http_proxy_total_client_connections{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}) -
sum(tdef_http_proxy_handoff_client_connections{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}))
```

HTTP Proxy - Transaction Rate

Description

HTTP Proxy - Transaction Rate

Expression

```
rate(tdef_http_proxy_client_requests_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])
```

HTTP Proxy - Transaction Success Rate

Description

HTTP Proxy - Transaction Success Rate

Expression

```
(sum(rate(tdef_http_proxy_client_requests_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))  
-  
sum(rate(tdef_http_proxy_client_requests_invalid{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))  
-  
sum(rate(tdef_http_proxy_service_failed{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])))
```

HTTP Proxy - Transaction Failure Rate

Description

HTTP Proxy - Transaction Failure Rate

Expression

```
(sum(rate(tdef_http_proxy_client_requests_invalid{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))  
+  
sum(rate(tdef_http_proxy_service_failed{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])))
```

HTTP Proxy - Transaction Success Percentage

Description

HTTP Proxy - Transaction Success Percentage

Expression

```
(
  (sum(tdef_http_proxy_client_requests_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"
}) -
  sum(tdef_http_proxy_client_requests_invalid{pod_id=~"$pod_id",proxy_id=~"$proxy_id"
}))
  - sum(tdef_http_proxy_service_failed{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
)
  /
  (
    sum(tdef_http_proxy_client_requests_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"
    })
    +
    sum(tdef_http_proxy_failures_client_connections{pod_id=~"$pod_id",proxy_id=~"$pro
xy_id"}) ) ) * 100
```

HTTP Proxy - Average Service Time

Description

HTTP Proxy - Average Service Time

Expression

```
sum((tdef_http_proxy_client_http_service_time_avg{pod_id=~"$pod_id",proxy_id=~"$pr
oxy_id"}))
```

HTTP Proxy - Client Downlink Throughput (Mbps)

Description

HTTP Proxy - Client Downlink Throughput (Mbps)

Expression

```
sum(rate(tdef_http_proxy_client_response_bytes_total{pod_id=~"$pod_id",proxy_id=~"
$proxy_id"}[5m])) / 1024/1024 * 8
```

HTTP Proxy - Client Uplink Throughput (Mbps)

Description

HTTP Proxy - Client Uplink Throughput (Mbps)

Expression

```
sum(rate(tdef_http_proxy_client_request_bytes_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

HTTP Proxy - Server Downlink Throughput (Mbps)

Description

HTTP Proxy - Server Downlink Throughput (Mbps)

Expression

```
sum(rate(tdef_http_proxy_server_response_bytes_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

HTTP Proxy - Server Uplink Throughput (Mbps)

Description

HTTP Proxy - Server Uplink Throughput (Mbps)

Expression

```
sum(rate(tdef_http_proxy_server_request_bytes_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

TCP-Splicing Success Percentage

Description

TCP-Splicing Success Percentage

Expression

```
(sum(rate(num_tcp_splicing_summary_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))
```

```
/sum(rate(num_tcp_splicing_summary_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) ) * 100
```


TCP-Splicing Failure Percentage

Description

TCP-Splicing Failure Percentage

Expression

```
(sum(num_tcp_splicing_summary_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}))
```

/sum(num_tcp_splicing_summary_attempts{pod_id=~"\$pod_id",proxy_id=~"\$proxy_id"}))*100

TCP-Splicing Transaction Rate

Description

TCP-Splicing Transaction Rate

Expression

```
sum(rate(num_tcp_splicing_summary_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))
```

TCP-Splicing Client Downlink Throughput (Mbps)

Description

TCP-Splicing Client Downlink Throughput (Mbps)

Expression

```
sum(rate(num_tcp_splicing_client_connections_bytes_sent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

TCP-Splicing Client Uplink Throughput (Mbps)

Description

TCP-Splicing Client Uplink Throughput (Mbps)

Expression

```
sum(rate(num_tcp_splicing_client_connections_bytes_received{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

TCP-Splicing Server Downlink Throughput (Mbps)

Description

TCP-Splicing Server Downlink Throughput (Mbps)

Expression

```
sum(rate(num_tcp_splicing_server_connections_bytes_received{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

TCP-Splicing Server Uplink Throughput (Mbps)

Description

TCP-Splicing Server Uplink Throughput (Mbps)

Expression

```
sum(rate(num_tcp_splicing_server_connections_bytes_sent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

TCP-Splicing Client Uplink Average Throughput (Mbps)

Description

TCP-Splicing Client Uplink Average Throughput (Mbps)

Expression

```
sum(rate(num_tcp_splicing_server_connections_bytes_sent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))
```

```
/sum(num_tcp_splicing_client_connections_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}) / 1024/1024 * 8
```

TCP-Splicing client connections

Description

TCP-Splicing client connections

Expression

```
(num_tcp_splicing_client_connections_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}))
```

UDP-Proxy - Transaction rate

Description

UDP-Proxy - Transaction rate

Expression

```
sum(rate(tdef_udp_proxy_summary_num_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))
```

UDP-Proxy - Transaction Success Percentage

Description

UDP-Proxy - Transaction Success Percentage

Expression

```
(  
sum(tdef_udp_proxy_summary_num_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

```
/ ( sum(tdef_udp_proxy_summary_num_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}) +  
sum(tdef_udp_proxy_summary_num_aborts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}) +  
sum(tdef_udp_proxy_summary_num_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}) ) ) * 100
```

UDP-Proxy - Upstream Throughput

Description

UDP-Proxy - Upstream Throughput

Expression

```
sum(rate(tdef_udp_proxy_num_client_connections_bytes_received{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

UDP-Proxy - Downstream Throughput

Description

UDP-Proxy - Downstream Throughput

Expression

```
sum(rate(tdef_udp_proxy_num_server_connections_bytes_received{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

UDP-Proxy - QUIC Transaction rate

Description

UDP-Proxy - QUIC Transaction rate

Expression

```
sum(rate(tdef_udp_proxy_num_quic_connections_current{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m]))
```

UDP-Proxy - QUIC Upstream Throughput

Description

UDP-Proxy - QUIC Upstream Throughput

Expression

```
sum(rate(tdef_udp_proxy_num_quic_bytes_upstream{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

UDP-Proxy - QUIC Downstream Throughput

Description

UDP-Proxy - QUIC Downstream Throughput

Expression

```
sum(rate(tdef_udp_proxy_num_quic_bytes_downstream{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}[5m])) / 1024/1024 * 8
```

TDEF SHIP API Dashboard

Description

TDEF SHIP API Dashboard

Panels

Summary Attempts

Description

Total Number of Attempts

Expression

```
tdef_ship_api_server_num_summary_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Successes

Description

Total Number of Successes

Expression

```
tdef_ship_api_server_num_summary_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
}
```

Summary Failures

Description

Total Number of Failures

Expression

```
tdef_ship_api_server_num_summary_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP OK Responses

Description

Number of HTTP OK Responses

Expression

```
tdef_ship_api_server_num_detail_http_ok_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP Error Responses

Description

Number of HTTP Error Responses

Expression

```
tdef_ship_api_server_num_detail_http_error_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Invalid Requests

Description

Number of invalid requests sent to server

Expression

```
tdef_ship_api_server_num_failures_invalid_requests{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Internal Failures

Description

Number of Internal Failures

Expression

```
tdef_ship_api_server_num_failures_internal_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

No Consent

Description

Number of requests with invalid/missing consent ID

Expression

```
tdef_ship_api_server_num_failures_no_consent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

TDEF SHIP Redirect Dashboard

Description

TDEF SHIP Redirect Dashboard

Panels

Summary Attempts

Description

Total Number of Attempts

Expression

```
tdef_ship_redirect_server_num_summary_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Successes

Description

Total Number of Successes

Expression

```
tdef_ship_redirect_server_num_summary_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Failures

Description

Total Number of Failures

Expression

```
tdef_ship_redirect_server_num_summary_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP OK Responses

Description

Number of HTTP OK Responses

Expression

```
tdef_ship_redirect_server_num_detail_http_ok_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP Error Responses

Description

Number of HTTP Error Responses

Expression

```
tdef_ship_redirect_server_num_detail_http_error_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Invalid Requests

Description

Number of invalid requests sent to server

Expression

```
tdef_ship_redirect_server_num_failures_invalid_requests{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Internal Failures

Description

Number of Internal Failures

Expression

```
tdef_ship_redirect_server_num_failures_internal_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

No Consent

Description

Number of requests with invalid/missing consent ID

Expression

```
tdef_ship_redirect_server_num_failures_no_consent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

TDEF Secure Entitlement Dashboard

Description

TDEF Secure Entitlement Dashboard

Panels

Summary Attempts

Description

Total Number of Attempts

Expression

```
tdef_secure_entitlement_server_num_summary_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Successes

Description

Total Number of Successes

Expression

```
tdef_secure_entitlement_server_num_summary_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Failures

Description

Total Number of Failures

Expression

```
tdef_secure_entitlement_server_num_summary_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP OK Responses

Description

Number of HTTP OK Responses

Expression

```
tdef_secure_entitlement_server_num_detail_http_ok_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP Error Responses

Description

Number of HTTP Error Responses

Expression

```
tdef_secure_entitlement_server_num_detail_http_error_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Invalid Requests

Description

Number of invalid requests sent to server

Expression

```
tdef_secure_entitlement_server_num_failures_invalid_requests{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Internal Failures

Description

Number of Internal Failures

Expression

```
tdef_secure_entitlement_server_num_failures_internal_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

No Consent

Description

Number of requests with invalid/missing consent ID

Expression

```
tdef_secure_entitlement_server_num_failures_no_consent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Description

TDEF Simple Entitlement Dashboard

Panels

Summary Attempts

Description

Total Number of Attempts

Expression

```
tdef_simple_entitlement_server_num_summary_attempts{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Successes

Description

Total Number of Successes

Expression

```
tdef_simple_entitlement_server_num_summary_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Summary Failures

Description

Total Number of Failures

Expression

```
tdef_simple_entitlement_server_num_summary_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP OK Responses

Description

Number of HTTP OK Responses

Expression

```
tdef_simple_entitlement_server_num_detail_http_ok_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

HTTP Error Responses

Description

Number of HTTP Error Responses

Expression

```
tdef_simple_entitlement_server_num_detail_http_error_responses{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Invalid Requests

Description

Number of invalid requests sent to server

Expression

```
tdef_simple_entitlement_server_num_failures_invalid_requests{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

Internal Failures

Description

Number of Internal Failures

Expression

```
tdef_simple_entitlement_server_num_failures_internal_failures{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

No Consent

Description

Number of requests with invalid/missing consent ID

Expression

```
tdef_simple_entitlement_server_num_failures_no_consent{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}
```

TDEF Video Optimization Dashboard

Description

TDEF Secure Entitlement Dashboard

Panels

HTTP Shaping - Segments

Description

HTTP Shaping - Segments

Expression

```
sum(tdef_video_opt_http_shaping_num_segment_total_hls{pod_id=~"$pod_id",proxy_id=~"$proxy_id"}) +
sum(tdef_video_opt_http_shaping_num_segment_total_dash{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

- sum(tdef_video_opt_http_shaping_num_segment_total_hds{pod_id=~"\$pod_id",proxy_id=~"\$proxy_id"}) +
sum(tdef_video_opt_http_shaping_num_segment_total_other{pod_id=~"\$pod_id",proxy_id=~"\$proxy_id"})

HTTP Proxy - Transaction Rate

Description

HTTP Proxy - Transaction Rate

Expression

```
sum(tdef_video_opt_http_shaping_num_session_total{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

HTTPS Shaping - Successes

Description

HTTPS Shaping - Successes

Expression

```
sum(tdef_video_opt_tcp_shaping_num_summary_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

HTTPS Shaping - Bytes

Description

HTTPS Shaping - Bytes

Expression

```
sum(tdef_video_opt_tcp_shaping_num_bytes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

HTTPS Shaping - Opted Out by Subscriber

Description

HTTPS Shaping - Opted Out by Subscriber

Expression

```
sum(tdef_video_opt_tcp_shaping_num_optedout_by_subscriber{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

HTTPS Shaping - Bypasses - Disabled

Description

HTTPS Shaping - Bypasses - Disabled

Expression

```
sum(tdef_video_opt_tcp_shaping_num_disabled{pod_id=~"$pod_id",proxy_id=~"$proxy_id"
})
```

UDP Shaping - Successes

Description

UDP Shaping - Successes

Expression

```
sum(tdef_video_opt_udp_shaping_summary_num_successes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

UDP Shaping - Bytes

Description

UDP Shaping - Bytes

Expression

```
sum(tdef_video_opt_udp_shaping_total_num_bytes{pod_id=~"$pod_id",proxy_id=~"$proxy_id"})
```

TVS Dashboard

Description

TVS Dashboard

Panels

TVS Shaped throughput (in Mbps)

Description

Shaped throughput from CPPE

Expression


```
round( sum(rate(cppe_scheduler_bytes_transmitted[$Interval]) / 1024/1024 * 8) by
(pod), 0.001)
```

TVS-CF Session Rate

Description

TVS-CF Session Rate

Expression

```
round( sum(rate(tvs_cf_total_shapers[$Interval])) by (pod), 0.001)
```

TVS-CF Flow Rate

Description

TVS-CF Flow Rate

Expression

```
round( sum(rate(tvs_cf_total_flows[$Interval])) by (pod), 0.001)
```

TVS Shapers

Description

Total and Deleted TVS Shapers

Expression

```
1 sum(tvs_cf_total_shapers) by (pod)

2 sum(tvs_cf_deleted_shapers) by (pod)
```

TVS-CF Total Pkt Rcvd

Description

Total Pkts received by TVS-CF

Expression

```
sum(tvs_cf_total_pkts_rx) by (pod)
```

TVS FLoWs

Description

Total and deleted TVS-CF flows

Expression

```
1 sum(tvs_cf_total_flows) by (pod)

2 sum(tvs_cf_deleted_flows) by (pod)
```

UPF DPI Application Statistics Dashboard

Description

UPF DPI Application Statistics Dashboard

Panels

Total packets seen from the application classified by attribute detected by DPI

Description

Total packets seen from the application classified by attribute detected by DPI

Expression

```
sum(application_total_packets{group_name=~"$group_name",
application_name=~"$application_name", attribute_name=~"$attribute_name"})
```

Total bytes seen from the application classified by attribute detected by DPI

Description

Total bytes seen from the application classified by attribute detected by DPI

Expression

```
sum(application_total_bytes{group_name=~"$group_name",  
application_name=~"$application_name", attribute_name=~"$attribute_name"})
```

Uplink packets seen from the application classified by attribute detected by DPI

Description

Uplink packets seen from the application classified by attribute detected by DPI

Expression

```
sum(application_uplink_packets{group_name=~"$group_name",  
application_name=~"$application_name", attribute_name=~"$attribute_name"})
```

Uplink bytes seen from the application classified by attribute detected by DPI

Description

Uplink bytes seen from the application classified by attribute detected by DPI

Expression

```
sum(application_uplink_bytes{group_name=~"$group_name",  
application_name=~"$application_name", attribute_name=~"$attribute_name"})
```

Downlink packets seen from the application classified by attribute detected by DPI

Description

Downlink packets seen from the application classified by attribute detected by DPI

Expression

```
sum(application_downlink_packets{group_name=~"$group_name",  
application_name=~"$application_name", attribute_name=~"$attribute_name"})
```

Downlink bytes seen from the application classified by attribute detected by DPI

Description

Downlink bytes seen from the application classified by attribute detected by DPI

Expression

```
sum(application_downlink_bytes{group_name=~"$group_name",
application_name=~"$application_name", attribute_name=~"$attribute_name"})
```

UPF DPI General Statistics Dashboard

Description

UPF DPI General Statistics Dashboard

Panels

Total connections processed by DPI detection

Description

Total connections processed by DPI detection

Expression

```
sum(total_connections{stats_group=~"dpi_general_stats"})
```

Total packets on flows processed by DPI detection

Description

Total packets on flows processed by DPI detection

Expression

```
sum(detection_total_packets{stats_group=~"dpi_general_stats"})
```

Total bytes on flows processed by DPI detection

Description

Total bytes on flows processed by DPI detection

Expression

```
sum(detection_total_bytes{stats_group=~"dpi_general_stats"})
```

Uplink packets processed by DPI detection

Description

Uplink packets processed by DPI detection

Expression

```
sum(detection_uplink_packets{stats_group=~"dpi_general_stats"})
```

Uplink bytes processed by DPI detection

Description

Uplink bytes processed by DPI detection

Expression

```
sum(detection_uplink_bytes{stats_group=~"dpi_general_stats"})
```

Downlink packets processed by DPI detection

Description

Downlink packets processed by DPI detection

Expression

```
sum(detection_downlink_packets{stats_group=~"dpi_general_stats"})
```

Downlink bytes processed by DPI detection

Description

Downlink bytes processed by DPI detection

Expression

```
sum(detection_downlink_bytes{stats_group=~"dpi_general_stats"})
```

UPF DPI Protocol Statistics Dashboard

Description

UPF DPI Protocol Statistics Dashboard

Panels

Total packets seen from the protocol classified by attribute detected by DPI

Description

Total packets seen from the protocol classified by attribute detected by DPI

Expression

```
sum(protocol_total_packets{group_name=~"$group_name",
protocol_name=~"$protocol_name", attribute_name=~"$attribute_name"})
```

Total bytes seen from the protocol classified by attribute detected by DPI

Description

Total bytes seen from the protocol classified by attribute detected by DPI

Expression

```
sum(protocol_total_bytes{group_name=~"$group_name",
protocol_name=~"$protocol_name", attribute_name=~"$attribute_name"})
```

Uplink packets seen from the protocol classified by attribute detected by DPI

Description

Uplink packets seen from the protocol classified by attribute detected by DPI

Expression

```
sum(protocol_uplink_packets{group_name=~"$group_name",
protocol_name=~"$protocol_name", attribute_name=~"$attribute_name"})
```

Uplink bytes seen from the protocol classified by attribute detected by DPI

Description

Uplink bytes seen from the protocol classified by attribute detected by DPI

Expression

```
sum(protocol_uplink_bytes{group_name=~"$group_name",
protocol_name=~"$protocol_name", attribute_name=~"$attribute_name"})
```

Downlink packets seen from the protocol classified by attribute detected by DPI

Description

Downlink packets seen from the protocol classified by attribute detected by DPI

Expression

```
sum(protocol_downlink_packets{group_name=~"$group_name",
protocol_name=~"$protocol_name", attribute_name=~"$attribute_name"})
```

Downlink bytes seen from the protocol classified by attribute detected by DPI

Description

Downlink bytes seen from the protocol classified by attribute detected by DPI

Expression

```
sum(protocol_downlink_bytes{group_name=~"$group_name",
protocol_name=~"$protocol_name", attribute_name=~"$attribute_name"})
```

UPF Interface Statistics Dashboard

Description

UPF Interface Statistics Dashboard

Panels

Rx Data Rate (Mbps)

Description

Bytes Received on a network context

Expression

```
sum(rate(upf_intf_bytes_received{networkContextName=~"$networkContextName"} [1m]))
/ 1024/1024 * 8
```

Rx Data Rate (in Mbps)

Description

Bytes Received on a network context

Expression

```
round( sum(rate(upf_intf_bytes_received{networkContextName=~"$networkContextName"}
[1m])) / 1024/1024 * 8) by (pod), 0.001)
```

Tx Data Rate (Mbps)

Description

Bytes Transmitted on a network context

Expression

```
sum(rate(upf_intf_bytes_transmitted{networkContextName=~"$networkContextName"}
[1m])) / 1024/1024 * 8
```

Tx Data Rate (in Mbps)

Description

Bytes Transmitted on a network context

Expression


```
round(  
  sum(rate(upf_intf_bytes_transmitted{networkContextName=~"$networkContextName"}  
    [1m])) / 1024/1024 * 8) by (pod), 0.001)
```

Tx Drop Rate (Mbps)

Description

Tx Bytes dropped on a network context

Expression

```
sum(rate(upf_intf_tx_drops{networkContextName=~"$networkContextName"}[1m])) /  
1024/1024 * 8
```

Packet Drop Data Rate (in Mbps)

Description

Bytes dropped in a network context

Expression

```
round( sum(rate(upf_intf_tx_drops{networkContextName=~"$networkContextName"}[1m])  
  / 1024/1024 * 8) by (pod), 0.001)
```

Rx Pkt Rate (in Mpps)

Description

Packets Received on a network context

Expression

```
sum(rate(upf_intf_packets_received{networkContextName=~"$networkContextName"}  
  [1m])) /1024/1024
```

Rx Pkt Rate (in Mpps)

Description

Packets Received on a network context

Expression

```
round(
  sum(rate(upf_intf_packets_received{networkContextName=~"$networkContextName"}[1m])
    / 1024/1024) by (pod), 0.001)
```

Tx Pkt Rate (in Mpps)

Description

Packets Transmitted on a network context

Expression

```
sum(rate(upf_intf_packets_transmitted{networkContextName=~"$networkContextName"}
  [1m])) /1024/1024
```

Tx Pkt Rate (in Mpps)

Description

Packets Transmitted on a network context

Expression

```
round(
  sum(rate(upf_intf_packets_transmitted{networkContextName=~"$networkContextName"}
    [1m]) / 1024/1024) by (pod), 0.001)
```

Tx Drop Pkt Rate (in Mpps)

Description

Packets Dropped on a network context

Expression

```
sum(rate(upf_intf_tx_drops{networkContextName=~"$networkContextName"}[1m]))
  /1024/1024
```

Drop Pkt Rate (in Mpps)

Description

Packets Dropped on a network context

Expression

```
round( sum(rate(upf_intf_tx_drops{networkContextName=~"$networkContextName"}[1m])
/ 1024/1024) by (pod), 0.001)
```

UPF NAT Module Statistics Dashboard

Description

UPF NAT Module Statistics Dashboard

Panels

Total sessions created by NAT

Description

Total sessions created by NAT

Expression

```
sum(total_nat_sessions_created{stats_group=~"nat_module_stats"})
```

Total sessions deleted by NAT

Description

Total sessions deleted by NAT

Expression

```
sum(total_nat_sessions_deleted{stats_group=~"nat_module_stats"})
```

Total connections created by NAT

Description

Total connections created by NAT

Expression

```
sum(total_nat_connections_created{stats_group=~"nat_module_stats"})
```

Total connections deleted by NAT

Description

Total connections deleted by NAT

Expression

```
sum(total_nat_connections_deleted{stats_group=~"nat_module_stats"})
```

Total connection create failures due to NAT pool not found

Description

Total connection create failures due to natpool not found

Expression

```
sum(connection_create_failure_natpool_not_found{stats_group=~"nat_module_stats"})
```

Total connection create failures due to NAT pool is pending deletion

Description

Total connection create failures due to natpool is pending deletion

Expression

```
sum(connection_create_failure_natpool_pending_delete{stats_group=~"nat_module_stats"})
```

Total connection create failures due to NAT IP pool not configured

Description

Total connection create failures due to nat ip pool not configured

Expression

```
sum(connection_create_failure_nat_ip_pool_not_configured{stats_group=~"nat_module_stats"})
```

Total connection create failures due to session NAT connections quota exhaustion

Description

Total connection create failures due to session nat connections quota exhaustion

Expression

```
sum(connection_create_failure_exceeded_per_session_nat_connections_quota{stats_group=~"nat_module_stats"})
```

Total connection create failures due to NAT port allocation failures

Description

Total connection create failures due to nat port allocation failures

Expression

```
sum(connection_create_failure_nat_port_allocation_failures{stats_group=~"nat_module_stats"})
```

Total NAT port allocation success

Description

Total nat port allocation success

Expression

```
sum(nat_port_allocation_success{stats_group=~"nat_module_stats"})
```

Total NAT port deallocation success

Description

Total nat port deallocation success

Expression

```
sum(nat_port_deallocation_success{stats_group=~"nat_module_stats"})
```

Total NAT port allocation failures due to internal error

Description

Total nat port allocation failures due to internal error

Expression

```
sum(nat_port_allocation_failure_internal_error{stats_group=~"nat_module_stats"})
```

Total NAT port allocation failures due to port unavailability

Description

Total nat port allocation failures due to port unavailability

Expression

```
sum(nat_port_allocation_failure_resource_exhaustion{stats_group=~"nat_module_stats"})
```

Total NAT port allocation failures due to unsupported protocol

Description

Total nat port allocation failures due to unsupported protocol

Expression

```
sum(nat_port_allocation_failure_unsupported_protocol{stats_group=~"nat_module_stats"})
```

Total NAT port deallocation failures due to internal error

Description

Total nat port deallocation failures due to internal error

Expression

```
sum(nat_port_deallocation_failure_internal_error{stats_group=~"nat_module_stats"})
```

Total NAT port deallocation failures due to unsupported protocol

Description

Total nat port deallocation failures due to unsupported protocol

Expression

```
sum(nat_port_deallocation_failure_unsupported_protocol{stats_group=~"nat_module_stats"})
```

UPF NAT Pool Statistics Dashboard

Description

UPF NAT Pool Statistics Dashboard

Panels

Total sessions for which NAT resources were allocated from nat-pool

Description

Total sessions for which NAT resources were allocated from nat-pool

Expression

```
sum(total_natpool_sessions_created{stats_group="nat_natpool_stats", networkcontext_name=~"$networkcontext_name", natpool_name=~"$natpool_name"})
```

Total sessions for which NAT resources were deallocated from nat-pool

Description

Total sessions for which NAT resources were deallocated from nat-pool

Expression

```
sum(total_natpool_sessions_deleted{stats_group=~"nat_natpool_stats",  
networkcontext_name=~"$networkcontext_name", natpool_name=~"$natpool_name"})
```

Total connections created using NAT resources from nat-pool

Description

Total connections created using NAT resources from nat-pool

Expression

```
sum(total_natpool_connections_created{stats_group=~"nat_natpool_stats",  
networkcontext_name=~"$networkcontext_name", natpool_name=~"$natpool_name"})
```

Total connections deleted using NAT resources from nat-pool

Description

Total connections deleted using NAT resources from nat-pool

Expression

```
sum(total_natpool_connections_deleted{stats_group=~"nat_natpool_stats",  
networkcontext_name=~"$networkcontext_name", natpool_name=~"$natpool_name"})
```

Total connection create failures due to nat port allocation failures

Description

Total connection create failures due to nat port allocation failures

Expression

```
sum(natpool_connection_create_failure_nat_port_allocation_failures{stats_group=~"n  
at_natpool_stats", networkcontext_name=~"$networkcontext_name",  
natpool_name=~"$natpool_name"})
```

Total connection create failures due to nat ip pool not configured

Description

Total connection create failures due to nat ip pool not configured

Expression

```
sum(natpool_connection_create_failure_nat_ip_pool_not_configured{stats_group=~"nat_natpool_stats", networkcontext_name=~"$networkcontext_name", natpool_name=~"$natpool_name"})
```

Total connection create failures due to session nat connections quota exhaustion

Description

Total connection create failures due to session nat connections quota exhaustion

Expression

```
sum(natpool_connection_create_failure_exceeded_per_session_nat_connections_quota{stats_group=~"nat_natpool_stats", networkcontext_name=~"$networkcontext_name", natpool_name=~"$natpool_name"})
```

UPF PPE Performance Dashboard

Description

UPF PPE Performance Dashboard

Panels

Rx Data Rate (Mbps)

Description

Bytes Received on an interface

Expression

```
sum(rate(upf_intf_bytes_received{interfaceName=~"$interfaceName", coreId=~"$coreId"}[1m])) / 1024/1024 * 8
```

Rx Data Rate (in Mbps)

Description

Bytes Received on an interface

Expression

```
round( sum(rate(upf_intf_bytes_received{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m]) / 1024/1024 * 8) by (pod), 0.001)
```

Tx Data Rate (Mbps)

Description

Bytes Transmitted on an interface

Expression

```
sum(rate(upf_intf_bytes_transmitted{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m])) / 1024/1024 * 8
```

Tx Data Rate (in Mbps)

Description

Bytes Transmitted on an interface

Expression

```
round( sum(rate(upf_intf_bytes_transmitted{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m]) / 1024/1024 * 8) by (pod), 0.001)
```

Rx Pkt Rate (in Mpps)

Description

Packets Received on an interface

Expression

```
sum(rate(upf_intf_packets_received{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m])) /1024/1024
```

Rx Pkt Rate (in Mpps)

Description

Packets Received on an interface

Expression

```
round( sum(rate(upf_intf_packets_received{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m]) / 1024/1024) by (pod), 0.001)
```

Tx Pkt Rate (in Mpps)

Description

Packets Transmitted on an interface

Expression

```
sum(rate(upf_intf_packets_transmitted{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m])) /1024/1024
```

Tx Pkt Rate (in Mpps)

Description

Packets Transmitted on an interface

Expression

```
round( sum(rate(upf_intf_packets_transmitted{interfaceName=~"$interfaceName",
coreId=~"$coreId"}[1m]) / 1024/1024) by (pod), 0.001)
```

UPF Session Duration

Description

UPF Sessions Dashboard

Panels

Session Create Duration

Description

Total Session Create Duration in milliseconds

Expression

{{le}}

```
sum by (le) (upf_session_create_total_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Create VPP Duration

Description

Session Create VPP Duration

Expression

{{le}}

```
sum by (le) (upf_session_create_vpp_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session DB Create Duration

Description

Duration of database operation to successfully add a session to CRUD database

Expression

{{le}}

```
sum by (le) (upf_session_db_creation_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Modify Duration

Description

Total Session Modify Duration in milliseconds

Expression

{{le}}

```
sum by (le) (upf_session_modify_total_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Modify VPP Duration

Description

Histogram of the duration, in milliseconds, of all modify vpp requests sent and response received for a PFCP sessModReq message.

Expression

{{le}}

```
sum by (le) (upf_session_modify_vpp_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session DB Modify Duration

Description

Duration of database operation to successfully modify a session to CRUD database

Expression

{{le}}

```
sum by (le) (upf_session_db_modify_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Delete Duration

Description

Total Session Delete Duration in milliseconds

Expression

{{le}}

```
sum by (le) (upf_session_delete_total_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Delete VPP Duration

Description

Histogram of the duration, in milliseconds, of all delete vpp requests sent and response received for a PFCP sessDelReq message.

Expression

{{le}}

```
sum by (le) (upf_session_delete_vpp_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session DB Delete Duration

Description

Duration of database operation to successfully delete a session to CRUD database

Expression

{{le}}

```
sum by (le) (upf_session_db_delete_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Create Proxy Duration

Description

Session Create Proxy Duration

Expression

{{le}}

```
sum by (le) (upf_session_create_proxy_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Modify Proxy Duration

Description

Histogram of the duration, in milliseconds, of all create vpp requests sent and response received for a PFCP sessEstReq message

Expression

{{le}}

```
sum by (le) (upf_session_modify_proxy_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Delete Proxy Duration

Description

Histogram of the duration, in milliseconds, of all delete proxy requests sent and response received for a PFCP sessDelReq message

Expression

{{le}}

```
sum by (le) (upf_session_delete_proxy_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Restore Proxy Duration

Description

Histogram of the duration, in milliseconds, of all delete proxy requests sent and response received for a PFCP sessDelReq message.

Expression

{{le}}

```
sum by (le) (upf_session_restore_proxy_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Report PFCP Duration

Description

Histogram of the duration, in milliseconds, of all session report requests sent to PFCP

Expression

{{le}}

```
sum by (le) (upf_session_report_pfcpc_duration_milliseconds_bucket{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

UPF Session Errors

Description

UPF Sessions Errors Dashboard

Panels

Session Create Failure

Description

Number of Session creates failed

Expression

```
sum(upf_session_creation_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Create Fail - No Resource Blocks

Description

Number of Session creates failed due to no resource block present.

Expression

```
sum(upf_session_creation_request_failure_resource_block_not_present{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Create Fail - UE IP Out of Range

Description

Number of Session creates failed due to ue ip out of range

Expression

```
sum(upf_session_creation_request_failure_ue_ip_out_of_range{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Create Fail - Punt Failure

Description

Number of Session Create requests that failed to punt to the right ppe

Expression

```
sum(upf_session_creation_request_punt_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Duplicate Session Establishment Failures

Description

Number of Duplicate Session Establishment Failures

Expression

```
sum(upf_session_dup_est_req_failures
```

```
{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Ignored Duplicate Session Establishment Resquests

Description

Number of Duplicate Session Establishment Requests Ignored

Expression

```
sum(upf_session_dup_est_req_ignored
```

```
{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Session Modify Failure

Description

Number of Session modify requests failed

Expression

```
sum(upf_session_modify_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Modify Fail - No Resource Blocks

Description

Number of Session modify failed due to no resource block present.

Expression

```
sum(upf_session_modify_request_failure_resource_block_not_present{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Modify Fail - Session ID Out of Range

Description

Number of Session modify failed due to upf session id out of range

Expression

```
sum(upf_session_modify_request_failure_upf_session_id_out_of_range{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Modify Fail - Punt Failure

Description

Number of Session Modify requests that failed to punt to the right ppe

Expression

```
sum(upf_session_modify_request_punt_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Ignored Duplicate Session Modify Requests

Description

Number of Duplicate Session Modify Requests Ignored

Expression

```
sum(upf_session_dup_modify_req_ignored
```

```
{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Session Delete Failure

Description

Number of Session delete requests failed

Expression

```
sum(upf_session_delete_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Delete Fail - No Resource Blocks

Description

Number of Session delete failed due to no resource block present.

Expression

```
sum(upf_session_delete_request_failure_resource_block_not_present{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Delete Fail - Session ID Out of Range

Description

Number of Session delete failed due to upf session id out of range

Expression

```
sum(upf_session_delete_request_failure_upf_session_id_out_of_range{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Delete Fail - Punt Failure

Description

Number of Session Delete requests that failed to punt to the right ppe

Expression

```
sum(upf_session_delete_request_punt_failure{cause=~"$cause", pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Ignored Duplicate Session Delete Requests

Description

Number of Duplicate Session Delete Requests Ignored

Expression

```
sum(upf_session_dup_del_req_ignored
```

```
{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Session Report Request Failure

Description

Number of Session reports failed

Expression

```
sum(upf_session_report_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Session Report Request Timeout

Description

Number of Session reports timeout

Expression

```
sum(upf_session_report_req_timeout{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Bad Session Report Requests

Description

Number of Bad Session Report Requests

Expression

```
sum(upf_session_bad_report_req_msgs
```

```
{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"}))
```

Session Report Response Failure

Description

Number of Session reports failed

Expression

```
sum(upf_session_report_response_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Session Restore Failure

Description

Number of Session restore failed

Expression

```
sum(upf_session_restore_request_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Session Local Deletes

Description

Number of Local Session Deletes

Expression

```
sum(upf_session_local_deletes{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

Local Delete Report Response Error

Description

Number of Local Delete Report Response Error

Expression

```
sum(upf_session_local_deletes_report_resp_error
```

{pod=~"\$pod", sm_instance_id=~"\$sm_instance_id", cause=~"\$cause"})

Session Clear

Description

Number of Session Clear Requests

Expression

```
sum(upf_session_num_session_clears
```

{pod=~"\$pod", sm_instance_id=~"\$sm_instance_id", cause=~"\$cause"})

Idle Session Clear Request

Description

Number of Requests to clear idle sessions

Expression

```
sum(upf_session_report_idle_session_clear_req
```

{pod=~"\$pod", sm_instance_id=~"\$sm_instance_id", cause=~"\$cause"})

Session Clear Pacer Attempts

Description

Number of clear pacer initiated

Expression

```
sum(upf_session_clear_pacer_initiated
```

{pod=~"\$pod", sm_instance_id=~"\$sm_instance_id", cause=~"\$cause"})

Session Clear Pacer Completed

Description

Number of clear pacer completed

Expression

```
sum(upf_session_clear_pacer_completed
```

{pod=~"\$pod", sm_instance_id=~"\$sm_instance_id", cause=~"\$cause"})

UPF Sessions Detailed View - Per DNN

Description

UPF Sessions Dashboard

Panels

Active Session Count

Description

Active Session Count

Expression

```
sum(upf_num_active_sessions{})
```

UPF Active Sessions

Description

UPF Active Sessions Per Pod

Expression

Active Sessions-{{pod}}

```
(sum by (pod) (rate(upf_num_active_sessions{}[1m])))
```

UPF Active Session Rate per pod per Sec

Description

UPF Active Session Rate per pod per Sec

Expression

Active Sessions-{{pod}}

```
(sum by (pod) (rate(upf_num_active_sessions{}[1m])))
```

Session Creation Request - Attempts

Description

Number of Session Create Attempts

Expression

```
sum(upf_session_creation_request{apn_dnn=~"$apn_dnn"})
```

Session Create Success %

Description

Session Create Success %

Expression


```
((sum(upf_session_creation_request_success{apn_dnn=~"$apn_dnn"}))/(sum(upf_session_creation_request{apn_dnn=~"$apn_dnn"})))*100
```

Session Creation Request - Successes

Description

Number of Session creates succeeded in net_ctx network context.

Expression

```
sum(upf_session_creation_request_success{apn_dnn=~"$apn_dnn"})
```

Session Creation Request - Failures

Description

Number of Session creates failed

Expression

```
sum(upf_session_creation_request_failure{apn_dnn=~"$apn_dnn", cause=~"$cause"})
```

UPF Session Creation Rate per pod per Sec

Description

UPF Session Creation Rate per pod per Sec

Expression

Create Requests-{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request{apn_dnn=~"$apn_dnn"}[1m])))
```

Create Success-{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_success{apn_dnn=~"$apn_dnn"}[1m])))
```

Create Failure-{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_failure{apn_dnn=~"$apn_dnn"}
[1m])))
```

Session Modify Request - Attempts

Description

Number of Session modify requests

Expression

```
sum(upf_session_modify_request{apn_dnn=~"$apn_dnn"})
```

Session Modify Success %

Description

Expression

```
((sum(upf_session_modify_request_success{apn_dnn=~"$apn_dnn"}))/(sum(upf_session_m
odify_request{apn_dnn=~"$apn_dnn"})))*100
```

Session Modify Request - Successes

Description

Number of Session modify requests succeeded

Expression

```
sum(upf_session_modify_request_success{apn_dnn=~"$apn_dnn"})
```

Session Modify Request - Failures

Description

Number of Session modify requests failed

Expression

```
sum(upf_session_modify_request_failure{apn_dnn=~"$apn_dnn", cause=~"$cause"})
```

UPF Session Modify Rate Per Pod

Description

UPF Session Modify Rate Per Pod

Expression

Modify Attempts--{{pod}}

```
(sum by (pod) (rate(upf_session_modify_request{apn_dnn=~"$apn_dnn"}[1m])))
```

Modify Success--{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_success{apn_dnn=~"$apn_dnn"}[1m])))
```

Modify Failure--{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_failure{apn_dnn=~"$apn_dnn"}[1m])))
```

Session Delete Request - Attempts

Description

Number of Session delete requests

Expression

```
sum(upf_session_delete_request{apn_dnn=~"$apn_dnn"})
```

Session Delete Success %

Description

Session Delete Success %

Expression

```
((sum(upf_session_delete_request_success{apn_dnn=~"$apn_dnn"}))/(sum(upf_session_delete_request{apn_dnn=~"$apn_dnn"})))*100
```

Session Delete Request - Successes

Description

Number of Session delete requests

Expression

```
sum(upf_session_delete_request_success{apn_dnn=~"$apn_dnn"})
```

Session Delete Request - Failures

Description

Number of Session delete requests failed

Expression

```
sum(upf_session_delete_request_failure{apn_dnn=~"$apn_dnn", cause=~"$cause"})
```

UPF Session Delete Rate Per Pod

Description

UPF Session Delete Rate Per Pod

Expression

Delete Attempt- `{{pod}}`

```
(sum by (pod) (rate(upf_session_delete_request{apn_dnn=~"$apn_dnn"}[1m])))
```

Delete Success--`{{pod}}`

```
(sum by (pod) (rate(upf_session_delete_request_success{apn_dnn=~"$apn_dnn"}[1m])))
```

Delete Failure-{{pod}}

(sum by (pod) (rate(upf_session_delete_request_failure{apn_dnn=~"\$apn_dnn"}[1m])))

Session Report Request - Attempts

Description

Number of Session report requests

Expression

sum(upf_session_report_request{apn_dnn=~"\$apn_dnn"})

Session Report Response - Attempts

Description

Number of Session report responses

Expression

sum(upf_session_report_response{apn_dnn=~"\$apn_dnn"})

Session Report Success %

Description

Session Report Success %

Expression

((sum(upf_session_report_response_success{apn_dnn=~"\$apn_dnn"}))/(sum(upf_session_report_request_success{apn_dnn=~"\$apn_dnn"})))*100

Session Report Failure%

Description

Number of Session Report Request Failed

Expression

```
(1-
((sum(upf_session_report_response_success{apn_dnn=~"$apn_dnn"}))/(sum(upf_session_report_request_success{apn_dnn=~"$apn_dnn"}))))*100
```

UPF Session Report Request Rate Per Pod

Description

UPF Session Report Request Rate Per Pod

Expression

Report Request-{{pod}}

```
(sum by (pod) (rate(upf_session_report_request{apn_dnn=~"$apn_dnn"}[1m])))
```

Report Request Success-{{pod}}

```
(sum by (pod) (rate(upf_session_report_request_success{apn_dnn=~"$apn_dnn"}[1m])))
```

Report Request Failure-{{pod}}

```
(sum by (pod) (rate(upf_session_report_request_failure{apn_dnn=~"$apn_dnn"}[1m])))
```

Bad Request-{{pod}}

```
(sum by (pod) (rate(upf_session_bad_report_req_msgs{apn_dnn=~"$apn_dnn"}[1m])))
```

Request Timeout-{{pod}}

```
(sum by (pod) (rate(upf_session_report_req_timeout{apn_dnn=~"$apn_dnn"}[1m])))
```

dldr - {{pod}}

```
(sum by (pod) (rate(upf_session_report_req_dldr{apn_dnn=~"$apn_dnn"}[1m])))
```

Usage Report - {{pod}}

```
(sum by (pod) (rate(upf_session_report_req_usage_report{apn_dnn=~"$apn_dnn"}[1m])))
```

Error Indication-{{pod}}

```
(sum by (pod) (rate(upf_session_report_req_error_ind{apn_dnn=~"$apn_dnn"}[1m])))
```

Inactivity-{{pod}}

```
(sum by (pod) (rate(upf_session_report_req_inactivity_report{apn_dnn=~"$apn_dnn"}[1m])))
```

Session Restore Request - Attempts

Description

Number of Session restore requests

Expression

```
sum(upf_session_restore_request{apn_dnn=~"$apn_dnn"})
```

Session Restore Success %

Description

Session Restore Success %

Expression

```
((sum(upf_session_restore_request_success{apn_dnn=~"$apn_dnn"}))/(sum(upf_session_restore_request{apn_dnn=~"$apn_dnn"})))*100
```

Session Restore Request - Successes

Description

Number of Session restore requests succeeded in net_ctx network context.

Expression

```
sum(upf_session_restore_request_success{apn_dnn=~"$apn_dnn"})
```

Session Restore Request - Failures

Description

Number of Session restore requests failed

Expression

```
sum(upf_session_restore_request_failure{apn_dnn=~"$apn_dnn", cause=~"$cause"})
```

UPF Session Report Response Rate Per Pod

Description

UPF Session Report Response Rate Per Pod

Expression

Report Response-{{pod}}

```
(sum by (pod) (rate(upf_session_report_response{apn_dnn=~"$apn_dnn"}[1m])))
```

Report Response Success-{{pod}}

```
(sum by (pod) (rate(upf_session_report_response_success{apn_dnn=~"$apn_dnn"}[1m])))
```

Report Response Failure-{{pod}}


```
(sum by (pod) (rate(upf_session_report_response_failure{apn_dnn=~"$apn_dnn"}
[1m])))
```

Wrong PPE-{{pod}}

```
(sum by (pod) (rate(upf_session_report_resp_wrong_ppe{apn_dnn=~"$apn_dnn"}[1m])))
```

Session NotFound-{{pod}}

```
(sum by (pod) (rate(upf_session_report_resp_session_notfound{apn_dnn=~"$apn_dnn"}
[1m])))
```

UPF Session Restore Rate Per Pod

Description

UPF Session Restore Rate Stats Per Pod

Expression

Restore Request Attempt--{{pod}}

```
(sum by (pod) (rate(upf_session_restore_request{apn_dnn=~"$apn_dnn"}[1m])))
```

Restore Request Success--{{pod}}

```
(sum by (pod) (rate(upf_session_restore_request_success{apn_dnn=~"$apn_dnn"}
[1m])))
```

Restore Request Failure--{{pod}}

```
(sum by (pod) (rate(upf_session_restore_request_failure{apn_dnn=~"$apn_dnn"}
[1m])))
```

UPF Sessions Detailed View - Per PPE

Description

UPF Sessions Dashboard

Panels

Active Session Count

Description

Number of Active Sessions

Expression

```
sum(upf_num_active_sessions{pod=~"$pod",sm_instance_id=~"$sm_instance_id"})
```

UPF Active Sessions

Description

UPF Active Sessions

Expression

Active Sessions-{{pod}}

```
sum by (pod) (upf_num_active_sessions{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

UPF Active Session Rate per pod per Sec

Description

UPF Active Session Rate per pod per Sec

Expression

Active Sessions-{{pod}}

```
(sum by (pod) (rate(upf_num_active_sessions{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Session Creation Request - Attempts

Description

Number of Session Create Attempts

Expression

```
sum(upf_session_creation_request{pod=~"$pod",sm_instance_id=~"$sm_instance_id"})
```

Session Create Success %

Description

Session Create Success %

Expression

```
((sum(upf_session_creation_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}))/(sum(upf_session_creation_request{pod=~"$pod"  
, sm_instance_id=~"$sm_instance_id"})))*100
```

Session Creation Request - Successes

Description

Number of Session creates succeeded in net_ctxt network context.

Expression

```
sum(upf_session_creation_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Session Creation Request - Failures

Description

Number of Session creates failed

Expression

```
sum(upf_session_creation_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

UPF Session Creation Rate per pod per Sec

Description

UPF Session Creation Rate per pod per Sec

Expression

Create Requests-{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Create Success-{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Create Failure-{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Session Modify Request - Attempts

Description

Number of Session modify requests in net_ctxt network context.

Expression

```
sum(upf_session_modify_request{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})
```

Session Modify Success %

Description

Expression

```
((sum(upf_session_modify_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}))/sum(upf_session_modify_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})))*100
```

Session Modify Request - Successes

Description

Number of Session modify requests succeeded in net_ctxt network context.

Expression

```
sum(upf_session_modify_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Modify Request - Failures

Description

Number of Session modify requests failed in net_ctxt network context.

Expression

```
sum(upf_session_modify_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

UPF Session Modify Rate Per Pod

Description

UPF Session Modify Rate Per Pod

Expression

Modify Attempts--{{pod}}

```
(sum by (pod) (rate(upf_session_modify_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Modify Success--{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Modify Failure--{{pod}}

```
(sum by (pod) (rate(upf_session_creation_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Session Delete Request - Attempts

Description

Number of Session delete requests

Expression

```
sum(upf_session_delete_request{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})
```

Session Delete Success %

Description

Session Delete Success %

Expression

```
((sum(upf_session_delete_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}))/(sum(upf_session_delete_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}))*100
```

Session Delete Request - Successes

Description

Number of Session delete requests succeeded in net_ctxt network context.

Expression

```
sum(upf_session_delete_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Delete Request - Failures

Description

Number of Session delete requests failed

Expression

```
sum(upf_session_delete_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

UPF Session Delete Rate Per Pod

Description

UPF Session Delete Rate Per Pod

Expression

Delete Attempt- `{{pod}}`

```
(sum by (pod) (rate(upf_session_delete_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Delete Success- `{{pod}}`

```
(sum by (pod) (rate(upf_session_delete_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Delete Failure- `{{pod}}`

```
(sum by (pod) (rate(upf_session_delete_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Session Report Request - Attempts

Description

Number of Session report requests

Expression

```
sum(upf_session_report_request{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})
```

Session Report Response - Attempts

Description

Number of Session report responses

Expression

```
sum(upf_session_report_response{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})
```

Session Report Success %

Description

Session Report Success %

Expression

```
((sum(upf_session_report_response_success{pod=~"$pod", sm_instance_id=~"$sm_instance_id"}))/(sum(upf_session_report_request_success{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})))*100
```

Session Report Failure %

Description

Session Report Failure %

Expression

```
(1-((sum(upf_session_report_response_success{pod=~"$pod", sm_instance_id=~"$sm_instance_id"}))/(sum(upf_session_report_request_success{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})))*100
```

UPF Session Report Request Rate Per Pod

Description

UPF Session Report Request Rate Per Pod

Expression

Report Request-{{pod}}

```
(sum by (pod) (rate(upf_session_report_request{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))  
  
2 (sum by (pod) (rate(upf_session_report_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))  
  
3 (sum by (pod) (rate(upf_session_report_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Bad Request-{{pod}}

```
(sum by (pod) (rate(upf_session_bad_report_req_msgs{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Request Timeout-{{pod}}

```
(sum by (pod) (rate(upf_session_report_req_timeout{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

dldr - {{pod}}

```
(sum by (pod) (rate(upf_session_report_req_dldr{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Usage Report - {{pod}}

```
(sum by (pod) (rate(upf_session_report_req_usage_report{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Error Indication-{{pod}}

```
(sum by (pod) (rate(upf_session_report_req_error_ind{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Inactivity-{{pod}}

```
(sum by (pod) (rate(upf_session_report_req_inactivity_report{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m]))) * 100
```

Session Restore Request - Attempts

Description

Number of Session restore requests

Expression

```
sum(upf_session_restore_request{pod=~"$pod", sm_instance_id=~"$sm_instance_id"})
```

Session Restore Success %

Description

Session Restore Success %

Expression

```
((sum(upf_session_restore_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})) / (sum(upf_session_restore_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}))) * 100
```

Session Restore Request - Successes

Description

Number of Session restore requests succeeded in net_ctx network context.

Expression

```
sum(upf_session_restore_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session Restore Request - Failures

Description

Number of Session restore requests failed

Expression

```
sum(upf_session_restore_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause"})
```

UPF Session Report Response Rate Per Pod

Description

UPF Session Report Response Rate Per Pod

Expression

Report Response-{{pod}}

```
(sum by (pod) (rate(upf_session_report_response{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Report Response Success-{{pod}}

```
(sum by (pod) (rate(upf_session_report_response_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Report Response Failure-{{pod}}

```
(sum by (pod) (rate(upf_session_report_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Wrong PPE-{{pod}}

```
(sum by (pod) (rate(upf_session_report_resp_wrong_ppe{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Session NotFound-{{pod}}

```
(sum by (pod) (rate(upf_session_report_resp_session_notfound{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

UPF Session Restore Rate Per Pod

Description

UPF Session Restore Rate Stats Per Pod

Expression

Restore Request Attempt--{{pod}}

```
(sum by (pod) (rate(upf_session_restore_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Restore Request Success--{{pod}}

```
(sum by (pod) (rate(upf_session_restore_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

Restore Request Failure--{{pod}}

```
(sum by (pod) (rate(upf_session_restore_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"}[1m])))
```

UPF Sessions Internal Errors

Description

UPF Sessions Dashboard

Panels

VPP Send Failures

Description

Session Failures due to VPP Send Failures

Expression

Create

```
sum (upf_session_create_vpp_send_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_modify_vpp_send_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_delete_vpp_send_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Restore

```
sum (upf_session_restore_vpp_send_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

VPP Response Failures

Description

Session Failures due to VPP Response Failures

Expression

Create

```
sum (upf_session_create_vpp_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_modify_vpp_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_delete_vpp_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Restore

```
sum (upf_session_restore_vpp_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

VPP Response Timeouts**Description**

Session Failures due to VPP Response Timeout

Expression**Create**

```
sum (upf_session_create_vpp_response_timeout{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_modify_vpp_response_timeout{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_delete_vpp_response_timeout{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Restore

```
sum (upf_session_restore_vpp_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Session DB Failures

Description

Session Failures due to DB failures

Expression

Create

```
sum (upf_session_db_creation_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_db_modify_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_db_delete_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Update

```
sum (upf_session_db_update_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Group Read

```
sum (upf_session_db_group_read_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Single Read

```
sum (upf_session_db_single_read_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Group Delete

```
sum (upf_session_db_group_delete_best_effort_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Proxy Send Failures

Description

Session Failures due to Proxy Send Failures

Expression

Create

```
sum (upf_session_create_proxy_send_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_modify_proxy_send_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_delete_proxy_send_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```


Restore

```
sum (upf_session_restore_proxy_send_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Proxy Response Failures

Description

Session Failures due to Proxy Response Failures

Expression**Create**

```
sum (upf_session_create_proxy_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_modify_proxy_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_delete_proxy_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Restore

```
sum (upf_session_restore_proxy_response_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id"})
```

Proxy Response Timeouts

Description

Session Failures due to Proxy Timeout

Expression

Create

```
sum (upf_session_create_proxy_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Modify

```
sum (upf_session_modify_proxy_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Delete

```
sum (upf_session_delete_proxy_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id"})
```

Restore

```
sum
(upf_session_restore_proxy_response_timeout{pod=~"$pod",sm_instance_id=~"$sm_insta
nce_id"})
```

UPF Sessions Overview

Description

UPF Sessions Overview

Panels

CPU Utilisation

Average CPU

Description

Average CPU utilization per PPE

Expression

IO

```
sum(upf_vpp_thread_cpu_utilization{thread_type="IO", pod=~"$pod"})/count(upf_vpp_thread_cpu_utilization{thread_type="IO", pod=~"$pod"})
```

Worker

```
sum(upf_vpp_thread_cpu_utilization{thread_type="Worker", pod=~"$pod"})/count(upf_vpp_thread_cpu_utilization{thread_type="Worker", pod=~"$pod"})
```

Max CPU

Description

Max CPU utilitsation per PPE

Expression

IO

```
max(upf_vpp_thread_cpu_utilization{thread_type="IO", pod=~"$pod"})
```

Worker

```
max(upf_vpp_thread_cpu_utilization{thread_type="Worker", pod=~"$pod"})
```

UPF Level Flows

Total Flows Created

Description

Total Flows Created at UPF level

Expression

Total Flows Created

```
sum (total_connections_created)
```

Total Flow Rate

Description

Total Flow Rate

Expression

Total Flow Rate

```
sum (rate (total_connections_created[$rate_interval]))
```

Total TCP Flows

Description

Total TCP flows at UPF level

Expression

TCP FLoWs

```
sum(total_tcp_flows_created)
```

Total UDP Flows

Description

Total UDP Flows at UPF Level

Expression

UDP FLoWs

```
sum(total_udp_flows_created)
```

Total CGNAT Flows

Description

Total CGNAT Flows at UPF Level

Expression

CGNAT Flows

```
sum(total_cgnat_flows_created)
```

Total CGNAT Flows rate

Description

Total CGNAT Flow rate at UPF Level

Expression

CGNAT Flow Rate

```
sum (rate (total_cgnat_flows_created[$rate_interval]))
```

Total QUIC Flows

Description

Total QUIC Flows at UPF Level

Expression

QUIC Flows

```
sum(total_udp_quic_flows_created)
```

Total HTTPS Flows

Description

Total HTTPS Flows at UPF Level

Expression

HTTPS Flows

```
sum(total_tcp_https_flows_created)
```

Total DNS Flows

Description

Total DNS Flows at UPF Level

Expression

DNS Flows

```
sum(total_dns_flows_created)
```

Total DNS flow Rate

Description

Total DNS Flow rate at UPF Level

Expression

DNS Flow Rate

```
sum (rate (total_dns_flows_created[$rate_interval]))
```

Total Proxy Flows

Description

Total Proxy flows at UPF Level (total_proxy_tcp_flows_created + total_proxy_udp_flows_created + total_proxy_tcp_443_flows_created + total_proxy_udp_443_flows_created)

Expression

Proxy Flows

```
sum(total_proxy_tcp_flows_created + total_proxy_udp_flows_created +  
total_proxy_tcp_443_flows_created + total_proxy_udp_443_flows_created)
```

Total DPI Detected Flows

Description

Total connections processed by DPI detection

Expression

DPI Flows

```
sum (total_connections)
```

Total DPI Un-detected Flows

Description

Total flows excluded from DPI-Engine

Expression

Non-DPI Flows

```
sum (no_dpi_flows)
```

UPF Level Total Throughput & Packet Rate

Total Throughput (in Mbps)

Description

Total throughput at UPF Level (bytes received and transmitted). Results can be fetched based on the selected networkContextName.

Expression

Total Throughput

```
round(
  (sum(rate(upf_intf_bytes_received{networkContextName=~"$networkContextName"}
    [$rate_interval]) +
    rate(upf_intf_bytes_transmitted{networkContextName=~"$networkContextName"}
    [$rate_interval]))/1000/1000 * 8), 0.001)
```

Total Rx Throughput (in Mbps)

Description

Total Rx throughput at UPF level (total bytes received). Results can be fetched based on the selected networkContextName.

Expression

Total rx throughput

```
round( sum(rate(upf_intf_bytes_received{networkContextName=~"$networkContextName"}
[$rate_interval]) / 1000/1000 * 8), 0.001)
```

Total Tx Throughput (in Mbps)

Description

Total throughput at UPF Level (total bytes transmitted). Results can be fetched based on the selected networkContextName.

Expression

Total Tx throughput

```
round(
sum(rate(upf_intf_bytes_transmitted{networkContextName=~"$networkContextName"}
[$rate_interval]) / 1000/1000 * 8), 0.001)
```

Total Packet Rate (in Mpps)

Description

Total Packet rate at UPF Level (received + transmitted). Results can be fetched based on the selected networkContextName.

Expression

Total Packet Rate

```
round(
(sum(rate(upf_intf_packets_received{networkContextName=~"$networkContextName"}
[$rate_interval]) +
rate(upf_intf_packets_transmitted{networkContextName=~"$networkContextName"}
[$rate_interval])) / 1000/1000) , 0.001)
```


Total Rx Packet Rate (in Mpps)

Description

Total Rx packet rate at UPF level (packets received). Results can be fetched based on the selected networkContextName.

Expression

Total rx packet rate

```
round(sum(rate(upf_intf_packets_received{networkContextName=~"$networkContextName"}[$rate_interval])/1000/1000), 0.001)
```

Total Tx Packet Rate (in Mpps)

Description

Total Tx packet rate at UPF level (packets transmitted). Results can be fetched based on the selected networkContextName.

Expression

Total tx packet rate

```
round(sum(rate(upf_intf_packets_transmitted{networkContextName=~"$networkContextName"}[$rate_interval])/1000/1000), 0.001)
```

Total Drops

Description

Total drops at UPF level- RX MISS + NO BUFF + QUEUE DROPS. Results can be fetched based on the selected networkContextName.

Expression

Total drops

```
sum( upf_intf_rx_miss{networkContextName=~"$networkContextName"} +  
upf_intf_rx_nobuff{networkContextName=~"$networkContextName"} +  
upf_intf_tx_drops{networkContextName=~"$networkContextName"})
```

UPF Level Session Overview

Session Creation Request - Attempts

Description

Number of Session Create Attempts

Expression

```
sum(upf_session_creation_request{})
```

Session Create Success %

Description

Session Create Success %

Expression

```
((sum(upf_session_creation_request_success{}))/(sum(upf_session_creation_request{})))*100
```

Session Creation Request - Successes

Description

Number of Session creates succeeded

Expression

```
sum(upf_session_creation_request_success{})
```

Session Creation Request - Failures

Description

Number of Session creates failed

Expression

```
sum(upf_session_creation_request_failure{cause=~"$cause"})
```

UPF Session Creation Rate

Description

UPF Session Creation Rate

Expression

Create Requests

```
(sum(rate(upf_session_creation_request{}[1m])))
```

Create Success

```
(sum(rate(upf_session_creation_request_success{}[1m])))
```

Create Failure

```
(sum(rate(upf_session_creation_request_failure{}[1m])))
```

Session Modify Request - Attempts

Description

Number of Session modify requests

Expression

```
sum(upf_session_modify_request{})
```

Session Modify Success %

Description

Expression

```
((sum(upf_session_modify_request_success{}))/(sum(upf_session_modify_request{})))*  
100
```

Session Modify Request - Successes

Description

Number of Session modify requests succeeded

Expression

```
sum(upf_session_modify_request_success{})
```

Session Modify Request - Failures

Description

Number of Session modify requests failed

Expression

```
sum(upf_session_modify_request_failure{cause=~"$cause"})
```

UPF Session Modify Rate

Description

UPF Session Modify Rate

Expression

Modify Requests

```
(sum(rate(upf_session_modify_request{}[1m])))
```

Modify Success

```
(sum(rate(upf_session_modify_request_success{}[1m])))
```

Modify Failure

```
(sum(rate(upf_session_modify_request_failure{}[1m])))
```

Session Delete Request - Attempts

Description

Number of Session delete requests

Expression

```
sum(upf_session_delete_request{})
```

Session Delete Success %

Description

Session Delete Success %

Expression

```
((sum(upf_session_delete_request_success{}))/(sum(upf_session_delete_request{})))*  
100
```

Session Delete Request - Successes

Description

Number of Session delete requests succeeded

Expression

```
sum(upf_session_delete_request_success{})
```

Session Delete Request - Failures

Description

Number of Session delete requests failed

Expression

```
sum(upf_session_delete_request_failure{cause=~"$cause"})
```

UPF Session Delete Rate

Description

UPF Session Delete Rate

Expression

Delete Requests

```
(sum(rate(upf_session_delete_request{}[1m])))
```

Delete Success

```
(sum(rate(upf_session_delete_request_success{}[1m])))
```

Delete Failure

```
(sum(rate(upf_session_delete_request_failure{}[1m])))
```

Active Session Count

Description

Number of Active Sessions

Expression

```
sum(upf_num_active_sessions{})
```

UPF Active Sessions

Description

UPF Active Sessions

Expression

Active Sessions

```
sum (upf_num_active_sessions{})
```

UPF Active Session Rate

Description

UPF Active Session Rate

Expression

Active Sessions

```
(sum(rate(upf_num_active_sessions{}[1m])))
```

UPF Standby Session Errors

Description

UPF Standby Sessions Errors Dashboard

Panels

Standby Session Create Failure

Description

Number of Standby Session creates failed

Expression

```
sum(upf_standby_session_creation_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Create Fail - No Resource Blocks

Description

Number of Standby Session creates failed due to no resource block present.

Expression

```
sum(upf_standby_session_creation_request_failure_resource_block_not_present{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Create Fail - UE IP Out of Range

Description

Number of Standby Session creates failed due to ue ip out of range

Expression

```
sum(upf_standby_session_creation_request_failure_ue_ip_out_of_range{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Create Fail - Invalid Length

Description

Number of Standby Session Create Failures due to Invalid Length

Expression

```
sum(upf_standby_session_creation_request_invalid_length_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Create Fail - Invalid Message

Description

Number of Standby Session Create Failures due to Invalid Message

Expression

```
sum(upf_standby_session_creation_request_invalid_message_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Duplicate Standby Session Create - Ignored

Description

Number of Duplicate Standby Session Create Requests Ignored

Expression

```
sum(upf_standby_session_dup_est_req_ignored{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Duplicate Standby Session Create - Failures

Description

Number of Duplicate Standby Session Establishment Failures

Expression

```
sum(upf_standby_session_creation_duplicate_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Session Create Fail- No Association

Description

Number of Standby Session Create Requests Failed due to no association

Expression

```
sum(upf_standby_session_creation_request_failure_no_association{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Session Modify Failure

Description

Number of Standby Session modify requests failed

Expression

```
sum(upf_standby_session_modify_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Modify Fail - No Resource Blocks

Description

Number of Standby Session modify failed due to no resource block present.

Expression

```
sum(upf_standby_session_modify_request_failure_resource_block_not_present{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Modify Fail - Session ID Out of Range

Description

Number of Standby Session modify failed due to upf session id out of range

Expression

```
sum(upf_standby_session_modify_request_failure_upf_session_id_out_of_range{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Modify Fail - Invalid Length

Description

Number of Standby Session Modify Failures due to Invalid Length

Expression

```
sum(upf_standby_session_modify_request_invalid_length_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Modify Fail - Invalid Message

Description

Number of Standby Session Modify Failures due to Invalid Message

Expression

```
sum(upf_standby_session_modify_request_invalid_message_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Session Modify Fail - No Session

Description

Number of Standby Session Modify Requests Failed due to no session

Expression

```
sum(upf_standby_session_modify_request_no_session_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Modify Request Fail - Deleting Session**Description**

Number of Standby Session Modify Request Failed to Delete Session

Expression

```
sum(upf_standby_session_modify_request_fail_deleting_session{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Modify Request Fail - Session Not In Vpp**Description**

Number of Standby Session Modify Request Failed due to Session Not In Vpp

Expression

```
sum(upf_standby_session_modify_request_fail_session_not_in_vpp{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Session Delete Failure**Description**

Number of Standby Session delete requests failed

Expression

```
sum(upf_standby_session_delete_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Delete Fail - No Resource Blocks

Description

Number of Standby Session delete failed due to no resource block present.

Expression

```
sum(upf_standby_session_delete_request_failure_resource_block_not_present{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Delete Fail - Session ID Out of Range

Description

Number of Standby Session delete failed due to upf session id out of range

Expression

```
sum(upf_standby_session_delete_request_failure_upf_session_id_out_of_range{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Duplicate Standby Session Delete Requests - Ignored

Description

Number of Duplicate Standby Session Delete Requests Ignored

Expression

```
sum(upf_standby_session_dup_del_req_ignored{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Session Delete Fail - Invalid Session

Description

Number of Standby Session Delete Requests Failed due to invalid session

Expression

```
sum(upf_standby_session_delete_request_invalid_session_failure{pod=~"$pod", sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

Standby Session Delete Fail - Invalid Message

Description

Number of Standby Session Delete Requests Failed due to invalid Message

Expression

```
sum(upf_standby_session_delete_request_invalid_message_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

UPF Standby Sessions Detailed View

Description

UPF Standby Sessions Detailed View Dashboard

Panels

Standby Session Count

Description

Number of Standby Sessions

Expression

```
sum(upf_num_standby_sessions{pod=~"$pod",sm_instance_id=~"$sm_instance_id",resourc
e_block_id=~"$resource_block_id",rb_id=~"$rb_id"})
```

UPF Standby Sessions

Description

UPF Standby Sessions

Expression

Standby Sessions-{{pod}}

```
sum by (pod) (upf_num_standby_sessions{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

UPF Standby Session Rate per Sec

Description

UPF Standby Session Rate per Sec

Expression

Standby Sessions-{{pod}}

```
(sum by (pod) (rate(upf_num_standby_sessions{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Standby Session Creation Request - Attempts

Description

Number of Standby Session Create Attempts

Expression

```
sum(upf_standby_session_creation_request{pod=~"$pod",sm_instance_id=~"$sm_instance  
_id",rb_id=~"$rb_id"})
```

Standby Session Create Success %

Description

Standby Session Create Success %

Expression

```
((sum(upf_standby_session_creation_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}))/(sum(upf_standby_session_crea  
tion_request{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})))*100
```

Standby Session Creation Request - Successes

Description

Number of Standby Session creates succeeded.

Expression

```
sum(upf_standby_session_creation_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Standby Session Creation Request - Failures

Description

Number of Standby Session creates failed

Expression

```
sum(upf_standby_session_creation_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

UPF Standby Session Creation Rate per pod per Sec

Description

UPF Standby Session Creation Rate per pod per Sec

Expression

Create Requests-{{pod}}

```
(sum by (pod) (rate(upf_standby_session_creation_request{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Create Success-{{pod}}

```
(sum by (pod) (rate(upf_standby_session_creation_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Create Failure-{{pod}}

```
(sum by (pod) (rate(upf_standby_session_creation_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Standby Session Modify Request - Attempts

Description

Number of Standby Session modify requests in net_ctxt network context.

Expression

```
sum(upf_standby_session_modify_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Standby Session Modify Success %

Description

Expression

```
((sum(upf_standby_session_modify_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}))/(sum(upf_standby_session_modi
fy_request{pod=~"$pod", sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})))*100
```

Standby Session Modify Request - Successes

Description

Number of Standby Session modify requests succeeded.

Expression

```
sum(upf_standby_session_modify_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Standby Session Modify Request - Failures

Description

Number of Standby Session modify requests failed in net_ctxt network context.

Expression


```
sum(upf_standby_session_modify_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

UPF Standby Session Modify Rate Per Pod

Description

UPF Standby Session Modify Rate Per Pod

Expression

Modify Attempts--{{pod}}

```
(sum by (pod) (rate(upf_standby_session_modify_request{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Modify Success--{{pod}}

```
(sum by (pod) (rate(upf_standby_session_modify_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Modify Failure--{{pod}}

```
(sum by (pod) (rate(upf_standby_session_modify_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Standby Session Delete Request - Attempts

Description

Number of Standby Session delete requests

Expression

```
sum(upf_standby_session_delete_request{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Standby Session Delete Success %

Description

Standby Session Delete Success %

Expression

```
((sum(upf_standby_session_delete_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}))/(sum(upf_standby_session_delete_request{pod=~"$pod", sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})))*100
```

Standby Session Delete Request - Successes

Description

Number of Standby Session delete requests succeeded.

Expression

```
sum(upf_standby_session_delete_request_success{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Standby Session Delete Request - Failures

Description

Number of Standby Session delete requests failed

Expression

```
sum(upf_standby_session_delete_request_failure{pod=~"$pod",  
sm_instance_id=~"$sm_instance_id", cause=~"$cause",rb_id=~"$rb_id"})
```

UPF Standby Session Delete Rate Per Pod

Description

UPF Standby Session Delete Rate Per Pod

Expression

Delete Attempt- -{{pod}}

```
(sum by (pod) (rate(upf_standby_session_delete_request{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Delete Success--{{pod}}

```
(sum by (pod) (rate(upf_standby_session_delete_request_success{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

Delete Failure--{{pod}}

```
(sum by (pod) (rate(upf_standby_session_delete_request_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"}[1m])))
```

UPF Standby Sessions Internal Errors

Description

UPF Standby Sessions Internal Erros Dashboard

Panels

VPP Send Failures

Description

Standby Session Failures due to VPP Send Failures

Expression

Create

```
sum (upf_standby_session_create_vpp_send_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Modify

```
sum (upf_standby_session_modify_vpp_send_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Delete

```
sum (upf_standby_session_delete_vpp_send_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

VPP Response Failures

Description

Standby Session Failures due to VPP Response Failures

Expression

Create

```
sum (upf_standby_session_create_vpp_response_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Modify

```
sum (upf_standby_session_modify_vpp_response_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Delete

```
sum (upf_standby_session_delete_vpp_response_failure{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

VPP Response Timeouts

Description

Standby Session Failures due to VPP Response Timeout

Expression

Create

```
sum (upf_standby_session_create_vpp_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Modify

```
sum (upf_standby_session_modify_vpp_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

Delete

```
sum (upf_standby_session_delete_vpp_response_timeout{pod=~"$pod",
sm_instance_id=~"$sm_instance_id",rb_id=~"$rb_id"})
```

vProbe Statistics Dashboard

Description

vProbe Statistics Dashboard

Panels

vProbe Statistics Dashboard

Description

vProbe statistics dashboard

Rate of Start Flow EDRs Generated

Description

Rate of Start Flow EDRs Generated

Expression

Starts

```
irate(total_start_flow_edrs_generated_total[1m])
```

Start Flow EDR Sent Success

Description

Start Flow EDR Sent Success

Expression

```
(sum(total_start_flow_edrs_successfully_sent_to_external_consumer_total) /
(sum(total_start_flow_edrs_successfully_sent_to_external_consumer_total +
total_start_flow_edrs_dropped_due_to_service_unavailable_total +
total_start_flow_edrs_failed_to_send_to_external_consumer_total)) ) * 100
```

Rate of Start Flow EDR Errors

Description

Rate of Start Flow EDR Errors

Expression

Service Unavailable

```
irate(total_start_flow_edrs_dropped_due_to_service_unavailable_total[1m])
```

Parse Failure

```
irate(total_start_flow_edrs_failed_to_parse_total[1m])
```

Failed to Send to External

```
irate(total_start_flow_edrs_failed_to_send_to_external_consumer_total[1m])
```

Rate of Interim Flow EDRs Generated

Description

Rate of Interim Flow EDRs Generated

Expression

Interims

```
irate(total_interim_flow_edrs_generated_total[1m])
```

Interim Flow EDR Sent Success

Description

Interim Flow EDR Sent Success

Expression

```
(sum( total_interim_flow_edrs_successfully_sent_to_external_consumer_total) / (sum
(total_interim_flow_edrs_successfully_sent_to_external_consumer_total +
total_interim_flow_edrs_dropped_due_to_service_unavailable_total +
total_interim_flow_edrs_failed_to_send_to_external_consumer_total))) * 100
```

Rate of Interim Flow EDR Errors

Description

Rate of Interim Flow EDR Errors

Expression

Service Unavailable

```
irate(total_interim_flow_edrs_dropped_due_to_service_unavailable_total[1m])
```

Parse Failure

```
irate(total_interim_flow_edrs_failed_to_parse_total[1m])
```

Failed to Send to External

```
irate(total_interim_flow_edrs_failed_to_send_to_external_consumer_total[1m])
```

Rate of Stop Flow EDRs Generated

Description

Rate of Stop Flow EDRs Generated

Expression

Stops

```
irate(total_stop_flow_edrs_generated_total[1m])
```

Stop Flow EDR Sent Success

Description

Stop Flow EDR Sent Success

Expression

```
(sum( total_stop_flow_edrs_successfully_sent_to_external_consumer_total) / (sum
(total_stop_flow_edrs_successfully_sent_to_external_consumer_total +
total_stop_flow_edrs_dropped_due_to_service_unavailable_total +
total_stop_flow_edrs_failed_to_send_to_external_consumer_total))) * 100
```

Rate of Stop Flow EDR Errors

Description

Rate of Stop Flow EDR Errors

Expression

Service Unavailable

```
irate(total_stop_flow_edrs_dropped_due_to_service_unavailable_total[1m])
```

Parse Failure

```
irate(total_stop_flow_edrs_failed_to_parse_total[1m])
```

Failed to Send to External


```
irate(total_stop_flow_edrs_failed_to_send_to_external_consumer_total[1m])
```

Rate of Start-Stop Flow EDRs Generated

Description

Rate of Start-Stop Flow EDRs Generated

Expression

Start-Stops

```
irate(total_start_stop_flow_edrs_generated_total[1m])
```

Start Stop Flow EDR Sent Success

Description

Start Stop Flow EDR Sent Success

Expression

```
(sum( total_start_stop_flow_edrs_successfully_sent_to_external_consumer_total) /  
(sum (total_start_stop_flow_edrs_successfully_sent_to_external_consumer_total +  
total_start_stop_flow_edrs_dropped_due_to_service_unavailable_total +  
total_start_stop_flow_edrs_failed_to_send_to_external_consumer_total))) * 100
```

Rate of Start Stop Flow EDR Errors

Description

Rate of Start Stop Flow EDR Errors

Expression

Service Unavailable

```
irate(total_start_stop_flow_edrs_dropped_due_to_service_unavailable_total[1m])
```

Parse Failure

```
irate(total_start_stop_flow_edrs_failed_to_parse_total[1m])
```

Failed to Send to External

```
irate(total_start_stop_flow_edrs_failed_to_send_to_external_consumer_total[1m])
```

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Alertmanager Dashboard	Total Critical
Alertmanager Dashboard	Total Warnings
Alertmanager Dashboard	Notifications Failed Total
Alertmanager Dashboard	Number of active alerts
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Alertmanager Dashboard

Description

Alertmanager dashboard

Panels

Total number of alerts received

Description

Total alerts

Expression

```
sum(alertmanager_alerts_received_total) without ( version,status )
```

Total Critical

Description

Total Critical

Expression

```
sum(ALERTS{alertname!="DeadMansSwitch",alertstate="firing",severity="critical"} > 0) OR on() vector(0)
```

Total Warnings

Description

Total Warnings

Expression

```
sum(ALERTS{alertname!="DeadMansSwitch",alertstate="firing",severity="warning"} > 0) OR on() vector(0)
```

Notifications Failed Total

Description

Notifications Failed Total

Expression

{{integration}}

```
alertmanager_notifications_failed_total
```

Number of active alerts

Description

Max number of active alerts

Expression

```
alertmanager_alerts{state="active"}
```

Prometheus Alerts

Description

Dashboard showing Prometheus Alerts (both pending and firing) for alerts adjustment and debugging

Panels

Alerts History

Description

Alerts History

Expression

{{alertname}} {{instance}} {{alertstate}} {{datacenter}}

```
ALERTS{instance=~"$instance", alertname=~"$alertname", alertstate=~"$alertstate"}
```

help_series_x_axes_in_center

```
1
```

Alerts Start List

Description

Alerts list for all alerts in a selected time range. For filling this table **Alerts annotations** have to be switched on.

Expression

Total Alert Counts

Description

Shows how many times was particular alert started in a defined time range. Alert can be started either directly as **firing** or as a **pending**. Pending alerts wait for a defined time before it flips to a **firing** alert. This is specified with the **FOR** clause in a Prometheus **rules** file.

Expression

```
changes( ALERTS_FOR_STATE{instance=~"$instance",alertname=~"$alertname"}[$__range]
) + 1
```

prometheus Dashboard

Description

prometheus grafana dashboard

Panels

prometheus Uptime

Description

prometheus instance up time

Expression

{{pod}}

```
(time() - process_start_time_seconds{namespace="$namespace"})
```

TSDB storage

Description

Time series database storage

Expression

{{pod}}

```
prometheus_tsdb_storage_blocks_bytes
```

Discovered targets

Description

Number of discovered targets

Expression

```
sum(up)
```

Scrape Duration(S)

Description

Scrape duration in seconds

Expression

{{pod}}

```
rate(scrape_duration_seconds[$interval])
```

Samples Scraped

Description

Number of samples scraped

Expression

{{pod}}

scrape_samples_scraped

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Summary

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Redis Server	Net Rate of Commands Processed

Geo Redundancy

Description

A quickstart to setup the Prometheus Redis Syncer with preconfigured dashboards, alerting rules, and recording rules.

Panels

Overall Status Per Sites

Description

It indicates the overall status of sites when the last sync happened, duration of sync, number of commands processed , discards while initial sync replication and describes the total number of commands sent during ongoing replication.

Expression

_auto

(

```
sum by (pod) (initial_sync_commands_parsed{siteName=~"$siteName",
pod=~"$Pod", cmdType="moduleDataCmds"}) )
```

```
( sum by (pod) (initial_sync_commands_discards{siteName=~"$siteName", pod=~"$Pod", reason = "filtered"})
or vector(0) ) * on(pod) group_left() (redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

_auto

```
redis_context_commands_sent{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
(redis_instance_info{namespace="fed-redis-cluster", role="master"})

3 last_initial_sync_start_time{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

```
4 last_initial_sync_duration{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

```
5 last_initial_sync_status{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

```
6 last_initial_sync_type{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

```
7 site_state{siteName=~"$siteName", pod=~"$Pod"} * on(pod) group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Latency of Commands

Description

Duration of roundtrip of commands sent to remote sites with command name and site name.

Expression

P50

```
histogram_quantile(0.50,
sum(irate(redis_context_cmds_latency_bucket{siteName=~"$siteName",pod=~ "$Pod"}
[1m]))) by (le))
```

P95

```
histogram_quantile(0.95,
sum(irate(redis_context_cmds_latency_bucket{siteName=~"$siteName",pod=~ "$Pod"}
[1m]))) by (le))
```

P99

```
histogram_quantile(0.99,
sum(irate(redis_context_cmds_latency_bucket{siteName=~"$siteName",pod=~ "$Pod"}
[1m]))) by (le))
```

Total Number Of Commands Sent

Description

It determines the number of commands sent to remote sites. It includes all commands eg ping, hset, udsf etc

Expression

```
{{pod}}-{{siteName}}-{{cmdName}}
```

```
redis_context_commands_sent{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Number of Commands Discarded

Description

It indicates the number of commands discarded for sites with the command name discarded.

Expression

```
{{pod}}-{{siteName}}-{{cmdName}}
```

```
site_commands_discards{siteName=~"$siteName", pod=~"$Pod"} * on(pod) group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Initial Sync Commands discarded

Description

It indicates the total number of commands discarded during initial sync.

Expression

```
_auto
```

```
initial_sync_commands_discards{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Rate Of Incoming Traffic

Description

it describes the rate of traffic incoming from redis per minute

Expression

```
sum(rate(local_redis_read_bytes{siteName=~"$siteName", pod=~"$Pod"}[5m]) * on(pod)
group_left())
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"}))
```

Rate Of Commands Sent To Remote Sites

Description

It describes the rate of Commands sent to Remote Site per minute.

Expression

```
sum(rate(redis_context_commands_sent{siteName=~"$siteName", pod=~"$Pod"}[1m])*  
on(pod) group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Rate of Responses for the commands

Description

Rate of Responses for the commands received.

Expression

```
sum(rate(redis_context_command_responses{siteName=~"$siteName", pod=~"$Pod"}[1m])  
* on(pod) group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Commands in the buffer backlog

Description

Pending commands in the buffer backlog describes the commands which are in buffer.

Expression

```
redis_context_commands_buffered{siteName=~"$siteName", pod=~"$Pod"} * on(pod)  
group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Site Consuming/Paused Status

Description

It indicates the status of sites paused for consuming data.

Expression

{{pod}}-{{siteName}}

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"}) * on(pod)
group_right() site_consuming_paused{siteName=~"$siteName", pod=~"$Pod"}
```

Send Failures Of Commands Sent to Remote Site

Description

Failures Of Commands Sent to Remote Site

Expression

```
redis_context_command_send_failures{siteName=~"$siteName", pod=~"$Pod"} * on(pod)
group_left()
```

(redis_instance_info{namespace="fed-redis-cluster", role="master"})

Redis Command Read Failures

Description

it describes the errors received which are failures.

Expression

```
site_parse_callbacks{callbackType="eParseError",siteName=~"$siteName",
pod=~"$Pod"} * on(pod) group_left()
```

(redis_instance_info{namespace="fed-redis-cluster", role="master"})

Error Responses of Commands Sent to Remote

Description

Error Responses of Commands Sent where responses are failure

Expression

```
rate(redis_context_command_responses{status!="success", siteName=~"$siteName",
pod=~"$Pod"}[1m]) * on(pod) group_left()
```

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"})
```

Local Redis Connection Status

Description

It indicates whether the redis connection is up/down for local.

Expression

{{job}}-{{siteName}}

```
(redis_instance_info{namespace="fed-redis-cluster", role="master"}) * on(pod)
group_right() local_redis_connection_status{siteName=~"$siteName", pod=~"$Pod"}
```

Redis Latency and Resources

Description

A quickstart to setup the Prometheus Redis Exporter with preconfigured dashboards, alerting rules, and recording rules.

Panels

Latency

Redis Resources

Redis Server

Description

A quickstart to setup the Prometheus Redis Exporter with preconfigured dashboards, alerting rules, and recording rules.

Panels

Cluster Overview

Performance Metrics for Redis Server

Primary Nodes with Connected Replicas Count

Description

This metric will only be non-zero if the instance is a master

Expression

{{instance}}

```
redis_connected_slaves{instance=~"$instance"}
```

Clients Statistics

Description

Sum taken across instances

Expression

Max Clients

```
sum(redis_config_maxclients{instance=~"$instance"})
```

Connected

```
sum(redis_connected_clients{instance=~"$instance"})
```

Blocked

```
sum(redis_blocked_clients{instance=~"$instance"})
```

Rejected

```
sum(redis_rejected_connections_total{instance=~"$instance"})
```

Network Input / Output

Expression

Input Bytes

```
avg(rate(redis_net_input_bytes_total{instance=~"$instance"}[5m]))
```

Output Bytes

```
avg(rate(redis_net_output_bytes_total{instance=~"$instance"}[5m]))
```

Total Key Count

Description

Sum taken across instances

Expression

```
sum (redis_db_keys{instance=~"$instance"})
```

Redis Up Time

Expression

{{instance}}

```
redis_uptime_in_seconds{instance=~"$instance"}
```

Total Connections Accepted by Server

Description

Total connections received

Expression

```
sum(redis_connections_received_total{instance=~"$instance"})
```

Memory Utilization Per Instance

Expression

Memory Max Bytes Configured


```
max(redis_memory_max_bytes{instance=~"$instance"})
```

Usage-{{instance}}

```
redis_memory_used_bytes{instance=~"$instance"}
```

Peak-{{instance}}

```
redis_memory_used_peak_bytes{instance=~"$instance"}
```

Remaining Memory Percentage

Expression

Remaining-{{instance}}-{{pod}}

```
(redis_memory_max_bytes{instance=~"$instance"} -  
redis_memory_used_bytes{instance=~"$instance"})/redis_memory_max_bytes{instance=~"  
$instance"}
```

Command Latency by command and instance

Expression

P{{quantile}}-{{cmd}}-{{instance}}

```
redis_latency_percentiles_usec{instance=~"$instance",  
quantile=~"$latency_quantile", cmd=~"$cmd"}
```

Rate of Commands Processed Per Instance

Description

Rate of commands processed by the server

Expression

{{instance}}

```
sum by (instance) (rate(redis_commands_processed_total{instance=~"$instance"}
[5m]))
```

Rate of commands processed by command

Description

Rate of commands processed by selected instances by command

Expression

{{cmd}}

```
sum(rate(redis_commands_total{instance=~"$instance", cmd=~"$cmd"}[5m])) by (cmd)
```

Hits/Miss Rate

Expression

Hits

```
sum(rate(redis_keyspace_hits_total{instance=~"$instance"}[5m]))
```

Miss

```
sum(rate(redis_keyspace_misses_total{instance=~"$instance"}[5m]))
```

Rejected Commands Rate

Expression

Rejected Commands Rate

```
sum(rate(redis_commands_rejected_calls_total{instance=~"$instance"}[5m]))
```

SlowLog Queries Length

Expression

{{instance}}-{{pod}}

```
redis_slowlog_length
```

Performance Metrics for Primary Instances Panels

Total Key Count

Description

Sum taken across instances

Expression

```
sum (redis_db_keys{instance=~"$primary"})
```

Rate of Commands processed by command

Description

Rate of commands processed for selected primaries by command

Expression

{{cmd}}

```
sum(rate(redis_commands_total{instance=~"$primary"}[5m])) by (cmd)
```

Command Latency by command and instance

Expression

P{{quantile}}-{{cmd}}-{{instance}}

```
redis_latency_percentiles_usec{instance=~"$primary",  
quantile=~"$latency_quantile", cmd=~"$cmd"}
```

Net Rate of Commands Processed

Description

Total number of commands processed by the server

Expression

{{instance}}-{{pod}}

```
rate(redis_commands_processed_total{instance=~"$primary"}[5m])
```

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Summary

Dashboard Name	Panel Name
Redis Operator	Number of reconciliations per second
Redis Operator	Reconciliation time
Redis Operator	Reconcile errors

Redis Operator

Description

A quickstart to setup the Prometheus with preconfigured dashboards, alerting rules, and recording rules.

Panels

Number of reconciliations per second

Description

Expression

{{result}}

```
sum(rate(controller_runtime_reconcile_total[1m]))
```

Reconciliation time

Expression

```
sum(rate(controller_runtime_reconcile_time_seconds_count[1m]))
```

Reconcile errors

Expression

errors

```
sum(rate(controller_runtime_reconcile_errors_total[1m]))
```

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Summary

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Dashboard Name	Panel Name
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Rook Ceph Cluster Dashboard	Avg OSD Op Latency
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Rook Ceph Cluster Dashboard

Description

Ceph Cluster overview.

Panels

CLUSTER STATE

Description

cluster state

Ceph Health Status

Description

the status of the ceph cluster

Expression

```
sum without (instance) (ceph_health_status)
```

Write Throughput

Description

the write throughput

Expression

```
sum without (instance, ceph_daemon) (irate(ceph_osd_op_w_in_bytes[5m]))
```

Read Throughput

Description

the read throughput

Expression

```
sum without (ceph_daemon, instance) (irate(ceph_osd_op_r_out_bytes[5m]))
```

Cluster Capacity

Description

the total capacity of the cluster

Expression

```
sum without (instance) (ceph_cluster_total_bytes)
```

Available Capacity

Description

available capacity on the cluster

Expression

```
sum without (instance) ((ceph_cluster_total_bytes-  
ceph_cluster_total_used_bytes)/ceph_cluster_total_bytes)
```

Number of Objects

Description

number of objects

Expression


```
sum without (instance, pool_id) (ceph_pool_objects)
```

Bytes Written

Description

bytes written

Expression

```
sum without (instance, ceph_daemon) (ceph_osd_op_w_in_bytes)
```

Bytes Read

Description

bytes read

Expression

```
sum without (instance, ceph_daemon) (ceph_osd_op_r_out_bytes)
```

Alerts

Description

prometheus alerts

Expression

```
count(ALERTS{alertstate="firing",alertname=~"^Ceph.*"}) OR vector(0)
```

Write IOPS

Description

Write IOPS

Expression

```
sum without (ceph_daemon, instance) (irate(ceph_osd_op_w[5m]))
```

Read IOPS

Description

Read IOPS

Expression

```
sum without (ceph_daemon, instance) (irate(ceph_osd_op_r[5m]))
```

Used Capacity

Description

used capacity on the cluster

Expression

```
sum without (instance) (ceph_cluster_total_used_bytes)
```

Mon Session Num

Description

mon session number

Expression

```
sum without (instance, ceph_daemon) (ceph_mon_num_sessions)
```

Monitors In Quorum

Description

monitors in quorum

Expression

```
count without (instance, ceph_daemon) (ceph_mon_quorum_status)
```

OSD STATE

OSDs OUT

Description

OSDs out

Expression

```
count without (instance, ceph_daemon) (ceph_osd_up) - count without (instance, ceph_daemon) (ceph_osd_in)
```

OSDs DOWN

Description

OSDs down

Expression

```
count(ceph_osd_up == 0.0) OR vector(0)
```

OSDs UP

Description

OSDs up

Expression

```
sum without (instance, ceph_daemon) (ceph_osd_up)
```

OSDs IN

Description

OSDs in

Expression

```
sum without (instance, ceph_daemon) (ceph_osd_in)
```

Avg PGs

Description

average pgs

Expression

```
avg without (instance, ceph_daemon) (ceph_osd_numpg)
```

Avg Apply Latency

Description

average apply latency

Expression

```
avg without (instance, ceph_daemon) (ceph_osd_apply_latency_ms)
```

Avg Commit Latency

Description

average commit latency

Expression

```
avg(ceph_osd_commit_latency_ms)
```

Avg Op Write Latency

Description

average op write latency

Expression

```
avg without (instance, ceph_daemon) (rate(ceph_osd_op_w_latency_sum[5m]) /  
rate(ceph_osd_op_w_latency_count[5m])) >= 0)
```

Avg Op Read Latency

Description

average op read latency

Expression

```
avg without (instance, ceph_daemon)  
(rate(ceph_osd_op_r_latency_sum[5m])/rate(ceph_osd_op_r_latency_count[5m])) >= 0)
```

Alerts

Alerts from CephThanos

Description

prometheus alerts

Expression

```
ALERTS{ alertstate="firing"}
```

Top Sluggish OSDs

Description

sluggish osds

Expression

```
topk(5,sort_desc(ceph_osd_apply_latency_ms + ceph_osd_commit_latency_ms))
```

Down OSDs

Description

down OSDs

Expression

```
ceph_osd_up == 0.0
```

Ceph Versions

Description

ceph versions

CLUSTER

Capacity

Description

the total capacity of the cluster

Expression

Total Capacity

```
sum without (instance) (ceph_cluster_total_bytes)
```

Available

```
sum without (instance) (ceph_cluster_total_bytes-ceph_cluster_total_used_bytes)
```

Used

```
sum without (instance) (ceph_cluster_total_used_bytes)
```

IOPS

Description

IOPS

Expression

Write

```
sum without (instance, ceph_daemon) (irate(ceph_osd_op_w[5m]))
```

Read

```
sum without (instance, ceph_daemon) (irate(ceph_osd_op_r[5m]))
```

Throughput

Description

throughput

Expression

Write

```
sum without (instance, ceph_daemon) (irate(ceph_osd_op_w_in_bytes[5m]))
```

Read

```
sum without (instance, ceph_daemon) (irate(ceph_osd_op_r_out_bytes[5m]))
```

Pool Bytes Recovered

Description

Pool Bytes Recovered

Expression

{{name}}

```
sum((ceph_pool_num_bytes_recovered) *on (instance, pool_id) group_left(name)
(ceph_pool_metadata)) by (name)
```

Pool Objects Recovered

Description

Pool Objects Recovered

Expression

{{name}}

```
sum without (instance, pool_id) ((ceph_pool_num_objects_recovered) *on (instance, pool_id) group_left(name)(ceph_pool_metadata))
```

Objects Per Pool

Description

objects per pool

Expression

{{name}}

```
sum without (instance) ((ceph_pool_objects) *on (instance, pool_id) group_left(name)(ceph_pool_metadata))
```

Pool Quota Bytes

Description

pool quota bytes

Expression

{{name}}

```
sum without (instance) ((ceph_pool_quota_bytes) *on (instance, pool_id) group_left(name)(ceph_pool_metadata))
```

Pool Objects Quota

Description

pool objects quota

Expression

{{name}}

```
sum without (instance) (ceph_pool_quota_objects) *on (instance, pool_id)
group_left(name)(ceph_pool_metadata)
```

OSD Type Count

Description

OSD type count

Expression

BlueStore

```
count without (instance, ceph_daemon) (ceph_bluestore_commit_lat_count)
```

FileStore

```
count without (instance, ceph_daemon) (ceph_filestore_journal_latency_count)
```

OBJECTS

Description

objects

Objects in the Cluster

Description

objects in the cluster

Expression

Total

```
sum without (instance, pool_id) (ceph_pool_objects)
```

PGs State

Description

PGs state

Expression

Active

```
sum without (instance, pool_id) (ceph_pg_active)
```

Clean

```
sum without (instance, pool_id) (ceph_pg_clean)
```

Peering

```
sum without (instance, pool_id) (ceph_pg_peering)
```

Degraded

```
sum without (instance, pool_id) (ceph_pg_degraded)
```

Stale

```
sum without (instance, pool_id) (ceph_pg_stale)
```

Unclean

```
sum without (instance, pool_id) (ceph_unclean_pgs)
```

Undersized

```
sum without (instance, pool_id) (ceph_pg_undersized)
```

Incomplete

```
sum without (instance, pool_id) (ceph_pg_incomplete)
```

Forced Backfill

```
sum without (instance, pool_id) (ceph_pg_forced_backfill)
```

Inconsistent

```
sum without (instance, pool_id) (ceph_pg_inconsistent)
```

Forced Recovery

```
sum without (instance, pool_id) (ceph_pg_forced_recovery)
```

Creating

```
sum without (instance, pool_id) (ceph_pg_creating)
```

Wait Backfill

```
sum without (instance, pool_id) (ceph_pg_wait_backfill)
```

Deep

```
sum without (instance, pool_id) (ceph_pg_deep)
```

Scrubbing

```
sum without (instance, pool_id) (ceph_pg_scrubbing)
```

Recovering

```
sum without (instance, pool_id) (ceph_pg_recovering)
```

Repair

```
sum without (instance, pool_id) (ceph_pg_repair)
```

Down

```
sum without (instance, pool_id) (ceph_pg_down)
```

Peered

```
sum without (instance, pool_id) (ceph_pg_peered)
```

Backfill

```
sum without (instance, pool_id) (ceph_pg_backfill)
```

Remapped

```
sum without (instance, pool_id) (ceph_pg_remapped)
```

Backfill Toofull

```
sum without (instance, pool_id) (ceph_pg_backfill_toofull)
```

Stuck PGs

Description

stuck PGs

Expression

Degraded

```
sum without (instance, pool_id) (ceph_pg_degraded)
```

Stale

```
sum without (instance, pool_id) (ceph_pg_stale)
```

Undersized

```
sum without (instance, pool_id) (ceph_pg_undersized)
```

Recovery Operations

Description

recovery operations

Expression

OPS

```
sum without (instance, ceph_daemon)
(irate(ceph_osd_recovery_ops[$__rate_interval]))
```

LATENCY

Description

latency

OSD Apply Latency Distribution

Expression

```
ceph_osd_apply_latency_ms
```

OSD Commit Latency Distribution

Description

OSD commit latency distribution

Expression

```
ceph_osd_commit_latency_ms
```

OSD Read Op Latency Distribution

Description

OSD read op latency distribution

Expression

```
rate(ceph_osd_op_r_latency_sum[5m]) / rate(ceph_osd_op_r_latency_count[5m]) >= 0
```

OSD Write Op Latency Distribution

Expression

```
rate(ceph_osd_op_w_latency_sum[5m]) / rate(ceph_osd_op_w_latency_count[5m]) >= 0
```

Avg OSD Op Latency

Description

average OSD op latency

Expression

read

```
avg without (instance,ceph_daemon) (rate(ceph_osd_op_r_latency_sum[5m]) /  
rate(ceph_osd_op_r_latency_count[5m]) >= 0)
```

write

```
avg without (instance, ceph_daemon) (rate(ceph_osd_op_w_latency_sum[5m]) /
rate(ceph_osd_op_w_latency_count[5m]) >= 0)
```

AVG OSD Apply + Commit Latency

Description

average osd apply and commit latency

Expression

apply

```
avg without (instance, ceph_daemon) (ceph_osd_apply_latency_ms)
```

commit

```
avg without (instance, ceph_daemon) (ceph_osd_commit_latency_ms)
```

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DB Client Sidecar Dashboard

Description

The DB Client sidecar is present in the cna-infra container within application worker PODs. Application workers communicate with their local DB Client when they need to carry out database operations.

Panels

Database Requests

Description

Total database requests per second from workers to the database.

Expression

DB Client Requests to the database

```
sum (rate(dbclient_database_request_duration_seconds_count{db_type="mongo_RAM"}[5m]))
```

Database Workload

Description

Database workload broken into individual CRUD (Create, Read, Update, Delete) operations to assist in understanding database latency on a per-operation basis.

Expression

```
1 sum (irate(dbclient_database_request_duration_seconds_count[5m])) by (crud_op)

2 (histogram_quantile(0.50,
sum(irate(dbclient_database_request_duration_seconds_bucket{}[5m])) by (le,
crud_op))) > 0

3 (histogram_quantile(0.95,
sum(irate(dbclient_database_request_duration_seconds_bucket{}[5m])) by (le,
crud_op))) > 0
```

HTTP Server Request Handling

Description

HTTP Server Request duration handling percentiles to assist in understanding database operation latency from the moment the request is received at DB Client to the moment that the operation is completed and DB Client responds to the worker.

Expression

P50

```
histogram_quantile(0.50,
sum(irate(http_server_request_handling_duration_seconds_bucket{app_type="dbClient"}[5m])) by (le, peer))
```

P95

```
histogram_quantile(0.95,  
sum(irate(http_server_request_handling_duration_seconds_bucket{app_type="dbClient"}[5m]))) by (le, peer))
```

P99

```
histogram_quantile(0.99,  
sum(irate(http_server_request_handling_duration_seconds_bucket{app_type="dbClient"}[5m]))) by (le, peer))
```

Database Request Duration Percentiles

Description

Database request duration percentiles to assist in understanding database operation latency.

Expression

P50

```
histogram_quantile(0.50,  
sum(irate(dbclient_database_request_duration_seconds_bucket{}[5m]))) by (le,  
db_type))
```

P95

```
histogram_quantile(0.95,  
sum(irate(dbclient_database_request_duration_seconds_bucket{}[5m]))) by (le,  
db_type))
```

P99

```
histogram_quantile(0.99,  
sum(irate(dbclient_database_request_duration_seconds_bucket{}[5m]))) by (le,  
db_type))
```

Thread Pool Operations

Description

The number of concurrent database operations in progress from a worker to its local DB Client sidecar.

Expression

```
topk(5, sum by (namespace, pod) (rate(dbclient_thread_pool_ops[2m])))
```

Error Status Code

Description

Tracks certain HTTP error status codes returned to a worker from its local DB Client sidecar that indicate probable error situations. Some error codes are not tracked, such as "record not found" which is HTTP status code 404, since this may be expected so the application layer might track independently.

Expression

```
sum by (namespace, pod, http_status_code)
(rate(dbclient_http_requests{http_status_code!="200",http_status_code!="201",http_
status_code!="404",http_status_code!="409"}[5m]))
```

Overloaded Try Later

Description

The number of HTTP 500 status code errors that are due to the worker overloading its local DB Client with database operations when the operation thread pool is at capacity. This may indicate that the worker thread pool should be increased, or that the worker is generating inordinate amounts of database operations that are unexpected, or that the database itself is experiencing increased latency.

Expression

```
topk(5, sum by (namespace, pod)
(rate(dbclient_internal{internal_counter="opres_overloadedTryLater"}[5m])))
```

FlexBlock Sidecar Dashboard

Description

The Flexblock sidecar metrics.

Ingress Gateway

Description

The dashboards for Ingress Gateway

Panels

Upstream Request Response Latency

Expression

{{pod}}

```
(histogram_quantile(${percentiles:raw},sum(irate(envoy_cluster_upstream_rq_time_bucket[5m])) by (le , pod))) > 0
```

Downstream Request Response Latency

Expression

{{pod}}

```
(histogram_quantile(${percentiles:raw},sum(irate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_time_bucket[5m])) by (le, pod))) > 0
```

Incoming HTTP/2 requests rate

Expression

{{pod}}

```
rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_http2_total[5m])
```

Total requests that resulted in a response

Expression

{{pod}}

```
rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_completed[5m])
```

Active Requests

Expression

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_rq_active)
```

Active HTTP/2 Connections

Expression

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_http2_active)
```

Downstream response

Expression

`_auto`

```
sum by (response_code_class)
(rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq[5m]))
```

Upstream response

Expression

`{{response_code_class}}`

```
sum by (response_code_class)
(rate(envoy_cluster_upstream_rq{cluster_name="outbound|80||smfcc.fed-
smf.svc.cluster.local", response_code_class!~""}[5m]))
```

Downstream HTTP/2 Connections

Expression

`{{pod}}`

```
sum by (pod) (envoy_http_outbound_0_0_0_0_8080_downstream_cx_http2_active)
```

Total Connections

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_http2_active)
```

Errors**Expression****Total requests closed due to Envoy overload**

```
sum(rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_overload_close[2m]))
```

Total requests resulting in a 413 due to an overly large body

```
sum(rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_too_large[2m]))
```

Total number of invalid received frames violating HTTP2

```
sum(rate(envoy_http2_rx_messaging_error[2m]))
```

Total number of invalid frames received by Envoy with a REFUSED_STREAM error code

```
sum(rate(envoy_http2_stream_refused_errors[2m]))
```

Total protocol errors

```
sum(rate(envoy_http_outbound_0_0_0_0_8080_downstream_cx_protocol_error[2m]))
```

Downstream Connections Closed**Expression****Total connection closed due to max requests per connection**

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_max_requests_reached)
```

Total connections closed due to max connection duration

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_max_duration_reached)
```

Total connections closed due to idle timeout

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_idle_timeout)
```

Total connections closed due to draining

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_drain_close)
```

Total connections destroyed due to local close

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_destroy_local)
```

Total connections destroyed due to remote close

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_destroy_remote)
```

Total number of connections reset due to the headers being larger than the configured value.

```
sum(envoy_http2_header_overflow)
```

Total number of connections terminated for exceeding the limit on outbound frames of all types.

```
sum(envoy_http2_outbound_flood)
```

Total Connections Destroyed

```
sum(envoy_http_outbound_0_0_0_0_8080_downstream_cx_destroy)
```

Total HTTP/2 RST STEAMS

Expression

Total Request reset sent

```
sum(rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_tx_reset[5m]))
```

Total Request reset received

```
sum(rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_rx_reset[5m]))
```

{{pod}}

```
rate(envoy_http_outbound_0_0_0_0_8080_downstream_rq_rx_reset[5m])
```

UDSF DbClient Dashboard

Description

Application workers communicate with their local DB Client when they need to carry out database operations.

Panels

UDSF Request Rate

Description

UDSF request rate at which applications invoke.

Expression

{{app_name}}

```
sum (rate(udsf_db_operation_request_total{app_name=~"$app_name"}[5m]))
```

UDSF Response Rate

Description

Database transaction status response segregated on the basis of status. status = SUCCESS in case of successful API response else status = for failure scenario.

Expression

{{status}}

```
sum(rate(udsf_db_operation_response_total{app_name=~"$app_name"}[5m])) by (status)
```

Database Transaction Latency

Description

Database workload broken into individual CRUD (Create_or_modify, Modify, Delete, Search) operations to assist in understanding database latency on a per-operation basis.

Expression

P\$percentiles={{crud_op}}

```
(histogram_quantile(${percentiles:raw},  
sum(irate(udsf_db_transaction_duration_bucket{app_name=~"$app_name"}[5m])) by (le,  
crud_op))) > 0
```

Database Cpu Execution Latency

Description

Database workload broken into individual CRUD (Create_or_modify, Modify, Delete, Search) operations to assist in understanding database latency on a per-operation basis.

Expression

P\$percentiles={{operation_type}}

```
(histogram_quantile(${percentiles:raw},  
sum(irate(udsf_db_execution_duration_bucket{app_name=~"$app_name"}[5m])) by (le,  
operation_type))) > 0
```

Database Transaction Rate

Description

The rate at which transactions are issued towards Redis.

Expression

{{crud_op}}

```
sum (rate(udsf_db_transaction_counter{app_name=~"$app_name"}[5m])) by (crud_op)
```

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Summary

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Converged Charging	Overview-CHF Create Session rates
Converged Charging	Overview-CHF Create Session result %
Converged Charging	Overview-Current CHF Create Session success rate
Converged Charging	Overview-CHF Update Session rates
Converged Charging	Overview-CHF Update Session result %
Converged Charging	Overview-Current CHF Update Session success rate

Dashboard Name	Panel Name
Converged Charging	Overview-CHF Release Session rates
Converged Charging	Overview-CHF Release Session result %
Converged Charging	Overview-Current CHF Release Session success rate
Converged Charging	Overview-CHF One-Time Report rates
Converged Charging	Overview-CHF One-Time Report result %
Converged Charging	Overview-Current CHF One-Time Report success rate
Converged Charging	Rates per DNN and SNSSAI-CHF Create Session rates per DNN
Converged Charging	Rates per DNN and SNSSAI-CHF Create Session rates per SNSSAI
Converged Charging	Rates per DNN and SNSSAI-CHF Update Session rates per DNN
Converged Charging	Rates per DNN and SNSSAI-CHF Update Session rates per SNSSAI
Converged Charging	Rates per DNN and SNSSAI-CHF Release Session rates per DNN
Converged Charging	Rates per DNN and SNSSAI-CHF Release Session rates per SNSSAI
Converged Charging	Errors-CHF Create Session error rates by cause
Converged Charging	Errors-CHF Update Session error rates by cause
Converged Charging	Errors-CHF Release Session error rates by cause
GTP Messages	GTP Messages
GTP Messages Enterprise	GTP Messages
HTTP Client and Server Stats Dashboard	HTTP server incoming request rate over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP server outgoing responses rate over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP Client outgoing request rate over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP server transaction success percentage over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP client transaction success percentage over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP Client incoming response rate over last 15 minutes
HTTP punting dashboard	HTTP punting outgoing request rate over last 15 min

Dashboard Name	Panel Name
HTTP punting dashboard	HTTP punting incoming response rate over last 15 min
HTTP punting dashboard	Punting response latency over last 15 min
HTTP punting dashboard	HTTP punting response forward rate over last 15 min
Handover Procedures	Handover Procedures
Policy Control	Overview-Create SmPolicy Control Service rates
Policy Control	Overview-Create SmPolicy Control Service result %
Policy Control	Overview-Current Create SmPolicy Control Service success rate
Policy Control	Overview-Update SmPolicy Control Service rates
Policy Control	Overview-Update SmPolicy Control Service result %
Policy Control	Overview-Current Update SmPolicy Control Service success rate
Policy Control	Overview-Delete SmPolicy Control Service rates
Policy Control	Overview-Delete SmPolicy Control Service result %
Policy Control	Overview-Current Delete SmPolicy Control Service success rate
Policy Control	Overview-SmPolicy Control Service Notify rates
Policy Control	Overview-SmPolicy Control Service Notify result %
Policy Control	Overview-Current SmPolicy Control Service Notify success rate
Policy Control	Rates per DNN and SNSSAI-Create SmPolicy Control Service rates
Policy Control	Rates per DNN and SNSSAI-Create SmPolicy Control Service rates
Policy Control	Rates per DNN and SNSSAI-Update SmPolicy Control Service rates
Policy Control	Rates per DNN and SNSSAI-Update SmPolicy Control Service rates
Policy Control	Rates per DNN and SNSSAI-Delete SmPolicy Control Service rates
Policy Control	Rates per DNN and SNSSAI-Delete SmPolicy Control Service rates
Policy Control	Policy Control Internals-PCC Rule Install rates
Policy Control	Policy Control Internals-PCC Rule Install result %
Policy Control	Policy Control Internals-Current PCC Rule Install success rate
Policy Control	Policy Control Internals-PCC Rule Modify rates
Policy Control	Policy Control Internals-PCC Rule Modify result %
Policy Control	Policy Control Internals-Current PCC Rule Modify success rate
Policy Control	Policy Control Internals-PCC Rule PCF initiated Update rates

Dashboard Name	Panel Name
Policy Control	Policy Control Internals-PCC Rule PCF initiated Update result %
Policy Control	Policy Control Internals-Current PCC Rule PCF initiated Update success rate
Policy Control	Policy Control Internals-PCC Rule Remove rates
Policy Control	Policy Control Internals-PCC Rule Remove result %
Policy Control	Policy Control Internals-Current PCC Rule Remove success rate
Policy Control	PCC Rule Rates per PCC Rule-PCC Rule Install rates per PCC RULE
Policy Control	PCC Rule Rates per PCC Rule-PCC Rule Modify rates per PCC RULE
Policy Control	PCC Rule Rates per PCC Rule-PCC Rule PCF initiated Update rates per PCC RULE
Policy Control	PCC Rule Rates per PCC Rule-PCC Rule Remove rates per PCC RULE
Policy Control	Errors-Create SmPolicy Control Service error rates by cause
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Policy Control	Errors-PCC Rule Modify error rates by cause
Policy Control	Errors-PCC Rule Remove error rates by cause
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Policy Control Enterprise	Overview-Create SmPolicy Control Service result %
Policy Control Enterprise	Overview-Current Create SmPolicy Control Service success rate
Policy Control Enterprise	Overview-Update SmPolicy Control Service rates
Policy Control Enterprise	Overview-Update SmPolicy Control Service result %
Policy Control Enterprise	Overview-Current Update SmPolicy Control Service success rate
Policy Control Enterprise	Overview-Delete SmPolicy Control Service rates
Policy Control Enterprise	Overview-Delete SmPolicy Control Service result %
Policy Control Enterprise	Overview-Current Delete SmPolicy Control Service success rate
Policy Control Enterprise	Overview-SmPolicy Control Service Notify rates
Policy Control Enterprise	Overview-SmPolicy Control Service Notify result %
Policy Control Enterprise	Overview-Current SmPolicy Control Service Notify success rate
Policy Control Enterprise	Rates per DNN and SNSSAI-Create SmPolicy Control Service rates

Dashboard Name	Panel Name
Policy Control Enterprise	Rates per DNN and SNSSAI-Create SmPolicy Control Service rates
Policy Control Enterprise	Rates per DNN and SNSSAI-Update SmPolicy Control Service rates
Policy Control Enterprise	Rates per DNN and SNSSAI-Update SmPolicy Control Service rates
Policy Control Enterprise	Rates per DNN and SNSSAI-Delete SmPolicy Control Service rates
Policy Control Enterprise	Rates per DNN and SNSSAI-Delete SmPolicy Control Service rates
Policy Control Enterprise	Policy Control Internals-PCC Rule Install rates
Policy Control Enterprise	Policy Control Internals-PCC Rule Install result %
Policy Control Enterprise	Policy Control Internals-Current PCC Rule Install success rate
Policy Control Enterprise	Policy Control Internals-PCC Rule Modify rates
Policy Control Enterprise	Policy Control Internals-PCC Rule Modify result %
Policy Control Enterprise	Policy Control Internals-Current PCC Rule Modify success rate
Policy Control Enterprise	Policy Control Internals-PCC Rule PCF initiated Update rates
Policy Control Enterprise	Policy Control Internals-PCC Rule PCF initiated Update result %
Policy Control Enterprise	Policy Control Internals-Current PCC Rule PCF initiated Update success rate
Policy Control Enterprise	Policy Control Internals-PCC Rule Remove rates
Policy Control Enterprise	Policy Control Internals-PCC Rule Remove result %
Policy Control Enterprise	Policy Control Internals-Current PCC Rule Remove success rate
Policy Control Enterprise	PCC Rule Rates per PCC Rule-PCC Rule Install rates per PCC RULE
Policy Control Enterprise	PCC Rule Rates per PCC Rule-PCC Rule Modify rates per PCC RULE
Policy Control Enterprise	PCC Rule Rates per PCC Rule-PCC Rule PCF initiated Update rates per PCC RULE
Policy Control Enterprise	PCC Rule Rates per PCC Rule-PCC Rule Remove rates per PCC RULE
Policy Control Enterprise	Errors-Create SmPolicy Control Service error rates by cause
Policy Control Enterprise	Errors-Update SmPolicy Control Service error rates by cause
Policy Control Enterprise	Errors-Delete SmPolicy Control Service error rates by cause
Policy Control Enterprise	Errors-PCC Rule Install error rates by cause
Policy Control Enterprise	Errors-PCC Rule Modify error rates by cause
Policy Control Enterprise	Errors-PCC Rule Remove error rates by cause
SM Context	Overview-PDU Session Create SM Context rates

Dashboard Name	Panel Name
SM Context	Overview-PDU Session Create SM Context result %
SM Context	Overview-Current PDU Session Create SM Context success rate
SM Context	Overview-PDU Session Update SM Context rates
SM Context	Overview-PDU Session Update SM Context result %
SM Context	Overview-Current PDU Session Update SM Context success rate
SM Context	Overview-PDU Session Retrieve SM Context rates
SM Context	Overview-PDU Session Retrieve SM Context result %
SM Context	Overview-Current PDU Session Retrieve SM Context success rate
SM Context	Overview-PDU Session Notify SM Context rates
SM Context	Overview-PDU Session Notify SM Context result %
SM Context	Overview-Current PDU Session Notify SM Context success rate
SM Context	Overview-PDU Session Release SM Context rates
SM Context	Overview-PDU Session Release SM Context result %
SM Context	Overview-Current PDU Session Release SM Context success rate
SM Context	Overview-N1N2 Message Transfer rates
SM Context	Overview-N1N2 Message Transfer result %
SM Context	Overview-Current N1N2 Message Transfer success rate
SM Context	Overview-NAMF Assign EBI Message rates
SM Context	Overview-NAMF Assign EBI Message result %
SM Context	Overview-Current NAMF Assign EBI Message success rate
SM Context	Overview-UE Initiated Modify QoS attempts rates
SM Context	Overview-UE Initiated Modify QoS attempts result %
SM Context	Overview-Current UE Initiated Modify QoS attempts success rate
SM Context	Rates per DNN and SNSSAI-PDU Session Create SM Context rates per DNN
SM Context	Rates per DNN and SNSSAI-PDU Session Create SM Context rates per SNSSAI
SM Context	Rates per DNN and SNSSAI-PDU Session Update SM Context rates per DNN
SM Context	Rates per DNN and SNSSAI-PDU Session Update SM Context rates per SNSSAI

Dashboard Name	Panel Name
SM Context	Rates per DNN and SNSSAI-PDU Session Release SM Context rates per DNN
SM Context	Rates per DNN and SNSSAI-PDU Session Release SM Context rates per SNSSAI
SM Context	Rates per DNN and SNSSAI-N1N2 Message Transfer rates per DNN
SM Context	Rates per DNN and SNSSAI-N1N2 Message Transfer rates per SNSSAI
SM Context	SM Context Internals-Internal Analyzer Classifier SM Context rates
SM Context	SM Context Internals-Internal Analyzer Classifier SM Context result %
SM Context	SM Context Internals-Current Internal Analyzer Classifier SM Context success rate
SM Context	SM Context Internals-DB Client State Store HTTP transactions Rate
SM Context	Errors-PDU Session Create SM Context error rates by cause
SM Context	Errors-PDU Session Update SM Context error rates by cause
SM Context	Errors-PDU Session Retrieve SM Context error rates by cause
SM Context	Errors-PDU Session Notify SM Context error rates by cause
SM Context	Errors-PDU Session Release SM Context error rates by cause
SM Context	Errors-N1N2 Message Transfer error rates by cause
SM Context	Errors-NAMF Assign EBI Message error rates by cause
SM Context	Errors-UE Initiated Modify QoS attempts error rates by cause
SMF Call Control	SM Context-PDU Session Create SM Context rates
SMF Call Control	SM Context-PDU Session Create SM Context result %
SMF Call Control	SM Context-Current PDU Session Create SM Context success rate
SMF Call Control	SM Context-PDU Session Update SM Context rates
SMF Call Control	SM Context-PDU Session Update SM Context result %
SMF Call Control	SM Context-Current PDU Session Update SM Context success rate
SMF Call Control	SM Context-PDU Session Retrieve SM Context rates
SMF Call Control	SM Context-PDU Session Retrieve SM Context result %
SMF Call Control	SM Context-Current PDU Session Retrieve SM Context success rate
SMF Call Control	SM Context-PDU Session Notify SM Context rates
SMF Call Control	SM Context-PDU Session Notify SM Context result %

Dashboard Name	Panel Name
SMF Call Control	SM Context-Current PDU Session Notify SM Context success rate
SMF Call Control	SM Context-PDU Session Release SM Context rates
SMF Call Control	SM Context-PDU Session Release SM Context result %
SMF Call Control	SM Context-Current PDU Session Release SM Context success rate
SMF Call Control	SM Context-N1N2 Message Transfer rates
SMF Call Control	SM Context-N1N2 Message Transfer result %
SMF Call Control	SM Context-Current N1N2 Message Transfer success rate
SMF Call Control	SM Context-NAMF Assign EBI Message rates
SMF Call Control	SM Context-NAMF Assign EBI Message result %
SMF Call Control	SM Context-Current NAMF Assign EBI Message success rate
SMF Call Control	SM Context-UE Initiated Modify QoS attempts rates
SMF Call Control	SM Context-UE Initiated Modify QoS attempts result %
SMF Call Control	SM Context-Current UE Initiated Modify QoS attempts success rate
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SMF Call Control	Policy Control-Create SmPolicy Control Service result %
SMF Call Control	Policy Control-Current Create SmPolicy Control Service success rate
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SMF Call Control	Policy Control-Update SmPolicy Control Service result %
SMF Call Control	Policy Control-Current Update SmPolicy Control Service success rate
SMF Call Control	Policy Control-Delete SmPolicy Control Service rates
SMF Call Control	Policy Control-Delete SmPolicy Control Service result %
SMF Call Control	Policy Control-Current Delete SmPolicy Control Service success rate
SMF Call Control	Policy Control-SmPolicy Control Service Notify rates
SMF Call Control	Policy Control-SmPolicy Control Service Notify result %
SMF Call Control	Policy Control-Current SmPolicy Control Service Notify success rate
SMF Call Control	Converged Charging-CHF Create Session rates
SMF Call Control	Converged Charging-CHF Create Session result %
SMF Call Control	Converged Charging-Current CHF Create Session success rate
SMF Call Control	Converged Charging-CHF Update Session rates
SMF Call Control	Converged Charging-CHF Update Session result %

Dashboard Name	Panel Name
SMF Call Control	Converged Charging-Current CHF Update Session success rate
SMF Call Control	Converged Charging-CHF Release Session rates
SMF Call Control	Converged Charging-CHF Release Session result %
SMF Call Control	Converged Charging-Current CHF Release Session success rate
SMF Call Control	Converged Charging-CHF One-Time Report rates
SMF Call Control	Converged Charging-CHF One-Time Report result %
SMF Call Control	Converged Charging-Current CHF One-Time Report success rate
SMF Call Control	UPF Sessions-Session Establishment rates
SMF Call Control	UPF Sessions-Session Establishment result %
SMF Call Control	UPF Sessions-Current Session Establishment success rate
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SMF Call Control	UPF Sessions-Session Modification result %
SMF Call Control	UPF Sessions-Current Session Modification success rate
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SMF Call Control	UPF Sessions-Session Deletion result %
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SMF Call Control	Unified Data Management-SMF-UDM Deregistration rates
SMF Call Control	Unified Data Management-SMF-UDM Deregistration result %
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SMF Call Control	NF Registration and Discovery-NRF Registration rates

Dashboard Name	Panel Name
SMF Call Control	NF Registration and Discovery-NRF Registration result %
SMF Call Control	NF Registration and Discovery-Current NRF Registration success rate
SMF Call Control	NF Registration and Discovery-NRF De-registration rates
SMF Call Control	NF Registration and Discovery-NRF De-registration result %
SMF Call Control	NF Registration and Discovery-Current NRF De-registration success rate
SMF Call Control	NF Registration and Discovery-NRF Update rates
SMF Call Control	NF Registration and Discovery-NRF Update result %
SMF Call Control	NF Registration and Discovery-Current NRF Update success rate
SMF Call Control	NF Registration and Discovery-NRF Discovery rates
SMF Call Control	NF Registration and Discovery-NRF Discovery result %
SMF Call Control	NF Registration and Discovery-Current NRF Discovery success rate
SMF Call Control	NF Registration and Discovery-NRF Subscription rates
SMF Call Control	NF Registration and Discovery-NRF Subscription result %
SMF Call Control	NF Registration and Discovery-Current NRF Subscription success rate
SMF Call Control	Individual Counter of PDU Session Procedures-Session Establishment counts
SMF Call Control	Individual Counter of PDU Session Procedures-UE Initiated Session Release counts
SMF Call Control	Individual Counter of PDU Session Procedures-UE Deregistration counts
SMF Call Control	GTP Messages
SMF Call Control	Handover Procedures
SMF Call Control Enterprise	SM Context-PDU Session Create SM Context rates
SMF Call Control Enterprise	SM Context-PDU Session Create SM Context result %
SMF Call Control Enterprise	SM Context-Current PDU Session Create SM Context success rate
SMF Call Control Enterprise	SM Context-PDU Session Update SM Context rates
SMF Call Control Enterprise	SM Context-PDU Session Update SM Context result %
SMF Call Control Enterprise	SM Context-Current PDU Session Update SM Context success rate
SMF Call Control Enterprise	SM Context-PDU Session Retrieve SM Context rates
SMF Call Control Enterprise	SM Context-PDU Session Retrieve SM Context result %
SMF Call Control Enterprise	SM Context-Current PDU Session Retrieve SM Context success rate

Dashboard Name	Panel Name
SMF Call Control Enterprise	SM Context-PDU Session Notify SM Context rates
SMF Call Control Enterprise	SM Context-PDU Session Notify SM Context result %
SMF Call Control Enterprise	SM Context-Current PDU Session Notify SM Context success rate
SMF Call Control Enterprise	SM Context-PDU Session Release SM Context rates
SMF Call Control Enterprise	SM Context-PDU Session Release SM Context result %
SMF Call Control Enterprise	SM Context-Current PDU Session Release SM Context success rate
SMF Call Control Enterprise	SM Context-N1N2 Message Transfer rates
SMF Call Control Enterprise	SM Context-N1N2 Message Transfer result %
SMF Call Control Enterprise	SM Context-Current N1N2 Message Transfer success rate
SMF Call Control Enterprise	SM Context-NAMF Assign EBI Message rates
SMF Call Control Enterprise	SM Context-NAMF Assign EBI Message result %
SMF Call Control Enterprise	SM Context-Current NAMF Assign EBI Message success rate
SMF Call Control Enterprise	SM Context-UE Initiated Modify QoS attempts rates
SMF Call Control Enterprise	SM Context-UE Initiated Modify QoS attempts result %
SMF Call Control Enterprise	SM Context-Current UE Initiated Modify QoS attempts success rate
SMF Call Control Enterprise	Policy Control-Create SmPolicy Control Service rates
SMF Call Control Enterprise	Policy Control-Create SmPolicy Control Service result %
SMF Call Control Enterprise	Policy Control-Current Create SmPolicy Control Service success rate
SMF Call Control Enterprise	Policy Control-Update SmPolicy Control Service rates
SMF Call Control Enterprise	Policy Control-Update SmPolicy Control Service result %
SMF Call Control Enterprise	Policy Control-Current Update SmPolicy Control Service success rate
SMF Call Control Enterprise	Policy Control-Delete SmPolicy Control Service rates
SMF Call Control Enterprise	Policy Control-Delete SmPolicy Control Service result %
SMF Call Control Enterprise	Policy Control-Current Delete SmPolicy Control Service success rate
SMF Call Control Enterprise	Policy Control-SmPolicy Control Service Notify rates
SMF Call Control Enterprise	Policy Control-SmPolicy Control Service Notify result %
SMF Call Control Enterprise	Policy Control-Current SmPolicy Control Service Notify success rate
SMF Call Control Enterprise	Converged Charging-CHF Create Session rates
SMF Call Control Enterprise	Converged Charging-CHF Create Session result %
SMF Call Control Enterprise	Converged Charging-Current CHF Create Session success rate

Dashboard Name	Panel Name
SMF Call Control Enterprise	Converged Charging-CHF Update Session rates
SMF Call Control Enterprise	Converged Charging-CHF Update Session result %
SMF Call Control Enterprise	Converged Charging-Current CHF Update Session success rate
SMF Call Control Enterprise	Converged Charging-CHF Release Session rates
SMF Call Control Enterprise	Converged Charging-CHF Release Session result %
SMF Call Control Enterprise	Converged Charging-Current CHF Release Session success rate
SMF Call Control Enterprise	Converged Charging-CHF One-Time Report rates
SMF Call Control Enterprise	Converged Charging-CHF One-Time Report result %
SMF Call Control Enterprise	Converged Charging-Current CHF One-Time Report success rate
SMF Call Control Enterprise	UPF Sessions-Session Establishment rates
SMF Call Control Enterprise	UPF Sessions-Session Establishment result %
SMF Call Control Enterprise	UPF Sessions-Current Session Establishment success rate
SMF Call Control Enterprise	UPF Sessions-Session Modification rates
SMF Call Control Enterprise	UPF Sessions-Session Modification result %
SMF Call Control Enterprise	UPF Sessions-Current Session Modification success rate
SMF Call Control Enterprise	UPF Sessions-Session Deletion rates
SMF Call Control Enterprise	UPF Sessions-Session Deletion result %
SMF Call Control Enterprise	UPF Sessions-Current Session Deletion success rate
SMF Call Control Enterprise	UPF Sessions-Session Report rates
SMF Call Control Enterprise	UPF Sessions-Session Report result %
SMF Call Control Enterprise	UPF Sessions-Current Session Report success rate
SMF Call Control Enterprise	Unified Data Management-Get SM Data rates
SMF Call Control Enterprise	Unified Data Management-Get SM Data result %
SMF Call Control Enterprise	Unified Data Management-Current Get SM Data success rate
SMF Call Control Enterprise	Unified Data Management-SMF-UDM Registration rates
SMF Call Control Enterprise	Unified Data Management-SMF-UDM Registration result %
SMF Call Control Enterprise	Unified Data Management-Current SMF-UDM Registration success rate
SMF Call Control Enterprise	Unified Data Management-SMF-UDM Deregistration rates
SMF Call Control Enterprise	Unified Data Management-SMF-UDM Deregistration result %

Dashboard Name	Panel Name
SMF Call Control Enterprise	Unified Data Management-Current SMF-UDM Deregistration success rate
SMF Call Control Enterprise	NF Registration and Discovery-NRF Registration rates
SMF Call Control Enterprise	NF Registration and Discovery-NRF Registration result %
SMF Call Control Enterprise	NF Registration and Discovery-Current NRF Registration success rate
SMF Call Control Enterprise	NF Registration and Discovery-NRF De-registration rates
SMF Call Control Enterprise	NF Registration and Discovery-NRF De-registration result %
SMF Call Control Enterprise	NF Registration and Discovery-Current NRF De-registration success rate
SMF Call Control Enterprise	NF Registration and Discovery-NRF Update rates
SMF Call Control Enterprise	NF Registration and Discovery-NRF Update result %
SMF Call Control Enterprise	NF Registration and Discovery-Current NRF Update success rate
SMF Call Control Enterprise	NF Registration and Discovery-NRF Discovery rates
SMF Call Control Enterprise	NF Registration and Discovery-NRF Discovery result %
SMF Call Control Enterprise	NF Registration and Discovery-Current NRF Discovery success rate
SMF Call Control Enterprise	NF Registration and Discovery-NRF Subscription rates
SMF Call Control Enterprise	NF Registration and Discovery-NRF Subscription result %
SMF Call Control Enterprise	NF Registration and Discovery-Current NRF Subscription success rate
SMF Call Control Enterprise	Individual Counter of PDU Session Procedures-Session Establishment counts
SMF Call Control Enterprise	Individual Counter of PDU Session Procedures-UE Initiated Session Release counts
SMF Call Control Enterprise	Individual Counter of PDU Session Procedures-UE Deregistration counts
SMF Call Control Enterprise	GTP Messages
SMF Call Control Enterprise	Handover Procedures
SMF Service Dashboard	Current Pdu Sessions
SMF Service Dashboard	Current Pdu Sessions Per dnn
SMF Service Dashboard	Current Pdu Sessions Per Rat Type
SMF Service Dashboard	Current Pdu Sessions Per UPF
SMF Service Dashboard	Simultaneously Attached Users
SMF Service Dashboard	Simultaneously Attached Active Users

Dashboard Name	Panel Name
SMF Service Dashboard	Current Pdu Sessions Using Default Policy
SMF Service Dashboard	Current number of Qos Flows
SMF Service Dashboard	Current Pdu Sessions Using Default Quota
SMF Service Dashboard	Current Bearers per fiveQi, arpPI in SMF+PGWC
SMF Service Dashboard	Current qos flows per fiveQi, arpPI in SMF
SMF Service Dashboard	Current Pdu Sessions With Charging Disabled
SMF Service Dashboard	Current Pdu Sessions Using Udm Allocated Static Ip Address
SMF Service Dashboard	Current number of bearers in SMF+PGWC
SMF Service Dashboard	Current Inbound Roaming Home Routed Sessions
SMF Service Dashboard	Current Outbound Roaming Sessions
SMF Service Dashboard	Current Pdu Sessions Per Mapped Dnn
SMF Service Dashboard	Current SGW Idle Pdu Sessions
SMF Service Dashboard	Current SGW Idle Subscribers
SMF Service Dashboard	Sgw Simultaneously Attached Active Users
SMF Service Dashboard	SMFCC Gauge Request Received
SMF Service Dashboard	SMFCC Gauge Requests Succeeded
SMF Service Enterprise	Current Pdu Sessions
SMF Service Enterprise	Current Pdu Sessions Per dnn
SMF Service Enterprise	Current Pdu Sessions Per Rat Type
SMF Service Enterprise	Current Pdu Sessions Per UPF
SMF Service Enterprise	Simultaneously Attached Users
SMF Service Enterprise	Simultaneously Attached Active Users
SMF Service Enterprise	Current Pdu Sessions Using Default Policy
SMF Service Enterprise	Current number of Qos Flows
SMF Service Enterprise	Current Pdu Sessions Using Default Quota
SMF Service Enterprise	Current Bearers per fiveQi, arpPI in SMF+PGWC
SMF Service Enterprise	Current qos flows per fiveQi, arpPI in SMF
SMF Service Enterprise	Current Pdu Sessions With Charging Disabled
SMF Service Enterprise	Current Pdu Sessions Using Udm Allocated Static Ip Address

Dashboard Name	Panel Name
SMF Service Enterprise	Current number of bearers in SMF+PGWC
SMF Service Enterprise	Current Inbound Roaming Home Routed Sessions
SMF Service Enterprise	Current Outbound Roaming Sessions
SMF Service Enterprise	Current Pdu Sessions Per Mapped Dnn
SMF Service Enterprise	Current SGW Idle Pdu Sessions
SMF Service Enterprise	Current SGW Idle Subscribers
SMF Service Enterprise	Sgw Simultaneously Attached Active Users
SMF Service Enterprise	SMFCC Gauge Request Received
SMF Service Enterprise	SMFCC Gauge Requests Succeeded
Session Management Overview	PDU Sessions and QoS Flows-Number of PDU sessions
Session Management Overview	PDU Sessions and QoS Flows-Current PDU sessions
Session Management Overview	PDU Sessions and QoS Flows-Number of QoS flows
Session Management Overview	PDU Sessions and QoS Flows-Current QoS flows
Session Management Overview	PDU Session Procedures-Session Establishment rates
Session Management Overview	PDU Session Procedures-Session Establishment result %
Session Management Overview	PDU Session Procedures-Current Session Establishment success rate
Session Management Overview	PDU Session Procedures-UE Initiated Session Release rates
Session Management Overview	PDU Session Procedures-UE Initiated Session Release result %
Session Management Overview	PDU Session Procedures-Current UE Initiated Session Release success rate
Session Management Overview	PDU Session Procedures-UE Deregistration rates
Session Management Overview	PDU Session Procedures-UE Deregistration result %

Dashboard Name	Panel Name
Session Management Overview	PDU Session Procedures-Current UE Deregistration success rate
Session Management Overview	Connection Management Procedures-Service Request rates
Session Management Overview	Connection Management Procedures-Service Request result %
Session Management Overview	Connection Management Procedures-Current Service Request success rate
Session Management Overview	Connection Management Procedures-AN UE Context Release rates
Session Management Overview	Connection Management Procedures-AN UE Context Release result %
Session Management Overview	Connection Management Procedures-Current AN UE Context Release success rate
Session Management Overview	Connection Management Procedures-Network Initiated Service Request rates
Session Management Overview	Connection Management Procedures-Network Initiated Service Request result %
Session Management Overview	Connection Management Procedures-Current Network Initiated Service Request success rate
Session Management Overview	Connection Management Procedures-Network Initiated UE Context Release rates
Session Management Overview	Connection Management Procedures-Network Initiated UE Context Release result %
Session Management Overview	Connection Management Procedures-Current Network Initiated UE Context Release success rate
Session Management Overview	Mobility Management Procedures-XN Handover rates
Session Management Overview	Mobility Management Procedures-XN Handover result %
Session Management Overview	Mobility Management Procedures-Current XN Handover success rate
Session Management Overview	Mobility Management Procedures-N2 Handover rates
Session Management Overview	Mobility Management Procedures-N2 Handover result %

Dashboard Name	Panel Name
Session Management Overview	Mobility Management Procedures-Current N2 Handover success rate
Session Management Overview Enterprise	PDU Sessions and QoS Flows-Number of PDU sessions
Session Management Overview Enterprise	PDU Sessions and QoS Flows-Current PDU sessions
Session Management Overview Enterprise	PDU Sessions and QoS Flows-Number of QoS flows
Session Management Overview Enterprise	PDU Sessions and QoS Flows-Current QoS flows
Session Management Overview Enterprise	PDU Session Procedures-Session Establishment rates
Session Management Overview Enterprise	PDU Session Procedures-Session Establishment result %
Session Management Overview Enterprise	PDU Session Procedures-Current Session Establishment success rate
Session Management Overview Enterprise	PDU Session Procedures-UE Initiated Session Release rates
Session Management Overview Enterprise	PDU Session Procedures-UE Initiated Session Release result %
Session Management Overview Enterprise	PDU Session Procedures-Current UE Initiated Session Release success rate
Session Management Overview Enterprise	PDU Session Procedures-UE Deregistration rates
Session Management Overview Enterprise	PDU Session Procedures-UE Deregistration result %
Session Management Overview Enterprise	PDU Session Procedures-Current UE Deregistration success rate
Session Management Overview Enterprise	Connection Management Procedures-Service Request rates
Session Management Overview Enterprise	Connection Management Procedures-Service Request result %
Session Management Overview Enterprise	Connection Management Procedures-Current Service Request success rate

Dashboard Name	Panel Name
Session Management Overview Enterprise	Connection Management Procedures-AN UE Context Release rates
Session Management Overview Enterprise	Connection Management Procedures-AN UE Context Release result %
Session Management Overview Enterprise	Connection Management Procedures-Current AN UE Context Release success rate
Session Management Overview Enterprise	Connection Management Procedures-Network Initiated Service Request rates
Session Management Overview Enterprise	Connection Management Procedures-Network Initiated Service Request result %
Session Management Overview Enterprise	Connection Management Procedures-Current Network Initiated Service Request success rate
Session Management Overview Enterprise	Connection Management Procedures-Network Initiated UE Context Release rates
Session Management Overview Enterprise	Connection Management Procedures-Network Initiated UE Context Release result %
Session Management Overview Enterprise	Connection Management Procedures-Current Network Initiated UE Context Release success rate
Session Management Overview Enterprise	Mobility Management Procedures-XN Handover rates
Session Management Overview Enterprise	Mobility Management Procedures-XN Handover result %
Session Management Overview Enterprise	Mobility Management Procedures-Current XN Handover success rate
Session Management Overview Enterprise	Mobility Management Procedures-N2 Handover rates
Session Management Overview Enterprise	Mobility Management Procedures-N2 Handover result %
Session Management Overview Enterprise	Mobility Management Procedures-Current N2 Handover success rate
UPF Sessions	Overview-Session Establishment rates
UPF Sessions	Overview-Session Establishment result %
UPF Sessions	Overview-Current Session Establishment success rate
UPF Sessions	Overview-Session Modification rates

Dashboard Name	Panel Name
UPF Sessions	Overview-Session Modification result %
UPF Sessions	Overview-Current Session Modification success rate
UPF Sessions	Overview-Session Deletion rates
UPF Sessions	Overview-Session Deletion result %
UPF Sessions	Overview-Current Session Deletion success rate
UPF Sessions	Overview-Session Report rates
UPF Sessions	Overview-Session Report result %
UPF Sessions	Overview-Current Session Report success rate
UPF Sessions	Rates per DNN and NSSAI-Session Establishment rates per DNN
UPF Sessions	Rates per DNN and NSSAI-Session Establishment rates per NSSAI
UPF Sessions	Rates per DNN and NSSAI-Session Modification rates per DNN
UPF Sessions	Rates per DNN and NSSAI-Session Modification rates per NSSAI
UPF Sessions	Rates per DNN and NSSAI-Session Deletion rates per DNN
UPF Sessions	Rates per DNN and NSSAI-Session Deletion rates per NSSAI
UPF Sessions	Rates per DNN and NSSAI-Session Report rates per DNN
UPF Sessions	Rates per DNN and NSSAI-Session Report rates per NSSAI
UPF Sessions	Errors-Session Establishment error rates by cause
UPF Sessions	Errors-Session Modification error rates by cause
UPF Sessions	Errors-Session Deletion error rates by cause
UPF Sessions	Errors-Session Report error rates by cause
UPF Sessions	DBCLIENT_HTTP-DB Client State Store HTTP transactions Rate
Unified Data Management	Overview-Get SM Data rates
Unified Data Management	Overview-Get SM Data result %
Unified Data Management	Overview-Current Get SM Data success rate
Unified Data Management	Overview-SMF-UDM Registration rates
Unified Data Management	Overview-SMF-UDM Registration result %
Unified Data Management	Overview-Current SMF-UDM Registration success rate
Unified Data Management	Overview-SMF-UDM Deregistration rates
Unified Data Management	Overview-SMF-UDM Deregistration result %

Dashboard Name	Panel Name
Unified Data Management	Overview-Current SMF-UDM Deregistration success rate
Unified Data Management	Rates per DNN and SNSSAI-SDM_GET_SM_DATA rates per DNN
Unified Data Management	Rates per DNN and SNSSAI-SDM_GET_SM_DATA rates per SNSSAI
Unified Data Management	Rates per DNN and SNSSAI-UDM_REGISTRATION rates per DNN
Unified Data Management	Rates per DNN and SNSSAI-UDM_REGISTRATION rates per SNSSAI
Unified Data Management	Rates per DNN and SNSSAI-UDM_DEREGISTRATION rates per DNN
Unified Data Management	Rates per DNN and SNSSAI-UDM_DEREGISTRATION rates per SNSSAI
Unified Data Management	Errors-SDM_GET_SM_DATA error rates by cause
Unified Data Management	Errors-UDM_REGISTRATION error rates by cause
Unified Data Management	Errors-UDM_DEREGISTRATION error rates by cause

Converged Charging

Description

Statistics for SMF operations on the Converged Charging (CHF) interface

Overview Panels

CHF Create Session rates

Description

Rate of CHF Create Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_create_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_create_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Create Session result %

Description

CHF Create Session result %

Expression**Success**

```
((sum(rate(smfcc_nchf_create_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_create_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Create Session success rate

Description

Current CHF Create Session success rate

Expression

```
(sum(rate(smfcc_nchf_create_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Update Session rates

Description

Rate of CHF Update Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_update_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_update_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Update Session result %

Description

CHF Update Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_update_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_update_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Update Session success rate

Description

Current CHF Update Session success rate

Expression

```
(sum(rate(smfcc_nchf_update_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Release Session rates

Description

Rate of CHF Release Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_release_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_release_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Release Session result %

Description

CHF Release Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_release_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_release_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Release Session success rate

Description

Current CHF Release Session success rate

Expression

```
(sum(rate(smfcc_nchf_release_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF One-Time Report rates

Description

Rate of CHF One-Time Report requests

Expression

Attempts

```
sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_one_time_report_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_one_time_report_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF One-Time Report result %

Description

CHF One-Time Report result %

Expression

Success

```
((sum(rate(smfcc_nchf_one_time_report_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_one_time_report_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF One-Time Report success rate

Description

Current CHF One-Time Report success rate

Expression

```
(sum(rate(smfcc_nchf_one_time_report_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Rates per DNN and SNSSAI Panels

CHF Create Session rates per DNN

Description

Rate of CHF Create Session requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_nchf_create_session_attempts{}[$rate_interval]))
```

CHF Create Session rates per SNSSAI

Description

Rate of CHF Create Session requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_nchf_create_session_attempts{}[$rate_interval]))
```

CHF Update Session rates per DNN

Description

Rate of CHF Update Session requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_nchf_update_session_attempts{}[$rate_interval]))
```

CHF Update Session rates per SNSSAI

Description

Rate of CHF Update Session requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_nchf_update_session_attempts{}[$rate_interval]))
```

CHF Release Session rates per DNN

Description

Rate of CHF Release Session requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_nchf_release_session_attempts{}[$rate_interval]))
```

CHF Release Session rates per SNSSAI

Description

Rate of CHF Release Session requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_nchf_release_session_attempts{}[$rate_interval]))
```

Errors Panels

CHF Create Session error rates by cause

Description

Rate of CHF Create Session errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_nchf_create_session_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

CHF Update Session error rates by cause

Description

Rate of CHF Update Session errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_nchf_update_session_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

CHF Release Session error rates by cause

Description

Rate of CHF Release Session errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_nchf_release_session_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

GTP Messages

Description

Statistics for GTP Messages

Panels

GTP Messages

GTP Messages Enterprise

Description

Statistics for GTP Messages

Panels

GTP Messages

HTTP Client and Server Stats Dashboard

Description

Dashboard for HTTP client and server stats over the last 15 minutes.

Panels

HTTP server incoming request rate over last 15 minutes

Description

Rate of incoming http requests at server from different clients over last 15 minutes.

Expression

```
rate(http_server_request_received[15m])
```

HTTP server outgoing responses rate over last 15 minutes

Description

Rate of outgoing http responses from server to different clients over last 15 minutes.

Expression

```
rate(http_server_response_sent[15m])
```

HTTP Client outgoing request rate over last 15 minutes

Description

Rate of outgoing HTTP client requests to various servers over last 15 minutes.

Expression

```
rate(http_async_client_request_enqueue_success[15m])
```

HTTP server transaction success percentage over last 15 minutes

Description

Percentage of HTTP server incoming requests received for which the application responds with 2xx status code over last 15 minutes.

Expression

```
sum without (http_status_code)
((sum_over_time(http_server_response_sent{http_status_code=~"2.."}[15m]) /
ignoring(http_status_code) group_left
sum_over_time(http_server_request_received[15m]))) * 100)
```

HTTP client transaction success percentage over last 15 minutes

Description

Percentage of HTTP client outgoing requests sent for which the server responds with 2xx status code over last 15 minutes.

Expression

```
sum without (http_status_code)
((sum_over_time(http_async_client_response_received{http_status_code=~"2.."}[15m])
/ ignoring(http_status_code) group_left
sum_over_time(http_async_client_request_enqueue_success[15m]))) * 100)
```

HTTP Client incoming response rate over last 15 minutes

Description

Rate of HTTP client incoming response messages from various servers over the last 15 minutes.

Expression

```
rate(http_async_client_response_received[15m])
```

HTTP punting dashboard

Description

Dashboard for HTTP punting metrics stats over the last 15 minutes

Panels

HTTP punting outgoing request rate over last 15 min

Description

Rate of outgoing punting requests from applications over last 15 min

Expression

```
rate(punting_http_requests_sent[15m])
```

HTTP punting incoming response rate over last 15 min

Description

Rate of incoming punting responses for punted requests over last 15 min

Expression

```
rate(punting_http_responses_received[15m])
```

Punting response latency over last 15 min

Description

95th percentile point of punting response latency over last 15 min

Expression

```
histogram_quantile(0.95,  
sum(rate(punting_http_request_duration_seconds_bucket[15m])) by (le,  
canaryVersion, fed_unique_name, job, pod))
```

HTTP punting response forward rate over last 15 min

Description

Rate of forwarding punting responses to client over last 15 min

Expression

```
rate(punting_http_responses_forwarded[15m])
```

Handover Procedures

Description

Statistics for different Handover Procedures

Panels

Handover Procedures

Policy Control

Description

Statistics for SMF operations on the Policy Control interface

Overview Panels

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_success{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures{})[$rate_interval]))
```

Create SmPolicy Control Service result %

Description

Create SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_success{}
```



```
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{
```



```
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures{}
```



```
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{
```



```
[$rate_interval]))))*100
```

Current Create SmPolicy Control Service success rate

Description

Current Create SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_create_service_success{}
```



```
OR on() vector(0)) /
```



```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{}
```



```
[$rate_interval]))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_success{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures{}[$rate_interval]))
```

Update SmPolicy Control Service result %

Description

Update SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_success{}[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures{}[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{}[$rate_interval]))))*100
```

Current Update SmPolicy Control Service success rate

Description

Current Update SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_service_success{}[$rate_interval]))  
OR on() vector(0)) /  
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{}[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures{}[$rate_interval]))
```

Delete SmPolicy Control Service result %

Description

Delete SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{  
}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{  
[$rate_interval]})))*100
```

Current Delete SmPolicy Control Service success rate

Description

Current Delete SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success{[$rate_interval]})  
OR on() vector(0)) /  
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{[$rate_interval]}))
```

SmPolicy Control Service Notify rates

Description

Rate of SmPolicy Control Service Notify requests

Expression**Attempts**

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success{[$rate_interval]}))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures{[$rate_interval]}))
```

SmPolicy Control Service Notify result %

Description

SmPolicy Control Service Notify result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{  
[$rate_interval]})))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{  
[$rate_interval]})))*100
```

Current SmPolicy Control Service Notify success rate

Description

Current SmPolicy Control Service Notify success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success{[$rate_interval]}) OR  
on() vector(0)) / sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{  
[$rate_interval]}))
```

Rates per DNN and SNSSAI Panels

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{[$rate_interval]}))
```

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

```
sum by (snssai) (rate(smfcc_nsmf_smpolicy_control_create_service_attempts{}  
[$rate_interval]))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{}[$rate_interval]))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

```
sum by (snssai) (rate(smfcc_nsmf_smpolicy_control_update_service_attempts{}  
[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{}[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

```
sum by (snssai) (rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{ }
[$rate_interval]))
```

Policy Control Internals Panels

PCC Rule Install rates

Description

Rate of PCC Rule Install requests

Expression

Attempts

```
sum(rate(smfcc_pcc_rule_add_attempts{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pcc_rule_add_successes{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pcc_rule_add_failures{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Install result %

Description

PCC Rule Install result %

Expression

Success

```
((sum(rate(smfcc_pcc_rule_add_successes{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}  
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_add_attempts{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pcc_rule_add_failures{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}  
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_add_attempts{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule Install success rate

Description

Current PCC Rule Install success rate

Expression

```
(sum(rate(smfcc_pcc_rule_add_successes{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /  
sum(rate(smfcc_pcc_rule_add_attempts{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Modify rates

Description

Rate of PCC Rule Modify requests

Expression

Attempts

```
sum(rate(smfcc_pcc_rule_modify_attempts{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pcc_rule_modify_successes{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pcc_rule_modify_failures{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Modify result %

Description

PCC Rule Modify result %

Expression**Success**

```
((sum(rate(smfcc_pcc_rule_modify_successes{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}  
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_modify_attempts{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pcc_rule_modify_failures{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}  
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_modify_attempts{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule Modify success rate

Description

Current PCC Rule Modify success rate

Expression

```
(sum(rate(smfcc_pcc_rule_modify_successes{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /  
sum(rate(smfcc_pcc_rule_modify_attempts{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule PCF initiated Update rates

Description

Rate of PCC Rule PCF initiated Update requests

Expression

Attempts

```
sum(rate(smfcc_process_pcf_update_policy_rules_attempts{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_process_pcf_update_policy_rules_successes{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_process_pcf_update_policy_rules_failures{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule PCF initiated Update result %

Description

PCC Rule PCF initiated Update result %

Expression

Success

```
((sum(rate(smfcc_process_pcf_update_policy_rules_successes{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

```
[$rate_interval]))/(sum(rate(smfcc_process_pcf_update_policy_rules_attempts{dnn=~
"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_process_pcf_update_policy_rules_failures{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}
[$rate_interval]))/(sum(rate(smfcc_process_pcf_update_policy_rules_attempts{dnn=~
"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule PCF initiated Update success rate

Description

Current PCC Rule PCF initiated Update success rate

Expression

```
(sum(rate(smfcc_process_pcf_update_policy_rules_successes{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_process_pcf_update_policy_rules_attempts{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Remove rates

Description

Rate of PCC Rule Remove requests

Expression

Attempts

```
sum(rate(smfcc_pcc_rule_delete_attempts{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pcc_rule_delete_successes{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pcc_rule_delete_failures{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Remove result %

Description

PCC Rule Remove result %

Expression**Success**

```
((sum(rate(smfcc_pcc_rule_delete_successes{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}
[$rate_interval]))/(sum(rate(smfcc_pcc_rule_delete_attempts{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pcc_rule_delete_failures{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}
[$rate_interval]))/(sum(rate(smfcc_pcc_rule_delete_attempts{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule Remove success rate

Description

Current PCC Rule Remove success rate

Expression

```
(sum(rate(smfcc_pcc_rule_delete_successes{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_pcc_rule_delete_attempts{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Rates per PCC Rule Panels

PCC Rule Install rates per PCC RULE

Description

Rate of PCC Rule Install requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_pcc_rule_add_attempts{}[$rate_interval]))
```

PCC Rule Modify rates per PCC RULE

Description

Rate of PCC Rule Modify requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_pcc_rule_modify_attempts{}[$rate_interval]))
```

PCC Rule PCF initiated Update rates per PCC RULE

Description

Rate of PCC Rule PCF initiated Update requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_process_pcf_update_policy_rules_attempts{}  
[$rate_interval]))
```

PCC Rule Remove rates per PCC RULE

Description

Rate of PCC Rule Remove requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_pcc_rule_delete_attempts{}[$rate_interval]))
```

Errors Panels

Create SmPolicy Control Service error rates by cause

Description

Rate of Create SmPolicy Control Service errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_nsmf_smpolicy_control_create_service_failures{}[$rate_interval]))
```

Update SmPolicy Control Service error rates by cause

Description

Rate of Update SmPolicy Control Service errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_nsmf_smpolicy_control_update_service_failures{}[$rate_interval]))
```

Delete SmPolicy Control Service error rates by cause

Description

Rate of Delete SmPolicy Control Service errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_nsmf_smpolicy_control_delete_service_failures{  
[$rate_interval]})
```

PCC Rule Install error rates by cause

Description

Rate of PCC Rule Install errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_pcc_rule_add_failures{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Modify error rates by cause

Description

Rate of PCC Rule Modify errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_pcc_rule_modify_failures{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Remove error rates by cause

Description

Rate of PCC Rule Remove errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_pcc_rule_delete_failures{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```


Policy Control Enterprise

Description

Statistics for SMF operations on the Policy Control interface

Overview Panels

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_total{}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_success_total{}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures_total{}
[$rate_interval]))
```

Create SmPolicy Control Service result %

Description

Create SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_success_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_
total{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_
total{}[$rate_interval]))))*100
```

Current Create SmPolicy Control Service success rate

Description

Current Create SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_create_service_success_total{}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_total{}
[$rate_interval]))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_total{}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_success_total{}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures_total{}  
[$rate_interval]))
```

Update SmPolicy Control Service result %

Description

Update SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_success_total{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_  
total{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures_total{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_  
total{}[$rate_interval]))))*100
```

Current Update SmPolicy Control Service success rate

Description

Current Update SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_service_success_total{}  
[$rate_interval])) OR on() vector(0)) /  
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_total{}  
[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_total{}  
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success_total{}  
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures_total{}  
[$rate_interval]))
```

Delete SmPolicy Control Service result %

Description

Delete SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success_total{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_  
total{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures_total{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_  
total{}[$rate_interval]))))*100
```

Current Delete SmPolicy Control Service success rate

Description

Current Delete SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success_total{}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_total{}
[$rate_interval]))
```

SmPolicy Control Service Notify rates

Description

Rate of SmPolicy Control Service Notify requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_total{}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success_total{}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures_total{}
[$rate_interval]))
```

SmPolicy Control Service Notify result %

Description

SmPolicy Control Service Notify result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_t
otal{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_t
otal{}[$rate_interval]))))*100
```

Current SmPolicy Control Service Notify success rate

Description

Current SmPolicy Control Service Notify success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success_total{}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_total{}
[$rate_interval]))
```

Rates per DNN and SNSSAI Panels

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_total{}
[$rate_interval]))
```

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

```
sum by (snssai) (rate(smfcc_nsmf_smpolicy_control_create_service_attempts_total{}  
[$rate_interval]))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_total{}  
[$rate_interval]))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

```
sum by (snssai) (rate(smfcc_nsmf_smpolicy_control_update_service_attempts_total{}  
[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_total{}  
[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

```
sum by (snssai) (rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_total{  
  [$rate_interval]})
```

Policy Control Internals Panels

PCC Rule Install rates

Description

Rate of PCC Rule Install requests

Expression

Attempts

```
sum(rate(smfcc_pcc_rule_add_attempts_total{dnn=~"$dnn", snssai=~"$snssai",  
  pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pcc_rule_add_successes_total{dnn=~"$dnn", snssai=~"$snssai",  
  pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pcc_rule_add_failures_total{dnn=~"$dnn", snssai=~"$snssai",  
  pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Install result %

Description

PCC Rule Install result %

Expression

Success

```
((sum(rate(smfcc_pcc_rule_add_successes_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_add_attempts_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pcc_rule_add_failures_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_add_attempts_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule Install success rate

Description

Current PCC Rule Install success rate

Expression

```
(sum(rate(smfcc_pcc_rule_add_successes_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_pcc_rule_add_attempts_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Modify rates

Description

Rate of PCC Rule Modify requests

Expression

Attempts

```
sum(rate(smfcc_pcc_rule_modify_attempts_total{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pcc_rule_modify_successes_total{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pcc_rule_modify_failures_total{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Modify result %

Description

PCC Rule Modify result %

Expression

Success

```
((sum(rate(smfcc_pcc_rule_modify_successes_total{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}  
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_modify_attempts_total{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pcc_rule_modify_failures_total{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}  
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_modify_attempts_total{dnn=~"$dnn",  
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule Modify success rate

Description

Current PCC Rule Modify success rate

Expression

```
(sum(rate(smfcc_pcc_rule_modify_successes_total{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_pcc_rule_modify_attempts_total{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule PCF initiated Update rates

Description

Rate of PCC Rule PCF initiated Update requests

Expression

Attempts

```
sum(rate(smfcc_process_pcf_update_policy_rules_attempts_total{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_process_pcf_update_policy_rules_successes_total{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_process_pcf_update_policy_rules_failures_total{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule PCF initiated Update result %

Description

PCC Rule PCF initiated Update result %

Expression

Success

```
((sum(rate(smfcc_process_pcf_update_policy_rules_successes_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}
[$rate_interval])))/(sum(rate(smfcc_process_pcf_update_policy_rules_attempts_total
{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_process_pcf_update_policy_rules_failures_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}
[$rate_interval])))/(sum(rate(smfcc_process_pcf_update_policy_rules_attempts_total
{dnn=~"$dnn", snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule PCF initiated Update success rate

Description

Current PCC Rule PCF initiated Update success rate

Expression

```
(sum(rate(smfcc_process_pcf_update_policy_rules_successes_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_process_pcf_update_policy_rules_attempts_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Remove rates

Description

Rate of PCC Rule Remove requests

Expression

Attempts

```
sum(rate(smfcc_pcc_rule_delete_attempts_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pcc_rule_delete_successes_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pcc_rule_delete_failures_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Remove result %

Description

PCC Rule Remove result %

Expression

Success

```
((sum(rate(smfcc_pcc_rule_delete_successes_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_delete_attempts_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pcc_rule_delete_failures_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}
[$rate_interval])))/(sum(rate(smfcc_pcc_rule_delete_attempts_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))))*100
```

Current PCC Rule Remove success rate

Description

Current PCC Rule Remove success rate

Expression

```
(sum(rate(smfcc_pcc_rule_delete_successes_total{dnn=~"$dnn", snssai=~"$snssai",
pccrule=~"$pccrule"}[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(smfcc_pcc_rule_delete_attempts_total{dnn=~"$dnn", snssai=~"$snssai",  
pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Rates per PCC Rule Panels

PCC Rule Install rates per PCC RULE

Description

Rate of PCC Rule Install requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_pcc_rule_add_attempts_total{}[$rate_interval]))
```

PCC Rule Modify rates per PCC RULE

Description

Rate of PCC Rule Modify requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_pcc_rule_modify_attempts_total{}[$rate_interval]))
```

PCC Rule PCF initiated Update rates per PCC RULE

Description

Rate of PCC Rule PCF initiated Update requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_process_pcf_update_policy_rules_attempts_total{  
[$rate_interval]))
```

PCC Rule Remove rates per PCC RULE

Description

Rate of PCC Rule Remove requests per PCC Rule

Expression

{{pccrule}}

```
sum by (pccrule) (rate(smfcc_pcc_rule_delete_attempts_total{}[$rate_interval]))
```

Errors Panels

Create SmPolicy Control Service error rates by cause

Description

Rate of Create SmPolicy Control Service errors by cause

Expression

{{causecode}}

```
sum by (causecode)
(rate(smfcc_nsmf_smpolicy_control_create_service_failures_total{}
[$rate_interval]))
```

Update SmPolicy Control Service error rates by cause

Description

Rate of Update SmPolicy Control Service errors by cause

Expression

{{causecode}}

```
sum by (causecode)
(rate(smfcc_nsmf_smpolicy_control_update_service_failures_total{}
[$rate_interval]))
```

Delete SmPolicy Control Service error rates by cause

Description

Rate of Delete SmPolicy Control Service errors by cause

Expression

{{causecode}}

```
sum by (causecode)
(rate(smfcc_nsmf_smpolicy_control_delete_service_failures_total{
[$rate_interval]))
```

PCC Rule Install error rates by cause

Description

Rate of PCC Rule Install errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_pcc_rule_add_failures_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Modify error rates by cause

Description

Rate of PCC Rule Modify errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_pcc_rule_modify_failures_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

PCC Rule Remove error rates by cause

Description

Rate of PCC Rule Remove errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_pcc_rule_delete_failures_total{dnn=~"$dnn",
snssai=~"$snssai", pccrule=~"$pccrule"}[$rate_interval]))
```

SM Context

Description

Statistics for SM Context operations on the interface to AMF

Overview Panels

PDU Session Create SM Context rates

Description

Rate of PDU Session Create SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Create SM Context result %

Description

PDU Session Create SM Context result %

Expression**Success**

```
((sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_create_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Create SM Context success rate

Description

Current PDU Session Create SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context rates

Description

Rate of PDU Session Update SM Context requests

Expression**Attempts**

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context result %

Description

PDU Session Update SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_update_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Update SM Context success rate

Description

Current PDU Session Update SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Retrieve SM Context rates

Description

Rate of PDU Session Retrieve SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Retrieve SM Context result %

Description

PDU Session Retrieve SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{
dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{
dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Retrieve SM Context success rate

Description

Current PDU Session Retrieve SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context rates

Description

Rate of PDU Session Notify SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context result %

Description

PDU Session Notify SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_atte
mpts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_atte
mpts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Notify SM Context success rate

Description

Current PDU Session Notify SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context rates

Description

Rate of PDU Session Release SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context result %

Description

PDU Session Release SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{d
nn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_release_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{d
nn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Release SM Context success rate

Description

Current PDU Session Release SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer rates

Description

Rate of N1N2 Message Transfer requests

Expression**Attempts**

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures


```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer result %

Description

N1N2 Message Transfer result %

Expression

Success

```
((sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~
"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_namf_comm_n1n2_message_transfer_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~
"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current N1N2 Message Transfer success rate

Description

Current N1N2 Message Transfer success rate

Expression

```
(sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message rates

Description

Rate of NAMF Assign EBI Message requests

Expression

Attempts

```
sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_namf_comm_assign_ebi_message_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_namf_comm_assign_ebi_message_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message result %

Description

NAMF Assign EBI Message result %

Expression

Success

```
((sum(rate(smfcc_namf_comm_assign_ebi_message_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$d
nn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_namf_comm_assign_ebi_message_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$d
nn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current NAMF Assign EBI Message success rate

Description

Current NAMF Assign EBI Message success rate

Expression

```
(sum(rate(smfcc_namf_comm_assign_ebi_message_successes{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

UE Initiated Modify QoS attempts rates

Description

Rate of UE Initiated Modify QoS attempts requests

Expression

Attempts

```
sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_ue_initiated_modify_qos_failures{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

UE Initiated Modify QoS attempts result %

Description

UE Initiated Modify QoS attempts result %

Expression

Success

```
((sum(rate(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_initiated_modify_qos_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current UE Initiated Modify QoS attempts success rate

Description

Current UE Initiated Modify QoS attempts success rate

Expression

```
(sum(rate(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Rates per DNN and SNSSAI Panels

PDU Session Create SM Context rates per DNN

Description

Rate of PDU Session Create SM Context requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{}  
[$rate_interval]))
```

PDU Session Create SM Context rates per SNSSAI

Description

Rate of PDU Session Create SM Context requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{}  
[$rate_interval]))
```

PDU Session Update SM Context rates per DNN

Description

Rate of PDU Session Update SM Context requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{}  
[$rate_interval]))
```

PDU Session Update SM Context rates per SNSSAI

Description

Rate of PDU Session Update SM Context requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{}  
[$rate_interval]))
```

PDU Session Release SM Context rates per DNN

Description

Rate of PDU Session Release SM Context requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{  
[$rate_interval]))
```

PDU Session Release SM Context rates per SNSSAI

Description

Rate of PDU Session Release SM Context requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{  
[$rate_interval]))
```

N1N2 Message Transfer rates per DNN

Description

Rate of N1N2 Message Transfer requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_namf_comm_n1n2_message_transfer_attempts{  
[$rate_interval]))
```

N1N2 Message Transfer rates per SNSSAI

Description

Rate of N1N2 Message Transfer requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_namf_comm_n1n2_message_transfer_attempts{  
  [$rate_interval]})
```

SM Context Internals Panels

Internal Analyzer Classifier SM Context rates

Description

Rate of Internal Analyzer Classifier SM Context requests

Expression

Attempts

```
sum(rate(analyzer_instance_query_attempt{][$rate_interval]))
```

Successes

```
sum(rate(analyzer_instance_query_success{][$rate_interval]))
```

Failures

```
sum(rate(analyzer_instance_query_failure{][$rate_interval]))
```

Internal Analyzer Classifier SM Context result %

Description

Internal Analyzer Classifier SM Context result %

Expression

Success

```
((sum(rate(analyzer_instance_query_success{
[$rate_interval]})))/(sum(rate(analyzer_instance_query_attempt{
[$rate_interval]})))*100
```

Failure

```
((sum(rate(analyzer_instance_query_failure{
[$rate_interval]})))/(sum(rate(analyzer_instance_query_attempt{
[$rate_interval]})))*100
```

Current Internal Analyzer Classifier SM Context success rate

Description

Current Internal Analyzer Classifier SM Context success rate

Expression

```
(sum(rate(analyzer_instance_query_success{[$rate_interval]}) OR on() vector(0)) /
sum(rate(analyzer_instance_query_attempt{[$rate_interval]}))
```

DB Client State Store HTTP transactions Rate

Description

DB Client HTTP Transaction Rate by result

Expression

{{http_status_code}}

```
sum by (http_status_code) (rate(dbclient_http_requests{namespace="fed-smf",
db_type="mongo_RAM", db="smfccDB"}[$rate_interval]))
```

Errors Panels

PDU Session Create SM Context error rates by cause

Description

Rate of PDU Session Create SM Context errors by cause

Expression**{{causecode}}**

```
sum by (causecode)
(rate(smfcc_nsmf_pdu_session_create_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context error rates by cause

Description

Rate of PDU Session Update SM Context errors by cause

Expression**{{causecode}}**

```
sum by (causecode)
(rate(smfcc_nsmf_pdu_session_update_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Retrieve SM Context error rates by cause

Description

Rate of PDU Session Retrieve SM Context errors by cause

Expression**{{causecode}}**

```
sum by (causecode)
(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context error rates by cause

Description

Rate of PDU Session Notify SM Context errors by cause

Expression

{{causecode}}

```
sum by (causecode)
(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context error rates by cause

Description

Rate of PDU Session Release SM Context errors by cause

Expression

{{causecode}}

```
sum by (causecode)
(rate(smfcc_nsmf_pdu_session_release_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer error rates by cause

Description

Rate of N1N2 Message Transfer errors by cause

Expression

{{causecode}}

```
sum by (causecode)
(rate(smfcc_namf_comm_n1n2_message_transfer_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message error rates by cause

Description

Rate of NAMF Assign EBI Message errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_namf_comm_assign_ebi_message_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

UE Initiated Modify QoS attempts error rates by cause

Description

Rate of UE Initiated Modify QoS attempts errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_ue_initiated_modify_qos_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

SMF Call Control

Description

Summary information for operations on all SMF interfaces

SM Context Panels

PDU Session Create SM Context rates

Description

Rate of PDU Session Create SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Create SM Context result %

Description

PDU Session Create SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_create_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Create SM Context success rate

Description

Current PDU Session Create SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context rates

Description

Rate of PDU Session Update SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context result %

Description

PDU Session Update SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_update_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Update SM Context success rate

Description

Current PDU Session Update SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Retrieve SM Context rates

Description

Rate of PDU Session Retrieve SM Context requests

Expression**Attempts**

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Retrieve SM Context result %

Description

PDU Session Retrieve SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{
dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{
dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Retrieve SM Context success rate

Description

Current PDU Session Retrieve SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context rates

Description

Rate of PDU Session Notify SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context result %

Description

PDU Session Notify SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_atte  
mpts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_atte  
mpts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```


Current PDU Session Notify SM Context success rate

Description

Current PDU Session Notify SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context rates

Description

Rate of PDU Session Release SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context result %

Description

PDU Session Release SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{d
nn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_release_sm_context_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{d
nn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Release SM Context success rate

Description

Current PDU Session Release SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer rates

Description

Rate of N1N2 Message Transfer requests

Expression

Attempts

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_failures{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer result %

Description

N1N2 Message Transfer result %

Expression

Success

```
((sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~  
"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_namf_comm_n1n2_message_transfer_failures{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~  
"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current N1N2 Message Transfer success rate

Description

Current N1N2 Message Transfer success rate

Expression

```
(sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message rates

Description

Rate of NAMF Assign EBI Message requests

Expression

Attempts

```
sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_namf_comm_assign_ebi_message_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_namf_comm_assign_ebi_message_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message result %

Description

NAMF Assign EBI Message result %

Expression

Success

```
((sum(rate(smfcc_namf_comm_assign_ebi_message_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_namf_comm_assign_ebi_message_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current NAMF Assign EBI Message success rate

Description

Current NAMF Assign EBI Message success rate

Expression

```
(sum(rate(smfcc_namf_comm_assign_ebi_message_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_namf_comm_assign_ebi_message_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

UE Initiated Modify QoS attempts rates

Description

Rate of UE Initiated Modify QoS attempts requests

Expression

Attempts

```
sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_ue_initiated_modify_qos_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

UE Initiated Modify QoS attempts result %

Description

UE Initiated Modify QoS attempts result %

Expression

Success

```
((sum(rate(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_initiated_modify_qos_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current UE Initiated Modify QoS attempts success rate

Description

Current UE Initiated Modify QoS attempts success rate

Expression

```
(sum(rate(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Policy Control Panels

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_success{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures{}[$rate_interval]))
```

Create SmPolicy Control Service result %

Description

Create SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_success{}[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{  
[$rate_interval]})))*100
```

Current Create SmPolicy Control Service success rate

Description

Current Create SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_create_service_success{[$rate_interval]})  
OR on() vector(0)) /  
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{[$rate_interval]}))
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_success{[$rate_interval]}))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures{[$rate_interval]}))
```

Update SmPolicy Control Service result %

Description

Update SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_success{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{  
[$rate_interval]})))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{  
[$rate_interval]})))*100
```

Current Update SmPolicy Control Service success rate

Description

Current Update SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_service_success{[$rate_interval]})  
OR on() vector(0)) /  
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{[$rate_interval]})
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{[$rate_interval]})
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success{})[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures{})[$rate_interval]))
```

Delete SmPolicy Control Service result %

Description

Delete SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success{}
```



```
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{
```



```
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures{}
```



```
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{
```



```
[$rate_interval]))))*100
```

Current Delete SmPolicy Control Service success rate

Description

Current Delete SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success{})[$rate_interval]))
```



```
OR on() vector(0)) /
```



```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{})[$rate_interval]))
```

SmPolicy Control Service Notify rates

Description

Rate of SmPolicy Control Service Notify requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures{}[$rate_interval]))
```

SmPolicy Control Service Notify result %

Description

SmPolicy Control Service Notify result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{}  
[$rate_interval]))))*100
```

Current SmPolicy Control Service Notify success rate

680 / 1037

Description

Current SmPolicy Control Service Notify success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success{})[$rate_interval])) OR  
on() vector(0)) / sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts{}  
[$rate_interval]))
```

Converged Charging Panels

CHF Create Session rates

Description

Rate of CHF Create Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}  
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_create_session_success{dnn=~"$dnn", snssai=~"$snssai"}  
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_create_session_failures{dnn=~"$dnn", snssai=~"$snssai"}  
[$rate_interval]))
```

CHF Create Session result %

Description

CHF Create Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_create_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_create_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Create Session success rate

Description

Current CHF Create Session success rate

Expression

```
(sum(rate(smfcc_nchf_create_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_create_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Update Session rates

Description

Rate of CHF Update Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_update_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_update_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Update Session result %

Description

CHF Update Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_update_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_update_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Update Session success rate

Description

Current CHF Update Session success rate

Expression

```
(sum(rate(smfcc_nchf_update_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_update_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Release Session rates

Description

Rate of CHF Release Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_release_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_release_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Release Session result %

Description

CHF Release Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_release_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_release_session_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Release Session success rate

Description

Current CHF Release Session success rate

Expression

```
(sum(rate(smfcc_nchf_release_session_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_release_session_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF One-Time Report rates

Description

Rate of CHF One-Time Report requests

Expression

Attempts

```
sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_one_time_report_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_one_time_report_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```


CHF One-Time Report result %

Description

CHF One-Time Report result %

Expression

Success

```
((sum(rate(smfcc_nchf_one_time_report_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_one_time_report_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF One-Time Report success rate

Description

Current CHF One-Time Report success rate

Expression

```
(sum(rate(smfcc_nchf_one_time_report_success{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_one_time_report_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

UPF Sessions Panels

Session Establishment rates

Description

Rate of Session Establishment requests

Expression

Attempts

```
sum(rate(pfcg_session_establishment_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcg_session_establishment_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcg_session_establishment_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Establishment result %

Description

Session Establishment result %

Expression**Success**

```
((sum(rate(pfcg_session_establishment_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcg_session_establishment_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcg_session_establishment_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcg_session_establishment_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Establishment success rate

Description

Current Session Establishment success rate

Expression

```
(sum(rate(pfcpsessionestablishmentsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionestablishmentattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Modification rates

Description

Rate of Session Modification requests

Expression

Attempts

```
sum(rate(pfcpsessionmodificationattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionmodificationsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionmodificationfailure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Modification result %

Description

Session Modification result %

Expression

Success

```
((sum(rate(pfcpsessionmodification_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionmodification_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionmodification_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionmodification_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Modification success rate

Description

Current Session Modification success rate

Expression

```
(sum(rate(pfcpsessionmodification_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionmodification_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Deletion rates

Description

Rate of Session Deletion requests

Expression

Attempts

```
sum(rate(pfcpsessiondeletion_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessiondeletion_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsession_deletion_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Deletion result %

Description

Session Deletion result %

Expression**Success**

```
((sum(rate(pfcpsession_deletion_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsession_deletion_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsession_deletion_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsession_deletion_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Deletion success rate

Description

Current Session Deletion success rate

Expression

```
(sum(rate(pfcpsession_deletion_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsession_deletion_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Report rates

Description

Rate of Session Report requests

Expression

Attempts

```
sum(rate(pfcpsessionreportattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionreportsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionreportfailure{}[$rate_interval]))
```

Session Report result %

Description

Session Report result %

Expression

Success

```
((sum(rate(pfcpsessionreportsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionreportattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionreportfailure{}
[$rate_interval])))/(sum(rate(pfcpsessionreportattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Report success rate

Description

Current Session Report success rate

Expression

```
(sum(rate(pfcpsessionreportsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionreportattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Unified Data Management Panels

Get SM Data rates

Description

Rate of Get SM Data requests

Expression

Attempts

```
sum(rate(smfccsdmgetsmdataattempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfccsdmgetsmdatasuccesses{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfccsdmgetsmdatafailures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Get SM Data result %

Description

Get SM Data result %

Expression

Success

```
((sum(rate(smfcc_sdm_get_sm_data_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_sdm_get_sm_data_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current Get SM Data success rate

Description

Current Get SM Data success rate

Expression

```
(sum(rate(smfcc_sdm_get_sm_data_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Registration rates

Description

Rate of SMF-UDM Registration requests

Expression

Attempts

```
sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```


Successes

```
sum(rate(smfcc_udm_registration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_udm_registration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Registration result %

Description

SMF-UDM Registration result %

Expression**Success**

```
((sum(rate(smfcc_udm_registration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_udm_registration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current SMF-UDM Registration success rate

Description

Current SMF-UDM Registration success rate

Expression

```
(sum(rate(smfcc_udm_registration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Deregistration rates

Description

Rate of SMF-UDM Deregistration requests

Expression

Attempts

```
sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_udm_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_udm_deregistration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Deregistration result %

Description

SMF-UDM Deregistration result %

Expression

Success

```
((sum(rate(smfcc_udm_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_udm_deregistration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current SMF-UDM Deregistration success rate

Description

Current SMF-UDM Deregistration success rate

Expression

```
(sum(rate(smfcc_udm_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

NF Registration and Discovery Panels

NRF Registration rates

Description

Rate of NRF Registration requests

Expression**Attempts**

```
sum(rate(nrfa_nnrnf_nfm_registration_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrnf_nfm_registration_response{status=~"2.*"}[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnrnf_nfm_registration_response{status!~"2.*"}[$rate_interval]))
```

NRF Registration result %

Description

NRF Registration result %

Expression

Success

```
(sum(rate(nrfa_nnrif_nfm_registration_response{status=~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrif_nfm_registration_response[$rate_interval])
))*100
```

Failure

```
(sum(rate(nrfa_nnrif_nfm_registration_response{status!~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrif_nfm_registration_response[$rate_interval])
))*100
```

Current NRF Registration success rate

Description

Current NRF Registration success rate

Expression

```
(sum(rate(nrfa_nnrif_nfm_registration_response{status=~"2.*"}[$rate_interval])) OR
on() vector(0)) / sum(rate(nrfa_nnrif_nfm_registration_request[$rate_interval]))
```

NRF De-registration rates

Description

Rate of NRF De-registration requests

Expression

Attempts

```
sum(rate(nrfa_nnrfa_nfm_deregistration_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrfa_nfm_deregistration_response{status=~"2.*"}[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnrfa_nfm_deregistration_response{status!~"2.*"}[$rate_interval]))
```

NRF De-registration result %

Description

NRF De-registration result %

Expression

Success

```
(sum(rate(nrfa_nnrfa_nfm_deregistration_response{status=~"2.*"}  
[$rate_interval])))/(sum(rate(nrfa_nnrfa_nfm_deregistration_response[$rate_interval  
])))*100
```

Failure

```
(sum(rate(nrfa_nnrfa_nfm_deregistration_response{status!~"2.*"}  
[$rate_interval])))/(sum(rate(nrfa_nnrfa_nfm_deregistration_response[$rate_interval  
])))*100
```

Current NRF De-registration success rate

Description

Current NRF De-registration success rate

Expression

```
(sum(rate(nrfa_nnrnf_nfm_deregistration_response{status=~"2.*"}[$rate_interval]))  
OR on() vector(0)) /  
sum(rate(nrfa_nnrnf_nfm_deregistration_request[$rate_interval]))
```

NRF Update rates

Description

Rate of NRF Update requests

Expression

Attempts

```
sum(rate(nrfa_nnrnf_nfm_update_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrnf_nfm_update_response{status=~"2.*"}[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnrnf_nfm_update_response{status!~"2.*"}[$rate_interval]))
```

NRF Update result %

Description

NRF Update result %

Expression

Success

```
(sum(rate(nrfa_nnrnf_nfm_update_response{status=~"2.*"}  
[$rate_interval])))/(sum(rate(nrfa_nnrnf_nfm_update_response[$rate_interval])))*100
```

Failure

```
(sum(rate(nrfa_nnr_fnm_update_response{status!="2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnr_fnm_update_response[$rate_interval])))*100
```

Current NRF Update success rate

Description

Current NRF Update success rate

Expression

```
(sum(rate(nrfa_nnr_fnm_update_response{status=~"2.*"}[$rate_interval])) OR on()
vector(0)) / sum(rate(nrfa_nnr_fnm_update_request[$rate_interval]))
```

NRF Discovery rates

Description

Rate of NRF Discovery requests

Expression

Attempts

```
sum(rate(nrfa_nnr_fdisc_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnr_fdisc_response{status=~"2.*"}[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnr_fdisc_response{status!="2.*"}[$rate_interval]))
```

NRF Discovery result %

Description

NRF Discovery result %

Expression

Success

```
(sum(rate(nrfa_nnrfdisc_response{status=~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrfdisc_response[$rate_interval])))*100
```

Failure

```
(sum(rate(nrfa_nnrfdisc_response{status!~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrfdisc_response[$rate_interval])))*100
```

Current NRF Discovery success rate

Description

Current NRF Discovery success rate

Expression

```
(sum(rate(nrfa_nnrfdisc_response{status=~"2.*"}[$rate_interval])) OR on()
vector(0)) / sum(rate(nrfa_nnrfdisc_request[$rate_interval]))
```

NRF Subscription rates

Description

Rate of NRF Subscription requests

Expression

Attempts

```
sum(rate(nrfa_nnrfnfm_subscription_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrfnfm_subscription_response{status=~"2.*"}[$rate_interval]))
```


Failures

```
sum(rate(nrfa_nnrf_nfm_subscription_response{status!~"2.*"}[$rate_interval]))
```

NRF Subscription result %

Description

NRF Subscription result %

Expression

Success

```
(sum(rate(nrfa_nnrf_nfm_subscription_response{status=~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnrf_nfm_subscription_response[$rate_interval])
))*100
```

Failure

```
(sum(rate(nrfa_nnrf_nfm_subscription_response{status!~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnrf_nfm_subscription_response[$rate_interval])
))*100
```

Current NRF Subscription success rate

Description

Current NRF Subscription success rate

Expression

```
(sum(rate(nrfa_nnrf_nfm_subscription_response{status=~"2.*"}[$rate_interval])) OR
on() vector(0)) / sum(rate(nrfa_nnrf_nfm_subscription_request[$rate_interval]))
```

Individual Counter of PDU Session Procedures Panels

Session Establishment counts

Description

Number of Session Establishment requests

Expression

Attempts

```
sum(smfcc_pdu_session_establishment_attempts{}) -  
sum(smfcc_pdu_session_establishment_attempts{} offset $rate_interval)
```

Successes

```
sum(smfcc_pdu_session_establishment_successes{}) -  
sum(smfcc_pdu_session_establishment_successes{} offset $rate_interval)
```

Failures

```
sum(smfcc_pdu_session_establishment_failures{}) -  
sum(smfcc_pdu_session_establishment_failures{} offset $rate_interval)
```

UE Initiated Session Release counts

Description

Number of UE Initiated Session Release requests

Expression

Attempts

```
sum(smfcc_ue_initiated_pdu_session_release_attempts{}) -  
sum(smfcc_ue_initiated_pdu_session_release_attempts{} offset $rate_interval)
```

Successes

```
sum(smfcc_ue_initiated_pdu_session_release_successes{}) -  
sum(smfcc_ue_initiated_pdu_session_release_successes{} offset $rate_interval)
```

Failures

```
sum(smfcc_ue_initiated_pdu_session_release_failures{}) -  
sum(smfcc_ue_initiated_pdu_session_release_failures{} offset $rate_interval)
```

UE Deregistration counts

Description

Number of UE Deregistration requests

Expression

Attempts

```
sum(smfcc_ue_deregistration_attempts{}) - sum(smfcc_ue_deregistration_attempts{}  
offset $rate_interval)
```

Successes

```
sum(smfcc_ue_deregistration_successes{}) - sum(smfcc_ue_deregistration_successes{}  
offset $rate_interval)
```

Failures

```
sum(smfcc_ue_deregistration_failures{}) - sum(smfcc_ue_deregistration_failures{}  
offset $rate_interval)
```

Panels

GTP Messages

Handover Procedures

SMF Call Control Enterprise

Description

Summary information for operations on all SMF interfaces

SM Context Panels

PDU Session Create SM Context rates

Description

Rate of PDU Session Create SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_failures_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Create SM Context result %

Description

PDU Session Create SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes_total{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts_to  
tal{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_create_sm_context_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts_to
tal{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Create SM Context success rate

Description

Current PDU Session Create SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context rates

Description

Rate of PDU Session Update SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_update_sm_context_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Update SM Context result %

Description

PDU Session Update SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes_total{dnn=~"$dnn", snssai=~"$snssai"} [$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts_total{dnn=~"$dnn", snssai=~"$snssai"} [$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_update_sm_context_failures_total{dnn=~"$dnn", snssai=~"$snssai"} [$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts_total{dnn=~"$dnn", snssai=~"$snssai"} [$rate_interval]))))*100
```

Current PDU Session Update SM Context success rate

Description

Current PDU Session Update SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_update_sm_context_successes_total{dnn=~"$dnn", snssai=~"$snssai"} [$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts_total{dnn=~"$dnn", snssai=~"$snssai"} [$rate_interval]))
```

PDU Session Retrieve SM Context rates

Description

Rate of PDU Session Retrieve SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Retrieve SM Context result %

Description

PDU Session Retrieve SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes_total{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts_  
total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_failures_total{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts_  
total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Retrieve SM Context success rate

Description

Current PDU Session Retrieve SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_retrieve_sm_context_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context rates

Description

Rate of PDU Session Notify SM Context requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Notify SM Context result %

Description

PDU Session Notify SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_failures_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Notify SM Context success rate**Description**

Current PDU Session Notify SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_successes_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_nsmf_pdu_session_sm_context_status_notify_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context rates**Description**

Rate of PDU Session Release SM Context requests

Expression**Attempts**

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Release SM Context result %

Description

PDU Session Release SM Context result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts_t
otal{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_release_sm_context_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts_t
otal{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Release SM Context success rate

Description

Current PDU Session Release SM Context success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_release_sm_context_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer rates

Description

Rate of N1N2 Message Transfer requests

Expression

Attempts

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_namf_comm_n1n2_message_transfer_failures_total{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

N1N2 Message Transfer result %

Description

N1N2 Message Transfer result %

Expression

Success

```
((sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes_total{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts_total  
{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_namf_comm_n1n2_message_transfer_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts_total
{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current N1N2 Message Transfer success rate

Description

Current N1N2 Message Transfer success rate

Expression

```
(sum(rate(smfcc_namf_comm_n1n2_message_transfer_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_namf_comm_n1n2_message_transfer_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message rates

Description

Rate of NAMF Assign EBI Message requests

Expression**Attempts**

```
sum(rate(smfcc_namf_comm_assign_ebi_message_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_namf_comm_assign_ebi_message_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_namf_comm_assign_ebi_message_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

NAMF Assign EBI Message result %

Description

NAMF Assign EBI Message result %

Expression

Success

```
((sum(rate(smfcc_namf_comm_assign_ebi_message_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_namf_comm_assign_ebi_message_attempts_total{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_namf_comm_assign_ebi_message_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_namf_comm_assign_ebi_message_attempts_total{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current NAMF Assign EBI Message success rate

Description

Current NAMF Assign EBI Message success rate

Expression

```
(sum(rate(smfcc_namf_comm_assign_ebi_message_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_namf_comm_assign_ebi_message_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

UE Initiated Modify QoS attempts rates

Description

Rate of UE Initiated Modify QoS attempts requests

Expression

Attempts

```
sum(rate(smfcc_ue_initiated_modify_qos_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_ue_initiated_modify_qos_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_ue_initiated_modify_qos_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

UE Initiated Modify QoS attempts result %

Description

UE Initiated Modify QoS attempts result %

Expression

Success

```
((sum(rate(smfcc_ue_initiated_modify_qos_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_modify_qos_attempts_total{dnn=~"$
dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_initiated_modify_qos_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_modify_qos_attempts_total{dnn=~"$
dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current UE Initiated Modify QoS attempts success rate

Description

Current UE Initiated Modify QoS attempts success rate

Expression

```
(sum(rate(smfcc_ue_initiated_modify_qos_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_ue_initiated_modify_qos_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Policy Control Panels

Create SmPolicy Control Service rates

Description

Rate of Create SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_total{ }
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_success_total{ }
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures_total{ }
[$rate_interval]))
```

Create SmPolicy Control Service result %

Description

Create SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_success_total{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_  
total{[$rate_interval]})))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_create_service_failures_total{  
[$rate_interval]})))/(sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_  
total{[$rate_interval]})))*100
```

Current Create SmPolicy Control Service success rate

Description

Current Create SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_create_service_success_total{  
[$rate_interval]}) OR on() vector(0)) /  
sum(rate(smfcc_nsmf_smpolicy_control_create_service_attempts_total{  
[$rate_interval]})
```

Update SmPolicy Control Service rates

Description

Rate of Update SmPolicy Control Service requests

Expression

Attempts


```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_total{}  
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_success_total{}  
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures_total{}  
[$rate_interval]))
```

Update SmPolicy Control Service result %

Description

Update SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_success_total{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_  
total{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_service_failures_total{}  
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_  
total{}[$rate_interval]))))*100
```

Current Update SmPolicy Control Service success rate

Description

Current Update SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_service_success_total{}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_smpolicy_control_update_service_attempts_total{}
[$rate_interval]))
```

Delete SmPolicy Control Service rates

Description

Rate of Delete SmPolicy Control Service requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_total{}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success_total{}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures_total{}
[$rate_interval]))
```

Delete SmPolicy Control Service result %

Description

Delete SmPolicy Control Service result %

Expression

Success

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_
total{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_delete_service_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_
total{}[$rate_interval]))))*100
```

Current Delete SmPolicy Control Service success rate

Description

Current Delete SmPolicy Control Service success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_delete_service_success_total{}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts_total{}
[$rate_interval]))
```

SmPolicy Control Service Notify rates

Description

Rate of SmPolicy Control Service Notify requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_total{}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success_total{}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures_total{}
[$rate_interval]))
```

SmPolicy Control Service Notify result %

Description

SmPolicy Control Service Notify result %

Expression**Success**

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_t
otal{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_smpolicy_control_update_notify_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_t
otal{}[$rate_interval]))))*100
```

Current SmPolicy Control Service Notify success rate

Description

Current SmPolicy Control Service Notify success rate

Expression

```
(sum(rate(smfcc_nsmf_smpolicy_control_update_notify_success_total{}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_smpolicy_control_update_notify_attempts_total{}
[$rate_interval]))
```

Converged Charging Panels

CHF Create Session rates

Description

Rate of CHF Create Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_create_session_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_create_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_create_session_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Create Session result %

Description

CHF Create Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_create_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_create_session_attempts_total{dnn=~"$dnn"
, snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_create_session_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_create_session_attempts_total{dnn=~"$dnn"
, snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Create Session success rate

Description

Current CHF Create Session success rate

Expression

```
(sum(rate(smfcc_nchf_create_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_create_session_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Update Session rates

Description

Rate of CHF Update Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_update_session_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_update_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_update_session_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Update Session result %

Description

CHF Update Session result %

Expression

Success

```
((sum(rate(smfcc_nchf_update_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_update_session_attempts_total{dnn=~"$dnn"
, snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_update_session_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_nchf_update_session_attempts_total{dnn=~"$dnn"
, snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Update Session success rate

Description

Current CHF Update Session success rate

Expression

```
(sum(rate(smfcc_nchf_update_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_update_session_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Release Session rates

Description

Rate of CHF Release Session requests

Expression

Attempts

```
sum(rate(smfcc_nchf_release_session_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_release_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_release_session_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF Release Session result %

Description

CHF Release Session result %

Expression**Success**

```
((sum(rate(smfcc_nchf_release_session_success_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_release_session_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_release_session_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_release_session_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF Release Session success rate

Description

Current CHF Release Session success rate

Expression

```
(sum(rate(smfcc_nchf_release_session_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_release_session_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF One-Time Report rates

Description

Rate of CHF One-Time Report requests

Expression

Attempts

```
sum(rate(smfcc_nchf_one_time_report_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nchf_one_time_report_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nchf_one_time_report_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

CHF One-Time Report result %

Description

CHF One-Time Report result %

Expression

Success

```
((sum(rate(smfcc_nchf_one_time_report_success_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_one_time_report_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nchf_one_time_report_failures_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_nchf_one_time_report_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current CHF One-Time Report success rate

Description

Current CHF One-Time Report success rate

Expression

```
(sum(rate(smfcc_nchf_one_time_report_success_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nchf_one_time_report_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

UPF Sessions Panels

Session Establishment rates

Description

Rate of Session Establishment requests

Expression**Attempts**

```
sum(rate(pfcps_session_establishment_attempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionestablishmentsuccess_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionestablishmentfailure_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Establishment result %

Description

Session Establishment result %

Expression**Success**

```
((sum(rate(pfcpsessionestablishmentsuccess_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionestablishmentattempt_total{nssai=~"$nssai", dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionestablishmentfailure_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionestablishmentattempt_total{nssai=~"$nssai", dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Establishment success rate

Description

Current Session Establishment success rate

Expression

```
(sum(rate(pfcpsessionestablishmentsuccess_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(pfcpsessionestablishmentattempttotal{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Modification rates

Description

Rate of Session Modification requests

Expression

Attempts

```
sum(rate(pfcpsessionmodificationattempttotal{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionmodificationsuccess_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionmodificationfailure_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Modification result %

Description

Session Modification result %

Expression

Success

```
((sum(rate(pfcpsessionmodificationsuccess_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionmodificationattempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionmodificationfailure_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionmodification_attempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))))*100
```

Current Session Modification success rate

Description

Current Session Modification success rate

Expression

```
(sum(rate(pfcpsessionmodification_success_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionmodification_attempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Deletion rates

Description

Rate of Session Deletion requests

Expression**Attempts**

```
sum(rate(pfcpsessiondeletion_attempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessiondeletion_success_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsession_deletion_failure_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Deletion result %

Description

Session Deletion result %

Expression

Success

```
((sum(rate(pfcpsession_deletion_success_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsession_deletion_attempt_total{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsession_deletion_failure_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsession_deletion_attempt_total{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Deletion success rate

Description

Current Session Deletion success rate

Expression

```
(sum(rate(pfcpsession_deletion_success_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsession_deletion_attempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Report rates

Description

Rate of Session Report requests

Expression

Attempts

```
sum(rate(pfcpsessionreportattempttotal{nssai=~"$nssai", dnn=~"$dnn"}
[$rateinterval]))
```

Successes

```
sum(rate(pfcpsessionreportsuccesstotal{nssai=~"$nssai", dnn=~"$dnn"}
[$rateinterval]))
```

Failures

```
sum(rate(pfcpsessionreportfailuretotal{}[$rateinterval]))
```

Session Report result %

Description

Session Report result %

Expression

Success

```
((sum(rate(pfcpsessionreportsuccesstotal{nssai=~"$nssai", dnn=~"$dnn"}
[$rateinterval])))/(sum(rate(pfcpsessionreportattempttotal{nssai=~"$nssai",
dnn=~"$dnn"}[$rateinterval]))))*100
```

Failure

```
((sum(rate(pfcpsessionreportfailuretotal{}
[$rateinterval])))/(sum(rate(pfcpsessionreportattempttotal{nssai=~"$nssai",
dnn=~"$dnn"}[$rateinterval]))))*100
```

Current Session Report success rate

Description

Current Session Report success rate

Expression

```
(sum(rate(pfcpsessionreportsuccess_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionreportattempt_total{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Unified Data Management Panels

Get SM Data rates

Description

Rate of Get SM Data requests

Expression

Attempts

```
sum(rate(smfcc_sdm_get_sm_data_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_sdm_get_sm_data_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_sdm_get_sm_data_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Get SM Data result %

Description

Get SM Data result %

Expression

Success

```
((sum(rate(smfcc_sdm_get_sm_data_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_sdm_get_sm_data_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_sdm_get_sm_data_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_sdm_get_sm_data_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current Get SM Data success rate

Description

Current Get SM Data success rate

Expression

```
(sum(rate(smfcc_sdm_get_sm_data_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_sdm_get_sm_data_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Registration rates

Description

Rate of SMF-UDM Registration requests

Expression**Attempts**

```
sum(rate(smfcc_udm_registration_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_udm_registration_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_udm_registration_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Registration result %

Description

SMF-UDM Registration result %

Expression

Success

```
((sum(rate(smfcc_udm_registration_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_udm_registration_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_udm_registration_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_udm_registration_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current SMF-UDM Registration success rate

Description

Current SMF-UDM Registration success rate

Expression

```
(sum(rate(smfcc_udm_registration_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_udm_registration_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Deregistration rates

Description

Rate of SMF-UDM Deregistration requests

Expression

Attempts

```
sum(rate(smfcc_udm_deregistration_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_udm_deregistration_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_udm_deregistration_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Deregistration result %

Description

SMF-UDM Deregistration result %

Expression

Success

```
((sum(rate(smfcc_udm_deregistration_successes_total{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_udm_deregistration_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_udm_deregistration_failures_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_deregistration_attempts_total{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current SMF-UDM Deregistration success rate

Description

Current SMF-UDM Deregistration success rate

Expression

```
(sum(rate(smfcc_udm_deregistration_successes_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_udm_deregistration_attempts_total{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

NF Registration and Discovery Panels

NRF Registration rates

Description

Rate of NRF Registration requests

Expression

Attempts

```
sum(rate(nrfa_nnr_fnm_registration_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnr_fnm_registration_response_total{status=~"2.*"}
[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnr_fnm_registration_response_total{status!~"2.*"}
[$rate_interval]))
```

NRF Registration result %

Description

NRF Registration result %

Expression

Success

```
(sum(rate(nrfa_nnrif_nfm_registration_response_total{status=~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrif_nfm_registration_response_total[$rate_inte
rval])))*100
```

Failure

```
(sum(rate(nrfa_nnrif_nfm_registration_response_total{status!~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrif_nfm_registration_response_total[$rate_inte
rval])))*100
```

Current NRF Registration success rate

Description

Current NRF Registration success rate

Expression

```
(sum(rate(nrfa_nnrif_nfm_registration_response_total{status=~"2.*"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(nrfa_nnrif_nfm_registration_request[$rate_interval]))
```

NRF De-registration rates

Description

Rate of NRF De-registration requests

Expression

Attempts

```
sum(rate(nrfa_nnrf_nfm_deregistration_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrf_nfm_deregistration_response_total{status=~"2.*"}
[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnrf_nfm_deregistration_response_total{status!~"2.*"}
[$rate_interval]))
```

NRF De-registration result %

Description

NRF De-registration result %

Expression

Success

```
(sum(rate(nrfa_nnrf_nfm_deregistration_response_total{status=~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnrf_nfm_deregistration_response_total[$rate_in
terval])))*100
```

Failure

```
(sum(rate(nrfa_nnrf_nfm_deregistration_response_total{status!~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnrf_nfm_deregistration_response_total[$rate_in
terval])))*100
```

Current NRF De-registration success rate

Description

Current NRF De-registration success rate

Expression

```
(sum(rate(nrfa_nnrnf_nfm_deregistration_response_total{status=~"2.*"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(nrfa_nnrnf_nfm_deregistration_request[$rate_interval]))
```

NRF Update rates

Description

Rate of NRF Update requests

Expression

Attempts

```
sum(rate(nrfa_nnrnf_nfm_update_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrnf_nfm_update_response_total{status=~"2.*"}[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnrnf_nfm_update_response_total{status!~"2.*"}[$rate_interval]))
```

NRF Update result %

Description

NRF Update result %

Expression

Success

```
(sum(rate(nrfa_nnrnf_nfm_update_response_total{status=~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrnf_nfm_update_response_total[$rate_interval])
))*100
```

Failure

```
(sum(rate(nrfa_nnr_fnm_update_response_total{status!~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnr_fnm_update_response_total[$rate_interval])
))*100
```

Current NRF Update success rate

Description

Current NRF Update success rate

Expression

```
(sum(rate(nrfa_nnr_fnm_update_response_total{status=~"2.*"}[$rate_interval])) OR
on() vector(0)) / sum(rate(nrfa_nnr_fnm_update_request[$rate_interval]))
```

NRF Discovery rates

Description

Rate of NRF Discovery requests

Expression

Attempts

```
sum(rate(nrfa_nnr_fdisc_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnr_fdisc_response_total{status=~"2.*"}[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnr_fdisc_response_total{status!~"2.*"}[$rate_interval]))
```

NRF Discovery result %

Description

NRF Discovery result %

Expression

Success

```
(sum(rate(nrfa_nnrfdisc_response_total{status=~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrfdisc_response_total[$rate_interval])))*100
```

Failure

```
(sum(rate(nrfa_nnrfdisc_response_total{status!~"2.*"}
[$rate_interval]))/(sum(rate(nrfa_nnrfdisc_response_total[$rate_interval])))*100
```

Current NRF Discovery success rate

Description

Current NRF Discovery success rate

Expression

```
(sum(rate(nrfa_nnrfdisc_response_total{status=~"2.*"}[$rate_interval])) OR on()
vector(0)) / sum(rate(nrfa_nnrfdisc_request[$rate_interval]))
```

NRF Subscription rates

Description

Rate of NRF Subscription requests

Expression

Attempts

```
sum(rate(nrfa_nnrfnfm_subscription_request[$rate_interval]))
```

Successes

```
sum(rate(nrfa_nnrnf_nfm_subscription_response_total{status=~"2.*"}
[$rate_interval]))
```

Failures

```
sum(rate(nrfa_nnrnf_nfm_subscription_response_total{status!~"2.*"}
[$rate_interval]))
```

NRF Subscription result %

Description

NRF Subscription result %

Expression

Success

```
(sum(rate(nrfa_nnrnf_nfm_subscription_response_total{status=~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnrnf_nfm_subscription_response_total[$rate_inte
rval])))*100
```

Failure

```
(sum(rate(nrfa_nnrnf_nfm_subscription_response_total{status!~"2.*"}
[$rate_interval])))/(sum(rate(nrfa_nnrnf_nfm_subscription_response_total[$rate_inte
rval])))*100
```

Current NRF Subscription success rate

Description

Current NRF Subscription success rate

Expression

```
(sum(rate(nrfa_nnrnf_nfm_subscription_response_total{status=~"2.*"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(nrfa_nnrnf_nfm_subscription_request[$rate_interval]))
```

Individual Counter of PDU Session Procedures Panels

Session Establishment counts

Description

Number of Session Establishment requests

Expression

Attempts

```
sum(smfcc_pdu_session_establishment_attempts_total{}) -  
sum(smfcc_pdu_session_establishment_attempts_total{} offset $rate_interval)
```

Successes

```
sum(smfcc_pdu_session_establishment_successes_total{}) -  
sum(smfcc_pdu_session_establishment_successes_total{} offset $rate_interval)
```

Failures

```
sum(smfcc_pdu_session_establishment_failures_total{}) -  
sum(smfcc_pdu_session_establishment_failures_total{} offset $rate_interval)
```

UE Initiated Session Release counts

Description

Number of UE Initiated Session Release requests

Expression

Attempts

```
sum(smfcc_ue_initiated_pdu_session_release_attempts_total{}) -  
sum(smfcc_ue_initiated_pdu_session_release_attempts_total{} offset $rate_interval)
```

Successes

```
sum(smfcc_ue_initiated_pdu_session_release_successes_total{}) -  
sum(smfcc_ue_initiated_pdu_session_release_successes_total{} offset  
$rate_interval)
```

Failures

```
sum(smfcc_ue_initiated_pdu_session_release_failures_total{}) -  
sum(smfcc_ue_initiated_pdu_session_release_failures_total{} offset $rate_interval)
```

UE Deregistration counts

Description

Number of UE Deregistration requests

Expression

Attempts

```
sum(smfcc_ue_deregistration_attempts_total{}) -  
sum(smfcc_ue_deregistration_attempts_total{} offset $rate_interval)
```

Successes

```
sum(smfcc_ue_deregistration_successes_total{}) -  
sum(smfcc_ue_deregistration_successes_total{} offset $rate_interval)
```

Failures

```
sum(smfcc_ue_deregistration_failures_total{}) -  
sum(smfcc_ue_deregistration_failures_total{} offset $rate_interval)
```

Panels

GTP Messages

Handover Procedures

SMF Service Dashboard

Description

SMF Call-Control Service Dashboard

Panels

Current Pdu Sessions

Description

Total number of sessions present in SMF

Expression

```
sum(smf_current_pdu_sessions{})
```

Current Pdu Sessions Per dnn

Description

Current Pdu sessions per dnn

Expression

```
{{dnn}}

sum(smf_current_pdu_sessions_per_dnn{dnn=~"$dnn"})
```

Current Pdu Sessions Per Rat Type

Description

Current Pdu sessions per rat type

Expression

```
{{ratType}}

sum(smf_current_pdu_sessions_per_rat_type{ratType=~"$ratType"})
```

Current Pdu Sessions Per UPF

Description

Total number of pdu sessions per upf

Expression

{{ upf_node_id }}

```
sum(smf_current_pdu_sessions_per_upf{upf_node_id=~"$upf_node_id"})
```

Simultaneously Attached Users**Description**

Total number of unique subscribers attached to the network

Expression

```
sum(smf_simultaneous_attached_users{})
```

Simultaneously Attached Active Users**Description**

Total number of unique subscribers attached to the network with active session.

Expression

```
sum(smf_simultaneous_attached_active_users{})
```

Current Pdu Sessions Using Default Policy**Description**

Total number of pdu sessions using locally configured default policy

Expression

```
sum(smf_current_pdu_sessions_using_default_policy{})
```

Current number of Qos Flows

Description

Current number of cumulative qos flows present in SMF

Expression

```
smf_current_cumulative_qos_flows
```

Current Pdu Sessions Using Default Quota

Description

Total number of pdu sessions using locally configured default quota.

Expression

```
sum(smf_current_pdu_sessions_using_default_quota{})
```

Current Bearers per fiveQi, arpPI in SMF+PGWC

Description

Current number of bearers present in SMF+PGWC per fiveQi and arpPI combination

Expression

{{fiveQi}}:{{arpPI}}

```
sum(smf_pgw_current_bearers{fiveQi=~"$fiveQi",arpPI=~"$arpPI"})
```

Current qos flows per fiveQi, arpPI in SMF

Description

Current number qos flows present in SMF per fiveQi and arpPI combination

Expression

{{fiveQi}}:{{arpPI}}

```
sum(smf_current_qos_flows{fiveQi=~"$fiveQi",arpPI=~"$arpPI"})
```

Current Pdu Sessions With Charging Disabled

Description

Total number of pdu sessions with charging disabled in SMF

Expression

```
sum(smf_current_pdu_sessions_without_charging{})
```

Current Pdu Sessions Using Udm Allocated Static Ip Address

Description

Total Pdu Sessions using udm allocated static ue ip address.

Expression

```
sum(smf_current_pdu_sessions_using_udm_allocated_static_ue_ip_address{})
```

Current number of bearers in SMF+PGWC

Description

Current number of bearers in SMF+PGW-C

Expression

```
sum(smf_pgw_current_cumulative_bearers{})
```

Current Inbound Roaming Home Routed Sessions

Description

Current Inbound Roaming Home Routed Sessions

Expression

```
sum(smf_current_inbound_roaming_home_routed_sessions{})
```

Current Outbound Roaming Sessions

Description

Current Outbound Roaming Sessions

Expression

```
sum(smf_current_outbound_roaming_sessions{})
```

Current Pdu Sessions Per Mapped Dnn

Description

Current Pdu Sessions Per Mapped Dnn

Expression

```
sum(smf_current_pdu_sessions_per_mapped_dnn{mappedDnn=~"$mappedDnn"})
```

Current SGW Idle Pdu Sessions

Description

Total number of SGW Idle sessions present in SMF

Expression

```
sum(smf_sgw_current_idle_pdu_sessions{})
```

Current SGW Idle Subscribers

Description

Total number of SGW Idle Subscribers present in SMF

Expression

```
sum(smf_sgw_idle_subscribers{})
```

Sgw Simultaneously Attached Active Users

Description

Total number of unique Sgw subscribers attached to the network with active session.

Expression

```
sum(smf_sgw_simultaneous_attached_active_users{})
```

SMFCC Gauge Request Received

Description

Number of get smfcc gauge request messages received

Expression

```
sum(smfcc_get_smfcc_gauge_attempts{gauge=~"$gauge"})
```

SMFCC Gauge Requests Succeeded

Description

SMFCC Gauge Requests Succeeded

Expression

```
sum(smfcc_get_smfcc_gauge_successes{gauge=~"$gauge"})
```

SMF Service Enterprise

Description

SMF Call-Control Service Dashboard

Panels

Current Pdu Sessions

Description

Total number of sessions present in SMF

Expression

```
sum(smf_current_pdu_sessions{})
```

Current Pdu Sessions Per dnn

Description

Current Pdu sessions per dnn

Expression

{{dnn}}

```
sum(smf_current_pdu_sessions_per_dnn{dnn=~"$dnn"})
```

Current Pdu Sessions Per Rat Type

Description

Current Pdu sessions per rat type

Expression

{{ratType}}

```
sum(smf_current_pdu_sessions_per_rat_type{ratType=~"$ratType"})
```

Current Pdu Sessions Per UPF

Description

Total number of pdu sessions per upf

Expression

{{ upf_node_id }}

```
sum(smf_current_pdu_sessions_per_upf{upf_node_id=~"$upf_node_id"})
```

Simultaneously Attached Users

Description

Total number of unique subscribers attached to the network

Expression

```
sum(smf_simultaneous_attached_users{})
```

Simultaneously Attached Active Users

Description

Total number of unique subscribers attached to the network with active session.

Expression

```
sum(smf_simultaneous_attached_active_users{})
```

Current Pdu Sessions Using Default Policy

Description

Total number of pdu sessions using locally configured default policy

Expression

```
sum(smf_current_pdu_sessions_using_default_policy{})
```

Current number of Qos Flows

Description

Current number of cumulative qos flows present in SMF

Expression

```
smf_current_cumulative_qos_flows
```

Current Pdu Sessions Using Default Quota

Description

Total number of pdu sessions using locally configured default quota.

Expression

```
sum(smf_current_pdu_sessions_using_default_quota{})
```

Current Bearers per fiveQi, arpPI in SMF+PGWC

Description

Current number of bearers present in SMF+PGWC per fiveQi and arpPI combination

Expression

{{fiveQi}}:{{arpPI}}

```
sum(smf_pgw_current_bearers{fiveQi=~"$fiveQi",arpPI=~"$arpPI"})
```

Current qos flows per fiveQi, arpPI in SMF

Description

Current number qos flows present in SMF per fiveQi and arpPI combination

Expression

{{fiveQi}}:{{arpPI}}

```
sum(smf_current_qos_flows{fiveQi=~"$fiveQi",arpPI=~"$arpPI"})
```

Current Pdu Sessions With Charging Disabled

Description

Total number of pdu sessions with charging disabled in SMF

Expression

```
sum(smf_current_pdu_sessions_without_charging{})
```

Current Pdu Sessions Using Udm Allocated Static Ip Address

Description

Total Pdu Sessions using udm allocated static ue ip address.

Expression

```
sum(smf_current_pdu_sessions_using_udm_allocated_static_ue_ip_address{})
```

Current number of bearers in SMF+PGWC

Description

Current number of bearers in SMF+PGW-C

Expression

```
sum(smf_pgw_current_cumulative_bearers{})
```

Current Inbound Roaming Home Routed Sessions

Description

Current Inbound Roaming Home Routed Sessions

Expression

```
sum(smf_current_inbound_roaming_home_routed_sessions{})
```

Current Outbound Roaming Sessions

Description

Current Outbound Roaming Sessions

Expression

```
sum(smf_current_outbound_roaming_sessions{})
```

Current Pdu Sessions Per Mapped Dnn

Description

Current Pdu Sessions Per Mapped Dnn

Expression

```
sum(smf_current_pdu_sessions_per_mapped_dnn{mappedDnn=~"$mappedDnn"})
```

Current SGW Idle Pdu Sessions

Description

Total number of SGW Idle sessions present in SMF

Expression

```
sum(smf_sgw_current_idle_pdu_sessions{})
```

Current SGW Idle Subscribers

Description

Total number of SGW Idle Subscribers present in SMF

Expression

```
sum(smf_sgw_idle_subscribers{})
```

Sgw Simultaneously Attached Active Users

Description

Total number of unique Sgw subscribers attached to the network with active session.

Expression

```
sum(smf_sgw_simultaneous_attached_active_users{})
```

SMFCC Gauge Request Received

Description

Number of get smfcc gauge request messages received

Expression

```
sum(smfcc_get_smfcc_gauge_attempts{gauge=~"$gauge"})
```

SMFCC Gauge Requests Succeeded

Description

SMFCC Gauge Requests Succeeded

Expression

```
sum(smfcc_get_smfcc_gauge_successes{gauge=~"$gauge"})
```

Session Management Overview

Description

High level information about the operation of the Session Management Function (SMF)

PDU Sessions and QoS Flows Panels

Number of PDU sessions

Description

Number of PDU sessions

Expression

```
sum(smfc_current_pdu_sessions{})
```

Current PDU sessions

Description

Current PDU sessions

Expression

```
sum(smfc_current_pdu_sessions{}) OR on() vector(0)
```


Number of QoS flows

Description

Number of QoS flows

Expression

```
sum(smf_current_cumulative_qos_flows{})
```

Current QoS flows

Description

Current QoS flows

Expression

```
sum(smf_current_cumulative_qos_flows{}) OR on() vector(0)
```

PDU Session Procedures Panels

Session Establishment rates

Description

Rate of Session Establishment requests

Expression

Attempts

```
sum(rate(smfcc_pdu_session_establishment_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pdu_session_establishment_successes{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pdu_session_establishment_failures{})[$rate_interval]))
```

Session Establishment result %

Description

Session Establishment result %

Expression

Success

```
((sum(rate(smfcc_pdu_session_establishment_successes{}  
[$rate_interval]))/(sum(rate(smfcc_pdu_session_establishment_attempts{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_pdu_session_establishment_failures{}  
[$rate_interval]))/(sum(rate(smfcc_pdu_session_establishment_attempts{}  
[$rate_interval]))))*100
```

Current Session Establishment success rate

Description

Current Session Establishment success rate

Expression

```
(sum(rate(smfcc_pdu_session_establishment_successes{}[$rate_interval])) OR on()  
vector(0)) / sum(rate(smfcc_pdu_session_establishment_attempts{}[$rate_interval]))
```

UE Initiated Session Release rates

Description

Rate of UE Initiated Session Release requests

Expression

Attempts

```
sum(rate(smfcc_ue_initiated_pdu_session_release_attempts{})[$rate_interval]))
```

Successes

```
sum(rate(smfcc_ue_initiated_pdu_session_release_successes{})[$rate_interval]))
```

Failures

```
sum(rate(smfcc_ue_initiated_pdu_session_release_failures{})[$rate_interval]))
```

UE Initiated Session Release result %

Description

UE Initiated Session Release result %

Expression

Success

```
((sum(rate(smfcc_ue_initiated_pdu_session_release_successes{}
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_pdu_session_release_attempts{}
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_initiated_pdu_session_release_failures{}
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_pdu_session_release_attempts{}
[$rate_interval]))))*100
```

Current UE Initiated Session Release success rate

Description

Current UE Initiated Session Release success rate

Expression

```
(sum(rate(smfcc_ue_initiated_pdu_session_release_successes{}[$rate_interval])) OR  
on() vector(0)) / sum(rate(smfcc_ue_initiated_pdu_session_release_attempts{}  
[$rate_interval]))
```

UE Deregistration rates

Description

Rate of UE Deregistration requests

Expression

Attempts

```
sum(rate(smfcc_ue_deregistration_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_ue_deregistration_successes{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_ue_deregistration_failures{}[$rate_interval]))
```

UE Deregistration result %

Description

UE Deregistration result %

Expression

Success

```
((sum(rate(smfcc_ue_deregistration_successes{}  
[$rate_interval])))/(sum(rate(smfcc_ue_deregistration_attempts{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_deregistration_failures{}
[$rate_interval])))/(sum(rate(smfcc_ue_deregistration_attempts{}
[$rate_interval]))))*100
```

Current UE Deregistration success rate

Description

Current UE Deregistration success rate

Expression

```
(sum(rate(smfcc_ue_deregistration_successes{}[$rate_interval])) OR on() vector(0))
/ sum(rate(smfcc_ue_deregistration_attempts{}[$rate_interval]))
```

Connection Management Procedures Panels

Service Request rates

Description

Rate of Service Request requests

Expression

Attempts

```
sum(rate(smfcc_service_request_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_service_request_successes{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_service_request_failures{}[$rate_interval]))
```

Service Request result %

Description

Service Request result %

Expression**Success**

```
((sum(rate(smfcc_service_request_successes{  
[$rate_interval]})))/(sum(rate(smfcc_service_request_attempts{  
[$rate_interval]})))*100
```

Failure

```
((sum(rate(smfcc_service_request_failures{  
[$rate_interval]})))/(sum(rate(smfcc_service_request_attempts{  
[$rate_interval]})))*100
```

Current Service Request success rate

Description

Current Service Request success rate

Expression

```
(sum(rate(smfcc_service_request_successes{[$rate_interval]}) OR on() vector(0)) /  
sum(rate(smfcc_service_request_attempts{[$rate_interval]}))
```

AN UE Context Release rates

Description

Rate of AN UE Context Release requests

Expression**Attempts**

```
sum(rate(smfcc_an_release_attempts{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_an_release_successes{})[$rate_interval]))
```

Failures

```
sum(rate(smfcc_an_release_failures{})[$rate_interval]))
```

AN UE Context Release result %

Description

AN UE Context Release result %

Expression**Success**

```
((sum(rate(smfcc_an_release_successes{}  
[$rate_interval])))/(sum(rate(smfcc_an_release_attempts{})[$rate_interval])))*100
```

Failure

```
((sum(rate(smfcc_an_release_failures{}  
[$rate_interval])))/(sum(rate(smfcc_an_release_attempts{})[$rate_interval])))*100
```

Current AN UE Context Release success rate

Description

Current AN UE Context Release success rate

Expression

```
(sum(rate(smfcc_an_release_successes{})[$rate_interval])) OR on() vector(0)) /  
sum(rate(smfcc_an_release_attempts{})[$rate_interval]))
```

Network Initiated Service Request rates

Description

Rate of Network Initiated Service Request requests

Expression

Attempts

```
sum(rate(smfcc_network_initiated_service_request_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_network_initiated_service_request_successes{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_network_initiated_service_request_failures{}[$rate_interval]))
```

Network Initiated Service Request result %

Description

Network Initiated Service Request result %

Expression

Success

```
((sum(rate(smfcc_network_initiated_service_request_successes{}  
[$rate_interval])))/(sum(rate(smfcc_network_initiated_service_request_attempts{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_network_initiated_service_request_failures{}  
[$rate_interval])))/(sum(rate(smfcc_network_initiated_service_request_attempts{}  
[$rate_interval]))))*100
```

Current Network Initiated Service Request success rate

Description

Current Network Initiated Service Request success rate

Expression

```
(sum(rate(smfcc_network_initiated_service_request_successes{}[$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_network_initiated_service_request_attempts{}[$rate_interval]))
```

Network Initiated UE Context Release rates

Description

Rate of Network Initiated UE Context Release requests

Expression

Attempts

```
sum(rate(smfcc_nw_initiated_release_attempts{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nw_initiated_release_successes{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nw_initiated_release_failures{}[$rate_interval]))
```

Network Initiated UE Context Release result %

Description

Network Initiated UE Context Release result %

Expression

Success

```
((sum(rate(smfcc_nw_initiated_release_successes{}
```

```
[$rate_interval])))/(sum(rate(smfcc_nw_initiated_release_attempts{}
```

```
[$rate_interval])))*100
```

Failure

```
((sum(rate(smfcc_nw_initiated_release_failures{}
```

```
[$rate_interval])))/(sum(rate(smfcc_nw_initiated_release_attempts{}
```

```
[$rate_interval])))*100
```

Current Network Initiated UE Context Release success rate

Description

Current Network Initiated UE Context Release success rate

Expression

```
(sum(rate(smfcc_nw_initiated_release_successes{][$rate_interval])) OR on()
```

```
vector(0)) / sum(rate(smfcc_nw_initiated_release_attempts{][$rate_interval]))
```

Mobility Management Procedures Panels

XN Handover rates

Description

Rate of XN Handover requests

Expression

Attempts

```
sum(rate(smfcc_xn_handover_attempts{][$rate_interval]))
```

Successes

```
sum(rate(smfcc_xn_handover_successes{][$rate_interval]))
```

Failures

```
sum(rate(smfcc_xn_handover_failures{}[$rate_interval]))
```

XN Handover result %

Description

XN Handover result %

Expression

Success

```
((sum(rate(smfcc_xn_handover_successes{}
[$rate_interval])))/(sum(rate(smfcc_xn_handover_attempts{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_xn_handover_failures{}
[$rate_interval])))/(sum(rate(smfcc_xn_handover_attempts{}[$rate_interval]))))*100
```

Current XN Handover success rate

Description

Current XN Handover success rate

Expression

```
(sum(rate(smfcc_xn_handover_successes{}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_xn_handover_attempts{}[$rate_interval]))
```

N2 Handover rates

Description

Rate of N2 Handover requests

Expression

Attempts

```
sum(rate(smfcc_n2_handover_attempts{})[$rate_interval]))
```

Successes

```
sum(rate(smfcc_n2_handover_successes{})[$rate_interval]))
```

Failures

```
sum(rate(smfcc_n2_handover_failures{})[$rate_interval]))
```

N2 Handover result %

Description

N2 Handover result %

Expression

Success

```
((sum(rate(smfcc_n2_handover_successes{}
[$rate_interval])))/(sum(rate(smfcc_n2_handover_attempts{})[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_n2_handover_failures{}
[$rate_interval])))/(sum(rate(smfcc_n2_handover_attempts{})[$rate_interval]))))*100
```

Current N2 Handover success rate

Description

Current N2 Handover success rate

Expression

```
(sum(rate(smfcc_n2_handover_successes{}[$rate_interval])) OR on() vector(0)) /  
sum(rate(smfcc_n2_handover_attempts{}[$rate_interval]))
```

Session Management Overview Enterprise

Description

High level information about the operation of the Session Management Function (SMF)

PDU Sessions and QoS Flows Panels

Number of PDU sessions

Description

Number of PDU sessions

Expression

```
sum(smfc_current_pdu_sessions{})
```

Current PDU sessions

Description

Current PDU sessions

Expression

```
sum(smfc_current_pdu_sessions{}) OR on() vector(0)
```

Number of QoS flows

Description

Number of QoS flows

Expression

```
sum(smfc_current_cumulative_qos_flows{})
```

Current QoS flows

Description

Current QoS flows

Expression

```
sum(smf_current_cumulative_qos_flows{ }) OR on() vector(0)
```

PDU Session Procedures Panels

Session Establishment rates

Description

Rate of Session Establishment requests

Expression

Attempts

```
sum(rate(smfcc_pdu_session_establishment_attempts_total{ }[$rate_interval]))
```

Successes

```
sum(rate(smfcc_pdu_session_establishment_successes_total{ }[$rate_interval]))
```

Failures

```
sum(rate(smfcc_pdu_session_establishment_failures_total{ }[$rate_interval]))
```

Session Establishment result %

Description

Session Establishment result %

Expression

Success

```
((sum(rate(smfcc_pdu_session_establishment_successes_total{  
[$rate_interval]})))/(sum(rate(smfcc_pdu_session_establishment_attempts_total{  
[$rate_interval]})))*100
```

Failure

```
((sum(rate(smfcc_pdu_session_establishment_failures_total{  
[$rate_interval]})))/(sum(rate(smfcc_pdu_session_establishment_attempts_total{  
[$rate_interval]})))*100
```

Current Session Establishment success rate

Description

Current Session Establishment success rate

Expression

```
(sum(rate(smfcc_pdu_session_establishment_successes_total{[$rate_interval]})) OR  
on() vector(0)) / sum(rate(smfcc_pdu_session_establishment_attempts_total{  
[$rate_interval]}))
```

UE Initiated Session Release rates

Description

Rate of UE Initiated Session Release requests

Expression

Attempts

```
sum(rate(smfcc_ue_initiated_pdu_session_release_attempts_total{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_ue_initiated_pdu_session_release_successes_total{  
[$rate_interval]}))
```

Failures

```
sum(rate(smfcc_ue_initiated_pdu_session_release_failures_total{}[$rate_interval]))
```

UE Initiated Session Release result %

Description

UE Initiated Session Release result %

Expression

Success

```
((sum(rate(smfcc_ue_initiated_pdu_session_release_successes_total{}  
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_pdu_session_release_attempts_total{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_initiated_pdu_session_release_failures_total{}  
[$rate_interval])))/(sum(rate(smfcc_ue_initiated_pdu_session_release_attempts_total{}  
[$rate_interval]))))*100
```

Current UE Initiated Session Release success rate

Description

Current UE Initiated Session Release success rate

Expression

```
(sum(rate(smfcc_ue_initiated_pdu_session_release_successes_total{}  
[$rate_interval])) OR on() vector(0)) /  
sum(rate(smfcc_ue_initiated_pdu_session_release_attempts_total{}[$rate_interval]))
```

UE Deregistration rates

Description

Rate of UE Deregistration requests

Expression

Attempts

```
sum(rate(smfcc_ue_deregistration_attempts_total{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_ue_deregistration_successes_total{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_ue_deregistration_failures_total{}[$rate_interval]))
```

UE Deregistration result %

Description

UE Deregistration result %

Expression

Success

```
((sum(rate(smfcc_ue_deregistration_successes_total{}
[$rate_interval])))/(sum(rate(smfcc_ue_deregistration_attempts_total{}
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_ue_deregistration_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_ue_deregistration_attempts_total{}
[$rate_interval]))))*100
```

Current UE Deregistration success rate

Description

Current UE Deregistration success rate

Expression

```
(sum(rate(smfcc_ue_deregistration_successes_total{}[$rate_interval])) OR on()  
vector(0)) / sum(rate(smfcc_ue_deregistration_attempts_total{}[$rate_interval]))
```

Connection Management Procedures Panels

Service Request rates

Description

Rate of Service Request requests

Expression**Attempts**

```
sum(rate(smfcc_service_request_attempts_total{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_service_request_successes_total{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_service_request_failures_total{}[$rate_interval]))
```

Service Request result %

Description

Service Request result %

Expression**Success**

```
((sum(rate(smfcc_service_request_successes_total{}  
[$rate_interval]))/(sum(rate(smfcc_service_request_attempts_total{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_service_request_failures_total{  
[$rate_interval]})))/(sum(rate(smfcc_service_request_attempts_total{  
[$rate_interval]})))*100
```

Current Service Request success rate

Description

Current Service Request success rate

Expression

```
(sum(rate(smfcc_service_request_successes_total{[$rate_interval]}) OR on(  
vector(0)) / sum(rate(smfcc_service_request_attempts_total{[$rate_interval]}))
```

AN UE Context Release rates

Description

Rate of AN UE Context Release requests

Expression

Attempts

```
sum(rate(smfcc_an_release_attempts_total{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_an_release_successes_total{[$rate_interval]}))
```

Failures

```
sum(rate(smfcc_an_release_failures_total{[$rate_interval]}))
```

AN UE Context Release result %

Description

AN UE Context Release result %

Expression

Success

```
((sum(rate(smfcc_an_release_successes_total{}  
[$rate_interval])))/(sum(rate(smfcc_an_release_attempts_total{}  
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_an_release_failures_total{}  
[$rate_interval])))/(sum(rate(smfcc_an_release_attempts_total{}  
[$rate_interval]))))*100
```

Current AN UE Context Release success rate

Description

Current AN UE Context Release success rate

Expression

```
(sum(rate(smfcc_an_release_successes_total{][$rate_interval])) OR on() vector(0))  
/ sum(rate(smfcc_an_release_attempts_total{][$rate_interval]))
```

Network Initiated Service Request rates

Description

Rate of Network Initiated Service Request requests

Expression

Attempts

```
sum(rate(smfcc_network_initiated_service_request_attempts_total{}  
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_network_initiated_service_request_successes_total{}  
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_network_initiated_service_request_failures_total{}  
[$rate_interval]))
```

Network Initiated Service Request result %

Description

Network Initiated Service Request result %

Expression

Success

```
((sum(rate(smfcc_network_initiated_service_request_successes_total{}  
[$rate_interval])))/(sum(rate(smfcc_network_initiated_service_request_attempts_tot  
al{}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_network_initiated_service_request_failures_total{}  
[$rate_interval])))/(sum(rate(smfcc_network_initiated_service_request_attempts_tot  
al{}[$rate_interval]))))*100
```

Current Network Initiated Service Request success rate

Description

Current Network Initiated Service Request success rate

Expression

```
(sum(rate(smfcc_network_initiated_service_request_successes_total{  
[$rate_interval]}) OR on() vector(0)) /  
sum(rate(smfcc_network_initiated_service_request_attempts_total{  
[$rate_interval]})
```

Network Initiated UE Context Release rates

Description

Rate of Network Initiated UE Context Release requests

Expression

Attempts

```
sum(rate(smfcc_nw_initiated_release_attempts_total{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_nw_initiated_release_successes_total{[$rate_interval]}))
```

Failures

```
sum(rate(smfcc_nw_initiated_release_failures_total{[$rate_interval]}))
```

Network Initiated UE Context Release result %

Description

Network Initiated UE Context Release result %

Expression

Success

```
((sum(rate(smfcc_nw_initiated_release_successes_total{  
[$rate_interval]})))/(sum(rate(smfcc_nw_initiated_release_attempts_total{  
[$rate_interval]})))*100
```

Failure

```
((sum(rate(smfcc_nw_initiated_release_failures_total{  
[$rate_interval]})))/(sum(rate(smfcc_nw_initiated_release_attempts_total{  
[$rate_interval]})))*100
```

Current Network Initiated UE Context Release success rate

Description

Current Network Initiated UE Context Release success rate

Expression

```
(sum(rate(smfcc_nw_initiated_release_successes_total{[$rate_interval]}) OR on(  
vector(0)) / sum(rate(smfcc_nw_initiated_release_attempts_total{  
[$rate_interval]}))
```

Mobility Management Procedures Panels

XN Handover rates

Description

Rate of XN Handover requests

Expression

Attempts

```
sum(rate(smfcc_xn_handover_attempts_total{[$rate_interval]}))
```

Successes

```
sum(rate(smfcc_xn_handover_successes_total{[$rate_interval]}))
```

Failures

```
sum(rate(smfcc_xn_handover_failures_total{}[$rate_interval]))
```

XN Handover result %

Description

XN Handover result %

Expression

Success

```
((sum(rate(smfcc_xn_handover_successes_total{}
[$rate_interval])))/(sum(rate(smfcc_xn_handover_attempts_total{}
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_xn_handover_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_xn_handover_attempts_total{}
[$rate_interval]))))*100
```

Current XN Handover success rate

Description

Current XN Handover success rate

Expression

```
(sum(rate(smfcc_xn_handover_successes_total{}[$rate_interval])) OR on() vector(0))
/ sum(rate(smfcc_xn_handover_attempts_total{}[$rate_interval]))
```

N2 Handover rates

Description

Rate of N2 Handover requests

Expression

Attempts

```
sum(rate(smfcc_n2_handover_attempts_total{}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_n2_handover_successes_total{}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_n2_handover_failures_total{}[$rate_interval]))
```

N2 Handover result %

Description

N2 Handover result %

Expression

Success

```
((sum(rate(smfcc_n2_handover_successes_total{}
[$rate_interval])))/(sum(rate(smfcc_n2_handover_attempts_total{}
[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_n2_handover_failures_total{}
[$rate_interval])))/(sum(rate(smfcc_n2_handover_attempts_total{}
[$rate_interval]))))*100
```

Current N2 Handover success rate

Description

Current N2 Handover success rate

Expression

```
(sum(rate(smfcc_n2_handover_successes_total{}[$rate_interval])) OR on() vector(0))  
/ sum(rate(smfcc_n2_handover_attempts_total{}[$rate_interval]))
```

UPF Sessions

Description

Statistics for SMF operations on the interface to UPF

Overview Panels

Session Establishment rates

Description

Rate of Session Establishment requests

Expression

Attempts

```
sum(rate(pfcpsessionestablishmentattempt{nssai=~"$nssai", dnn=~"$dnn"}  
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionestablishmentsuccess{nssai=~"$nssai", dnn=~"$dnn"}  
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionestablishmentfailure{nssai=~"$nssai", dnn=~"$dnn"}  
[$rate_interval]))
```

Session Establishment result %

Description

Session Establishment result %

Expression**Success**

```
((sum(rate(pfcpsessionestablishmentsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionestablishmentattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionestablishmentfailure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessionestablishmentattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Establishment success rate

Description

Current Session Establishment success rate

Expression

```
(sum(rate(pfcpsessionestablishmentsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionestablishmentattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Modification rates

Description

Rate of Session Modification requests

Expression**Attempts**

```
sum(rate(pfcpsessionmodificationattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionmodification_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionmodification_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Modification result %

Description

Session Modification result %

Expression

Success

```
((sum(rate(pfcpsessionmodification_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsessionmodification_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionmodification_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsessionmodification_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Modification success rate

Description

Current Session Modification success rate

Expression

```
(sum(rate(pfcpsessionmodification_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionmodification_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Deletion rates

Description

Rate of Session Deletion requests

Expression

Attempts

```
sum(rate(pfcpsessiondeletionattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessiondeletion_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessiondeletion_failure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Deletion result %

Description

Session Deletion result %

Expression

Success

```
((sum(rate(pfcpsessiondeletion_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsessiondeletion_attempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessiondeletionfailure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])))/(sum(rate(pfcpsessiondeletionattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Deletion success rate

Description

Current Session Deletion success rate

Expression

```
(sum(rate(pfcpsessiondeletion_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessiondeletion_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Session Report rates

Description

Rate of Session Report requests

Expression

Attempts

```
sum(rate(pfcpsessionreport_attempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Successes

```
sum(rate(pfcpsessionreport_success{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Failures

```
sum(rate(pfcpsessionreport_failure{}[$rate_interval]))
```

Session Report result %

Description

Session Report result %

Expression

Success

```
((sum(rate(pfcpsessionreportsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))/(sum(rate(pfcpsessionreportattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(pfcpsessionreportfailure{
[$rate_interval]))/(sum(rate(pfcpsessionreportattempt{nssai=~"$nssai",
dnn=~"$dnn"}[$rate_interval]))))*100
```

Current Session Report success rate

Description

Current Session Report success rate

Expression

```
(sum(rate(pfcpsessionreportsuccess{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(pfcpsessionreportattempt{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

Rates per DNN and NSSAI Panels

Session Establishment rates per DNN

Description

Rate of Session Establishment requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(pfcpsessionestablishmentattempt{}[$rateinterval]))
```

Session Establishment rates per NSSAI

Description

Rate of Session Establishment requests per NSSAI

Expression

{{nssai}}

```
sum by (nssai) (rate(pfcpsessionestablishmentattempt{}[$rateinterval]))
```

Session Modification rates per DNN

Description

Rate of Session Modification requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(pfcpsessionmodificationattempt{}[$rateinterval]))
```

Session Modification rates per NSSAI

Description

Rate of Session Modification requests per NSSAI

Expression

{{nssai}}

```
sum by (nssai) (rate(pfcpsessionmodificationattempt{}[$rateinterval]))
```

Session Deletion rates per DNN

Description

Rate of Session Deletion requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(pfcpsessiondeletionattempt{}[$rateinterval]))
```

Session Deletion rates per NSSAI

Description

Rate of Session Deletion requests per NSSAI

Expression

{{nssai}}

```
sum by (nssai) (rate(pfcpsessiondeletionattempt{}[$rateinterval]))
```

Session Report rates per DNN

Description

Rate of Session Report requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(pfcpsessionreportattempt{}[$rateinterval]))
```

Session Report rates per NSSAI

Description

Rate of Session Report requests per NSSAI

Expression

{{nssai}}

```
sum by (nssai) (rate(pfcpsessionreportattempt{{}}[$rate_interval]))
```

Errors Panels

Session Establishment error rates by cause

Description

Rate of Session Establishment errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(pfcpsessionestablishmentfailure{nssai=~"$nssai",  
dnn=~"$dnn"}[$rate_interval]))
```

Session Modification error rates by cause

Description

Rate of Session Modification errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(pfcpsessionmodificationfailure{nssai=~"$nssai",  
dnn=~"$dnn"}[$rate_interval]))
```

Session Deletion error rates by cause

Description

Rate of Session Deletion errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(pfcpsessiondeletionfailure{nssai=~"$nssai", dnn=~"$dnn"}  
[$rate_interval]))
```

Session Report error rates by cause

Description

Rate of Session Report errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(pfcpsessionreportfailure{nssai=~"$nssai", dnn=~"$dnn"}
[$rate_interval]))
```

DBCLIENT_HTTP Panels

DB Client State Store HTTP transactions Rate

Description

DB Client HTTP Transaction Rate by result

Expression

{{http_status_code}}

```
sum by (http_status_code) (rate(dbclienthttprequests{namespace="fed-smf",
db_type="mongo_RAM", db="pfcpsdb"}[$rate_interval]))
```

Unified Data Management

Description

Statistics for SMF operations on the Unified Data Management (UDM) interface

Overview Panels

Get SM Data rates

Description

Rate of Get SM Data requests

Expression

Attempts

```
sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_sdm_get_sm_data_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_sdm_get_sm_data_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Get SM Data result %

Description

Get SM Data result %

Expression**Success**

```
((sum(rate(smfcc_sdm_get_sm_data_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_sdm_get_sm_data_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current Get SM Data success rate

Description

Current Get SM Data success rate

Expression

```
(sum(rate(smfcc_sdm_get_sm_data_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_sdm_get_sm_data_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Registration rates

Description

Rate of SMF-UDM Registration requests

Expression

Attempts

```
sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_udm_registration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_udm_registration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Registration result %

Description

SMF-UDM Registration result %

Expression

Success

```
((sum(rate(smfcc_udm_registration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_udm_registration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current SMF-UDM Registration success rate

Description

Current SMF-UDM Registration success rate

Expression

```
(sum(rate(smfcc_udm_registration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_udm_registration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Deregistration rates

Description

Rate of SMF-UDM Deregistration requests

Expression

Attempts

```
sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_udm_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_udm_deregistration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

SMF-UDM Deregistration result %

Description

SMF-UDM Deregistration result %

Expression

Success

```
((sum(rate(smfcc_udm_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_udm_deregistration_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current SMF-UDM Deregistration success rate

Description

Current SMF-UDM Deregistration success rate

Expression

```
(sum(rate(smfcc_udm_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_udm_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Rates per DNN and SNSSAI Panels

SDM GET SM DATA rates per DNN

Description

Rate of SDM_GET_SM_DATA requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_sdm_get_sm_data_attempts{}[$rate_interval]))
```

SDM GET SM DATA rates per SNSSAI

Description

Rate of SDM_GET_SM_DATA requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_sdm_get_sm_data_attempts{}[$rate_interval]))
```

UDM REGISTRATION rates per DNN

Description

Rate of UDM_REGISTRATION requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_udm_registration_attempts{}[$rate_interval]))
```

UDM REGISTRATION rates per SNSSAI

Description

Rate of UDM_REGISTRATION requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_udm_registration_attempts{}[$rate_interval]))
```

UDM DEREGISTRATION rates per DNN

Description

Rate of UDM_DEREGISTRATION requests per DNN

Expression

{{dnn}}

```
sum by (dnn) (rate(smfcc_udm_deregistration_attempts{}[$rate_interval]))
```

UDM DEREGISTRATION rates per SNSSAI

Description

Rate of UDM_DEREGISTRATION requests per SNSSAI

Expression

{{snssai}}

```
sum by (snssai) (rate(smfcc_udm_deregistration_attempts{}[$rate_interval]))
```

Errors Panels

SDM GET SM DATA error rates by cause

Description

Rate of SDM_GET_SM_DATA errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_sdm_get_sm_data_failures{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))
```

UDM REGISTRATION error rates by cause

Description

Rate of UDM_REGISTRATION errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_udm_registration_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

UDM DEREGISTRATION error rates by cause

Description

Rate of UDM_DEREGISTRATION errors by cause

Expression

{{causecode}}

```
sum by (causecode) (rate(smfcc_udm_deregistration_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

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Summary

Dashboard Name	Panel Name
HSMF Call Control	Pdu Session
HSMF Call Control	PDU Session Create Rates
HSMF Call Control	PDU Session Create Result %
HSMF Call Control	Current PDU Session Create Success Rate
HSMF Call Control	PDU Session Modification Rates
HSMF Call Control	PDU Session Modification Result %
HSMF Call Control	Current PDU Session Modification Success Rate
HSMF Call Control	Vsmf Init Pdu Session Release Rates
HSMF Call Control	Vsmf Init Pdu Session Release Result %

Dashboard Name	Panel Name
HSMF Call Control	Current Vsmf Init Pdu Session Release Success Rate
HSMF Call Control	Hsmf Init Pdu Session Release Rates
HSMF Call Control	Hsmf Init Pdu Session Release Result %
HSMF Call Control	Current Hsmf Init Pdu Session Release Success Rate
HSMF Call Control	Ue Init Pdu Session Modify Rates
HSMF Call Control	Ue Init PDU Session Modify Result %
HSMF Call Control	Current Ue Init Pdu Session Modify success rate
HSMF Call Control	Ue Init Pdu Session Release Rates
HSMF Call Control	Ue Init Pdu Session Release result %
HSMF Call Control	Ue Init Pdu Session Release success rate
HSMF Call Control	Handover Procedures
HSMF Call Control	Policy Control
HSMF Call Control	Converged Charging
HSMF Call Control	SMF Interface
HSMF Call Control	Pdu Session Create Rates
HSMF Call Control	Pdu Session Create Result %
HSMF Call Control	Current Pdu Session Create Success Rate
HSMF Call Control	Hsmf Init Pdu Session Update Rates
HSMF Call Control	Hsmf Init Pdu Session Update Result %
HSMF Call Control	Current Hsmf Init Pdu Session Update Success Rate
HSMF Call Control	Pdu Session Release Rates
HSMF Call Control	Pdu Session Release Result %
HSMF Call Control	Current Pdu Session Release Success Rate
HSMF Call Control	Vsmf Init Pdu Session Update Rates
HSMF Call Control	Vsmf Init Pdu Session Update Result %
HSMF Call Control	Current Vsmf Init Pdu Session Update Success Rate
HSMF Call Control	Pdu Session Status Notify Rates
HSMF Call Control	Pdu Session Status Notify Result %
HSMF Call Control	Pdu Session Status Notify Success Rate

Dashboard Name	Panel Name
HSMF Procedures Dashboard	
HSMF Procedures Dashboard	PDU Session Establishment - Attempts
HSMF Procedures Dashboard	PDU Session Establishment - Success %
HSMF Procedures Dashboard	PDU Session Establishment - Successes
HSMF Procedures Dashboard	Pdu Session Modification - Attempts
HSMF Procedures Dashboard	Pdu Session Modification - Success %
HSMF Procedures Dashboard	Pdu Session Modification - Successes
HSMF Procedures Dashboard	Vsmf Init Pdu Session Release - Attempts
HSMF Procedures Dashboard	Vsmf Init Pdu Session Release - Success %
HSMF Procedures Dashboard	Vsmf Init Pdu Session Release - Successes
HSMF Procedures Dashboard	Hsmf Initiated Pdu Session Release - Attempts
HSMF Procedures Dashboard	Hsmf Initiated Pdu Session Release - Success %
HSMF Procedures Dashboard	Hsmf Initiated Pdu Session Release - Successes
HSMF Procedures Dashboard	Ue Init Pdu Session Modification - Attempts
HSMF Procedures Dashboard	Ue Init Pdu Session Modification - Success %
HSMF Procedures Dashboard	Ue Init Pdu Session Modification - Successes
HSMF Procedures Dashboard	Ue Init Pdu Session Release - Attempts
HSMF Procedures Dashboard	Ue Init Pdu Session Release - Success %
HSMF Procedures Dashboard	Ue Init Pdu Session Release - Successes
HSMF Procedures Dashboard	5GS To EPS Handover - Attempts
HSMF Procedures Dashboard	5GS To EPS Handover - Success %
HSMF Procedures Dashboard	5GS To EPS Handover - Successes
HSMF Procedures Dashboard	EPS To 5GS Idle Mode Handover - Attempts
HSMF Procedures Dashboard	EPS To 5GS Idle Mode Handover - Success %
HSMF Procedures Dashboard	EPS To 5GS Idle Mode Handover - Successes
HSMF Procedures Dashboard	Pcf Update Policy Rules - Attempts
HSMF Procedures Dashboard	Pcf Update Policy Rules - Success %
HSMF Procedures Dashboard	Pcf Update Policy Rules - Successes
HSMF Procedures Dashboard	Pcf Init Session Terminate Request - Attempts
HSMF Procedures Dashboard	Pcf Init Session Terminate Request - Success %

Dashboard Name	Panel Name
HSMF Procedures Dashboard	Pcf Init Session Terminate Request - Successes
HSMF Procedures Dashboard	Chf Server Notification - Attempts
HSMF Procedures Dashboard	Chf Server Notification - Success %
HSMF Procedures Dashboard	Chf Server Notification - Successes
HSMF Procedures Dashboard	Session Idle Procedure - Attempts
HSMF Procedures Dashboard	Session Idle Procedure - Success %
HSMF Procedures Dashboard	Session Idle Procedure - Successes
HSMF Procedures Dashboard	Vsmf Error Indication - Attempts
HSMF Procedures Dashboard	Vsmf Error Indication - Success %
HSMF Procedures Dashboard	Vsmf Error Indication - Successes
PGWC Errors Dashboard	PGW Initial Attach Procedures failed
PGWC Errors Dashboard	PGW Delete Session Procedures failed
PGWC Errors Dashboard	NW Initiated Release Procedures failed
PGWC Errors Dashboard	Modify Bearer Request Procedures failed
PGWC Errors Dashboard	PCF Initiated Policy Modification Procedures failed
PGWC Errors Dashboard	Pcf Update Policy Rules Procedures failed
PGWC Errors Dashboard	PCF Initiated Terminate failed
PGWC Errors Dashboard	PCF Initiated Dedicated Bearer Activation Procedures failed
PGWC Errors Dashboard	PCF Initiated Dedicated Bearer Update Procedures failed
PGWC Errors Dashboard	PCF Initiated Dedicated Bearer Delete Procedures failed
PGWC Errors Dashboard	PGW Initiated Dedicated Bearer Activation Procedure Post IRAT HO failed
PGWC Errors Dashboard	HSS Initiated Modify Bearer Command Procedures failed
PGWC Errors Dashboard	MME Initiated Delete Bearer Command Procedures failed
PGWC Errors Dashboard	Error Indication Procedures failed
PGWC Roaming Errors Dashboard	
PGWC Roaming Errors Dashboard	PGW Initial Attach Procedures failed
PGWC Roaming Errors Dashboard	PGW Delete Session Procedures failed

Dashboard Name	Panel Name
PGWC Roaming Errors Dashboard	NW Initiated Release Procedures failed
PGWC Roaming Errors Dashboard	Modify Bearer Request Procedures failed
PGWC Roaming Errors Dashboard	PCF Initiated Policy Modification Procedures failed
PGWC Roaming Errors Dashboard	Pcf Update Policy Rules Procedures failed
PGWC Roaming Errors Dashboard	PCF Initiated Terminate failed
PGWC Roaming Errors Dashboard	PCF Initiated Dedicated Bearer Activation Procedures failed
PGWC Roaming Errors Dashboard	PCF Initiated Dedicated Bearer Update Procedures failed
PGWC Roaming Errors Dashboard	PCF Initiated Dedicated Bearer Delete Procedures failed
PGWC Roaming Errors Dashboard	PGW Initiated Dedicated Bearer Activation Procedure Post IRAT HO failed
PGWC Roaming Errors Dashboard	HSS Initiated Modify Bearer Command Procedures failed
PGWC Roaming Errors Dashboard	MME Initiated Delete Bearer Command Procedures failed
PGWC Roaming Errors Dashboard	Error Indication Procedures failed
SGW Procedures dashboard	
SGW Procedures dashboard	Initial Attach - Attempts
SGW Procedures dashboard	Initial Attach - Success %
SGW Procedures dashboard	Initial Attach - Successful
SGW Procedures dashboard	UE Initiated Detach - Attempts
SGW Procedures dashboard	UE Initiated Detach - Success %
SGW Procedures dashboard	UE Initiated Detach - Successful
SGW Procedures dashboard	NW Initiated Release - Attempts
SGW Procedures dashboard	NW Initiated Release - Success %
SGW Procedures dashboard	NW Initiated Release - Successful
SGW Procedures dashboard	NW Initiated Dedicated Bearer Activation - Attempts
SGW Procedures dashboard	NW Initiated Dedicated Bearer Activation - Success %
SGW Procedures dashboard	NW Initiated Dedicated Bearer Activation Procedures - Successful
SGW Procedures dashboard	Modify Bearer Request - Attempts
SGW Procedures dashboard	Modify Bearer Request - Success %

Dashboard Name	Panel Name
SGW Procedures dashboard	Modify Bearer Request - Successful
SGW Procedures dashboard	Create Bearer Request - Attempts
SGW Procedures dashboard	Create Bearer Request - Success %
SGW Procedures dashboard	Create Bearer Request - Successful
SGW Procedures dashboard	Delete Bearer Request - Attempts
SGW Procedures dashboard	Delete Bearer Request - Success %
SGW Procedures dashboard	Delete Bearer Request - Successful
SGW Procedures dashboard	HSS Initiated Modify Bearer Command - Attempts
SGW Procedures dashboard	HSS Initiated Modify Bearer Command - Success %
SGW Procedures dashboard	HSS Initiated Modify Bearer Command - Successful
SGW Procedures dashboard	NW Initiated Dedicated Bearer Delete - Attempts
SGW Procedures dashboard	NW Initiated Dedicated Bearer Delete - Success %
SGW Procedures dashboard	NW Initiated Dedicated Bearer Delete - Successful
SGW Procedures dashboard	NW Initiated Dedicated Bearer Update - Attempts
SGW Procedures dashboard	NW Initiated Dedicated Bearer Update - Success %
SGW Procedures dashboard	NW Initiated Dedicated Bearer Update - Successful
SGW Procedures dashboard	MME Initiated Delete Bearer Command - Attempts
SGW Procedures dashboard	MME Initiated Delete Bearer Command - Success %
SGW Procedures dashboard	MME Initiated Delete Bearer Command - Successful
SMF AMF Events Interface Dashboard	
SMF AMF Events Interface Dashboard	AMF Event Subscription Create - Attempts
SMF AMF Events Interface Dashboard	AMF Event Subscription Create - Success %
SMF AMF Events Interface Dashboard	AMF Event Subscription Create - Successes
SMF AMF Events Interface Dashboard	AMF Event Subscription Modify - Attempts
SMF AMF Events Interface Dashboard	AMF Event Subscription Modify - Success %
SMF AMF Events Interface Dashboard	AMF Event Subscription Modify - Successes
SMF AMF Events Interface Dashboard	AMF Event Subscription Delete - Attempts
SMF AMF Events Interface Dashboard	AMF Event Subscription Delete - Success %
SMF AMF Events Interface Dashboard	AMF Event Subscription Delete - Successes

Dashboard Name	Panel Name
SMF AMF Events Interface Dashboard	AMF Event Notification - Attempts
SMF AMF Events Interface Dashboard	AMF Event Notification - Success %
SMF AMF Events Interface Dashboard	AMF Event Notification - Successes
SMF AMF Events Interface Dashboard	SMF-AMF EVTS Transaction Rate per Pod
SMF AMF Events Interface Dashboard	SMF-AMF EVTS Transaction Rate Per Service
SMF AMF Events Interface Dashboard	SMF-AMF EVTS Event Subscription Stats per Pod
SMF CHF Interface Procedure Dashboard	CHF Create Session - Attempts
SMF CHF Interface Procedure Dashboard	CHF Create Session - Success %
SMF CHF Interface Procedure Dashboard	CHF Create Session - Successes
SMF CHF Interface Procedure Dashboard	CHF Update Session - Attempts
SMF CHF Interface Procedure Dashboard	CHF Update Session - Success %
SMF CHF Interface Procedure Dashboard	CHF Update Session - Successes
SMF CHF Interface Procedure Dashboard	CHF Release Session - Attempts
SMF CHF Interface Procedure Dashboard	CHF Release Session - Success %
SMF CHF Interface Procedure Dashboard	CHF Release Session - Successes
SMF ETCD Interfaces Dashboard	ETCD Gtpc Abort lease grant requests
SMF ETCD Interfaces Dashboard	ETCD Gtpc Abort lease grant - Success %
SMF ETCD Interfaces Dashboard	ETCD Gtpc Abort lease grant responses
SMF ETCD Interfaces Dashboard	ETCD Gtpc Abort put with lease grant requests
SMF ETCD Interfaces Dashboard	ETCD Gtpc Abort put with lease grant - Success %
SMF ETCD Interfaces Dashboard	ETCD Gtpc Abort put with lease grant responses
SMF Errors Dashboard	PDU Session Create SM Context Failures
SMF Errors Dashboard	PDU Session Update SM Context Failures
SMF Errors Dashboard	PDU Session Release SM Context Failures
SMF Errors Dashboard	PDU Session Create Failures
SMF Errors Dashboard	PDU Session Update Failures
SMF Errors Dashboard	PDU Session Release Failures
SMF Errors Dashboard	PDU Session Create Client Failures
SMF Errors Dashboard	PDU Session Update Client Failures

Dashboard Name	Panel Name
SMF Errors Dashboard	PDU Session Release Client Failures
SMF Errors Dashboard	PDU Session Establishment Failures
SMF Errors Dashboard	UE Deregistration Failures
SMF Errors Dashboard	AN Release Failures
SMF Errors Dashboard	Service Request Failures
SMF Errors Dashboard	NW Initiated Release Failures
SMF Errors Dashboard	Xn Handover With No UPF change Failures
SMF Errors Dashboard	SmPolicy Control Service Create Failures
SMF Errors Dashboard	SmPolicy Control Service Update Failures
SMF Errors Dashboard	SmPolicy Control Service Delete Failures
SMF Errors Dashboard	CHF Create Session - Failures
SMF Errors Dashboard	CHF Update Session - Failures
SMF Errors Dashboard	CHF Release Session - Failures
SMF Errors Dashboard	UE Initiated Qos Modify - Failures
SMF Errors Dashboard	PCF Initiated Qos Modify - Failures
SMF Errors Dashboard	UE Initiated PDU Session Release - Failures
SMF Errors Dashboard	Network Initiated Service Request - Failures
SMF Errors Dashboard	PCF Initiated Terminate - Failures
SMF Errors Dashboard	Network of Usage Report - Failures
SMF Errors Dashboard	N1N2 Message Transfer Requests Failures
SMF Handover Procedures Dashboard	
SMF Handover Procedures Dashboard	Xn Handover Initiated Release - Attempts
SMF Handover Procedures Dashboard	Xn Handover Initiated Release - Success %
SMF Handover Procedures Dashboard	Xn Handover Initiated Release - Successes
SMF Handover Procedures Dashboard	Xn Handover Without UPF Relocation - Attempts
SMF Handover Procedures Dashboard	Xn Handover Without UPF Relocation - Successes
SMF Handover Procedures Dashboard	Xn Handover With UPF Relocation - Attempts
SMF Handover Procedures Dashboard	Xn Handover With UPF Relocation - Success %
SMF Handover Procedures Dashboard	Xn Handover With UPF Relocation - Successes

Dashboard Name	Panel Name
SMF Handover Procedures Dashboard	N2 Handover Without UPF Relocation - Attempts
SMF Handover Procedures Dashboard	N2 Handover Without UPF Relocation - Success %
SMF Handover Procedures Dashboard	N2 Handover Without UPF Relocation - Successes
SMF Handover Procedures Dashboard	N2 Handover With UPF Relocation - Attempts
SMF Handover Procedures Dashboard	N2 Handover With UPF Relocation - Success %
SMF Handover Procedures Dashboard	N2 Handover With UPF Relocation - Successes
SMF Handover Procedures Dashboard	N2 Handover Without AMF Mobility - Attempts
SMF Handover Procedures Dashboard	N2 Handover Without AMF Mobility - Success %
SMF Handover Procedures Dashboard	N2 Handover Without AMF Mobility - Successes
SMF Handover Procedures Dashboard	N2 Handover With AMF Mobility - Attempts
SMF Handover Procedures Dashboard	N2 Handover With AMF Mobility - Success %
SMF Handover Procedures Dashboard	N2 Handover With AMF Mobility - Successes
SMF Handover Procedures Dashboard	5GS to EPS Handover with N26 interface - Attempts
SMF Handover Procedures Dashboard	5GS to EPS Handover with N26 interface - Success %
SMF Handover Procedures Dashboard	5GS to EPS Handover with N26 interface - Successes
SMF Handover Procedures Dashboard	5GS to EPS Idle Mode Mobility with N26 interface - Attempts
SMF Handover Procedures Dashboard	5GS to EPS Idle Mode Mobility with N26 interface - Success %
SMF Handover Procedures Dashboard	5GS to EPS Idle Mode Mobility with N26 interface - Successes
SMF Handover Procedures Dashboard	EPS to 5GS Handover with N26 interface without I-UPF Relocation - Attempts
SMF Handover Procedures Dashboard	EPS to 5GS Handover with N26 interface without I-UPF Relocation - Success %
SMF Handover Procedures Dashboard	EPS to 5GS Handover with N26 interface without I-UPF Relocation - Successes
SMF Handover Procedures Dashboard	EPS to 5GS Handover with N26 interface with I-UPF Relocation - Attempts
SMF Handover Procedures Dashboard	EPS to 5GS Handover with N26 interface with I-UPF Relocation - Success %
SMF Handover Procedures Dashboard	EPS to 5GS Handover with N26 interface with I-UPF Relocation - Successes

Dashboard Name	Panel Name
SMF Handover Procedures Dashboard	EPS to 5GS Idle Mode Mobility with N26 interface - Attempts
SMF Handover Procedures Dashboard	EPS to 5GS Idle Mode Mobility with N26 interface - Success %
SMF Handover Procedures Dashboard	EPS to 5GS Idle Mode Mobility with N26 interface - Successes
SMF Handover Procedures Dashboard	EPS to ePDG Handover - Attempts
SMF Handover Procedures Dashboard	EPS to ePDG Handover - Success %
SMF Handover Procedures Dashboard	EPS to ePDG Handover - Successes
SMF Handover Procedures Dashboard	5GS to ePDG Handover - Attempts
SMF Handover Procedures Dashboard	5GS to ePDG Handover - Success %
SMF Handover Procedures Dashboard	5GS to ePDG Handover - Successes
SMF Handover Procedures Dashboard	ePDG to 5GS Handover - Attempts
SMF Handover Procedures Dashboard	ePDG to 5GS Handover - Success %
SMF Handover Procedures Dashboard	ePDG to 5GS Handover - Successes
SMF Handover Procedures Dashboard	ePDG to 5GS Idle Mode Mobility - Attempts
SMF Handover Procedures Dashboard	ePDG to 5GS Idle Mode Mobility- Success %
SMF Handover Procedures Dashboard	ePDG to 5GS Idle Mode Mobility - Successes
SMF Handover Procedures Dashboard	5GS to ePDG Idle Mode Mobility - Attempts
SMF Handover Procedures Dashboard	5GS to ePDG Idle Mode Mobility- Success %
SMF Handover Procedures Dashboard	5GS to ePDG Idle Mode Mobility - Successes
SMF Handover Procedures Dashboard	ePDG to EPS Handover - Attempts
SMF Handover Procedures Dashboard	ePDG to EPS Handover - Success %
SMF Handover Procedures Dashboard	ePDG to EPS Handover - Successes
SMF Handover Procedures Dashboard	4G TAU X2 HO with SGW Change - Attempts
SMF Handover Procedures Dashboard	4G TAU X2 HO with SGW Change - Success %
SMF Handover Procedures Dashboard	4G TAU X2 HO with SGW Change - Successes
SMF Handover Procedures Dashboard	3G RAU HO with SGW Change - Attempts
SMF Handover Procedures Dashboard	3G RAU HO with SGW Change - Success %
SMF Handover Procedures Dashboard	3G RAU HO with SGW Change - Successes
SMF Handover Procedures Dashboard	5GS to EPS Handover without N26 interface - Attempts

Dashboard Name	Panel Name
SMF Handover Procedures Dashboard	5GS to EPS Handover without N26 interface - Success %
SMF Handover Procedures Dashboard	5GS to EPS Handover without N26 interface - Successes
SMF Handover Procedures Dashboard	5GS to EPS Handover without N26 Resource Release- Attempts
SMF Handover Procedures Dashboard	5GS to EPS Handover without N26 Resource Release - Success %
SMF Handover Procedures Dashboard	5GS to EPS Handover without N26 Resource Release - Successes
SMF Handover Procedures Dashboard	5GS to EPS Idle Mode Mobility without N26 interface - Attempts
SMF Handover Procedures Dashboard	5GS to EPS Idle Mode Mobility without N26 interface - Success %
SMF Handover Procedures Dashboard	5GS to EPS Idle Mode Mobility without N26 interface - Successes
SMF Handover Procedures Dashboard	EPS to 5GS Handover without N26 interface without I-UPF Relocation - Attempts
SMF Handover Procedures Dashboard	EPS to 5GS Handover without N26 interface without I-UPF Relocation - Success %
SMF Handover Procedures Dashboard	EPS to 5GS Handover without N26 interface without I-UPF Relocation - Successes
SMF Handover Procedures Dashboard	EPS to 5GS Handover without N26 interface with I-UPF Relocation - Attempts
SMF Handover Procedures Dashboard	EPS to 5GS Handover without N26 interface with I-UPF Relocation - Success %
SMF Handover Procedures Dashboard	EPS to 5GS Handover without N26 interface with I-UPF Relocation - Successes
SMF Handover Procedures Dashboard	3G to 4G Handover - Attempts
SMF Handover Procedures Dashboard	3G to 4G Handover - Success %
SMF Handover Procedures Dashboard	3G to 4G Handover - Successes
SMF Handover Procedures Dashboard	4G to 3G Handover - Attempts
SMF Handover Procedures Dashboard	4G to 3G Handover - Success %
SMF Handover Procedures Dashboard	4G to 3G Handover- Successes
SMF Handover Procedures Dashboard	Inter SGSN Handover - Attempts
SMF Handover Procedures Dashboard	Inter SGSN Handover - Success %

Dashboard Name	Panel Name
SMF Handover Procedures Dashboard	Inter SGSN Handover Success
SMF Handover Procedures Dashboard	Inter Plmn Handover with Roaming Status Change - Attempts
SMF Handover Procedures Dashboard	Inter Plmn Handover with Roaming Status Change - Success %
SMF Handover Procedures Dashboard	Inter Plmn Handover With Roaming Status Change - Successful
SMF Interfaces Dashboard	PDU Session Create SM Context - Attempts
SMF Interfaces Dashboard	SM Context Create - Success %
SMF Interfaces Dashboard	PDU Session Create SM Context - Successes
SMF Interfaces Dashboard	PDU Session Update SM Context - Attempts
SMF Interfaces Dashboard	SM Context Update - Success %
SMF Interfaces Dashboard	PDU Session Update SM Context - Successes
SMF Interfaces Dashboard	PDU Session Release SM Context - Attempts
SMF Interfaces Dashboard	SM Context Release - Success %
SMF Interfaces Dashboard	PDU Session Release SM Context - Successes
SMF Interfaces Dashboard	PDU Session Create - Attempts
SMF Interfaces Dashboard	PDU Session Create - Success %
SMF Interfaces Dashboard	PDU Session Create - Successes
SMF Interfaces Dashboard	PDU Session Update - Attempts
SMF Interfaces Dashboard	PDU Session Update - Success %
SMF Interfaces Dashboard	PDU Session Update - Successes
SMF Interfaces Dashboard	PDU Session Release - Attempts
SMF Interfaces Dashboard	PDU Session Release - Success %
SMF Interfaces Dashboard	PDU Session Release - Successes
SMF Interfaces Dashboard	PDU Session Create Client - Attempts
SMF Interfaces Dashboard	PDU Session Create Client - Success %
SMF Interfaces Dashboard	PDU Session Create Client - Successes
SMF Interfaces Dashboard	PDU Session Update Client - Attempts
SMF Interfaces Dashboard	PDU Session Update Client - Success %

Dashboard Name	Panel Name
SMF Interfaces Dashboard	PDU Session Update Client - Successes
SMF Interfaces Dashboard	PDU Session Release Client - Attempts
SMF Interfaces Dashboard	PDU Session Release Client - Success %
SMF Interfaces Dashboard	PDU Session Release Client - Successes
SMF Interfaces Dashboard	N1N2 Message Transfer Requests - Attempts
SMF Interfaces Dashboard	N1N2 Message Transfer Requests - Success %
SMF Interfaces Dashboard	N1N2 Message Transfer Requests - Successes
SMF Interfaces Dashboard	SMF-CC Transaction Rate per Pod
SMF Interfaces Dashboard	SMF-CC Transaction Rate Per Service
SMF Interfaces Dashboard	SMFCC Session Activation Stats per Pod
SMF PCF Interfaces Dashboard	Create SmPolicy Control Service - Attempts
SMF PCF Interfaces Dashboard	Create SmPolicy Control Service - Success %
SMF PCF Interfaces Dashboard	Create SmPolicy Control Service - Successes
SMF PCF Interfaces Dashboard	Update SmPolicy Control Service - Attempts
SMF PCF Interfaces Dashboard	Update SmPolicy Control Service - Success %
SMF PCF Interfaces Dashboard	Update SmPolicy Control Service - Successes
SMF PCF Interfaces Dashboard	Delete SmPolicy Control Service - Attempts
SMF PCF Interfaces Dashboard	Delete SmPolicy Control Service - Success %
SMF PCF Interfaces Dashboard	Delete SmPolicy Control Service - Successes
SMF PCF Interfaces Dashboard	SMF-PCF Transaction Rate per Pod
SMF PCF Interfaces Dashboard	SMF-PCF Transaction Rate Per Service
SMF PCF Interfaces Dashboard	SMF-PCF Session Activation Stats per Pod
SMF Procedures Dashboard	PDU Session Establishment - Attempts
SMF Procedures Dashboard	PDU Session Establishment - Success %
SMF Procedures Dashboard	PDU Session Establishment - Successes
SMF Procedures Dashboard	UE Deregistration - Attempts
SMF Procedures Dashboard	UE Deregistration - Success %
SMF Procedures Dashboard	UE Deregistration - Successes
SMF Procedures Dashboard	AN Release - Attempts

Dashboard Name	Panel Name
SMF Procedures Dashboard	AN Release - Success %
SMF Procedures Dashboard	AN Release - Successes
SMF Procedures Dashboard	Service Request - Attempts
SMF Procedures Dashboard	Service Request - Success %
SMF Procedures Dashboard	Service Request - Successes
SMF Procedures Dashboard	NW Initiated Release - Attempts
SMF Procedures Dashboard	NW Initiated Release - Success %
SMF Procedures Dashboard	NW Initiated Release - Successes
SMF Procedures Dashboard	Xn Handover With No UPF Change - Attempts
SMF Procedures Dashboard	Xn Handover With No UPF Change - Success %
SMF Procedures Dashboard	Xn Handover With No UPF Change - Successes
SMF Procedures Dashboard	UE Initiated Qos Modify - Attempts
SMF Procedures Dashboard	UE Initiated Qos Modify - Success %
SMF Procedures Dashboard	UE Initiated Qos Modify - Successes
SMF Procedures Dashboard	PCF Initiated Qos Modify - Attempts
SMF Procedures Dashboard	PCF Initiated Qos Modify - Success %
SMF Procedures Dashboard	PCF Initiated Qos Modify - Successes
SMF Procedures Dashboard	UE Initiated PDU Session Release - Attempts
SMF Procedures Dashboard	UE Initiated PDU Session Release - Success %
SMF Procedures Dashboard	UE Initiated PDU Session Release - Successes
SMF Procedures Dashboard	Network Initiated Service Request - Attempts
SMF Procedures Dashboard	Network Initiated Service Request - Success %
SMF Procedures Dashboard	Network Initiated Service Request - Successes
SMF Procedures Dashboard	PCF Initiated Terminate - Attempts
SMF Procedures Dashboard	PCF Initiated Terminate - Success %
SMF Procedures Dashboard	PCF Initiated Terminate - Successes
SMF Procedures Dashboard	Network of Usage Report - Attempts
SMF Procedures Dashboard	Network of Usage Report - Success %
SMF Procedures Dashboard	Network of Usage Report - Successes
SMF Procedures Dashboard	AMF Event Notification - Attempts

Dashboard Name	Panel Name
SMF Procedures Dashboard	AMF Event Notification - Success %
SMF Procedures Dashboard	AMF Event Notification - Successes
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	ETCD add Inband Radius Tunnel Attempts
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	ETCD add Inband Radius Tunnel Success
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	ETCD add Inband Radius Tunnel Failure
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	ETCD del Inband Radius Tunnel Attempts
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	ETCD del Inband Radius Tunnel Success
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	ETCD del Inband Radius Tunnel Failure
SMF Radius InBand Tunnel ETCD Add/Delete DashBoard	SMFCC RADIUS INBAND TUNNEL TIMER DETAILS
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Tunnel Create Proc Attempts
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Tunnel Create Proc Success
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Tunnel Create Proc Failure
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Gtpu Error Indication Request
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Tunnel Delete Proc Attempts
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Tunnel Delete Proc Success
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Tunnel Delete Proc Failure
SMF Radius InBand Tunnel Procedure DashBoard	Inband Radius Gtpu Error Indication Response Sent
SMF Radius InBand Tunnel Procedure DashBoard	SMFCC RADIUS INBAND TUNNEL TIMER DETAILS

Dashboard Name	Panel Name
SMF Radius InBand Tunnel Timer DashBoard	Inband Radius Tunnel Timer Start Attempts
SMF Radius InBand Tunnel Timer DashBoard	Inband Radius Tunnel Timer Start Success
SMF Radius InBand Tunnel Timer DashBoard	Inband Radius Tunnel Timer Start Failure
SMF Radius InBand Tunnel Timer DashBoard	Inband Radius Tunnel Timer Fire
SMF Radius InBand Tunnel Timer DashBoard	Inband Radius Tunnel Timer Restarts
SMF Radius InBand Tunnel Timer DashBoard	SMFCC RADIUS INBAND TUNNEL TIMER DETAILS
SMFCC Current Pdu Session Count by RAT Type	PDU Sessions by RAT Type
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Start Request Sent
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Start Request Send Failed
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Start Response Success
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Start Response Failures
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Start Request Timeouts
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Start Message Details
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Interim Request Sent
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Interim Request Send Failed
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Interim Response Success
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Interim Response Failures

Dashboard Name	Panel Name
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Interim Request Timeouts
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Interim Message Details
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Stop Request Sent
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Stop Request Send Failed
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Stop Response Success
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Stop Response Failures
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Stop Request Timeouts
SMFCC RADIUS [ACCOUNTING] INTERFACE STATS	Radius Accounting Stop Message Details
SMFCC RADIUS [AUTHENTICATION] INTERFACE STATS	Radius Access Request Sent
SMFCC RADIUS [AUTHENTICATION] INTERFACE STATS	Radius Access Request Send Failed
SMFCC RADIUS [AUTHENTICATION] INTERFACE STATS	Radius Access Accept Received
SMFCC RADIUS [AUTHENTICATION] INTERFACE STATS	Radius Access Reject Received
SMFCC RADIUS [AUTHENTICATION] INTERFACE STATS	Radius Access Request Sent Vs Radius Access Accept Received vs Radius Reject Received
SMFCC-PFCP Session Bundling Dashboard	Failure %
SMFCC-PFCP Session Bundling Dashboard	PFCP number of bundled message sent failure - \$cause
SMFCC-PFCP Session Bundling Dashboard	Failure %
SMFCC-PFCP Session Bundling Dashboard	PFCP number of bundled message decode failure - \$causeR
SMFCC-PFCP Session Bundling Dashboard	Pfcp number of bundled messages sent \$bundle_trigger
SMFCC-PFCP Session Bundling Dashboard	Pfcp number of bundled messages sent \$peer_ip
SMFCC-PFCP Session Bundling Dashboard	PFCP number of bundled messages received \$peer_ip

Dashboard Name	Panel Name
SMFCC-PFCP Session Bundling Dashboard	\$peer_ip
SMFCC-PFCP Session Bundling Dashboard	Message Size in Bundle
SMFCC-PFCP Session Bundling Dashboard	\$peer_ip
SMFCC-PFCP Session Bundling Dashboard	Number of messages in Bundle
SMFCC-PFCP Session Bundling Dashboard	\$peer_ip
SMFCC-PFCP Session Bundling Dashboard	Duration of Bundling in Sent Bundle
SMFCC-PFCP Session Bundling Dashboard	\$peer_ip
SMFCC-PFCP Session Bundling Dashboard	Message Size in Received Bundle
SMFCC-PFCP Session Bundling Dashboard	\$peer_ip
SMFCC-PFCP Session Bundling Dashboard	Number of Messages in Bundle Received
SMFCC-PFCP Sessions Procedures Dashboard	Session Establishment - Attempts
SMFCC-PFCP Sessions Procedures Dashboard	Session Establishment - Success %
SMFCC-PFCP Sessions Procedures Dashboard	Session Establishment - Successes
SMFCC-PFCP Sessions Procedures Dashboard	Session Establishment - Failures
SMFCC-PFCP Sessions Procedures Dashboard	Session Modification - Attempts
SMFCC-PFCP Sessions Procedures Dashboard	Session Modification - Success %
SMFCC-PFCP Sessions Procedures Dashboard	Session Modification - Successes
SMFCC-PFCP Sessions Procedures Dashboard	Session Modification - Failures
SMFCC-PFCP Sessions Procedures Dashboard	Session Deletion - Attempts
SMFCC-PFCP Sessions Procedures Dashboard	Session Deletion - Success %
SMFCC-PFCP Sessions Procedures Dashboard	Session Deletion - Successes

Dashboard Name	Panel Name
SMFCC-PFCP Sessions Procedures Dashboard	Session Deletion - Failures
SMFCC-PFCP Sessions Protocols Dashboard	Session Establishment - Requests Sent
SMFCC-PFCP Sessions Protocols Dashboard	Session Establishment - Retransmitted Requests Sent
SMFCC-PFCP Sessions Protocols Dashboard	Session Establishment - Responses Received
SMFCC-PFCP Sessions Protocols Dashboard	Session Modification - Requests Sent
SMFCC-PFCP Sessions Protocols Dashboard	Session Modification - Retransmitted Requests Sent
SMFCC-PFCP Sessions Protocols Dashboard	Session Modification - Responses Received
SMFCC-PFCP Sessions Protocols Dashboard	Session Deletion - Requests Sent
SMFCC-PFCP Sessions Protocols Dashboard	Session Deletion - Retransmitted Requests Sent
SMFCC-PFCP Sessions Protocols Dashboard	Session Deletion - Responses Received
Smfcc Interface Round Trip Duration	\$interface
Smfcc Interface Round Trip Duration	Round Trip Time
Smfcc Interface Timeout Count	Total failures due to timeout
Smfcc Interface Timeout Count	AMF Timeout count
Smfcc Interface Timeout Count	PCF Timeout count
Smfcc Interface Timeout Count	CHF Timeout count
Smfcc Interface Timeout Count	UDM Timeout count
Smfcc Interface Timeout Count	GTPC Timeout Count
Smfcc Procedure Duration Statistics	\$procedure
Smfcc Procedure Duration Statistics	SUCCESS
Smfcc Procedure Duration Statistics	FAILURE

Dashboard Name	Panel Name
Smfcc Procedure Duration Statistics	DROPPED
Smfcc Radius Inband Tunnel ETCD put/delete range Dashboard	ETCD put/delete range summary-Inband Radius Tunnel Put Request
Smfcc Radius Inband Tunnel ETCD put/delete range Dashboard	ETCD put/delete range summary-Inband Radius Tunnel Put Response
Smfcc Radius Inband Tunnel ETCD put/delete range Dashboard	ETCD put/delete range summary-Inband Radius Tunnel Delete Range Request
Smfcc Radius Inband Tunnel ETCD put/delete range Dashboard	ETCD put/delete range summary-Inband Radius Tunnel Delete Range Response
Smfcc Radius Inband Tunnel ETCD put/delete range Dashboard	ETCD put/delete range details-Panel Title
Smfcc SGW GTPC Interface Dashboard	Create Session - Attempts
Smfcc SGW GTPC Interface Dashboard	Create Session - Success %
Smfcc SGW GTPC Interface Dashboard	Create Session - Successful
Smfcc SGW GTPC Interface Dashboard	Delete Session - Attempts
Smfcc SGW GTPC Interface Dashboard	Delete Session - Success %
Smfcc SGW GTPC Interface Dashboard	Delete Session - Successful
Smfcc SGW GTPC Interface Dashboard	Modify Bearer Request - Attempts
Smfcc SGW GTPC Interface Dashboard	Modify Bearer Request - Success %
Smfcc SGW GTPC Interface Dashboard	Modify Bearer Request - Successful
Smfcc SGW GTPC Interface Dashboard	Create Bearer Request - Attempts
Smfcc SGW GTPC Interface Dashboard	Create Bearer Request - Success %
Smfcc SGW GTPC Interface Dashboard	Create Bearer Request - Successful
Smfcc SGW GTPC Interface Dashboard	Create Bearer Request Forwarded to MME - Attempts
Smfcc SGW GTPC Interface Dashboard	Create Bearer Request forwarded to MME - Success %
Smfcc SGW GTPC Interface Dashboard	Create Bearer Request Forwarded to MME - Successful
Smfcc SGW GTPC Interface Dashboard	Delete Bearer Request - Attempts
Smfcc SGW GTPC Interface Dashboard	Delete Bearer Request - Success %
Smfcc SGW GTPC Interface Dashboard	Delete Bearer Request - Successful
VSMF Call Control	SM Context

Dashboard Name	Panel Name
VSMF Call Control	SMF Interface
VSMF Call Control	Handover Procedures
VSMF Call Control	PDU Session
VSMF Call Control	UPF Sessions
VSMF Procedures Dashboard	PDU Session Establishment - Attempts
VSMF Procedures Dashboard	PDU Session Establishment - Success %
VSMF Procedures Dashboard	PDU Session Establishment - Successes
VSMF Procedures Dashboard	UE Deregistration - Attempts
VSMF Procedures Dashboard	UE Deregistration - Success %
VSMF Procedures Dashboard	UE Deregistration - Successes
VSMF Procedures Dashboard	AN Release - Attempts
VSMF Procedures Dashboard	AN Release - Success %
VSMF Procedures Dashboard	AN Release - Successes
VSMF Procedures Dashboard	Service Request - Attempts
VSMF Procedures Dashboard	Service Request - Success %
VSMF Procedures Dashboard	Service Request - Successes
VSMF Procedures Dashboard	NW Initiated Release - Attempts
VSMF Procedures Dashboard	NW Initiated Release - Success %
VSMF Procedures Dashboard	NW Initiated Release - Successes
VSMF Procedures Dashboard	Xn Handover - Attempts
VSMF Procedures Dashboard	Xn Handover - Success %
VSMF Procedures Dashboard	Xn Handover - Successes
VSMF Procedures Dashboard	UE Initiated Modify - Attempts
VSMF Procedures Dashboard	UE Initiated Modify - Success %
VSMF Procedures Dashboard	UE Initiated Modify - Successes
VSMF Procedures Dashboard	HSMF Initiated Qos Modify - Attempts
VSMF Procedures Dashboard	HSMF Initiated Qos Modify - Success %
VSMF Procedures Dashboard	HSMF Initiated Qos Modify - Successes
VSMF Procedures Dashboard	UE Initiated PDU Session Release - Attempts

Dashboard Name	Panel Name
VSMF Procedures Dashboard	UE Initiated PDU Session Release - Success %
VSMF Procedures Dashboard	UE Initiated PDU Session Release - Successes
VSMF Procedures Dashboard	Network Initiated Service Request - Attempts
VSMF Procedures Dashboard	Network Initiated Service Request - Success %
VSMF Procedures Dashboard	Network Initiated Service Request - Successes
VSMF Procedures Dashboard	HSMF Initiated Release - Attempts
VSMF Procedures Dashboard	HSMF Initiated Release - Success %
VSMF Procedures Dashboard	HSMF Initiated Release - Successes
VSMF Procedures Dashboard	EPS to 5G Idle Handover - Attempts
VSMF Procedures Dashboard	EPS to 5G Idle Handover - Success %
VSMF Procedures Dashboard	EPS to 5G Idle Handover - Successes
VSMF Procedures Dashboard	Network of Usage Report - Attempts
VSMF Procedures Dashboard	Network of Usage Report - Success %
VSMF Procedures Dashboard	Network of Usage Report - Successes
VSMF Procedures Dashboard	Error Indication - Attempts
VSMF Procedures Dashboard	Error Indication - Success %
VSMF Procedures Dashboard	Error Indication - Successes
VSMF Procedures Dashboard	Session Idle - Attempts
VSMF Procedures Dashboard	Session Idle - Success %
VSMF Procedures Dashboard	Session Idle - Successes

GGSN GTPC Interface Dashboard

Description

Dashboard showing GGSN GTPC interface messages statistics

GGSN Procedures dashboard

Description

Dashboard showing GGSN Procedure statistics

GGSN Roaming Procedures dashboard

Description

Dashboard showing GGSN Roaming Procedures Statistics

HSMF Call Control

Description

Summary information for operations on all HSMF interfaces

Panels

Pdu Session

PDU Session Create Rates

Description

Rate of PDU Session Create requests

Expression

Attempts

```
sum(rate(smfcc_hsmf_pdu_session_establishment_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_hsmf_pdu_session_establishment_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_hsmf_pdu_session_establishment_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Create Result %

Description

PDU Session Create result %

Expression

Success

```
((sum(rate(smfcc_hsmf_pdu_session_establishment_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_pdu_session_establishment_attempts{dnn=~"
$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_hsmf_pdu_session_establishment_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_pdu_session_establishment_attempts{dnn=~"
$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Create Success Rate

Description

Current PDU Session Create Success rate

Expression

```
(sum(rate(smfcc_hsmf_pdu_session_establishment_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_hsmf_pdu_session_establishment_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Modification Rates

Description

Rate of PDU Session Modification requests

Expression

Attempts

```
sum(rate(smfcc_hsmf_pdu_session_modification_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_hsmf_pdu_session_modification_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_hsmf_pdu_session_modification_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

PDU Session Modification Result %

Description

PDU Session Modification result %

Expression

Success

```
((sum(rate(smfcc_hsmf_pdu_session_modification_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_hsmf_pdu_session_modification_attempts{dnn=~"$
dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_hsmf_pdu_session_modification_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_hsmf_pdu_session_modification_attempts{dnn=~"$
dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current PDU Session Modification Success Rate

Description

Current PDU Session Modification success rate

Expression

```
(sum(rate(smfcc_hsmf_pdu_session_modification_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_hsmf_pdu_session_modification_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Vsmf Init Pdu Session Release Rates

Description

Rate of vsmf init pdu session release requests

Expression

Attempts

```
sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Vsmf Init Pdu Session Release Result %

Description

Vsmf Init Pdu Session Release result %

Expression

Success

```
((sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current Vsmf Init Pdu Session Release Success Rate

Description

Current Vsmf Init Pdu Session Release success rate

Expression

```
(sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Hsmf Init Pdu Session Release Rates

Description

Rate of Hsmf Init Pdu Session Release requests

Expression

Attempts

```
sum(rate(smfcc_hsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_hsmf_initiated_pdu_session_release_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Hsmf Init Pdu Session Release Result %

Description

Hsmf Init Pdu Session Release result %

Expression

Success

```
((sum(rate(smfcc_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_initiated_pdu_session_release_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_hsmf_initiated_pdu_session_release_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_initiated_pdu_session_release_attempts{dn
n=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Current Hsmf Init Pdu Session Release Success Rate

Description

Current Hsmf Init Pdu Session Release success rate

Expression

```
(sum(rate(smfcc_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
```

```
sum(rate(smfcc_hsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))
```

Ue Init Pdu Session Modify Rates

Description

Rate of Ue Init Pdu Session Modification requests

Expression

Attempts

```
sum(rate(smfcc_hsmf_ue_initiated_modify_attempts{dnn=~"$dnn", snssai=~"$snssai"}  
[$rate_interval]))
```

Successes

```
sum(rate(smfcc_hsmf_ue_initiated_modify_successes{dnn=~"$dnn", snssai=~"$snssai"}  
[$rate_interval]))
```

Failures

```
sum(rate(smfcc_hsmf_ue_initiated_modify_failures{dnn=~"$dnn", snssai=~"$snssai"}  
[$rate_interval]))
```

Ue Init PDU Session Modify Result %

Description

Ue Init Pdu Session Modify result %

Expression

Success

```
((sum(rate(smfcc_hsmf_ue_initiated_modify_successes{dnn=~"$dnn",  
snssai=~"$snssai"}  
[$rate_interval])))/(sum(rate(smfcc_hsmf_ue_initiated_modify_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_hsmf_ue_initiated_modify_failures{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])))/(sum(rate(smfcc_hsmf_ue_initiated_modify_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))))*100
```

Current Ue Init Pdu Session Modify success rate

Description

Current Ue Init Pdu Session Modify success rate

Expression

```
(sum(rate(smfcc_hsmf_ue_initiated_modify_successes{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_hsmf_ue_initiated_modify_attempts{dnn=~"$dnn", snssai=~"$snssai"}
[$rate_interval]))
```

Ue Init Pdu Session Release Rates

Description

Rate of Ue Init Pdu Session Release requests

Expression**Attempts**

```
sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Failures


```
sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_failures{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Ue Init Pdu Session Release result %

Description

Ue Init Pdu Session Release result %

Expression

Success

```
((sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_attempts
{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_failures{dnn=~"$dnn",
snssai=~"$snssai"}
[$rate_interval]))/(sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_attempts
{dnn=~"$dnn", snssai=~"$snssai"}[$rate_interval]))))*100
```

Ue Init Pdu Session Release success rate

Description

Ue Init Pdu Session Release success rate

Expression

```
(sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_hsmf_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[$rate_interval]))
```

Handover Procedures

Policy Control

Converged Charging

SMF Interface

Pdu Session Create Rates

Description

Rate of Pdu Session Create Requests

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_create_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_create_successes{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_create_failures{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Pdu Session Create Result %

Description

Pdu Session Create result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_create_successes{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Success

```
((sum(rate(smfcc_nsmf_pdu_session_create_failures{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_create_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Current Pdu Session Create Success Rate**Description**

Current Pdu Session Create Success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_create_successes{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_nsmf_pdu_session_create_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Hsmf Init Pdu Session Update Rates**Description**

Rate of Hsmf Init Pdu Session Update Requests

Expression**Attempts**

```
sum(rate(smfcc_nsmf_pdu_session_update_tx_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_update_tx_successes{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_update_tx_failures{dnn=~"$dnn", snssai=~"$snssai",
smfSessionType=~"HSMF"}[$rate_interval]))
```

Hsmf Init Pdu Session Update Result %

Description

Hsmf Init Pdu Session Update result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_update_tx_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_tx_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_update_tx_failures{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_tx_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Current Hsmf Init Pdu Session Update Success Rate

Description

Current Hsmf Init Pdu Session Update Success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_update_tx_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_update_tx_attempts{dnn=~"$dnn", snssai=~"$snssai",
smfSessionType=~"HSMF"}[$rate_interval]))
```

Pdu Session Release Rates

Description

Pdu Session Release Rates

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_release_request_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_release_request_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_release_request_failures{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Pdu Session Release Result %

Description

Pdu Session Release result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_release_request_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_request_attempts{dnn=
~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_release_request_failures{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_release_request_attempts{dnn=
~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Current Pdu Session Release Success Rate

Description

Current Pdu Session Release Success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_release_request_successes{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])) OR on() vector(0)) / sum(rate(smfcc_nsmf_pdu_session_release_request_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Vsmf Init Pdu Session Update Rates

Description

Vsmf Init Pdu Session Update Rates

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_update_rx_attempts{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_update_rx_successes{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_update_rx_failures{dnn=~"$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Vsmf Init Pdu Session Update Result %

Description

Vsmf Init Pdu Session Update result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_update_rx_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_rx_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_update_rx_failures{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_update_rx_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Current Vsmf Init Pdu Session Update Success Rate

Description

Current Vsmf Init Pdu Session Update Success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_update_rx_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_update_rx_attempts{dnn=~"$dnn", snssai=~"$snssai",
smfSessionType=~"HSMF"}[$rate_interval]))
```

Pdu Session Status Notify Rates

Description

Pdu Session Status Notify Rates

Expression

Attempts

```
sum(rate(smfcc_nsmf_pdu_session_status_notify_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Successes

```
sum(rate(smfcc_nsmf_pdu_session_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Failures

```
sum(rate(smfcc_nsmf_pdu_session_status_notify_failures{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

Pdu Session Status Notify Result %

Description

Pdu Session Status Notify result %

Expression

Success

```
((sum(rate(smfcc_nsmf_pdu_session_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_status_notify_attempts{dnn=~"
$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(smfcc_nsmf_pdu_session_status_notify_failures{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}
[$rate_interval])))/(sum(rate(smfcc_nsmf_pdu_session_status_notify_attempts{dnn=~"
$dnn", snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))))*100
```

Pdu Session Status Notify Success Rate

Description

Current Pdu Session Status Notify Success rate

Expression

```
(sum(rate(smfcc_nsmf_pdu_session_status_notify_successes{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval])) OR on() vector(0)) /
sum(rate(smfcc_nsmf_pdu_session_status_notify_attempts{dnn=~"$dnn",
snssai=~"$snssai", smfSessionType=~"HSMF"}[$rate_interval]))
```

HSMF Procedures Dashboard

Description

SMF Procedures Dashboard

Panels

PDU Session Establishment - Attempts

Description

Number of PDU Session Establishment Procedures attempted

Expression

```
sum(smfcc_hsmf_pdu_session_establishment_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Establishment - Success %

Description

PDU Session Establishment - Success %

Expression

```
((sum(smfcc_hsmf_pdu_session_establishment_successes{dnn=~"$dnn",
snssai=~"$snssai"}))/sum(smfcc_hsmf_pdu_session_establishment_attempts{dnn=~"$dnn",
snssai=~"$snssai"})))*100
```

PDU Session Establishment - Successes

Description

Number of PDU Session Establishment Procedures successful

Expression

```
sum(smfcc_hsmf_pdu_session_establishment_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Pdu Session Modification - Attempts

Description

Number of Pdu Session Modification Procedures attempted

Expression

```
sum(smfcc_hsmf_pdu_session_modification_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Pdu Session Modification - Success %

Description

Pdu Session Modification - Success %

Expression

```
((sum(smfcc_hsmf_pdu_session_modification_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_hsmf_pdu_session_modification_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

Pdu Session Modification - Successes

Description

Number of Pdu Session Modification Procedures successful

Expression

```
sum(smfcc_hsmf_pdu_session_modification_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Vsmf Init Pdu Session Release - Attempts

Description

Number of Vsmf Init Pdu Session Release Procedures attempted

Expression

```
sum(smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Vsmf Init Pdu Session Release - Success %

Description

Vsmf Init Pdu Session Release - Success %

Expression

```
((sum(smfcc_hsmf_vsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_hsmf_vsmf_initiated_pdu_session_release_attempts{d  
nn=~"$dnn", snssai=~"$snssai"})))*100
```

Vsmf Init Pdu Session Release - Successes

Description

Number of Vsmf Init Pdu Session Release Procedures successful

Expression

```
sum(smfcc_hsmf_vsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Hsmf Initiated Pdu Session Release - Attempts

Description

Number of Hsmf Init Pdu Session Release Procedures attempted

Expression

```
sum(smfcc_hsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Hsmf Initiated Pdu Session Release - Success %

Description

Hsmf Initiated Pdu Session Release - Success %

Expression

```
((sum(smfcc_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn", snssai=~"$snssai"})))/(sum(smfcc_hsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn", snssai=~"$snssai"}))*100
```

Hsmf Initiated Pdu Session Release - Successes

Description

Hsmf Initiated Pdu Session Release Procedures successful

Expression

```
sum(smfcc_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Ue Init Pdu Session Modification - Attempts

Description

Number of Ue Initiated Pdu Session Modification Procedures attempted

Expression

```
sum(smfcc_hsmf_ue_initiated_modify_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Ue Init Pdu Session Modification - Success %

Description

Ue Init Pdu Session Modification - Success %

Expression

```
((sum(smfcc_hsmf_ue_initiated_modify_successes{dnn=~"$dnn", snssai=~"$snssai"})))/(sum(smfcc_hsmf_ue_initiated_modify_attempts{dnn=~"$dnn", snssai=~"$snssai"}))*100
```

Ue Init Pdu Session Modification - Successes

Description

Number of Ue Init Pdu Session Modification Procedures successful

Expression

```
sum(smfcc_hsmf_ue_initiated_modify_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Ue Init Pdu Session Release - Attempts

Description

Number of Ue Init Pdu Session Release Procedures attempted

Expression

```
sum(smfcc_hsmf_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Ue Init Pdu Session Release - Success %

Description

Ue Init Pdu Session Release - Success %

Expression

```
((sum(smfcc_hsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_hsmf_ue_initiated_pdu_session_release_attempts{dnn  
=~"$dnn", snssai=~"$snssai"})))*100
```

Ue Init Pdu Session Release - Successes

Description

Ue Init Pdu Session Release Procedures - Success

Expression

```
sum(smfcc_hsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

5GS To EPS Handover - Attempts

Description

Number of 5GS To EPS Procedures attempted

Expression

```
sum(smfcc_hsmf_5gs_to_eps_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS To EPS Handover - Success %

Description

5GS To EPS Handover - Success %

Expression

```
((sum(smfcc_hsmf_5gs_to_eps_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_hsmf_5gs_to_eps_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

5GS To EPS Handover - Successes

Description

Number of 5GS To EPS Handover Procedures successful

Expression

```
sum(smfcc_hsmf_5gs_to_eps_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS To 5GS Idle Mode Handover - Attempts

Description

Number of EPS To 5Gs Procedures attempted

Expression

```
sum(smfcc_hsmf_eps_to_5gs_idle_mobility_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS To 5GS Idle Mode Handover - Success %

Description

EPS To 5GS Idle Mode Handover - Success %

Expression

```
((sum(smfcc_hsmf_eps_to_5gs_idle_mobility_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_hsmf_eps_to_5gs_idle_mobility_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

EPS To 5GS Idle Mode Handover - Successes

Description

Number of EPS To 5GS Idle Mode Handover Procedures successful

Expression

```
sum(smfcc_hsmf_eps_to_5gs_idle_mobility_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Pcf Update Policy Rules - Attempts

Description

Number of Pcf Update Policy Rules Procedures attempted

Expression

```
sum(smfcc_process_hsmf_pcf_update_policy_rules_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Pcf Update Policy Rules - Success %

Description

Pcf Update Policy Rules - Success %

Expression

```
((sum(smfcc_process_hsmf_pcf_update_policy_rules_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_process_hsmf_pcf_update_policy_rules_attempts{dnn=  
~"$dnn", snssai=~"$snssai"}))) * 100
```

Pcf Update Policy Rules - Successes

Description

Number of Pcf Update Policy Rules Procedures successful

Expression

```
sum(smfcc_process_hsmf_pcf_update_policy_rules_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Pcf Init Session Terminate Request - Attempts

Description

Number of Pcf Init Session Terminate Request Procedures attempted

Expression

```
sum(smfcc_hsmf_pcf_initiated_terminate_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Pcf Init Session Terminate Request - Success %

Description

Pcf Init Session Terminate Request - Success %

Expression

```
((sum(smfcc_hsmf_pcf_initiated_terminate_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_hsmf_pcf_initiated_terminate_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))) * 100
```

Pcf Init Session Terminate Request - Successes

Description

Number of Pcf Init Session Terminate Request Procedures successful

Expression

```
sum(smfcc_hsmf_pcf_initiated_terminate_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Chf Server Notification - Attempts

Description

Number of Chf Server Notification Procedures attempted

Expression

```
sum(smfcc_hsmf_chf_server_notification_procedure_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Chf Server Notification - Success %

Description

Chf Server Notification - Success %

Expression

```
((sum(smfcc_hsmf_chf_server_notification_procedure_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_hsmf_chf_server_notification_procedure_attempts{dn  
n=~"$dnn", snssai=~"$snssai"})))*100
```

Chf Server Notification - Successes

Description

Chf Server Notification Procedures successful

Expression

```
sum(smfcc_hsmf_chf_server_notification_procedure_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Session Idle Procedure - Attempts

Description

Number of Session Idle Procedure attempted

Expression

```
sum(smfcc_hsmf_session_idle_procedure_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Session Idle Procedure - Success %

Description

Session Idle Procedure - Success %

Expression

```
((sum(smfcc_hsmf_session_idle_procedure_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_hsmf_session_idle_procedure_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))*100
```

Session Idle Procedure - Successes

Description

Number of Session Idle Procedure successful

Expression

```
sum(smfcc_hsmf_session_idle_procedure_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Vsmf Error Indication - Attempts

Description

Number of vsmf error indication procedure attempts

Expression

```
sum(smfcc_hsmf_error_indication_procedure_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Vsmf Error Indication - Success %

Description

Vsmf Error Indication - Success %

Expression

```
((sum(smfcc_hsmf_error_indication_procedure_successes{dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_hsmf_error_indication_procedure_attempts{dnn=~"$dnn", snssai=~"$snssai"})))*100
```

Vsmf Error Indication - Successes

Description

Number of Vsmf Error Indication successes

Expression

```
sum(smfcc_hsmf_error_indication_procedure_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

PGW Internal dashboard

Description

PGW Internal Dashboard

PGW Procedures dashboard

Description

PGW Procedures Dashboard

PGW Roaming Procedures dashboard

Description

PGW Roaming Procedures Dashboard

PGWC Errors Dashboard

Description

PGWC Errors Dashboard

Panels

PGW Initial Attach Procedures failed

Description

Number of Initial Attach Procedures failed

Expression

```
sum(smfcc_pgw_initial_attach_failures{dnn=~"$dnn", snssai=~"$snssai",
ratType=~"$ratType", causecode=~"$causecode"})
```

PGW Delete Session Procedures failed

Description

Number of PGW Delete Session Procedures failed

Expression

```
sum(smfcc_pgw_ue_initiated_detach_failures{dnn=~"$dnn", snssai=~"$snssai",
ratType=~"$ratType", causecode=~"$causecode"})
```

NW Initiated Release Procedures failed

Description

Number of NW Initiated Release Procedures failed

Expression

```
sum(smfcc_pgw_nw_initiated_release_failures{dnn=~"$dnn", snssai=~"$snssai",
ratType=~"$ratType", causecode=~"$causecode"})
```

Modify Bearer Request Procedures failed

Description

Number of Modify Bearer Request Procedures failed

Expression

```
sum(smfcc_pgw_modify_bearer_request_failures{dnn=~"$dnn", snssai=~"$snssai",
ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Policy Modification Procedures failed

Description

Number of PCF Initiated Policy Modification Procedures failed

Expression

```
sum(smfcc_pgw_pcf_initiated_policy_modification_failures{dnn=~"$dnn",
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

Pcf Update Policy Rules Procedures failed

Description

Number of Process Pcf Update Policy Rules Procedures failed

Expression

```
sum(smfcc_pgw_process_pcf_update_policy_rules_failures{dnn=~"$dnn",
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Terminate failed

Description

Number of PCF Initiated Terminate Procedures failed

Expression

```
sum(smfcc_pgw_pcf_initiated_terminate_failures{dnn=~"$dnn", snssai=~"$snssai",
ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Dedicated Bearer Activation Procedures failed

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures failed

Expression

```
sum(smfcc_pgw_pcf_initiated_dedicated_bearer_activation_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Dedicated Bearer Update Procedures failed

Description

Number of PCF Initiated Dedicated Bearer Update Procedures failed

Expression

```
sum(smfcc_pgw_pcf_initiated_dedicated_bearer_update_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Dedicated Bearer Delete Procedures failed

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures failed

Expression

```
sum(smfcc_pgw_pcf_initiated_dedicated_bearer_delete_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PGW Initiated Dedicated Bearer Activation Procedure Post IRAT HO failed

Description

Number of PGW Initiated Dedicated Bearer Activation Procedure Post IRAT HO failed

Expression

```
sum(smfcc_pgw_initiated_legacy_dedicated_bearer_activation_failures_ho{dnn=~"$dnn"  
, snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

HSS Initiated Modify Bearer Command Procedures failed

Description

Number of HSS Initiated Modify Bearer Command Procedures failed

Expression

```
sum(smfcc_pgw_hss_initiated_modify_bearer_command_failures{dnn=~"$dnn",
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

MME Initiated Delete Bearer Command Procedures failed

Description

Number of MME Initiated Delete Bearer Command Procedures failed

Expression

```
sum(smfcc_pgw_mme_initiated_delete_bearer_command_failures{dnn=~"$dnn",
snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

Error Indication Procedures failed

Description

Number of Error Indication Procedures failed

Expression

```
sum(smfcc_pgw_error_indication_procedure_failures{dnn=~"$dnn", snssai=~"$snssai",
ratType=~"ratType", causecode=~"$causecode"})
```

PGWC Roaming Errors Dashboard

Description

PGWC Roaming Errors Dashboard

Panels

PGW Initial Attach Procedures failed

Description

Number of Initial Attach Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_initial_attach_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

PGW Delete Session Procedures failed

Description

Number of PGW Delete Session Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_ue_initiated_detach_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

NW Initiated Release Procedures failed

Description

Number of NW Initiated Release Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_nw_initiated_release_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

Modify Bearer Request Procedures failed

Description

Number of Modify Bearer Request Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_modify_bearer_request_failures{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

PCF Initiated Policy Modification Procedures failed

Description

Number of PCF Initiated Policy Modification Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_pcf_initiated_policy_modification_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

Pcf Update Policy Rules Procedures failed

Description

Number of Process Pcf Update Policy Rules Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_process_pcf_update_policy_rules_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Terminate failed

Description

Number of PCF Initiated Terminate Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_pcf_initiated_terminate_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Dedicated Bearer Activation Procedures failed

Description

Number of PCF Initiated Dedicated Bearer Activation Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_activation_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

PCF Initiated Dedicated Bearer Update Procedures failed

Description

Number of PCF Initiated Dedicated Bearer Update Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_update_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

PCF Initiated Dedicated Bearer Delete Procedures failed

Description

Number of PCF Initiated Dedicated Bearer Delete Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_pcf_initiated_dedicated_bearer_delete_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

PGW Initiated Dedicated Bearer Activation Procedure Post IRAT HO failed

Description

Number of PGW Initiated Dedicated Bearer Activation Procedure Post IRAT HO failed

Expression

```
sum(smfcc_pgw_initiated_legacy_dedicated_bearer_activation_roaming_session_failure_s_ho{dnn=~"$dnn", snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

HSS Initiated Modify Bearer Command Procedures failed

Description

Number of HSS Initiated Modify Bearer Command Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_hss_initiated_modify_bearer_command_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"$ratType", causecode=~"$causecode"})
```

MME Initiated Delete Bearer Command Procedures failed

Description

Number of MME Initiated Delete Bearer Command Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_mme_initiated_delete_bearer_command_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

Error Indication Procedures failed

Description

Number of Error Indication Procedures failed

Expression

```
sum(smfcc_pgw_roaming_session_error_indication_procedure_failures{dnn=~"$dnn", snssai=~"$snssai", ratType=~"ratType", causecode=~"$causecode"})
```

SAEGW GTPC Interface Dashboard

Description

Dashboard showing SAEGW GTPC interface messages statistics

SAEGW Procedures dashboard

Description

SAEGW Procedures Dashboard

SGW Procedures dashboard

Description

SGW Procedures Dashboard

Panels

Description

Expression

Initial Attach - Attempts

Description

Number of Initial Attach Procedures attempted

Expression

```
sum(smfcc_sgw_initial_attach_attempts{dnn=~"$dnn", ratType=~"$ratType"})
```

Initial Attach - Success %

Description

Meter showing percentage of Initial Attach procedure which are successful

Expression

```
sum(smfcc_sgw_initial_attach_successes{dnn=~"$dnn", ratType=~"$ratType"})/sum(smfcc_sgw_initial_attach_attempts{dnn=~"$dnn", ratType=~"$ratType"})*100
```

Initial Attach - Successful

Description

Number of Initial Attach Procedures which are Successful

Expression

```
sum(smfcc_sgw_initial_attach_successes{dnn=~"$dnn", ratType=~"$ratType"})
```

UE Initiated Detach - Attempts

Description

Number of UE initiated Detach Procedures attempted

Expression

```
sum(smfcc_sgw_ue_initiated_detach_attempts{dnn=~"$dnn", ratType=~"$ratType"})
```

UE Initiated Detach - Success %

Description

Meter showing percentage of UE Initiated Detach procedure which are successful

Expression

```
((sum(smfcc_sgw_ue_initiated_detach_successes{dnn=~"$dnn",  
ratType=~"$ratType"})))/(sum(smfcc_sgw_ue_initiated_detach_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})))*100
```

UE Initiated Detach - Successful

Description

Number of UE initiated Detach Procedures which are Successful

Expression

```
sum(smfcc_pgw_ue_initiated_detach_successes{dnn=~"$dnn", ratType=~"$ratType"})
```

NW Initiated Release - Attempts

Description

Number of Network Initiated Release Procedures attempted

Expression

```
sum(smfcc_sgw_nw_initiated_release_attempts{dnn=~"$dnn", ratType=~"$ratType"})
```

NW Initiated Release - Success %

Description

Meter showing percentage of NW Initiated Release procedure which are successful

Expression

```
((sum(smfcc_sgw_nw_initiated_release_successes{dnn=~"$dnn",  
ratType=~"$ratType"}))/sum(smfcc_sgw_nw_initiated_release_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})))*100
```

NW Initiated Release - Successful

Description

Number of Network Initiated Release Procedures which are Successful

Expression

```
sum(smfcc_sgw_nw_initiated_release_successes{dnn=~"$dnn",ratType=~"$ratType"})
```

NW Initiated Dedicated Bearer Activation - Attempts

Description

Number of NW Initiated Dedicated Bearer Activation Procedures attempted

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_activation_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})
```

NW Initiated Dedicated Bearer Activation - Success %

Description

Meter showing percentage of NW Initiated Dedicated Bearer Activation procedure which are successful

Expression

```
((sum(smfcc_sgw_nw_initiated_dedicated_bearer_activation_successes{dnn=~"$dnn",  
ratType=~"$ratType"}))/sum(smfcc_sgw_nw_initiated_dedicated_bearer_activation_att  
empts{dnn=~"$dnn", ratType=~"$ratType"})))*100
```

NW Initiated Dedicated Bearer Activation Procedures - Successful

Description

Number of NW Initiated Dedicated Bearer Activation Procedures which are Successful

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_activation_successes{dnn=~"$dnn",ratType=~"$ratType"})
```

Modify Bearer Request - Attempts**Description**

Number of Modify Bearer Request Procedures attempted

Expression

```
sum(smfcc_sgw_modify_bearer_request_attempts{dnn=~"$dnn", ratType=~"$ratType"})
```

Modify Bearer Request - Success %**Description**

Meter showing percentage of Modify Bearer Request procedure which are Successful

Expression

```
sum(smfcc_sgw_modify_bearer_request_successes{dnn=~"$dnn",ratType=~"$ratType"})/sum(smfcc_sgw_modify_bearer_request_attempts{dnn=~"$dnn",ratType=~"$ratType"})*100
```

Modify Bearer Request - Successful**Description**

Number of Modify Bearer Request Procedures Successful

Expression

```
sum(smfcc_sgw_modify_bearer_request_successes{dnn=~"$dnn", ratType=~"$ratType"})
```

Create Bearer Request - Attempts

Description

Number of Create Bearer Request Procedures attempted

Expression

```
sum(smfcc_sgw_create_bearer_request_attempts{dnn=~"$dnn", ratType=~"$ratType"})
```

Create Bearer Request - Success %

Description

Meter showing percentage of Create Bearer Request procedure which are Successful

Expression

```
sum(smfcc_sgw_create_bearer_request_successes{dnn=~"$dnn",  
ratType=~"$ratType"})/sum(smfcc_sgw_create_bearer_request_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})*100
```

Create Bearer Request - Successful

Description

Number of Create Bearer Request Procedures Successful

Expression

```
sum(smfcc_sgw_create_bearer_request_successes{dnn=~"$dnn", ratType=~"$ratType"})
```

Delete Bearer Request - Attempts

Description

Number of Delete Bearer Request Procedures attempted

Expression

```
sum(smfcc_sgw_delete_bearer_request_attempts{dnn=~"$dnn", ratType=~"$ratType"})
```

Delete Bearer Request - Success %

Description

Meter showing percentage of Delete Bearer Request procedure which are Successful

Expression

```
sum(smfcc_sgw_delete_bearer_request_successes{dnn=~"$dnn",  
ratType=~"$ratType"})/sum(smfcc_sgw_delete_bearer_request_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})*100
```

Delete Bearer Request - Successful

Description

Number of Delete Bearer Request Procedures Successful

Expression

```
sum(smfcc_sgw_delete_bearer_request_successes{dnn=~"$dnn", ratType=~"$ratType"})
```

HSS Initiated Modify Bearer Command - Attempts

Description

HSS Initiated Modify Bearer Command Procedures attempted

Expression

```
sum(smfcc_sgw_hss_initiated_modify_bearer_command_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})
```

HSS Initiated Modify Bearer Command - Success %

Description

Meter showing percentage of HSS Initiated Modify Bearer Command procedure which are successful

Expression

```
sum(smfcc_sgw_hss_initiated_modify_bearer_command_successes{dnn=~"$dnn",  
ratType=~"$ratType"})/sum(smfcc_sgw_hss_initiated_modify_bearer_command_attempts{d  
nn=~"$dnn", ratType=~"$ratType"})*100
```

HSS Initiated Modify Bearer Command - Successful

Description

Number of HSS Initiated Modify Bearer Command Procedures which are Successful

Expression

```
sum(smfcc_sgw_hss_initiated_modify_bearer_command_successes{dnn=~"$dnn",  
ratType=~"$ratType"})
```

NW Initiated Dedicated Bearer Delete - Attempts

Description

Number of NW Initiated Dedicated Bearer Delete Procedures attempted

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_delete_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})
```

NW Initiated Dedicated Bearer Delete - Success %

Description

Meter showing percentage of NW Initiated Dedicated Bearer Delete procedure which are successful

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_delete_successes{dnn=~"$dnn",  
ratType=~"$ratType"})/sum(smfcc_sgw_nw_initiated_dedicated_bearer_delete_attempts{  
dnn=~"$dnn", ratType=~"$ratType"})*100
```

NW Initiated Dedicated Bearer Delete - Successful

Description

Number of NW Initiated Dedicated Bearer Delete Procedures which are Successful

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_delete_successes{dnn=~"$dnn",  
ratType=~"$ratType"})
```

NW Initiated Dedicated Bearer Update - Attempts

Description

Number of NW Initiated Dedicated Bearer Update Procedures attempted

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_update_attempts{dnn=~"$dnn",  
ratType=~"$ratType"})
```

NW Initiated Dedicated Bearer Update - Success %

Description

Meter showing percentage of Number of NW Initiated Dedicated Bearer Update procedure which are successful

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_update_successes{dnn=~"$dnn",  
ratType=~"$ratType"})/sum(smfcc_sgw_nw_initiated_dedicated_bearer_update_attempts{  
dnn=~"$dnn", ratType=~"$ratType"})*100
```

NW Initiated Dedicated Bearer Update - Successful

Description

Number of NW Initiated Dedicated Bearer Update Procedures which are Successful

Expression

```
sum(smfcc_sgw_nw_initiated_dedicated_bearer_update_successes{dnn=~"$dnn",  
ratType=~"$ratType"})
```

MME Initiated Delete Bearer Command - Attempts

Description

Number of MME Initiated Delete Bearer Command Procedures attempted

Expression

```
sum(smfcc_sgw_mme_initiated_delete_bearer_command_attempts{dnn=~"$dnn",
ratType=~"$ratType"})
```

MME Initiated Delete Bearer Command - Success %

Description

Meter showing percentage of MME Initiated Delete Bearer Command procedure which are successful

Expression

```
sum(smfcc_sgw_mme_initiated_delete_bearer_command_successes{dnn=~"$dnn",
ratType=~"$ratType"})/sum(smfcc_sgw_mme_initiated_delete_bearer_command_attempts{d
nn=~"$dnn", ratType=~"$ratType"})*100
```

MME Initiated Delete Bearer Command - Successful

Description

Number of MME Initiated Delete Bearer Command Procedures which are Successful

Expression

```
sum(smfcc_sgw_mme_initiated_delete_bearer_command_successes{dnn=~"$dnn",
ratType=~"$ratType"})
```

SMF AMF Events Interface Dashboard

Description

SMF AMF Events Interface Dashboard

Panels

AMF Event Subscription Create - Attempts

Description

Number of AMF Event Subscription Create requests attempted

Expression

```
sum(smfcc_namf_evts_subscr_create_attempts{dnn=~"$dnn", snssai=~"$snssai",  
peerid=~"$peerid"})
```

AMF Event Subscription Create - Success %

Description

AMF Event Subscription Create - Success %

Expression

```
((sum(smfcc_namf_evts_subscr_create_successes{dnn=~"$dnn", snssai=~"$snssai",  
peerid=~"$peerid"})))/(sum(smfcc_namf_evts_subscr_create_attempts{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid"})))*100
```

AMF Event Subscription Create - Successes

Description

Number of AMF Event Subscription Create requests successful

Expression

```
sum(smfcc_namf_evts_subscr_create_successes{dnn=~"$dnn", snssai=~"$snssai",  
peerid=~"$peerid"})
```

AMF Event Subscription Modify - Attempts

Description

Number of AMF Event Subscription Modify requests attempted

Expression

```
sum(smfcc_namf_evts_subscr_modify_attempts{dnn=~"$dnn", snssai=~"$snssai",  
peerid=~"$peerid"})
```

AMF Event Subscription Modify - Success %

Description

AMF Event Subscription Modify - Success %

Expression

```
((sum(smfcc_namf_evts_subscr_modify_successes{dnn=~"$dnn", snssai=~"$snssai", peerid=~"$peerid"})))/(sum(smfcc_namf_evts_subscr_modify_attempts{dnn=~"$dnn", snssai=~"$snssai", peerid=~"$peerid"}))*100
```

AMF Event Subscription Modify - Successes

Description

Number of AMF Event Subscription Modify requests successful

Expression

```
sum(smfcc_namf_evts_subscr_modify_successes{dnn=~"$dnn", snssai=~"$snssai", peerid=~"$peerid"})
```

AMF Event Subscription Delete - Attempts

Description

Number of AMF Event Subscription Delete requests attempted

Expression

```
sum(smfcc_namf_evts_subscr_delete_attempts{dnn=~"$dnn", snssai=~"$snssai", peerid=~"$peerid"})
```

AMF Event Subscription Delete - Success %

Description

AMF Event Subscription Delete - Success %

Expression

```
((sum(smfcc_namf_evts_subscr_delete_successes{dnn=~"$dnn", snssai=~"$snssai", peerid=~"$peerid"})))/(sum(smfcc_namf_evts_subscr_delete_attempts{dnn=~"$dnn", snssai=~"$snssai", peerid=~"$peerid"}))*100
```

AMF Event Subscription Delete - Successes

Description

Number of AMF Event Subscription Delete requests successful

Expression

```
sum(smfcc_namf_evts_subscr_delete_successes{dnn=~"$dnn", snssai=~"$snssai",  
peerid=~"$peerid"})
```

AMF Event Notification - Attempts

Description

Number of AMF Event Notification requests attempted

Expression

```
sum(smfcc_namf_evts_event_notification_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

AMF Event Notification - Success %

Description

AMF Event Notification - Success %

Expression

```
((sum(smfcc_namf_evts_event_notification_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_namf_evts_event_notification_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

AMF Event Notification - Successes

Description

Number of AMF Event Notification requests successful

Expression

```
sum(smfcc_namf_evts_event_notification_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

SMF-AMF EVTS Transaction Rate per Pod

Description

SMF-AMF EVTS Transaction Rate per Pod

Expression

```
round(sum by (pod)(rate(smfcc_namf_evts_subscr_create_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m])) + sum by (pod)
(rate(smfcc_namf_evts_subscr_modify_attempts{dnn=~"$dnn", snssai=~"$snssai",
peerid=~"$peerid"}[1m])) + sum by (pod)
(rate(smfcc_namf_evts_subscr_delete_attempts{dnn=~"$dnn", snssai=~"$snssai",
peerid=~"$peerid"}[1m])))
```

SMF-AMF EVTS Transaction Rate Per Service

Description

SMF-AMF EVTS Transaction Rate Per Service

Expression

smfcc-{{version}}

```
round(sum by (version)(rate(smfcc_namf_evts_subscr_create_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m])) + sum by (version)
(rate(smfcc_namf_evts_subscr_modify_attempts{dnn=~"$dnn", snssai=~"$snssai",
peerid=~"$peerid"}[1m])) + sum by (version)
(rate(smfcc_namf_evts_subscr_delete_attempts{dnn=~"$dnn", snssai=~"$snssai",
peerid=~"$peerid"}[1m])))
```

SMF-AMF EVTS Event Subscription Stats per Pod

Description

SMF-AMF EVTS Event Subscription Stats per Pod

Expression

Session Activation Attempt-{{pod}}


```
sum(smfcc_namf_evts_subscr_create_attempts{dnn=~"$dnn", snssai=~"$snssai",
peerid=~"$peerid"}) by (pod)
```

Session Activation Failure-{{pod}}

```
sum(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}) by (pod) -
sum(smfcc_nsmf_smpolicy_control_create_service_success{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}) by (pod)
```

SMF CHF Interface Procedure Dashboard

Description

SMF CHF Interface Procedure Dashboard

Panels

CHF Create Session - Attempts

Description

Number of CHF Create Session attempted

Expression

```
sum(smfcc_nchf_create_session_attempts{peerid=~"$peerid", dnn=~"$dnn",
snssai=~"$snssai"})
```

CHF Create Session - Success %

Description

CHF Create Session - Success %

Expression

```
((sum(smfcc_nchf_create_session_success{peerid=~"$peerid", dnn=~"$dnn",
snssai=~"$snssai"}))/(sum(smfcc_nchf_create_session_attempts{peerid=~"$peerid",
dnn=~"$dnn", snssai=~"$snssai"})))*100
```

CHF Create Session - Successes

Description

Number of CHF Create Session successful

Expression

```
sum(smfcc_nchf_create_session_success{peerid=~"$peerid", dnn=~"$dnn", snssai=~"$snssai"})
```

CHF Update Session - Attempts

Description

Number of CHF Update Session successful

Expression

```
sum(smfcc_nchf_update_session_attempts{peerid=~"$peerid", dnn=~"$dnn", snssai=~"$snssai"})
```

CHF Update Session - Success %

Description

CHF Update Session - Success %

Expression

```
((sum(smfcc_nchf_update_session_success{peerid=~"$peerid", dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_nchf_update_session_attempts{peerid=~"$peerid", dnn=~"$dnn", snssai=~"$snssai"})))*100
```

CHF Update Session - Successes

Description

Number of CHF Update Session successful

Expression

```
sum(smfcc_nchf_update_session_success{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai"})
```

CHF Release Session - Attempts

Description

Number of CHF Release Session attempted

Expression

```
sum(smfcc_nchf_release_session_attempts{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai"})
```

CHF Release Session - Success %

Description

CHF Release Session - Success %

Expression

```
((sum(smfcc_nchf_release_session_success{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_nchf_release_session_attempts{peerid=~"$peerid",  
dnn=~"$dnn", snssai=~"$snssai"}))*100
```

CHF Release Session - Successes

Description

Number of CHF Release Session successful

Expression

```
sum(smfcc_nchf_release_session_success{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai"})
```

SMF ETCD Interfaces Dashboard

Description

SMF ETCD Interfaces Dashboard

Panels

ETCD Gtpc Abort lease grant requests

Description

Number of ETCD Gtpc Abort lease grant requests sent

Expression

```
sum(smfcc_etcd_gtpc_abort_lease_grant_request)
```

ETCD Gtpc Abort lease grant - Success %

Description

ETCD Gtpc Abort lease grant - Success %

Expression

```
((sum(smfcc_etcd_gtpc_abort_lease_grant_request))/(sum(smfcc_etcd_gtpc_abort_lease_grant_response)))*100
```

ETCD Gtpc Abort lease grant responses

Description

Number of ETCD Gtpc Abort lease grant responses

Expression

```
sum(smfcc_etcd_gtpc_abort_lease_grant_response)
```

ETCD Gtpc Abort put with lease grant requests

Description

Number of ETCD Gtpc Abort put with lease grant requests sent

Expression

```
sum(smfcc_etcd_gtpc_abort_put_with_lease_grant_request)
```

ETCD Gtpc Abort put with lease grant - Success %

Description

ETCD Gtpc Abort put with lease grant - Success %

Expression

```
((sum(smfcc_etcd_gtpc_abort_put_with_lease_grant_request))/(sum(smfcc_etcd_gtpc_abort_put_with_lease_grant_response)))*100
```

ETCD Gtpc Abort put with lease grant responses

Description

Number of ETCD Gtpc Abort put with lease grant responses

Expression

```
sum(smfcc_etcd_gtpc_abort_put_with_lease_grant_response)
```

SMF Errors Dashboard

Description

SMF Errors Dashboard

Panels

PDU Session Create SM Context Failures

Description

Number of Create SM Context request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_create_sm_context_failures{dnn=~"$dnn", snssai=~"$snssai", causecode=~"$causecode"})
```

PDU Session Update SM Context Failures

Description

Number of Update SM Context request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_update_sm_context_failures{dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

PDU Session Release SM Context Failures

Description

Number of Release SM Context request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_release_sm_context_failures{dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

PDU Session Create Failures

Description

Number of Create request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_create_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PDU Session Update Failures

Description

Number of Update request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_update_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PDU Session Release Failures

Description

Number of Release request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_release_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PDU Session Create Client Failures

Description

Number of Create request messages failed at the client

Expression

```
sum(smfcc_nsmf_pdu_session_create_client_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PDU Session Update Client Failures

Description

Number of Update request messages received at the client

Expression

```
sum(smfcc_nsmf_pdu_session_update_client_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PDU Session Release Client Failures

Description

Number of Release request messages failed

Expression

```
sum(smfcc_nsmf_pdu_session_release_client_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PDU Session Establishment Failures

Description

Number of PDU Session Establishment Procedures failed

Expression

```
sum(smfcc_pdu_session_establishment_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

UE Deregistration Failures

Description

Number of UE Deregistration Procedures failed

Expression

```
sum(smfcc_ue_deregistration_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

AN Release Failures

Description

Number of AN Release Procedures failed

Expression

```
sum(smfcc_an_release_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

Service Request Failures

Description

Number of Service Request Procedures failed

Expression

```
sum(smfcc_service_request_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

NW Initiated Release Failures

Description

Number of NW Initiated Release Procedures failed

Expression

```
sum(smfcc_nw_initiated_release_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

Xn Handover With No UPF change Failures

Description

Number of Xn Handover Procedures with No UPF change failed

Expression

```
sum(smfcc_xn_handover_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode", iupfrelocation=~"false"})
```

SmPolicy Control Service Create Failures

Description

Number of Create SmPolicyControl Service requests failed

Expression

```
sum(smfcc_nsmf_smpolicy_control_create_service_failures{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid", causecode=~"$causecode"})
```

SmPolicy Control Service Update Failures

Description

Number of Create SmPolicyControl Service requests failed

Expression

```
sum(smfcc_nsmf_smpolicy_control_update_service_failures{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid", causecode=~"$causecode"})
```

SmPolicy Control Service Delete Failures**Description**

Number of Delete SmPolicyControl Service requests failed

Expression

```
sum(smfcc_nsmf_smpolicy_control_delete_service_failures{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid", causecode=~"$causecode"})
```

CHF Create Session - Failures**Description**

Number of CHF Create Session failed

Expression

```
sum(smfcc_nchf_create_session_failures{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

CHF Update Session - Failures**Description**

Number of CHF Update Session failed

Expression

```
sum(smfcc_nchf_update_session_failures{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

CHF Release Session - Failures

Description

Number of CHF Release Session failed

Expression

```
sum(smfcc_nchf_release_session_failures{peerid=~"$peerid", dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

UE Initiated Qos Modify - Failures

Description

Number of UE Initiated Modify Qos Procedures failed

Expression

```
sum(smfcc_ue_initiated_modify_qos_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

PCF Initiated Qos Modify - Failures

Description

Number of PCF Initiated Modify Qos Procedures failed

Expression

```
sum(smfcc_pcf_initiated_modify_qos_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

UE Initiated PDU Session Release - Failures

Description

Number of UE Initiated PDU Session Release Procedures failed

Expression

```
sum(smfcc_ue_initiated_pdu_session_release_failures{dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

Network Initiated Service Request - Failures

Description

Number of Network Initiated Service Request Procedures failed

Expression

```
sum(smfcc_network_initiated_service_request_failures{dnn=~"$dnn",  
snssai=~"$snssai", causecode=~"$causecode"})
```

PCF Initiated Terminate - Failures

Description

Number of PCF Initiated Terminate Procedures failed

Expression

```
sum(smfcc_pcf_initiated_terminate_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

Network of Usage Report - Failures

Description

Number of Usage Report Processing Procedures failed

Expression

```
sum(smfcc_upf_usage_report_procedure_failures{dnn=~"$dnn", snssai=~"$snssai",  
causecode=~"$causecode"})
```

N1N2 Message Transfer Requests Failures

Description

Number of N1N2 Message Transfer Request messages failed

Expression

```
sum(smfcc_namf_comm_n1n2_message_transfer_failures{dnn=~"$dnn", snssai=~"$snssai", causecode=~"$causecode"})
```

SMF Handover Procedures Dashboard

Description

SMF Handovers Procedures Dashboard

Panels

Description

Xn Handover Initiated Release - Attempts

Description

Number of Xn Handover Initiated Release Procedures attempted

Expression

```
sum(smfcc_xn_handover_initiated_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Xn Handover Initiated Release - Success %

Description

Xn Handover Initiated Release - Success %

Expression

```
((sum(smfcc_xn_handover_initiated_release_successes{dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_xn_handover_initiated_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})))*100
```

Xn Handover Initiated Release - Successes

Description

Number of Xn Handover Initiated Release Procedures successful

Expression

```
sum(smfcc_xn_handover_initiated_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Xn Handover Without UPF Relocation - Attempts

Description

Number of Xn Handover Without UPF Relocation Procedures attempted

Expression

```
sum(smfcc_xn_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

Xn Handover Without UPF Relocation - Successes

Description

Number of Xn Handover Without UPF Relocation Procedures Successes

Expression

```
sum(smfcc_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

Xn Handover With UPF Relocation - Attempts

Description

Number of Xn Handover With UPF Relocation Procedures attempted

Expression

```
sum(smfcc_xn_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})
```

Xn Handover With UPF Relocation - Success %

Description

Xn Handover With UPF Relocation - Success %

Expression

```
((sum(smfcc_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})))/(sum(smfcc_xn_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"true"})))*100
```

Xn Handover With UPF Relocation - Successes

Description

Number of Xn Handover With UPF Relocation Procedures successful

Expression

```
sum(smfcc_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})
```

N2 Handover Without UPF Relocation - Attempts

Description

Number of N2 Handover Without UPF Relocation Procedures attempted

Expression

```
sum(smfcc_n2_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

N2 Handover Without UPF Relocation - Success %

Description

N2 Handover Without UPF Relocation - Success %

Expression

```
((sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})))/(sum(smfcc_n2_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"false"})))*100
```

N2 Handover Without UPF Relocation - Successes

Description

Number of N2 Handover Without UPF Relocation Procedures Successes

Expression

```
sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

N2 Handover With UPF Relocation - Attempts

Description

Number of N2 Handover With UPF Relocation Procedures attempted

Expression

```
sum(smfcc_n2_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})
```

N2 Handover With UPF Relocation - Success %

Description

N2 Handover With UPF Relocation - Success %

Expression

```
((sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})))/(sum(smfcc_n2_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"true"}))*100
```

N2 Handover With UPF Relocation - Successes

Description

Number of N2 Handover Without UPF Relocation Procedures Successes

Expression

```
sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})
```

N2 Handover Without AMF Mobility - Attempts

Description

Number of N2 Handover Without AMF Mobility Procedures attempted

Expression

```
sum(smfcc_n2_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
amfmobility=~"false"})
```

N2 Handover Without AMF Mobility - Success %

Description

N2 Handover Without AMF Mobility - Success %

Expression

```
((sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
amfmobility=~"false"}))/(sum(smfcc_n2_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", amfmobility=~"false"})))*100
```

N2 Handover Without AMF Mobility - Successes

Description

Number of N2 Handover Without AMF Mobility Procedures Successes

Expression

```
sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
amfmobility=~"false"})
```

N2 Handover With AMF Mobility - Attempts

Description

Number of N2 Handover With AMF Mobility Procedures attempted

Expression

```
sum(smfcc_n2_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
amfmobility=~"true"})
```

N2 Handover With AMF Mobility - Success %

Description

N2 Handover With AMF Mobility - Success %

Expression

```
((sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai", amfmobility=~"true"}))/sum(smfcc_n2_handover_attempts{dnn=~"$dnn", snssai=~"$snssai", amfmobility=~"true"}))*100
```

N2 Handover With AMF Mobility - Successes

Description

Number of N2 Handover With AMF Mobility Procedures Successes

Expression

```
sum(smfcc_n2_handover_successes{dnn=~"$dnn", snssai=~"$snssai", amfmobility=~"true"})
```

5GS to EPS Handover with N26 interface - Attempts

Description

Number of 5GS to EPS Handover with N26 interface Procedures attempted

Expression

```
sum(smfcc_5gs_to_eps_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to EPS Handover with N26 interface - Success %

Description

5GS to EPS Handover with N26 interface - Success %

Expression

```
((sum(smfcc_5gs_to_eps_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_5gs_to_eps_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

5GS to EPS Handover with N26 interface - Successes

Description

Number of 5GS to EPS Handover with N26 interface Procedures successful

Expression

```
sum(smfcc_5gs_to_eps_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to EPS Idle Mode Mobility with N26 interface - Attempts

Description

Number of 5GS to EPS Idle Mode Mobility with N26 interface Procedures attempted

Expression

```
sum(smfcc_5gs_to_eps_idle_mobility_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to EPS Idle Mode Mobility with N26 interface - Success %

Description

5GS to EPS Idle Mode Mobility with N26 interface - Success %

Expression

```
((sum(smfcc_5gs_to_eps_idle_mobility_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_5gs_to_eps_idle_mobility_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

5GS to EPS Idle Mode Mobility with N26 interface - Successes

Description

Number of 5GS to EPS Idle Mode Mobility with N26 interface Procedures successful

Expression

```
sum(smfcc_5gs_to_eps_idle_mobility_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS to 5GS Handover with N26 interface without I-UPF Relocation - Attempts

Description

Number of EPS to 5GS Handover with N26 interface without I-UPF Relocation Procedures attempted

Expression

```
sum(smfcc_eps_to_5gs_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

EPS to 5GS Handover with N26 interface without I-UPF Relocation - Success %

Description

EPS to 5GS Handover with N26 interface without I-UPF Relocation - Success %

Expression

```
((sum(smfcc_eps_to_5gs_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})))/(sum(smfcc_eps_to_5gs_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"false"})))*100
```

EPS to 5GS Handover with N26 interface without I-UPF Relocation - Successes

Description

Number of EPS to 5GS Handover with N26 interface without I-UPF Relocation Procedures successful

Expression

```
sum(smfcc_eps_to_5gs_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

EPS to 5GS Handover with N26 interface with I-UPF Relocation - Attempts

Description

Number of EPS to 5GS Handover with N26 interface with I-UPF Relocation Procedures attempted

Expression

```
sum(smfcc_eps_to_5gs_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})
```

EPS to 5GS Handover with N26 interface with I-UPF Relocation - Success %

Description

EPS to 5GS Handover with N26 interface with I-UPF Relocation - Success %

Expression

```
((sum(smfcc_eps_to_5gs_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"}))/(sum(smfcc_eps_to_5gs_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"true"})))*100
```

EPS to 5GS Handover with N26 interface with I-UPF Relocation - Successes

Description

Number of EPS to 5GS Handover with N26 interface with I-UPF Relocation Procedures successful

Expression

```
sum(smfcc_5gs_to_eps_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"true"})
```

EPS to 5GS Idle Mode Mobility with N26 interface - Attempts

Description

Number of EPS to 5GS Idle Mode Mobility with N26 interface Procedures attempted

Expression

```
sum(smfcc_eps_to_5gs_idle_mobility_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS to 5GS Idle Mode Mobility with N26 interface - Success %

Description

EPS to 5GS Idle Mode Mobility with N26 interface - Success %

Expression

```
((sum(smfcc_eps_to_5gs_idle_mobility_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_eps_to_5gs_idle_mobility_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

EPS to 5GS Idle Mode Mobility with N26 interface - Successes

Description

Number of EPS to 5GS Idle Mode Mobility with N26 interface Procedures successful

Expression

```
sum(smfcc_eps_to_5gs_idle_mobility_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS to ePDG Handover - Attempts

Description

Number of EPS to ePDG Handover Procedures attempted

Expression

```
sum(smfcc_4g_to_epdg_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS to ePDG Handover - Success %

Description

EPS to ePDG Handover - Success %

Expression

```
((sum(smfcc_4g_to_epdg_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_4g_to_epdg_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

EPS to ePDG Handover - Successes

Description

Number of EPS to ePDG Handover Procedures successful

Expression

```
sum(smfcc_4g_to_epdg_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to ePDG Handover - Attempts

Description

Number of 5GS to ePDG Handover Procedures attempted

Expression

```
sum(smfcc_5gs_to_epdg_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to ePDG Handover - Success %

Description

5GS to ePDG Handover - Success %

Expression

```
((sum(smfcc_5gs_to_epdg_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_5gs_to_epdg_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

5GS to ePDG Handover - Successes

Description

Number of 5GS to ePDG Handover Procedures successful

Expression

```
sum(smfcc_5gs_to_epdg_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

ePDG to 5GS Handover - Attempts

Description

Number of ePDG to 5GS Handover Procedures attempted

Expression

```
sum(smfcc_non3gpp_to_5gs_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

ePDG to 5GS Handover - Success %

Description

ePDG to 5GS Handover - Success %

Expression

```
((sum(smfcc_non3gpp_to_5gs_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_non3gpp_to_5gs_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))*100
```

ePDG to 5GS Handover - Successes

Description

Number of ePDG to 5GS Handover Procedures successful

Expression

```
sum(smfcc_non3gpp_to_5gs_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

ePDG to 5GS Idle Mode Mobility - Attempts

Description

Number of ePDG to 5GS Idle Mode Mobility Procedures attempted

Expression

```
sum(smfcc_non3gpp_to_5gs_idle_mobility_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```


ePDG to 5GS Idle Mode Mobility- Success %

Description

ePDG to 5GS Idle Mode Mobility - Success %

Expression

```
((sum(smfcc_non3gpp_to_5gs_idle_mobility_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_non3gpp_to_5gs_idle_mobility_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

ePDG to 5GS Idle Mode Mobility - Successes

Description

Number of ePDG to 5GS Idle Mode Mobility Procedures successful

Expression

```
sum(smfcc_non3gpp_to_5gs_idle_mobility_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to ePDG Idle Mode Mobility - Attempts

Description

Number of 5GS to ePDG Idle Mode Mobility Procedures attempted

Expression

```
sum(smfcc_5gs_to_epdg_idle_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

5GS to ePDG Idle Mode Mobility- Success %

Description

5GS to ePDG Idle Mode Mobility - Success %

Expression

```
((sum(smfcc_5gs_to_epdg_idle_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_5gs_to_epdg_idle_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

5GS to ePDG Idle Mode Mobility - Successes

Description

Number of 5GS to ePDG Idle Mode Mobility Procedures successful

Expression

```
sum(smfcc_5gs_to_epdg_idle_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

ePDG to EPS Handover - Attempts

Description

Number of ePDG to EPS Handover Procedures attempted

Expression

```
sum(smfcc_epdg_to_eps_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

ePDG to EPS Handover - Success %

Description

ePDG to EPS Handover - Success %

Expression

```
((sum(smfcc_epdg_to_eps_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_epdg_to_eps_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

ePDG to EPS Handover - Successes

Description

Number of ePDG to EPS Handover Procedures successful

Expression

```
sum(smfcc_epdg_to_eps_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

4G TAU X2 HO with SGW Change - Attempts

Description

Number of 4G TAU X2 HO with SGW Change Procedures attempted

Expression

```
sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"EUTRA"})
```

4G TAU X2 HO with SGW Change - Success %

Description

4G TAU X2 HO with SGW Change - Success %

Expression

```
((sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai",  
ratType=~"EUTRA"}))/(sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_attempts{dnn  
=~"$dnn", snssai=~"$snssai", ratType=~"EUTRA"})))*100
```

4G TAU X2 HO with SGW Change - Successes

Description

Number of 4G TAU X2 HO with SGW Change Procedures successful

Expression

```
sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"EUTRA"})
```

3G RAU HO with SGW Change - Attempts

Description

Number of 3G RAU HO with SGW Change Procedures attempted

Expression

```
sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"UTRA"})
```

3G RAU HO with SGW Change - Success %

Description

3G RAU HO with SGW Change - Success %

Expression

```
((sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai",  
ratType=~"UTRA"})))/(sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_attempts{dnn=  
~"$dnn", snssai=~"$snssai", ratType=~"UTRA"}))*100
```

3G RAU HO with SGW Change - Successes

Description

Number of 3G RAU HO with SGW Change Procedures successful

Expression

```
sum(smfcc_pgw_mbr_tau_rau_x2_sgw_change_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai", ratType=~"UTRA"})
```

5GS to EPS Handover without N26 interface - Attempts

Description

Number of 5GS to EPS Handover without N26 interface Procedures attempted

Expression

```
sum(smfcc_5gs_to_eps_handover_without_n26_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

5GS to EPS Handover without N26 interface - Success %

Description

5GS to EPS Handover without N26 interface - Success %

Expression

```
((sum(smfcc_5gs_to_eps_handover_without_n26_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_5gs_to_eps_handover_without_n26_attempts{dnn=~"$dnn", snssai=~"$snssai"}))*100
```

5GS to EPS Handover without N26 interface - Successes

Description

Number of 5GS to EPS Handover without N26 interface Procedures successful

Expression

```
sum(smfcc_5gs_to_eps_handover_without_n26_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

5GS to EPS Handover without N26 Resource Release- Attempts

Description

Number of 5GS to EPS Handover without N26 interface Resource Release Procedures attempted

Expression

```
sum(smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

5GS to EPS Handover without N26 Resource Release - Success %

Description

5GS to EPS Handover without N26 interface Resource Release - Success %

Expression

```
((sum(smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_attempts{  
dnn=~"$dnn", snssai=~"$snssai"}))*100
```

5GS to EPS Handover without N26 Resource Release - Successes

Description

Number of 5GS to EPS Handover without N26 interface Resource Release Procedures successful

Expression

```
sum(smfcc_5gs_to_eps_handover_without_n26_rsrc_rel_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

5GS to EPS Idle Mode Mobility without N26 interface - Attempts

Description

Number of 5GS to EPS Idle Mode Mobility without N26 interface Procedures attempted

Expression

```
sum(smfcc_5gs_to_eps_idle_handover_without_n26_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

5GS to EPS Idle Mode Mobility without N26 interface - Success %

Description

5GS to EPS Idle Mode Mobility without N26 interface - Success %

Expression

```
((sum(smfcc_5gs_to_eps_idle_handover_without_n26_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_5gs_to_eps_idle_handover_without_n26_attempts{dnn=  
~"$dnn", snssai=~"$snssai"})))*100
```

5GS to EPS Idle Mode Mobility without N26 interface - Successes

Description

Number of 5GS to EPS Idle Mode Mobility without N26 interface Procedures successful

Expression

```
sum(smfcc_5gs_to_eps_idle_handover_without_n26_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

EPS to 5GS Handover without N26 interface without I-UPF Relocation - Attempts

Description

Number of EPS to 5GS Handover without N26 interface without I-UPF Relocation Procedures attempted

Expression

```
sum(smfcc_eps_to_5gs_handover_without_n26_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

EPS to 5GS Handover without N26 interface without I-UPF Relocation - Success %

Description

EPS to 5GS Handover without N26 interface without I-UPF Relocation - Success %

Expression

```
((sum(smfcc_eps_to_5gs_handover_without_n26_successes{dnn=~"$dnn",  
snssai=~"$snssai",  
iupfrelocation=~"false"})))/(sum(smfcc_eps_to_5gs_handover_without_n26_attempts{dnn  
=~"$dnn", snssai=~"$snssai", iupfrelocation=~"false"})))*100
```

EPS to 5GS Handover without N26 interface without I-UPF Relocation - Successes

Description

Number of EPS to 5GS Handover without N26 interface without I-UPF Relocation Procedures successful

Expression

```
sum(smfcc_eps_to_5gs_handover_without_n26_successes{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"false"})
```

EPS to 5GS Handover without N26 interface with I-UPF Relocation - Attempts

Description

Number of EPS to 5GS Handover without N26 interface with I-UPF Relocation Procedures attempted

Expression

```
sum(smfcc_eps_to_5gs_handover_without_n26_attempts{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"true"})
```

EPS to 5GS Handover without N26 interface with I-UPF Relocation - Success %

Description

EPS to 5GS Handover without N26 interface with I-UPF Relocation - Success %

Expression

```
((sum(ssmfcc_eps_to_5gs_handover_without_n26_successes{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"true"}))/(sum(smfcc_eps_to_5gs_handover_without_n26_attempts{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"true"})))*100
```

EPS to 5GS Handover without N26 interface with I-UPF Relocation - Successes

Description

Number of EPS to 5GS Handover without N26 interface with I-UPF Relocation Procedures successful

Expression

```
sum(smfcc_eps_to_5gs_handover_without_n26_attempts{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"true"})
```

3G to 4G Handover - Attempts

Description

Number of 3G to 4G Handover Procedures attempted

Expression

```
sum(smfcc_ggsn_3g_to_4g_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

3G to 4G Handover - Success %

Description

3G to 4G Handover - Success %

Expression

```
((sum(smfcc_ggsn_3g_to_4g_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_ggsn_3g_to_4g_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

3G to 4G Handover - Successes

Description

Number of 3G to 4G Handover Procedures successful

Expression

```
sum(smfcc_ggsn_3g_to_4g_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

4G to 3G Handover - Attempts

Description

Number of 4G to 3G Handover Procedures attempted

Expression

```
sum(smfcc_ggsn_4g_to_3g_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

4G to 3G Handover - Success %

Description

4G to 3G Handover - Success %

Expression

```
((sum(smfcc_ggsn_4g_to_3g_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_ggsn_4g_to_3g_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

4G to 3G Handover- Successes

Description

Number of 4G to 3G Handover Procedures successful

Expression

```
sum(smfcc_ggsn_4g_to_3g_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Inter SGSN Handover - Attempts

Description

Number of Inter SGSN Handovers attempted

Expression

```
sum(smfcc_ggsn_inter_sgsn_handover_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Inter SGSN Handover - Success %

Description

Inter SGSN Handover - Success %

Expression

```
((sum(smfcc_ggsn_inter_sgsn_handover_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_ggsn_inter_sgsn_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

Inter SGSN Handover Success

Description

Number of Inter SGSN Handover Procedures successful

Expression

```
sum(smfcc_ggsn_inter_sgsn_handover_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Inter Plmn Handover with Roaming Status Change - Attempts

Description

Number of Inter Plmn Handover with Roaming Status Change Attempted

Expression

```
sum(smfcc_pgw_interplmn_handover_with_subscriber_type_change_attempts{dnn=~"$dnn",  
snssai=~"$snssai", newstatus=~"$roamingStatus"})
```

Inter Plmn Handover with Roaming Status Change - Success %

Description

Meter showing success rate for inter plmn handover with roaming status change

Expression

```
((sum(smfcc_pgw_interplmn_handover_with_subscriber_type_change_successes{dnn=~"$dn  
n",  
snssai=~"$snssai",newstatus=~"$roamingStatus"})))/(sum(smfcc_pgw_interplmn_handover  
_with_subscriber_type_change_attempts{dnn=~"$dnn", snssai=~"$snssai",  
newstatus=~"$roamingStatus"}))) * 100
```

Inter Plmn Handover With Roaming Status Change - Successful

Description

Number Inter Plmn Handovers With Roaming Status Change - Successes

Expression

```
sum(smfcc_pgw_interplmn_handover_with_subscriber_type_change_successes{dnn=~"$dnn"  
,snssai=~"$snssai", newstatus=~"$roamingStatus"})
```

SMF Interfaces Dashboard

Description

SMF Interfaces Dashboard

Panels

PDU Session Create SM Context - Attempts

Description

Number of Create SM Context request messages received

Expression

```
sum(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

SM Context Create - Success %

Description

SM Context Create - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"  
$dnn", snssai=~"$snssai"})))*100
```

PDU Session Create SM Context - Successes

Description

Number of Create SM Context request messages successful

Expression

```
sum(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

PDU Session Update SM Context - Attempts

Description

Number of Update SM Context request messages received

Expression

```
sum(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

SM Context Update - Success %

Description

SM Context Update - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"  
$dnn", snssai=~"$snssai"})))*100
```

PDU Session Update SM Context - Successes

Description

Number of Update SM Context request messages successful

Expression

```
sum(smfcc_nsmf_pdu_session_update_sm_context_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

PDU Session Release SM Context - Attempts

Description

Number of Release SM Context request messages received

Expression

```
sum(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

SM Context Release - Success %

Description

SM Context Release - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn", snssai=~"$snssai"})))/(sum(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn", snssai=~"$snssai"}))*100
```

PDU Session Release SM Context - Successes

Description

Number of Release SM Context request messages successful

Expression

```
sum(smfcc_nsmf_pdu_session_release_sm_context_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Create - Attempts

Description

Number of Create request messages received

Expression

```
sum(smfcc_nsmf_pdu_session_create_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Create - Success %

Description

PDU Session Create - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_create_successes{dnn=~"$dnn", snssai=~"$snssai"})))/(sum(smfcc_nsmf_pdu_session_create_attempts{dnn=~"$dnn", snssai=~"$snssai"}))*100
```

PDU Session Create - Successes

Description

Number of Create request messages successful

Expression

```
sum(smfcc_nsmf_pdu_session_create_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Update - Attempts

Description

Number of Update request messages received

Expression

```
sum(smfcc_nsmf_pdu_session_update_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Update - Success %

Description

PDU Session Update - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_update_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_nsmf_pdu_session_update_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

PDU Session Update - Successes

Description

Number of Update request messages successful

Expression

```
sum(smfcc_nsmf_pdu_session_update_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Release - Attempts

Description

Number of Release request messages received

Expression

```
sum(smfcc_nsmf_pdu_session_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Release - Success %

Description

PDU Session Release - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_nsmf_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

PDU Session Release - Successes

Description

Number of Release request messages successful

Expression

```
sum(smfcc_nsmf_pdu_session_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Create Client - Attempts

Description

Number of Create request messages received at the client

Expression

```
sum(smfcc_nsmf_pdu_session_create_client_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Create Client - Success %

Description

PDU Session Create Client - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_create_client_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_nsmf_pdu_session_create_client_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

PDU Session Create Client - Successes

Description

Number of Create request messages successful at the client

Expression

```
sum(smfcc_nsmf_pdu_session_create_client_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

PDU Session Update Client - Attempts

Description

Number of Update request messages received at the client

Expression

```
sum(smfcc_nsmf_pdu_session_update_client_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Update Client - Success %

Description

PDU Session Update Client - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_update_client_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_nsmf_pdu_session_update_client_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

PDU Session Update Client - Successes

Description

Number of Update request messages successful at the client

Expression

```
sum(smfcc_nsmf_pdu_session_update_client_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

PDU Session Release Client - Attempts**Description**

Number of Release request messages received at the client

Expression

```
sum(smfcc_nsmf_pdu_session_release_client_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

PDU Session Release Client - Success %**Description**

PDU Session Release Client - Success %

Expression

```
((sum(smfcc_nsmf_pdu_session_release_client_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_nsmf_pdu_session_release_client_attempts{dnn=~"$dn  
n", snssai=~"$snssai"})))*100
```

PDU Session Release Client - Successes**Description**

Number of Release request messages successful at the client

Expression

```
sum(smfcc_nsmf_pdu_session_release_client_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

N1N2 Message Transfer Requests - Attempts

Description

Number of N1N2 Message Transfer Request messages sent

Expression

```
sum(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

N1N2 Message Transfer Requests - Success %

Description

N1N2 Message Transfer Requests - Success %

Expression

```
((sum(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_namf_comm_n1n2_message_transfer_attempts{dnn=~"$dn  
n", snssai=~"$snssai"}))*100
```

N1N2 Message Transfer Requests - Successes

Description

Number of N1N2 Message Transfer Request messages successful

Expression

```
sum(smfcc_namf_comm_n1n2_message_transfer_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

SMF-CC Transaction Rate per Pod

Description

SMF-CC Transaction Rate per Pod

Expression

{{pod}}

```
round(sum(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])) by (pod), 0.001)
```

```
2 round(sum by (pod)
(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])) + sum by (pod)
(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])) + sum by (pod)
(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])))
```

SMF-CC Transaction Rate Per Service

Description

SMF-CC Transaction Rate Per Service

Expression

smfcc-{{version}}

```
round(sum by (version)
(rate(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])) + sum by (version)
(rate(smfcc_nsmf_pdu_session_update_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])) + sum by (version)
(rate(smfcc_nsmf_pdu_session_release_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}[1m])))
```

SMFCC Session Activation Stats per Pod

Description

SMFCC Session Activation Stats per Pod

Expression

Session Activation Attempt-{{pod}}

```
sum(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}) by (pod)
```

Session Activation Failure-{{pod}}

```
sum(smfcc_nsmf_pdu_session_create_sm_context_attempts{dnn=~"$dnn",
snssai=~"$snssai"}) by (pod) -
sum(smfcc_nsmf_pdu_session_create_sm_context_successes{dnn=~"$dnn",
snssai=~"$snssai"}) by (pod)
```

SMF PCF Interfaces Dashboard

Description

SMF PCF Interfaces Dashboard

Panels

Create SmPolicy Control Service - Attempts

Description

Number of Create SmPolicy Control Service requests attempted

Expression

```
sum(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"})
```

Create SmPolicy Control Service - Success %

Description

Create SmPolicy Control Service - Success %

Expression

```
((sum(smfcc_nsmf_smpolicy_control_create_service_success{dnn=~"$dnn",
snssai=~"$snssai",
peerid=~"$peerid"}))/(sum(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=
~"$dnn", snssai=~"$snssai", peerid=~"$peerid"})))*100
```

Create SmPolicy Control Service - Successes

Description

Number of Create SmPolicy Control Service requests successful

Expression

```
sum(smfcc_nsmf_smpolicy_control_create_service_success{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid"})
```

Update SmPolicy Control Service - Attempts

Description

Number of Update SmPolicy Control Service requests attempted

Expression

```
sum(smfcc_nsmf_smpolicy_control_update_service_attempts{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid"})
```

Update SmPolicy Control Service - Success %

Description

Update SmPolicy Control Service - Success %

Expression

```
((sum(smfcc_nsmf_smpolicy_control_update_service_success{dnn=~"$dnn",  
snssai=~"$snssai",  
peerid=~"$peerid"}))/(sum(smfcc_nsmf_smpolicy_control_update_service_attempts{dnn=  
~"$dnn", snssai=~"$snssai", peerid=~"$peerid"}))) * 100
```

Update SmPolicy Control Service - Successes

Description

Number of Update SmPolicy Control Service requests successful

Expression

```
sum(smfcc_nsmf_smpolicy_control_update_service_success{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid"})
```

Delete SmPolicy Control Service - Attempts

Description

Number of Delete SmPolicy Control Service requests attempted

Expression

```
sum(smfcc_nsmf_smpolicy_control_delete_service_attempts{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid"})
```

Delete SmPolicy Control Service - Success %

Description

Delete SmPolicy Control Service - Success %

Expression

```
((sum(smfcc_nsmf_smpolicy_control_delete_service_success{dnn=~"$dnn",  
snssai=~"$snssai",  
peerid=~"$peerid"}))/(sum(smfcc_nsmf_smpolicy_control_delete_service_attempts{dnn=  
~"$dnn", snssai=~"$snssai", peerid=~"$peerid"})))*100
```

Delete SmPolicy Control Service - Successes

Description

Number of Delete SmPolicy Control Service requests successful

Expression

```
sum(smfcc_nsmf_smpolicy_control_delete_service_success{dnn=~"$dnn",  
snssai=~"$snssai", peerid=~"$peerid"})
```

SMF-PCF Transaction Rate per Pod

Description

SMF-PCF Transaction Rate per Pod

Expression

```
round(sum by (pod)  
(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=~"$dnn",
```

```
snssai=~"$snssai", peerid=~"$peerid"}[1m])) + sum by (pod)
(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m])) + sum by (pod)
(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m]))))
```

SMF-PCF Transaction Rate Per Service

Description

SMF-PCF Transaction Rate Per Service

Expression

smfcc-{{version}}

```
round(sum by (version)
(rate(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m])) + sum by (version)
(rate(smfcc_nsmf_smpolicy_control_update_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m])) + sum by (version)
(rate(smfcc_nsmf_smpolicy_control_delete_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}[1m]))))
```

SMF-PCF Session Activation Stats per Pod

Description

SMF-PCF Session Activation Stats per Pod

Expression

Session Activation Attempt-{{pod}}

```
sum(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}) by (pod)
```

Session Activation Failure-{{pod}}

```
sum(smfcc_nsmf_smpolicy_control_create_service_attempts{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}) by (pod) -
sum(smfcc_nsmf_smpolicy_control_create_service_success{dnn=~"$dnn",
snssai=~"$snssai", peerid=~"$peerid"}) by (pod)
```


SMF Procedures Dashboard

Description

SMF Procedures Dashboard

Panels

PDU Session Establishment - Attempts

Description

Number of PDU Session Establishment Procedures attempted

Expression

```
sum(smfcc_pdu_session_establishment_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Establishment - Success %

Description

PDU Session Establishment - Success %

Expression

```
((sum(smfcc_pdu_session_establishment_successes{dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_pdu_session_establishment_attempts{dnn=~"$dnn", snssai=~"$snssai"})))*100
```

PDU Session Establishment - Successes

Description

Number of PDU Session Establishment Procedures successful

Expression

```
sum(smfcc_pdu_session_establishment_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Deregistration - Attempts

Description

Number of UE Deregistration Procedures attempted

Expression

```
sum(smfcc_ue_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Deregistration - Success %

Description

UE Deregistration - Success %

Expression

```
((sum(smfcc_ue_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_ue_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"})))*100
```

UE Deregistration - Successes

Description

Number of UE Deregistration Procedures successful

Expression

```
sum(smfcc_ue_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

AN Release - Attempts

Description

Number of AN Release Procedures attempted

Expression

```
sum(smfcc_an_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

AN Release - Success %

Description

AN Release - Success %

Expression

```
((sum(smfcc_an_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_an_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))*100
```

AN Release - Successes

Description

Number of AN Release Procedures successful

Expression

```
sum(smfcc_an_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Service Request - Attempts

Description

Number of Service Request Procedures attempted

Expression

```
sum(smfcc_service_request_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Service Request - Success %

Description

Service Request - Success %

Expression

```
((sum(smfcc_service_request_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_service_request_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))*100
```

Service Request - Successes

Description

Number of Service Request Procedures successful

Expression

```
sum(smfcc_service_request_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

NW Initiated Release - Attempts

Description

Number of NW Initiated Release Procedures attempted

Expression

```
sum(smfcc_nw_initiated_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

NW Initiated Release - Success %

Description

NW Initiated Release - Success %

Expression

```
((sum(smfcc_nw_initiated_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_nw_initiated_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

NW Initiated Release - Successes

Description

Number of NW Initiated Release Procedures successful

Expression

```
sum(smfcc_nw_initiated_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Xn Handover With No UPF Change - Attempts

Description

Number of Xn Handover Procedures with No UPF change attempted

Expression

```
sum(smfcc_xn_handover_attempts{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

Xn Handover With No UPF Change - Success %

Description

Xn Handover With No UPF Change - Success %

Expression

```
((sum(smfcc_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"}))/(sum(smfcc_xn_handover_attempts{dnn=~"$dnn",  
snssai=~"$snssai", iupfrelocation=~"false"})))*100
```

Xn Handover With No UPF Change - Successes

Description

Number of Xn Handover Procedures with No UPF change successful

Expression

```
sum(smfcc_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai",  
iupfrelocation=~"false"})
```

UE Initiated Qos Modify - Attempts

Description

Number of UE Initiated Modify Qos Procedures attempted

Expression

```
sum(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Initiated Qos Modify - Success %

Description

UE Initiated Qos Modify - Success %

Expression

```
((sum(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_ue_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"})))*100
```

UE Initiated Qos Modify - Successes

Description

Number of UE Initiated Modify Qos Procedures successful

Expression

```
sum(smfcc_ue_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

PCF Initiated Qos Modify - Attempts

Description

Number of PCF Initiated Modify Qos Procedures attempted

Expression

```
sum(smfcc_pcf_initiated_modify_qos_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PCF Initiated Qos Modify - Success %

Description

PCF Initiated Qos Modify - Success %

Expression

```
((sum(smfcc_pcf_initiated_modify_qos_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_pcf_initiated_modify_qos_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

PCF Initiated Qos Modify - Successes

Description

Number of PCF Initiated Modify Qos Procedures successful

Expression

```
sum(smfcc_pcf_initiated_modify_qos_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Initiated PDU Session Release - Attempts

Description

Number of UE Initiated PDU Session Release Procedures attempted

Expression

```
sum(smfcc_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

UE Initiated PDU Session Release - Success %

Description

UE Initiated PDU Session Release - Success %

Expression

```
((sum(smfcc_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

UE Initiated PDU Session Release - Successes

Description

Number of UE Initiated PDU Session Release Procedures successful

Expression

```
sum(smfcc_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Network Initiated Service Request - Attempts

Description

Number of Network Initiated Service Request Procedures attempted

Expression

```
sum(smfcc_network_initiated_service_request_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Network Initiated Service Request - Success %

Description

Network Initiated Service Request - Success %

Expression

```
((sum(smfcc_network_initiated_service_request_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_network_initiated_service_request_attempts{dnn=~"$  
dnn", snssai=~"$snssai"})))*100
```

Network Initiated Service Request - Successes

Description

Number of Network Initiated Service Request Procedures successful

Expression

```
sum(smfcc_network_initiated_service_request_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

PCF Initiated Terminate - Attempts

Description

Number of PCF Initiated Terminate Procedures attempted

Expression

```
sum(smfcc_pcf_initiated_terminate_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PCF Initiated Terminate - Success %

Description

PCF Initiated Terminate - Success %

Expression

```
((sum(smfcc_pcf_initiated_terminate_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_pcf_initiated_terminate_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

PCF Initiated Terminate - Successes

Description

Number of PCF Initiated Terminate Procedures successful

Expression

```
sum(smfcc_pcf_initiated_terminate_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Network of Usage Report - Attempts

Description

Number of Usage Report Processing Procedure attempted

Expression

```
sum(smfcc_upf_usage_report_procedure_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Network of Usage Report - Success %

Description

Network of Usage Report - Success %

Expression

```
((sum(smfcc_upf_usage_report_procedure_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_upf_usage_report_procedure_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

Network of Usage Report - Successes

Description

Number of Usage Report Processing Procedure successful

Expression

```
sum(smfcc_upf_usage_report_procedure_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

AMF Event Notification - Attempts

Description

Number of AMF Event Notification attempts

Expression

```
sum(smfcc_amf_event_notification_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

AMF Event Notification - Success %

Description

AMF Event Notification - Success %

Expression

```
((sum(smfcc_amf_event_notification_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_amf_event_notification_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

AMF Event Notification - Successes

Description

Number of AMF Event Notification successes

Expression

```
sum(smfcc_amf_event_notification_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

SMF Radius InBand Tunnel ETCD Add/Delete DashBoard

Description

This will display the etcd add/delete operation for inband tunnel

SMFCC RADIUS INBAND TUNNEL PROCEDURE SUMMARY Panels

Panels

ETCD add Inband Radius Tunnel Attempts

Description

Number of etcd add inband radius tunnel attempts

Expression

`__auto`

```
sum(smfcc_etcd_add_inband_radius_tunnel_attempts{inbandProfile=~"$inbandProfile", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"})
```

ETCD add Inband Radius Tunnel Success

Description

Number of etcd add inband radius tunnel Success

Expression

`__auto`

```
sum(smfcc_etcd_add_inband_radius_tunnel_successes{inbandProfile=~"$inbandProfile",  
peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"})
```

ETCD add Inband Radius Tunnel Failure

Description

Number of etcd add inband radius tunnel Failure

Expression

`_auto`

```
sum(smfcc_etcd_add_inband_radius_tunnel_failures{inbandProfile=~"$inbandProfile",  
peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"})
```

ETCD del Inband Radius Tunnel Attempts

Description

Number of etcd del inband radius tunnel attempts

Expression

`_auto`

```
sum(smfcc_etcd_del_inband_radius_tunnel_attempts{inbandProfile=~"$inbandProfile",  
peerUpfRole=~"$peerUpfRole"})
```

ETCD del Inband Radius Tunnel Success

Description

Number of etcd del inband radius tunnel Success

Expression

`_auto`

```
sum(smfcc_etcd_del_inband_radius_tunnel_successes{inbandProfile=~"$inbandProfile",  
peerUpfRole=~"$peerUpfRole"})
```

ETCD del Inband Radius Tunnel Failure

Description

Number of etcd del inband radius tunnel Failure

Expression

`_auto`

```
sum(smfcc_etcd_del_inband_radius_tunnel_failures{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"})
```

SMFCC RADIUS INBAND TUNNEL PROCEDURE DETAILS Panels

SMFCC RADIUS INBAND TUNNEL TIMER DETAILS

Expression

Etcd Add Radius Inband Tunnel Attempts {{{inbandProfile}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_etcd_add_inband_radius_tunnel_attempts{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Etcd Add Radius Inband Tunnel Success {{{inbandProfile}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_etcd_add_inband_radius_tunnel_successes{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Etcd Add Radius Inband Tunnel Failures {{{inbandProfile}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_etcd_add_inband_radius_tunnel_failures{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Etcd Del Radius Inband Tunnel Attempts {{{inbandProfile}}} {{{peerUpfRole}}}

```
smfcc_etcd_del_inband_radius_tunnel_attempts{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole"}
```

Etcd Del Radius Inband Tunnel Success {{{inbandProfile}}} {{{peerUpfRole}}}

```
smfcc_etcd_del_inband_radius_tunnel_successes{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole"}
```

Etcd Del Radius Inband Tunnel Failure {{{inbandProfile}}} {{{peerUpfRole}}}

```
smfcc_etcd_del_inband_radius_tunnel_failures{inbandProfile=~"$inbandProfile",
peerUpfRole=~"$peerUpfRole"}
```

SMF Radius InBand Tunnel Procedure DashBoard

Description

This will display the radius inband tunnel proc details

SMFCC RADIUS INBAND TUNNEL PROCEDURE SUMMARY Panels

Panels

Inband Radius Tunnel Create Proc Attempts

Description

Number of Radius Inband Tunnel Create Proc Attempts

Expression

_auto

```
sum(smfcc_inband_radius_tunnel_create_attempts{inbandProfile=~"$inbandProfile",
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Tunnel Create Proc Success

Description

Number of Radius Inband Tunnel Create Proc Success

Expression

_auto

```
sum(smfcc_inband_radius_tunnel_create_successes{inbandProfile=~"$inbandProfile",  
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",  
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Tunnel Create Proc Failure

Description

Number of Radius Inband Tunnel Create Proc failures

Expression

_auto

```
sum(smfcc_inband_radius_tunnel_create_failures{inbandProfile=~"$inbandProfile",  
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",  
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Gtpu Error Indication Request

Description

Number of inband radius gtpu error indication request received

Expression

_auto

```
sum(smfcc_inband_radius_gtpu_error_indications_received{inbandProfile=~"$inbandProfile",  
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",  
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Tunnel Delete Proc Attempts

Description

Number of Radius Inband Tunnel Delete Proc Attempts

Expression

_auto

```
sum(smfcc_inband_radius_tunnel_delete_attempts{inbandProfile=~"$inbandProfile",  
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",  
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Tunnel Delete Proc Success

Description

Number of Radius Inband Tunnel Delete Proc Success

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_delete_successes{inbandProfile=~"$inbandProfile",  
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",  
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Tunnel Delete Proc Failure

Description

Number of Radius Inband Tunnel Delete Proc failures

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_delete_failures{inbandProfile=~"$inbandProfile",  
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole",  
peerUpfIp=~"$peerUpfIp"})
```

Inband Radius Gtpu Error Indication Response Sent

Description

Number of inband radius gtpu error indication request received

Expression

`_auto`


```
sum(smfcc_inband_radius_gtpu_error_indications_response_sent{inbandProfile=~"$inbandProfile", peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"})
```

SMFCC RADIUS INBAND TUNNEL PROCEDURE DETAILS Panels

SMFCC RADIUS INBAND TUNNEL TIMER DETAILS

Expression

Radius Inband Tunnel Create Proc Attempts {{{inbandProfile}}} {{{peerUpfName}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_inband_radius_tunnel_create_attempts{inbandProfile=~"$inbandProfile", peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Radius Inband Tunnel Create Proc Success {{{inbandProfile}}} {{{peerUpfName}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_inband_radius_tunnel_create_successes{inbandProfile=~"$inbandProfile", peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Radius Inband Tunnel Create Proc Failures {{{inbandProfile}}} {{{peerUpfName}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_inband_radius_tunnel_create_failures{inbandProfile=~"$inbandProfile", peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Radius Inband Tunnel Delete Proc Attempts {{{inbandProfile}}} {{{peerUpfName}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_inband_radius_tunnel_delete_attempts{inbandProfile=~"$inbandProfile", peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Radius Inband Tunnel Delete Proc Success {{{inbandProfile}}} {{{peerUpfName}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_inband_radius_tunnel_delete_successes{inbandProfile=~"$inbandProfile", peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

Radius Inband Tunnel Delete Proc Failure {{{inbandProfile}}} {{{peerUpfName}}} {{{peerUpfRole}}} {{{peerUpfIp}}}

```
smfcc_inband_radius_tunnel_delete_failures{inbandProfile=~"$inbandProfile",
peerUpfName=~"$peerUpfName", peerUpfRole=~"$peerUpfRole", peerUpfIp=~"$peerUpfIp"}
```

SMF Radius InBand Tunnel Timer DashBoard

Description

This will display the radius inband tunnel timer create/delete stats

SMFCC RADIUS INBAND TUNNEL TIMER SUMMARY Panels

Panels

Inband Radius Tunnel Timer Start Attempts

Description

Number of Inband Radius Tunnel Create/Delete Timer Start Request messages sent

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_timer_start_attempts)
```

Inband Radius Tunnel Timer Start Success

Description

Number of Inband Radius Tunnel Create/Delete Timer Start Request Create Successful

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_timer_start_successes)
```

Inband Radius Tunnel Timer Start Failure

Description

Number of Inband Radius Tunnel Create/Delete Timer Start Request Failed

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_timer_start_failures{causecode=~"$causecode"})
```

Inband Radius Tunnel Timer Fire

Description

Number of Inband Radius Tunnel Create/Delete Timer Fire

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_timer_fires)
```

Inband Radius Tunnel Timer Restarts

Description

Number of Inband Radius Tunnel Create/Delete Timer Restart

Expression

`_auto`

```
sum(smfcc_inband_radius_tunnel_timer_restarts)
```

SMFCC RADIUS INBAND TUNNEL TIMER DETAILS Panels

SMFCC RADIUS INBAND TUNNEL TIMER DETAILS

Expression

Radius Inband Tunnel Timer Start Attempts

```
smfcc_inband_radius_tunnel_timer_start_attempts
```

Radius Inband Tunnel Timer Start Success

```
smfcc_inband_radius_tunnel_timer_start_successes
```

Radius Inband Tunnel Timer Start Failure {{{causecode}}}

```
smfcc_inband_radius_tunnel_timer_start_failures{causecode=~"$causecode"}
```

Radius Inband Tunnel Timer Fire

```
smfcc_inband_radius_tunnel_timer_fires
```

Radius Inband Tunnel Timer Restarts

```
smfcc_inband_radius_tunnel_timer_restarts
```

SMFCC Current Pdu Session Count by RAT Type

Description

SMFCC Current Pdu Session Count by RAT Type

Panels

PDU Sessions by RAT Type

Description

Guages showing the PDU Current Pdu Sessions count by RAT Type

Expression

`_auto`

```
sum(smf_current_pdu_sessions_per_rat_type{ratType="NR"})
```

`_auto`

```
sum(smf_current_pdu_sessions_per_rat_type{ratType="EUTRA"})
```

_auto

```
sum(smf_current_pdu_sessions_per_rat_type{ratType="UTRAN"})
```

_auto

```
sum(smf_current_pdu_sessions_per_rat_type{ratType="GERAN"})
```

_auto

```
sum(smf_current_pdu_sessions_per_rat_type{ratType="VIRTUAL"})
```

_auto

```
sum(smf_current_pdu_sessions_per_rat_type{ratType="WLAN"})
```

SMFCC RADIUS [ACCOUNTING] INTERFACE STATS

Description

This will display the number of radius accounting messages sent towards radius server

Radius Accounting Start Request Messages Summary Panels

Radius Accounting Start Request Sent

Description

This will display the number of accounting start request sent towards radius server via radius agent

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_start_requests_sent{dnn=~"$dnn", ratype=~"$ratype"})
```

Radius Accounting Start Request Send Failed

Description

This will display the number of accounting start request message send failed towards radius server

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_start_requests_send_failed{dnn=~"$dnn",  
ratype=~"$ratype"})
```

Radius Accounting Start Response Success

Description

This will display the number of accounting start response success received from radius server

Expression

```
sum(smfcc_radius_accounting_start_response_successes{dnn=~"$dnn",  
ratype=~"$ratype"})
```

Radius Accounting Start Response Failures

Description

This will display the number of accounting start response failures.

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_start_response_failures{dnn=~"$dnn",  
ratype=~"$ratype"})
```

Radius Accounting Start Request Timeouts

Description

This will display the number of accounting start request timeouts.

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_start_request_timeouts{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Start Request Messages Details Panels

Radius Accounting Start Message Details

Expression

Radius Accounting Start Request Sent {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_start_requests_sent{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Start Request Send Failed {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_start_requests_send_failed{dnn=~"$dnn",  
rattype=~"$rattype"}
```

Radius Accounting Start Response Success {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_start_response_successes{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Start Response Failure {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_start_response_failures{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Start Request Timeout {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_start_request_timeouts{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Interim Request Messages Summary Panels

Radius Accounting Interim Request Sent

Description

This will display the number of accounting Interim request sent towards radius server via radius agent

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_interim_requests_sent{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Interim Request Send Failed

Description

This will display the number of accounting interim request message send failed towards radius server

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_interim_requests_send_failed{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Interim Response Success

Description

This will display the number of accounting Interim response success received from radius server

Expression

```
sum(smfcc_radius_accounting_interim_response_successes{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Interim Response Failures

Description

This will display the number of accounting interim response failures.

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_interim_response_failures{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Interim Request Timeouts

Description

This will display the number of accounting Interim request timeouts.

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_interim_request_timeouts{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Interim Request Messages Details Panels

Radius Acccounting Interim Message Details

Expression

Radius Accounting Interim Request Sent {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_interim_requests_sent{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Interim Request Send Failed {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_interim_requests_send_failed{dnn=~"$dnn",  
rattype=~"$rattype"}
```

Radius Accounting Interim Response Success {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_interim_response_successes{dnn=~"$dnn",  
rattype=~"$rattype"}
```

Radius Accounting Interim Response Failure [{{dnn}}] [{{rattype}}]

```
smfcc_radius_accounting_interim_response_failures{dnn=~"$dnn",  
rattype=~"$rattype"}
```

Radius Accounting Interim Request Timeout [{{dnn}}] [{{rattype}}]

```
smfcc_radius_accounting_interim_request_timeouts{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Stop Request Messages Summary Panels

Radius Accounting Stop Request Sent

Description

This will display the number of accounting Stop request sent towards radius server via radius agent

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_stop_requests_sent{dnn=~"$dnn", rattype=~"$rattype"})
```

Radius Accounting Stop Request Send Failed

Description

This will display the number of accounting stop request message send failed towards radius server

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_stop_requests_send_failed{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Stop Response Success

Description

This will display the number of accounting stop response success received from radius server

Expression

```
sum(smfcc_radius_accounting_stop_response_successes{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Stop Response Failures

Description

This will display the number of accounting stop response failures.

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_stop_response_failures{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Stop Request Timeouts

Description

This will display the number of accounting Stop request timeouts.

Expression

Access Request Sent

```
sum(smfcc_radius_accounting_stop_request_timeouts{dnn=~"$dnn",  
rattype=~"$rattype"})
```

Radius Accounting Stop Request Messages Details Panels

Radius Accounting Stop Message Details

Expression

Radius Accounting Stop Request Sent {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_stop_requests_sent{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Stop Request Send Failed {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_stop_requests_send_failed{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Stop Response Success {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_stop_response_successes{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Stop Response Failure {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_stop_response_failures{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Accounting Stop Request Timeout {{{dnn}}} {{{rattype}}}

```
smfcc_radius_accounting_stop_request_timeouts{dnn=~"$dnn", rattype=~"$rattype"}
```

SMFCC RADIUS [AUTHENTICATION] INTERFACE STATS

Description

This will display the number of radius authentication messages sent towards radius server

Radius Authentication Messages Summary Panels

Panels

Radius Access Request Sent

Description

This will display the number of access request sent towards radius server via radius agent

Expression

Access Request Sent

```
sum(smfcc_radius_access_requests_sent{dnn=~"$dnn", ratype=~"$ratype"})
```

Radius Access Request Send Failed

Description

This will display the number of access request sent failed towards radius agent.

Expression

Access Request Sent

```
sum(smfcc_radius_access_requests_send_failed{dnn=~"$dnn", ratype=~"$ratype"})
```

Radius Access Accept Received

Description

This will display the number of access accept received from radius server via radius agent

Expression

```
sum(smfcc_radius_access_accepts{dnn=~"$dnn", ratype=~"$ratype"})
```

Radius Access Reject Received

Description

This will display the number of access reject received from radius server via radius agent.

Expression

Access Request Sent

```
sum(smfcc_radius_access_rejects{dnn=~"$dnn", ratype=~"$ratype"})
```

Radius Authentication Messages Comparison Panels

Radius Access Request Sent Vs Radius Access Accept Received vs Radius Reject Received

Expression

Radius Access Request Sent {{{dnn}}}[{{{rattype}}}]

```
smfcc_radius_access_requests_sent{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Access Accept Received {{{dnn}}}[{{{rattype}}}]

```
smfcc_radius_access_accepts{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Access Rejects Received {{{dnn}}}[{{{rattype}}}][{{{cause}}}]

```
smfcc_radius_access_rejects{dnn=~"$dnn", rattype=~"$rattype"}
```

Radius Access Request Send Failed {{{dnn}}}[{{{rattype}}}]

```
smfcc_radius_access_requests_send_failed{dnn=~"$dnn", rattype=~"$rattype"}
```

SMFCC-PFCP Session Bundling Dashboard

Description

SMFCC-PFCP Session Bundling Statistics Dashboard

PFCP Sent Failures - \$peer_ip Panels

Panels

Failure %

Expression

```
((sum(pfcip_bundle_message_sent_failure{namespace=~"$namespace",  
job="smfcc",cause=~"$cause"}))/sum(pfcip_bundle_message_sent{namespace=~"$namespac  
e", job="smfcc",cause=~"$cause"})))*100
```

PCFP number of bundled message sent failure - \$cause

Description

Number of PCFP bundle messages sent failures to a particular peer(represented by peer_ip and peer_port)

Expression

```
sum(pfcf_bundle_message_sent_failure{namespace=~"$namespace",
job="smfcc",cause=~"$cause"} OR on() vector(0))
```

PCFP Receive Failures - \$peer_ip Panels

Failure %

Expression

```
((sum(pfcf_bundle_message_received_failure{namespace=~"$namespace",
job="smfcc",cause=~"$cause"}))/(sum(pfcf_bundle_message_received{namespace=~"$name
space", job="smfcc",cause=~"$cause"}))) * 100
```

PCFP number of bundled message decode failure - \$causeR

Description

Number of PCFP bundle messages decode failures, cause represents failure reason(like invalid packet)

Expression

```
sum(pfcf_bundle_message_received_failure{namespace=~"$namespace",
job="smfcc",cause=~"$causeR"} OR on() vector(0))
```

PCFP Number of Sent Messages Per Bundle Trigger - \$peer_ip Panels

Pfcf number of bundled messages sent \$bundle trigger

Description

Number of PCFP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PCFP messages in a single bundle)

Expression

`_auto`

```
sum(pfcip_bundle_message_sent{namespace=~"$namespace",
job="smfcc",bundle_trigger=~"$bundle_trigger"} OR on() vector(0))
```

PFCP Total Number of Sent and Received Messages Per Peer Panels

Pfcip number of bundled messages sent \$peer ip

Description

Number of PFCP bundle messages sent to a particular peer(represented by peer_ip and peer_port) due to a particular trigger(timeout or max buffer size or max number of PFCP messages in a single bundle

Expression

`_auto`

```
sum(pfcip_bundle_message_sent{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"} OR on() vector(0))
```

PFCP number of bundled messages received \$peer ip

Description

Number of PFCP bundle messages received from a particular peer(represented by peer_ip and peer_port).

Expression

```
sum(pfcip_bundle_message_received{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"} OR on() vector(0))
```

PFCP Sent Message Size in Bundle Panels

\$peer ip

Expression

`{{peer_ip}}`

```
sum by (peer_ip)
((pfcip_size_of_message_of_bundled_request_sent_sum{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})/(pfcip_size_of_message_of_bundled_request_sent_co
unt{namespace=~"$namespace", job="smfcc",peer_ip=~"$peer_ip"}))
```


Message Size in Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfcp_size_of_message_of_bundled_request_sent_bucket{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})
```

PFCP Sent Message Count in Bundle Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcp_number_of_messages_in_bundled_request_sent_sum{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})/(pfcp_number_of_messages_in_bundled_request_sent
_count{namespace=~"$namespace", job="smfcc",peer_ip=~"$peer_ip"}))
```

Number of messages in Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfcp_number_of_messages_in_bundled_request_sent_bucket{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})
```

PFCP Sent Message Bundling Duration Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfc_p_duration_of_bundling_in_bundled_request_sent_sum{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})/(pfc_p_duration_of_bundling_in_bundled_request_se
nt_count{namespace=~"$namespace", job="smfcc",peer_ip=~"$peer_ip"}))
```

Duration of Bundling in Sent Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfc_p_duration_of_bundling_in_bundled_request_sent_bucket{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})
```

PFCP Received Message Size Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfc_p_size_of_message_of_bundled_request_received_sum{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})/(pfc_p_size_of_message_of_bundled_request_receive
d_count{namespace=~"$namespace", job="smfcc",peer_ip=~"$peer_ip"}))
```

Message Size in Received Bundle

Expression

{{le}}

```
sum by (procedure,le)
(pfc_p_size_of_message_of_bundled_request_received_bucket{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})
```

PFCP Received Messages Count Panels

\$peer ip

Expression

{{peer_ip}}

```
sum by (peer_ip)
((pfcip_number_of_messages_in_bundled_request_received_sum{namespace=~"$namespace",
job="smfcc",peer_ip=~"$peer_ip"})/(pfcip_number_of_messages_in_bundled_request_rece
ived_count{namespace=~"$namespace", job="smfcc",peer_ip=~"$peer_ip"}))
```

Number of Messages in Bundle Received

Expression

{{le}}

```
sum by (procedure,le)
(pfcip_number_of_messages_in_bundled_request_received_bucket{namespace=~"$namespace
", job="smfcc",peer_ip=~"$peer_ip"})
```

SMFCC-PFCP Sessions Procedures Dashboard

Description

SMFCC-PFCP Sessions Procedures Dashboard

Panels

Session Establishment - Attempts

Description

The number of session establishment attempts.

Expression

```
sum(pfcip_session_establishment_attempt{namespace=~"$namespace", job="smfcc",
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Establishment - Success %

Description

Session Establishment - Success %

Expression

```
((sum(pfcpsessionestablishmentsuccess{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))/sum(pfcpsessionestablishmentattempt{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))*100
```

Session Establishment - Successes

Description

The number of session establishment successes.

Expression

```
sum(pfcpsessionestablishmentsuccess{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Establishment - Failures

Description

The number of session establishment failures.

Expression

```
sum(pfcpsessionestablishmentfailure{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Modification - Attempts

Description

The number of session modification attempts.

Expression

```
sum(pfcpsessionmodificationattempt{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Modification - Success %

Description

Session Modification - Success %

Expression

```
((sum(pfcpsessionmodification_success{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})))/(sum(pfcpsessionmodification_attempt{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"}))*100
```

Session Modification - Successes

Description

The number of session modification successes.

Expression

```
sum(pfcpsessionmodification_success{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Modification - Failures

Description

The number of session modification failures.

Expression

```
sum(pfcpsessionmodification_failure{namespace=~"$namespace", job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

Session Deletion - Attempts

Description

The number of session deletion attempts.

Expression

```
sum(pfcpc_session_deletion_attempt{namespace=~"$namespace", job="smfcc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Deletion - Success %

Description

Session Deletion - Success %

Expression

```
((sum(pfcpc_session_deletion_success{namespace=~"$namespace", job="smfcc",  
peer_id=~"$peer_id", nssai=~"$nssai",  
dnn=~"$dnn"})))/(sum(pfcpc_session_deletion_attempt{namespace=~"$namespace",  
job="smfcc", peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})))*100
```

Session Deletion - Successes

Description

The number of session deletion successes.

Expression

```
sum(pfcpc_session_deletion_success{namespace=~"$namespace", job="smfcc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn"})
```

Session Deletion - Failures

Description

The number of session deletion failures.

Expression

```
sum(pfcpc_session_deletion_failure{namespace=~"$namespace", job="smfcc",  
peer_id=~"$peer_id", nssai=~"$nssai", dnn=~"$dnn", cause=~"$cause"})
```

SMFCC-PFCP Sessions Protocols Dashboard

Description

SMFCC-PFCP Sessions Protocols Dashboard

Panels

Session Establishment - Requests Sent

Description

The number of session establishment request messages sent.

Expression

```
sum(pfc_session_establishment_request_sent{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip"})
```

Session Establishment - Retransmitted Requests Sent

Description

The number of session establishment retransmitted request messages sent.

Expression

```
sum(pfc_session_establishment_retransmitted_request_sent{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip"})
```

Session Establishment - Responses Received

Description

The number of session establishment response messages received.

Expression

```
sum(pfc_session_establishment_response_received{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip", pfc_cause=~"$pfc_cause"})
```

Session Modification - Requests Sent

Description

The number of session modification request messages sent.

Expression

```
sum(pfcpsessionmodificationrequest_sent{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip"})
```

Session Modification - Retransmitted Requests Sent

Description

The number of session modification retransmitted request messages sent.

Expression

```
sum(pfcpsessionmodificationretransmitted_request_sent{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip"})
```

Session Modification - Responses Received

Description

The number of session modification response messages received.

Expression

```
sum(pfcpsessionmodificationresponse_received{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip", pfcpcause=~"$pfcpcause"})
```

Session Deletion - Requests Sent

Description

The number of session deletion request messages sent.

Expression

```
sum(pfcpsessiondeletionrequest_sent{namespace=~"$namespace", job="smfcc", peer_ip=~"$peer_ip"})
```

Session Deletion - Retransmitted Requests Sent

Description

The number of session deletion retransmitted request messages sent.

Expression

```
sum(pfcsp_session_deletion_retransmitted_request_sent{namespace=~"$namespace",
job="smfcc", peer_ip=~"$peer_ip"})
```

Session Deletion - Responses Received

Description

The number of session deletion response messages received.

Expression

```
sum(pfcsp_session_deletion_response_received{namespace=~"$namespace", job="smfcc",
peer_ip=~"$peer_ip", pfcsp_cause=~"$pfcsp_cause"})
```

Smfcc GTPC Interface Dashboard

Description

Dashboard showing SMFCC GTPC interface messages statistics

Smfcc Interface Round Trip Duration

Description

Displays interface round trip duration for SMFCC initiated messages

Panels

\$interface

Description

Displays interface round trip duration for SMFCC initiated messages

Expression

{{interface}}

```
sum by (interface)
((smfcc_interface_round_trip_duration_sum{interface=~"$interface"})/(smfcc_interfa
ce_round_trip_duration_count{interface=~"$interface"}))
```

Round Trip Time

Description

Round Trip Time

Expression

{{le}}

```
sum by (procedure,le)
(smfcc_interface_round_trip_duration_bucket{interface=~"$interface"})
```

Smfcc Interface Timeout Count

Description

Dashboard for Timeout count on every smfcc interface

Panels

Total failures due to timeout

AMF Timeout count

PCF Timeout count

CHF Timeout count

UDM Timeout count

GTPC Timeout Count

Smfcc Procedure Duration Statistics

Description

Smfcc Procedure Duration Statistics

Panels

\$procedure

Description

Total Procedure Count

Expression

{{procedure}}

```
sum by (procedure) (smfcc_3gpp_procedure_duration_count{procedure=~"$procedure"})
```

SUCCESS

Description

Total Success Procedure Count

Expression

{{le}}

```
sum by (procedure,le)
(smfcc_3gpp_procedure_duration_bucket{procedure=~"$procedure",status="SUCCESS"})
```

FAILURE

Description

Total Failure Procedure Count

Expression

{{le}}

```
sum by (procedure,le)
(smfcc_3gpp_procedure_duration_bucket{procedure=~"$procedure",status="FAILURE"})
```

DROPPED

Description

Total Dropped Procedure Count

Expression

{{le}}

```
sum by (procedure,le)
(smfcc_3gpp_procedure_duration_bucket{procedure=~"$procedure",status="DROPPED"})
```

Smfcc Radius Inband Tunnel ETCD put/delete range Dashboard

Description

This will display the radius inband etcd put/response stats

ETCD put/delete range summary Panels

Inband Radius Tunnel Put Request

Expression

`_auto`

```
sum(smfcc_etcd_inband_radius_tunnel_put_request)
```

Inband Radius Tunnel Put Response

Expression

`_auto`

```
sum(smfcc_etcd_inband_radius_tunnel_put_response{responsecode=~"$responsecode"})
```

Inband Radius Tunnel Delete Range Request

Expression

`_auto`

```
sum(smfcc_etcd_inband_radius_tunnel_delete_range_request)
```

Inband Radius Tunnel Delete Range Response

Expression

`_auto`

```
sum(smfcc_etcd_inband_radius_tunnel_delete_range_response{responsecode=~"$response  
code"})
```

ETCD put/delete range details Panels

Panel Title

Expression

Inband Radius Tunnel Put Request

```
smfcc_etcd_inband_radius_tunnel_put_request
```

Inband Radius Tunnel Put Response {{{responsecode}}}

```
smfcc_etcd_inband_radius_tunnel_put_response{responsecode=~"$responsecode"}
```

Inband Radius Tunnel Delete Range Request

```
smfcc_etcd_inband_radius_tunnel_delete_range_request
```

Inband Radius Tunnel Delete Range Response {{{responsecode}}}

```
smfcc_etcd_inband_radius_tunnel_delete_range_response{responsecode=~"$responsecode  
"}
```

Description

Dashboard showing SMFCC SGW GTPC interface messages statistics

Panels

Create Session - Attempts

Description

Number of Create Session attempted

Expression

```
sum(smfcc_gtpc_sgw_create_session_attempts{dnn=~"$dnn"})
```

Create Session - Success %

Description

Meter showing percentage of Create Session attempts which are successful

Expression

```
sum(smfcc_gtpc_sgw_create_session_accepts{dnn=~"$dnn"})/sum(smfcc_gtpc_sgw_create_session_attempts{dnn=~"$dnn"})*100
```

Create Session - Successful

Description

Number of Create Session attempts successful

Expression

```
sum(smfcc_gtpc_sgw_create_session_accepts{dnn=~"$dnn"})
```

Delete Session - Attempts

Description

Number of Delete Session request attempted

Expression

```
sum(smfcc_gtpc_sgw_delete_session_attempts{dnn=~"$dnn"})
```

Delete Session - Success %

Description

Meter showing percentage of Delete Session attempts which are successful

Expression

```
sum(smfcc_gtpc_sgw_delete_session_successes{dnn=~"$dnn"})/sum(smfcc_gtpc_sgw_delete_session_attempts{dnn=~"$dnn"})*100
```

Delete Session - Successful

Description

Number of Delete Session attempts successful

Expression

```
sum(smfcc_gtpc_sgw_delete_session_successes{dnn=~"$dnn"})
```

Modify Bearer Request - Attempts

Description

Number of GTPC SGW Modify Bearer Request attempted

Expression

```
sum(smfcc_gtpc_sgw_modify_bearer_request_attempts{dnn=~"$dnn"})
```

Modify Bearer Request - Success %

Description

Meter showing percentage of Modify Bearer Request attempts which are successful

Expression

```
sum(smfcc_gtpc_sgw_modify_bearer_request_successes{dnn=~"$dnn"})/sum(smfcc_gtpc_sgw_modify_bearer_request_attempts{dnn=~"$dnn"})*100
```

Modify Bearer Request - Successful

Description

Number of Modify Bearer Request attempts successful

Expression

```
sum(smfcc_gtpc_sgw_modify_bearer_request_successes{dnn=~"$dnn"})
```

Create Bearer Request - Attempts

Description

Number of GTPC SGW Create Bearer Request attempted

Expression

```
sum(smfcc_gtpc_sgw_create_bearer_attempts{dnn=~"$dnn",ratType=~"$ratType"})
```

Create Bearer Request - Success %

Description

Meter showing percentage of Create Bearer Request attempts which are successful

Expression

```
sum(smfcc_gtpc_sgw_create_bearer_accepts{dnn=~"$dnn",ratType=~"$ratType"})/sum(smfcc_gtpc_sgw_create_bearer_attempts{dnn=~"$dnn",ratType=~"$ratType"})*100
```

Create Bearer Request - Successful

Description

Number of Create Bearer Request attempts successful

Expression

```
sum(smfcc_gtpc_sgw_create_bearer_accepts{dnn=~"$dnn",ratType=~"$ratType"})
```

Create Bearer Request Forwarded to MME - Attempts

Description

Number of GTPC SGW Create Bearer Request forwarded by SGW to MME attempted

Expression

```
sum(smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_attempts{dnn=~"$dnn",ratType=~"$ratType"})
```

Create Bearer Request forwarded to MME - Success %

Description

Meter showing percentage of Create Bearer Request forwarded to MME attempts which are successful

Expression

```
sum(smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_accepts{dnn=~"$dnn",ratType=~"$ratType"})/sum(smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_attempts{dnn=~"$dnn",ratType=~"$ratType"})*100
```

Create Bearer Request Forwarded to MME - Successful

Description

Number of Create Bearer Request Forwarded to MME attempts successful

Expression

```
sum(smfcc_gtpc_sgw_create_bearer_request_sent_to_mme_accepts{dnn=~"$dnn",ratType=~"$ratType"})
```

Delete Bearer Request - Attempts

Description

Number of GTPC SGW Delete Bearer Request attempted

Expression

```
sum(smfcc_gtpc_sgw_delete_bearer_request_attempts{dnn=~"$dnn"})
```

Delete Bearer Request - Success %

Description

Meter showing percentage of Delete Bearer Request attempts which are successful

Expression

```
sum(smfcc_gtpc_sgw_delete_bearer_request_successes{dnn=~"$dnn"})/sum(smfcc_gtpc_sgw_delete_bearer_request_attempts{dnn=~"$dnn"})*100
```

Delete Bearer Request - Successful

Description

Number of Delete Bearer Request attempts successful

Expression

```
sum(smfcc_gtpc_sgw_delete_bearer_request_successes{dnn=~"$dnn"})
```

VSMF Call Control

Description

Summary information for operations on all VSMF interfaces

Panels

SM Context

SMF Interface

Handover Procedures

PDU Session

VSMF Procedures Dashboard

Description

VSMF Procedures Dashboard

Panels

PDU Session Establishment - Attempts

Description

Number of PDU Session Establishment Procedures attempted

Expression

```
sum(smfcc_vsmf_pdu_session_establishment_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

PDU Session Establishment - Success %

Description

PDU Session Establishment - Success %

Expression

```
((sum(smfcc_vsmf_pdu_session_establishment_successes{dnn=~"$dnn", snssai=~"$snssai"})) / (sum(smfcc_vsmf_pdu_session_establishment_attempts{dnn=~"$dnn", snssai=~"$snssai"}))) * 100
```

PDU Session Establishment - Successes

Description

Number of PDU Session Establishment Procedures successful

Expression

```
sum(smfcc_vsmf_pdu_session_establishment_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Deregistration - Attempts

Description

Number of UE Deregistration Procedures attempted

Expression

```
sum(smfcc_vsmf_ue_deregistration_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Deregistration - Success %

Description

UE Deregistration - Success %

Expression

```
((sum(smfcc_vsmf_ue_deregistration_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_vsmf_ue_deregistration_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

UE Deregistration - Successes

Description

Number of UE Deregistration Procedures successful

Expression

```
sum(smfcc_vsmf_ue_deregistration_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

AN Release - Attempts

Description

Number of AN Release Procedures attempted

Expression

```
sum(smfcc_vsmf_an_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

AN Release - Success %

Description

AN Release - Success %

Expression

```
((sum(smfcc_vsmf_an_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_vsmf_an_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))*100
```

AN Release - Successes

Description

Number of AN Release Procedures successful

Expression

```
sum(smfcc_vsmf_an_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Service Request - Attempts

Description

Number of Service Request Procedures attempted

Expression

```
sum(smfcc_vsmf_service_request_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Service Request - Success %

Description

Service Request - Success %

Expression

```
((sum(smfcc_vsmf_service_request_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/sum(smfcc_vsmf_service_request_attempts{dnn=~"$dnn",  
snssai=~"$snssai"}))*100
```

Service Request - Successes

Description

Number of Service Request Procedures successful

Expression

```
sum(smfcc_vsmf_service_request_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

NW Initiated Release - Attempts

Description

Number of NW Initiated Release Procedures attempted

Expression

```
sum(smfcc_vsmf_initiated_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

NW Initiated Release - Success %

Description

NW Initiated Release - Success %

Expression

```
((sum(smfcc_vsmf_initiated_release_successes{dnn=~"$dnn", snssai=~"$snssai"}))/(sum(smfcc_vsmf_initiated_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})))*100
```

NW Initiated Release - Successes

Description

Number of NW Initiated Release Procedures successful

Expression

```
sum(smfcc_vsmf_initiated_release_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Xn Handover - Attempts

Description

Number of Xn Handover Procedures with No UPF change attempted

Expression

```
sum(smfcc_vsmf_xn_handover_attempts{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"false"})
```

Xn Handover - Success %

Description

Xn Handover With No UPF Change - Success %

Expression

```
((sum(smfcc_vsmf_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"false"}))/(sum(smfcc_vsmf_xn_handover_attempts{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"false"})))*100
```

Xn Handover - Successes

Description

Number of Xn Handover Procedures with No UPF change successful

Expression

```
sum(smfcc_vsmf_xn_handover_successes{dnn=~"$dnn", snssai=~"$snssai", iupfrelocation=~"false"})
```

UE Initiated Modify - Attempts

Description

Number of UE Initiated Modify Procedures attempted

Expression

```
sum(smfcc_vsmf_ue_initiated_modify_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Initiated Modify - Success %

Description

UE Initiated Modify - Success %

Expression

```
((sum(smfcc_vsmf_ue_initiated_modify_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_vsmf_ue_initiated_modify_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

UE Initiated Modify - Successes

Description

Number of UE Initiated Modify Procedures successful

Expression

```
sum(smfcc_vsmf_ue_initiated_modify_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

HSMF Initiated Qos Modify - Attempts

Description

Number of HSMF Initiated Modify Qos Procedures attempted

Expression

```
sum(smfcc_vsmf_hsmf_initiated_pdu_session_modification_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

HSMF Initiated Qos Modify - Success %

Description

HSMF Initiated Qos Modify - Success %

Expression


```
((sum(smfcc_vsmf_hsmf_initiated_pdu_session_modification_successes{dnn=~"$dnn", snssai=~"$snssai"})))/(sum(smfcc_vsmf_hsmf_initiated_pdu_session_modification_attempts{dnn=~"$dnn", snssai=~"$snssai"}))) * 100
```

HSMF Initiated Qos Modify - Successes

Description

Number of HSMF Initiated Modify Qos Procedures successful

Expression

```
sum(smfcc_vsmf_hsmf_initiated_pdu_session_modification_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Initiated PDU Session Release - Attempts

Description

Number of UE Initiated PDU Session Release Procedures attempted

Expression

```
sum(smfcc_vsmf_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

UE Initiated PDU Session Release - Success %

Description

UE Initiated PDU Session Release - Success %

Expression

```
((sum(smfcc_vsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn", snssai=~"$snssai"})))/(sum(smfcc_vsmf_ue_initiated_pdu_session_release_attempts{dnn=~"$dnn", snssai=~"$snssai"}))) * 100
```

UE Initiated PDU Session Release - Successes

Description

Number of UE Initiated PDU Session Release Procedures successful

Expression

```
sum(smfcc_vsmf_ue_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Network Initiated Service Request - Attempts

Description

Number of Network Initiated Service Request Procedures attempted

Expression

```
sum(smfcc_vsmf_network_initiated_service_request_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Network Initiated Service Request - Success %

Description

Network Initiated Service Request - Success %

Expression

```
((sum(smfcc_vsmf_network_initiated_service_request_successes{dnn=~"$dnn",  
snssai=~"$snssai"})))/(sum(smfcc_vsmf_network_initiated_service_request_attempts{dn  
n=~"$dnn", snssai=~"$snssai"}))*100
```

Network Initiated Service Request - Successes

Description

Number of Network Initiated Service Request Procedures successful

Expression

```
sum(smfcc_vsmf_network_initiated_service_request_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

HSMF Initiated Release - Attempts

Description

Number of HSMF Initiated Release Procedures attempted

Expression

```
sum(smfcc_vsmf_hsmf_initiated_pdu_session_release_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

HSMF Initiated Release - Success %

Description

HSMF Initiated Release - Success %

Expression

```
((sum(smfcc_vsmf_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_vsmf_hsmf_initiated_pdu_session_release_attempts{d  
nn=~"$dnn", snssai=~"$snssai"})))*100
```

HSMF Initiated Release - Successes

Description

Number of HSMF Initiated Release Procedures successful

Expression

```
sum(smfcc_vsmf_hsmf_initiated_pdu_session_release_successes{dnn=~"$dnn",  
snssai=~"$snssai"})
```

EPS to 5G Idle Handover - Attempts

Description

Number of EPS to 5G Idle Handover Procedures attempted

Expression

```
sum(smfcc_vsmf_eps_to_5gs_idle_mobility_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

EPS to 5G Idle Handover - Success %

Description

EPS to 5G Idle Handover - Success %

Expression

```
((sum(smfcc_vsmf_eps_to_5gs_idle_mobility_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_vsmf_eps_to_5gs_idle_mobility_attempts{dnn=~"$dnn"  
, snssai=~"$snssai"})))*100
```

EPS to 5G Idle Handover - Successes

Description

Number of EPS to 5G Idle Handover Procedures successful

Expression

```
sum(smfcc_vsmf_eps_to_5gs_idle_mobility_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Network of Usage Report - Attempts

Description

Number of Usage Report Processing Procedure attempted

Expression

```
sum(smfcc_upf_usage_report_procedure_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Network of Usage Report - Success %

Description

Network of Usage Report - Success %

Expression

```
((sum(smfcc_upf_usage_report_procedure_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_upf_usage_report_procedure_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})))*100
```

Network of Usage Report - Successes

Description

Number of Usage Report Processing Procedure successful

Expression

```
sum(smfcc_upf_usage_report_procedure_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

Error Indication - Attempts

Description

Number of Error Indication Procedure attempted

Expression

```
sum(smfcc_vsmf_error_indication_procedure_attempts{dnn=~"$dnn",  
snssai=~"$snssai"})
```

Error Indication - Success %

Description

Error Indication - Success %

Expression

```
((sum(smfcc_vsmf_error_indication_procedure_successes{dnn=~"$dnn",  
snssai=~"$snssai"}))/(sum(smfcc_vsmf_error_indication_procedure_attempts{dnn=~"$dn  
n", snssai=~"$snssai"}))) * 100
```

Error Indication - Successes

Description

Number of Error Indication Procedure successful

Expression

```
sum(smfcc_vsmf_error_indication_procedure_successes{dnn=~"$dnn",
snssai=~"$snssai"})
```

Session Idle - Attempts

Description

Number of Session Idle Procedure attempted

Expression

```
sum(smfcc_vsmf_session_idle_procedure_attempts{dnn=~"$dnn", snssai=~"$snssai"})
```

Session Idle - Success %

Description

Session Idle - Success %

Expression

```
((sum(smfcc_vsmf_session_idle_procedure_successes{dnn=~"$dnn",
snssai=~"$snssai"}))/(sum(smfcc_vsmf_session_idle_procedure_attempts{dnn=~"$dnn",
snssai=~"$snssai"})))*100
```

Session Idle - Successes

Description

Number of Session Idle Procedure successful

Expression

```
sum(smfcc_vsmf_session_idle_procedure_successes{dnn=~"$dnn", snssai=~"$snssai"})
```

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Chronos Dashboard

Description

Chronos Dashboard

Cluster health metrics Panels

Panels

Count of primary replicas per pod

Description

Counts the number of primary replicas each pod owns.

Expression

{{namespace}} - {{pod}}

```
sum(_experimental_chronos_timer_count_by_replica_position{namespace=~"[[namespace]]", pod=~"$pod", position="0"}) by (namespace, pod)
```

Timer callback results across namespaces

Expression

{{namespace}} - Success

```
sum(rate(_experimental_chronos_callback_success{namespace=~"[[namespace]]"}[1m])) by (namespace)
```

{{namespace}} - Failure

```
sum(rate(_experimental_chronos_callback_failure{namespace=~"[[namespace]]"}[1m])) by (namespace)
```

Count of Chronos replicas per pod

Description

This includes non-primary replicas. Thus, in a deployment with timers that have a replication factor of 2, each Chronos pod will hold both primary and backup replicas.

Expression

{{namespace}} - {{pod}}

```
sum(_experimental_chronos_timer_count_total{namespace=~"[[namespace]]", type="active", pod=~"$pod"}) by (namespace, pod)
```

Rejected timer requests

Description

Total number of timers requests have been rejected by Chronos

Expression

{{failure_type}}

```
chronos_failed_to_create_timer{namespace="[[namespace]]"}
```

Replicas by tag (includes duplicates)

Description

This counts the number of replicas across the deployment which have a particular tag attached to them. Timers can have multiple tags attached at once.

Expression

{{namespace}} - {{caller_tag}}

```
sum(chronos_timer_count_by_tag{namespace=~"[[namespace]]", pod=~"$pod"}) by (namespace, caller_tag)
```

Deployment breakdown Panels

Count of total Chronos timers

Description

Including timers that have been "tombstoned"

Expression

{{namespace}} - {{pod}} - {{type}}

```
sum(_experimental_chronos_timer_count_total{namespace=~"[[namespace]]", pod=~"$pod"}) by (namespace, pod, type)
```

_auto

Total replica counts

Expression**Replica - {{position}}**

```
sum(_experimental_chronos_timer_count_by_replica_position{namespace=~"[[namespace]]"}) by (position)
```

Rate of incoming Chronos requests

Description

Rate of incoming Chronos requests.

Expression**{{pod}} - {{request_type}}**

```
sum (rate(_experimental_incoming_requests{namespace=~"[[namespace]]"}[ $__rate_interval])) by (namespace, pod, request_type)
```

Rate of incoming Chronos create requests

Description

Average incoming POST messages received.

Expression**{{namespace}} - {{pod}}**

```
sum (rate(_experimental_incoming_requests{namespace=~"[[namespace]]", request_type="POST"}[ $__rate_interval])) by (namespace, pod)
```

Split of ownership across Chronos pods

Description

e.g. "chronos-X - 0" tracks the number of primary replicas that Chronos X owns

Expression**{{pod}} - {{position}}**

```
sum(_experimental_chronos_timer_count_by_replica_position{namespace=~"[[namespace]]", pod=~"$pod"}) by (namespace, pod, position)
```

Total Chronos incoming requests

Expression

{{pod}}

```
sum (rate(_experimental_incoming_requests{namespace=~"[[namespace]]"}[30s])) by (namespace, pod)
```

Callbacks Panels

Successful callback resolutions

Expression

{{pod}} - {{result}}

```
rate(_experimental_chronos_callback_success_resolution{namespace=~"[[namespace]]"}[2m])
```

Failed callback resolution

Description

Chronos failed to complete a callback for a timer. Upon completing the callback, Chronos took the follow action:

- missing_timer - There was no record of the original timer that had popped! (indicates an error condition).
- timer_tombstoned - The original timer was found, and is tombstoned with the expectation a replica will take responsibility for it.
- already_tombstoned - The original timer was already a tombstone (indicates a client has updated the timer during callback processing).
- not_local - The original timer no longer belonged to the node (indicates that a resync operation has moved the timer during callback processing).
- timer_untouched - The original timer was reset for a short amount of time to allow a retry.
- timer_dropped - The original timer was dropped because it failed too many callbacks.

Expression

{{namespace}} - {{result}}

```
sum(rate(_experimental_chronos_callback_failure_resolution{namespace=~"
[[namespace]]"}[2m])) by (namespace, result)
```

Callback results - by URL

Expression

{{pod}} - {{callback_type}} (Success)

```
rate(_experimental_chronos_callback_success_by_url{namespace=~"[[namespace]]"}
[1m])
```

{{pod}} - {{callback_type}} (Failure)

```
rate(_experimental_chronos_callback_failure_by_url{namespace=~"[[namespace]]"}
[1m])
```

Callback results - by tag

Expression

{{pod}} - {{caller_tag}} (Success)

```
rate(_experimental_chronos_callback_success_by_tag{namespace=~"[[namespace]]"}
[1m])
```

{{pod}} - {{caller_tag}} (Failure)

```
rate(_experimental_chronos_callback_failure_by_tag{namespace=~"[[namespace]]"}
[1m])
```

Resynchronization Panels

Failed resync count

Expression

{{pod}}

```
sum(_experimental_chronos_failed_current_resyncs_total{namespace=~"[[namespace]]"}) by (namespace, pod)
```

Count of resync events

Expression**{{pod}} - Resync events**

```
sum(_experimental_chronos_resync_event_count{namespace=~"[[namespace]]"}) by (namespace, pod)
```

{{pod}} - Resync Pending nodes

```
sum(_experimental_chronos_resync_pending_nodes{namespace=~"[[namespace]]"}) by (pod)
```

{{pod}} - Resync Timers processed

```
sum(_experimental_chronos_resync_timers_processed{namespace=~"[[namespace]]"}) by (pod)
```

Queues and Latency Panels

Per-Callback Delay Incurred Waiting for Hash Ring Construction

Description**Expression****_auto**

```
sum(rate(_experimental_hash_ring_cache_delay_ms_bucket[$__rate_interval])) by (le) / ignoring (le) group_left  
sum(rate(_experimental_hash_ring_cache_delay_ms_count[$__rate_interval])) by (le)
```

Queue lengths

Expression

{{pod}} - {{queue_name}}

```
sum(_experimental_queue_length{namespace=~"[[namespace]]", pod=~"$pod",
queue_name=~"timers_pending_callback|timers_pending_replication|gr_replication"})
by (namespace, pod, queue_name)
```

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Summary

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HTTP Client and Server Stats Dashboard	HTTP server outgoing responses rate over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP Client outgoing request rate over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP server transaction success percentage over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP client transaction success percentage over last 15 minutes
HTTP Client and Server Stats Dashboard	HTTP Client incoming response rate over last 15 minutes
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Dashboard Name	Panel Name
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UPF Overview	UPF Network Information-UPF Downlink - bytes

HTTP Client and Server Stats Dashboard

Description

Dashboard for HTTP client and server stats over the last 15 minutes.

Panels

HTTP server incoming request rate over last 15 minutes

Description

Rate of incoming http requests at server from different clients over last 15 minutes.

Expression

```
rate(http_server_request_received[15m])
```

HTTP server outgoing responses rate over last 15 minutes

Description

Rate of outgoing http responses from server to different clients over last 15 minutes.

Expression

```
rate(http_server_response_sent[15m])
```


HTTP Client outgoing request rate over last 15 minutes

Description

Rate of outgoing HTTP client requests to various servers over last 15 minutes.

Expression

```
rate(http_async_client_request_enqueue_success[15m])
```

HTTP server transaction success percentage over last 15 minutes

Description

Percentage of HTTP server incoming requests received for which the application responds with 2xx status code over last 15 minutes.

Expression

```
sum without (http_status_code)
((sum_over_time(http_server_response_sent{http_status_code=~"2.."}[15m]) /
ignoring(http_status_code) group_left
sum_over_time(http_server_request_received[15m])) * 100)
```

HTTP client transaction success percentage over last 15 minutes

Description

Percentage of HTTP client outgoing requests sent for which the server responds with 2xx status code over last 15 minutes.

Expression

```
sum without (http_status_code)
((sum_over_time(http_async_client_response_received{http_status_code=~"2.."}[15m])
/ ignoring(http_status_code) group_left
sum_over_time(http_async_client_request_enqueue_success[15m])) * 100)
```

HTTP Client incoming response rate over last 15 minutes

Description

Rate of HTTP client incoming response messages from various servers over the last 15 minutes.

Expression

```
rate(http_async_client_response_received[15m])
```

HTTP punting dashboard

Description

Dashboard for HTTP punting metrics stats over the last 15 minutes

Panels

HTTP punting outgoing request rate over last 15 min

Description

Rate of outgoing punting requests from applications over last 15 min

Expression

```
rate(punting_http_requests_sent[15m])
```

HTTP punting incoming response rate over last 15 min

Description

Rate of incoming punting responses for punted requests over last 15 min

Expression

```
rate(punting_http_responses_received[15m])
```

Punting response latency over last 15 min

Description

95th percentile point of punting response latency over last 15 min

Expression

```
histogram_quantile(0.95,
sum(rate(punting_http_request_duration_seconds_bucket[15m])) by (le,
canaryVersion, fed_unique_name, job, pod))
```

HTTP punting response forward rate over last 15 min

Description

Rate of forwarding punting responses to client over last 15 min

Expression

```
rate(punting_http_responses_forwarded[15m])
```

UPF Detailed View

Description

Detailed metrics for the operation of the UPF NF

DBCLIENT_HTTP Panels

DB Client State Store HTTP transactions Rate

Description

DB Client HTTP Transaction Rate by result

Expression

{{http_status_code}}

```
sum by (http_status_code) (rate(dbclient_http_requests{namespace="fed-upf",
db_type="mongo_RAM"}[$rate_interval]))
```

UPF Session Panels

Total UPF Session Create Request rates

Description

Rate of Total UPF Session Create Request requests

Expression**Attempts**

```
sum(rate(upf_session_db_creation_request[$rate_interval]))
```

Cancels

```
sum(rate(upf_session_db_creation_request_cancel[$rate_interval]))
```

Successes

```
sum(rate(upf_session_db_creation_request_success[$rate_interval]))
```

Failures

```
sum(rate(upf_session_db_creation_request_failure[$rate_interval]))
```

Total UPF Session Create Request result %

Description

Total UPF Session Create Request result %

Expression**Success**

```
((sum(rate(upf_session_db_creation_request_success[$rate_interval])))/(sum(rate(upf_session_db_creation_request[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_db_creation_request_failure[$rate_interval])))/(sum(rate(upf_session_db_creation_request[$rate_interval]))))*100
```

Current Total UPF Session Create Request success rate

Description

Current Total UPF Session Create Request success rate

Expression

```
(sum(rate(upf_session_db_creation_request_success[$rate_interval])) OR on()  
vector(0)) / sum(rate(upf_session_db_creation_request[$rate_interval]))
```

Total UPF Session Modify Request rates

Description

Rate of Total UPF Session Modify Request requests

Expression

Attempts

```
sum(rate(upf_session_db_modify_request[$rate_interval]))
```

Cancels

```
sum(rate(upf_session_db_modify_request_cancel[$rate_interval]))
```

Successes

```
sum(rate(upf_session_db_modify_request_success[$rate_interval]))
```

Failures

```
sum(rate(upf_session_db_modify_request_failure[$rate_interval]))
```

Total UPF Session Modify Request result %

Description

Total UPF Session Modify Request result %

Expression

Success

```
((sum(rate(upf_session_db_modify_request_success[$rate_interval])))/(sum(rate(upf_session_db_modify_request[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_db_modify_request_failure[$rate_interval])))/(sum(rate(upf_session_db_modify_request[$rate_interval]))))*100
```

Current Total UPF Session Modify Request success rate

Description

Current Total UPF Session Modify Request success rate

Expression

```
(sum(rate(upf_session_db_modify_request_success[$rate_interval])) OR on() vector(0)) / sum(rate(upf_session_db_modify_request[$rate_interval]))
```

Total UPF Session Delete Request rates

Description

Rate of Total UPF Session Delete Request requests

Expression

Attempts

```
sum(rate(upf_session_db_delete_request[$rate_interval]))
```

Cancels

```
sum(rate(upf_session_db_delete_request_cancel[$rate_interval]))
```

Successes

```
sum(rate(upf_session_db_delete_request_success[$rate_interval]))
```

Failures

```
sum(rate(upf_session_db_delete_request_failure[$rate_interval]))
```

Total UPF Session Delete Request result %

Description

Total UPF Session Delete Request result %

Expression

Success

```
((sum(rate(upf_session_db_delete_request_success[$rate_interval])))/(sum(rate(upf_session_db_delete_request[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_db_delete_request_failure[$rate_interval])))/(sum(rate(upf_session_db_delete_request[$rate_interval]))))*100
```

Current Total UPF Session Delete Request success rate

Description

Current Total UPF Session Delete Request success rate

Expression

```
(sum(rate(upf_session_db_delete_request_success[$rate_interval])) OR on()  
vector(0)) / sum(rate(upf_session_db_delete_request[$rate_interval]))
```

Total UPF Session Update Request rates

Description

Rate of Total UPF Session Update Request requests

Expression

Attempts

```
sum(rate(upf_session_db_update_request[$rate_interval]))
```

Successes

```
sum(rate(upf_session_db_update_request_success[$rate_interval]))
```

Failures

```
sum(rate(upf_session_db_update_request_failure[$rate_interval]))
```

Total UPF Session Update Request result %

Description

Total UPF Session Update Request result %

Expression

Success

```
((sum(rate(upf_session_db_update_request_success[$rate_interval])))/(sum(rate(upf_session_db_update_request[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_db_update_request_failure[$rate_interval])))/(sum(rate(upf_session_db_update_request[$rate_interval]))))*100
```

Current Total UPF Session Update Request success rate

Description

Current Total UPF Session Update Request success rate

Expression

```
(sum(rate(upf_session_db_update_request_success[$rate_interval])) OR on()  
vector(0)) / sum(rate(upf_session_db_update_request[$rate_interval]))
```

DPI Information Panels

L4 Protocol Statistics - bytes

Description

L4 Protocol Statistics - bytes

Expression

Total - {{protocol_name}}

```
sum by (protocol_name) (rate(protocol_total_bytes{pod=~"$pod"}[$rate_interval]))
```

Uplink - {{protocol_name}}

```
sum by (protocol_name) (rate(protocol_uplink_bytes{pod=~"$pod"}[$rate_interval]))
```

Downlink - {{protocol_name}}

```
sum by (protocol_name) (rate(protocol_downlink_bytes{pod=~"$pod"}  
[$rate_interval]))
```

L7 Protocol Statistics - bytes

Description

L7 Protocol Statistics - bytes

Expression

Total - {{application_name}}

```
sum by (application_name) (rate(application_total_bytes{pod=~"$pod"}
[$rate_interval]))
```

Uplink - {{application_name}}

```
sum by (application_name) (rate(application_uplink_bytes{pod=~"$pod"}
[$rate_interval]))
```

Downlink - {{application_name}}

```
sum by (application_name) (rate(application_downlink_bytes{pod=~"$pod"}
[$rate_interval]))
```

UPF NAT Information Panels

NAT Connections

Description

NAT Connections

Expression**Created**

```
sum(rate(total_nat_connections_created{}[$rate_interval]))
```

Failed to create - NAT Pool not found

```
sum(rate(connection_create_failure_natpool_not_found{}[$rate_interval]))
```

Failed to create - NAT Pool pending delete

```
sum(rate(connection_create_failure_natpool_pending_delete{}[$rate_interval]))
```

Failed to create - NAT IP Pool not configured

```
sum(rate(connection_create_failure_nat_ip_pool_not_configured{}[$rate_interval]))
```

Failed to create - Exceeded per-session NAT quota

```
sum(rate(connection_create_failure_exceeded_per_session_nat_connections_quota{}[$rate_interval]))
```

Failed to create - NAT Port allocation failed

```
sum(rate(connection_create_failure_nat_port_allocation_failures{}[$rate_interval]))
```

NAT Port Allocations**Description**

NAT Port Allocations

Expression**Allocated**

```
sum(rate(nat_port_allocation_success{}[$rate_interval]))
```

De-allocated

```
sum(rate(nat_port_deallocation_success{}[$rate_interval]))
```

Failed to allocate - Internal Error

```
sum(rate(nat_port_allocation_failure_internal_error{}[$rate_interval]))
```

Failed to allocate - Resource Exhaustion

```
sum(rate(nat_port_allocation_failure_resource_exhaustion{}[$rate_interval]))
```

Failed to allocate - Unsupported Protocol

```
sum(rate(nat_port_allocation_failure_unsupported_protocol{}[$rate_interval]))
```

Failed to de-allocate - Internal Error

```
sum(rate(nat_port_deallocation_failure_internal_error{}[$rate_interval]))
```

Failed to de-allocate - Unsupported Protocol

```
sum(rate(nat_port_deallocation_failure_unsupported_protocol{}[$rate_interval]))
```

NAT Connections from NAT-Pool

Description

NAT Connections from NAT-Pool

Expression

Created

```
sum(rate(total_natpool_connections_created{}[$rate_interval]))
```

Deleted

```
sum(rate(total_natpool_connections_deleted{}[$rate_interval]))
```

Failed to create - Port allocation failed

```
sum(rate(natpool_connection_create_failure_nat_port_allocation_failures{}[$rate_interval]))
```

Failed to create - NAT IP Pool not configured

```
sum(rate(natpool_connection_create_failure_nat_ip_pool_not_configured{  
[$rate_interval]))
```

Failed to create - Exceeded per-session NAT quota

```
sum(rate(natpool_connection_create_failure_exceeded_per_session_nat_connections_quota{  
[$rate_interval]))
```

Errors Panels

Session Creation Requests error rates by cause

Description

Rate of Session Creation Requests errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(upf_session_creation_request_failure{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Modification Requests error rates by cause

Description

Rate of Session Modification Requests errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(upf_session_modify_request_failure{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Deletion Requests error rates by cause

Description

Rate of Session Deletion Requests errors by cause

Expression

{{cause}}

```
sum by (cause) (rate(upf_session_delete_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

UPF Overview

Description

High level information about the operation of the UPF NF

UPF Sessions Panels

Session Creation Requests rates

Description

Rate of Session Creation Requests requests

Expression

Attempts

```
sum(rate(upf_session_creation_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Successes

```
sum(rate(upf_session_creation_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Failures

```
sum(rate(upf_session_creation_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Creation Requests result %

Description

Session Creation Requests result %

Expression**Success**

```
((sum(rate(upf_session_creation_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_creation_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_creation_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_creation_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Current Session Creation Requests success rate

Description

Current Session Creation Requests success rate

Expression

```
(sum(rate(upf_session_creation_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval])) OR on()
vector(0)) / sum(rate(upf_session_creation_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Modification Requests rates

Description

Rate of Session Modification Requests requests

Expression**Attempts**

```
sum(rate(upf_session_modify_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Successes

```
sum(rate(upf_session_modify_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Failures

```
sum(rate(upf_session_modify_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Modification Requests result %

Description

Session Modification Requests result %

Expression

Success

```
((sum(rate(upf_session_modify_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_modify_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_modify_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_modify_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Current Session Modification Requests success rate

Description

Current Session Modification Requests success rate

Expression

```
(sum(rate(upf_session_modify_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval])) OR on()
vector(0)) / sum(rate(upf_session_modify_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Deletion Requests rates

Description

Rate of Session Deletion Requests requests

Expression

Attempts

```
sum(rate(upf_session_delete_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Successes

```
sum(rate(upf_session_delete_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Failures

```
sum(rate(upf_session_delete_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Deletion Requests result %

Description

Session Deletion Requests result %

Expression

Success

```
((sum(rate(upf_session_delete_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_delete_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_delete_request_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_delete_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Current Session Deletion Requests success rate

Description

Current Session Deletion Requests success rate

Expression

```
(sum(rate(upf_session_delete_request_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval])) OR on()
vector(0)) / sum(rate(upf_session_delete_request{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Report Responses rates

Description

Rate of Session Report Responses requests

Expression

Attempts

```
sum(rate(upf_session_report_response{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Successes

```
sum(rate(upf_session_report_response_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Failures

```
sum(rate(upf_session_report_response_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Session Report Responses result %

Description

Session Report Responses result %

Expression

Success

```
((sum(rate(upf_session_report_response_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_report_response{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Failure

```
((sum(rate(upf_session_report_response_failure{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}
[$rate_interval])))/(sum(rate(upf_session_report_response{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))))*100
```

Current Session Report Responses success rate

Description

Current Session Report Responses success rate

Expression

```
(sum(rate(upf_session_report_response_success{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval])) OR on()
```

```
vector(0)) / sum(rate(upf_session_report_response{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

UPF Network Information Panels

UPF Uplink - packets

Description

Rate of UPF Uplink - packets

Expression

Received

```
sum(rate(upf_session_uplink_packets_received{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Transmitted

```
sum(rate(upf_session_uplink_packets_transmitted{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Discarded

```
sum(rate(upf_session_uplink_packets_discarded{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

UPF Uplink - bytes

Description

Rate of UPF Uplink - bytes

Expression

Received

```
sum(rate(upf_session_uplink_bytes_received{net_ctxt=~"$net_ctxt",  
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Transmitted

```
sum(rate(upf_session_uplink_bytes_transmitted{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Discarded

```
sum(rate(upf_session_uplink_bytes_discarded{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

UPF Downlink - packets

Description

Rate of UPF Downlink - packets

Expression**Received**

```
sum(rate(upf_session_downlink_packets_received{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Transmitted

```
sum(rate(upf_session_downlink_packets_transmitted{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Discarded

```
sum(rate(upf_session_downlink_packets_discarded{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

UPF Downlink - bytes

Description

Rate of UPF Downlink - bytes

Expression

Received

```
sum(rate(upf_session_downlink_bytes_received{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Transmitted

```
sum(rate(upf_session_downlink_bytes_transmitted{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

Discarded

```
sum(rate(upf_session_downlink_bytes_discarded{net_ctxt=~"$net_ctxt",
remote_node_id=~"$remote_node_id", pod=~"$pod"}[$rate_interval]))
```

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Summary

Dashboard Name	Panel Name
URM DB Dashboard	Create Session Record Requests
URM DB Dashboard	Create Session Record Responses
URM DB Dashboard	Read and Lock Session Record Requests
URM DB Dashboard	Read and Lock Session Record Responses
URM DB Dashboard	Delete Session Record Requests
URM DB Dashboard	Delete Session Record Responses

Dashboard Name	Panel Name
URM DB Dashboard	Create Deleted Pool Requests
URM DB Dashboard	Create Deleted Pool Responses
URM DB Dashboard	Read Session Record Requests
URM DB Dashboard	Read Session Record Responses
URM DB Dashboard	Remove Deleted Pool Requests
URM DB Dashboard	Remove Deleted Pool Responses
URM DB Dashboard	Update Deleted Pool Requests
URM DB Dashboard	Update Deleted Pool Responses
URM DB Dashboard	Get Sessions From Deleted Pool Requests
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URM DB Dashboard	Audit Session Record Requests
URM DB Dashboard	Audit Session Record Responses
URM DB Dashboard	Inband TEID Record Requests
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URM Dashboard	URM Resource Requests
URM Dashboard	URM Resource Responses
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Dashboard Name	Panel Name
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Dashboard Name	Panel Name
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URM Flexblock Dashboard	Resource Pool Create Requests
URM Flexblock Dashboard	Resource Pool Create Responses
URM Flexblock Dashboard	Resource Type Create Requests
URM Flexblock Dashboard	Resource Type Create Responses
URM Flexblock Dashboard	Resource Type Delete Requests
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URM Resource Pool Dashboard	TEID Resource Allocation Successes
URM Resource Pool Dashboard	TEID Resource Allocation Failures
URM Resource Pool Dashboard	TEID Resource Release Successes
URM Resource Pool Dashboard	TEID Resource Release Failures

URM DB Dashboard

Description

URM DB Dashboard

Panels

Create Session Record Requests

Description

Create session record request sent to CRUD DB

Expression

```
sum(urm_db_create_session_record_req{method=~"$method"})
```

Create Session Record Responses

Description

Create session record response received

Expression

```
sum(urm_db_create_session_record_rsp{statusCode=~"$statuscode"})
```

Read and Lock Session Record Requests

Description

Read and Lock session record request sent to CRUD DB

Expression

```
sum(urm_db_read_and_lock_session_record_req{method=~"$method"})
```

Read and Lock Session Record Responses

Description

Read and Lock session record response received

Expression

```
sum(urm_db_read_and_lock_session_record_rsp{statusCode=~"$statuscode"})
```

Delete Session Record Requests

Description

Delete session record request sent to CRUD DB

Expression

```
sum(urm_db_delete_session_record_req{method=~"$method"})
```

Delete Session Record Responses**Description**

Delete session record response received

Expression

```
sum(urm_db_delete_session_record_rsp{statusCode=~"$statusCode"})
```

Create Deleted Pool Requests**Description**

Create and Lock deleted pool request sent to CRUD DB

Expression

```
sum(urm_db_create_deleted_pool_req{method=~"$method"})
```

Create Deleted Pool Responses**Description**

Create and Lock deleted pool response received

Expression

```
sum(urm_db_create_deleted_pool_rsp{statusCode=~"$statusCode"})
```

Read Session Record Requests**Description**

Read session record request sent to CRUD DB

Expression

```
sum(urm_db_read_session_record_req{method=~"$method"})
```

Read Session Record Responses

Description

Read session record response received

Expression

```
sum(urm_db_read_session_record_rsp{statusCode=~"$statuscode"})
```

Remove Deleted Pool Requests

Description

Remove deleted pool request sent to CRUD DB

Expression

```
sum(urm_db_remove_deleted_pool_req{method=~"$method"})
```

Remove Deleted Pool Responses

Description

Remove deleted pool response received

Expression

```
sum(urm_db_remove_deleted_pool_rsp{statusCode=~"$statuscode"})
```

Update Deleted Pool Requests

Description

Update deleted pool request sent to CRUD DB

Expression

```
sum(urm_db_update_deleted_pool_req{method=~"$method"})
```

Update Deleted Pool Responses

Description

Update deleted pool response received

Expression

```
sum(urm_db_update_deleted_pool_rsp{statusCode=~"$statusCode"})
```

Get Sessions From Deleted Pool Requests

Description

Get Sessions from deleted pool request sent to CRUD DB

Expression

```
sum(urm_db_get_deleted_pool_sessions_req{method=~"$method"})
```

Get Sessions From Deleted Pool Responses

Description

Get Sessions from Deleted pool response

Expression

```
sum(urm_db_get_deleted_pool_sessions_rsp{statusCode=~"$statusCode"})
```

Audit Session Record Requests

Description

Audit session record request sent to CRUD DB

Expression

```
sum(urm_db_audit_session_record_req{method=~"$method"})
```

Audit Session Record Responses

Description

Audit session record response received

Expression

```
sum(urm_db_audit_session_record_rsp{statusCode=~"$statusCode"})
```

Inband TEID Record Requests

Description

Read session record based on inband teid request sent to CRUD DB

Expression

```
sum(urm_db_read_inband_teid_session_record_req{method=~"$method"})
```

Inband TEID Record Responses

Description

Read session record based on inband teid response received

Expression

```
sum(urm_db_read_inband_teid_session_record_rsp{method=~"$method"})
```

URM DTM Dashboard

Description

URM DTM Dashboard

Panels

Start Timer Requests

Description

Start timer request sent to DTM

Expression

```
sum(urm_dtm_start_timer_req{method=~"$method"})
```

Start Timer Responses

Description

Start timer response received

Expression

```
sum(urm_dtm_start_timer_rsp{statusCode=~"$statusCode"})
```

Stop Timer Requests

Description

Stop timer request sent to DTM

Expression

```
sum(urm_dtm_stop_timer_req{method=~"$method"})
```

Stop Timer Responses

Description

Stop timer response received

Expression

```
sum(urm_dtm_stop_timer_rsp{statusCode=~"$statusCode"})
```

Update Timer Requests

Description

Update timer request sent to DTM

Expression

```
sum(urm_dtm_update_timer_req{method=~"$method"})
```

Update Timer Responses

Description

Update timer response received

Expression

```
sum(urm_dtm_update_timer_rsp{statusCode=~"$statuscode"})
```

Delete Pool Timer Expiry Notifications

Description

Delete Pool timer expiry notification from DTM

Expression

```
sum(urm_dtm_delete_pool_timer_expiry_notification{method=~"$method"})
```

Delete Pool Timer Expiry Notification ACKs

Description

Delete Pool timer notification ack

Expression

```
sum(urm_dtm_delete_pool_timer_expiry_notification_ack{statusCode=~"$statuscode"})
```

Unsolicited RA Timer Expiry Notifications

Description

UnSolicited RA timer expiry notification from DTM

Expression

```
sum(urm_dtm_unsolicited_ra_timer_expiry_notification{method=~"$method"})
```

Unsolicited RA Timer Expiry Notification ACKs

Description

UnSolicited RA timer notification ack

Expression

```
sum(urm_dtm_unsolicited_ra_timer_expiry_notification_ack{statusCode=~"$statuscode"})
```

URM Dashboard

Description

URM Dashboard

Panels

URM Resource Requests

Description

Resource requests received by urm

Expression

```
sum(urm_resources_req{method=~"$method",dnn=~"$dnn",slice=~"$slice"})
```

URM Resource Responses

Description

Resource responses sent by urm

Expression

```
sum(urm_resources_rsp{method=~"$method", dnn=~"$dnn", slice=~"$slice",  
upfId=~"$upfid", statusCode=~"$statusCode", cause=~"$cause"})
```

URM Status Requests

Description

Status request statistics

Expression

```
sum(urm_status_req{method=~"$method"})
```

URM Status Responses

Description

Status response statistics

Expression

```
sum(urm_status_rsp{method=~"$method", statusCode=~"$statusCode"})
```

URM User Planes Requests

Description

Configured user planes requests statistics

Expression

```
sum(urm_user_planes_req{method=~"$method"})
```

URM User Plane Responses

Description

Configured user planes response statistics

Expression

```
sum(urm_user_planes_rsp{method=~"$method", statusCode=~"$statusCode"})
```

Session Release Notification

Description

urm to smfcc session release notification

Expression

```
sum(urm_smfcc_release_session_notify{cause=~"$cause"})
```

Acknowledgement for Session Release Notification

Description

acknowledgement from smfcc for session release notification

Expression

```
sum(urm_smfcc_release_session_notify_ack{statusCode=~"$statusCode"})
```

Inband GTPU Data Indication

Description

inband gtpu data indication statistics

Expression

```
sum(urm_inband_data_ind{method=~"$method"})
```

Inband GTPU Data Indication Responses

Description

inband gtpu data indication response statistics

Expression

```
sum(urm_inband_data_ind_rsp{statusCode=~"$statusCode"})
```

Router Solicit Received

Description

router solicit received statistics

Expression

```
sum(urm_router_solicit_received{method=~"$method"})
```

Router Advertisements Sent

Description

router advertisement sent statistics

Expression

```
sum(urm_router_advertisement_sent{method=~"$method"})
```

Unsolicited RA Trigger Indications

Description

unsolicited ra trigger indication statistics

Expression

```
sum(urm_unsolicited_ra_trigger_ind{method=~"$method"})
```

Unsolicited RA Trigger Indication Responses

Description

unsolicited ra trigger indication response statistics

Expression

```
sum(urm_unsolicited_ra_trigger_ind_rsp{statusCode=~"$statusCode"})
```

URM Resource Allocation Rate per Pod

Description

URM Resource Allocation Rate per Pod

Expression

URM Resource Allocation Attempt-{{pod}}

```
round(sum(rate(urm_resources_req{method=~"$method",dnn=~"$dnn",slice=~"$slice"}[1m])) by (pod),0.0001)
```

URM Resource Allocation Stats per Pod

Description

URM Resource Allocation Stats per Pod

Expression

URM Resource Allocation Attempt-{{pod}}

```
sum(urm_resources_req{method=~"$method",dnn=~"$dnn",slice=~"$slice"}) by (pod)
```

URM ETCD Dashboard

Description

URM ETCD Dashboard

Panels

IP-POOL Table Watch Requests

Description

ip_pool_watch request sent to ETCD DB

Expression

```
sum(urm_etcd_ip_pool_table_watch_req{})
```

IP-POOL Table Watch Responses

Description

IP Pool Table Watch Response

Expression

```
sum(urm_etcd_ip_pool_table_watch_rsp{statusCode=~"$statuscode"})
```

IP-POOL Table Range Requests

Description

IP Pool Range request sent to ETCD DB

Expression

```
sum(urm_etcd_ip_pool_table_range_req{})
```

IP-POOL Table Range Responses

Description

IP Pool Table Range Response

Expression

```
sum(urm_etcd_ip_pool_table_range_rsp{statusCode=~"$statuscode"})
```

UP Resource Info Table Watch Requests

Description

UP Resource Info Table watch request sent to ETCD DB

Expression

```
sum(urm_etcd_up_rsc_info_table_watch_req{})
```

UP Resource Info Table Watch Responses

Description

UP Resource Info Table Watch Response

Expression

```
sum(urm_etcd_up_rsc_info_table_watch_rsp{statusCode=~"$statuscode"})
```

UP Resource Info Table Range Requests

Description

UP Resource Info Table Range request sent to ETCD DB

Expression

```
sum(urm_etcd_up_rsc_info_table_range_req{})
```

UP Resource Info Table Range Responses

Description

UP Resource Info Table Range Response

Expression

```
sum(urm_etcd_up_rsc_info_table_range_rsp{statusCode=~"$statuscode"})
```

PFCP Association Table Watch Requests

Description

Pfcp Association Table watch request sent to ETCD DB

Expression

```
sum(urm_etcd_pfcg_association_table_watch_req{})
```

PFCP Association Table Watch Responses

Description

Pfcg Association Table Watch Response

Expression

```
sum(urm_etcd_pfcg_association_table_watch_rsp{statusCode=~"$statusCode"})
```

IP-POOL Table Watch Events

Description

IP Pool Table Table Watch Event

Expression

```
sum(urm_etcd_ip_pool_table_watch_event{upfId=~"$upfid", event=~"$event"})
```

UP Resource Info Table Watch Events

Description

UP Resource Info Table Watch Event

Expression

```
sum(urm_etcd_up_rsc_info_table_watch_event{upfId=~"$upfid", event=~"$event"})
```

PFCP Association Table Watch Events

Description

Pfcg Association Table Watch Event

Expression


```
sum(urm_etcd_pfcg_association_table_watch_event{upfId=~"$upfid", event=~"$event"})
```

URM Flexblock Dashboard

Description

URM Flexblock Dashboard

Panels

Flexblock Resource Allocation Requests

Description

urm flexblock resource allocation request

Expression

```
sum(urm_fb_resource_allocation_req{poolId=~"$poolid"})
```

Flexblock Resource Allocation Responses

Description

urm flexblock resource allocation rsp

Expression

```
sum(urm_fb_resource_allocation_rsp{poolId=~"$poolid", statusCode=~"$statusCode"})
```

Flexblock Resource Release Requests

Description

urm flexblock resource release request

Expression

```
sum(urm_fb_resource_release_req{poolId=~"$poolid"})
```

Flexblock Resource Release Responses

Description

urm flexblock resource release rsp

Expression

```
sum(urm_fb_resource_release_rsp{poolId=~"$poolid", statusCode=~"$statusCode"})
```

Get Resource Pool Info Requests

Description

urm flexblock get resource pool info request

Expression

```
sum(urm_fb_get_resource_pool_info_req{poolId=~"$poolid"})
```

Get Resource Pool Info Responses

Description

urm flexblock get resource pool info rsp

Expression

```
sum(urm_fb_get_resource_pool_info_rsp{poolId=~"$poolid",  
statusCode=~"$statusCode"})
```

Resource Pool Create Requests

Description

urm flexblock unaffined resource pool create request

Expression

```
sum(urm_fb_unaffined_resource_pool_create_req{poolId=~"$poolid"})
```

Resource Pool Create Responses

Description

urm flexblock unaffined resource pool create response

Expression

```
sum(urm_fb_unaffined_resource_pool_create_rsp{poolId=~"$poolid",  
statusCode=~"$statusCode"})
```

Resource Type Create Requests

Description

urm flexblock unaffined resource type create request

Expression

```
sum(urm_fb_unaffined_resource_type_create_req{poolId=~"$poolid"})
```

Resource Type Create Responses

Description

urm flexblock unaffined resource type create response

Expression

```
sum(urm_fb_unaffined_resource_type_create_rsp{poolId=~"$poolid",  
statusCode=~"$statusCode"})
```

Resource Type Delete Requests

Description

urm flexblock unaffined resource type delete request

Expression

```
sum(urm_fb_unaffined_resource_type_delete_req{poolId=~"$poolid"})
```

Resource Type Delete Responses

Description

urm flexblock unaffined resource type delete response

Expression

```
sum(urm_fb_unaffined_resource_type_delete_rsp{poolId=~"$poolid",  
statusCode=~"$statusCode"})
```

Bulk Resource Allocation Requests

Description

urm flexblock bulk/cache resource allocation request

Expression

```
sum(urm_fb_bulk_resource_allocation_req{poolId=~"$poolid"})
```

Bulk Resource Allocation Responses

Description

urm flexblock bulk/cache resource allocation rsp

Expression

```
sum(urm_fb_bulk_resource_allocation_rsp{poolId=~"$poolid",  
statusCode=~"$statusCode"})
```

Bulk Resource Release Requests

Description

urm flexblock bulk/cache resource release request

Expression

```
sum(urm_fb_bulk_resource_release_req{poolId=~"$poolid"})
```

Bulk Resource Release Responses

Description

urm flexblock bulk/cache resource release rsp

Expression

```
sum(urm_fb_bulk_resource_release_rsp{poolId=~"$poolid",
statusCode=~"$statusCode"})
```

Audit Request Notifications

Description

urm audit request notification from flexblock

Expression

```
sum(urm_fb_audit_notify_count{poolId=~"$poolid"})
```

URM Resource Pool Dashboard

Description

URM Resource Pool Dashboard

Panels

IPv4 Resource Allocation Successes

Description

urm ipv4 resource allocation success

Expression

```
sum(urm_ipv4_resource_allocation_success{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv4 Resource Allocation Failures

Description

urm ipv4 resource allocation failure

Expression

```
sum(urm_ipv4_resource_allocation_failure{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv4 Resource Release Successes

Description

urm ipv4 resource release success

Expression

```
sum(urm_ipv4_resource_release_success{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv4 Resource Release Failures

Description

urm ipv4 resource release failure

Expression

```
sum(urm_ipv4_resource_release_failure{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv6 Resource Allocation Successes

Description

urm ipv6 resource allocation success

Expression

```
sum(urm_ipv6_resource_allocation_success{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv6 Resource Allocation Failures

Description

urm ipv6 resource allocation failure

Expression

```
sum(urm_ipv6_resource_allocation_failure{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv6 Resource Release Successes

Description

urm ipv6 resource release success

Expression

```
sum(urm_ipv6_resource_release_success{upfId=~"$upfid", poolId=~"$poolid"})
```

IPv6 Resource Release Failures

Description

urm ipv6 resource release failure

Expression

```
sum(urm_ipv6_resource_release_failure{upfId=~"$upfid", poolId=~"$poolid"})
```

TEID Resource Allocation Successes

Description

urm teid resource allocation success

Expression

```
sum(urm_teid_resource_allocation_success{upfId=~"$upfid", poolId=~"$poolid"})
```

TEID Resource Allocation Failures

Description

urm teid resource allocation failure

Expression

```
sum(urm_teid_resource_allocation_failure{upfId=~"$upfid", poolId=~"$poolid"})
```

TEID Resource Release Successes

Description

urm teid resource release success

Expression

```
sum(urm_teid_resource_release_success{upfId=~"$upfid", poolId=~"$poolid"})
```

TEID Resource Release Failures

Description

urm teid resource release failure

Expression

```
sum(urm_teid_resource_release_failure{upfId=~"$upfid", poolId=~"$poolid"})
```